Continuum Structural Complexity: Target-Aligned Source Algebra, Physical Expenditure, and Singular Reachability

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Abstract

A public continuum presentation consists of a weak PDE or flow, its datum and boundary conditions, a rational approximation grammar, its continuation and native-operation syntax, a canonical weak test core, and the formal physical carrier identity. No target-relative estimate, compactness statement, rigidity theorem, or continuation conclusion is input. The presentation generates an Archimedean ordered cylinder algebra. Its state space contains ordinary solutions, every approximation limit, and the first graph value at which an equation, carrier, chart, trace, or covariance relation fails.

The continuum extension preserves the five source families of Part I. They are $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}$, and Bdry. Limits, concentrations, oscillations, defect measures, tails, traces, and blow-up germs retain the ancestry of the records that generate them. A primitive-role theorem partitions every current path of the public weak expression graph into canonical bulk form/directed, quotient, lift, and boundary roles. The free totalized constructor algebra has a unique provenance morphism into the five-letter support semilattice. Thus every target-active atomic current occurrence belongs to one of the $2^5 - 1 = 31$ possible nonempty provenance cells, while a total current retains their joint vector, including every nonordinary graph value and sourcewise limit. The residual is taken only after this sourcewise completion and is exactly the common kernel of the prospective target functionals. A target-visible residual source is therefore impossible. Finite-cylinder target shells carry an exact derived Stieltjes-current identity. Contact forces unit shell progress, hence a positive contribution from at least one inherited source on every sufficiently small pre-contact shell. A canonical disjoint cascade then converts finite total physical expenditure into either divergent source price, which excludes contact, or a zero-price target-aligned successor path.

A distinguished nonreversible subfamily applies a Wick transform to the self-adjoint Abs quotient field of an existing Part I compensated record and leaves its real Geom and authorized Caus fields unchanged. The transformed causal bridge produces an exact Feshbach self-energy, a Mori–Zwanzig memory kernel, and a completed-square structural action. Its positive response kernel transfers the effective-dimension, Rademacher, PAC–Bayes, sequential, control, and continuum-learning bounds of Part I. Empirical trace terms use the finite Part I evaluation carriers; the population effective dimension is the extended supremum of those finite compressions. Structure-preserving likelihood, inverse-response kernel ridge, and action-safe fitted Bellman/Gibbs losses train the dynamics, unrestricted bridge, and RL branches, respectively. The inverse-response norm is the exact minimum energy of a latent response lift. The Part I exact-cell Rademacher union and learned-operator cover, or the cell-mixture PAC–Bayes divergence and coefficient-selection charge, select the model family without an additional simplicity criterion. The learner restricts the Part I saturated usefulness profile to records at a requested risk tolerance, then returns the Pareto-minimal points of its strict inverse at the inherited visibility threshold. Minimum-capacity covering readout remains an inner coordinate within each fixed geometry; it does not replace saturated-profile selection.

The remaining path problem is evaluated by an explicit ordered calculus. Every identity is first restricted to the same-origin target/source stratum that invoked it and decomposed into signed five-channel remainders, carrier coboundary, admissible physical production, exterior input, target-null current, and total-graph defects. No equation-wide identity has an independent reachability consequence. For each source support and rational target floor, a quadratic module contains the weak equation, signed carrier balance, total graph relations, same-origin successor relations, nonnegative physical production, and shift coboundaries. Exactly one algebraic alternative holds: either a finite rational identity represents -1 in the ordered cone and excludes that target-aligned cell, or an Archimedean separation theorem produces a positive shift-invariant functional satisfying all retained relations. A nested cylinder-moment hierarchy enumerates the finite identities and reconstructs the positive functional from compatible finite moments. Hamiltonian resolvents, signed carrier cohomology, entropy cocycles, commutator squares, and compactness–rigidity formulae are consequence generators inside this calculus, not additional hypotheses. A complementary positive functional is an equality successor: it re-enters the target-cyclic compiler generated by the selected shell covector. The compiler restricts every operation to the same-origin target cell. Projection-first saturation proves that quotienting by the common target kernel commutes with every constructor; a failed descent becomes a projected graph defect. The resulting target-cyclic spine assembles one feeding–storage current before a canonical zero/signed-dyadic/graph-boundary polarity split. A new coordinate increases resolved-prefix rank; an equivalent positive cell is followed by its induced first-return law. On each recurrent cell, a primal–dual circulation hierarchy computes the sharp physical price of unit target return. Positive price produces a carrier-potential identity; zero price reconstructs the invariant circulation that generates the next target-cyclic refinement. Adjoint transport, constraint descent, defect polarization, commutator production, and covariant scale valuation are internal operations of this one queue. Its contact-seeded outputs are divergent physical charge, target-nullity, or a joint-saturated five-source witness for a nonzero saturated prospective source cell. Every cofinal refinement is cumulatively joined, closed under mixed operations, and reprojected before this last output. Origin polarity prevents that hypothetical witness from being recycled as singular existence. An independently generated ordinary contacting orbit is a separate constructive output. A formal model, equality kernel, failed analytic property, or equation-wide estimate is an internal coordinate rather than a terminal.

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Introduction

Let u be an ordinary maximal branch of a fixed continuum evolution and let Σ be a target in its state or germ space. The central question is whether the equation can reach Σ while spending only a finite amount of the physical quantity controlled by its evolution law. Each orbit carries one nonnegative countably additive expenditure measure μ_u on a generated event space $\mathfrak{Q}_u$. The finite-budget stratum is

$$\mu_u(\mathfrak{Q}_u) \leq B_{\mathrm{phys}} < \infty.$$

The public carrier identity selects, among action, dissipation, entropy production, forcing work, and boundary flux terms, the mutually compatible positive variations that constitute this one measure. Signed internal currents are glued first, and only carrier-dominated target-local production may spend it. Energy and conservation identities then bound its total mass before the ceiling B_{phys} is set.

The source alphabet, ancestry discipline, saturation rule, and target-relative residual extend the structural algebra of Part I [2]. The continuum construction below derives its physical budget, source-preserving limit closure, prospective contact identity, signed carrier calculus, successor recurrence, Hamiltonian formulation, and saturation equations within this paper. The remaining citations identify standard analytic results used at explicitly named steps. These references and Part I constitute the complete logical dependency set.

The structural source alphabet records how target-directed behavior is generated. The native dissipative or reversible mechanism is Geom; the fixed directed or nonreversible transport is Caus; quotient and symmetry reduction are Abs; charts, renormalizations, and augmented state variables are Lift; and boundary, terminal, killing, and exterior-flux records are Bdry. A composite continuum object may carry several labels, but every elementary source belongs to this alphabet.

The completion is sourcewise. A limit object is placed in the closure of the cell that generated it. Thus a concentration measure retains the ancestry of the concentrating quantity, a blow-up profile retains the ancestry of the physical orbit together with its $\mathsf{Abs}/\mathsf{Lift}$ chart, and a terminal tail trace acquires Bdry ancestry. Every source cell has an exact structural row value; its complementary value remains in the same cell and does not create an additional continuum source.

The target-relative residual is correspondingly strict. Let $\mathscr{N}_\Sigma$ be the common kernel of the target shell derivatives, target fluxes, and other declared prospective contact functionals. The residual coordinate is contained in $\mathscr{N}_\Sigma$. Hence

$$\boxed{\text{target-visible} \Longrightarrow \text{ancestry in } \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}; \qquad J_{\text{res}} \Longrightarrow \text{target-null.}}$$

Every target-visible limit therefore retains ancestry in one of the five source families.

The central target-relative question is therefore not whether the entire solution is large, but which source current contributes positively to the formation of the singularity. For a normalized shell crossing I_r, set

$$Q_{c,r} := \int_{I_r} \langle D\Phi_r(u_t), J_t^c \rangle \, dt.$$

The compiled current identity gives the exact pre-contact identity on the finite-value ordinary shell branch

$$1 = \sum_{c \in \mathfrak{C}} Q_{c,r}, \qquad \sum_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq 1, \qquad \max_{c \in \mathfrak{C}} (Q_{c,r})_+ \geq \frac{1}{5}.$$

On an infinite-variation value, the joint shell graph instead retains a source owner o with $\overline{Q}^o(1) \geq 1/5$, without forming an extended signed sum. The residual source term vanishes. Thus contact itself supplies the unit progress and proves the existence of target-aligned structure. If all source currents vanish in the target-active quotient, their magnitude is irrelevant: every shell observable is constant along the orbit, so contact is excluded without compactness, a physical-price gap, or a source-transverse estimate.

The same accounting is stable under every generated continuum composition, not by separate estimates for time, scale, nonlinearities, or limits, but by the expanded-derivation rule inherited from Part I. Before a residual is formed, the complete derivation DAG of the entire same-origin prefix is joined and closed under all mixed public operations. Every physical quantity remains attached to its owner and domination graph. Normalized values retain their numerator, denominator, boundary cell, and unnormalized target cocycle. The target projection on these increasing joint algebras has orthogonal increments d_n, so

$$\|P_\infty y_\Phi\|_2^2 = \sum_n \|d_n\|_2^2.$$

For actual crossings, the canonical source-current frontier gives a two-record contrast of squared norm $1/4$. The finite occurrence direct sum therefore satisfies the exact identities

$$\frac{N}{4} = \sum_{j<N} \sum_\ell \Delta_{j,\ell}, \qquad N = \sum_{j<N} \sum_\ell \widehat{\Delta}_{j,\ell}.$$

No increment is discarded for being small. The same N crossings retain the unnormalized cocycle

$$N = \sum_{c \in \mathfrak{C}} Q^c_{\Phi,N}, \qquad \max_c (Q^c_{\Phi,N})_+ \geq N/5.$$

Thus a countable accumulation of locally target-null-looking operations is either genuinely null after joint saturation or reappears as cumulative target-aligned structure with one of the same five ancestries. It cannot become a target-bearing residual at the limit.

This is the root no-emergence theorem. Scale collapse, long-time accumulation, weak defects, oscillation, concentration, recurrent averaging, tail escape, boundary passage, and nonlinear cross terms are constructor instances, not separate loopholes. Individually regular inputs may have a target-aligned joint effect, just as individually uninformative observables may have informative joint fibers in Part I; the joint effect is retained as an existing source cell and cannot be declared free merely because it was absent from each input separately.

The aligned branch is simplified before any size estimate is attempted. A target-null or gauge direction is passed to the generated quotient and never reconsidered. Projection-first saturation generates every descendant inside the target-cyclic spine and retains each failed descent as a graph coordinate. For every surviving total-graph value the algebra forms its positive target-current measure, compact activity lift, and witness-indexed successors. Every subsequent analytic identity is restricted to that cell and compiled into (110) before it can close or charge the branch. The topological sets $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ describe whether an occurrence can sustain one more target-retaining zero-price step. Proof is supplied separately by the ordered cone: a checked finite identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ excludes the active cell, while the complementary order-unit functional is a stationary equality successor satisfying every generated relation at that depth. Its first unprocessed local coordinate is split into zero, signed dyadic, and graph-boundary cells; finite local identities prune target-null cells, while positive-moment cells re-enter source selection. New coordinates increase resolved-prefix rank; recurrent equivalent cells use their induced return maps. The target-aligned recurrent closure compiler performs all of these transitions internally and returns on a contact-seeded execution only divergent physical charge, a target-null path, or a saturated prospective source-profile witness. An independently generated ordinary orbit may separately realize the same structural atom and thereby establish a singular mechanism. An infinite rank chain is first joined with its complete transcript; every unprocessed joint increment re-enters the five-source compiler. Semantic emptiness is never used as a proof object.

The main expenditure argument then uses the same measure μ_u. A singular contact compiler produces infinitely many pairwise disjoint pre-contact events and selects a persistent aligned source cell. A divergent-price value contradicts countable additivity and $\mu_u(\mathfrak{Q}_u) \leq B_{\text{phys}}$. A finite-price value produces vanishing-price windows and activates the ordered successor calculus. It is therefore a reduction step, not a terminal singularity class.

$$\boxed{\begin{gathered}\text{contact} \xrightarrow{\text{normalized shell}} \text{target-aligned source,}\\[1ex] \text{target-aligned source} \xrightarrow{\text{one physical measure}} \begin{cases}\text{divergent expenditure,}\\ \text{zero-price successor path,}\end{cases}\\[1ex] \text{first branch contradicts } B_{\text{phys}}; \qquad Z_{n+1} = \mathcal{V}_\eta(Z_n) \text{ evaluates the second;}\\ \text{a nonpruned fixed atom remains prospective.}\end{gathered}}$$

Hamiltonian, resolvent, commutator, constraint-tail, scale, and entropy identities are functorial refinements generated when their typed expressions occur in the public equation. The symmetric form yields a spectral Hamiltonian and resistance; the full fixed-Caus generator retains its nonnormal current; signed carrier gluing cancels internal transport and exposes scale-feed cohomology; and a covariant entropy cocycle telescopes production across changing charts. A failed identity is lifted as a successor with the same ancestry. These modules operate on the five source families; they neither enlarge the public presentation nor alter the source algebra.

The Wick-rotated compensated-quotient module treats a self-adjoint Abs generator in spectral time and a real Geom generator in dissipative time. The actual selected Caus bridge determines both the Feshbach response and the structural action. The resulting positive response kernel imports the learning theory of Part I without diagonalizing a nonnormal generator, while free, hybrid, and imposed authority remain distinct optimization domains. The unrestricted branch optimizes spectral kernel ridge risk on the zero-waste bridge fiber; the RL branch uses a frozen

Bellman target, safe response compression, and a Gibbs actor calibrated to the same action-valued critic.

Scope discipline. The paper uses exhaustive quantification only to construct the closed five-source algebra: enumeration of tests, total graphs, sourcewise closures, and the common target kernel. Such a construction statement has no reachability conclusion. An analytic identity becomes a reachability fact only after contact has selected a same-origin active branch $\mathfrak{b}$, the identity has been restricted to $X_{\mathfrak{b}}$, and its terms have been placed in the target-local normal form (110). A physical consequence additionally requires the charging rule (98). Thus full-algebra quantifiers provide source exhaustion, whereas every exclusion, price, successor, and continuation inference is target-local and source-resolved. This distinction is formalized in Theorem II.2.18 and Corollary II.2.20.

Part I
Closed Continuum Structural Algebra

I.1 Physical continuum presentations

Definition I.1.1 (Canonical physical continuum presentation)**.** A canonical physical continuum presentation is the public tuple

$$\mathfrak{P}_{\text{phys}} = \big(\mathsf{Eq}, u_0, \mathcal{D}, \mathsf{Appr}, \mathsf{Cont}, \mathsf{Op}, \mathcal{K}\big)$$

with the following data, all transcribed directly from a public textbook or standard reference statement of the initial–boundary value problem. A nonliteral entry below is admissible only when it is target-blind and already part of the standard local theory of the displayed equation; establishing a new estimate or theorem is not part of the specialization datum.

(P1) Eq is the displayed differential equation with its domain, coefficients, boundary conditions, and declared regularity class; u_0 is its datum; and $\mathcal{D} = \{\varphi_n\}_{n\geq 1}$ is the canonical rational smooth test core. A standard local solution theorem may identify an ordinary component of the generated solution graph, but it is not needed to define that graph. Any local theorem used for this identification is target-blind in the sense of Definition I.1.12.

(P2) The *total local-solution graph* contains every maximal path $u : [0, T_*(u)) \to X_{\text{reg}}$ satisfying Eq against every φ_n, with the prescribed datum and physical boundary data. If the fiber over u_0 is empty, its value is the explicit record $\perp_{\text{loc}}$. Nonuniqueness retains all maximal paths. Consequently an empty solution fiber is a form defect and can never be read as target avoidance. The approximation graph below independently tests whether the ordinary fiber is nonempty.

(P3) Appr is the least countable grammar generated by rational Galerkin projections, rational mollifiers, rational implicit time steps, rational domain exhaustions, and the typed gauge or boundary operations in Op. A grammar word is retained with its finite-dimensional or finite-step residual relation against the first finitely many tests and with the signed carrier identity obtained by applying the same word to $\mathcal{K}$. Every compatible grammar word is enumerated; no convergence property or preferred scheme belongs to the presentation.

(P4) Cont is the syntactic extension relation in the declared regularity class: two local paths are related when their restrictions agree. No continuation theorem is supplied. Its ordinary values are actual regular extensions and its graph-boundary values are terminal germs. A terminal germ is stored together with its orbit-origin tag and the cofinal family of restrictions of that same orbit which generates it. In the completed restriction-record space, each member of this cofinal family carries the same origin and terminal-status tags; by definition the family converges to its terminal germ. These tags belong to the retrospective proof record and do not alter the PDE current on the preterminal interval. Thus the continuation graph is closed fiberwise over the orbit origin: two distinct orbits are not identified merely because some evaluation coordinates agree. The singular target is generated from this same-origin graph by Definition III.1.1; it is not an independently chosen favorable subset.

(P5) $\mathsf{Op} = \{O_n\}_{n\geq 1}$ is a countable typed list of the restriction, rational truncation, quotient, chart, augmentation, trace, and boundary operations syntactically available for Eq. An operation acts as a symmetry or gauge only on the ordinary value of its total covariance graph, where it preserves the weak equation, the carrier, and the continuation target. Thus the generated gauge family is the equalizer of these three actions; it is not declared independently.

(P6)

$$\mathfrak{C} = \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}$$

is the inherited source alphabet of Definition I.2.1. Native terms are routed by the sector/gradient compiler and subsequent orbit-record operations acquire their source words by the constructor grammar; this is proved in Theorem I.4.5.

(P7) $\mathcal{K}$ is the native textbook energy, action, or entropy density displayed with the equation before the continuation target is generated. It is not selected after a singular scenario has been inspected. Only its formal multiplier identity—obtained by testing Eq, integration by parts, and cancellation of equal internal faces—belongs to the public presentation. Testing Eq, summing with its native signs, and cancelling common internal faces generates a total signed carrier relation. Positivity, a lower carrier bound, finite exterior input, and finite total expenditure are values of that relation, not presentation data.

(P8) Let $\mathcal{A}_{\mathbb{Q}}(\mathfrak{P})$ be the unital commutative rational cylinder algebra generated by bounded truncations of all weak pairings, continuation coordinates, approximation residuals, carrier coordinates, and input–output coordinates of the typed operations in (P1)–(P7). For each typed operation it also contains graph-defect, oscillation, concentration, and exterior-tail coordinates. Let $M_{\mathfrak{P}}$ be the quadratic module generated by squares, the bounds $1 - e_n^2$, nonnegative defect masses, carrier-production atoms, and the positive and negative forms of every exact rational cylinder identity. The graph-complete state space is

$$X_{\mathrm{gc}} := \{L : \mathcal{A}_{\mathbb{Q}}(\mathfrak{P}) \to \mathbb{R} : L(1) = 1, \quad L(M_{\mathfrak{P}}) \subseteq [0, \infty)\}\,. \tag{1}$$

It carries pointwise convergence on the countable cylinder algebra. Ordinary states and germs embed by evaluation. The notation X_{ev} below means X_{gc}, and

$$d_{\mathrm{ev}}(L_1, L_2) := \sum_{n\geq 1} 2^{-n} |L_1(e_n) - L_2(e_n)| \tag{2}$$

for a fixed enumeration of normalized generators and rational products.

The presentation is therefore fixed by public equation data and a formal multiplier calculation. The specialization step consists only of recording these statements and their public references; it contains no new mathematical argument. It may not contain a compactness theorem, coercive estimate, price lower bound, Liouville statement, target-exclusion lemma, special blow-up theorem, or exact invariant value. Such an insertion changes the equation signature and is rejected by the compiler.

Theorem I.1.2 (Graph-complete compactness and separation)**.** *The module $M_{\mathfrak{P}}$ is Archimedean. Its state space X_{gc} is compact and metrizable. The cylinder coordinates separate its points, and every ordinary state, germ, orbit segment, or approximation record defines an evaluation state in X_{gc}. If $X_{\mathrm{gc}} = \varnothing$, then $-1 \in M_{\mathfrak{P}}$; this finite algebraic inconsistency is the total form-defect value.*

Proof. Every cylinder polynomial contains only finitely many normalized generators. The relations $1 - e_n^2 \in M_{\mathfrak{P}}$, together with closure of a quadratic module under addition and multiplication by squares, give $N \pm a \in M_{\mathfrak{P}}$ for some rational N and every $a \in \mathcal{A}_{\mathbb{Q}}(\mathfrak{P})$. Thus the module is Archimedean. Positivity gives $|L(a)| \le N$ whenever $N \pm a \in M_{\mathfrak{P}}$. The state space is therefore a closed subset of a countable product of compact intervals; it is compact and metrizable. Pointwise equality on the cylinder algebra is equality of states. Ordinary records satisfy every defining graph and positivity relation, so evaluation is positive and normalized.

For the final assertion, the Archimedean separation theorem for quadratic modules says that a proper Archimedean module admits a normalized positive linear functional [13]. Hence an empty state space forces $M_{\mathfrak{P}}$ to be improper, equivalently $-1 \in M_{\mathfrak{P}}$. Membership uses one finite algebraic expression and hence only finitely many presentation relations. □

Definition I.1.3 (Exhaustive approximation tower)**.** An approximation record is a tuple

$$a = (g, N, h, \mathsf{r}_a, \mathsf{k}_a),$$

where g is a word of Appr, N is the number of tested weak coordinates, $h \in \mathbb{Q}_+$ is its resolution, r_a is the vector of equation and operation-graph residuals, and k_a is its exact signed carrier record. Write $a' \preceq a$ when a' is obtained from a by forgetting tests, coarsening resolution, or restricting its time or spatial domain. The exhaustive approximation tower is the countable inverse graph

$$\mathfrak{A}(\mathfrak{P}) := \{a : a \text{ is generated by } \mathsf{Appr}\}$$

with all maps $a \mapsto a'$ for $a' \preceq a$. A compatible tower path is *zero defect* when every fixed residual coordinate tends to zero and the signed carrier coordinates converge in the graph-complete topology. For each enumerated normalized residual $r_k \in [-1, 1]$, the grammar also contains the monotone tail coordinate

$$d_{k,N} := \sup\{|r_k(a)| : a \text{ occurs after refinement level } N\}, \qquad d_k := \inf_N d_{k,N}.$$

Suprema are totalized through bounded rational upper-envelope coordinates. Thus $d_k = \limsup |r_k|$, and failure of convergence to zero is a nonzero closed graph coordinate rather than an unrecorded oscillation.

Theorem I.1.4 (Approximation relaxation and first failure)**.** *Every zero-defect compatible tower path has a graph-complete positive-functional limit satisfying the public weak equation and native carrier relation in every rational test coordinate. This is a relaxed equation state; it is an ordinary solution only after the Dirac-realization gate below. Conversely, every ordinary weak solution embeds in X_{gc} by evaluation and is a zero-defect value of the total weak solution graph. For any compatible tower path which is not zero defect, there is a least operation index whose residual, oscillation, concentration, tail, or carrier-defect coordinate is nonzero. That coordinate is the first-failure record of the path.*

Proof. Compactness gives a convergent subnet, and metrizability gives a convergent subsequence. Fix a rational test coordinate. Compatibility makes that coordinate present on every sufficiently refined record, while zero defect makes its residual tend to zero. Pointwise convergence of states therefore passes the tested weak equation and the signed carrier relation to the limit. Here satisfaction of the public weak equation means satisfaction on the declared separating rational test core. When the conventional formulation uses all smooth tests, the approximation grammar also records the standard distribution-order seminorm bound; on its ordinary graph that bound extends the identity by density, while failure of the bound is its typed form coordinate. An ordinary weak solution satisfies these relations directly, so its evaluation state lies on the zero-defect total graph.

The tail-limsup failure coordinates are countable and nonnegative. If the path is not zero defect, at least one d_k is nonzero. The fixed enumeration therefore has a least nonzero coordinate. Its typed graph already contains its input records and constructor ancestry, which makes it a first-failure record rather than an untyped remainder. □

Lemma I.1.5 (Generator variances determine a Dirac cylinder state)**.** *Let L be a normalized positive functional on the Archimedean real cylinder algebra $\mathcal{A}_{\mathbb{Q}}(\mathfrak{P})$, and let $(e_n)_{n\geq 1}$ be its bounded algebra generators. If*

$$L(e_n^2) = L(e_n)^2 \qquad (n \geq 1),$$

then, for every cylinder polynomial p,

$$L(p(e_1, \ldots, e_m)) = p(L(e_1), \ldots, L(e_m)). \tag{3}$$

Consequently $L(fg) = L(f)L(g)$ for all cylinder polynomials f, g.

Proof. Use the GNS seminorm $\|a\|_L^2 = L(a^2)$. The variance hypothesis gives $\|e_n - L(e_n)1\|_L = 0$. The null space of this seminorm is a left ideal: for every cylinder polynomial b, Archimedean boundedness supplies $C_b > 0$ with $C_b^2 - b^2 \in M_{\mathfrak{P}}$. Indeed, from $C_b \pm b \in M_{\mathfrak{P}}$, the quadratic-module identity

$$2C_b(C_b^2 - b^2) = (C_b - b)(C_b + b)^2 + (C_b + b)(C_b - b)^2$$

gives the assertion after increasing C_b to a positive rational value. Consequently

$$\|ba\|_L^2 = L(b^2a^2) \leq C_b^2 L(a^2) = 0$$

whenever $\|a\|_L = 0$. Therefore, in the GNS quotient, $e_n = L(e_n)1$ for every generator. Substitution through a finite product gives

$$p(e_1, \ldots, e_m) = p(L(e_1), \ldots, L(e_m))1$$

in that quotient. Pairing with the cyclic vector proves (3); applying it to fg gives multiplicativity. □

Theorem I.1.6 (Dirac realization gate)**.** *Let $L \in X_{\mathrm{gc}}$ be a zero-defect limit of one compatible approximation-tower path. Enumerate the real state and path generators by $(e_n)_{n\geq 1}$, and put*

$$\mathrm{Var}_L(e_n) := L(e_n^2) - L(e_n)^2 \geq 0.$$

Exactly one of the following occurs.

(R1) *Some least variance, operation-graph, distribution-order, origin, or continuation coordinate is nonordinary or nonzero. That coordinate, its detecting cylinder word, and its inherited five-source ancestry form the next realization successor.*

(R2) *Every variance vanishes, every public operation graph has its ordinary value, the distribution-order bounds are finite, and the approximation indices have independent origin. Then L is evaluation on one ordinary same-origin weak path satisfying the public carrier law. It is a singular terminal exactly when its ordinary continuation rows cross a cofinal target shell family.*

In particular, a positive functional, invariant law, compatible moment family, or zero-defect relaxed state is not an ordinary or singular realization before this gate.

Proof. Apply Lemma I.1.5; the generator-variance rows make L multiplicative on the entire cylinder algebra. Its linear test coordinates, together with the finite distribution-order bounds, define one distributional path; ordinary values of all typed operation graphs identify every nonlinear and boundary coordinate with the corresponding public operation on that path. Zero approximation residuals give the weak equation and carrier law. The retained origin indices give independence, and the ordinary continuation-shell rows give contact. If any clause fails, countability of the fixed enumeration selects its least coordinate. Total-graph and source-preservation rules make it the successor in (R1). □

Theorem I.1.7 (Approximation–orbit realization criterion)**.** *Let $\mathfrak{a}$ be any graph-complete limit record, positive-functional state, carrier-polar state, or joint-saturated prospective source atom. If there is one compatible path $(a_N)_{N\geq 1}$ in the public approximation tower satisfying*

(D1) *every fixed equation, carrier, initial-data, operation-graph, trace, chart, and continuation residual tends to zero along this same path;*

(D2) *the path evaluations converge to $\mathfrak{a}$ on every rational cylinder coordinate;*

(D3) *the limiting positive functional has zero variance on every real state and path generator; and*

(D4) *every distribution-order and domain coordinate has its ordinary finite value.*

then $\mathfrak{a}$ represents one ordinary public PDE orbit. Conversely, an ordinary orbit claimed as a limit of the public approximation grammar has such a path if and only if that approximation claim satisfies (D1)–(D4)*. An ordinary orbit already present in the total solution graph is realized directly by its evaluation record and need not be assumed approximable by this particular grammar.*

The tower-reconstructed orbit has independent origin precisely when the tower path is generated without a target-contact or noncontinuation premise. It reaches the continuation target precisely when its ordinary continuation coordinates cross every member of the generated cofinal shell family. Thus neither a compatible moment table, a measure on the graph compactification, a polar separator, a nonordinary total-graph value, nor a contact-derived tower is a singular-existence terminal.

Proof. Conditions (D1)–(D2) give the relaxed weak equation and carrier law by Theorem I.1.4. Because the convergence occurs along one compatible path, all coordinates have the same origin; no coordinatewise change of subsequence is permitted. By (D3), Cauchy–Schwarz makes the limit functional multiplicative on the cylinder algebra. Condition (D4) and the ordinary values in (D1) then invoke the Dirac realization gate Theorem I.1.6, reconstructing one distributional path and identifying every nonlinear, boundary, chart, and continuation coordinate with the corresponding public operation. The zero residuals give the public weak equation and initial datum. Origin polarity is inherited from the one tower path, and shell contact is exactly the stated ordinary continuation condition. This proves sufficiency.

If an ordinary orbit is asserted to arise through the approximation grammar, the restrictions of that one asserted approximating path necessarily satisfy (D1)–(D2); multiplicativity of its ordinary evaluation gives (D3), and ordinary domain values give (D4). The reverse implication is the sufficiency just proved. A direct evaluation record from the total solution graph already is the orbit it represents, so no density or convergence property of Appr is inferred from its existence. □

Corollary I.1.8 (No compactification-to-existence promotion)**.** *Every point of the graph-complete compactification that is neither a direct ordinary evaluation from the total solution graph nor reconstructed by a tower satisfying* (D1)–(D4) *remains an internal typed successor. Compactness may be used to exclude a target cell in the larger relaxed space, but a point of that space proves physical attainability only through Theorem I.1.7. In particular, graph completeness proves that no coordinate was forgotten; it does not prove that an arbitrary compatible coordinate family is dynamically attainable.*

Definition I.1.9 (Continuum extension functor)**.** Let $\mathfrak{C}$ send a finite cylinder record to its evaluation state, a finite current to its distributional cylinder pairings, a finite target probe to its pullback on X_{gc}, and a constructor word to the closure of its typed graph. It sends a convergent family to its sourcewise graph-complete closure and sends stopping, restriction, signed addition, composition, and physical-ledger addition to the corresponding closed graph operations.

Theorem I.1.10 (Part-I preservation under continuum completion)**.** *The functor $\mathfrak{C}$ preserves the five source tags, constructor composition, signed sums, restriction, first-contact stopping, and additive physical ledgers. Quotient, chart, and terminal operations append only* Abs*,* Lift*, and* Bdry*, respectively. For every cylinder record R,*

$$\operatorname{Anc}(\mathfrak{C}R) = \mathrm{cl}_{\mathrm{src}} \operatorname{Anc}(R), \qquad \Pi_\Sigma \mathfrak{C}R = \mathfrak{C}(\Pi_\Sigma R). \tag{4}$$

Consequently sourcewise limit closure creates no new primitive source and the target-null kernel commutes with continuum completion. On a finite state space, $\mathfrak{C}$ is the identity and prospective contact accountability reduces to the Part-I first-hit identity.

Proof. Each assertion is first checked on an atomic constructor. Native currents retain their assigned source. A quotient changes only provenance and appends Abs; a chart or augmentation appends Lift; a terminal restriction or exterior trace appends Bdry. The graph of a composite is the composition of the typed graphs, so induction on constructor length proves the assertions for finite records. Sourcewise closure is taken separately inside each tagged graph, proving the first identity in (4). Target probes are cylinder functionals and therefore commute with pointwise graph closure, proving the second identity. Their common kernel is consequently closed under $\mathfrak{C}$. Finite state spaces have no additional analytic boundary: their evaluation image is already closed, so the construction returns the original finite algebra and its first-hit decomposition. □

Proposition I.1.11 (Public-presentation input contract)**.** *After* Eq*, its public datum class, and its native formal carrier identity are fixed, every object used by the target-reachability track is generated functorially: the test core by rational enumeration, the solution and continuation graphs by satisfaction of the weak relation, the operation list by the typed syntax, the topology by bounded evaluation coordinates, the target by failed continuation, the source words by normal-form compilation, and the physical measure by the positive Jordan part of the signed carrier identity. A PDE specialization therefore supplies no additional problem-specific regularity input, proof object, favorable constant, or target-relative lemma.*

Proof. Each construction is the one stated in items (P1)–(P8). All subsequent objects are defined by graph closure, target quotient, Jordan decomposition, normalized shell formation, compact activity lifting, or the successor operator. These operations consume only the preceding generated objects. Inspection of the dependency chain shows that none has an input slot for any of the forbidden conclusions listed above. □

Definition I.1.12 (Target-blind public input)**.** Let $\mathsf{Pre}(\mathsf{Eq})$ be the signature formed before the continuation target, target shells, normalized bad windows, source quotients, or physical-price strata are constructed. It contains the displayed initial–boundary value problem, its declared local regularity class, weak test core, optional standard local solution theorem, native algebraic symmetries and scalings, and formal multiplier identities. It may use a fixed universal library of target-independent results such as distribution theory, integration by parts, Sobolev-space definitions, Jordan decomposition, compactness of countable products of compact metric spaces, weak compactness of probability measures on such spaces, Archimedean ordered-algebra separation, and the spectral theorem.

A fact is an admissible public input precisely when it is expressible in $\mathsf{Pre}(\mathsf{Eq})$ and remains unchanged if the continuation target and all target observables are deleted. In particular, the following are not public inputs:

$$\begin{gathered}\text{compactness of a target-approaching sequence, an epsilon-regularity or continuation conclusion,}\\ \text{a target price gap, a blow-up Liouville theorem, absence of a scale-soliton,}\\ \text{vanishing of a target quotient, or any decomposition selected after singular shells are formed.}\end{gathered} \tag{5}$$

All objects in (5) must be generated and processed by the successor algebra.

Theorem I.1.13 (Syntactic no-smuggling criterion)**.** *The dependency graph of the target-reachability compiler has exactly one edge from user data: the inclusion of* $\mathsf{Pre}(\mathsf{Eq})$ *into the canonical presentation. Every target, source, price, compactness, equality, and recurrence object lies strictly downstream of that edge. Hence no target-relative conclusion can enter as a public input. Adding any fact in* (5) *changes the signature and does not count as an evaluation of the original presentation.*

Proof. The target is generated from the continuation graph, the shells from the target distance in X_{ev}, the source cells from normal-form syntax, the price from the native carrier, and the recurrent equations from the successor fixed point. These constructions form a directed acyclic dependency chain beginning at $\mathsf{Pre}(\mathsf{Eq})$. None has a backward input slot. Every forbidden fact mentions a downstream object and therefore cannot be a term of the pre-target signature. $\square$

Theorem I.1.14 (Closed target-aligned proof discipline)**.** *Every target, source, invariant, physical-price, or reachability assertion in the continuum structural calculus is a statement in the following generated proof algebra:*

$$\mathsf{Proof}_\Sigma(\mathfrak{P}) := \mathrm{Cl}_{\mathrm{fin,lim}}\big(\mathsf{Pre}(\mathsf{Eq}),\ \textit{public weak and total graphs},\ \textit{target-shell covectors},\ \textit{five-source constructors},\ \textit{joi}$$

Here $\mathrm{Cl}_{\mathrm{fin,lim}}$ *means finite typed construction, source-preserving graph completion, and monotone completion of finite same-origin stopped records. In particular:*

(P1) *every premise is a public-presentation coordinate or the conclusion of an earlier typed construction in* $\mathsf{Proof}_\Sigma(\mathfrak{P})$*;*

(P2) *every target-visible term has a complete joint frontier before source projection and is assigned one nonempty support* $A \subseteq \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}$*;*

(P3) *every positive target coordinate has a finite same-origin detecting occurrence, while a zero coordinate deletes its target-incidence edge;*

(P4) *every physical assertion is the evaluation of the unique physical carrier coordinate of that occurrence, with equality re-entering the same joint source word rather than creating a new conclusion.*

Thus no theorem in the calculus takes a whole-orbit estimate, a target-side compactness statement, a continuation conclusion, a rigidity assertion, or a problem-specific certificate as an input. Such a statement can occur only when it has first been compiled into a target-aligned coordinate of the public presentation.

Proof. The input contract and Theorem I.1.13 give (P1). Structural induction over the closed constructor grammar gives (P2): joining precedes projection, while quotient, lift, and boundary constructors append only their declared letters. The finite-record zero/positive equivalence (560)–(561) gives (P3); a target-visible limit has a detecting finite prefix and therefore re-enters the same source cell. Finally, Theorem III.10.44 gives (P4). These are precisely the permitted proof rules, so induction on a derivation in the displayed closure proves the claim. □

Corollary I.1.15 (Uniform proof rule for the thirty-one target supports)**.** *For each nonempty support $A \subseteq \mathfrak{C}$, every proof concerning the contribution of A is obtained by restricting $\mathsf{Proof}_\Sigma(\mathfrak{P})$ to its complete source word, forming its joint detector before any coordinate is discarded, and applying the zero/positive and carrier/equality rules of Theorem I.1.14. The resulting statement is either target absence on that cell or an actual finite public-presentation occurrence carrying that support and its physical charge. Since the nonempty supports are exactly the thirty-one nonzero subsets of $\mathfrak{C}$, this is a proof rule for each structural source cell, not a separate global PDE argument.*

Proof. Apply (P2)–(P4) of Theorem I.1.14 to the fixed word A. The five-constructor grammar has no other nonempty word. □

Corollary I.1.16 (Textbook-only specialization interface)**.** *The external specialization datum ends with* $\mathsf{Pre}(\mathsf{Eq})$*. Its mathematical content consists of the displayed equation, its native formal carrier identity, and any cited target-blind local theorem already attached to its standard presentation. In particular, no later row has an argument slot for a PDE-specific compactness proof, shell estimate, price bound, equality analysis, continuation criterion, or prescribed value of Z_n. The compiler itself generates the total graphs, the ordered cones, their finite proof identities, and their complementary positive functionals.*

Proof. The first sentence is Definition I.1.12. The dependency statement follows from Theorem I.1.13. Totalization and the successor relation generate both values of every downstream relation; the proof/model alternative is Theorem II.2.47. None of these constructions adds an input edge to the dependency graph. □

Proposition I.1.17 (Non-vacuity of the total solution graph)**.** *For every datum, exactly one of the following occurs: the total local-solution fiber contains at least one maximal orbit, or it contains the nonzero form defect $\perp_{\mathrm{loc}}$. In particular, no reachability or regularity functional defined from the presentation assigns the value zero because the orbit fiber is empty.*

Proof. The total graph is the disjoint union of its ordinary solution values and its explicit failure value. The empty ordinary fiber therefore selects $\perp_{\mathrm{loc}}$. Subsequent quotients retain that record unless its target projection is proved to vanish. □

Definition I.1.18 (Generated physical expenditure and budget)**.** Let u be an ordinary value of the total solution graph. Evaluate the total signed carrier relation generated by $\mathcal{K}$. On its ordinary branch, signed gluing gives

$$K_u(t) + \mu_u((s,t]) + W_u^-((s,t]) = K_u(s) + W_u^+((s,t]), \qquad 0 \le s < t < T_*(u), \tag{6}$$

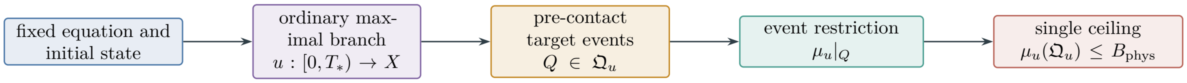

Figure 1: One physical-budget architecture. The equation generates one countably additive orbit measure, while target approach selects pre-contact events in its measurable event algebra. A selected event pays for target progress only after the target-local carrier factorization proves the charging rule; the measure's total mass is bounded by B_{phys}.

where $W_u^{\pm}$ are the Jordan parts of true exterior work and μ_u is the sum, with the native coefficients of the identity, of all generated nonnegative production and balance-defect measures. Internal transport, chart, gauge, and interface currents have already cancelled and do not enter μ_u. Every surviving summand retains its five-source ancestry. This orbitwise construction bounds the available physical measure; it does not yet assign any portion of that measure to target progress. Such an assignment is made only by Definition II.2.3 on an active branch.

If $K_u \ge K_- > -\infty$ and $W_u^+([0, T_*(u))) < \infty$, define the single orbit budget

$$B_{\text{phys}}(u_0, W^+) := K_u(0) - K_- + W_u^+([0, T_*(u))) < \infty. \tag{7}$$

Then (6) derives

$$\mu_u(\mathfrak{Q}_u) \le B_{\text{phys}}(u_0, W^+). \tag{8}$$

Failure of the carrier identity, nonnegativity, the lower bound, or finite exterior input is the corresponding total-graph value. Its target-visible part enters the activity lift and ordered successor algebra. A finite ordered identity may exclude that value; otherwise it remains in the positive structural model. It is neither discarded nor misread as finite-budget regularity. The finite-budget reachability theorems apply only to the ordinary branch on which (8) has been derived.

The measure is localizable. For pairwise disjoint $Q_j \in \mathcal{A}_u$,

$$\sum_j \mu_u(Q_j) = \mu_u\left(\bigsqcup_j Q_j\right) \le B_{\text{phys}}. \tag{9}$$

No target-dependent expenditure may be appended. A Hamiltonian, entropy, commutator, or localized identity refines μ_u only after it has been proved to be an algebraic decomposition of (6). If the basic carrier assigns finite price to a singular cascade, that cascade proceeds to the zero-cost successor calculus; the budget is not strengthened after the fact.

Theorem I.1.19 (Universal physical-quantity ownership)**.** *Let $F(u)$ be any scalar, vector, tensor, measure, flux, action, entropy, energy, enstrophy, moment, capacity, frequency, scale, or localized quantity generated from the public equation. Every use of F in target accounting is evaluated through the complete expanded derivation of F and its balance graph. Exactly one of the following holds on a target-active cell:*

(O1) *its target projection vanishes;*

(O2) *its owner-tagged positive part is algebraically dominated by the single measure μ_u through a finite identity in $\mathscr{D}_{\mathfrak{b}}^{\text{alg}}$;*

(O3) *its owner belongs to $\mathscr{D}_{\mathfrak{b}}^{\text{phys}} \setminus \mathscr{D}_{\mathfrak{b}}^{\text{alg}}$; the approximating rows and all target-shell errors form the cone-boundary successor, and no charge is declared;*

(O4) *its owner lies outside $\mathscr{D}_{\mathfrak{b}}^{\text{phys}}$; finite-cylinder separation produces the carrier-polar successor with the same five-source ancestry.*

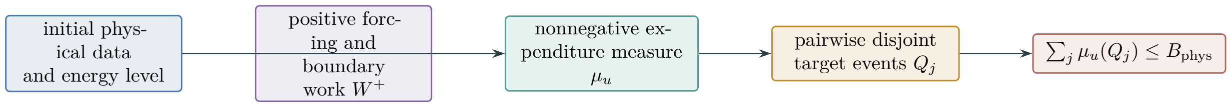

Figure 2: Physical expenditure accounting. The initial state and legitimate positive input bound one countably additive measure. Pairwise disjoint target events have disjoint restrictions of that measure. Only restrictions admitted by the local target-aligned charging rule pay for shell progress.

Finite, conserved, monotone, coercive, scale-invariant, or summable values of the constituents do not imply any of these conclusions for a composite. They are predicates inside its expanded joint record. In particular, two physical quantities may cancel or amplify only through their signed joint balance, and that joint balance is processed before either positive parts or normalization are taken. The theorem creates no second budget.

Proof. Form $\mathrm{Exp}(F)$, retain every intermediate and owner-tagged restriction, and apply Theorem III.4.58. Target-nullity gives (O1). Algebraic membership of the resulting target owner gives (O2). Closure-only membership gives the total cone-boundary graph in (O3). On the complement, finite-cylinder separation and Theorem II.2.32 give (O4). Signed gluing precedes the cone test, so cancellation is retained exactly. Since only domination by μ_u authorizes charge, no auxiliary physical quantity becomes another budget. $\square$

Example I.1.20 (Energy–work budget)**.** On the zero-defect branch of the lower-energy, dissipation-balance, and finite-work rows, an energy $\mathcal{E}$, lower value E_-, and nonnegative dissipation measure μ_{diss} satisfy

$$\mathcal{E}(u(t)) + \mu_{\mathrm{diss}}([0,t]) \le \mathcal{E}(u_0) + W^+([0,t]),$$

and $W^+([0,\infty)) < \infty$. Then

$$\mu_{\mathrm{diss}}([0,\infty)) \le \mathcal{E}(u_0) - E_- + W^+([0,\infty)).$$

Taking $\mu_u = \mu_{\mathrm{diss}}$ gives the single budget

$$B_{\mathrm{phys}} = \mathcal{E}(u_0) - E_- + W^+([0,\infty)).$$

If this physical measure assigns finite total price to a singular cascade, the cascade enters the zero-price successor calculus. No stronger measure or second budget is introduced after the target has been inspected.

Definition I.1.21 (Fixed-Caus gradient-flow presentation)**.** Let $\mathcal{V} \hookrightarrow \mathcal{H} \hookrightarrow \mathcal{V}^*$ be a Gelfand triple. A fixed-Caus gradient-flow presentation consists of a differentiable energy $\mathcal{E}$, a symmetric nonnegative mobility $\mathbb{K}(u) : \mathcal{V}^* \to \mathcal{V}$, a fixed constitutive transport $A_{\mathsf{Caus}}(u) \in \mathcal{V}^*$, and an exterior input $f_{\mathsf{Bdry}} \in \mathcal{V}^*$, with equation

$$\dot{u} + A_{\mathsf{Caus}}(u) = -\mathbb{K}(u) D\mathcal{E}(u) + f_{\mathsf{Bdry}}. \tag{10}$$

The Caus term is energy-skew when

$$\langle D\mathcal{E}(u), A_{\mathsf{Caus}}(u) \rangle = 0. \tag{11}$$

Write

$$\left\| \mathbb{K}(u)^{1/2} \xi \right\|^2 := \langle \xi, \mathbb{K}(u) \xi \rangle$$

for $\xi \in \mathcal{V}^*$; this notation does not assume a bounded inverse or a globally defined square root on $\mathcal{V}^*$. Here Geom is the mobility current $-\mathbb{K}D\mathcal{E}$, Caus is the declared transport, and Bdry contains the exterior input and boundary flux. Quotients and symmetry reductions applied to this orbit have Abs ancestry; charts, blow-up rescalings, and augmented variables have Lift ancestry.

Proposition I.1.22 (Fixed-Caus expenditure bound)**.** *Let u satisfy Eqs.* (10) *and* (11) *on* $[0,T]$, *with the chain rule for $\mathcal{E}(u)$. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{E}(u(t)) = -\left\|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\right\|^2 + \langle D\mathcal{E}(u), f_{\mathsf{Bdry}}\rangle. \tag{12}$$

If $\mathcal{E} \geq E_-$, then

$$\int_0^T \left\|\mathbb{K}(u)^{1/2}D\mathcal{E}(u)\right\|^2 \mathrm{d}t \leq \mathcal{E}(u_0) - E_- + \int_0^T \langle D\mathcal{E}(u), f_{\mathsf{Bdry}}\rangle_+ \,\mathrm{d}t. \tag{13}$$

For a weak solution with an energy inequality, the same statement holds with an additional nonnegative energy-defect measure on the left.

Proof. Pair (10) with $D\mathcal{E}(u)$. The chain rule gives the left side of (12), the Caus pairing vanishes by (11), and positivity of $\mathbb{K}$ gives the squared norm. Integrate, discard the terminal energy above E_-, and replace exterior work by its positive part. The weak form is the increment formula for the energy inequality; its defect is nonnegative. □

I.2 The five-source orbit algebra

Definition I.2.1 (Continuum source alphabet)**.** The source alphabet is

$$\mathfrak{C} = \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}.$$

A raw continuum record is an orbit occurrence together with a finite ancestry word $\mathrm{Anc}(r) \in \mathfrak{C}^*$. Its source support $\mathrm{Src}(r) \subseteq \mathfrak{C}$ is the set of letters appearing in that word. The primitive and closure rules are:

- (S1) symmetric mobility, Dirichlet-form, reversible, and dissipative current records start in Geom;
- (S2) after the canonical native form split, the remaining oriented bulk evolution current starts in Caus; this includes fixed transport, reaction, nonvariational forcing, memory, and a raw bulk current whose attempted form realization is nonordinary;
- (S3) quotient, symmetry reduction, gauge removal, and coarse variables append Abs;
- (S4) charts, renormalizations, blow-up variables, auxiliary-state extensions, and Young- or defect-variable lifts append Lift;
- (S5) boundary traces, exterior fluxes, killing, terminal values, and tails read at the boundary of a compactification append Bdry;
- (S6) finite algebraic combination, restriction, and concatenation take the union of the input source supports.

The symbol J_{res} is absent from this source alphabet. It is introduced only as the target-null coordinate of Definition I.4.2.

Remark I.2.2 (Channel and source terminology). The channels Geom and Abs carry intrinsic reusable structure, while Caus, Lift, and Bdry orient, reorganize, and read out that structure. A *source cell* is a provenance cell in the complete five-channel normal form; this terminology does not promote the three derived channels to independent creators of intrinsic structure.

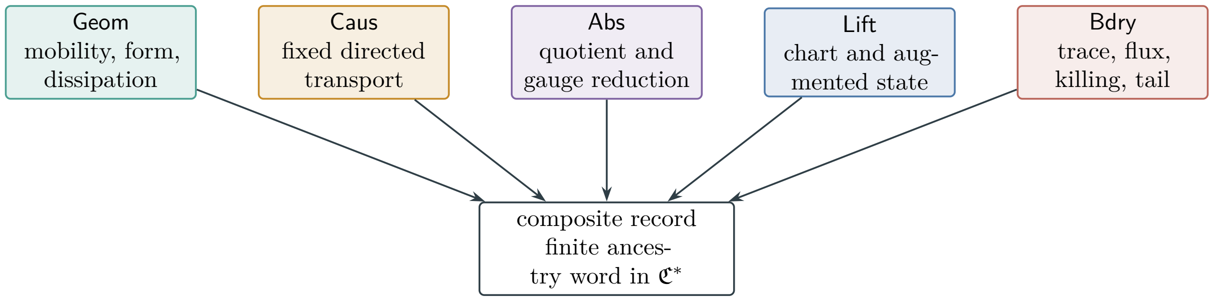

Figure 3: The five continuum source roles. The canonical native form split routes its positive symmetric output to Geom and its complete oriented bulk remainder to Caus. Abs and Lift record quotient and presentation currents, and Bdry records interface, exterior, and terminal currents. Composite records retain the union of their ancestries.

Remark I.2.3 (Forcing and constraints). The source compiler routes a volume force according to its constitutive role. A fixed directed body force may be Caus, while exterior work or a boundary supply is Bdry. A Lagrange multiplier implementing a quotient constraint has Abs ancestry. The five labels are assigned from the declared weak equation, not from the sign of a later estimate.

Definition I.2.4 (Operational weak-PDE signature)**.** An operational weak-PDE signature is formed downstream from the public presentation. It consists of a Gelfand triple $V \hookrightarrow H \hookrightarrow V^*$, a separating test core $\mathcal{D} \subset V$, an initial datum, the continuation target Σ generated by Definition III.1.1, and a weak evolution relation

$$\langle \dot{u}, \varphi \rangle_{V^*,V} = \mathcal{A}(u;\varphi) + \mathcal{F}_{\mathrm{ext}}(u;\varphi), \qquad \varphi \in \mathcal{D}. \tag{14}$$

The native spatial relation is entered before any realization property is asserted. Its totalized realization relation has the following ordinary values.

(W1) A fixed-Caus gradient realization:

$$\mathcal{A}(u;\varphi) = -\langle \mathbb{K}(u) D\mathcal{E}(u), \varphi \rangle - \langle A_{\mathsf{Caus}}(u), \varphi \rangle,$$

with $\mathbb{K}(u)$ symmetric and nonnegative.

(W2) A sector-form realization on a real L^2-space:

$$\mathcal{E} = \mathcal{E}^{\mathrm{s}} + \mathcal{E}^{\mathrm{a}}, \qquad \mathcal{E}^{\mathrm{s}}(v,w) = \tfrac{1}{2}\{\mathcal{E}(v,w) + \mathcal{E}(w,v)\}, \quad \mathcal{E}^{\mathrm{a}}(v,w) = \tfrac{1}{2}\{\mathcal{E}(v,w) - \mathcal{E}(w,v)\},$$

where $\mathcal{E}^{\mathrm{s}}$ is a Dirichlet form and $\mathcal{E}^{\mathrm{a}}$ satisfies the sector bound.

If neither ordinary value exists, the realization relation returns $\perp_{\mathrm{nat}}$. The raw fixed bulk current remains a Caus occurrence and the failed attempted Geom–Caus split is a form-defect record with support $\{\mathsf{Geom}, \mathsf{Caus}\}$. Its target-active quotient is evaluated by the form row. Thus absence of a gradient, Dirichlet, positivity, or sector realization is admitted as structure rather than excluded from the signature. The signature also declares the symmetry, quotient, chart, auxiliary-state, and trace operations that may be applied to (14). It contains no assertion of compactness, regularity, target avoidance, positive physical price, or continuation beyond the declared graph.

Definition I.2.5 (Continuum source compiler)**.** Let $\mathsf{Syn}(\mathfrak{P})$ be the smallest typed syntax containing the native terms of (14) and closed under every finite algebraic or differential operation appearing in its expression graph, signed sum, composition, restriction, localization, polarization, commutator, quotient, chart, state augmentation, trace, total-graph completion, and sourcewise weak limit. The source compiler

$$\mathsf{C}_{\mathfrak{P}} : \mathsf{Syn}(\mathfrak{P}) \longrightarrow 2^{\mathfrak{C}} \setminus \{\varnothing\}$$

is defined recursively by

$$\begin{aligned} \mathsf{C}_{\mathfrak{P}}(-\mathbb{K}D\mathcal{E}) &= \{\mathsf{Geom}\}, & \mathsf{C}_{\mathfrak{P}}(\mathcal{E}^{\mathrm{s}}_{\mathrm{local/jump}}) &= \{\mathsf{Geom}\}, \\ \mathsf{C}_{\mathfrak{P}}(-A_{\mathsf{Caus}}) &= \{\mathsf{Caus}\}, & \mathsf{C}_{\mathfrak{P}}(\mathcal{E}^{\mathrm{a}}) &= \{\mathsf{Caus}\}, \\ \mathsf{C}_{\mathfrak{P}}(\mathcal{F}_{\mathrm{ext}}) &= \{\mathsf{Bdry}\}, & \mathsf{C}_{\mathfrak{P}}(\perp_{\mathrm{nat}}) &= \{\mathsf{Geom}, \mathsf{Caus}\}. \end{aligned} \tag{15}$$

Here A_{Caus} denotes the complete oriented bulk remainder returned by the canonical native form split; on the native failure branch it denotes the raw bulk current. It is not restricted to a first-order or antisymmetric operator. Finite sums take unions. Quotients append Abs; charts and auxiliary variables append Lift; traces, killing, terminal values, and compactification boundaries append Bdry. Restriction, concatenation, and sourcewise weak closure retain support. A constraint multiplier is compiled through the operation that generates it: quotient constraints append Abs, while an exterior constraint supply appends Bdry.

Definition I.2.6 (Primitive dynamical-role partition)**.** Expand the displayed weak equation into its finite expression graph before forming target shells. A displayed integral, series, or nonlocal operator is one bulk leaf unless an exact public identity expands it. Exact decomposition identities are applied in their fixed public enumeration, so different valid presentations remain distinct ancestry records. A *primitive dynamical occurrence* is a leaf together with its first edge that contributes to a weak current pairing. The totalized native form-split graph of Definition I.2.4 is then applied. Its ordinary value may return both a positive symmetric occurrence and an oriented remainder; its nonordinary value returns the raw bulk current and the typed form defect. The following ordered role test assigns each resulting occurrence.

(R1) A generated positive symmetric form, mobility, reversible quadratic form, carré du champ, or its polarization is Geom.

(R2) Every oriented remainder and every raw bulk evolution relation on the nonordinary split branch is Caus. This is the complement of (R1) in the canonical native form split, not an assumption that the current is antisymmetric or has a gradient representation.

(R3) An identification of states, constraints, gauges, symmetries, or coarse variables appends Abs.

(R4) A change or enlargement of the state presentation—including a chart, localization state, history state, stochastic path, modulation variable, blow-up variable, or weak-product variable—appends Lift.

(R5) A spatial, temporal, exterior, killed, terminal, or compactification interface appends Bdry.

Scalar coefficients, datum values, and exact storage terms carry no independent source: they acquire the ancestry of the current-producing edge in which they occur. A path meeting several roles retains their union. The test is ordered only to make the primitive assignment disjoint; it does not erase composite ancestry.

Lemma I.2.7 (Exhaustion of primitive current paths)**.** *Every path from a resulting primitive occurrence of a canonical physical presentation to a weak current pairing has one of the following forms:*

(E1) *an interior bulk edge, assigned by exactly one of* (R1) *and* (R2)*, followed by zero or more quotient, lift, or boundary edges;*

(E2) *a constraint, gauge-reaction, or identification current, assigned* Abs*, retaining every upstream bulk role;*

(E3) *a chart velocity, auxiliary-state, modulation, or presentation-defect current, assigned* Lift*, retaining every upstream role;*

(E4) *a prescribed exterior, trace, or terminal current, assigned* Bdry*, retaining every upstream role.*

Thus every primitive current occurrence has one head in $\mathfrak{C}$ and a nonempty support containing that head. An expression leaf with no path to a weak current pairing contributes zero to every prospective shell current.

Proof. An edge of the weak expression graph either changes the evolving state in the interior, identifies states, changes the state presentation, or reads an interface value. These are respectively the bulk, quotient, lift, and boundary cases in Definition I.2.6. The cases are disjoint after the ordered test.

For a bulk edge, apply the canonical totalized form-split graph. On its ordinary branch, polarization gives the Geom occurrence and its algebraic remainder is the oriented Caus occurrence; either may be zero. On its nonordinary branch the raw bulk current remains Caus, while the failed form graph has support $\{\mathsf{Geom}, \mathsf{Caus}\}$. Thus no positivity, closedness, sector, or gradient property is required for the split, and the fixed enumeration makes the ancestry of each occurrence unique. Quotient, lift, and boundary edges append their corresponding letters. Their horizontal or base descendants retain the upstream head; their vertical, interface, or defective current outputs have the head declared in (E2)–(E4). A prescribed exterior or terminal current begins with Bdry.

A coefficient or datum with no path through a current-producing edge is not a summand of the weak evolution current. It therefore contributes zero to $\frac{\mathrm{d}}{\mathrm{d}t}\Phi(u_t)$, even when it depends explicitly on space or time. When it enters a bulk or boundary current, it acquires that edge's ancestry. Exact storage contributes only through the endpoints of an already sourced current identity and acquires that identity's ancestry. □

Definition I.2.8 (Free provenance algebra)**.** Let $\mathfrak{T}(\mathfrak{P})$ be the free totalized term algebra on the primitive occurrences of Definition I.2.6, with constructors

$$+,\ \circ,\ \mathrm{res},\ q,\ \ell,\ \mathrm{tr},\ \mathrm{Graph},\ \mathrm{cl}_{\mathrm{src}}\,.$$

They denote signed finite combination, composition, restriction, quotient, lift/chart, trace/terminal readout, total-graph completion, and sourcewise closure. Its provenance semilattice is

$$\mathfrak{P}_5 := (2^{\mathfrak{C}}, \cup, \varnothing).$$

The provenance morphism $\mathrm{Prov} : \mathfrak{T}(\mathfrak{P}) \to \mathfrak{P}_5$ maps primitive roles to their letters, maps finite constructors to union, appends $\mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$ under q, ℓ, tr, and preserves support under restriction, totalization, and sourcewise closure. Before Graph is applied, typed elaboration inserts the interface constructor of the least operation being totalized: for example a weak-product state inserts ℓ, a failed quotient inserts q, and a trace failure inserts tr. Thus Graph preserves the already complete typed support; it cannot suppress a failure-interface letter. The empty value is reserved for current-free constants. Exact cancellation does not erase provenance; it places a nonempty sourced record in $\mathscr{N}_\Sigma$.

Every atomic current term also has a *head* $\mathrm{hd}(T) \in \mathfrak{C}$, namely the output-current sort of the typed term. A native occurrence has the sort assigned by the primitive role partition. A composition has the sort of its outer current-producing constructor and retains the provenance of every input. A signed sum of different current sorts is a total current rather than an atomic term and is expanded into its typed summands. Ordinary restriction graphs preserve the head. The horizontal quotient descendant and ordinary lift-base descendant preserve it; quotient intertwining/vertical/cokernel/graph coordinates have head Abs, while lift interface/fiber/concentration/oscillation/scale/graph coordinates have head Lift. Every boundary-current readout, ordinary or defective, has head Bdry; and a native form-split occurrence has head $\mathsf{Geom}, \mathsf{Caus}$, or Bdry as assigned above. A signed sum has no head until it is expanded into its top-level atomic summands. Always $\mathrm{hd}(T) \in \mathrm{Prov}(T)$.

Theorem I.2.9 (Universal five-source provenance theorem)**.** *The morphism* Prov *is the unique provenance assignment which agrees with the primitive dynamical-role partition and respects all constructors of the closed continuum algebra. More generally, let $(S, \vee, 0)$ be an idempotent join semilattice and let $\tau : \mathfrak{T}(\mathfrak{P}) \to S$ respect the same constructors, taking the value s_c on a primitive role c. There is a unique join homomorphism*

$$\overline{\tau} : \mathfrak{P}_5 \to S, \qquad \overline{\tau}(A) = \bigvee_{c \in A} s_c, \qquad \tau = \overline{\tau} \circ \mathrm{Prov}\,. \tag{16}$$

If a finite generated weak current is written as its top-level signed sum $J = \sum_{i=1}^{m} \mathrm{ev}(T_i)$, then

$$q_\Sigma J = \sum_{c \in \mathfrak{C}} q_\Sigma J^c(J) = \sum_{\varnothing \neq A \subseteq \mathfrak{C}} q_\Sigma J_A(J), \qquad J^c(J) := \sum_{\mathrm{hd}(T_i) = c} \mathrm{ev}(T_i), \quad J_A(J) := \sum_{\mathrm{Prov}(T_i) = A} \mathrm{ev}(T_i). \tag{17}$$

Every graph-complete or sourcewise limit retains the joint *vector of head components and exact-support components together with the finite factorization graph. Thus no component is discarded before closure and no sum of independently compactified extended values is asserted. As a space of provenance-tagged atomic occurrences,*

$$\mathscr{J}_\Sigma^{\mathrm{dyn}}(E) = \bigsqcup_{\varnothing \neq A \subseteq \mathfrak{C}} q_\Sigma \mathscr{J}_A^{\mathrm{phys,sat}}(E), \tag{18}$$

where $\mathscr{J}_\Sigma^{\mathrm{dyn}}(E)$ is the target-active atomic current-occurrence space generated by all public orbit records of physical ceiling E, and $\mathscr{J}_A^{\mathrm{phys,sat}}(E)$ is the current evaluation of the saturated record cell with exact support A. Analytically equal currents with different histories remain distinct in the disjoint union.

Proof. Existence follows by recursion. Primitive existence and disjointness are Lemma I.2.7. Signed addition and composition retain the union of the source leaves occurring in the expression graph. Quotient, lift, and trace constructors append their letters. Restriction changes no source leaf. A total-graph value stores the same input expression graph together with its first nonordinary output, so it has the same ancestry, with the interface letter already appended. Sourcewise closure is taken after the tag is fixed and therefore preserves it.

For uniqueness, let P be another assignment with these properties. It agrees with Prov on primitive leaves. Structural induction on the term tree shows agreement under each finite constructor. Total-graph and closure compatibility then gives agreement on completed terms. This is the universal property of the free totalized term algebra. The same induction gives a unique head because every current-valued constructor in the typed syntax has one output sort. A graph defect changes that sort only to the declared interface-failure sort. This typing does not alter the complete provenance support.

For the semilattice assertion, the values s_c determine a unique join homomorphism on the free finite-subset semilattice by the displayed formula. Constructor induction gives $\tau(T) = \bigvee_{c \in \mathrm{Prov}(T)} s_c$, which is precisely (16). Thus a sixth irreducible value would require either a sixth primitive role or a source-creating constructor; Lemma I.2.7 and the declared constructor list exclude both.

To prove (17), group the top-level summands first by their heads and then by the nonempty support of their complete term trees. Each is a partition of the same finite summand list. Evaluation is linear on weak pairings, so both grouped identities hold before the target quotient and hence after applying q_Σ. Terms with empty provenance are current-free constants and vanish in the quotient; sourced exact cancellations retain their ancestry and also vanish there. Every bounded target readout is a structural-test coordinate; it therefore commutes with graph-complete sourcewise limits. Taking the closure of the joint finite factorization graph and retaining the exact provenance tags proves (18). □

Corollary I.2.10 (Audit of derived and singularity-scale occurrences). *Each commonly arising continuum occurrence is routed as follows:*

Occurrence	*Forced provenance route*
nonlinear product or coefficient degeneration	*ordinary output-current head with the union of its bulk-input ancestries; a nonordinary weak-product state appends* Lift*, while a coefficient-domain, chart, or interface failure receives the head and appended letter of its least failed typed operation; every case retains the input union*
commutator or polarized multilinear descendant	*the output head dictated by its signed weak-current role, with the union of all input ancestries; a positive square generated by polarization is a* Geom*-headed production descendant, while a failed polarization graph is* Lift*-headed*
reaction, nonvariational body term, or nonlocal bulk operator	Caus *after the canonical split, with its ordinary positive symmetric component in* Geom
time derivative, conservation law, or exact storage	*the shell derivative is replaced by the sourced right-hand-side current; storage endpoints retain the ancestry of their carrier identity*
initial datum, coefficient, parameter, or target-shell observable	*no source by itself; it acquires the ancestry of the current-producing edge through which it affects shell progress*
higher-order time equation	Lift *for its first-order phase-space presentation, followed by the* Geom/Caus *split of the resulting bulk evolution*
memory or delay	Caus *as a bulk relation, with* Lift *appended when represented on a history state*
constraint multiplier, pressure, gauge, or symmetry reduction	*input ancestry together with* Abs
moving frame, modulation, localization state, or blow-up microscope	*input ancestry together with* Lift
weak defect, Reynolds stress, concentration, oscillation, or Young measure	*sourcewise limit of the nonlinear input ancestry, represented in the* Lift *normal form*
stochastic drift, covariance, and sample path	*drift ancestry;* Geom *for positive quadratic variation;* Lift *for the path augmentation*

moving physical boundary or free interface	Lift *for its chart/state motion and* Bdry *for its oriented trace*
shock, internal discontinuity, or topology-changing interface	*input ancestry together with* Bdry *for its trace and* Lift *when the interface position is a state variable*
nonuniqueness of weak evolution	*no new source; every branch retains the sources of its own weak current and the total solution graph retains all branches*
spatial infinity, exhaustion tail, killing, initial endpoint, or terminal germ	*input ancestry together with* Bdry
loss of closed range, trace, chain rule, covariance, compactness, or continuation	*first nonordinary total-graph value with the ancestry of the operation whose graph failed*

The table is a consequence of the constructor theorem; it introduces no additional routing convention.

Proof. Apply Theorem I.2.9 to the indicated expression graph. Positive quadratic variation is the Geom branch of Proposition I.2.19; every other row is one of the finite, interface, total-graph, or closure constructors in the theorem. □

Theorem I.2.11 (No additional singularity source)**.** *Let u be a public orbit in a finite physical stratum, and let W be any atomic ordinary or graph-complete current occurrence in the canonical top-level expansion of the prospective derivative of a continuation-shell observable along u. Exactly one of the following target-relative alternatives holds:*

(N1) $q_\Sigma W = 0$, *so $W \in \mathscr{N}_\Sigma$ and the occurrence is residual for singular contact;*

(N2) $q_\Sigma W \neq 0$, *and the occurrence has a nonempty exact provenance support $A \subseteq \mathfrak{C}$ and a unique head $c = \mathrm{hd}(W) \in A$, hence belongs to one of the* 31 *saturated source cells and exactly one of the five head normal forms.*

If u contacts the complete singular target, at least one occurrence of type (N2) *appears on every normalized shell crossing. Therefore every singular contact uses target-aligned structure in* Geom, Caus, Abs, Lift, Bdry*, possibly with composite support, and no additional target-bearing source exists.*

Proof. The prospective derivative is obtained by pairing the shell covector with the public weak equation, its generated transformations, and their sourcewise graph completions. The totalized Stieltjes chain rule in Theorem III.2.3 exhausts its ordinary, jump, Cantor, concentration, oscillation, and variation-boundary values. By Theorem I.2.9, every occurrence in that pairing belongs to a saturated cell with exact support $A \subseteq \mathfrak{C}$, unless it is a current-free constant. Constants vanish under the orbit derivative. For a nonempty source cell, either its target quotient vanishes, giving (N1), or it does not, giving (N2). These cases are disjoint and exhaustive.

If contact occurs, the normalized shell observable has unit endpoint increment. On a finite-value record, the affine shell graph (39) therefore has a positive summand. On an infinite-variation record, the closed owner relation (41) gives a positive atomic occurrence directly. Thus some occurrence is of type (N2) without invoking an extended signed sum. Sourcewise completion covers weak defects and singularity-scale limits, while total graphs cover every failed operation. There are $2^5 - 1 = 31$ possible nonempty supports. Within a fixed head, the ordered refinements are exhaustive by Theorems III.5.3, III.6.3, III.7.3, III.8.3 and III.9.3. □

Remark I.2.12 (Taxonomy versus exclusion)*.* The source theorem classifies prospective singularity-forming structure; it does not by itself bound that structure. A contact-seeded formation-active point may persist in any of the 31 cells and then gives a lower-bound witness for the corresponding saturated source profile. It constitutes a singular outcome only if an independent public orbit realizes it. Regularity is obtained by the later target-local invariant and physical-price analysis that upper-bounds every active source cell below the normalized contact floor. The conclusion established here is that no target-bearing contribution lies outside the five-source algebra.

Corollary I.2.13 (Four-way audit of a proposed mechanism)**.** *For a fixed public presentation, every proposed singularity mechanism is resolved by the following exhaustive audit:*

(A1) *if it occurs in the weak current graph, its constructor path gives one of the* 31 *source supports;*

(A2) *if it is a limit or failed operation on that graph, sourcewise completion or totalization gives the same inherited support;*

(A3) *if every target-shell pairing vanishes, it lies in* $\mathscr{N}_\Sigma$*;*

(A4) *if it has neither a public current path nor a total-graph limit of such paths, it is not an occurrence of the displayed evolution.*

Thus an analytically named object cannot form an additional source merely by being singular, noncompact, nonlocal, distributional, or difficult to estimate.

Proof. The first two cases are Theorem I.2.9. The third is the definition of $\mathscr{N}_\Sigma$. The total solution graph and exhaustive approximation tower contain precisely the ordinary public paths and their graph-complete limits, so failure of both membership tests gives the fourth case. □

Corollary I.2.14 (No source created by a continuum microscope)**.** *Let two orbit descriptions be related by a generated quotient, chart, localization, modulation, or compactification graph. On its ordinary covariance cell, their target-active currents have the same native heads and ancestry, with only the declared* $\mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$ *interface letters appended. On every nonordinary covariance cell, the first descent or graph value has the corresponding interface head and retains the input ancestry. Hence a blow-up microscope, moving frame, weak compactification, or alternate gauge cannot create an additional singularity source.*

Proof. Apply the head and provenance recursion of Definition I.2.8 to the generated transformation graph. Ordinary descent preserves weak current pairings and the target quotient. The totalized nonordinary branch is one of the quotient, lift, or boundary coordinates specified in that recursion. □

Theorem I.2.15 (Operational native-source exhaustion)**.** *For every operational weak-PDE signature,* $\mathsf{C}_{\mathfrak{P}}$ *is defined on every finite dynamically generated record and has image in the five inherited source families. Its evaluation commutes with the weak equation: for every* $\varphi \in \mathcal{D}$*,*

$$\langle \dot{u}, \varphi \rangle = \sum_{c \in \mathfrak{C}} \langle J^c(u), \varphi \rangle. \tag{19}$$

The source word of a transformed record is the union of the native word and the letters appended by the transformation. There is no untyped native or finite-constructor remainder.

Proof. The primitive paths of the displayed weak relation are exhausted by Lemma I.2.7. In particular, every native bulk term outside an ordinary positive symmetric form is routed through the complementary Caus branch, and every direct exterior occurrence is Bdry.

On ordinary branch (W1), the displayed realization refines the native relation into the mobility, directed-transport, and exterior terms. Nonnegativity of $\mathbb{K}$ identifies the first as Geom; the fixed transport and exterior relation give Caus and Bdry. On ordinary branch (W2), polarization uniquely separates the symmetric and antisymmetric forms. The Beurling–Deny representation uniquely separates the symmetric form into strongly local, jump, and killing terms; the first two are Geom and killing is Bdry. The antisymmetric form is Caus.

On the failure branch, retain the weak bulk current itself as a fixed Caus occurrence, so (19) still holds, and record the failed attempted split with composite $\{\mathsf{Geom}, \mathsf{Caus}\}$-ancestry. The form row evaluates its target-active quotient without asserting the missing realization.

The universal structural induction of Theorem I.2.9 proves the remaining claim. Each constructor is covered by exactly one recursive clause in Definition I.2.5; finite combination takes unions and cannot erase an ancestor. Applying the evaluation map at each induction step gives (19). Weak closure is treated sourcewise, so it changes the analytic representative but not the cell label. □

Corollary I.2.16 (Failure is compiled structure)**.** *If an operation in* $\mathsf{Syn}(\mathfrak{P})$ *lacks a classical value on a limit record, its totalized graph defect has the source support assigned to the approximating graph, together with* Lift *or* Bdry *when the failure occurs at a chart or terminal interface. Consequently lack of closed range, compactness, a classical trace, or a strong nonlinear limit cannot create a sixth source.*

Proof. Apply the compiler before graph closure, then use the sourcewise closure rule. The graph defect is a limit of records in that tagged cell. The chart and terminal cases use the corresponding recursive clauses. □

Theorem I.2.17 (Sector-form source compiler)**.** *Let X be locally compact and separable, let m be a Radon measure of full support, and let $(\mathcal{E}^{\mathrm{s}}, D(\mathcal{E}))$ be a regular symmetric Dirichlet form on the real space $L^2(X, m)$. Write its Beurling–Deny decomposition, with normalization absorbed into J, as*

$$\mathcal{E}^{\mathrm{s}}(u, v) = \mathcal{E}^{\mathrm{c}}(u, v) + \int_{X\times X\setminus\mathrm{diag}} (u(x) - u(y))(v(x) - v(y))\, J(\,\mathrm{d}x,\, \mathrm{d}y) + \int_X uv\, \kappa(\,\mathrm{d}x). \tag{20}$$

Let $\mathcal{E}^{\mathrm{a}}$ be antisymmetric on the same domain and satisfy the sector bound

$$|\mathcal{E}^{\mathrm{a}}(u, v)| \leq C_{\mathrm{sec}}\, \mathcal{E}^{\mathrm{s}}_1(u, u)^{1/2}\mathcal{E}^{\mathrm{s}}_1(v, v)^{1/2}. \tag{21}$$

Then the form $\mathcal{E} = \mathcal{E}^{\mathrm{s}} + \mathcal{E}^{\mathrm{a}}$ is closed and sectorial on $D(\mathcal{E})$, and its source assignment is exhaustive: $\mathcal{E}^{\mathrm{c}}$ and the symmetric jump form are Geom, *$\mathcal{E}^{\mathrm{a}}$ is the fixed* Caus *form, and the killing measure κ is* Bdry. *The uniqueness of* (20) *leaves no additional symmetric Markovian primitive. Quotients and state-space enlargements append* Abs *and* Lift *as usual.*

Proof. The Beurling–Deny theorem gives (20) and uniqueness of its strongly local, jump, and killing measures [5]. Antisymmetry gives $\mathcal{E}^{\mathrm{a}}(u, u) = 0$, while (21) makes it continuous in the $\mathcal{E}^{\mathrm{s}}_1$-norm. Hence $\mathcal{E}$ has the same form norm completeness as $\mathcal{E}^{\mathrm{s}}$ and satisfies the sector condition. The stated routing is exactly the source assignment of Definition I.2.1. A quasi-regular form has the same conclusion after a declared quasi-homeomorphic regular realization. □

Proposition I.2.18 (Exact-drift symmetrization)**.** *Let M be a Riemannian domain, let $\Psi \in C^2(M)$. On the locally-finite-weight branch let $\pi(\,\mathrm{d}x) = e^{-\Psi(x)}\,\mathrm{d}x$ is a locally finite measure. For*

$$L_\Psi g = \Delta g - \nabla\Psi \cdot \nabla g,$$

totalize the integration-by-parts trace. On its zero boundary-defect branch, the resulting form core satisfies

$$-\langle f, L_\Psi g\rangle_{L^2(\pi)} = \int_M \nabla f \cdot \nabla g \, \mathrm{d}\pi. \tag{22}$$

Thus an exact drift is absorbed into a symmetric Geom *form by a weighted* Abs*-gauge. If an exact/nonexact decomposition of a general drift is proved on the declared domain, only its nonexact remainder retains independent* Caus *ancestry. No global boundedness of* $\mathrm{e}^{\pm\Psi}$ *is inferred; comparison with another physical measure is evaluated by its own totalized physical-transfer row.*

Proof. The weighted divergence identity

$$L_\Psi g = e^{\Psi} \operatorname{div}(e^{-\Psi}\nabla g)$$

and the stated trace condition give (22). Symmetry and nonnegativity of the right side prove the source statement; changing the reference measure is the declared gauge operation. □

Proposition I.2.19 (Wiener lift creates no additional source)**.** *Let H and U be separable Hilbert spaces and let a legal Itô evolution have drift $b : H \to H$ and Hilbert–Schmidt noise coefficient $\sigma : H \to \mathcal{L}_2(U, H)$. Totalize the generator relation on bounded C^2 cylinder functions. Its ordinary branch is*

$$LF(u) = \langle DF(u), b(u)\rangle + \frac{1}{2} \operatorname{Tr}\big[\sigma(u)\sigma(u)^* D^2 F(u)\big]. \tag{23}$$

Then the carré du champ is

$$\Gamma(F) := L(F^2) - 2FLF = \|\sigma^* DF\|_U^2 \geq 0. \tag{24}$$

Thus the diffusion covariance and predictable quadratic variation are Geom*; the drift retains its declared* Geom/Caus/Bdry *decomposition; and adjoining the Wiener path is* Lift*. Noise therefore creates no sixth continuum source.*

Proof. Apply the second-order chain rule to F^2 in (23); the drift terms cancel and the remaining trace is $\|\sigma^* DF\|_U^2$. Itô's formula gives the same integrand for the predictable quadratic variation of the martingale part. The source labels follow from positivity of Γ, the fixed drift decomposition, and the state-space augmentation rule in Definition I.2.1. □

I.3 Physical-budget-saturated sourcewise completion

The closed continuum algebra contains every mathematical limit generated by an ordinary finite-budget orbit. Its topology is generated canonically by the equation, target, physical balance, and constructor grammar. Strong convergence of a particular nonlinear term is subsequently read as a graph invariant in this completion.

Definition I.3.1 (Structural test system)**.** For each nonempty $A \subseteq \mathfrak{C}$ and derived physical ceiling B_{phys}, let $\mathscr{R}_A^0(B_{\mathrm{phys}})$ be the raw orbit records with source support equal to A. The structural test system $\mathfrak{F}_A = (\phi_n^A)_{n\geq 1}$ is generated recursively from the countable coordinates (2). At every finite grammar depth it contains:

- (T1) the weak PDE pairings and source-current pairings;
- (T2) the target shells and target fluxes;
- (T3) every truncation of the physical-expenditure and entropy-production coordinates generated by the fixed balance;

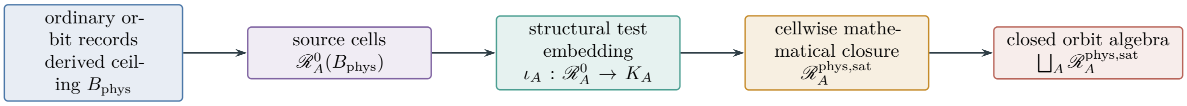

Figure 4: Sourcewise physical saturation. Closure is taken separately in each ancestry cell before the cells are assembled, and every cell retains the same physical ceiling B_{phys}.

(T4) every chart, quotient, boundary, and tail coordinate generated by the source grammar;
(T5) the total graphs of every finite algebraic operation generated by that grammar, encoded by rational distance-to-graph truncations.

An unbounded real-valued test is represented by its rational bounded truncations. Enumerating grammar depth and rational parameters gives one countable family with values in $[-1, 1]$. No map or topology may be appended after a row value is inspected. Failure of an operation to extend as an ordinary continuous map is recorded by the boundary of its total graph.

Definition I.3.2 (Physical-budget-saturated closed orbit algebra)**.** Let

$$\iota_A(r) = (\phi^A_n(r))_{n\geq 1} \in K_A := [-1, 1]^{\mathbb{N}}.$$

The space K_A is compact metrizable in the product metric. The sourcewise saturated cell and the full physical orbit algebra are

$$\mathscr{R}^{\text{phys,sat}}_A(B_{\text{phys}}) := \overline{\iota_A(\mathscr{R}^0_A(B_{\text{phys}}))}^{K_A}, \qquad \mathscr{R}^{\text{phys,sat}}(B_{\text{phys}}) := \bigsqcup_{\varnothing\neq A\subseteq\mathfrak{C}} \mathscr{R}^{\text{phys,sat}}_A(B_{\text{phys}}). \tag{25}$$

A point of the closure is a mathematical limit record. Its ancestry support is the cell label A. A finer ancestry word may also be retained when it is retained by the generated test system. The closure contains all sequential limits of ordinary orbit records seen by the generated tests. Here Src denotes provenance support, not the smallest support of an analytically simplified formula.

Remark I.3.3 (Why the disjoint union is retained)*.* The same analytic denotation can arise through different source histories. The closed algebra retains those histories as distinct records. Forgetting ancestry is an evaluation map from the disjoint union to the ambient analytic space; it is not part of the source decomposition.

Theorem I.3.4 (Source-preserving continuum completion)**.** *Every sequence (r_i) of ordinary orbit records with* $\mathrm{Src}(r_i) = A$ *that converges in the structural test topology has its limit record in* $\mathscr{R}^{\text{phys,sat}}_A(B_{\text{phys}})$*. Consequently*

$$\mathrm{Src}(\lim_i r_i) = A = \bigcup_i \mathrm{Src}(r_i). \tag{26}$$

If the sequence first passes through quotient or chart maps, Abs *or* Lift*, respectively, is appended before taking the limit. If the limit is read as a boundary or tail trace,* Bdry *is appended. Thus defect measures, oscillation variables, concentration profiles, closure directions, blow-up germs, and tails create no source outside* $\mathfrak{C}$*.*

Proof. By definition, $\mathscr{R}^{\text{phys,sat}}_A(B_{\text{phys}})$ is the sequential closure of the test image of the raw cell with provenance support A; compact metrizability makes this equal to its topological closure. Hence every limit of such a sequence belongs to that cell. The algebraic rules of Definition I.2.1

take unions of source supports and append the labels of quotient, chart, and boundary operations before the closure is formed. The new coordinates of a limit record are limits of tests on the same tagged sequence; they do not supply a new primitive letter. This proves (26) and each listed case. □

Corollary I.3.5 (Mixed sequences)**.** *Every sequence of raw records has source support in the finite set $2^{\mathfrak{C}} \setminus \{\varnothing\}$. It has a subsequence with constant source support. Each convergent such subsequence has its limit therefore contained in an existing saturated cell.*

Proof. There are only $2^5 - 1$ nonempty source supports. One support occurs infinitely often. Apply Theorem I.3.4 to that subsequence. □

Definition I.3.6 (Source realization map)**.** Let $\mathcal{P}_A$ and $\mathcal{J}_A$ be Hilbert spaces of potentials and currents for a fixed source support $A \subseteq \mathfrak{C}$. A *source realization map* is a densely defined closed operator

$$G_A : D(G_A) \subseteq \mathcal{P}_A \longrightarrow \mathcal{J}_A$$

whose range consists of the currents carrying an actual potential witness in that source cell. Its saturated current space is $\overline{\operatorname{Ran} G_A}^{\mathcal{J}_A}$. The closure has the same source support A, independently of whether every closure direction has a potential witness.

Theorem I.3.7 (Closed-range realization criterion)**.** *Let $G : D(G) \subseteq \mathcal{P} \to \mathcal{J}$ be a source realization map. The following conditions are equivalent.*
(R1) $\operatorname{Ran} G$ *is closed in* $\mathcal{J}$.
(R2) *There is* $\gamma > 0$ *such that*

$$\|G\phi\|_{\mathcal{J}}^2 \geq \gamma \operatorname{dist}_{\mathcal{P}}(\phi, \ker G)^2, \qquad \phi \in D(G). \tag{27}$$

(R3) *The restriction of* G^*G *to* $(\ker G)^\perp$ *has spectrum in* $[\gamma, \infty)$ *for some* $\gamma > 0$.
(R4) *The Moore–Penrose inverse* G^+ *is bounded on* $\operatorname{Ran} G$.
When these conditions hold, every saturated current in $\overline{\operatorname{Ran} G}$ *has a potential witness, unique modulo* $\ker G$. *When they fail, elements of* $\overline{\operatorname{Ran} G} \setminus \operatorname{Ran} G$ *remain limit records in the same source cell by Theorem I.3.4; failure of potential realization does not define another source family. In the reachability calculus this operator theorem is activated only after restriction to the selected target-local source cell; its complementary range value is then a polarity successor of that cell.*

Proof. The closed-range theorem for densely defined closed operators gives the equivalence of (R1), (R2), and boundedness of G^+. On $(\ker G)^\perp$, $\|G\phi\|^2 = \langle G^*G\phi, \phi\rangle$, so the spectral theorem makes (R2) equivalent to (R3). Closed range gives $\overline{\operatorname{Ran} G} = \operatorname{Ran} G$, and the kernel is exactly the ambiguity in a potential witness. The final assertion is Theorem I.3.4 applied to the source cell generated by G. □

Theorem I.3.8 (Target-relative gradient exhaustiveness)**.** *Let $G : D(G) \subseteq \mathcal{P} \to \mathcal{J}$ be a densely defined closed realization of one inherited source cell, and let $P_{\overline{G}}$ be the orthogonal projection onto $\overline{\operatorname{Ran} G}$. Then every $j \in \mathcal{J}$ has the unique split*

$$j = P_{\overline{G}} j + (I - P_{\overline{G}}) j, \qquad P_{\overline{G}} j \in \overline{\operatorname{Ran} G}, \quad (I - P_{\overline{G}}) j \in \ker G^*. \tag{28}$$

The first term belongs to the saturated cell generated by G, including when it lies in $\overline{\operatorname{Ran} G} \setminus \operatorname{Ran} G$. Relative to the target family Γ_Σ, the second term has the further canonical split into its class in $\ker G^* / (\ker G^* \cap \mathscr{N}_\Sigma)$ and its target-null part. Hence failure of closed range creates neither a new

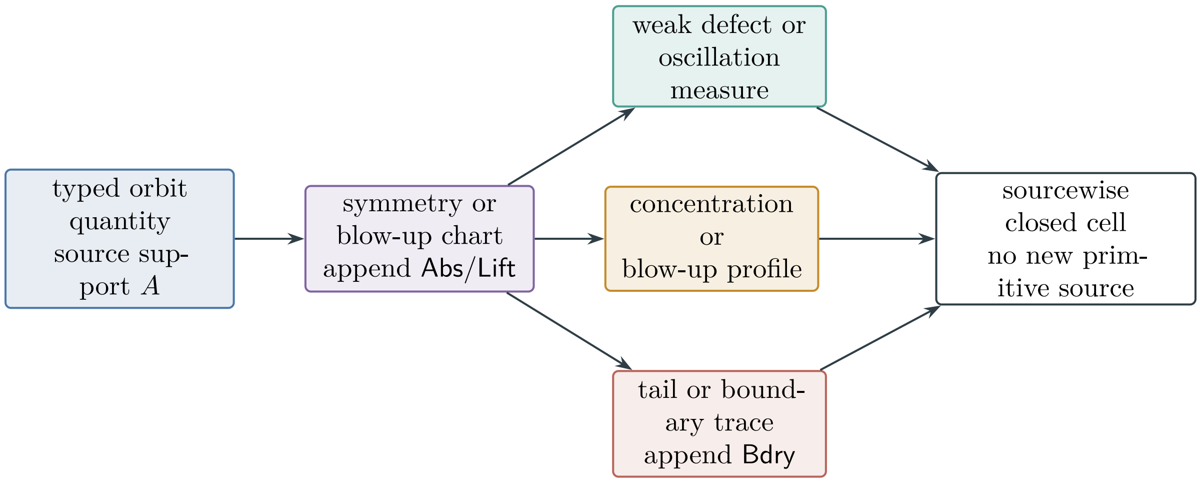

Figure 5: Continuum limit phenomena inherit ancestry. Normalization changes the source word in the declared Abs/Lift manner; a terminal trace appends Bdry. Weakness or concentration changes the analytic realization, not the source alphabet.

source nor a target-visible residual: it creates a closure direction in the inherited cell, while every orthogonal target-visible class remains a current of the native compiled equation and retains its native ancestry.

Proof. The range closure is a closed Hilbert subspace, so the projection theorem gives (28); the identity $(\operatorname{Ran} G)^{\perp} = \ker G^{*}$ identifies its complement. The definition of the sourcewise completion and Theorem I.3.4 assign $P_{\overline{G}}j$ to the same tagged cell as the approximating gradients. Quotienting the complementary space by its intersection with $\mathcal{N}_{\Sigma}$ is exact. Its kernel is target-null by definition, and a nonzero quotient class has a nonzero target pairing. Such a class cannot be renamed residual; its ancestry is the ancestry of the native current from which j was compiled by Theorem I.4.5. These alternatives exhaust $\mathcal{J}$. $\square$

Corollary I.3.9 (Regularized realization)**.** *For $j \in \mathcal{J}$ and $\lambda > 0$, the functional*

$$\phi \longmapsto \|G\phi - j\|_{\mathcal{J}}^{2} + \lambda \|\phi\|_{\mathcal{P}}^{2}$$

has a unique minimizer $\phi_\lambda \in D(G)$. The resulting current $G\phi_\lambda$ is a stable regularized realization, with the parameter λ recording its bias. Existence of this minimizer for every $\lambda > 0$ does not imply any of the equivalent conditions in Theorem I.3.7.

Proof. On $D(G)$, the form $\langle G\phi, G\psi\rangle + \lambda\langle \phi, \psi\rangle$ is closed and coercive in its form norm. The functional $\psi \mapsto \langle j, G\psi\rangle$ is continuous in that norm, so the Riesz representation theorem gives a unique minimizer. The final statement follows because adding λI supplies coercivity even when the unregularized range is not closed. $\square$

Remark I.3.10 (Structural and evaluator topologies)*.* The graph-complete state space closes every typed operation together with its defect coordinates. A stronger PDE topology is represented by adjoining its bounded seminorm and graph coordinates to the cylinder algebra. Compactness in that topology is then an ordered state-space assertion. Failure of the strong coordinate to be finite or closed is its typed graph value in the same source cell.

I.4 Target quotient and the null residual

Definition I.4.1 (Generated prospective target family)**.** Let $\mathscr{J}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})$ be the current-occurrence part of the closed orbit algebra. The finite-current locus is replaced, when necessary, by its Hausdorff locally convex envelope under the continuously extended addition and scalar multiplication; the kernel of the envelope map is retained as a tagged closure defect. Records at an infinite compactification endpoint are treated as boundary or tail records and are tested by their bounded readouts, not inserted into this vector space. If $\Sigma = \varnothing$, the target-shell algebra contains only constants and $\Gamma_\Sigma = \{0\}$. Otherwise let $\rho_\Sigma(x) = d_{\mathrm{ev}}(x, \Sigma)$, where Σ is the complete continuation boundary of Definition III.1.1. The *target-shell algebra* is the least countable algebra containing every rational ramp of ρ_Σ, its pullback by every ordinary generated covariance map, and the rational cylindrical approximants supplied by the coordinates (2).

For nonempty Σ, the prospective target family Γ_Σ is the closed normalized cone generated by every derived finite-cylinder shell current, together with the killed-flux and predictable-compensator differentials when their types occur in the fixed equation. In particular it contains

$$J \longmapsto \langle D\widehat{\phi}_{r,m,I}(z), J\rangle$$

on the ordinary branch and the corresponding Stieltjes, oscillation, concentration, exterior, or variation-boundary pairing on every other branch. Thus neither the probes nor their normalization are application data: they are generated by the continuation target, approximation tower, and verified covariance grammar.

Definition I.4.2 (Target-null residual)**.** The target-null space and target-active quotient are

$$\mathscr{N}_\Sigma := \bigcap_{\Lambda \in \Gamma_\Sigma} \ker \Lambda, \qquad \mathscr{J}_\Sigma := \mathscr{J}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})/\mathscr{N}_\Sigma. \tag{29}$$

The post-saturation residual is any closed current coordinate containing zero and satisfying

$$\mathscr{J}^{\mathrm{phys,sat}}_{J_{\mathrm{res}}} \subseteq \mathscr{N}_\Sigma. \tag{30}$$

It is an audit coordinate for target-invisible transport, not a sixth source family. A current is target-visible precisely when its quotient class in $\mathscr{J}_\Sigma$ is nonzero.

Theorem I.4.3 (Target quotient commutes with sourcewise completion)**.** *For every source support $A \subseteq \mathfrak{C}$,*

$$\overline{\mathscr{R}_A + \mathscr{N}_\Sigma}^{\,\mathrm{gc}}/\mathscr{N}_\Sigma = \overline{q_\Sigma(\mathscr{R}_A)}^{\,\mathscr{J}_\Sigma}.$$

The kernel $\mathscr{N}_\Sigma$ is closed under sourcewise limits, signed gluing, restriction, and every ordinary descendant map T whose covariance graph proves $T^*\Gamma_\Sigma \subseteq \Gamma_\Sigma$. Consequently a target-conditioned carrier, cohomology, moment, or pruning construction may be performed on $\mathscr{J}_\Sigma$ after ancestry has been fixed exactly on that descent cell. Every failed or noncovariant descent is retained instead as the target-visible defect of Theorem I.4.7.

Proof. Every element of Γ_Σ is a continuous cylinder-current coordinate on the graph-complete space. Its kernel is closed, and so is their intersection. The displayed identity is the standard quotient–closure identity for a continuous linear quotient by a closed kernel. Signed gluing and restriction preserve every shell pairing by their defining graph identities. An ordinary descendant map preserves the common kernel when $T^*\Gamma_\Sigma \subseteq \Gamma_\Sigma$: for $J \in \mathscr{N}_\Sigma$ and $\Lambda \in \Gamma_\Sigma$, $\Lambda(TJ) = (T^*\Lambda)(J) = 0$.

Without this covariance relation no such claim is made; the descent defect is a separate target-active coordinate. The displayed identity follows from the definition of the quotient topology after saturating the source cell by the kernel. Sourcewise completion precedes this saturation, so no ancestry tag is lost. □

Definition I.4.4 (Compiled continuum current)**.** A continuum current is *compiled* when it is generated from the native terms of the fixed equation by the finite constructor rules (S1)–(S6) of Definition I.2.1. Its evaluation has the normal form

$$J = J^{\mathsf{Geom}} + J^{\mathsf{Caus}} + J^{\mathsf{Abs}} + J^{\mathsf{Lift}} + J^{\mathsf{Bdry}} + J^{J_{\mathrm{res}}}, \qquad J^{J_{\mathrm{res}}} \in \mathscr{N}_\Sigma, \tag{31}$$

as an equality of weak current pairings. Empty components are allowed. The five displayed components are the head partition $J^c(J)$ of (17); within each component every atomic occurrence retains its exact ancestry support A. Thus the five additive current roles and the 31 provenance cells are compatible classifications, not competing decompositions. The residual is an optional regrouping in the common target kernel after the five source components have been evaluated; the choice $J^{J_{\mathrm{res}}} = 0$ is always admissible. It is not a constructor and carries no source letter.

Theorem I.4.5 (Continuum normal-form compilation)**.** *For every operational weak-PDE signature, totalize the native realization relation and adjoin the quotient, chart, auxiliary-state, trace, forcing, and boundary operations generated by Definition I.2.1. Every finite orbit-current expression is compiled and has* (31)*. A failed native realization additionally generates its target-active form-defect class with* $\{\mathsf{Geom}, \mathsf{Caus}\}$*-ancestry. After sourcewise completion, every finite target pairing has the same normal form. Thus five-channel exhaustion does not presume a gradient or sector realization.*

Proof. On the fixed-Caus ordinary branch, positivity of $\mathbb{K}$, the directed transport, and the exterior term give Geom, Caus, and Bdry, respectively. On the form ordinary branch, the sector-form compiler makes the same assignment to the symmetric mobility, antisymmetric directed, and killing terms. The universal induction in Theorem I.2.9 proves the claim for every orbit-current expression. A quotient appends Abs, a chart or auxiliary variable appends Lift, a trace appends Bdry, and finite combination takes the union of the already assigned supports. These exhaust the transformed ordinary branches.

On the native failure branch, retain the raw fixed bulk current in Caus and its totalized attempted split in the composite Geom–Caus form-defect cell, as in Theorem I.2.15. This additional case remains inside the five-source alphabet.

For a sourcewise convergent net, each prospective target functional is a coordinate of the structural test system. Its value therefore passes to the limit. By Theorem I.3.4, the limit remains in the same tagged source cell. The image in the quotient by $\mathscr{N}_\Sigma$ is therefore generated by the five cell images. Taking $J^{J_{\mathrm{res}}} = 0$, or regrouping any already identified target-null summand as $J^{J_{\mathrm{res}}}$, gives (31) without requiring a complemented kernel and without adding a primitive source. □

Theorem I.4.6 (No post-saturation target-visible residual)**.** *Let* $\mathfrak{P}_{\mathrm{phys}}$ *be generated by the compiled continuum algebra of Theorem I.4.5. Every target-visible current record in* $\mathscr{J}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})$ *has nonempty ancestry in* $\mathfrak{C}$*, and its target-active quotient is generated by the five head components, refined by the* 31 *possible nonempty exact-support cells of* (18)*. The residual source profile vanishes:*

$$G^{J_{\mathrm{res}},\mathrm{src,sat}}_{B_{\mathrm{phys}},\Sigma} := \sup_{J \in \mathscr{J}^{\mathrm{phys,sat}}_{J_{\mathrm{res}}}} \sup_{\Lambda \in \Gamma_\Sigma} (\Lambda J)_+ = 0. \tag{32}$$

If a provisional limit object pairs nontrivially with a target functional, then Theorem I.3.4 places it in the existing source cell inherited from its generating records. Thus every dynamically generated target-visible limit belongs to one of the five source families.

Proof. For raw occurrences, use (17) and apply the quotient map $q_\Sigma : \mathscr{J}^{\mathrm{phys,sat}} \to \mathscr{J}_\Sigma$ to (31). The residual term vanishes because it lies in $\mathscr{N}_\Sigma$. Continuity of every $\Lambda \in \Gamma_\Sigma$ passes the same identity to each sourcewise limit. The source-preserving theorem assigns every nonresidual limit term to the closure of its existing exact-support cell. Thus a nonzero quotient class lies in a five-head component and one of its 31 provenance refinements. Finally, every functional in Γ_Σ vanishes on $\mathscr{J}^{\mathrm{phys,sat}}_{J_{\mathrm{res}}}$, so both nested suprema in (32) are zero. $\square$

Theorem I.4.7 (Projection-first saturation)**.** *Let* G *be the continuum constructor grammar generated by the public equation, and let* $q_\Sigma : \mathscr{J}^{\mathrm{phys,sat}} \to \mathscr{J}_\Sigma$ *be the quotient by* $\mathscr{N}_\Sigma$*. Totalize the descent graph of every constructor* $T \in \mathsf{G}$*. On its ordinary descent cell set* $T_\Sigma(q_\Sigma J) := q_\Sigma(TJ)$*; on every other cell retain the first descent defect*

$$\mathcal{D}_{T,\Sigma}(J) := q_\Sigma(TJ) - T_\Sigma(q_\Sigma J) \tag{33}$$

with the ancestry of T *and the target covector that detects it. If* G_Σ *denotes the resulting projected grammar, then*

$$q_\Sigma(\mathrm{Sat}_{\mathsf{G}}\,\mathscr{J}) = \mathrm{Sat}_{\mathsf{G}_\Sigma}(q_\Sigma\mathscr{J})\,. \tag{34}$$

The identity holds sourcewise and with the physical carrier and same-origin coordinates retained. Thus the reachability calculus may quotient by $\mathscr{N}_\Sigma$ *before generating descendants. A failed commutation is a projected graph coordinate, so projection first loses no target-visible current.*

Proof. For a native current, (34) is the quotient of (31). Suppose it holds for the inputs of a constructor. On the ordinary descent graph, the definition of T_Σ gives the identity for its output. On the complementary graph, add $\mathcal{D}_{T,\Sigma}$; then $q_\Sigma TJ = T_\Sigma q_\Sigma J + \mathcal{D}_{T,\Sigma}(J)$ by (33). Source-preserving totalization retains the ancestry of the failed operation. This proves both inclusions for finite words by structural induction.

Every prospective target functional is a coordinate of the evaluation compactification. Hence q_Σ is continuous on each sourcewise total graph, and the two finite-word inclusions pass to sourcewise completion. Signed carrier gluing commutes with the quotient because it is performed before Jordan decomposition. Physical domination and same-origin relations are retained coordinates rather than quotient relations. This proves the completed identity. $\square$

Corollary I.4.8 (Source type and invariant value are independent)**.** *Each saturated source cell has a value in every applicable fast-track invariant: null or nonnull quotient, finite or infinite radius, positive or zero price, zero or nonzero equality remainder, zero or positive capacity, finite or infinite resistance, and divergent or finite cascade price. These values change neither the source word nor the target quotient. Failure of a compactness or resolvent invariant therefore returns the complementary invariant branch; it never creates an untyped or residual source.*

I.5 Prospective contact accountability

Target contact may produce a retrospective object, such as a singular germ or a completed hitting distribution. Source accountability is instead read from the evolution strictly before contact. The next three theorems give deterministic, semigroup, and stochastic versions of the same factorization.

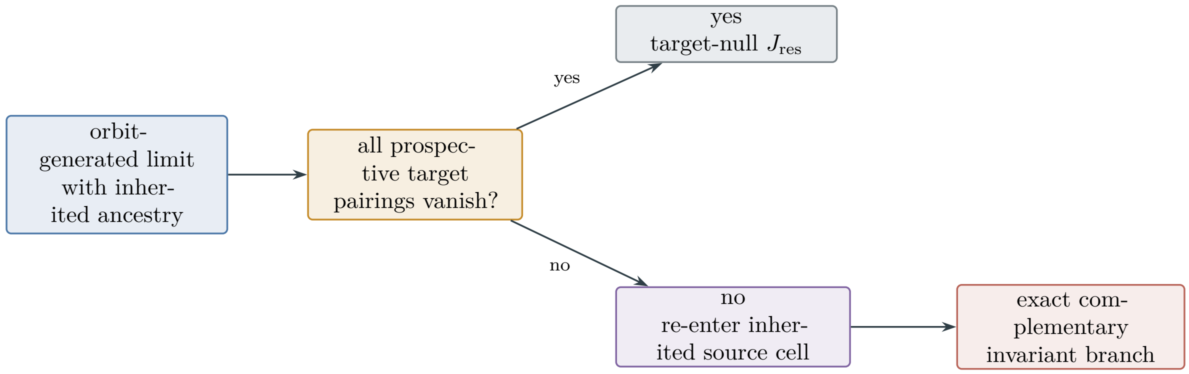

Figure 6: Target-visible limits retain their inherited source cell. The only residual output is target-null. A nonzero pairing fixes structural ancestry before any attempt is made to estimate the corresponding source profile.

Definition I.5.1 (Canonical target shells and total shell current)**.** If $\Sigma = \varnothing$, set $\tau_\Sigma = +\infty$, $\Gamma_\Sigma = \{0\}$, and generate no shell records. Suppose henceforth in this definition that $\Sigma \neq \varnothing$. In the compact metric space X_{ev}, put $\rho_\Sigma(x) = d_{\mathrm{ev}}(x, \Sigma)$, $U_r(\Sigma) = \{\rho_\Sigma < r\}$, and, for rational $r > 0$,

$$\Phi_r(x) = \begin{cases} 1, & \rho_\Sigma(x) \leq r, \\ 2 - \rho_\Sigma(x)/r, & r < \rho_\Sigma(x) < 2r, \\ 0, & \rho_\Sigma(x) \geq 2r. \end{cases} \tag{35}$$

Then Φ_r is continuous and r^{-1}-Lipschitz, $\bigcap_{r \in \mathbb{Q}_+} U_{2r}(\Sigma) = \Sigma$, and

$$\Phi_r = 0 \quad \text{on } X_{\mathrm{ev}} \setminus U_{2r}(\Sigma), \qquad \Phi_r = 1 \quad \text{on } U_r(\Sigma). \tag{36}$$

Adjoin the bounded coordinate Φ_r to the graph-complete algebra and its defining distance-ramp graph to the ordered relations. The rational cylinder algebra separates the resulting compact state space. By Stone–Weierstrass, for every $m \geq 1$ there is a rational cylinder polynomial $\phi_{r,m}$ such that

$$\|\phi_{r,m} - \Phi_r\|_\infty \leq 2^{-m}. \tag{37}$$

Choose the polynomial first with error $2^{-(m+2)}$. The two strictly positive cylinder polynomials $2^{-m} \pm (\phi_{r,m} - \Phi_r)$ then belong to the Archimedean quadratic module by the Archimedean Positivstellensatz [13]. The canonical choice is the first polynomial in the fixed enumeration accompanied by these two finite ordered identities. For a crossing interval $I = [a, b]$ and m with $\delta_{r,m}(I) := \phi_{r,m}(u_b) - \phi_{r,m}(u_a) > 0$, put

$$\widehat{\phi}_{r,m,I} := \frac{\phi_{r,m} - \phi_{r,m}(u_a)}{\delta_{r,m}(I)}.$$

The reciprocal is not inserted analytically: its total graph is generated by the bounded coordinate $h_{r,m,I}$ and the polynomial relations

$$\delta_{r,m}(I) h_{r,m,I} = 1, \qquad \widehat{\phi}_{r,m,I} = h_{r,m,I}(\phi_{r,m} - \phi_{r,m}(u_a)).$$

Since a crossing gives $\delta_{r,m}(I) \geq 1 - 2^{1-m} > 0$ for $m \geq 2$, this graph has one ordinary value and a uniform bound. The event-normalized cylinder shell therefore has endpoint increment one on every crossing record.

On every finite approximation record, the weak evolution and the typed graph chain rules derive a signed Stieltjes current

$$\mathrm{d}\widehat{\phi}_{r,m,I}(u_t) = \sum_{c\in\mathfrak{C}} \mathrm{d}Q^c_{r,m,I}(t). \tag{38}$$

Discrete time steps contribute their exact jump measures. A failed product, trace, inverse, chart, or covariance rule contributes the graph current of its least failed typed operation to the corresponding inherited source cell. There is no undifferentiated shell remainder.

The current coordinates are

$$Q^c_{r,m,I}(\psi) := \int_I \psi \, \mathrm{d}Q^c_{r,m,I}, \qquad \psi \in C^1_{\mathbb{Q}}(I).$$

Their sourcewise graph closure defines distributional shell currents on X_{gc}. On the bounded-variation branch they are finite signed Radon measures with their Lebesgue, jump, and Cantor parts retained. The provenance disintegration additionally retains oscillation, concentration, and exterior coordinates inherited from the approximating state records. If their total variation is unbounded, the normalized current and the compactified variation coordinate

$$\left(\frac{Q^c}{1+\|Q^c\|_{\mathrm{TV}}}, \frac{\|Q^c\|_{\mathrm{TV}}}{1+\|Q^c\|_{\mathrm{TV}}}\right)$$

give the typed variation-boundary value. The five totals are compactified *jointly*, rather than summed after separate compactification. Fix the enumeration of $\mathfrak{C}$, write

$$q = (q_c)_{c\in\mathfrak{C}}, \qquad q_c = Q^c_{r,m,I}(1), \qquad \iota(x) = \frac{x}{1+|x|},$$

and, on every finite approximation, retain the affine shell graph

$$\sum_{c\in\mathfrak{C}} q_c = 1, \qquad o(q) := \min\{c : q_c \geq 1/5\}. \tag{39}$$

The set defining $o(q)$ is nonempty because the five signed coordinates sum to one. The *total shell-current graph* is the closure of the joint records

$$((Q^c)_{c\in\mathfrak{C}}, (\iota(q_c))_{c\in\mathfrak{C}}, o(q)) \quad \text{subject to (39)}. \tag{40}$$

After passage to a subsequence the finite owner is constant. Consequently every value of the total graph has a source owner $o \in \mathfrak{C}$. Write $\overline{q}_c := \iota^{-1}(\lim \iota(q_c)) \in [-\infty, +\infty]$, using the continuous order-compactification inverse. Then

$$\overline{q}_o \in [1/5, +\infty], \qquad \text{equivalently} \qquad \iota(\overline{q}_o) \in [1/6, 1]. \tag{41}$$

If all five totals are finite, the closed affine relation also gives $\sum_c \overline{q}_c = 1$. At an infinite-variation endpoint no sum of extended signed quantities is formed; the retained owner relation (41) is the exact target-aligned information which survives closure. Thus cancellation cannot turn an $\infty - \infty$ expression into a fictitious residual current.

The contact time is $\tau_\Sigma(u) = \inf\{t : u(t) \in \Sigma\}$ for a state target and is $T_*(u)$ for the singular-event target when an ordinary same-origin maximal path has no ordinary regular extension. The latter event is physical even when its terminal analytic graph coordinate is nonordinary. An arbitrary source-complete terminal record remains nonphysical until its preterminal path and noncontinuation flag pass Theorem I.1.7. A completed physical contact record has the following exhaustive shellwise form. For each sufficiently small r, either the shell is entered before contact, in which case there are strictly pre-contact times

$$0 \le t_r^- < t_r^+ < \tau_\Sigma(u), \qquad u(t_r^-) \notin U_{2r}(\Sigma), \qquad u(t_r^+) \in U_r(\Sigma). \tag{42}$$

or it is not entered before contact, in which case the same relations hold with $t_r^+ = \tau_\Sigma(u)$, and the endpoint increment is stored as the Bdry-headed jump current of the completed path. For a source-complete terminal record not attached to an ordinary same-origin preterminal path, the same alternatives are statements only in $\widehat{\Sigma}_{\mathrm{sing}}$; they generate prospective currents but no physical-contact conclusion. For a realized singular event, the completed same-origin restrictions enter every sufficiently small event shell before or at the terminal time. The metric shells separate every point outside the closed target, while contact supplies the inner endpoint. A target at infinity is already a boundary point of X_{ev}; an additional ordinary normalization merely appends its $\mathsf{Abs}/\mathsf{Lift}$ ancestry.

Proposition I.5.2 (Shell cofinality and realization fidelity)**.** *The canonical metric shells are cofinal for the closed source-complete target $\widehat{\Sigma}_{\mathrm{sing}}$. Their restriction to a same-origin fiber of an ordinary maximal preterminal path detects physical noncontinuation exactly through its discrete continuation-status coordinate. Hence shell crossing in the source-complete graph space is sufficient for prospective source accounting, whereas physical singular contact additionally requires either that realized preterminal event or an ordinary realization of a state target. No graph-boundary current, unattached compactified germ, or shell limit supplies such realization by itself.*

Proof. Cofinality is the metric identity $\bigcap_{r\in\mathbb{Q}_+} U_{2r}(\widehat{\Sigma}_{\mathrm{sing}}) = \widehat{\Sigma}_{\mathrm{sing}}$. On a singular-event fiber the preterminal path and its origin are ordinary, and the discrete continuation-status coordinate separates extendible from nonextendible maximal paths; no claim is made that the terminal analytic graph coordinate is ordinary. For a state target, ordinary realization follows from Theorem I.1.7. Outside these fibers the same shells see only the source-preserving closure, so only prospective accountability is sound. □

Theorem I.5.3 (Total normalized shell-crossing accountability)**.** *Let u be an ordinary maximal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$, on the contact branch. For every sufficiently small r, choose the completed crossing interval $I_r = [t_r^-, t_r^+]$, where $t_r^+ < \tau_\Sigma$ on a pre-contact entry and $t_r^+ = \tau_\Sigma$ on a terminal entry, and then $m \ge 2$. Equation* (37) *gives $\delta_{r,m}(I_r) \ge 1 - 2^{1-m} > 0$. Write the physical current as*

$$J = \sum_{c\in\mathfrak{C}} J^c + J^{J_{\mathrm{res}}},$$

where $J^{J_{\mathrm{res}}} \in \mathscr{N}_\Sigma$. Evaluate the derived cylinder-shell current on I_r. Its ordinary signed source progress is

$$Q_{c,r,m}(u) := \int_{I_r} \langle D\widehat{\phi}_{r,m,I_r}(u_t), J_t^c \rangle \,\mathrm{d}t. \tag{43}$$

On every retained realization of this same-origin crossing branch, the shell observable has unit endpoint increment. If the five totals are finite, the exact accountability identity is

$$\boxed{1 = \widehat{\phi}_{r,m,I_r}(u(t_r^+)) - \widehat{\phi}_{r,m,I_r}(u(t_r^-)) = \sum_{c\in\mathfrak{C}} Q^c_{r,m,I_r}(1).} \tag{44}$$

On the ordinary branch,

$$\int_{I_r} \langle D\widehat{\phi}_{r,m,I_r}(u_t), J_t^{J_{\mathrm{res}}}\rangle \,\mathrm{d}t = 0. \tag{45}$$

On every finite-value realization of this same-origin crossing branch,

$$\sum_{c\in\mathfrak{C}} \big(Q^c_{r,m,I_r}(1)\big)_+ \geq 1, \qquad \max_{c\in\mathfrak{C}} \big(Q^c_{r,m,I_r}(1)\big)_+ \geq \frac{1}{5}. \tag{46}$$

On every realization, including an infinite-variation boundary value, the joint shell graph supplies a canonical owner $o_{r,m,I_r} \in \mathfrak{C}$ with

$$\overline{Q}^{\,o_{r,m,I_r}}_{r,m,I_r}(1) \in [1/5, +\infty]. \tag{47}$$

The positive contribution may be ordinary, jump, Cantor, oscillation, concentration, exterior, or variation-boundary current, but it always has the ancestry of one of the five source cells. Thus, without requiring a classical chain rule,

$$\boxed{\begin{gathered} u \textit{ contacts } \Sigma \quad \Longrightarrow \\ \forall r \textit{ sufficiently small } \exists \textit{ a five-channel source occurrence } R_r : \operatorname{Prog}_\Sigma(R_r) > 0. \end{gathered}} \tag{48}$$

Thus contact itself exposes a target-aligned occurrence in an existing source cell at every sufficiently small shell. It is strictly pre-contact when that shell is entered before contact and is a Bdry*-headed endpoint occurrence otherwise.*

Proof. The cylinder approximation gives the stated positivity of $\delta_{r,m}(I_r)$, and event normalization gives the unit increment. On every finite approximation, (39) gives the source sum. On the ordinary branch, insert the current normal form. The generated shell differential belongs to Γ_Σ, so residual target-nullity proves (45). Taking positive parts of a five-term signed sum equal to one proves (46). Along any convergent net of finite records, choose the canonical owner and then a subnet on which its value is constant. Closedness of $\iota(q_o) \geq 1/6$ proves (47) without forming an extended signed sum. The first-failure theorem and sourcewise closure assign the owner its inherited nonempty support. A terminal endpoint increment is the completed path's boundary jump and therefore has Bdry head. ☐

Corollary I.5.4 (No spontaneous deterministic contact)**.** *Let u be an ordinary maximal orbit generated by the compiled continuum algebra, with $u(0) \notin \Sigma$. Suppose that every ordinary and graph-boundary derived shell current is target-null on every completed crossing record, including a possible terminal boundary jump. On the ordinary approach branch this reads*

$$\langle D\widehat{\phi}_{r,m,I}(u_t), J_t^c\rangle = 0 \quad \textit{for a.e. } t < \tau_\Sigma(u), \quad c \in \mathfrak{C}. \tag{49}$$

Then u cannot contact Σ. Target avoidance also follows when every derived source-current mass is nonpositive. For a singularity at infinity, these statements apply to the canonical boundary shells in the generated evaluation compactification.

Proof. If contact occurred, Theorem I.5.3 would produce a positive source-resolved derived current on every sufficiently small shell. This contradicts the stated target-nullity. Under the stronger ordinary zero-pairing condition, the same conclusion follows directly from (39); on an infinite-variation endpoint it follows from the retained owner floor (41). ☐

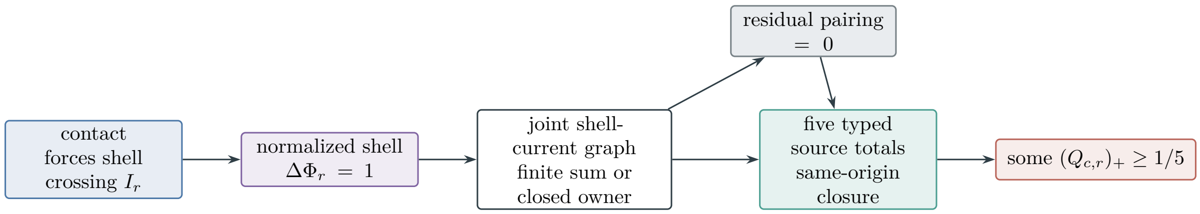

Figure 7: Normalized shell-crossing accountability. Contact forces a unit increase of the shell observable along the actual pre-contact current. The target-null residual contributes zero, so one of the five source integrals has positive part at least 1/5.

Corollary I.5.5 (Normalized contact at infinity)**.** *On the branch where singular contact is defined after generated normalizations $v_r = \mathcal{S}_r u$ into a state or germ compactification carrying normalized shells $\widetilde{\Phi}_r$. If the normalized current $\widetilde{J}_r$ has the compiled normal form, including every chart derivative with its* $\mathsf{Abs}/\mathsf{Lift}$ *ancestry, then every normalized crossing interval satisfies*

$$1 = \sum_{c\in\mathfrak{C}} \int_{I_r} \langle D\widetilde{\Phi}_r(v_r(t)), \widetilde{J}^c_{r,t}\rangle \,\mathrm{d}t, \qquad \max_{c\in\mathfrak{C}} \left(\int_{I_r} \langle D\widetilde{\Phi}_r(v_r(t)), \widetilde{J}^c_{r,t}\rangle \,\mathrm{d}t \right)_+ \geq \frac{1}{5}.$$

This holds when the five current totals are finite. On an infinite-variation value the joint normalized shell graph instead retains an owner $o \in \mathfrak{C}$ with compactified total $\overline{Q}^o(1) \geq 1/5$; no extended signed sum is used. Thus blow-up at infinity cannot occur without target-aligned structure in an existing source cell.

Proof. Apply Theorem I.5.3 in the normalized state space. Source preservation under the legal chart is Theorem I.3.4. □

Theorem I.5.6 (Resolvent source-response comparison)**.** *Let L and L_0 be closed killed generators on the same Banach space, and let $\lambda \in \rho(L) \cap \rho(L_0)$. Write*

$$R_L(\lambda) = (\lambda - L)^{-1}, \qquad R_0(\lambda) = (\lambda - L_0)^{-1}.$$

Totalize domain membership of $R_0(\lambda)k_\Sigma$ and the generator difference. On their ordinary branch,

$$L - L_0 = \sum_{c\in\mathfrak{C}} K^c + K^{J_{\mathrm{res}}} \tag{50}$$

on that vector. Then, for every continuous initial functional ℓ_0,

$$\begin{aligned} \langle \ell_0, (R_L(\lambda) - R_0(\lambda))k_\Sigma\rangle = &\sum_{c\in\mathfrak{C}} \langle R_L(\lambda)^*\ell_0, K^c R_0(\lambda)k_\Sigma\rangle \\ &+ \langle R_L(\lambda)^*\ell_0, K^{J_{\mathrm{res}}} R_0(\lambda)k_\Sigma\rangle. \end{aligned} \tag{51}$$

If the last pairing is target-null, it vanishes. If the real part p of the left side is positive, some existing source cell has real pairing at least $p/5$. Any domain or generator-split failure is returned by the form/realization row and appears only as the ordinary value of that specialized row. The factorization is used for reachability only on the retained shell/source cell; the five displayed summands inherit the source words of (50).

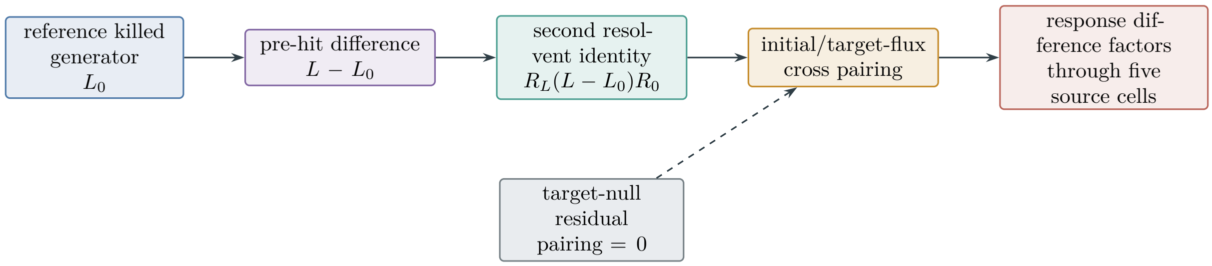

Figure 8: Resolvent source-response comparison. A discounted target-response difference is sandwiched around the pre-hit generator difference and therefore inherits its five-source decomposition. The no-spontaneous-contact theorem itself comes from normalized shell crossing.

Proof. The second resolvent identity gives

$$R_L(\lambda) - R_0(\lambda) = R_L(\lambda)(L - L_0)R_0(\lambda).$$

Pair with ℓ_0 and k_Σ, then insert (50). Target-nullity removes the residual term. A positive sum of five real numbers has a summand at least one fifth of that sum. $\square$

Remark I.5.7 (Arrival semantics). Equation (51) is an operator identity. It becomes a difference of discounted contact responses when the killed semigroups admit the target-flux realization of Theorem II.5.4. No probabilistic meaning is inferred from the algebraic resolvent alone.

Theorem I.5.8 (Stochastic no-spontaneous-contact accountability)**.** *Let τ_Σ be a stopping time and $N_t = \mathbf{1}_{\{\tau_\Sigma \le t\}}$. Totalize the Doob–Meyer and source-split relations. On the branch where $N - A$ is a martingale, the predictable compensator on $[0, T]$ has the finite-variation decomposition*

$$A = \sum_{c \in \mathfrak{C}} A^c + A^{J_{\mathrm{res}}}, \tag{52}$$

where all measures are supported on $\{t < \tau_\Sigma\}$, the signed source terms are integrable, every term starts from zero, $\mathbb{P}\{\tau_\Sigma = 0\} = 0$, and $\mathbb{E}A_T^{J_{\mathrm{res}}} = 0$. Then

$$\mathbb{P}\{\tau_\Sigma \le T\} = \sum_{c \in \mathfrak{C}} \mathbb{E}A_T^c. \tag{53}$$

If $p_T := \mathbb{P}\{\tau_\Sigma \le T\} > 0$, one pre-contact source compensator satisfies $\mathbb{E}A_T^c \ge p_T/5$. In particular, all five source expectations cannot vanish when contact has positive probability. Failure of the martingale, finite-variation, support, or integrability relation is its stochastic total-graph defect and remains in the fluctuation row.

Proof. Taking expectations of the integrable martingale $N - A$, using the stated zero initial values, gives $\mathbb{E}N_T = \mathbb{E}A_T$. Insert (52) and use the zero expectation of the residual term. The final assertion follows from the five-term pigeonhole argument. $\square$

Corollary I.5.9 (Continuum prospective source exhaustion)**.** *Normalized deterministic contact forces a positive completed-crossing occurrence with ancestry in $\mathfrak{C}$; it is pre-contact on the approach branch and* Bdry*-headed on a terminal jump. Positive-probability stochastic contact forces a predictable pre-contact occurrence. A positive killed-resolvent response difference has the same source-extraction property. The post-saturation residual contribution is zero in all three interfaces.*

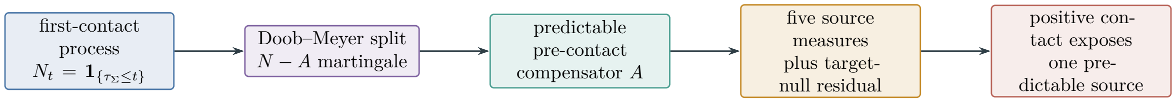

Figure 9: Stochastic no-spontaneous-contact accountability. First contact is charged by its predictable pre-contact compensator. A positive contact probability forces a positive contribution from one of the five source measures before the stopping time.

Definition I.5.10 (Saturated prospective source profile)**.** Fix one of the three prospective interfaces, restricted to records satisfying its stated target-null residual condition, and fix its normalization. Write $\mathrm{Prog}^c_\Sigma(R)$ for the signed c-source contribution: $Q_{c,r}$ for a deterministic shell, $\mathbb{E}A^c_T$ for a stochastic contact record, and the real c-pairing in (51) for a resolvent comparison. For a deterministic infinite-variation boundary record, use its joint shell owner o and set

$$\mathrm{Prog}^c_\Sigma(R) := \begin{cases} \overline{Q}^o(1), & c = o, \\ 0, & c \neq o. \end{cases} \tag{54}$$

This is a target-aligned readout of the closed affine graph, not a proposed sum of the underlying extended signed currents. For finite additive records set

$$\mathrm{Prog}_\Sigma(R) := \sum_{c \in \mathfrak{C}} \mathrm{Prog}^c_\Sigma(R). \tag{55}$$

For a boundary-owner record, use the same notation for the sum in (54), namely $\mathrm{Prog}_\Sigma(R) = \overline{Q}^o(1) \geq 1/5$. Define

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,c}(B_{\mathrm{phys}}) := \sup_{R \in \mathscr{R}^{\mathrm{phys,sat}}(B_{\mathrm{phys}})} \left(\mathrm{Prog}^c_\Sigma(R)\right)_+, \tag{56}$$

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma}(B_{\mathrm{phys}}) := \sum_{c \in \mathfrak{C}} \mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,c}(B_{\mathrm{phys}}). \tag{57}$$

The supremum is taken over the ordinary graph values and the owner-lifted boundary values of the selected interface and may equal $+\infty$. It is also saturated over every universal expanded composite record of Definition III.4.27: the profile sees the joint value before any postprocessing, normalization, averaging, or residual quotient. Inputs outside an ordinary graph enter the corresponding totalized defect cell, so every input and every target-visible cross term has a structural value in an inherited source cell. This is the direct continuum analogue of Part I's saturated profile over complete represented derivations.

Definition I.5.11 (Source-support and realization-mode profiles)**.** For each nonempty $A \subseteq \mathfrak{C}$, set

$$\mathrm{Prog}^A_\Sigma(R) := \sum_{c \in A} \mathrm{Prog}^c_\Sigma(R) \mathbf{1}_{\{\mathrm{Src}(R) = A\}}.$$

Every compiled record also carries the subset of realization-mode flags

$$\mathfrak{M} = \{\mathsf{exact}, \mathsf{closure}, \mathsf{harmonic}, \mathsf{concentration}, \mathsf{boundary}\}$$

determined by its construction:
(M1) exact: it has a witness in the range of its native realization map;
(M2) $\mathsf{closure}$: it lies in the sourcewise range closure, possibly outside the range;

(M3) harmonic: its class in $\ker G_A^*/(\ker G_A^* \cap \mathscr{N}_\Sigma)$ is nonzero;

(M4) concentration: it is read through a Young measure, defect measure, profile, or blow-up-state lift;

(M5) boundary: it is a trace, killing, tail, terminal germ, or graph-boundary record.

Flags may coexist; they refine ancestry and do not form new sources. For bookkeeping fix the order

$$\mathsf{exact} < \mathsf{closure} < \mathsf{harmonic} < \mathsf{concentration} < \mathsf{boundary}$$

and let $m_*(R)$ be the maximal flag carried by R. Thus the complete flag set is retained, while every record belongs to exactly one *primary-mode* cell. Generated operations only append flags, so m_* is monotone along a successor path and hence eventually constant. For $m \in \mathfrak{M}$, define

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,A,m}(B_{\mathrm{phys}}) := \sup_{\substack{R \in \mathscr{R}^{\mathrm{phys,sat}}_A(B_{\mathrm{phys}}) \\ m_*(R)=m}} \left(\mathrm{Prog}^A_\Sigma(R)\right)_+, \tag{58}$$

with supremum 0 for an empty class.

Theorem I.5.12 (Realization-mode exhaustion)**.** *Every target-active continuum record belongs to a nonempty source cell $A \subseteq \mathfrak{C}$, has at least one flag in $\mathfrak{M}$, and has exactly one primary mode. Moreover,*

$$\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma}(B_{\mathrm{phys}}) \le \sum_{\varnothing \ne A \subseteq \mathfrak{C}} \sum_{m \in \mathfrak{M}} \mathsf{T}^{\mathrm{phys,sat}}_{\Sigma,A,m}(B_{\mathrm{phys}}). \tag{59}$$

Thus exact currents, nonclosed-range limits, harmonic classes, concentrations, and boundary defects exhaust the analytic realization modes of target progress without enlarging the five-source alphabet.

Proof. Native records are exact. A sourcewise limit is in the closure mode. The orthogonal decomposition $\mathcal{J}_A = \overline{\operatorname{Ran} G_A} \oplus \ker G_A^*$, followed by the target quotient, assigns every target-active orthogonal record to the harmonic mode. The compiler marks every profile lift and every boundary operation with the last two flags. These cases cover the finite syntax and its sourcewise completion. Every positive source contribution is therefore bounded by at least one term on the right of (59); summing gives the result. Multiple flags only enlarge the right side. □

Theorem I.5.13 (Source-only prospective envelope)**.** *Every record in the domain of a prospective interface satisfies*

$$\left(\mathrm{Prog}_\Sigma(R)\right)_+ \le \mathsf{T}^{\mathrm{phys,sat}}_{\Sigma}(B_{\mathrm{phys}}). \tag{60}$$

The envelope consists solely of the five source profiles. Any finite upper bounds on those profiles give a complete prospective progress bound. A finite-value normalized deterministic contact record satisfies $\mathrm{Prog}_\Sigma(R) = 1$*; an infinite-variation contact record has owner progress at least* $1/5$*.*

Proof. On a finite additive record, each of Theorems I.5.3, I.5.6 and I.5.8 has, after removal of its target-null residual pairing, the form

$$\mathrm{Prog}_\Sigma(R) = \sum_{c \in \mathfrak{C}} \mathrm{Prog}^c_\Sigma(R).$$

Take the positive part, bound every signed summand by its positive part, and then apply (56). For a deterministic variation-boundary record, (54) has a single nonzero source coordinate, so the same bound follows without summing extended signed currents. Every other total-graph defect is evaluated in its inherited source cell by its bounded target readout, to which the same one-coordinate argument applies. □

Definition I.5.14 (Singularity-aligned structural support)**.** For a pre-contact record R in any prospective interface, define

$$\mathrm{Align}_\Sigma(R) := \{c \in \mathfrak{C} : \mathrm{Prog}^c_\Sigma(R) > 0\} \cup \begin{cases} \mathrm{Src}(R), & R \text{ is a total-graph defect with } \mathrm{Prog}_\Sigma(R) > 0, \\ \varnothing, & \text{otherwise.} \end{cases} \tag{61}$$

Equivalently, in the deterministic interface an occurrence is aligned when the singularity-descending covector $D\Phi_r(u_t)$ has positive integrated pairing with its source current on the crossing interval. The word "aligned" is target-relative: it does not rename a source as Caus, and it is evaluated only along the fixed equation.

Theorem I.5.15 (No-spontaneous-progress principle)**.** *For every prospective record, including total-graph defect records,*

$$\mathrm{Prog}_\Sigma(R) > 0 \quad \Longrightarrow \quad \mathrm{Align}_\Sigma(R) \neq \varnothing. \tag{62}$$

For an ordinary five-term additive record one also has

$$\max_{c\in\mathfrak{C}} \big(\mathrm{Prog}^c_\Sigma(R)\big)_+ \geq \frac{\mathrm{Prog}_\Sigma(R)}{5}. \tag{63}$$

If every ordinary source occurrence and every total-graph defect represented by R lies in $\mathscr{N}_\Sigma$, then

$$\mathrm{Prog}_\Sigma(R) = 0, \tag{64}$$

regardless of the size, oscillation, or compactness of those target-null currents. More generally, if every source contribution is nonpositive, then positive target progress is impossible. For a normalized deterministic shell crossing, the derived Stieltjes-current identity yields a positive contribution of at least $1/5$ in one inherited source cell on every ordinary or graph-boundary branch.

Proof. Equation (55) is the compiled five-term identity on finite additive records. A positive finite sum has a positive summand, and its largest positive part is at least one fifth of the sum. On a variation-boundary record, the owner convention (54) already supplies a positive summand with inherited nonempty support; no extended sum is asserted. Source-preserving graph closure assigns every other positive defect readout its inherited support. If every occurrence lies in the common target kernel, each ordinary or bounded defect readout vanishes. The shell assertion is Theorem I.5.3. $\square$

Corollary I.5.16 (No-spontaneous-contact regularity)**.** *Let the target have a normalized shell presentation, and let every legal orbit be generated by the compiled algebra on its regular pre-contact interval. If* $\mathrm{Align}_\Sigma(R) = \varnothing$ *for every legal shell record, then the target is unreachable. In particular, if all five source currents have zero class in $\mathscr{J}_\Sigma$ along every completed crossing record, their ancestry is retained while their target contribution lies in $\mathscr{N}_\Sigma$, and contact is impossible:*

$$\boxed{q_\Sigma(J^c) = 0 \textit{ on every completed target-shell crossing for all } c \in \mathfrak{C} \quad \Longrightarrow \quad \tau_\Sigma = \infty.} \tag{65}$$

This conclusion requires neither compactness of the full orbit nor a positive physical-price estimate.

Proof. If an orbit contacted Σ, Theorem I.5.3 would provide a normalized shell record with unit progress. By Theorem I.5.15, its aligned support would be nonempty, contrary to the displayed null-source condition. Under that quotient value, every shell source integral vanishes, so the same contradiction follows directly from (44). $\square$

Remark I.5.17 (Why the target quotient is stronger than a full norm bound). A classical energy estimate commonly controls every component of a current, including directions irrelevant to the singular target. The quotient $\mathscr{J}_\Sigma$ discards a current only after all declared prospective target pairings vanish. The discarded current may be large or noncompact; no estimate of it is needed for (64). Only the target-aligned quotient directions proceed to resistance, compactness, rigidity, or physical-price invariants. The quotient strictly reduces the regularity problem because target-null directions are absent from all downstream estimates.

Part II
Physical Structural-Invariant Calculus

II.1 Structural invariant reduction of analytic arguments

Remark II.1.1 (Ordinary-branch convention for analytic modules). Every conditional identity in this part describes the ordinary value of a total graph generated from the public signature. Its operator, form, chart, or multiplier must already occur in the typed syntax or in the fixed target-independent textbook library. The condition is never appended as specialization data: when it does not hold, its complementary graph value is sent to the activity lift and successor relation. Hamiltonian, commutator, compactness, tail, rigidity, and entropy modules can shorten a pruning transcript, but the core contact-to-successor theorem does not require their ordinary values.

The closed source algebra separates the origin of a continuum quantity from the estimate used to control it. Once a source family has been placed in a closed target-null quotient or removed from the admissible orbit family by a proved physical exclusion, later invariants are evaluated on the remaining reduced space. The analytic calculation then has only three possible jobs: close a remaining class, compare two nonnegative forms, or resolve the equality set of a sharp comparison.

Definition II.1.2 (Quotient-compatible structural reduction)**.** Let Y be a Hausdorff locally convex space of currents, profiles, or limit records, and let $K \subseteq Y$ be a closed linear subspace already annihilated by the downstream target and invariant maps. Write

$$\pi_K : Y \longrightarrow Y/K.$$

A downstream map, form, evaluation closure, or source transport is *compatible with the reduction* when it is constant on cosets of K and therefore factors continuously through π_K. A previously excluded closed family is removed from the admissible domain before the quotient is formed. Thus a reduction never discards a target-visible family without the proved invariant identity that excludes that family.

Theorem II.1.3 (Propagation of structural reductions)**.** *Let $K_0 \subseteq K_1 \subseteq Y$ be closed subspaces. Totalize every later map relative to both reductions. On the zero descent-defect branch, the canonical map is a topological linear isomorphism*

$$(Y/K_0)/(K_1/K_0) \cong Y/K_1. \tag{66}$$

On a nonzero descent-defect branch that defect is retained in its inherited source cell. Consequently a source direction closed at one stage may be propagated into the next quotient and omitted from every later compatible calculation. A later invariant computed after reduction has the same value as the corresponding invariant on Y after quotienting by all previously closed directions.

Proof. The quotient map $Y/K_0 \to Y/K_1$ is continuous, open, and has kernel K_1/K_0. The topological first isomorphism theorem gives (66). Every zero-defect downstream object factors through these quotient maps, so its reduced and unreduced evaluations agree under the displayed isomorphism. A failure to factor is the nonzero descent-defect branch stipulated in the theorem. ☐

Theorem II.1.4 (Target-quotient coercivity)**.** *Let Y be a Hilbert source space, let $K \subseteq Y$ be a closed target-null subspace, and let $T : Y \to Z_\Sigma$ be bounded with $K \subseteq \ker T$. Let $q : Y \to [0, \infty]$ be an extended nonnegative quadratic form with finite domain $D(q)$ satisfying $D(q) + K \subseteq D(q)$, and define its relaxed quotient form by*

$$q_{\mathrm{red}}([y]) := \inf_{k \in K} q(y + k), \qquad [y] \in Y/K. \tag{67}$$

If, on the required reduced domain,

$$\|[y]\|^2_{Y/K} \le C q_{\mathrm{red}}([y]), \tag{68}$$

then

$$\|Ty\|^2_{Z_\Sigma} \le \|\widetilde{T}\|^2 C q_{\mathrm{red}}([y]) \le \|\widetilde{T}\|^2 C q(y), \tag{69}$$

where $\widetilde{T} : Y/K \to Z_\Sigma$ is the descended target map. No bound for the full norm $\|y\|_Y$ follows or is needed. When invoked by the regularity calculus, Y/K, q_{red}, and T are first restricted to the contact-generated active source cell; failure of (68) *is its complementary local structural value.*

Proof. Because T annihilates K, the quotient universal property gives the bounded map $\widetilde{T}$ with $T = \widetilde{T}\pi_K$. Hence

$$\|Ty\| \le \|\widetilde{T}\| \, \|[y]\|_{Y/K}.$$

Apply (68); the final inequality follows by using $k = 0$ in (67). ☐

Example II.1.5 (Strict gain over full-state coercivity)**.** Let $Y = K \oplus Z_\Sigma$, let $T(k, z) = z$, and let $q(k, z) = \|z\|^2$. Then (68) holds with $C = 1$, while no inequality $\|(k, z)\|^2_Y \le C' q(k, z)$ can hold when $K \ne \{0\}$. The sequence $(nk_0, 0)$ may diverge in the full state norm but is exactly target-null. Thus the quotient theorem proves every target estimate available from q without controlling that divergent motion.

Definition II.1.6 (Structural evaluation invariants)**.** Work on a zero-descent-defect reduced space Y_{red}, with a continuous target projection $\Pi_\Sigma : Y_{\mathrm{red}} \to Z_\Sigma$. The space and every family below are restricted to one propagated same-origin target/source cell: a fixed nonempty source support, realization mode, target-shell floor, and all earlier zero transcript coordinates. No extremum in this definition ranges over a source-transverse state. The structural invariants used below have the following forms.

(I1) For a retained target/source family $\mathcal{F} \subseteq Y_{\mathrm{red}}$, its *target closure invariant* is

$$\mathfrak{H}^\Sigma_{\mathcal{F}} := \overline{\operatorname{span} \Pi_\Sigma(\mathcal{F})}^{Z_\Sigma}. \tag{70}$$

(I2) For nonnegative functionals D, N_+ on a reduced target/source family $\mathcal{P}$, the *sharp domination radius* is

$$\alpha_*(N_+ \mid D) := \sup_{\substack{V \in \mathcal{P} \\ D(V) > 0}} \frac{N_+(V)}{D(V)}. \tag{71}$$

The value is $+\infty$ when some V has $D(V) = 0 < N_+(V)$. For a normalized retained family $\mathcal{N} \subseteq \mathcal{P}$, the *physical-price gap* is

$$\delta_*(D \mid \mathcal{N}) := \inf_{V \in \mathcal{N}} D(V). \tag{72}$$

(I3) When $\alpha_* < \infty$, let $\widehat{\mathcal{E}}_{\alpha_*}$ be the closure in the canonical evaluation compactification of the normalized equality family

$$\mathcal{E}_{\alpha_*} := \{V \in \mathcal{P} : D(V) > 0,\ N_+(V) = \alpha_* D(V)\}.$$

Totalize the remainder relation generated by the equality identities. Its ordinary graph image has *equality–rigidity invariant*

$$\mathfrak{H}^{\Sigma}_{\mathrm{eq}} := \overline{\operatorname{span} \Pi_\Sigma \mathscr{R}_{\mathrm{eq}}(\widehat{\mathcal{E}}_{\alpha_*})}^{Z_\Sigma}. \tag{73}$$

and its failure and graph-boundary values enter $\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$. For $\alpha_* = +\infty$, the equality row is canonically inapplicable and the positive null-cost or unbounded-feeding class is already returned by the radius row. The equality identities may be saturation equations, conservation laws, modulation equations, Pohozaev identities, or Liouville constraints.

Each invariant is computed after all zero-descent-defect source, gauge, trace, and target-null reductions already proved have been propagated.

Theorem II.1.7 (Structural normal form for invariant evaluation)**.** *Within the interfaces of this paper, every analytic input to target exclusion has one of the following exact structural meanings.*

(N1) *A retained target/source family is target-null on the reduced interface exactly when $\mathfrak{H}^{\Sigma}_{\mathcal{F}} = 0$.*

(N2) *The domination inequality $N_+ \le aD$ holds on the retained target/source family $\mathcal{P}$ exactly when $\alpha_*(N_+ \mid D) \le a$.*

(N3) *The normalized target/source family $\mathcal{N}$ has a uniform positive physical price exactly when $\delta_*(D \mid \mathcal{N}) > 0$.*

(N4) *A sharp equality branch has no target-visible remainder exactly when $\mathfrak{H}^{\Sigma}_{\mathrm{eq}} = 0$.*

Source closure and target-null residual calculations use (N1)*; resolvent, flux, and carrier identities generate* (N2)*; physical prices generate* (N3)*; and zero-cost equations generate* (N4)*. None is an endpoint. Its nonclosing value supplies a normalized witness to the activity lift, after which the successor and viability operators continue the calculation.*

Proof. Item (N1) is the definition of the closed target projection in (70). For (N2), divide by $D(V)$ when it is positive; the convention for a positive D-null mode gives the remaining case. Item (N3) is the definition of the infimum in (72). Item (N4) follows from (73): its vanishing is precisely the vanishing of every target projection of the equality remainder. These are the four structural coordinates occurring in Theorems I.4.6 and I.5.13, Definition II.8.2, and Lemma II.8.3. □

Definition II.1.8 (Operational fast-track evaluator)**.** For a source/mode cell (A, m), propagate all previously vanished closed subspaces and let $Y^{\Sigma}_{A,m}$ be the resulting target-active quotient. Introduce the ordered row set

$$\begin{aligned}\mathscr{R}_{\mathrm{FT}} := \{&\mathrm{form}, \mathrm{grad}, \mathrm{chain}, \mathrm{bdry}, \mathrm{tail},\\ &\mathrm{target}, \mathrm{feed}, \mathrm{price}, \mathrm{route}, \mathrm{capacity},\\ &\mathrm{scale}, \mathrm{sat}, \mathrm{fluct}, \mathrm{cov}, \mathrm{succ}\}.\end{aligned} \tag{74}$$

Its fast-track value is the row-indexed tuple

$$\mathfrak{I}^{\Sigma}_{A,m} := (\mathfrak{I}^{\Sigma,r}_{A,m})_{r \in \mathscr{R}_{\mathrm{FT}}}, \tag{75}$$

where the first five and covariance entries are target-active total-graph defect quotients; $\mathfrak{H}^{\Sigma}_{A,m}$ is the target closure; α^* is the feeding radius; δ^* is the physical-price infimum; $\mathcal{R}$ is effective resistance; Cap is target capacity; $\mathscr{S}$ is the pulled-back nonnegative scale-price series; $\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$ is the projected saturation or equality remainder; and χ^{Σ} is the target-active covariance, bracket, or fluctuation radius when the evolution is stochastic. An inapplicable row has the singleton value $\{\perp_{\mathrm{n/a}}\}$ precisely when its defining type is absent from the fixed PDE signature, as for the fluctuation row of a deterministic equation. Applicability is not a choice. Every other entry is an exact quotient, graph defect, infimum, supremum, kernel, capacity, resistance, or series value. No favorable analytic property occurs in the definition.

Definition II.1.9 (Structural property compiler)**.** For a row $r \in \mathscr{R}_{\mathrm{FT}}$, let F^{tot}_r be its totalized interface restricted to the current presentation-generated target-aligned record cell and let $\mathcal{K}^{\Sigma}_r$ be the closed ledger already propagated into that cell. Every supremum, infimum, kernel, graph value, and carrier comparison below is taken on finite stopped records and their source-preserving closure; there is no ambient solution-space quantifier. The structural property compiler P_{Σ} assigns the following exact invariant statements:

$$\mathsf{P}_{\Sigma}[\text{native realization or local existence}] := \mathfrak{H}^{\Sigma}_{\mathrm{form},A,m} = 0, \tag{76}$$

$$\mathsf{P}_{\Sigma}[\text{target-relative closed gradient range}] := \mathfrak{H}^{\Sigma}_{\mathrm{grad},A,m} = 0, \tag{77}$$

$$\mathsf{P}_{\Sigma}[\text{target-relevant compactness of generated limits}] := \mathfrak{H}^{\Sigma}_{\mathrm{bdry},A,m} = 0, \tag{78}$$

$$\mathsf{P}_{\Sigma}[\text{pre-contact chain rule}] := \mathfrak{H}^{\Sigma}_{\mathrm{chain},A,m} = 0, \tag{79}$$

$$\mathsf{P}_{\Sigma}[\text{constraint-tail closure}] := \mathfrak{H}^{\Sigma}_{\mathrm{tail},A,m} = 0, \tag{80}$$

$$\mathsf{P}_{\Sigma}[\text{generated operation descends}] := \mathfrak{H}^{\Sigma}_{\mathrm{cov},A,m} = 0, \tag{81}$$

$$\mathsf{P}_{\Sigma}[N_+ \le aD] := \alpha^*_{A,m}(N_+ \mid D) \le a, \tag{82}$$

$$\mathsf{P}_{\Sigma}[\inf_{\mathcal{N}} D > 0] := \delta^*_{A,m}(D \mid \mathcal{N}) > 0, \tag{83}$$

$$\mathsf{P}_{\Sigma}[\text{routable target current}] := \mathcal{R}_{A,m}(\Sigma) < \infty, \tag{84}$$

$$\mathsf{P}_{\Sigma}[\text{target polar}] := \mathrm{Cap}_{A,m}(\Sigma) = 0, \tag{85}$$

$$\mathsf{P}_{\Sigma}[\text{finite-budget scale exclusion}] := \mathscr{S}^{\Sigma}_{A,m} = +\infty, \tag{86}$$

$$\mathsf{P}_{\Sigma}[\text{sharp equality or saturation closes}] := \mathfrak{H}^{\Sigma}_{\mathrm{sat},A,m} = 0, \tag{87}$$

$$\mathsf{P}_{\Sigma}[\text{fluctuation is target-null or physically routed}] := \chi^{\Sigma}_{A,m} = 0 \text{ or its resistance is finite.} \tag{88}$$

The successor row is the finite-depth sequence $Z_{n+1,A,m} = \mathcal{V}_{\eta}(Z_{n,A,m})$; its two values are finite pruning with its rational-cylinder transcript and cumulative joint saturation of every retained cofinal transcript. The latter is reprojected and is either target-null, has a least unprocessed finite joint increment, or is a primal–polar fixed structural atom under Theorems II.2.16 and II.2.32. Here every object is formed after quotienting by $\mathcal{K}^{\Sigma}_r$. Any algebraic identity derivable from the public weak equation may refine a row into nonnegative descendants. The compiler has no input slot for an external compactness, coercivity, Liouville, or realization statement.

Definition II.1.10 (Propagated physical fast-track ledger)**.** Order the rows as in (74). Begin with the closed target-null ledger $\mathcal{K}^{\Sigma}_0 = \mathscr{N}_{\Sigma}$. If row i has closed branch C_i and transport T_i into the next row, set

$$\mathcal{K}^{\Sigma}_{i+1} := \overline{\mathcal{K}^{\Sigma}_i + T_i(C_i)}. \tag{89}$$

All spaces, totalized interfaces, target projections, physical forms, and subsequent invariants are then replaced by their descended objects on the quotient by $\mathcal{K}^{\Sigma}_{i+1}$. If a map fails to descend, its nonzero commutator with the quotient map is the next row's target-active graph defect. Thus quotient compatibility is tested by the construction rather than imposed on it.

Theorem II.1.11 (Analytic-property compilation)**.** *Every analytic property used by the continuum argument to decide whether a presentation-generated source/mode cell can contribute to singular contact is equivalent to the corresponding structural invariant statement in Definition II.1.9. The opposite analytic property is equivalent to the complementary invariant value and produces a tagged activity profile and successor. In particular, the fast track never presupposes closed range, compactness, a chain rule, a coercive estimate, a positive price, rigidity, polarity, finite resistance, summable tails, fluctuation control, orbit realization, or continuation.*

Proof. Existence and regularity properties are membership statements in the graph of a partial map. Totalization makes their negations explicit graph values, and quotienting their closed graph boundary by $\mathscr{N}_\Sigma$ proves (76)–(80) and (81). In the compactness row only target-visible loss of compactness survives; target-null escape is already in J_{res}. Thus the invariant tests exactly the compactness needed downstream rather than ambient compactness of the orbit. The radius and gap equivalences follow directly from the supremum and infimum definitions, including positive D-null directions and zero-price sequences. The spectral theorem gives the resistance/range/harmonic trichotomy, with the harmonic part retained in the target quotient. Capacity zero is the definition of the polar branch; positive capacity is its complementary target-compatible class. A nonnegative scale-price series either diverges or is finite. Saturation equations define a remainder map, so closure of the equality branch is exactly vanishing of its target quotient. Finally, stochastic drift, bracket, covariance, and conditional-expectation contributions are currents in the lifted state space; quotienting their target-null part and applying the same radius/resistance construction gives (88). Every complementary value enters the successor row. Its finite-depth values and cumulative joint closure are exhaustive by Theorem II.2.16; compactness is used only to reconstruct the compatible joint record, not to declare that record terminal. These cases exhaust (74). □

Definition II.1.12 (Target-relevant analytic statement grammar)**.** Let $\mathsf{An}_\Sigma(\mathfrak{P})$ be the least grammar generated from the fixed PDE signature by:

(A1) membership in, or failure of, a native, limit, chain, trace, tail, normalization, or continuation graph;
(A2) vanishing or nonvanishing after the target quotient;
(A3) comparison of two nonnegative physical forms, including an infimum or sharp radius;
(A4) range, kernel, resistance, capacity, and harmonic statements for the generated operator or form;
(A5) convergence or divergence of a nonnegative physical-price series;
(A6) a saturation/equality remainder after sharp comparison;
(A7) covariance, bracket, compensator, or conditional-expectation statements in a generated stochastic lift;
(A8) membership in a finite-depth successor set, a cumulative joint-prefix state, or the fixed-atom locus of its graph-complete saturation.

The grammar is closed under finite conjunction, sourcewise limit, pullback by a generated chart, and descent through a generated quotient. These closure operations are themselves totalized.

Theorem II.1.13 (Exhaustive fast-track compilation)**.** *The property compiler extends uniquely to a total map*

$$\mathsf{P}_\Sigma : \mathsf{An}_\Sigma(\mathfrak{P}) \longrightarrow \prod_{\substack{\varnothing \neq A \subseteq \mathfrak{C} \\ m \in \mathfrak{M}}} \prod_{r \in \mathscr{R}_{\mathrm{FT}}} \mathrm{Val}\Big(\mathfrak{I}^{\Sigma,r}_{A,m}\Big)\,. \tag{90}$$

Every statement that can affect the contact identity or the physical expenditure contradiction occurs in exactly one primitive row, possibly with several source/mode tags. Its negation is the *complementary value of that same row. Hence no analytic property, compactness assertion, or equation-specific estimate is an unregistered premise of the fast track.*

Proof. Clauses (A1)–(A8) map respectively to the total-graph rows; target row; feeding/price rows; gradient/routing/capacity rows; scale row; saturation row; fluctuation row; and finite-depth successor row. Source and realization tags are assigned before evaluation by Theorems I.2.15 and I.5.12. Finite conjunction gives a product value. Sourcewise limits remain in their tagged closures. Chart pullback appends $\mathsf{Abs}/\mathsf{Lift}$, and quotient descent either factors or produces its totalized descent defect. Structural induction therefore defines P_Σ on the whole grammar and leaves no unmatched constructor.

By normalized shell accountability, contact affects the argument only through a target-active current or a totalized shell/interface defect. By the physical balance, exclusion affects it only through a nonnegative expenditure comparison or series. These are clauses (A1)–(A6); stochastic and successor-path semantics are exactly (A7)–(A8). Thus the grammar covers every input to the contact-versus-expenditure proof. ☐

Theorem II.1.14 (Closed-input rule)**.** *The value of $\mathfrak{I}^{\Sigma,r}_{A,m}$ is determined by the fixed PDE signature, target, physical balance, and the closures generated from them. An auxiliary lemma may prove that value but cannot be added as data defining it. Any proposed object not generated by this algebra either:*

(E1) *is derived from the fixed signature, in which case the source compiler and property compiler assign it to existing rows; or*

(E2) *changes the signature, in which case the conclusion concerns the enlarged equation/presentation and is not a conclusion about the original PDE.*

These alternatives exhaust the generated continuum algebra.

Proof. The evaluation compactification is generated by Definition I.3.1; the source syntax by Definition I.2.5; and the analytic statement syntax by Definition II.1.12. Membership in either syntax is therefore determined before an invariant is evaluated. Structural induction gives (E1). Failure of membership means that a new primitive map, topology, law, or balance has been appended, which is precisely a change of signature and gives (E2). ☐

Theorem II.1.15 (Ledger propagation and no re-estimation)**.** *At each fast-track row, exactly one of the closed and complementary branches is selected by its invariant value. A closed branch propagated by* (89) *remains zero in every later descended target quotient. If a downstream object does not descend, its descent defect is retained in the appropriate existing source/mode cell. Therefore later rows neither assume nor re-estimate any property closed by an earlier row.*

Proof. For closed subspaces the third isomorphism theorem gives $(Y/\mathcal{K}_i)/(\mathcal{K}_{i+1}/\mathcal{K}_i) \cong Y/\mathcal{K}_{i+1}$. Hence every propagated class is zero in all later quotients. The same statement for closed convex ledgers follows by passing to the associated quotient seminorm. Failure of descent is precisely a nonzero commutator between the downstream map and the quotient map, which is a totalized graph defect. Sourcewise closure retains its ancestry. The exhaustive two-branch statement follows from Theorem II.1.11. ☐

Theorem II.1.16 (Fast-track branch-to-successor completeness)**.** *For every target-active source/mode cell, the tuple $\mathfrak{I}^{\Sigma}_{A,m}$ returns one of the following exhaustive structural branches.*

(F1) *Each form, gradient, chain, boundary, tail, and covariance defect quotient vanishes or survives as a target-visible total-graph defect with the same source ancestry.*

(F2) $\mathfrak{H}^{\Sigma}_{A,m} = 0$*: the cell is target-null and closes. Otherwise its nonzero quotient is retained.*

(F3) $\alpha^*_{A,m} \le 1$*: feeding is dominated by the generated carrier. If* $\alpha^*_{A,m} > 1$*, or* $D = 0 < N_+$ *for a member of the cell, the maximizing sequence becomes a target-feeding activity profile.*

(F4) $\delta^*_{A,m} > 0$*: the cell has positive normalized price. If* $\delta^*_{A,m} = 0$*, a zero-price sequence exists in the closure and is retained with the same ancestry.*

(F5) $\mathfrak{H}^{\Sigma}_{\mathrm{eq},A,m} = 0$*: the sharp equality branch closes. Otherwise its target-visible equality remainder survives.*

(F6) $\mathrm{Cap}_{A,m}(\Sigma) = 0$*: the target is polar for that cell. Positive capacity supplies a normalized target-active activity profile.*

(F7) $\mathcal{R}_{A,m}(\Sigma) < \infty$*: the source can be routed through the generated geometry. Infinite resistance supplies its harmonic component to the activity lift.*

(F8) *For a scale cascade, the pulled-back series* $\sum_j r_j^{\kappa_{A,m}} \delta_{A,m,j}$ *either diverges, excluding that finite-budget cascade, or is finite, yielding vanishing-price windows for the ordered successor and recurrent-saturation compiler.*

(F9) *The saturation quotient* $\mathfrak{H}^{\Sigma}_{\mathrm{sat},A,m}$ *vanishes or retains the target-visible equality remainder. Every Pohozaev, Noether, virial, frequency, or entropy identity already generated by the equation refines its nonnegative descendants; the remaining equality record becomes a successor.*

(F10) *The fluctuation invariant* $\chi^{\Sigma}_{A,m}$ *is target-null, physically routed at finite resistance, or survives as a target-visible covariance/bracket class in the existing lifted source cell.*

(F11) *The total weak-solution graph retains ordinary values and every graph-boundary witness. Both enter the successor graph; persistent boundary values survive only through an infinite zero-price witness path.*

(F12) *An inapplicable row returns* $\{\perp_{\mathrm{n/a}}\}$*; failure of a type-present map to descend returns its graph defect and activates the same successor operator.*

Every complementary branch is a value of the same invariant used by the closing branch. The tuple is not a verdict: after each row, the canonical activity map sends every nonzero target quotient to $\mathcal{R}_\eta$*, and the viability sets* Z_n *either remove it at finite depth or construct its recurrent zero-price mechanism. Before either action, the row is converted to the target-local normal form* (110)*; hence no row acts on an equation-wide class or charges a source-transverse positive quantity.*

Proof. The graph-defect row follows from totalization. The target, radius, gap, and equality rows follow from Theorem II.1.7 and the definitions of the closed quotient, supremum, infimum, and equality remainder. In the radius row, the convention $\alpha^* = +\infty$ records a positive D-null feeding direction. If $\delta^* = 0$, the definition of the infimum gives a sequence with prices tending to zero; sourcewise closure retains its limit records.

Zero capacity is the polar branch of the associated form. Its complement is positive capacity and supplies an activity profile. The spectral theorem gives the finite resistance/range branch and the orthogonal harmonic remainder. Finally a nonnegative series either diverges or has finite sum. The physical pullback identity makes these precisely the excluded branch and the successor-activation branch. The latter is closed by Theorem II.2.58; it is not a terminal formation class. The saturation and fluctuation alternatives are Theorem II.1.11. The singleton inapplicable value and totalized descent defect exhaust the remaining interface cases. Applying the activity lift and Theorem III.2.9 to each nonzero quotient proves the final constructive statement. □

Corollary II.1.17 (Estimate compression after structural derivation)**.** *Once a graph relation, positive-form identity, target-null quotient, or zero-set relation has been derived from the symbolic presentation, subsequent rows use only that structural output. Formal identities and universal functional calculus refine the generated positive cone directly. When they do not remove a row, its* *graph-boundary, positive-null, or zero-price witnesses are generated and passed to the successor calculus. Closed outputs are propagated by Theorem II.1.15.*

Definition II.1.18 (Canonical symbolic structural transcript)**.** For the fixed PDE, complete continuation target, carrier, and derived ceiling, define

$$\Theta_\Sigma^{\mathrm{syn}}(\mathfrak{P}, B_{\mathrm{phys}}) := \Big(\mathsf{Syn}(\mathfrak{P}), \mathsf{Graph}^{\mathrm{tot}}(\mathfrak{P}), \mathsf{Pos}(\mathcal{K}), \Pi_\Sigma, \mathsf{Cov}(\mathsf{Op}), \mathsf{Anc}\Big)\,. \tag{91}$$

The entries are respectively the typed equation syntax, total graphs of its generated operations, the cone of nonnegative terms occurring in the proved carrier identity, the generated target morphism, the total covariance graphs, and the event/source ancestry maps. Exact extrema, capacities, resistances, realization images, contact orbits, continuation verdicts, and Liouville conclusions are not entries. They are semantic outputs of rows constructed from $\Theta_\Sigma^{\mathrm{syn}}$.

Theorem II.1.19 (Public-signature structural sufficiency)**.** *The symbolic transcript functorially determines the source cells, totalized interfaces, target-null quotient, carrier complex, activity measures, and successor relation used by the regularity argument:*

$$\mathsf{Compiler}_\Sigma = \overline{\mathsf{Compiler}}_\Sigma \circ \Theta_\Sigma^{\mathrm{syn}}. \tag{92}$$

Every row is constructed together with its ordinary and complementary witness graphs. No exact row value or actual-contact image is consulted in this construction. Two presentations with isomorphic symbolic transcripts therefore have isomorphic prospective successor and zero-cost fixed-point algebras. An analytic derivation can only prove a relation already present in the transcript or refine its generated positive cone; it cannot insert a favorable value.

Proof. Proceed by structural induction over the typed syntax and then by the fixed row order (74). Each constructor supplies its ordinary graph, explicit failure value, source transport, and event ancestry. Graph closure supplies the corresponding boundary witnesses. The target morphism forms the null quotient, and the positive carrier cone supplies the expenditure coordinate. Quotient descent either factors or produces its commutator graph. These operations construct both branches of every row without deciding which branch contains a given future orbit record.

The activity and successor constructions use only the generated metric, target shell, graph witnesses, and event ancestry. The prospective fixed-point construction below uses their closed images, not the image of actual contact orbits. This proves (92) and the isomorphism statement. □

Corollary II.1.20 (Closed structural input)**.** *A PDE specialization supplies only the target-blind public signature of Definition I.1.12. The continuum algebra generates* (91)*. Every identity obtainable by formal manipulation of that signature or by the fixed universal library is inserted automatically as a graph relation or positive-descendant refinement. The ordinary graph and every first-failure witness are inputs to the canonical activity lift and successor operator of Definitions II.2.40 and II.2.43. Their descending zero-cost closure is computed by Definition II.2.51. Thus every generated branch is reduced to a target-null class or to an explicit recurrent zero-cost mechanism; there is no unevaluated row and no additional invariant value to be supplied.*

Proposition II.1.21 (Positive-separator evaluation of an equality branch)**.** *Totalize the separator relation on $\widehat{\mathcal{E}}_{\alpha_*}$. On its zero graph-defect branch the identity is*

$$0 = Q_\Sigma(\Pi_\Sigma \mathscr{R}_{\mathrm{eq}}(Z)) + \sum_{j=1}^{m} q_j(Z), \tag{93}$$

where $Q_\Sigma \geq 0$ is positive definite on Z_Σ and every $q_j \geq 0$. Then $\mathfrak{H}_{\mathrm{eq}}^{\Sigma} = 0$.

Proof. Every term on the right of (93) is nonnegative, so each term vanishes. Positive definiteness gives $\Pi_\Sigma \mathscr{R}_{\mathrm{eq}}(Z) = 0$ for every Z. Taking the closed linear span proves the claim. $\square$

II.2 Coherent carrier passivity and structural successor closure

The scalar price infimum of Definition II.1.6 is not the terminal evaluator for a finite-price cascade. A finite measure on infinitely many disjoint target windows produces a vanishing-price subsequence. The continuum analogue of the residual closure in the discrete algebra is to follow that subsequence through all compact, concentrating, escaping, diffuse, gauge, and equality outputs. This section constructs that closure directly.

Definition II.2.1 (Coherent carrier complex)**.** Let u be one physical orbit. A *coherent carrier complex* consists of a locally finite rooted tree $\mathcal{T}$, cells Q_α, and nonnegative localization tests ψ_α with the following properties.

(C1) Every child is a subevent of its parent and, on the region where the local balance is evaluated,

$$\psi_\alpha = \sum_{\beta \in \mathrm{ch}(\alpha)} \psi_\beta. \tag{94}$$

(C2) All cells are restrictions, chart images, or normalized descendants of the same orbit. Their transition maps satisfy the cocycle composition law. Galilean, gauge, and quotient representatives are fixed compatibly on overlaps.

(C3) The weak equation gives a local carrier relation, linear in the test,

$$K_\psi(t) + D_I(\psi) + M_I(\psi) = K_\psi(s) + J_I(\psi) + E_I(\psi). \tag{95}$$

Here $D_I, M_I \geq 0$ are respectively the intrinsic production and the weak energy-defect measure, J_I is the signed transport current, and E_I is true exterior input. An inequality is written as equality by adjoining its nonnegative defect to M_I.

(C4) The normalized target observable is constant under the allowed transition maps. Consequently target retention is inherited by a child selected from a target-retaining parent.

The complex is generated from the shell cascade by taking the maximal locally finite family of compatible target-retaining descendants. A failure of local finiteness, the partition relation, transition coherence, or gauge compatibility is a totalized $\mathsf{Lift}/\mathsf{Bdry}$ successor and is not an additional admissibility condition.

Theorem II.2.2 (Signed carrier gluing)**.** *Let $\mathcal{T}_0 \subset \mathcal{T}$ be a finite rooted subtree. Sum* (95) *over $\mathcal{T}_0$ before taking positive parts. Every current through an internal face cancels with the same current with opposite orientation in the adjacent cell. In particular, conservative transport, pressure or constraint transport, chart translation, gauge motion, cutoff motion, and exact scale motion contribute only through the root, the leaves, or the true exterior boundary. The result has the form*

$$K_{\mathrm{leaf}}(\mathcal{T}_0) + \mu_{\mathsf{Geom}}(\mathcal{T}_0) + \mu_{\mathrm{def}}(\mathcal{T}_0) = K_{\mathrm{root}} + I_{\mathrm{ext}}(\mathcal{T}_0) + H_{\mathrm{sc}}(\mathcal{T}_0), \tag{96}$$

where the first two measures on the left are nonnegative. The term H_{sc} is the cohomology class of scale motion that is not an edge coboundary in the chosen scale-critical carrier. It has inherited $\mathsf{Caus}/\mathsf{Lift}$ *ancestry. It vanishes when the scale current admits a coherent carrier potential. No sum of the absolute values of the internal currents occurs in* (96)*.*

Proof. Linearity in ψ and (94) imply that the carrier, production, defect, transport, and exterior terms of a parent are the sums of the corresponding child terms. Give every common face the orientation induced by its parent. The child has the opposite induced orientation, so its signed flux cancels the parent flux. Pressure-gauge constants cancel because their weak pairing with the divergence-free or constraint-compatible test is zero. Transition coherence makes translation, gauge, and exact scale terms edge coboundaries; the sum of a coboundary over a finite tree is its root value minus its leaf values. The only possible scale term left after this cancellation is its class modulo the range of the edge coboundary, which is H_{sc}. The intrinsic production and weak balance defect retain their signs and add over the cells. This proves (96). $\square$

Definition II.2.3 (Target-aligned physical charge)**.** Let Q be a pre-contact event selected by a shell covector $D\Phi$. First glue every compatible internal face and form the signed target current on the finite-value branch,

$$q_{\Phi,Q} := \sum_{c\in\{\mathsf{Geom},\mathsf{Caus},\mathsf{Abs},\mathsf{Lift},\mathsf{Bdry}\}} \int_Q \langle D\Phi, J^c\rangle. \tag{97}$$

On a variation-boundary branch, replace this finite sum by the joint shell-current owner readout $q^{\mathrm{own}}_{\Phi,Q} := \overline{Q}^o(1) \geq 1/5$ and its increasing bounded rational truncations. Every charging relation is closed jointly with the owner coordinate; no extended signed sum is formed. A nonnegative measure P is an *admissible charge* for this event only if a generated local carrier identity proves

$$(q_{\Phi,Q})^+ \leq \Delta K_Q + P(Q) + I^+_{\mathrm{ext}}(Q) + n_Q, \qquad n_Q = \langle D\Phi, N_Q\rangle = 0, \quad N_Q \in \mathscr{N}_\Sigma, \tag{98}$$

with the carrier increments telescoping on same-origin concatenation and P dominated by the public physical ledger. Positive quantities not appearing in such an identity are structural coordinates, not expenditure. Positive parts of individual internal currents are never charged before signed gluing. On the boundary-owner branch, (98) denotes the compatible family of bounded truncation inequalities; monotone passage to the owner value is made only after the carrier domination has been established.

Equation (98) is an eventwise factorization, not yet a physical-price inequality. The measure P becomes an admissible target price only after the generated storage row either represents the cumulative ΔK_Q terms as a coboundary with a finite root–leaf contribution on the selected same-origin family, or retains every positive storage return as a target-aligned Caus-headed successor. No telescoping is inferred from disjointness alone.

Theorem II.2.4 (Local target-aligned charging rule)**.** *Every physical charge used by the successor algebra is obtained by Definition II.2.3. On a disjoint same-origin shell cascade, summing* (98) *cancels target-null terms and exposes storage increments, physical production, and true exterior input separately. Internal carrier increments cancel only after the intervening gaps are inserted into the same oriented carrier complex. The remaining root–leaf storage is evaluated by its coboundary/return row. Consequently repeated target progress is paid by the restriction of the public physical measure and true positive exterior input precisely on the zero-storage-transfer cell; every nonzero exact storage transfer is itself a target-aligned successor. If no generated identity dominates a nonzero target current, that current is likewise passed to the polarity stratifier; it is not assigned zero price and is not charged to an unrelated positive invariant.*

Proof. The shell identity exposes finite $q_{\Phi,Q}$, or its closed owner readout, before contact. Signed carrier gluing gives the aggregate current and the root–leaf form (96). Restriction to the target quotient annihilates N_Q, hence $n_Q = 0$. Add the identities over pairwise disjoint same-origin events and insert the oriented gap cells between consecutive events. The full complex telescopes, but removing the gaps leaves exactly their signed storage-transfer current; hence disjointness alone does not cancel it. Countable additivity bounds the P-terms by the single physical ledger and the exterior terms by its declared input component. Apply the generated coboundary graph to the remaining storage. Its zero or root–leaf-bounded value gives the admitted physical charge, while a nonzero signed return value gives the storage successor. The grammar admits no other rule for inserting a quantity into the ledger. Failure to derive the displayed domination has a nonzero total-graph coordinate, whose polarity split gives the stated successor. $\square$

Definition II.2.5 (Carrier cohomology and aggregate feeding)**.** Let $\mathscr{W}_\eta$ be the saturated closure of normalized windows whose target observable is at least $\eta > 0$, after all generated symmetry and gauge quotients. Include the values of the carrier, production, exterior input, and every totalized graph coordinate in the evaluation topology. Let S be the successor or unit-window shift on its ordinary graph.

On a coherent ordinary branch, write F_{sc} for the aggregate signed inward scale-space feed left after Theorem II.2.2. Its target-active carrier cohomology is

$$\mathfrak{H}^{\Sigma}_{\mathrm{car}} := \overline{\Pi_\Sigma F_{\mathrm{sc}}}/\overline{\Pi_\Sigma\{\Psi - \Psi \circ S : \Psi \in C(\mathscr{W}_\eta)\}}. \tag{99}$$

The quotient is totalized when S, the carrier, or a transition is partial. Its graph-boundary values retain their source ancestry. The ordinary carrier-deficit cocycle is

$$g_{\mathrm{car}} := F_{\mathrm{sc}} - D - M. \tag{100}$$

For a finite block, its sum is the aggregate inward feeding minus production, up to root–leaf storage and true exterior input. Coboundaries disappear in block averages and under invariant measures. If $[F_{\mathrm{sc}}] = 0$ in (99), then for every $\varepsilon > 0$ there is a continuous Ψ_ε whose coboundary approximates the feed class within ε; no exact correction is asserted.

Theorem II.2.6 (Target-relative passivity)**.** *Let $\mathcal{K}_\eta \subseteq \mathscr{W}_\eta$ be a compact S-invariant ordinary carrier class generated by the activity lift. On the ordinary signed-carrier branch, finite exterior input is already part of the derived physical budget. Then every S-invariant Borel probability measure ν on $\mathcal{K}_\eta$ satisfies*

$$\int_{\mathcal{K}_\eta} g_{\mathrm{car}}\, \mathrm{d}\nu \le 0. \tag{101}$$

Since every invariant measure annihilates continuous coboundaries, this is equivalent to

$$\int F_{\mathrm{sc}}\, \mathrm{d}\nu \le \int (D + M)\, \mathrm{d}\nu. \tag{102}$$

Equality in (101) *is the zero carrier-deficit branch. It enters the zero-density successor fixed point together with its unnormalized block ledger. It implies vanishing of a particular production term only when the generated consequence algebra contains a domination identity $p = q + r$ with that production as q. Even then, the conclusion is vanishing density on the recurrent support, not vanishing of the cumulative production or source cocycle of the originating path. Every measure and current in this statement is restricted to $\mathcal{K}_\eta$, generated by one retained target/source cell; the theorem has no consequence for invariant measures outside that cell.*

Proof. Apply (96) to the first N consecutive cells of a coherent path. Internal currents cancel and the carrier terms telescope. The carrier is bounded on the compact evaluation class. The exterior input has finite total mass, so its contribution divided by N tends to zero. Integrating the block identity against an invariant measure cancels the endpoint terms exactly and gives (101), which is (102). Continuous coboundaries and their uniform closure have zero integral against every invariant measure, justifying the cohomological quotient without producing an exact potential. This averaging proves only the stationary density identity. The finite-block exterior, production, and source totals remain in the augmented occurrence ledger and are restored before any target projection. □

Corollary II.2.7 (Totalized passivity branch)**.** *For every target-retaining carrier route, exactly one of the following is generated:*

(T1) *the coherent ordinary branch of Theorem II.2.6;*

(T2) *a first failure of local finiteness, carrier partition, transition coherence, gauge descent, compact retention, or carrier correction, together with its closed-graph witness descendants;*

(T3) *a target-null output removed by the generated quotient.*

The second output enters $\mathcal{R}_\eta$; the third closes. Positive exterior variation has Bdry *ancestry and is eligible for charge only through the local target-aligned carrier identity. Consequently compactness, coherence and carrier cohomology are processed by the successor operator.*

Proof. Totalize each partial relation in the order displayed in Definitions II.2.1 and II.2.5. If all ordinary values occur, Theorem II.2.6 applies. The first nonordinary value has a witness sequence in its closed total graph and therefore defines the successor in Definition II.2.51. Positive exterior variation enters the physical measure through Theorem II.2.4. The first-failure alternatives are exhaustive because all graph coordinates belong to the compact activity lift. □

Definition II.2.8 (Carrier saturation algebra)**.** The *carrier saturation algebra* is the closed target projection

$$\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}} := \overline{\operatorname{span}}\left\{ \Pi_\Sigma \operatorname{supp} \nu : \begin{array}{l} \nu \text{ is invariant on a generated compact successor core,} \\ \int g_{\mathrm{car}} \, \mathrm{d}\nu = 0 \end{array} \right\}. \tag{103}$$

The support is evaluated through all five source tags and all totalized zero-deficit equations. It may contain equilibria, relative equilibria, or nonstationary recurrent conservative dynamics. The subsequent zero-cost fixed point, rather than an assumed rigidity statement, decides its target projection.

Theorem II.2.9 (Ergodic strict passivity or saturation)**.** *Let $\mathcal{K}_\eta$ be as in Theorem II.2.6, and set*

$$\beta_\eta := \sup_{\nu \in \mathcal{P}_S(\mathcal{K}_\eta)} \int g_{\mathrm{car}} \, \mathrm{d}\nu, \tag{104}$$

where $\mathcal{P}_S(\mathcal{K}_\eta)$ is the compact set of S-invariant Borel probability measures. Then $\beta_\eta \leq 0$, and exactly one of the following holds.

(P1) *$\beta_\eta < 0$. There exist an integer $N_\eta \geq 1$ and $\delta_\eta > 0$ such that every N_η-block retained in $\mathcal{K}_\eta$ satisfies*

$$-\sum_{k=0}^{N_\eta - 1} g_{\mathrm{car}}(S^k w) \geq \delta_\eta. \tag{105}$$

(P2) $\beta_\eta = 0$. *The class $\mathcal{K}_\eta$ carries an invariant maximizing measure and its target projection belongs to $\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}}$. The maximizing measure is only a support selector; the original block cocycle remains attached without division by N.*

If $\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}} = 0$, only (P1) *is possible. The positive block price is therefore derived from the absence of a target-visible equality state; no contact margin or compactness gap is inserted.*

Proof. Let $h = g_{\mathrm{car}}$. Compactness of $\mathcal{P}_S(\mathcal{K}_\eta)$ and continuity of the integral give a maximizing measure. By Theorem II.2.6, $\beta_\eta \le 0$.

Suppose $\beta_\eta < 0$ but (P1) fails. Then there are $w_n \in \mathcal{K}_\eta$ with block length $N_n = n$ and total carrier deficit smaller than n^{-1}. In particular the mean of h has lower limit at least zero. Form the empirical measures

$$\nu_n := \frac{1}{N_n} \sum_{k=0}^{N_n-1} \delta_{S^k w_n}.$$

Compactness of probability measures on $\mathcal{K}_\eta$ gives a weakly convergent subsequence. The endpoint identity

$$\int f \circ S \, \mathrm{d}\nu_n - \int f \, \mathrm{d}\nu_n = \frac{f(S^{N_n} w_n) - f(w_n)}{N_n}$$

shows that its limit ν is invariant. Continuity of the evaluation coordinates and the lower bound give $\int h \, \mathrm{d}\nu \ge 0$, while Theorem II.2.6 gives the reverse inequality. Hence $\int h \, \mathrm{d}\nu = 0$, contradicting $\beta_\eta < 0$. Thus (P1) holds. If $\beta_\eta = 0$, a maximizing invariant measure is a saturation measure and gives (P2). Conversely, an invariant saturation measure contradicts any uniform positive block deficit after integration. Vanishing of $\mathfrak{H}^{\Sigma}_{\mathrm{eq,car}}$ excludes a target-retaining instance of (P2). □

II.2.1 Target-aligned dual-carrier compilation

The preceding passivity theorem is used only after contact has selected a source and a target covector. This is the continuum selector cut. It avoids an equation-wide classification of equality states.

Definition II.2.10 (Active branch signature)**.** An *active branch signature* is a tuple

$$\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R}), \tag{106}$$

where $A \neq \varnothing$ is a persistent source support, m is its eventual realization mode, Γ is a generated target-shell family, $\eta > 0$ is its retained source floor, and $\mathbf{R}$ is the same-origin cascade tag. The branch ideal $I_{\mathfrak{b}}$ is generated by the public equation and by the relations already established on this branch: source support A, mode m, shell pairing at least η, chart and gauge choices, event order, and every earlier zero graph coordinate. No relation outside this list is inserted into $I_{\mathfrak{b}}$.

Proposition II.2.11 (Branch-local consequence rule)**.** *Let $q = 0$, $q \ge 0$, or $q_1 \le q_2$ be derivable from the public weak equation modulo $I_{\mathfrak{b}}$. Then that relation holds on every ordinary or zero-defect realization of $\mathfrak{b}$. Its failure on a total-graph boundary is the next typed successor. Consequently a relation need only be proved on the active branch on which its antecedent has already been generated; no assertion about states outside that branch is required.*

Proof. Every generator of $I_{\mathfrak{b}}$ vanishes on an ordinary realization of the branch, so substitution in the finite weak-equation derivation proves the relation there. The closed total graph records the first generator that fails under passage to a limit. Source-preserving completion assigns that failure its inherited source and realization mode. These two cases exhaust the totalized branch. □

Definition II.2.12 (Target-local structural stratum)**.** For an active branch $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$, let $\mathbb{G}_\mathfrak{b}$ be the total evaluation graph generated by the public weak equation, its five source currents, the shell family, and all operations already used on the branch. Its *target-local structural stratum* is

$$X_\mathfrak{b} := \{z \in \mathbb{G}_\mathfrak{b} : I_\mathfrak{b}(z) = 0,\ Q_{A,\Phi}(z) \geq \eta,\ \mathrm{Origin}(z) = \mathbf{R}\}. \tag{107}$$

All consequence, compactness, production, and equality operations below are performed after restriction to $X_\mathfrak{b}$. Values outside this stratum are irrelevant to the contact branch and are never quantified over.

Definition II.2.13 (Generated polarity stratification)**.** Fix once and for all a countable enumeration $(F_j)_{j\geq 0}$ of the scalar rational-cylinder coordinates generated by the public equation and the target-cyclic grammar. Vector, tensor, measure, and operator coordinates enter through their rational scalar probes. Replace an unbounded real coordinate by $\widehat{F}_j = F_j/(1 + |F_j|)$, and retain nonordinary values in its total graph. On a target-local stratum, the j-th *polarity split* is the disjoint ordered partition

$$\{\widehat{F}_j = 0\}, \qquad \{s\widehat{F}_j \in [2^{-k}, 2^{-k+1})\}\ (s \in \{-1, 1\},\ k \geq 1), \qquad \partial_{\mathrm{gr}} F_j. \tag{108}$$

Endpoints are assigned to the leftmost displayed cell, and $\partial_{\mathrm{gr}} F_j$ is further split by the first failed public graph operation and its inherited source word. Intersect every cell with $X_\mathfrak{b}$. First dovetail the finite identities and cylinder moments of the parent local Archimedean cone. A finite derivation of -1 prunes the parent cell. Otherwise compatible moments reconstruct a positive local functional; disintegrate it over the countable partition (108) and select the least cell of positive target mass. That cell is a target-active successor with the same origin and with its shell current renormalized to the universal source floor. Thus no semantic emptiness or exact-zero oracle is used.

Thus both possession and failure of an analytic property are structural values. For example, closed descent is the zero cell of its cokernel coordinate; nonclosed descent is a signed dyadic or graph-boundary cell, not an omitted hypothesis.

Definition II.2.14 (Resolved-prefix rank)**.** The enumeration in Definition II.2.13 is dovetailed with the enumerations of weak words, source descendants, quotient operations, carrier rows, and total-graph defects. A branch has *resolved-prefix rank*

$$\rho(\mathfrak{b}) := sup\{N : \text{the polarity cell of every } F_j,\ j < N, \text{ belongs to } I_\mathfrak{b}\}. \tag{109}$$

When $\rho(\mathfrak{b}) < \infty$ and $F_{\rho(\mathfrak{b})}$ is split, every retained proper successor adjoins its selected cell to the branch ideal and therefore has larger resolved prefix. If that split generates new operations, they enter later in the same universal dovetailing. The rank is consequently constructed by the algebra; it is not presentation data and no well-foundedness premise is required.

Theorem II.2.15 (Target-local stratification exhaustiveness)**.** *Let $\mathfrak{b}$ be a same-origin active contact branch. At its next unprocessed coordinate exactly one of the following occurs:*

(L1) *a finite local ordered identity proves that the current target cell is empty or target-null;*

(L2) *a generated carrier identity charges the retained cell to the finite physical ledger;*

(L3) *a zero, signed dyadic, or graph-boundary cell has nonzero target projection and becomes a proper successor $\mathfrak{b}'$ with $\rho(\mathfrak{b}') > \rho(\mathfrak{b})$.*

The alternatives exhaust ordinary values, loss of every analytic property used by the derivation, and all limit values of the coordinate. No global estimate and no target-nullity statement is an input.

Proof. The total graph makes (108) exhaustive, and the endpoint convention makes it disjoint. Apply the Archimedean proof/model alternative to the parent target cell. Its finite -1-identity gives (L1). On the complementary cell, evaluate its target detector on the presentation-generated reachable domain. By Theorem III.10.41, positivity returns a finite stopped same-origin occurrence with its shell covector, complete frontier, and source word. The finite affine identity or closed owner relation acts on that occurrence and restores the fixed floor. If it is dominated by a public carrier row it is (L2). Otherwise its failed polarity relation is appended to the occurrence record, producing (L3) and increasing the resolved prefix. Total graph ancestry proves the assertion for failed analytic properties. □

Theorem II.2.16 (Diagonal target-aligned saturation)**.** *An infinite chain of nonempty proper successors selected from one contact branch is evaluated by first forming its cumulative same-origin joint record, closing it under every mixed public operation and total graph, and only then reapplying the target projection. Exactly one of the following occurs:*

(D1) *the cumulative target projection vanishes, so all induced source pairings with the retained shell covectors vanish along the retained path and shell contact is impossible;*

(D2) *a least unprocessed target increment occurs on a finite joint prefix; its polarity, complete mixed-operation word, and inherited five-channel ancestry form the next proper successor;*

(D3) *every finite increment is source-resolved, joint saturation is fixed, every source owner is fixed under recursive carrier-polar exhaustion, and the totalized ordinary realization row reconstructs a joint-saturated primal–polar five-channel prospective source atom retaining the cofinal shell family and its origin polarity.*

The third conclusion is a saturated source-profile witness. It is an explicit singular mechanism only in the independent-origin fiber of Theorem II.2.41. A compatible inverse limit, unbounded resolved-prefix rank, or raw ordinary path is an internal state and never a conclusion.

Proof. At depth n, join the first n successor records before forming any residual. The joint grammar adjoins every mixed product, polarization, constraint descent, carrier row, and total-graph value generated from that prefix. For any nonzero finite positive observer law ν on the common measurable same-origin evaluation space, let $\mathcal{A}_n$ be the sigma algebra of the depth-n joint record and let P_n be conditional expectation onto $L^2(\mathcal{A}_n, \nu)$. The observer algebras increase. Their conditional expectations of the centered shell contrast y_Φ therefore converge strongly, with the exact orthogonal increment decomposition

$$P_\infty y_\Phi = \sum_{n\geq 0} d_n, \qquad d_0 = P_0 y_\Phi, \quad d_n = (P_n - P_{n-1})y_\Phi, \qquad \|P_\infty y_\Phi\|_2^2 = \sum_{n\geq 0} \|d_n\|_2^2.$$

Consequently a target component of the graph-complete joint limit is the sum of finite joint increments; it cannot first appear outside all finite prefixes.

If the full projection vanishes, prospective accountability makes every source pairing vanish on the retained path and proves (D1). If a least unprocessed increment exists, source-preserving totalization supplies its five-channel ancestry and polarity cell, giving (D2). Notice that no lower bound on the increment is used. If every increment is processed, apply the joint closure once more. Failure of fixedness generates another mixed coordinate and returns to (D2). At a fixed point the unnormalized shell cocycle, common origin, complete source ancestry, and all public graph rows remain present. Before realization, apply Theorem II.2.32 to every canonical owner in the fixed enumeration. A charge, null value, or new polar coordinate returns to (D1) or (D2). Only simultaneous primal–polar fixedness reaches the totalized realization row; its nonordinary value is another proper successor and its ordinary value gives the prospective source atom in

(D3). Origin-polarity preservation prevents a contact-derived ordinary value from becoming an independent realization. Thus rank growth by itself proves nothing and no target-bearing structure is assigned to a residual limit. $\square$

Definition II.2.17 (Target-local structural fact)**.** Fix a same-origin active branch $\mathfrak{b}$, its shell covector $\lambda = D\Phi$, and its target-local stratum $X_{\mathfrak{b}}$. A relation is an *admissible target-local structural fact* when its complete expanded derivation has first been universally joined and saturated, and its restriction to $X_{\mathfrak{b}}$ has the signed form

$$Q_{\mathfrak{b},\lambda} = \delta K_{\mathfrak{b}} + \sum_{c \in \mathfrak{C}} R^{c}_{\mathfrak{b},\lambda} + P_{\mathfrak{b}} + B_{\mathfrak{b}} + N_{\mathfrak{b}} + E_{\mathfrak{b}}. \tag{110}$$

Here $Q_{\mathfrak{b},\lambda}$ is the canonical owner of the selected target progress in that expanded joint record; $R^{c}_{\mathfrak{b},\lambda}$ is the unabsorbed signed remainder with inherited c-ancestry; $\delta K_{\mathfrak{b}}$ is a same-origin carrier coboundary; $P_{\mathfrak{b}} \geq 0$ is admitted only through a carrier identity dominated by the public physical measure; $B_{\mathfrak{b}}$ is true exterior input; $N_{\mathfrak{b}} \in \mathscr{N}_{\Sigma}$; and $E_{\mathfrak{b}}$ is the ordered list of nonordinary or failed graph coordinates. Compatible internal currents are added with their signs before any positive part is formed, and each expanded term occurs exactly once: either in a signed remainder, in the generated production, in exterior input, in the null term, or in the defect list.

The fact may close or charge the branch only on $X_{\mathfrak{b}}$. A nonzero component of $E_{\mathfrak{b}}$, or a nonzero uncharged component of the signed source sum, is passed to the local polarity successor with the same origin and ancestry. No identity involving a sum, product, localization, normalization, limit, or other composite is admissible when applied termwise before this expanded joint factorization.

Theorem II.2.18 (Target-local factorization of continuum facts)**.** *Every relation used by the continuum reachability or regularity calculus is an admissible target-local structural fact after restriction to the active branch that invoked it. This applies to weak-equation, gradient, resolvent, Hamiltonian, commutator, compactness, constraint, entropy, covariance, carrier, and continuation relations. Consequently:*

(F1) *a relation on states outside $X_{\mathfrak{b}}$ has no effect on that branch;*

(F2) *a target-null term closes no nonnull source cell and contributes no physical charge;*

(F3) *a nonnegative quantity pays for target progress only through Definition II.2.3; and*

(F4) *failure of localization, descent, ancestry, signed gluing, carrier domination, or realization is a target-local polarity successor rather than an omitted antecedent.*

Proof. First apply universal saturated-composition accountability to the complete expanded derivation of the invoked relation. Pair its joint value with the selected shell covector and restrict every input and graph coordinate to $X_{\mathfrak{b}}$. The current normal form (31) gives the five source terms and the target-null term. Structural induction over restriction, quotient, chart, trace, adjoint, polarization, and limit constructors preserves their source words by Theorem I.3.4. The carrier constructor separates a coboundary, generated nonnegative production, and true exterior input; Theorem II.2.2 cancels compatible internal faces before taking positive parts. Totalization places the first failed operation in $E_{\mathfrak{b}}$. These are exactly the terms of (110). The local charging rule gives (F2)–(F3), while target-local polarity stratification gives (F1) and (F4). $\square$

Corollary II.2.19 (Activation rule for universal identities)**.** *An equation-wide functional-analytic identity is a library relation. It enters the regularity calculus only through its factorization* (110) *on a contact-generated active cell. Thus an equation-wide operator bound, compactness statement, spectral gap, entropy law, or coercive inequality has no independent regularity consequence; only its target-local source components and admissible physical charge do.*

Corollary II.2.20 (No unscoped reachability inference). *Every derivation that changes a physical reachability profile factors through an active branch signature* $\mathfrak{b}$ *and a structural fact of the form* (110). *More precisely, its final nonidentity step is exactly one of:*

(R1) *the target-null quotient annihilates every source remainder on* $X_{\mathfrak{b}}$*;*

(R2) *the signed local carrier identity charges the selected shell progress to the restriction of the public physical measure;*

(R3) *a nonordinary coordinate creates a same-origin polarity successor; or*

(R4) *after every cumulative source owner has passed recursive carrier–polar exhaustion, diagonal saturation reconstructs a primal–polar fixed same-origin path with actual target contact.*

Consequently no theorem about an unselected state family, no transverse positive form, and no equation-wide analytic estimate can remove, charge, or continue a contact branch.

Proof. The ordered reachability calculus consumes a relation only through the local cone generated by $I_{\mathfrak{b}}$, the retained shell floor, and the factorization (110). Its zero target projection is (R1). Its admitted production and exterior terms are processed by Theorem II.2.4, giving (R2). Every nonzero defect or unabsorbed signed remainder is sent by Theorem II.2.15 to (R3). Every cumulative owner is then processed by Theorem II.2.32: cone membership returns to (R2), target-nullity returns to (R1), and separation creates another instance of (R3). Compatible cofinal successors are processed by Theorem II.2.16; only their primal–polar fixed ordinary realization gives (R4). These are the complete outgoing edges of the target-local fact compiler, so there is no fifth inference rule with equation-wide scope. $\square$

Theorem II.2.21 (Canonical record-scale selector). *Suppose a same-origin target-retaining cascade carries scale coordinates* $r_n > 0$*. Exactly one of the following generated alternatives holds:*

(S1) $\inf_n r_n > 0$*, and the cascade remains in a fixed-scale stratum;*

(S2) $\inf_n r_n = 0$*, and it has a canonical subsequence* $n_0 < n_1 < \cdots$ *defined by*

$$n_{j+1} := \min\{n > n_j : r_n \le \tfrac{1}{2} r_{n_j}\}. \tag{111}$$

For the second alternative, interpolate $\log r_{n_j}$ *affinely along the ordered normalized windows. The resulting absolutely continuous chart scale satisfies*

$$a := \frac{\dot{r}}{r} \le 0 \quad \text{a.e.}, \qquad r(\tau) \downarrow 0. \tag{112}$$

The selected centers, charts, and transitions retain the origin tag $\mathbf{R}$*. Thus* (112) *is a consequence of scale collapse, not an analytic hypothesis on the original orbit.*

Proof. If the first alternative fails, choose n_0. Since the tail infimum is zero, the set in (111) is nonempty; its least element defines n_{j+1}. The selected scales decrease by at least a factor two and therefore tend to zero. Affine interpolation of their logarithms is nonincreasing, which gives (112). All coordinates are restrictions and chart images of the original cascade, so same-origin ancestry is preserved. $\square$

Definition II.2.22 (Source-adapted dual-carrier grammar). Fix $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$. For $\Phi \in \Gamma$, begin with its descending covector $D\Phi$ and generate the smallest countable rational test module $\mathscr{W}_{\mathfrak{b}}$ closed under:

(D1) adjoints of the source operators occurring in A, including killed generator and predictable-compensator pullbacks when present;

(D2) constraint projection, gauge quotient, chart pullback, record-scale restriction, and multiplication by the generated rational cutoff algebra;

(D3) signed parent–child gluing, temporal summation by parts, carrier coboundaries, and cancellation of target-null directions;

(D4) polarization and completion of squares using the native positive Geom form and every nonnegative defect generated by the weak equation.

For a word $w \in \mathscr{W}_{\mathfrak{b}}$, expansion against the weak equation produces a *dual-carrier row*

$$Q_{A,\Phi} = \delta K_w + P_w + B_w + N_w + E_w. \tag{113}$$

Here $Q_{A,\Phi}$ is the selected source pairing, δK_w is a bounded carrier coboundary, $P_w \geq 0$ is generated physical production, B_w is true exterior input, $N_w \in \mathscr{N}_\Sigma$, and E_w is the ordered list of total-graph defects. Paired internal currents are added with their signs before P_w is formed. A production row is admitted only when the public carrier identity proves that its total mass belongs to the single physical ledger.

Remark II.2.23 (Constructive enumeration). The grammar is generated from the public equation and the selected shell covector. At depth k, enumerate weak multipliers, adjoint pulls, cutoffs, partitions, chart words, and square completions of syntactic length at most k, then reduce them modulo $I_{\mathfrak{b}}$. This is the continuum counterpart of saturating all prospective selectors in Part I. A specialization supplies the equation and its textbook weak identities; it does not supply a dual carrier or an equality theorem.

Theorem II.2.24 (Target-aligned carrier duality)**.** *For every active zero-price successor branch $\mathfrak{b}$, the generated dual-carrier hierarchy has exactly one of the following outcomes.*

(C1) *A finite ordered identity shows that the active branch is target-null.*

(C2) *At a finite depth it derives a row* (113) *whose graph defects are zero on the branch and whose exterior positive variation is in the physical ledger. Repeated target progress is then charged only to P_w, up to a bounded telescoping carrier and finite exterior input.*

(C3) *The ordered dual cone has a positive normalized functional L with*

$$L(I_{\mathfrak{b}}) = 0, \qquad L(P_w) = 0 \quad (w \in \mathscr{W}_{\mathfrak{b}}), \qquad L(Q_{A,\Phi}) \geq \eta. \tag{114}$$

It is used only to select a finite dual coefficient word for the next carrier calculation attached to the already specified owner $Q_{A,\Phi}$, with its mode, target, chart, origin, and complete parent coordinates. The functional is not inserted into the source-evaluation domain and supplies neither a source occurrence nor an active branch. Source presence requires a finite same-origin pre-contact occurrence from the presentation-generated reachable domain.

No conclusion requires control of a source or equality state outside the target-active branch $\mathfrak{b}$.

Proof. Form the Archimedean ordered cylinder algebra generated by $I_{\mathfrak{b}}$, the nonnegative productions P_w, the exterior ledger, and the inequality $Q_{A,\Phi} \geq \eta$. If its quadratic module contains -1 at a finite level, it gives (C1). Otherwise dovetail the dual-carrier words. A word whose exact telescoping identity charges the retained crossings gives (C2). If neither finite event occurs, the ordered Hahn–Banach theorem applied to the cofinal cylinder cones gives a positive normalized functional annihilating the branch ideal. Since this is a zero-price successor branch and every admitted P_w is dominated by the physical ledger, it also annihilates every P_w, while the defining shell inequality retains the target floor. This is (C3). The separating functional is used to extract the next finite dual coefficient word. This word acts only on the source owner already present in the branch and cannot establish that an occurrence realizing that owner exists. Weak duality makes the alternatives disjoint after selecting the first finite closing row; the ordered proof/model theorem makes them exhaustive. □

Definition II.2.25 (Target-local carrier cone and polar successor)**.** For an active signature $\mathfrak{b}$, let $\mathscr{Y}_{\mathfrak{b}}^{\Sigma}$ be the closed locally convex space generated by the source-owned target-current cylinder coordinates on $X_{\mathfrak{b}}$, equipped with the initial topology of the countable family of bounded rational cylinder seminorms. Its target-effective space is $\mathscr{Y}_{\mathfrak{b}}^{\Sigma}/\overline{\mathscr{N}_{\Sigma}}$. Let $\mathscr{D}_{\mathfrak{b}}^{\mathrm{alg}}$ be the algebraic convex cone generated by finite rows of

$$\delta\mathscr{K}_{\mathfrak{b}} + \mathscr{P}_{\mathfrak{b}}^{\mathrm{phys}} + \mathscr{B}_{\mathfrak{b}}^{\mathrm{phys}} + \mathscr{N}_{\Sigma} \tag{115}$$

and put $\mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}} := \overline{\mathscr{D}_{\mathfrak{b}}^{\mathrm{alg}}}$ in this topology. Here $\delta\mathscr{K}_{\mathfrak{b}}$ is included with both signs, $\mathscr{P}_{\mathfrak{b}}^{\mathrm{phys}}$ contains only nonnegative production already dominated by the single public physical measure, and $\mathscr{B}_{\mathfrak{b}}^{\mathrm{phys}}$ contains only true exterior input already present in that ledger. Membership in $\mathscr{D}_{\mathfrak{b}}^{\mathrm{alg}}$ is a finite generated target-local charging identity. Membership only in its closure is a limiting domination value, not a finite proof: the total cone-closure graph retains the approximating rows and their cylinder errors, and closes an owner only after those errors vanish on every target shell used downstream.

If $Q \notin \mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}$, a *carrier-polar successor* is the first finite-cylinder separator in the fixed dual enumeration whose coefficient tuple is retained through its total graph, or the corresponding graph-boundary value, satisfying

$$\lambda(Q) > 0, \qquad \lambda|_{\delta\mathscr{K}_{\mathfrak{b}}+\mathscr{N}_{\Sigma}} = 0, \qquad \lambda(P+B) \le 0 \quad (P+B \in \mathscr{P}_{\mathfrak{b}}^{\mathrm{phys}} + \mathscr{B}_{\mathfrak{b}}^{\mathrm{phys}}). \tag{116}$$

Adjoin λ, every adjoint pullback of λ through the public source words, every polarization and square it generates, and all their total-graph values as dual coefficient data. Retain the canonical source owner, unnormalized shell cocycle, and same-origin tag only when they come from a finite presentation-generated occurrence; do not infer them from λ. Denote this dual operation by $\mathfrak{P}_{\Sigma}(\mathfrak{b}, Q)$.

Lemma II.2.26 (Finite-cylinder detection of strict carrier separation)**.** *If $Q \notin \mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}$, then there are a finite cylinder projection π_F, a real coefficient vector q_F, and a margin $\varepsilon > 0$ such that the cylinder functional $\lambda_F = q_F \cdot \pi_F$ satisfies*

$$\lambda_F(Q) \ge \varepsilon, \qquad \sup_{D \in \mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}} \lambda_F(D) \le 0.$$

The coefficient tuple and the verification that it is nonpositive on the projected cone are retained through the total dual graph. Rational enumeration is complete only on finite rational-polyhedral projections; no rationalization or boundary-membership claim is made in general.

Proof. The quotient by the closed target-null space is locally convex and Hausdorff. Strict separation of the point Q from the closed convex cone gives a continuous functional and a positive margin. Continuity for the initial cylinder topology makes that functional factor through a finite projection π_F. This proves the displayed statement. If the projected cone is rational polyhedral, ordinary finite-dimensional linear programming gives a rational separator after reducing the margin. Without that extra property, replacing the real coefficient vector by a rational vector need not preserve membership in the polar cone, which is why the total dual graph is retained. □

Definition II.2.27 (Finite-cylinder carrier comparison problem)**.** Fix an active same-origin branch $\mathfrak{b}$, a finite generated cylinder F, and a source-owned target current Q. Enlarge F, when necessary, so that it contains Q, the order unit 1, the physical ledger coordinate p_{phys}, and every coordinate occurring in the finite relations imposed at this stage. Let A_F be the resulting finite-dimensional real order-unit space modulo the projected equality ideal and target-null subspace. Write $C_F \subseteq A_F$

for the closed cone generated by the projected squares, admitted physical productions, true exterior inputs, and signed carrier coboundaries.

For $b \geq 0$, define the normalized finite-cylinder dual test set and its carrier upper bound by

$$\mathcal{S}_F(b) := \{L \in A_F^* : L(1) = 1,\ L(C_F) \geq 0,\ 0 \leq L(p_{\mathrm{phys}}) \leq b\}, \tag{117}$$

$$\mathsf{U}_{F,Q}(b) := \sup_{L \in \mathcal{S}_F(b)} L(Q), \tag{118}$$

with supremum $-\infty$ when $\mathcal{S}_F(b) = \varnothing$. The cylinder is called *checked* when every relation used to form A_F and C_F carries its finite derivation from the public equation. Thus $\mathcal{S}_F(b)$ is a dual test set generated by checked relations, not a source-evaluation domain. Its only use is to derive a finite upper bound or separating inequality for evaluations of actual occurrences.

Definition II.2.28 (Finite-cylinder occurrence presence)**.** Let $\mathcal{O}_F^{\mathrm{pre}}(\mathfrak{b})$ be the finite same-origin pre-contact occurrences obtained by restricting ordinary paths in the presentation-generated reachable domain of $\mathfrak{b}$ to the cylinder F. Each $R \in \mathcal{O}_F^{\mathrm{pre}}(\mathfrak{b})$ defines its actual evaluation functional $\mathrm{ev}_R \in \mathcal{S}_F(b_R)$. A source owner Q is *present above* η *on* F precisely when

$$\exists R \in \mathcal{O}_F^{\mathrm{pre}}(\mathfrak{b}) \quad \mathrm{ev}_R(Q) \geq \eta. \tag{119}$$

The occurrence, not an arbitrary member of $\mathcal{S}_F(b)$, carries the chronology, public initial datum, source ancestry, and pre-contact origin needed by prospective accountability. For a physical ceiling b, the finite-prefix source profile is

$$\mathsf{T}_{F,Q}^{\mathrm{occ}}(b) := \sup\{\mathrm{ev}_R(Q) : R \in \mathcal{O}_F^{\mathrm{pre}}(\mathfrak{b}),\ b_R \leq b\}. \tag{120}$$

This is the only source profile associated with F.

Theorem II.2.29 (Exact finite-cylinder carrier-bound duality)**.** *Let F be a checked cylinder. Then $\mathcal{S}_F(b)$ is compact and, whenever it is nonempty,*

$$\mathsf{U}_{F,Q}(b) = \inf_{\alpha \geq 0,\, \beta \in \mathbb{R}} \{\alpha b + \beta : \beta 1 + \alpha p_{\mathrm{phys}} - Q \in C_F\}. \tag{121}$$

If the infimum is attained only in $\overline{C_F^{\mathrm{alg}}} = C_F$, the approximating cone rows and their errors are retained as a cone-boundary successor. It is a finite charging proof precisely when the displayed element belongs to C_F^{alg}.

For every threshold η, exactly one of the following statements about the carrier upper bound holds:

(F1) *a checked dual row gives $\mathsf{U}_{F,Q}(b) < \eta$;*

(F2) *$\mathsf{U}_{F,Q}(b) = \eta$, and the exposed equality face*

$$\mathcal{E}_{F,Q}(b, \eta) := \{L \in \mathcal{S}_F(b) : L(Q) = \eta\} \tag{122}$$

is used only to identify the vanishing terms of an attaining or approximating dual row; those finite algebraic terms, rather than a member of the face, are adjoined to the carrier calculation;

(F3) *$\mathsf{U}_{F,Q}(b) > \eta$; this numerical value supplies no source presence statement.*

In (F2) *the equality face is processed by generated squares, polarizations, adjoints, symmetries, and the dyadic ordered-coercivity rule. Neither* (F2) *nor* (F3) *establishes source presence. The actual occurrence profile satisfies only the rigorously licensed inequality*

$$\mathsf{T}_{F,Q}^{\mathrm{occ}}(b) \leq \mathsf{U}_{F,Q}(b). \tag{123}$$

Presence is exactly (119)*.*

Proof. The order-unit normalization bounds every coordinate of A_F^*; positivity and the ledger inequality define a closed subset. Finite dimensionality therefore makes $\mathcal{S}_F(b)$ compact. Its support value at Q is the primal value in (118). The polar of the intersection of the normalized positive dual cone with $L(p_{\mathrm{phys}}) \le b$ consists exactly of the affine majorants $Q \le \alpha p_{\mathrm{phys}} + \beta 1$, with $\alpha \ge 0$. Finite-dimensional separation, applied to the epigraph of the support function, gives (121); this is the standard bipolar proof and requires neither semialgebraicity nor rational coefficients.

Compactness gives a maximizer of the dual carrier-bound problem; the maximizer is not retained as a source-evaluation value. Comparison of the maximum with η gives the three disjoint cases. Every actual evaluation ev_R is an admissible dual test, which proves (123). In the equality case, positivity implies that every nonnegative summand on which an attaining dual row has zero total value vanishes separately. For a nonattained boundary row the same statement is expressed by its approximating rows and errors, and no finite proof is asserted. The listed equality constructors preserve the same finite cylinder after adjoining their finitely many output coordinates; Theorem III.4.45 applies whenever its generated row is present. ☐

Theorem II.2.30 (Monotone cylinder carrier bounds)**.** *Let $F_1 \subseteq F_2 \subseteq \cdots$ be the canonical enumeration of the checked target-relative cylinders containing $Q, 1, p_{\mathrm{phys}}$, and let restriction map $\mathcal{S}_{F_{n+1}}(b)$ into $\mathcal{S}_{F_n}(b)$. Then*

$$\mathsf{U}_{F_1,Q}(b) \ge \mathsf{U}_{F_2,Q}(b) \ge \cdots, \qquad \mathsf{U}_{\infty,Q}(b) := \inf_n \mathsf{U}_{F_n,Q}(b). \tag{124}$$

For every actual finite same-origin occurrence whose restriction is defined on F_n,

$$\mathrm{ev}_R(Q) \le \mathsf{U}_{F_n,Q}(b_R). \tag{125}$$

Hence $\mathsf{U}_{F_n,Q}(b) < \eta$ at one finite depth excludes every actual occurrence of physical cost at most b and target value at least η. If no such finite depth is found, the carrier-bound calculation has produced no source-presence or terminal statement. Source presence and cofinal accumulation remain entirely in the actual occurrence domain of Definition II.2.28 and the prospective backward frontier.

Proof. Adding checked relations can only shrink the dual test set, proving monotonicity of its support bound. The actual evaluation functional of an occurrence belongs to every finite dual test set on which that occurrence is defined; (125) follows. The finite exclusion statement is its contrapositive. No inverse limit of dual test functionals is formed. ☐

Corollary II.2.31 (Part-I form of continuum envelope evaluation)**.** *Let a stopped first-contact readout have target floor $\eta > 0$. Apply the backward frontier before terminal passage, classify every conditional joint increment by its complete five-source ancestry, and evaluate each resulting owner through Theorems II.2.29 and II.2.30. Then a checked carrier ceiling below η excludes that contact. Source presence, when contact is the premise, is supplied only by its finite same-origin pre-contact occurrence and Definition II.2.28. The dual calculation neither adds nor replaces an occurrence. Fixedness, cone-boundary membership, an irrational separator, failure of semialgebraic elimination, or survival through every finite cylinder creates no additional source or terminal value. This is the continuum counterpart of bounding the saturated capability profile in Part I while keeping actual executions separate from dual bounds.*

Theorem II.2.32 (Universal recursive amplification exhaustion)**.** *Let Q be the canonical source owner of a nonzero target increment created by any vertex of a universal expanded derivation on* *an active same-origin branch $\mathfrak{b}$. This includes primitive, cross-term, localized, stopped, normalized, scale-transferred, recurrent, limiting, tail, and boundary owners. Exactly one of the following compiler transitions occurs.*

(P1) *The target projection of Q vanishes after the generated quotient, so the owner cannot contribute to shell contact.*

(P2) *On the complementary quotient cell, $Q \in \mathscr{D}_{\mathfrak{b}}^{\mathrm{alg}}$. Its finite cone identity telescopes storage, removes target-null terms, and charges the remaining positive mass to the public physical ledger.*

(P3) *The owner belongs to $\mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}} \setminus \mathscr{D}_{\mathfrak{b}}^{\mathrm{alg}}$. Its cone-closure approximation rows, target-shell errors, and inherited owner form the next typed cone-boundary successor. It is not declared charged until the limiting errors vanish on the retained shell cocycle.*

(P4) *The carrier-polar finite-prefix calculation $\mathfrak{P}_{\Sigma}(\mathfrak{b}, Q)$ is nonzero and is processed before circulation, recurrence, compact realization, or another shell is evaluated.*

A finite value of the universal amplification graph is merely another coordinate of the cone test; it closes the owner only when it supplies the displayed membership identity. An unbounded or zero-denominator value is processed by (P3) *or* (P4)*, according to its cone-closure value, with the numerator, denominator, source word, and unnormalized cocycle retained. The compiler applies this transition to every owner in the fixed enumeration of the expanded joint record before any postprocessing or realization row.*

A cofinal polar chain is joined to, but kept type-distinct from, its primal owner chain. Its joint comparison limit is charged, proves the primal owner target-null, exposes a finite unprocessed primal occurrence, or is fixed under the comparison grammar. Fixedness supplies no source atom. On a contact-seeded execution, prospective source presence comes solely from the retained finite same-origin occurrences; an independently constructed ordinary public orbit is still required for a singular-existence verdict.

Proof. The first cell is removed before the cone test. Algebraic membership, closure-only membership, and nonmembership are disjoint total-graph values. The first gives the finite charging row in (P2); the second gives the retained convergence graph in (P3). On their complement, geometric Hahn–Banach separation of Q from the closed convex cone $\mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}$, followed by Lemma II.2.26, gives λ satisfying (116); the coboundary and null summands are linear spaces and are therefore annihilated. The topology of $\mathscr{Y}_{\mathfrak{b}}^{\Sigma}$ is generated by its finite-cylinder seminorms, so a continuous strict separator is detected at finite cylinder depth. Its coefficient tuple is a value of the totalized dual graph: an ordinary value gives the separator and a nonordinary value is its typed dual graph boundary. Totalization retains the separator's association with the owner; it does not turn the separator into a source occurrence.

Apply the adjoint and polarization clauses of Definition II.2.22. A previously absent separator or descendant is a finite proper comparison refinement. For a cofinal chain, retain the cumulative primal occurrence record and its dual comparison record as distinct typed coordinates. Apply Theorem III.4.58 only to the primal occurrence coordinates: every nonzero limiting target component then occurs on a finite same-origin prefix. Dual fixedness supplies only a finite-prefix bound. This proves exhaustiveness of the four cone values and prevents a positive uncharged owner from being passed directly to realization. Because the proof used only algebraic membership, closure, and separation for an arbitrary generated owner, it applies verbatim to every constructor named in the theorem. Totalization of its amplification graph prevents division by a vanishing physical quantity or normalization factor from deleting the numerator. Dovetailing the countable owner enumeration before realization proves the universal assertion. $\square$

Corollary II.2.33 (No pre-polar realization)**.** *A polar functional, equality functional, or compatible dual inverse limit never enters an ordinary-orbit realization row. It is a finite-prefix bound record. Only a presentation-generated same-origin primal approximation branch may enter realization, after each of its canonical source owners has been processed by Theorem II.2.32. Comparison fixedness neither supplies nor manufactures a source occurrence.*

Corollary II.2.34 (No diffuse comparison-to-presence loophole)**.** *Let $(\mathcal{J}_n^{\mathrm{pr}})$ be the cumulative same-origin primal occurrence records of a cofinal compiler execution, and let (Λ_n) be the associated finite-prefix polar calculations. The target map acts on $\mathcal{J}_n^{\mathrm{pr}}$, not on Λ_n. If the primal limit has nonzero target projection, then some finite same-origin primal prefix has nonzero target projection. Equivalently,*

$$q_\Sigma(\mathcal{J}_n^{\mathrm{pr}}) = 0 \quad \textit{for every } n \quad \Longrightarrow \quad q_\Sigma(\mathcal{J}_\infty^{\mathrm{pr}}) = 0. \tag{126}$$

Hence no countable family of individually target-null or already canceled primal occurrences can create target-aligned structure only at the limit, and no collection of polar comparisons can create source presence at all. If the cumulative primal target contribution is nonzero, its canonical owner is exposed on a finite same-origin prefix and is processed again by Theorem II.2.32. Neither division by the number of returns nor passage to an average is permitted before this test.

Proof. Carrier–polar separators are coefficient records in the dual cylinder algebra. They may test a primal current but are outside the domain of the target projection. Apply Theorem III.4.58 to the cumulative primal occurrence algebra. Its exact orthogonal finite-prefix decomposition makes the target projection of the limit the sum of its actual finite occurrence increments. Thus a nonzero projection has a nonzero finite-prefix increment. The occurrence owner and its complete ancestry are retained by the cumulative primal join, so recursive carrier–polar exhaustion applies to that owner. The final sentence follows from the unnormalized cocycle coordinate $A_{\Phi,N} = N$, which is included before every recurrent normalization. □

Definition II.2.35 (Target-local equality stratum)**.** The equality kernel of $\mathfrak{b}$ is

$$\mathcal{E}_{\mathfrak{b}} := \{z : I_{\mathfrak{b}}(z) = 0,\ P_w(z) = 0 \text{ for all } w \in \mathscr{W}_{\mathfrak{b}}\}. \tag{127}$$

It is restricted immediately to the retained shell cell $Q_{A,\Phi} \geq \eta$. Its target-null cell is precisely

$$\mathcal{E}_{\mathfrak{b}} \cap \{Q_{A,\Phi} \geq \eta\} = \varnothing. \tag{128}$$

The finite-level quadratic modules are dovetailed with the generated polarity stratification. A derivation of -1 after adjoining $Q_{A,\Phi} - \eta$ closes the local cell; compatible positive functionals are used only to select further finite equality rows and are not retained in the source-evaluation domain. A target-retaining equality successor is present only when a finite same-origin pre-contact occurrence satisfies the equality rows. Thus (128) is an output of local saturation, not a property to be established separately and not a classification of all solutions of the equation.

Proposition II.2.36 (Covariant modulation conjugation)**.** *Suppose a dual-carrier word on an active moving-chart branch gives*

$$K' = -D + 2aK + R, \qquad K, D \geq 0, \tag{129}$$

where R is the remaining signed source current. Let $\theta > 0$ solve the scalar terminal or initial-value problem

$$\theta' + 2a\theta = -1. \tag{130}$$

Then the generated carrier $\mathcal{G} = \theta K$ satisfies the exact identity

$$\mathcal{G}' = -K - \theta D + \theta R. \tag{131}$$

No sign or integrability condition on a is used. If R is canceled by an adjoint word, a signed interface gluing, or a target-null quotient, then $K + \theta D$ is a nonnegative production generator in $\mathscr{W}_{\mathfrak{b}}$.

Proof. Differentiate θK and substitute Eqs. (129) and (130):

$$(\theta K)' = (-1 - 2a\theta)K + \theta(-D + 2aK + R) = -K - \theta D + \theta R.$$

All operations belong to the scalar adjoint and carrier-coboundary clauses of Definition II.2.22. $\square$

Theorem II.2.37 (Branch-local modulation stratification)**.** *Let $\mathfrak{b}$ be a record-scale active branch, so that $a \leq 0$ by Theorem II.2.21. Suppose its generated stationary equations contain*

$$\int (P - aO)\, \mathrm{d}\varpi = 0, \qquad P, O \geq 0, \tag{132}$$

Then every retained stationary law is supported on

$$\mathcal{E}_{\mathfrak{b}}^{P=0} := \mathcal{E}_{\mathfrak{b}} \cap \{P = 0\}. \tag{133}$$

Restrict the local ordered cone to $\mathcal{E}_{\mathfrak{b}}^{P=0}$. A finite -1-identity closes its retained target cell; otherwise its compatible positive moments select the next finite equality tests. If a finite same-origin pre-contact occurrence is present on this stratum, the shell covector, five source currents, and every generated operation of that occurrence are restricted to (133)*, its next polarity coordinate is recorded, and prospective accountability is rerun there. The positive moments alone do not create this successor. No property or semantic emptiness decision for the zero set is supplied as a premise.*

Proof. Both terms in $P - aO = P + (-a)O$ are nonnegative. Equation (132) therefore gives $P = 0$ ϖ-almost surely, proving (133). Apply the local Archimedean proof/model alternative after adjoining $P = 0$ and the retained shell floor. A finite contradiction closes the carrier test; otherwise the positive law is used only to select another finite coefficient row. Source presence and reselection use only an occurrence in $\mathcal{O}_F^{\mathrm{pre}}(\mathfrak{b})$ satisfying the same equality rows. Restriction, polarity refinement, and prospective reselection of that occurrence are generated operations in Definitions II.2.13 and II.2.22. $\square$

Theorem II.2.38 (Exhaustive loss-of-compactness routing)**.** *Let a sequence in an active branch fail to converge in an ordinary topology used by a dual-carrier row. Its image in the generated evaluation compactification has a subsequence with exactly one primary successor mode:*

$$\mathsf{atom},\ \mathsf{separated},\ \mathsf{nested},\ \mathsf{diffuse},\ \mathsf{tail},\ \mathsf{rough},\ \mathsf{gauge},\ \mathsf{constraint},\ \mathsf{equality}. \tag{134}$$

The compactified value selects the next finite cylinder test; it carries no source-presence or target-floor semantics. The target floor remains on the finite same-origin occurrences generating the sequence. Atom, separated, nested, and diffuse values are disintegrations of oscillation or concentration and append Lift*; tail values append* Bdry*; gauge values append* Abs*; rough equation and constraint values retain the ancestry of the failed operation. If a scale coordinate tends to zero, Theorem II.2.21 supplies the nonincreasing record chart before the next test is evaluated. If the mode's conditional increment vanishes on every generating occurrence, that occurrence cell closes in $\mathscr{N}_{\Sigma}$. Otherwise the first finite same-origin occurrence detecting the mode is processed by Theorem II.2.24 with the same owner.*

Proof. The normalized activity measures are probabilities on the compact observer space. Sequential compactness and disintegration give the mutually ordered readouts in (134); the first nonordinary total-graph coordinate fixes the primary mode. Completion identifies the next coordinate test but creates no occurrence. The prospective finite-prefix theorem preserves the target floor on the generating occurrence sequence. Applying the new test there gives either zero actual conditional increments or a first nonzero finite increment. The latter retains the failed owner's ancestry and re-enters the dual-carrier compiler. □

Theorem II.2.39 (Continuous selector-accountability closure)**.** *Let an ordinary finite-physical-budget orbit contact Σ. Then there is an active branch signature $\mathfrak{b}$ with $\eta \geq 1/5$. Along that branch, repeated contact has exactly one of the following consequences:*

(A1) *a generated dual carrier charges infinitely many disjoint crossings to the single finite physical ledger, a contradiction;*

(A2) *a generated target-null, regular, continuable, or class-incompatible successor contradicts retained contact;*

(A3) *target-local polarity stratification produces a proper successor with strictly larger resolved-prefix rank;*

(A4) *the retained finite same-origin occurrences continue cofinally after the carrier coefficient enumeration produces no further finite closing row. The occurrences, through their unnormalized shell cocycle, witness the lower bound in the saturated prospective source cell.*

In the fourth alternative, prospective source presence is the cofinal family of finite pre-contact occurrences. A singular-existence verdict still requires an independent-origin ordinary orbit. A stationary law, equality kernel, ordinary path, or failed analytic property is processed by the third alternative and is never a terminal conclusion. Hence contact forces a nonzero target-aligned cell of the fully saturated five-source prospective profile. An upper bound below its retained shell floor contradicts contact by Theorem I.5.13.

Proof. Shell accountability and finite source exhaustion select a persistent support and give the floor 1/5. Eventual constancy of the realization mode gives m, while the cascade supplies Γ and the origin tag. This constructs $\mathfrak{b}$. Apply Theorem II.2.24. A finite carrier row telescopes over the oriented complex containing both selected events and their gaps. On the zero/root–leaf-bounded storage and finite-exterior-input cell, the remaining term is nonnegative physical production, and infinite retained progress contradicts the physical ledger. A nonzero storage return or unabsorbed exterior-input coordinate is instead the next target-aligned successor. Every zero target projection or terminal regularity relation gives (A2). Otherwise Theorem II.2.38 produces the next active branch. Any stationary equality law is restricted to its target-local structural stratum and passed through Theorem II.2.15. A finite nonnull cell gives the proper successor in (A3). If proper successors continue cofinally, form their cumulative join. By Theorem III.4.58, it is target-null, exposes a least unprocessed finite joint increment, or is fixed under every generated operation after all increments have been source-resolved. The first gives (A2), the second returns to (A3), and the third is passed through Theorem II.2.32 for every canonical primal and polar owner. Target-nullity gives (A2), cone membership gives (A1), and a separator is a proper successor in (A3). Only simultaneous primal–polar fixedness retains the unit shell cocycle and complete five-channel ancestry and gives (A4). By Theorem II.2.41, its contact-derived origin remains hypothetical and cannot supply singular existence. Thus no equality test, analytic classification, or raw inverse limit remains as a premise. □

Definition II.2.40 (Canonical activity lift and ordered successor rows)**.** Let $\mathfrak{C}_\eta$ be the closure, in the countable product of the generated evaluation compactification, of all indexed normalized shell cascades

$$\mathbf{R} = ((R_n, Q_n, p_n))_{n\geq 1}, \qquad \mathrm{Prog}_\Sigma(R_n) \geq \eta,$$

that satisfy the total weak-equation, shell, and event-order graphs. On an ordinary contact value, the Q_n are the canonical pairwise disjoint physical events, $p_n = \mu_u(Q_n)$, and the progress is one. Here p_n is the raw event-expenditure coordinate. It becomes an admissible price for shell progress only when the target-local factorization satisfies (98); otherwise the carrier-domination graph value is retained in R_n as a successor coordinate. At a graph boundary, p_n is the lower-semicontinuous event-expenditure coordinate. Boundary points of $\mathfrak{C}_\eta$ retain the sequences that approximate them. The compact cascade coordinate $\mathbf{R}$ is an *origin tag*; every descendant below retains it, so successor chains cannot splice different orbits or cascades.

The tag also carries an origin polarity

$$\mathrm{pol}_{\mathrm{org}}(\mathbf{R}) \in \{\mathsf{Hyp}, \mathsf{Ind}\}. \tag{135}$$

A record extracted after assuming target contact has polarity Hyp. A record has polarity Ind only when its public orbit is generated from the declared PDE data without using target contact, noncontinuation, or any descendant of either statement as an input. Every restriction, limit, quotient, chart, adjoint, polarization, return, and joint-saturation constructor preserves origin polarity. In particular, there is no constructor $\mathsf{Hyp} \to \mathsf{Ind}$.

Within the successor algebra, "zero price" abbreviates zero restriction of the public physical measure on a retained target-aligned event. It does not assert a target-progress domination inequality. When that inequality is ordinary and admitted, zero price is also zero admissible target charge; when it fails, its total-graph coordinate remains part of the same zero-expenditure successor and is processed before any exclusion inference.

The generated compact observer space Ω_{ev} is the product of the state, spacetime, scale, source, chart, constraint, origin, and total-graph coordinates occurring in the symbolic transcript.

Theorem II.2.41 (Origin-polarity preservation and no recycled contact)**.** *The generated continuum algebra preserves* $\mathrm{pol}_{\mathrm{org}}$*. Consequently a contact-selected compiler execution and every one of its finite or cofinal descendants have polarity* Hyp*. Such a record may prove a lower bound on the saturated prospective source profile, but it cannot prove existence of a singular orbit. A singular-existence conclusion requires an ordinary public orbit with polarity* Ind*.*

Proof. The primitive contact selector is tagged Hyp. Every generated constructor copies the origin coordinate into its output, and cumulative joining is defined only for equal origin tags. Totalization adds a graph value without changing its input origin. Structural induction over the grammar therefore preserves Hyp. An independently generated orbit begins instead in the public-data fiber tagged Ind. Since the grammar contains no origin-changing constructor, the two fibers are disjoint. The target-profile lower bound uses only the retained shell cocycle and is valid in the hypothetical fiber; singular existence is an existential statement in the independent fiber. □

On the bounded-variation branch, let q_n^c be the finite signed Radon measure containing all ordinary, jump, Cantor, oscillation, concentration, and exterior parts of the derived c-source shell current. After pushing them to Ω_{ev}, set

$$a_n := \sum_{c\in\mathfrak{C}} (q_n^c)^+, \qquad \nu_n := \frac{a_n}{a_n(\Omega_{\mathrm{ev}})}. \tag{136}$$

The target floor and (39) imply $a_n(\Omega_{\mathrm{ev}}) \ge \eta$, and an ordinary normalized contact cascade has the stronger lower bound one. If a current has no finite Radon value, its normalized variation-boundary value and its owner from (41) are placed at the corresponding typed boundary coordinate; positive rational truncations of the owner current give the probability-valued lift without summing the remaining extended signed coordinates. The lifted record consists of

$$\widehat{R}_n = \left(\mathbf{R}, R_n, \nu_n, \frac{a_n(\Omega_{\mathrm{ev}})}{1 + a_n(\Omega_{\mathrm{ev}})}, \frac{p_n}{1 + p_n}, n\right). \tag{137}$$

Run the following total graphs in order on this lift:

$$\begin{gathered}
\text{form/constraint realization} \longrightarrow \text{chart/gauge/scale coherence,} \\
\text{activity-measure lift} \longrightarrow \text{atomic/non-atomic/boundary disintegration,} \\
\text{state/scale disintegration} \longrightarrow \text{Young, profile-tree, and tail coordinates,} \\
\text{carrier cohomology} \longrightarrow \text{strict passivity/saturation} \longrightarrow \text{zero-cost recurrent closure.}
\end{gathered} \tag{138}$$

The grammar includes the cofinal tail shift $(R_n, Q_n, p_n) \mapsto (R_{n+1}, Q_{n+1}, p_{n+1})$. Hence a target-retaining zero-price witness sequence always has a successor; an analytic failure is carried by its total-graph coordinate rather than by an artificial failure self-loop. The familiar realization modes

$$\mathsf{atom}, \quad \mathsf{separated}, \quad \mathsf{nested}, \quad \mathsf{diffuse}, \quad \mathsf{tail}, \quad \mathsf{rough}, \quad \mathsf{gauge}, \quad \mathsf{constraint}, \quad \mathsf{equality}. \tag{139}$$

are canonical readouts of ν_n and its coordinate disintegrations; they are not asserted to be mutually exclusive alternatives. Every ordinary or graph-boundary output retains its five-channel ancestry, original event index, and target floor. Hence a failed analytic relation is immediately another lifted successor input, not an unevaluated row.

Theorem II.2.42 (Compact activity-profile exhaustion)**.** *Every sequence of lifted records $\widehat{R}_n$ has a convergent subsequence in*

$$\widehat{X}_\Sigma := \mathfrak{C}_\eta \times X_{\mathrm{ev}} \times \mathcal{P}(\Omega_{\mathrm{ev}}) \times [0,1]^2 \times \overline{\mathbb{N}}, \tag{140}$$

where $\mathcal{P}(\Omega_{\mathrm{ev}})$ carries the narrow topology and $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$. Its limit retains the target floor (unit progress for a contact origin) and source ancestry. The limiting activity law has the canonical decomposition

$$\nu = \sum_{j \ge 1} m_j \delta_{z_j} + \nu_{\mathrm{na}}, \qquad m_j > 0, \tag{141}$$

and its projections to the compactified scale, spatial-boundary, state-law, and total-graph coordinates record, respectively, nested or separated profiles, tails, Young oscillation, and rough/gauge/constraint failures. These coordinates exhaust the limit because they are coordinates of the single probability law ν; no separate compactness input or remainder class is required.

Proof. The spaces $\mathfrak{C}_\eta$, X_{ev}, and Ω_{ev} are compact metric. Hence $\mathcal{P}(\Omega_{\mathrm{ev}})$ is compact metric, and so is the product in (140). Sequential compactness gives a convergent subsequence. Unit target progress and the finitely many source supports are closed coordinates; pass first to a constant-support subsequence.

Every probability measure has a unique at most countable atomic part and a nonatomic remainder, which proves (141). Coordinate projection and disintegration record all spatial, scale, state-law, and graph-boundary behavior. Boundary escape is mass on the compactification boundary, rather than failure of tightness. Since every generated observation is one of these countable coordinates, equality of all coordinates is equality of lifted records. Thus no unrecorded limit output remains. $\square$

Definition II.2.43 (Witness-indexed retained successor relation)**.** Let $\mathfrak{A}_\eta \subset \widehat{X}_\Sigma$ be the closed set of lifted records with target progress at least $\eta > 0$. An elementary successor witness is a tuple

$$(w, w', \mathbf{R}, \sigma, O), \tag{142}$$

where $\mathbf{R} = ((R_n, Q_n, p_n))_n \in \mathfrak{C}_\eta$, $\sigma : \mathbb{N} \to \mathbb{N}$ is strictly increasing, and O is one of the ordered total-graph operations in (138). Enumerate these operations and compactify their index by $\overline{\mathbb{N}}$; an operation-index boundary is represented by the corresponding total-graph limit. The two endpoints must be limits along the *same* selector:

$$w = \lim_{k\to\infty} \widehat{R}_{\sigma(k)}, \qquad w' = \lim_{k\to\infty} O^{\mathrm{tot}}(\widehat{R}_{\sigma(k)}), \tag{143}$$

and both retain the origin coordinate $\mathbf{R}$. Let $\mathfrak{W}_\eta^0$ be the set of these tuples. Compactify the selector by its graph in $\overline{\mathbb{N}}^{\mathbb{N}}$, retain the compactified operation index, take the closure $\mathfrak{W}_\eta = \overline{\mathfrak{W}_\eta^0}$, and define

$$\mathcal{R}_\eta := \pi_{w,w'}(\mathfrak{W}_\eta) \subseteq \mathfrak{A}_\eta \times \mathfrak{A}_\eta. \tag{144}$$

The witness space is compact, so $\mathcal{R}_\eta$ is closed and every edge has a nonempty compact witness fiber. Every witness carries the common origin, approximating sequence, source tags, unit target coordinate, and original event indices. A graph-boundary value is therefore another edge limit. It is never promoted to an actual orbit state without its witness sequence, and edges from different origin cascades cannot be concatenated.

Lemma II.2.44 (Diagonal event ancestry and no double counting)**.** *Let a finite or infinite chain of successors have one ordinary contact origin $\mathbf{R} = ((R_n, Q_n, p_n))_n$. For every k one can choose an original index N_k such that $N_{k+1} > N_k$ and the N_k-th original record, after the first k ordered operations in the witness chain, approximates the k-th successor coordinate within 2^{-k}. Consequently the physical events Q_{N_k} are pairwise disjoint and*

$$\sum_k \mu_u(Q_{N_k}) \le \mu_u\left(\bigsqcup_n Q_n\right) \le B_{\mathrm{phys}}. \tag{145}$$

Proof. Choose a witness from the compact fiber of every edge. At depth k, use the same-origin limits (143) and a diagonal approximation of the first k witness tuples by elementary witnesses. Their selectors have cofinal image in the one origin sequence. Choose $N_k > N_{k-1}$ sufficiently far out that the composed depth- k graph value is within 2^{-k} of w_k. No single event is used to approximate two different depths. The events therefore belong to distinct members of the original disjoint shell family, and countable additivity gives (145). □

Theorem II.2.45 (Successor recurrence)**.** *The relation $\mathcal{R}_\eta$ is closed in the compact metric lifted topology. If there is an infinite chain*

$$w_0 \mathcal{R}_\eta w_1 \mathcal{R}_\eta w_2 \mathcal{R}_\eta \cdots, \tag{146}$$

then the path space of all such chains is compact and invariant under the left shift. The orbit closure of the given path carries a shift-invariant Borel probability measure. Every point in the support of its coordinate projection is a limit generated from one retained cascade origin with a diagonal witness sequence, retains the target floor, and retains the union of source ancestries. No claim of pointwise realization by a new PDE orbit is used.

Proof. The witness bundle is compact and coordinate projection is continuous, so $\mathcal{R}_\eta$ is compact and hence closed. The set $\mathfrak{A}_\eta$ is compact in the lifted topology, so $\mathfrak{A}_\eta^{\mathbb{N}}$ is compact. The path space is the intersection, over all j, of the closed conditions $(w_j, w_{j+1}) \in \mathcal{R}_\eta$; it is therefore compact and is nonempty by (146). Empirical measures of the shifted given path have a weakly convergent subsequence, and the endpoint calculation proves shift invariance. Apply Lemma II.2.44 to obtain the diagonal original-event witnesses. Sourcewise closure preserves ancestry and the target floor. $\square$

Definition II.2.46 (Target-aligned ordered path calculus)**.** Fix an active branch $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$. Let $\mathfrak{A}_\mathfrak{b}$ be the closed lifted stratum with that source support, realization mode, shell family and floor, origin tag, and all prior zero transcript coordinates. Let $\mathcal{B}_\mathfrak{b}^{\mathbb{Q}}$ be the unital rational cylinder algebra generated by finitely supported path coordinates on $\mathfrak{A}_\mathfrak{b}^{\mathbb{N}}$, including:

(a) the graph-complete equation, carrier, operation-defect, and source coordinates at every rational path time;
(b) the closed successor-witness coordinates for every adjacent pair;
(c) the bounded price coordinate $\bar{p} = p/(1+p)$;
(d) the normalized target pairing and the positive sourcewise shell masses;
(e) the shift pullback S^*.

Generators annihilated by every element of Γ_Σ are removed before $\mathcal{B}_\mathfrak{b}^{\mathbb{Q}}$ is formed.

Let $I_\mathfrak{b}$ be the ideal generated by the weak equation, signed carrier, typed graph, same-origin, successor, zero-density, target-floor, and source-support equalities together with the complete active-branch signature. The zero-density generator acts only on the normalized support coordinate. The unnormalized occurrence cocycle, physical expenditure, and source currents are independent coordinates and no element of $I_\mathfrak{b}$ identifies them with zero. Let $M_\mathfrak{b}$ be the quadratic module generated by squares, normalized coordinate bounds, nonnegative production and defect masses, and the positive target-alignment coordinates. Finally, let

$$\mathcal{D}_S := \operatorname{span}_{\mathbb{Q}}\{f \circ S - f : f \in \mathcal{B}_\mathfrak{b}^{\mathbb{Q}}\}$$

and define the ordered proof cone

$$\mathcal{K}_\mathfrak{b} := I_\mathfrak{b} + M_\mathfrak{b} + \mathcal{D}_S. \tag{147}$$

A finite ordered derivation is an explicit equality

$$-1 = i + m + d, \qquad i \in I_\mathfrak{b}, \quad m \in M_\mathfrak{b}, \quad d \in \mathcal{D}_S,$$

written with rational cylinder polynomials.

Theorem II.2.47 (Ordered proof or stationary structural model)**.** *For every reachable active branch* $\mathfrak{b}$*, exactly one of the following holds.*

(O1) $-1 \in \mathcal{K}_\mathfrak{b}$*. Then a finite ordered derivation proves that no stationary target-aligned zero-density successor support in* $X_\mathfrak{b}$ *exists.*
(O2) *There is a linear functional* $L : \mathcal{B}_\mathfrak{b}^{\mathbb{Q}} \to \mathbb{R}$ *such that*

$$L(1) = 1, \qquad L(M_\mathfrak{b}) \geq 0, \qquad L(I_\mathfrak{b}) = L(\mathcal{D}_S) = 0. \tag{148}$$

Its continuous extension is integration against a shift-invariant Borel probability law on the graph-complete path space. That law has zero physical expenditure density and satisfies the full signature $\mathfrak{b}$*, including target pairing at least* η*, source support* A*, mode* m*, origin* $\mathbf{R}$*, and all same-origin successor coordinates. It is returned jointly with the unnormalized multiplicity ledger inherited from* $\mathbf{R}$*.*

The alternatives are exclusive. Every assertion in (O1) *is checked by finite rational algebra; every object in* (O2) *is reconstructed from the public graph relations.*

Proof. Pass first to the rational vector-space quotient by $I_{\mathfrak{b}} + \mathcal{D}_S$. The normalized coordinate bounds make the image of $M_{\mathfrak{b}}$ an Archimedean order cone with order unit [1]. If $-[1]$ is not in this cone, the order-unit separation theorem gives a positive linear functional with value one on [1]. Pulling it back gives (148). Positivity makes the functional bounded in the order-unit norm. The Archimedean representation theorem for quadratic modules, followed by the Riesz representation theorem, therefore represents its continuous extension by a Borel probability measure supported on the joint positivity set of the graph-complete path relations [13]. Annihilation of $\mathcal{D}_S$ is precisely shift invariance. Annihilation of $I_{\mathfrak{b}}$ imposes the equation, successor, same-origin, zero-density, target-floor, source, mode, origin, and prior-zero relations. This proves (O2). Since the cumulative coordinates are not generators of this zero-density ideal, separation cannot erase them.

Conversely, a functional satisfying (148) sends every $i + m + d \in \mathcal{K}_{\mathfrak{b}}$ to a nonnegative number. It therefore cannot coexist with $-1 \in \mathcal{K}_{\mathfrak{b}}$. If no such functional exists, order-unit separation gives $-1 \in \mathcal{K}_{\mathfrak{b}}$, proving exhaustiveness. An element of the algebraic sum is, by definition, a finite expression, so the derivation uses finitely many rational relations. □

Definition II.2.48 (Finite cylinder-moment hierarchy)**.** Fix an active branch $\mathfrak{b}$ and enumerations of its generators and proof rules. At level $\ell = (n, d, N)$, retain the first n generators, monomials of degree at most $2d$, and successor coordinates through time N. The primal cone $\mathsf{P}_{\mathfrak{b}}^{\ell}$ consists of truncated functionals whose moment and localizing matrices for $M_{\mathfrak{b}}$ are positive semidefinite and which annihilate the retained ideal and shift-difference rows. The dual cone $\mathsf{D}_{\mathfrak{b}}^{\ell}$ consists of rational identities

$$-1 = i_\ell + \sum_j s_j^2 m_j + d_\ell \tag{149}$$

using only the retained rows. Exact rational matrix arithmetic and rational interval enclosures check every dual identity.

Theorem II.2.49 (Proof-producing hierarchy completeness)**.** *For a fixed reachable active branch* $\mathfrak{b}$*:*

(i) *every feasible primal truncation gives valid lower moment bounds and every dual derivation is a sound exclusion of the source cell;*
(ii) *if* $-1 \in \mathcal{K}_{\mathfrak{b}}$*, then* $\mathsf{D}_{\mathfrak{b}}^{\ell}$ *contains a derivation at some finite level;*
(iii) *if every finite level is primal feasible, a diagonal subsequence of normalized moment tables converges to a functional satisfying* (148)*.*

Thus semantic membership and compact-set inclusion are absent from the pruning interface.

Proof. Weak duality follows by applying a feasible positive truncated functional to (149). A finite expression for $-1 \in \mathcal{K}_{\mathfrak{b}}$ contains finitely many generators, a finite degree, and a finite shift depth, so it occurs at some level, proving (ii). For (iii), the Archimedean bounds make each scalar moment sequence bounded. Repeated diagonal extraction gives a value on every cylinder monomial. Positivity of every finite moment and localizing matrix, and annihilation of every retained relation at all sufficiently large levels, pass to the limit. The resulting normalized positive functional satisfies (148); apply Theorem II.2.47. □

Proposition II.2.50 (Sound scope of the ordered hierarchy)**.** *The ordered hierarchy has the following exact logical contract.*

(S1) *A displayed finite identity* $-1 \in \mathcal{K}_{\mathfrak{b}}$ *is a checked exclusion of its stated finite target cell.*

(S2) *Feasibility of every finite relaxation reconstructs a positive functional on the completed ordered cylinder algebra. It does not by itself reconstruct a PDE orbit, an analytic continuation germ, or a singular solution.*
(S3) *Nonpolynomial maps, unbounded operators, distributional products, measures, and limiting relations are represented by their bounded total graphs. Finite semialgebraic elimination applies only to the finite polynomial moment relaxation of those graph coordinates.*
(S4) *Strict inequalities may be removed by an Archimedean ordered identity. A zero-margin equality cell remains a positive-functional successor unless a finite ideal/quadratic-module identity removes it or the approximation–orbit criterion realizes it.*
(S5) *Empty decreasing compact semantic sets imply an empty finite level, but a regularity proof uses that level only with the recorded finite deletion identities. Semantic emptiness is not a proof object.*
(S6) *A primal–polar fixed point and a cofinal cumulative successor are internal graph-complete states. They become physical terminals only through Theorem I.1.7; otherwise their next nonordinary, cone-boundary, or target-visible coordinate remains in the successor algebra.*

Consequently the hierarchy is proof-producing and sound, but failure to find a finite identity is never interpreted as a physical counterexample. No relative-completeness claim for all true PDE inequalities or real-radical equalities is used anywhere in a reachability conclusion.

Proof. Items (S1)–(S2) are Theorems II.2.47 and II.2.49. Item (S3) follows from total graph completion and the definition of the finite moment levels. For (S4), strict Archimedean positivity has a finite quadratic-module representation, whereas equality is imposed in the ideal and its complementary functional is retained. Nested compactness proves the semantic assertion in (S5); checking the concatenated finite identities supplies the only permitted proof step. Item (S6) is the Dirac realization gate and the no-compactification-promotion corollary. These implications establish the contract without assuming a complete calculus for the full analytic graph language. □

Definition II.2.51 (Prospective zero-density successor fixed point)**.** Fix a reachable active branch $\mathfrak{b}$. In this definition and the following pruning results, $\mathfrak{A}_\eta$, $\mathcal{R}_\eta$, Z_n, and Z_∞^Σ denote their restrictions to the closed same-origin stratum $\mathfrak{A}_\mathfrak{b}$; the branch index is suppressed only to keep the formulas readable. Let $p : \mathfrak{A}_\mathfrak{b} \to [0, \infty]$ be the lower-semicontinuous relaxation of the original-event price coordinate. For a closed $K \subseteq \mathfrak{A}_\mathfrak{b}$, define the viability operator

$$\mathcal{V}_\eta(K) := \{w \in K : p(w) = 0,\ \mathcal{R}_\eta(w) \cap K \neq \varnothing\}\,. \tag{150}$$

It maps closed sets to closed sets, because it is the projection of the compact set $\mathcal{R}_\eta \cap (K \times K) \cap (\{p = 0\} \times K)$. Starting with the target-active zero-density support set $Z_0 = \mathfrak{A}_\eta \cap \{p = 0\}$, set

$$Z_{n+1} = \mathcal{V}_\eta(Z_n), \qquad Z_\infty^\Sigma = \bigcap_{n=0}^{\infty} Z_n. \tag{151}$$

Thus Z_n consists exactly of the support records from which a target-retaining zero-density successor path of length n can be generated. Every such record remains paired with its finite-prefix multiplicity ledger. Compactness and closedness give

$$\mathcal{V}_\eta(Z_\infty^\Sigma) = Z_\infty^\Sigma. \tag{152}$$

Indeed, for $w \in Z_\infty^\Sigma$, choose $w_n \in Z_n$ with $w\mathcal{R}_\eta w_n$; a convergent subsequence has a limit in every Z_n, and the closed successor graph retains the edge. Consequently the countable finite-depth iteration is the complete topological description of viability. It is not, by itself, a proof procedure. If $Z_\infty^\Sigma = \varnothing$, nested compactness implies $Z_N = \varnothing$ for some finite N. This semantic fact becomes a checkable exclusion only when the ordered hierarchy below produces an explicit identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ on each active component. If every Z_n is nonempty, their intersection contains an infinite zero-density support path; this statement does not identify the accumulated cocycle with its normalized support law.

Let $\mathsf{Path}(Z_\infty^\Sigma)$ be the compact path space of $\mathcal{R}_\eta$ restricted to this fixed point. Let $\mathcal{P}^{0,\Sigma}_{\mathcal{R}_\eta}$ be all shift-invariant Borel probability laws on this path space. The prospective successor-saturation record associated with $\boldsymbol{\nu}$ is its joint law $\mathrm{Aug}(\boldsymbol{\nu})$ with the prefix functions

$$\Gamma_N = \sum_{j<N} \delta_{e_j}, \qquad A_{\Phi,N} = \sum_{j<N} a_\Phi(e_j), \qquad \Pi_N = \sum_{j<N} p(e_j), \qquad Q_N^c = \sum_{j<N} q^c(e_j).$$

These are random finite measures or scalar coordinates on the path space; their joint law is determined by $\boldsymbol{\nu}$ and is not extra data. The prospective successor-saturation algebra is

$$\mathfrak{H}^\Sigma_{\mathrm{succ}} := \overline{\mathrm{span}} \left\{ (\Pi_\Sigma(\pi_0)_\# \boldsymbol{\nu}, \mathrm{Aug}(\boldsymbol{\nu})) : \boldsymbol{\nu} \in \mathcal{P}^{0,\Sigma}_{\mathcal{R}_\eta} \right\}. \tag{153}$$

Every object in this definition is constructed prospectively from $\Theta_\Sigma^{\mathrm{syn}}$. Neither actual contact orbits nor exact invariant values occur. A total-graph failure contributes only when it belongs to an infinite target-retaining zero-density support path; naming the failure supplies no self-loop. The augmented ledger is part of every generator, so the closed span is taken in the product of the support and cumulative-cocycle spaces rather than in the support space alone.

Definition II.2.52 (Finite ordered pruning transcript)**.** Put

$$E_n := \mathfrak{A}_\eta \setminus Z_n. \tag{154}$$

A *finite ordered pruning transcript* for active branch $\mathfrak{b}$ is the complete finite expression

$$-1 = i + \sum_{j=1}^{M} s_j^2 m_j + d, \qquad i \in I_{\mathfrak{b}}, \quad m_j \in M_{\mathfrak{b}}^{\mathrm{gen}}, \quad d \in \mathcal{D}_S, \tag{155}$$

including the rational coefficients and the indices of every equation, carrier, graph, source, target-floor, same-origin, successor, positivity, and shift relation used. Here $M_{\mathfrak{b}}^{\mathrm{gen}}$ is the enumerated set of quadratic-module generators. Verification consists only of expanding (155), checking the cited ideal and shift rows, and checking that each multiplier is a square. The transcript is accepted only after this finite calculation succeeds. Its existence is never inferred from semantic emptiness of Z_n. Among valid transcripts use the first in the fixed enumeration.

Theorem II.2.53 (Ordered pruning or same-origin structural model)**.** *The elimination sets are open and satisfy*

$$E_{n+1} = E_n \cup \{ w \in Z_n : \mathcal{R}_\eta(w) \cap Z_n = \varnothing \}. \tag{156}$$

The recursion is the topological shadow of the ordered hierarchy. Exactly one of the following proof-producing alternatives holds for every reachable active branch $\mathfrak{b}$.

(P1) *At some finite moment level, the dual hierarchy produces $-1 \in \mathcal{K}_{\mathfrak{b}}$. The identity itself is the finite proof-valid pruning transcript and excludes the cell.*

(P2) $-1 \notin \mathcal{K}_{\mathfrak{b}}$. *Order-unit separation gives the positive functional of Theorem II.2.47, hence an infinite same-origin zero-price path law with target floor η. Equivalently, a compatible family of all finite moment restrictions reconstructs this law.*

Thus every pruning assertion is verified by the ordered proof kernel. A presently unsuccessful finite search is neither an exclusion nor a model; the model branch is the mathematical alternative $-1 \notin \mathcal{K}_{\mathfrak{b}}$. It is successor data: zero-price resaturation must still restrict its kernel, reselect its prospective source, and totalize its realization graph.

Proof. The set E_0 is open because Z_0 is closed. If E_n is open, then

$$\{w : \mathcal{R}_\eta(w) \cap Z_n \neq \varnothing\} = \pi_1(\mathcal{R}_\eta \cap (\mathfrak{A}_\eta \times Z_n))$$

is compact and hence closed. This proves (156) and openness of E_{n+1}.

Apply Theorem II.2.49. In its dual branch, the finite identity is exactly (155); applying any putative positive stationary functional gives $-1 \geq 0$, so the cell is excluded. In the other algebraic branch, Theorem II.2.47 supplies a normalized positive functional annihilating the successor and shift-difference relations and reconstructs its stationary path law. Same-origin coordinates belong to the ideal and hence hold almost surely. The alternatives are exclusive by weak duality. $\square$

Definition II.2.54 (Generated zero-density support dynamics)**.** Let $\boldsymbol{\nu}$ be a stationary target-retaining successor law with zero physical-expenditure density, paired with the unnormalized occurrence ledger $(\Gamma_N, A_{\Phi,N}, \Pi_N, (Q_N^c)_c)_N$. The density equality is an identity of the support selector; it imposes no zero relation on Π_N or Q_N^c. Let $\mathsf{Conseq}(\mathfrak{P})$ be the least countable consequence algebra containing the public weak equation and native carrier identity and closed under rational linear combination, localization by the generated test core, integration by parts already encoded in the weak formulation, polarization, adjoint formation, quotient descent, chart pullback, Jordan decomposition, completion of squares, composition of total graphs, and sourcewise graph closure. If a product, chain, trace, or covariance operation lacks an ordinary value, the algebra inserts its total-graph defect rather than the desired identity. This grammar is enumerated from $\mathsf{Pre}(\mathsf{Eq})$ and has no additional input slot.

Let $\mathscr{P}_{\text{gen}}$ be the countable family of all generated nonnegative functions q for which $\mathsf{Conseq}(\mathfrak{P})$ contains a domination identity

$$p = q + r_q, \qquad q \geq 0, \quad r_q \geq 0. \tag{157}$$

Include $q = p$ and $r_p = 0$. This definition takes the union of all valid positive descendants and does not require a choice of a maximal or unique decomposition. The family is generated by the fixed equation and carrier: mobility squares, weak energy defects, positive commutator squares, Fisher-information or Hessian squares, constraint defects, and target-ray defects occur only when their defining identity is already a relation in $\mathsf{Conseq}(\mathfrak{P})$. No extra positive term is inserted.

The *saturation equations* are

$$q = 0 \quad (q \in \mathscr{P}_{\text{gen}}), \qquad \dot{u} = J_{\mathsf{Geom}} + J_{\mathsf{Caus}} + J_{\mathsf{Abs}} + J_{\mathsf{Lift}} + J_{\mathsf{Bdry}}, \tag{158}$$

together with the totalized constitutive, constraint, chart, and covariance relations. No tangency to a symmetry orbit is asserted. Equilibria, relative equilibria, periodic or recurrent conservative orbits, scale breathers, and persistent graph-boundary dynamics are retained whenever they solve these relations. If $\mathcal{F}_{\text{gen}}$ is the generated finite or countable family of partial relations used on the current branch, let $\partial_\infty F$ denote the recurrent graph-boundary values obtained from infinite witness-successor paths for F, rather than the entire graph boundary. Define the totalized saturation set

$$\mathscr{Z}_{\text{sat}}^{\text{tot}} := \left(\bigcap_{q \in \mathscr{P}_{\text{gen}}} \ker q \cap \{\dot{u} = J_{\text{tot}}\} \right) \sqcup \bigsqcup_{F \in \mathcal{F}_{\text{gen}}} \partial_\infty F. \tag{159}$$

Thus a failed relation enters the saturation set only through a recurrent zero-density graph-boundary path retaining its multiplicity ledger. The definition imposes only identities already derived from the equation and does not convert stationarity of a law into stationarity of a state.

Theorem II.2.55 (Saturation-equation compiler)**.** *Every generator of $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$ is supported on the totalized solution set of* (158) *and on the greatest viable set Z^{Σ}_{∞}. Conversely, every stationary law on the prospective path space supported on this solution set generates a class in $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$. Hence*

$$\mathfrak{H}^{\Sigma}_{\mathrm{succ}} = \overline{\operatorname{span}}\, \Pi_{\Sigma} \left(Z^{\Sigma}_{\infty} \cap \mathscr{X}^{\mathrm{tot}}_{\mathrm{sat}} \right), \tag{160}$$

where the right side denotes the coordinate supports of stationary path laws on the displayed set. It is generated entirely by algebraic kernels, the closed successor graph, source quotients, and the descending viability operator.

Proof. For a stationary zero-density selector law, $\int p \, \mathrm{d}\boldsymbol{\nu} = 0$, so $p = 0$ almost surely. For every $q \in \mathscr{P}_{\mathrm{gen}}$, (157) gives $0 \le q \le p$, hence $q = 0$ almost surely on the ordinary branch. The full weak current identity remains one of the total graph coordinates, so its ordinary support solves $\dot{u} = J_{\mathrm{tot}}$ without any symmetry-tangency inference. On a graph-boundary branch, membership in the path fixed point supplies the recurrent value in $\partial_{\infty} F$. This proves the first inclusion in (160). Conversely, a stationary path law in the displayed ordinary joint kernel has $p = \mathcal{P}_0 = 0$; a recurrent graph-boundary law has the same zero price by membership in Z^{Σ}_{∞}. These are density statements on the selected support. The accompanying cumulative coordinates retain the exact identities $\Gamma_N = A_{\Phi,N} \gamma_N$, $\Pi_N = A_{\Phi,N} \int p \, \mathrm{d}\gamma_N$, and $Q^c_N = A_{\Phi,N} \int q^c \, \mathrm{d}\gamma_N$; they are not set to zero. Both are generators of the successor-saturation algebra. Taking target projections and closed spans proves equality. ☐

Corollary II.2.56 (Complete zero-cost structural reduction)**.** *The finite-depth recursion* (151) *has as its intersection the largest target-retaining zero-cost successor set. Its recurrent projection satisfies the exact alternative*

$$\mathfrak{H}^{\Sigma}_{\mathrm{succ}} = 0 \quad \Longleftrightarrow \quad \Pi_{\Sigma} \operatorname{supp} \boldsymbol{\nu} = 0 \textit{ for every stationary law on } Z^{\Sigma}_{\infty}. \tag{161}$$

At each successor stage the compiler applies the five-source quotient to the full zero-cost weak equation and to every recurrent graph-boundary coordinate. A target-null output is removed at the next stage. Any nonzero fixed-point output is a positive structural model of recurrent conservative, relative-equilibrium, profile, or graph-boundary dynamics satisfying the relations retained so far. It is not asserted to be an actual singular PDE orbit and is not terminal. It is passed to the zero-price resaturation operator of Definition III.4.13; only the exhaustive alternatives of Theorem III.4.82 may terminate that route. No compactness gap or separate rigidity datum appears in this reduction. A checked identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ closes a finite stage; the complementary positive functional supplies the next kernel and prospective-source calculation.

Theorem II.2.57 (Zero-density successor support with multiplicity retention)**.** *Let an infinite witness-indexed successor chain originate from the disjoint shell sequence of a finite-expenditure contact orbit. Then every coordinate of its diagonal path has relaxed original-event expenditure zero, and its path closure carries a stationary successor support law with zero physical expenditure density. The law is paired with the original cumulative records, which satisfy*

$$A_{\Phi,N} = N, \qquad \Pi_N = \sum_{j<N} p_j, \qquad \sum_c Q^c_N = N.$$

Consequently its augmented target projection lies in $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$. No cumulative physical or structural coordinate is declared zero.

Proof. By countable additivity, the original event expenditures $p_n = \mu_u(Q_n)$ are summable and hence tend to zero. Every successor uses a subsequence of these same indexed events. The diagonal selection in Lemma II.2.44 therefore approximates each path coordinate by records whose expenditures tend to zero. Lower semicontinuity and nonnegativity give zero expenditure at every path coordinate. Whenever that coordinate is used as target price, admissibility follows separately from Theorem II.2.4; otherwise its failed domination relation remains in the successor graph.

Theorem II.2.45 gives a compact shift-invariant path closure. A weak limit of its empirical path measures is shift invariant and has zero price density. The empirical normalization is stored jointly with the original occurrence ledger and the exact identities $\Gamma_N = A_{\Phi,N}\gamma_N$ and $\Pi_N = A_{\Phi,N}\int p \,\mathrm{d}\gamma_N$ restore multiplicity before target projection. Target retention and source ancestry are closed coordinates, so the augmented law is one of the generators in (153). □

Theorem II.2.58 (Estimate-free finite-budget contact routing)**.** *Let a contact orbit with finite physical expenditure generate infinitely many pairwise disjoint target-retaining windows I_j. The canonical activity lift and cofinal tail shift generate an infinite path in Z^{Σ}_{∞}, and*

$$\Pi_\Sigma Z^{\Sigma}_{\infty} \neq 0, \qquad \mathfrak{H}^{\Sigma}_{\mathrm{succ}} \neq 0. \tag{162}$$

Thus finite price is completely reduced to an explicit recurrent zero-density support with its full cumulative occurrence, physical, and source ledger; it is not a terminal scalar branch or an outstanding analytic row. No lower price, compactness estimate in the original state topology, coercivity constant, or Liouville input is used: the only numerical fact is countable additivity of the equation-generated physical measure.

Proof. Countable additivity gives $\sum_j \mu_u(I_j) < \infty$, hence after a subsequence $\mu_u(I_j) \to 0$. Apply the compact activity lift. Its limit has relaxed price zero, unit endpoint shell increment, the closed source-owner floor, and inherited source support. Apply each ordered total graph to cofinal tails. Ordinary and boundary values are both coordinates of the compact lift, and the tail shift supplies a successor at every stage. This gives an infinite target-retaining density-zero support path while retaining the unnormalized prefix ledger.

The descending viability recursion retains exactly the points admitting all finite continuations of such a path; compactness supplies an infinite path in their intersection. Hence the path lies in Z^{Σ}_{∞}. Theorem II.2.57 supplies a stationary zero-density support law with multiplicity restored. Its unit shell coordinate and positive owner coordinate make its target projection nonzero, proving (162). □

Theorem II.2.59 (Ordered prospective regularity alternative)**.** *For a fixed canonical presentation, run the total local-solution and carrier graphs, source compiler, canonical shells, signed carrier compiler, source-adapted dual-carrier grammar, activity lift, and zero-cost successor fixed point. Every target-visible local-solution or carrier defect is inserted into the same successor graph. Define the generated regularity mechanism algebra*

$$\mathfrak{H}^{\Sigma}_{\mathrm{out}} := \overline{\mathrm{span}}\{\Pi_\Sigma \mathfrak{a} : \mathfrak{a} \textit{ is a joint-saturated prospective source atom}\}, \tag{163}$$

where explicit realization is understood in the sense of Definition III.4.77. Put $\eta_0 = 1/5$. Exactly one of the following algebraic alternatives holds:

(R1) *every algebraically reachable active branch is eliminated by a local target-null, continuation, class-incompatibility, or divergent-charge relation produced by polarity refinement and resaturation; every ordinary maximal finite-expenditure branch then remains inside the declared continuation graph and avoids the complete target;*

(R2) *zero-price resaturation reconstructs a prospective source atom* $\mathfrak{a}$. *It supplies the closed reduced weak equation, source ancestry, vanishing productions, complete typed values for every auxiliary graph and their descendants, physical and origin ledgers, and a retained source-profile floor at least* η_0.

Independently, if the public local-solution or carrier graph has no ordinary value, its explicit first-failure graph coordinate is retained. This is an existence or carrier-realization output, not a regularity conclusion, and the regularity alternative is applied only to ordinary fibers. Every actual finite-expenditure contact orbit constructs a hypothetical output of type (R2) *with its own diagonal event witnesses; stationary positive laws and equality kernels encountered on the way are nonterminal resaturation states. Consequently*

$$\boxed{\begin{array}{c}\mathsf{T}_{\Sigma}^{\mathrm{phys,sat}}(B_{\mathrm{phys}}) < \eta_0 \\ \Longrightarrow \;\; \textit{every ordinary maximal finite-carrier branch avoids the declared target.}\end{array}} \tag{164}$$

This is the continuum prospective-profile bound corresponding to Part I's universal accountability inequality. A closing branch carries the finite transcript (155)*; the complementary branch is iterated by Theorem III.4.82 until it closes or produces the source-profile data listed in* (R2)*. Singular existence is asserted only by an independently realized member satisfying Definition III.4.77.*

Proof. Insert ordinary values of the total local-solution and carrier graphs into the activity lift. The form value $\perp_{\mathrm{loc}}$ is retained separately. On an ordinary finite-expenditure branch, suppose a maximal orbit contacts the complete target. Canonical shells give the disjoint unit increment cascade and its source-owner floor. A divergent generated lower-price series contradicts countable additivity; the finite-price tail instead enters the successor graph. Theorem II.2.58 constructs a stationary law in Z_∞^Σ with nonzero target coordinate and original shell-event witnesses. Theorem II.2.39 either closes its active branch or reprocesses its typed equality successor. Theorem III.4.82 turns every nonclosing compatible route into the source-profile witness (R2). Its origin remains hypothetical by Theorem II.2.41. Otherwise each finite local elimination in (R1) contradicts the retained target coordinate; diagonal saturation supplies the same conclusion for a cofinal target-null route. If the saturated profile is below η_0, (R2) is impossible; terminal coverage then keeps every ordinary maximal branch inside the declared continuation graph. This proves (164). ☐

Proposition II.2.60 (Covariant entropy as a carrier correction)**.** *Let a covariant entropy cocycle* $\mathcal{M}$ *satisfy the gluing law of Theorem II.10.2. If its local balance has the form*

$$\mathcal{M}(Sw) - \mathcal{M}(w) = F_{\mathrm{sc}}(w) - D(w) - M(w) - R(w), \qquad R \geq 0, \tag{165}$$

then $\Psi = -\mathcal{M}$ *is the canonical carrier correction of Definition II.2.5, and*

$$D + M - F_{\mathrm{sc}} + \Psi - \Psi \circ S = R \geq 0. \tag{166}$$

Every invariant saturation measure is supported on $R = 0$*. It is supported on* $D = M = 0$ *only when the generated carrier transcript supplies a nonnegative domination identities expressing the zero-price deficit with* D *and* M *as positive descendants. Thus a Perelman-type formula refines the carrier cohomology and zero-price equations. Its equality models are exactly the ordinary or recurrent graph-boundary solutions of those generated equations; relative equilibria are one possible subclass, not an automatic conclusion.*

Proof. Rearrange (165) to obtain (166). The entropy cocycle glues on one coherent orbit, so its increments telescope and define a carrier coboundary. For an invariant saturation measure the nonnegative right side has zero integral and therefore $R = 0$ almost surely. Any further vanishing follows only by applying the generated positive-descendant rule of Definition II.2.54; the full weak equation remains in the support coordinates. $\square$

Corollary II.2.61 (Gradient-kernel equality reduction)**.** *Let Ω be connected and let the ordinary saturation equations on a normalized successor core contain*

$$D_m(V) := \int_{K_m} |\nabla V|^2 \, \mathrm{d}y = 0, \qquad m \geq 1, \tag{167}$$

for an increasing compact exhaustion $K_m \Subset \Omega$. Then every ordinary zero-expenditure stationary law is supported on spatially constant states. If the generated gauge quotient sends spatial constants to $\mathscr{N}_\Sigma$, its ordinary equality contribution to $\mathfrak{H}^{\Sigma}_{\mathrm{succ}}$ vanishes. Every nonordinary value of the gradient, exhaustion, connected-core, or gauge relation remains a witness-successor and is reduced by $\widehat{\mathcal{R}}_\eta$.

Proof. For a zero-expenditure stationary law $\boldsymbol{\nu}$, nonnegativity and the saturation-equation compiler give $\int D_m(V) \, \mathrm{d}\boldsymbol{\nu} = 0$ for every m. Hence $D_m(V) = 0$ for $\boldsymbol{\nu}$-almost every state, simultaneously for all m. The exhaustion gives $\nabla V = 0$ almost everywhere on Ω. The weak fundamental theorem on connected domains makes V spatially constant. The target projection then vanishes by the stated gauge quotient. Totalization of each relation gives the last assertion. $\square$

II.3 Covariant action and the physical budget

The fixed-Caus formulation separates the current selected by the PDE from the metric used to measure irreversible expenditure. This separation is physical: the action below is the quantity appearing in the energy identity. The explicit inverse notation in this section is used only when the mobility has the Hilbert realization stated next. In a general Gelfand triple, the same identities are read at the level of the closed mobility form. These action identities remain library relations until a shell covector selects an active source cell. Their reachability content is exactly the target-local five-source factorization of Theorem II.2.18; only the resulting carrier-dominated production may enter the physical ledger.

Definition II.3.1 (Mobility current space)**.** Fix a state u, a real Hilbert tangent space $\mathcal{H}_u$, and a bounded self-adjoint nonnegative mobility $\mathbb{K}(u) \in \mathcal{L}(\mathcal{H}_u)$. Let $\mathbb{K}(u)^{-1/2}$ denote the Moore–Penrose inverse on $(\ker \mathbb{K}(u))^{\perp}$. The mobility current space is the Cameron–Martin space

$$\mathfrak{J}_u := \operatorname{Ran} \mathbb{K}(u)^{1/2}$$

with inner product and norm

$$(J_1, J_2)_{u,*} := \langle \mathbb{K}(u)^{-1/2} J_1, \mathbb{K}(u)^{-1/2} J_2 \rangle_{\mathcal{H}_u}, \qquad \|J\|^2_{u,*} := \|\mathbb{K}(u)^{-1/2} J\|^2_{\mathcal{H}_u}.$$

Equivalently, this is the completion of $\operatorname{Ran} \mathbb{K}(u)$ in the displayed norm. Covectors are represented in $\mathcal{H}_u$, with components in $\ker \mathbb{K}(u)$ identified when they act on currents. For a velocity v, put

$$W(u, v) := v + A_{\mathsf{Caus}}(u) - f_{\mathsf{Bdry}}(u).$$

The dissipation Lagrangian is

$$\mathscr{L}(u,v) := \begin{cases} \frac{1}{2}\|W(u,v)\|_{u,*}^2, & W(u,v) \in \mathfrak{J}_u, \\ +\infty, & \text{otherwise.} \end{cases} \tag{168}$$

Its physical action on an interval I is $\mathcal{A}_I(u) = \int_I 2\mathscr{L}(u,\dot{u})\,\mathrm{d}t$.

Definition II.3.2 (Covariant dissipation Hamiltonian)**.** For $p \in \mathcal{H}_u$, define

$$\mathscr{H}_{\mathrm{cov}}(u,p) := -\langle p, A_{\mathsf{Caus}}(u) - f_{\mathsf{Bdry}}(u)\rangle + \frac{1}{2}\|\mathbb{K}(u)^{1/2}p\|^2. \tag{169}$$

This is the Legendre–Fenchel dual of $v \mapsto \mathscr{L}(u,v)$.

Proposition II.3.3 (Fenchel equality and the selected current)**.** *Whenever the displayed quantities are finite,*

$$\mathscr{L}(u,v) + \mathscr{H}_{\mathrm{cov}}(u,p) - \langle p, v\rangle = \frac{1}{2}\|W(u,v) - \mathbb{K}(u)p\|_{u,*}^2 \geq 0. \tag{170}$$

Equality holds precisely when $W(u,v) = \mathbb{K}(u)p$ in the current quotient. For the fixed-Caus gradient flow (10)*, the selected velocity has*

$$W(u,\dot{u}) = -\mathbb{K}(u)D\mathcal{E}(u),$$

so equality holds at $p = -D\mathcal{E}(u)$, modulo the mobility kernel.

Proof. Substitute (168) and (169); since $v = W - A_{\mathsf{Caus}} + f_{\mathsf{Bdry}}$, the linear terms combine to $-\langle p, W\rangle$. Completing the square in the $\mathbb{K}^{-1}$-metric gives (170). The equality and selected-current statements follow directly. □

Definition II.3.4 (Physical carrier balance)**.** Let $u : [0,T_*) \to X$ be an ordinary maximal branch. Expand the native signed carrier graph of Definition I.1.18 and take the Jordan decomposition of its signed currents. Denote the resulting scalar by $\mathcal{K}_u$, the gross positive production by $\mathfrak{D}_u$, the returned internal feeding by Φ_u^+, and true positive exterior input by R_u^+. Its ordinary graph value is, for $0 \leq s < t < T_*$,

$$\mathcal{K}_u(t) + \mathfrak{D}_u((s,t]) \leq \mathcal{K}_u(s) + \Phi_u^+((s,t]) + R_u^+((s,t]). \tag{171}$$

Here all four quantities are generated from the one signed identity. The generated passivity radius is the least extended-real β for which the measure domination

$$\Phi_u^+ \leq \beta\,\mathfrak{D}_u. \tag{172}$$

holds. Whether $\beta \leq 1$, whether $\mathcal{K}_u$ has a lower bound, and whether the exterior input is finite are values of generated relations, not additional data. All terms are expressed in the physical clock. A normalized-clock identity is routed through the totalized clock-comparison relation in Proposition II.10.5.

Theorem II.3.5 (Carrier compiler for the single physical budget)**.** *If the generated values satisfy $\beta \leq 1$, $\inf_t \mathcal{K}_u(t) = K_- > -\infty$, and $R_u^+([0,T_*)) < \infty$, then*

$$\mu_u := \mathfrak{D}_u - \Phi_u^+ \tag{173}$$

is a nonnegative countably additive physical expenditure measure and

$$\mu_u([0,T_*)) \le \mathcal{K}_u(0) - K_- + R_u^+([0,T_*)) =: B_{\mathrm{phys}} < \infty. \tag{174}$$

If $\beta < 1$, then

$$\mu_u \ge (1-\beta)\mathfrak{D}_u. \tag{175}$$

For $\beta = 1$, μ_u is the exact flux-deficit measure. If any displayed relation has the opposite value, its total-graph witness is sent directly to the activity lift and successor recursion. No alternative expenditure or independent budget is appended later. This theorem constructs the available orbitwise physical measure. A target event spends that measure only after its source-resolved shell current satisfies the local charging rule (98)*.*

Proof. The domination $\Phi_u^+ \le \mathfrak{D}_u$ makes the difference in (173) a nonnegative Radon measure. Rearrange (171) with $s = 0$, use $\mathcal{K}_u(t) \ge K_-$, and let $t \uparrow T_*$. Monotone convergence gives (174). Finally, $\Phi_u^+ \le \beta\mathfrak{D}_u$ gives (175). $\square$

Definition II.3.6 (Reduced passivity radius)**.** Let $\mathcal{P}_{\mathrm{car}}^{\mathrm{red}}$ be a normalized carrier profile space obtained after zero-descent-defect source reductions. Let $D(V) \ge 0$ and $\Phi_+(V) \ge 0$ be the descended gross-expenditure and feeding-flux densities in the same physical units. The reduced passivity radius is

$$\beta_*^{\mathrm{red}} := \alpha_*(\Phi_+ \mid D) = \sup_{\substack{V \in \mathcal{P}_{\mathrm{car}}^{\mathrm{red}} \\ D(V)>0}} \frac{\Phi_+(V)}{D(V)}, \tag{176}$$

with the null-mode convention of Definition II.1.6. A bound $\beta_*^{\mathrm{red}} \le 1$ supplies (172) whenever the carrier identity is local in the normalized profiles and its physical-clock pullback is proved.

Proposition II.3.7 (Quadratic passivity is a spectral invariant)**.** *On the quadratic-form realization branch of the reduced Hilbert profile space,*

$$D(V) = \langle AV, V\rangle, \qquad \Phi_+(V) = \max\{\langle BV, V\rangle, 0\},$$

where $A \ge 0$, $B = B^*$, all D-null target-visible feeding modes have been separated, and B is bounded in the A-form norm. Let $S_{\mathrm{flux}} = A^{-1/2}BA^{-1/2}$ be the descended self-adjoint form operator on the completion modulo $\ker A$*. Then*

$$\beta_*^{\mathrm{red}} = \max\{\sup \sigma(S_{\mathrm{flux}}), 0\}. \tag{177}$$

Thus every multiplier or casewise proof of $\Phi_+ \le D$ is, on this reduced quadratic interface, an upper bound for one spectral radius.

Proof. The A-form completion is a Hilbert space after quotienting by $\ker A$. The form induced by B is represented there by S_{flux}. Rayleigh–Ritz identifies the supremum of $\langle BV, V\rangle / D(V)$ with $\sup \sigma(S_{\mathrm{flux}})$; taking the positive part gives (177). $\square$

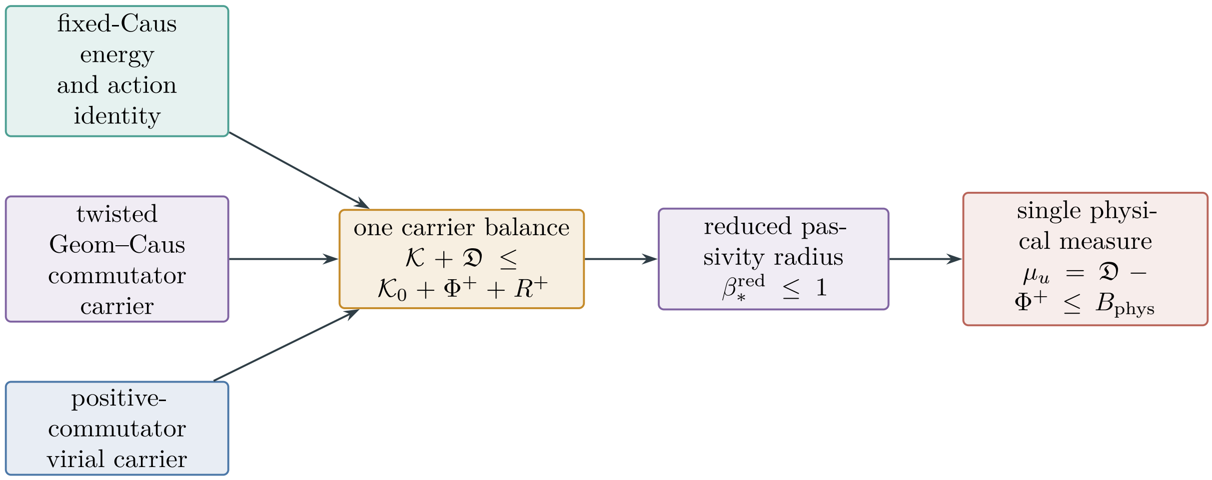

Figure 10: Carrier-to-budget compiler. Gradient-flow, hypocoercive, and positive-commutator identities are different ways to expose the same structural object: a lower-bounded carrier whose feeding flux is dominated by gross expenditure. The output is the sole physical budget measure.

II.4 Hamiltonian meanings and chart covariance

The constructions in this section are universal operator identities, not equation-wide regularity assertions. In a reachability argument they are activated only after restriction to a contact-generated stratum $X_{\mathfrak{b}}$, where Theorem II.2.18 separates their $\mathsf{Geom}/\mathsf{Caus}/\mathsf{Abs}/\mathsf{Lift}/\mathsf{Bdry}$ components, target-null part, carrier coboundary, and total-graph defects. Only that factorized local value may enter a physical charge or close a branch.

Definition II.4.1 (Geometric spectral Hamiltonian)**.** Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric form on a real Hilbert space $\mathcal{H}$. Its geometric Hamiltonian is the unique self-adjoint operator $H \geq 0$ satisfying

$$\mathfrak{q}(u,v) = \langle H^{1/2}u, H^{1/2}v\rangle, \qquad u,v \in D(\mathfrak{q}).$$

This Hamiltonian records only the Geom form; the fixed antisymmetric current remains in Caus.

Proposition II.4.2 (Separation of Hamiltonian meanings)**.** *The following constructions require distinct data.*

(i) *A closed symmetric form determines the spectral Hamiltonian H of Definition II.4.1.*

(ii) *A canonical conservative Hamiltonian vector field requires a Poisson tensor $\mathbb{J}$, a domain on which the Jacobi identity holds, and a functional $\mathcal{H}$; the field is $\mathbb{J}D\mathcal{H}$.*

(iii) *A positive mobility and a covariant connection determine the Fenchel Hamiltonian $\mathscr{H}_{\mathrm{cov}}$ of Definition II.3.2.*

An antisymmetric Caus form alone supplies directed transport but does not provide the Poisson or Jacobi data in item (ii)*.*

Proof. The representation theorem for closed nonnegative forms gives item (i) [7, Chapter VI]. Item (ii) is the definition of a Poisson Hamiltonian system; antisymmetry is only one of the Poisson requirements. The square completion in Proposition II.3.3 proves item (iii). □

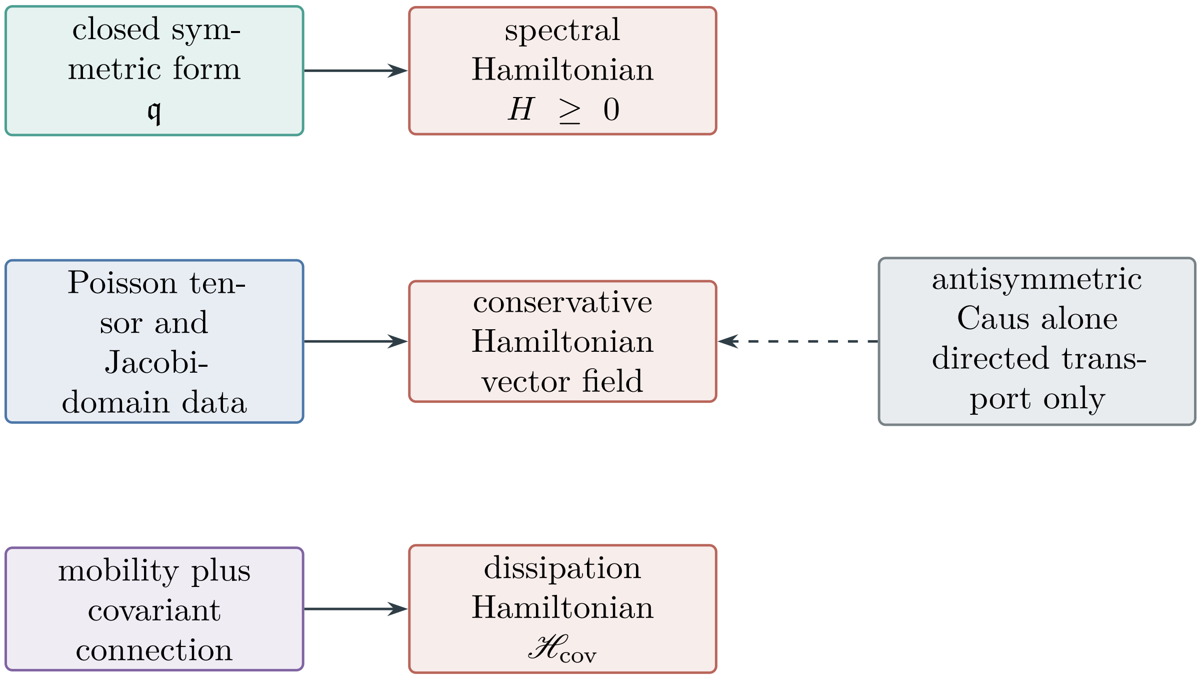

Figure 11: Three noninterchangeable Hamiltonian constructions. The dashed line marks a missing implication: skew transport alone does not establish a Poisson structure.

Theorem II.4.3 (Same-current covariance)**.** *Let $v = \Phi_t(u)$ be a generated C^1 chart relation. Totalize its diffeomorphism and tangent-map relations. On the branch where $T_t = D\Phi_t(u) : \mathcal{H}_u \to \mathcal{H}_v$ is a bounded linear isomorphism with bounded inverse, with every expression on the right evaluated at $u = \Phi_t^{-1}(v)$, define*

$$\widetilde{\mathbb{K}} = T_t \mathbb{K} T_t^*, \qquad \widetilde{A}_{\mathsf{Caus}} = T_t A_{\mathsf{Caus}}, \qquad A^{\Phi}_{\mathsf{Lift}} = -\partial_t \Phi_t, \qquad \widetilde{f}_{\mathsf{Bdry}} = T_t f_{\mathsf{Bdry}}. \tag{178}$$

Then

$$\dot{v} + \widetilde{A}_{\mathsf{Caus}}(v) + A^{\Phi}_{\mathsf{Lift}}(v) - \widetilde{f}_{\mathsf{Bdry}}(v) = T_t(\dot{u} + A_{\mathsf{Caus}}(u) - f_{\mathsf{Bdry}}(u)). \tag{179}$$

The map T_t is an isometry between the corresponding mobility-current spaces equipped with their transported metrics, so the Lagrangian and physical action are invariant. The chart has Lift *ancestry, and quotienting any declared symmetry additionally has* Abs *ancestry.*

Proof. Differentiate $v = \Phi_t(u)$ and substitute (178); the $\partial_t \Phi_t$ terms cancel, giving (179). For $\widetilde{J} = T_t J$, the variational identity

$$\|J\|^2_{\mathbb{K}^{-1}} = \sup_{p \in \mathcal{H}_u} \{2\langle p, J\rangle - \langle p, \mathbb{K}p\rangle\}$$

and the bijective change of covector $p = T_t^* \widetilde{p}$ give $\|\widetilde{J}\|_{\widetilde{\mathbb{K}}^{-1}} = \|J\|_{\mathbb{K}^{-1}}$. Integrating proves action invariance. The source labels follow from Definition I.2.1. □

Proposition II.4.4 (Chart covariance of the dissipation Hamiltonian)**.** *On the ordinary covariance-graph branch of Theorem II.4.3, let $p = T_t^* \widetilde{p}$, and define the transformed Hamiltonian using the total connection $\widetilde{A}_{\mathsf{Caus}} + A^{\Phi}_{\mathsf{Lift}}$. Then*

$$\widetilde{\mathscr{H}_{\mathsf{cov}}}(t, v, \widetilde{p}) = \mathscr{H}_{\mathsf{cov}}(u, p) + \langle \widetilde{p}, \partial_t \Phi_t(u)\rangle. \tag{180}$$

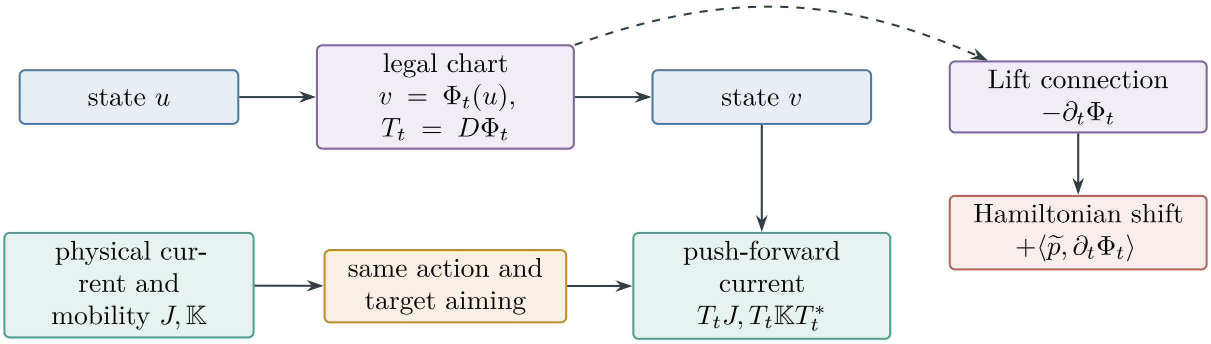

Figure 12: Same-current chart covariance. A chart transports the physical current and metric; its time dependence is a Lift connection with the exact Hamiltonian shift shown.

The additional term is the time-dependent Lift connection. It is not a new physical Caus term.

Proof. Insert (178) into (169). The quadratic term is invariant because $\widetilde{\mathbb{K}} = T_t\mathbb{K}T_t^*$, while

$$-\langle\widetilde{p}, T_t(A_{\mathsf{Caus}} - f_{\mathsf{Bdry}}) - \partial_t\Phi_t\rangle = -\langle p, A_{\mathsf{Caus}} - f_{\mathsf{Bdry}}\rangle + \langle\widetilde{p}, \partial_t\Phi_t\rangle.$$

□

Remark II.4.5 (A change of physical clock is additional data). Theorem II.4.3 keeps the physical time variable fixed. If a blow-up chart also introduces $\tau = \theta(t)$, the current, mobility, and action density must be transformed with the exact Jacobian $\mathrm{d}t/\mathrm{d}\tau$. Action invariance, or comparison with the original physical measure, is not automatic; the required one-sided comparison is the content of Proposition II.10.5.

II.5 Killed resolvents, resistance, and target realization

Every resolvent, capacity, and resistance relation below is a library relation in the sense of Corollary II.2.19. Its regularity use is confined to the active target/source cell selected by contact. A failed domain, closedness, killing, flux, or range relation is therefore a typed local successor; it is not an equation-wide obstruction and supplies no physical charge by itself.

Definition II.5.1 (Killed geometric Hamiltonian)**.** Let $\mathfrak{q}$ be a densely defined closed nonnegative symmetric Markovian form and let Σ be quasi-closed for a regular or quasi-regular realization. The killed form is the restriction

$$D(\mathfrak{q}_\Sigma) = \{u \in D(\mathfrak{q}) : \widetilde{u} = 0 \text{ q.e. on } \Sigma\}, \qquad \mathfrak{q}_\Sigma = \mathfrak{q}|_{D(\mathfrak{q}_\Sigma)},$$

and its self-adjoint operator, when the restricted form is closed, is $H_\Sigma \geq 0$. Adding an admitted nonnegative killing potential instead defines a *soft*-killed form. It is a separate construction unless convergence to the hard restriction is proved.

Proposition II.5.2 (Existence of the killed Hamiltonian)**.** *If $D(\mathfrak{q}_\Sigma)$ is closed in the form norm, then $\mathfrak{q}_\Sigma$ is closed and determines a unique self-adjoint $H_\Sigma \geq 0$.*

Proof. A form-norm closed subspace of the domain of a closed form is complete in that norm. The representation theorem for closed nonnegative forms gives H_Σ. □

Proposition II.5.3 (Committor is the canonical binary-contact quotient)**.** *Let A and B be disjoint quasi-closed sets for an associated right process, and let*

$$h(x) := \mathbb{P}_x\{\tau_A < \tau_B\}$$

be its quasi-continuous committor. Define $q_h(x) = h(x)$ outside the exceptional set. If a quotient $q : X \to Q$ preserves this binary contact law in the precise sense that $h = \bar{h} \circ q$ quasi-everywhere for some $\bar{h} : Q \to [0, 1]$, then

$$q_h = \bar{h} \circ q \qquad \text{quasi-everywhere.} \tag{181}$$

Hence q_h is the coarsest quotient preserving the scalar event $\{\tau_A < \tau_B\}$: every other such quotient is finer than its level-set partition. This assertion concerns the binary hitting probability, not the full hitting-time distribution.

Proof. The preservation branch is exactly (181). Therefore $q(x) = q(y)$ implies $h(x) = h(y)$, so every fiber of q lies in a fiber of q_h. Conversely, q_h itself preserves the event by the defining identity for h. □

Theorem II.5.4 (Killed-resolvent stopwatch)**.** *Let L_Σ generate a killed semigroup P_t^Σ with growth bound ω. Let ℓ_0 be an initial functional and let k_Σ be a terminal-flux vector such that*

$$\mathbb{P}_{\ell_0}\{\tau_\Sigma \in \mathrm{d}t, \tau_\Sigma < \infty\} = \langle \ell_0, P_t^\Sigma k_\Sigma\rangle \, \mathrm{d}t \tag{182}$$

as finite measures. Then, for $\lambda > \omega$,

$$\mathbb{E}_{\ell_0}\big[e^{-\lambda \tau_\Sigma}; \tau_\Sigma < \infty\big] = \langle \ell_0, (\lambda - L_\Sigma)^{-1} k_\Sigma\rangle. \tag{183}$$

Without (182)*, the right side remains a cross-resolvent pairing and has no automatic first-contact interpretation.*

Proof. Integrate (182) against $e^{-\lambda t}$. The Bochner Laplace transform of the semigroup is $\int_0^\infty e^{-\lambda t} P_t^\Sigma \, \mathrm{d}t = (\lambda - L_\Sigma)^{-1}$, which proves the identity. □

Theorem II.5.5 (Resolvent Fenchel dual and resistance)**.** *Let $K \geq 0$ be self-adjoint with closed form $\mathfrak{q}$, and let $e \in \mathcal{H}$. For $\lambda > 0$,*

$$\langle e, (K + \lambda)^{-1} e\rangle = \sup_{v \in D(\mathfrak{q})} \Big\{ 2\,\mathrm{Re}\langle e, v\rangle - \mathfrak{q}(v, v) - \lambda \|v\|^2 \Big\}. \tag{184}$$

If $P_{\ker K} e = 0$, then monotone convergence gives

$$\lim_{\lambda \downarrow 0} \langle e, (K + \lambda)^{-1} e\rangle = \int_{(0,\infty)} s^{-1} \, \mathrm{d}\nu_e(s) =: \mathcal{R}_K(e) \in [0, \infty]. \tag{185}$$

Finiteness of $\mathcal{R}_K(e)$ is equivalent to boundedness of $v \mapsto \langle e, v\rangle$ on the energy space with norm $\mathfrak{q}(v, v)^{1/2}$; in that case it has a unique weak energy representer modulo $\ker K$*.*

Proof. For $R_\lambda = (K + \lambda)^{-1}$, completing the square in the form norm of $K + \lambda$ shows that the supremum is attained at $v = R_\lambda e$ and equals $\langle e, R_\lambda e\rangle$. The spectral theorem gives $\langle e, R_\lambda e\rangle = \int (\lambda + s)^{-1} \, \mathrm{d}\nu_e(s)$. Monotone convergence proves (185). The last equivalence is the Riesz theorem on the completion of $D(\mathfrak{q})/\ker K$. □

Theorem II.5.6 (Harmonic defect equals resistance error)**.** *Let $K \geq 0$ be self-adjoint with form $\mathfrak{q}$, and let*

$$\mathcal{E}_K := \overline{D(\mathfrak{q})/\ker K}^{\mathfrak{q}}$$

be its energy space. For $\ell \in \mathcal{E}_K^*$, totalize the weak solution relation $\mathfrak{q}(x, \phi) = \ell(\phi)$ modulo $\ker K$. On its ordinary branch let x denote the solution and let $v \in D(\mathfrak{q})$ be a trial state with residual

$$e_v(\phi) := \mathfrak{q}(v, \phi) - \ell(\phi). \tag{186}$$

Then

$$\|e_v\|^2_{\mathcal{E}_K^*} = \|[v - x]\|^2_{\mathcal{E}_K} = \mathfrak{q}(v - x, v - x). \tag{187}$$

If e_v is represented by a vector $e \perp \ker K$ in $\mathcal{H}$, then

$$\|e_v\|^2_{\mathcal{E}_K^*} = \langle e, K^+ e\rangle = \mathcal{R}_K(e). \tag{188}$$

Thus an approximate committor, elliptic corrector, or resolvent state is controlled by one resistance invariant of its harmonic defect; no $\ker K$ component of the trial state enters the error. When that kernel is a closed gauge or target-null space, the corresponding discarded motion requires no estimate.

Proof. The weak equation and (186) give

$$e_v(\phi) = \mathfrak{q}(v - x, \phi).$$

The Riesz map of $\mathcal{E}_K$ is an isometry onto its dual, proving (187). When the residual is represented by e, the spectral theorem identifies its squared energy-dual norm with $\int_{(0,\infty)} s^{-1}\, \mathrm{d}\nu_e(s)$, which is (188); the middle pairing is understood as this spectral quadratic-form value. ☐

Corollary II.5.7 (Defect-aligned drift bound)**.** *If $e \perp \ker K$ and $\mathcal{R}_K(e) < \infty$, then for every $w \in D(\mathfrak{q})$,*

$$|\langle e, w\rangle| \leq \mathcal{R}_K(e)^{1/2} \mathfrak{q}(w, w)^{1/2}. \tag{189}$$

Only the component of w paired with the target-selected routable defect enters a reachability charge, and it does so through the local carrier rule.

Proof. Use the energy representer from Theorem II.5.5 and apply Cauchy–Schwarz in $\mathcal{E}_K$. ☐

Proposition II.5.8 (Routable and zero-mode resistance split)**.** *For arbitrary $e \in \mathcal{H}$, write*

$$e = e_0 + e_\perp, \qquad e_0 = P_{\ker K} e, \qquad e_\perp = P_{(\ker K)^\perp} e.$$

Then

$$\langle e, (K + \lambda)^{-1} e\rangle = \frac{\|e_0\|^2}{\lambda} + \int_{(0,\infty)} \frac{1}{s + \lambda}\, \mathrm{d}\nu_{e_\perp}(s). \tag{190}$$

Consequently the zero-discount resistance is finite precisely when $e_0 = 0$ and $\int_{(0,\infty)} s^{-1}\, \mathrm{d}\nu_{e_\perp}(s) < \infty$. A nonzero target-visible zero mode is an inherited Geom/Caus zero-frequency occurrence, not a residual coordinate. It is removed only when it belongs to a declared gauge or to $\mathscr{N}_\Sigma$; otherwise a separate carrier or price row evaluates it.

Proof. The spectral projection of K at 0 contributes $\lambda^{-1}\|e_0\|^2$; the remaining spectral measure is supported on $(0, \infty)$, which gives (190). Monotone convergence yields the finiteness criterion. The source statement follows from Theorems I.3.4 and I.4.6. ☐

Proposition II.5.9 (Capacity and committor realization). *Let A and B be disjoint quasi-closed sets for a regular or quasi-regular Dirichlet form. The condenser capacity*

$$\operatorname{Cap}(A,B)=\inf\{\mathfrak{q}(v,v):v\in D(\mathfrak{q}),\ \widetilde{v}=1 \text{ on } A,\ \widetilde{v}=0 \text{ on } B\}$$

is an analytic Geom invariant. On the zero realization-defect branch, if the minimizer is unique and the associated right process realizes the weak Dirichlet problem, then its quasi-continuous version is the committor $x\mapsto \mathbb{P}_x\{\tau_A<\tau_B\}$ quasi-everywhere. The probabilistic conclusion is the ordinary value of the right-process realization row; positive capacity alone is only its prospective value [5, 9].

Theorem II.5.10 (Polar-target capacity criterion). *Let $\mathfrak{q}$ be a symmetric quasi-regular Dirichlet form with associated right process, and let Σ be quasi-closed. Write*

$$\operatorname{Cap}_{\mathfrak{q}}(\Sigma):=\inf\{\mathfrak{q}(v,v)+\|v\|_2^2:v\in D(\mathfrak{q}),\ \widetilde{v}\geq 1 \text{ q.e. on a neighborhood of } \Sigma\}.$$

If $\operatorname{Cap}_{\mathfrak{q}}(\Sigma)=0$, then Σ is polar:

$$\mathbb{P}_x\{\tau_\Sigma<\infty\}=0 \qquad \text{for quasi-every } x. \tag{191}$$

This is a direct Geom nonreachability criterion for the Markov realization and requires no cascade argument. Positive capacity selects the complementary row but supplies no hitting lower bound without a starting-law or nonpolarity statement.

Proof. For a quasi-regular Dirichlet form, sets of zero 1-capacity are properly exceptional up to a capacity-zero enlargement. The associated right process avoids every properly exceptional set for quasi-every starting point, which gives (191) [5, 9]. □

Proposition II.5.11 (Quotient pullback of zero capacity). *Let $q:X\to Q$ identify a target $\Sigma=q^{-1}(\Sigma_Q)$ quasi-everywhere, and let $\mathfrak{q}_X,\mathfrak{q}_Q$ be symmetric quasi-regular Dirichlet forms. Totalize the lift relation which, for every admissible quotient potential f with $\widetilde{f}\geq 1$ near Σ_Q, there is a lifted potential ψ_f with $\widetilde{\psi}_f\geq 1$ near Σ and*

$$(\mathfrak{q}_X)_1(\psi_f,\psi_f)\leq C(\mathfrak{q}_Q)_1(f,f)+\varepsilon(f). \tag{192}$$

If a minimizing family for $\operatorname{Cap}_{\mathfrak{q}_Q}(\Sigma_Q)$ can be chosen with $\varepsilon(f)\to 0$, then

$$\operatorname{Cap}_{\mathfrak{q}_X}(\Sigma)\leq C\operatorname{Cap}_{\mathfrak{q}_Q}(\Sigma_Q). \tag{193}$$

In particular, zero quotient capacity pulls back to zero capacity on X.

Proof. Every ψ_f is an admissible competitor for the capacity of Σ. Apply (192) along the stated minimizing family and take the liminf. □

II.6 Full-generator response and physical aiming

Full-generator formulas enter the reachability calculus only through the same target-local activation rule. The input and output projections below are those generated by the active branch signature, and every differentiated or factorized term retains its five-source ancestry. Amplification outside that target-cyclic cell is transverse to the branch under evaluation.

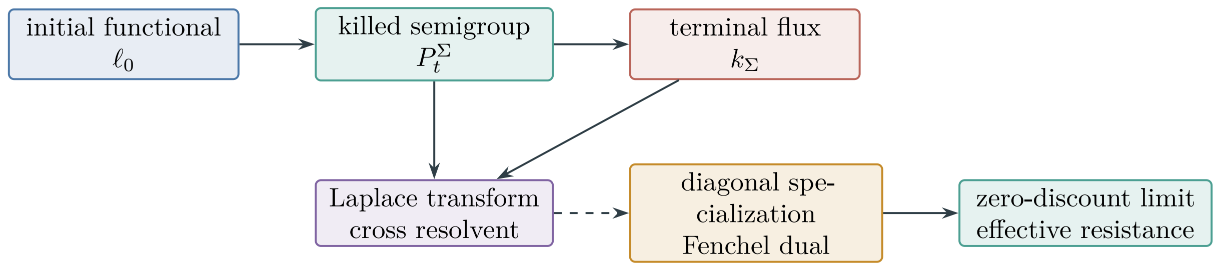

Figure 13: Killed arrival and resistance are related but distinct. Arrival is a cross pairing that requires a flux realization. The Fenchel and resistance identities concern a diagonal self-adjoint specialization.

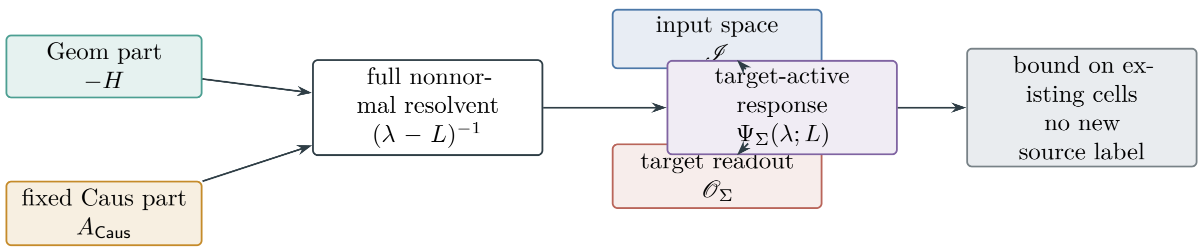

Figure 14: Full nonnormal response retains both Geom and fixed Caus. Input and target compression select a response coefficient; they do not change the operator's source ancestry.

Definition II.6.1 (Target-active full resolvent)**.** Let $L = -H + A_{\mathsf{Caus}} + B_{\mathsf{Bdry}}$ be a closed generator with a declared resolvent point λ. For closed mathematical input and target-output subspaces $\mathscr{I}$ and $\mathscr{O}_\Sigma$, with bounded projections $P_{\mathscr{I}}$ and $P_{\mathscr{O}}$, define

$$\Psi_\Sigma(\lambda; L) := \left\| P_{\mathscr{O}} (\lambda - L)^{-1} P_{\mathscr{I}} \right\|. \tag{194}$$

This quantity measures nonnormal amplification visible to the fixed target interface. It evaluates the existing Geom–Caus–Bdry generator, not an additional source.

Proposition II.6.2 (Nonnormal source invariance)**.** *Let L_0 be closed on a Banach space and let every K^c be bounded. Put $L_\theta = L_0 + \sum_c \theta_c K^c$ on $D(L_0)$. On the branch $\lambda \in \rho(L_\theta)$ locally in θ, this is the open ordinary value of the totalized resolvent relation; on it the resolvent is norm differentiable and*

$$\partial_{\theta_c} (\lambda - L_\theta)^{-1} = (\lambda - L_\theta)^{-1} K^c (\lambda - L_\theta)^{-1}. \tag{195}$$

Thus every first variation of target-active response is factored through the same source term K^c. Large pseudospectral response can expose a large Geom–Caus interaction, but it creates no new nonnormal source.

Proof. Differentiate $(\lambda - L_\theta)(\lambda - L_\theta)^{-1} = I$ on the common domain and solve for the derivative. Bounded input/output compression preserves the factorization. □

Definition II.6.3 (Target-gradient ray)**.** Let F be a differentiable target shell. Totalize membership of its mobility gradient

$$G_F := \mathbb{K}(u) DF(u)$$

in $\mathfrak{J}_u$. On the membership branch, for a physical current $J \in \mathfrak{J}_u$, define its positive target power and aiming contribution by

$$P_F(J) := (J, G_F)_{u,*}, \qquad \mathrm{Align}_F(J) := \frac{(P_F(J)_+)^2}{\|G_F\|^2_{u,*}}, \tag{196}$$

with value zero when $G_F = 0$. By the Cameron–Martin pairing,

$$P_F(J) = \langle DF(u), J\rangle$$

for every $J \in \mathfrak{J}_u$; components of DF in $\ker \mathbb{K}(u)$ annihilate that current space.

Theorem II.6.4 (Projection onto the target ray)**.** *For every $J, G \in \mathfrak{J}_u$,*

$$\|J\|^2_{u,*} = \frac{\left((J,G)_{u,*}\right)_+^2}{\|G\|^2_{u,*}} + \mathrm{dist}^2_{u,*}(J, \mathbb{R}_+ G), \tag{197}$$

with the first term zero when $G = 0$. Hence

$$(P_F(J)_+)^2 \le \|J\|^2_{u,*}\|G_F\|^2_{u,*}. \tag{198}$$

Proof. The orthogonal projection of J onto the closed ray $\mathbb{R}_+ G$ has coefficient $\left((J,G)_{u,*}\right)_+/\|G\|^2_{u,*}$. Pythagoras gives (197); dropping the nonnegative distance gives (198). □

Proposition II.6.5 (Integrated action–aiming bound)**.** *Let I be a pre-contact interval. On the finite ordinary branch of the two totalized squared-norm integrals below,*

$$\int_I P_F(J(t))_+ \,\mathrm{d}t \le \left(\int_I \|J(t)\|^2_{u(t),*}\,\mathrm{d}t\right)^{1/2} \left(\int_I \|G_F(u(t))\|^2_{u(t),*}\,\mathrm{d}t\right)^{1/2}. \tag{199}$$

For the selected Geom current of Definition I.1.21, the square of the first factor is the physical action and is bounded by the right-hand side of (13)*.*

Proof. Apply (198) and then Cauchy–Schwarz in time. The final assertion is Proposition I.1.22. □

II.7 Degenerate geometry and commutator-generated carriers

Direct resistance controls only the positive spectral subspace of the Geom operator. Fixed Caus transport may nevertheless move a Geom-null direction into a dissipative direction. The correct reduced invariant is then a positive commutator Gramian together with a carrier identity realizing that Gramian along the actual evolution. These relations are evaluated only on the contact-generated Geom–Caus cell. A commutator square becomes physical expenditure only after its target-local factorization proves domination by the public carrier; every complementary graph value remains a same-origin successor.

Definition II.7.1 (Degenerate Geom–Caus core)**.** Let $\mathcal{H}$ be a complex Hilbert space and

$$L = -S + A, \qquad S = S^* \ge 0, \qquad A^* = -A,$$

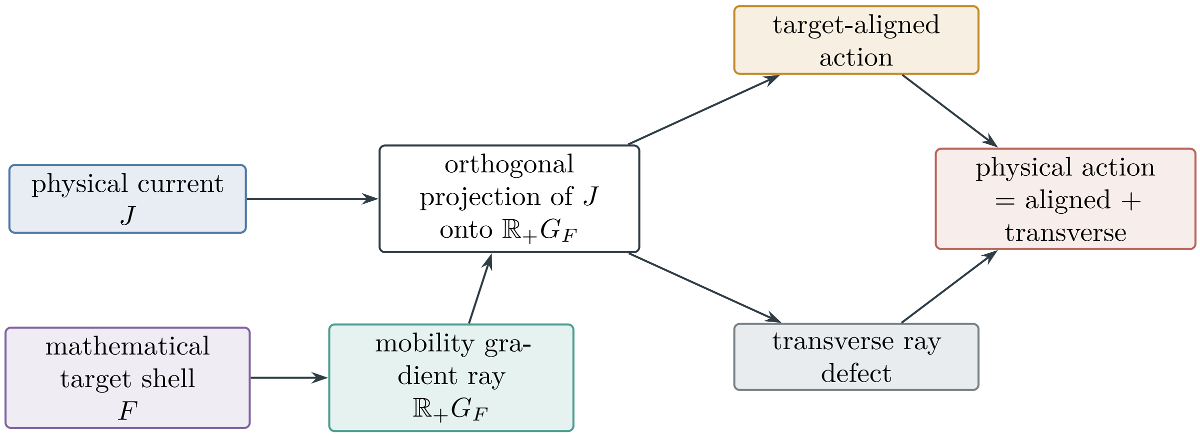

Figure 15: Physical aiming geometry. The target shell is a mathematical test of the already selected current. Projection splits the physical action into target-aligned and transverse expenditure.

on a common invariant core $\mathcal{D}$. Let $\mathcal{G} \subseteq \ker S$ be a closed A-invariant gauge subspace, and write $P_{\mathcal{G}}^{\perp}$ for its orthogonal complement. Put $C_0 = S^{1/2}$. Totalize the iterated commutator relations on $\mathcal{D}$; their ordinary graph branch sets

$$C_{k+1} = [A, C_k] \quad \text{on } \mathcal{D}.$$

For fixed K and $c_k > 0$, totalize the quadratic form

$$q_K(v) := \sum_{k=0}^{K} c_k \|C_k v\|^2. \tag{200}$$

On its closable graph branch its positive self-adjoint operator is denoted by Q_K, and its reduced bracket gap is

$$\gamma_K := \inf_{\substack{v \in D(q_K) \cap \mathcal{G}^{\perp} \\ \|v\|=1}} q_K(v). \tag{201}$$

Failure of a domain, closure, or gauge relation is its totalized graph defect; algebraic commutator span alone does not select the closing branch.

Proposition II.7.2 (Bracket resistance invariant)**.** *If $\gamma_K > 0$, then $Q_K \geq \gamma_K I$ on $\mathcal{G}^{\perp}$, and for every $e \in \mathcal{G}^{\perp}$,*

$$\mathcal{R}_K^{\mathrm{br}}(e) := \langle e, Q_K^{-1} e \rangle \leq \gamma_K^{-1} \|e\|^2. \tag{202}$$

The quotient kernel

$$\mathfrak{h}_K := \mathcal{G}^{\perp} \cap \ker Q_K$$

is the structural rank deficit when $\gamma_K = 0$. Every target-visible element of $\mathfrak{h}_K$ retains its Geom–Caus ancestry and remains an existing source occurrence for the routing, carrier, and price rows.

Proof. The first assertion is the definition of the form lower bound in (201); functional calculus gives (202). The kernel is the intersection of the kernels of the closed commutator forms in (200). Source preservation and the target-null residual theorem give the final assertion. □

Definition II.7.3 (Twisted-carrier realization evaluator)**.** A bracket Gramian Q_K has a twisted-carrier realization when there is a bounded self-adjoint operator M on $\mathcal{G}^{\perp}$, constants $0 < m_- \le m_+ < \infty$ and $c_M > 0$, and a common evolution core on which

$$m_-\|v\|^2 \le \langle Mv, v\rangle \le m_+\|v\|^2, \tag{203}$$

$$2\,\mathrm{Re}\langle Mv, Lv\rangle \le -c_M q_K(v). \tag{204}$$

For a forced equation, the positive part of $2\,\mathrm{Re}\langle Mu, f_{\mathsf{Bdry}}\rangle\,\mathrm{d}t$ is the candidate Bdry-input current. It enters R_u^+ only when the public carrier identity identifies it as true exterior input; otherwise its target-local projection is a boundary successor.

Theorem II.7.4 (Hypocoercive carrier produces physical expenditure)**.** *On the zero graph-defect branch of Definition II.7.3, the evolution and carrier chain-rule relations for an ordinary maximal branch have ordinary values $\dot u = Lu + f_{\mathsf{Bdry}}$ and $\mathrm{d}\langle Mu, u\rangle/\,\mathrm{d}t$. On this branch,*

$$\mathcal{K}_u(t) := \langle Mu(t), u(t)\rangle, \qquad \mathfrak{D}_u(\,\mathrm{d}t) := c_M q_K(u(t))\,\mathrm{d}t, \qquad \Phi_u^+ = 0$$

satisfy the carrier balance of Definition II.3.4. Hence

$$\mu_u([0,T_*)) \le \langle Mu_0, u_0\rangle + R_u^+([0,T_*)) \le m_+\|u_0\|^2 + R_u^+([0,T_*)) \tag{205}$$

after subtracting any known lower carrier level. If $\gamma_K > 0$, this expenditure controls every nongauge direction in $\mathcal{G}^{\perp}$. The commutator calculation is therefore a structural compiler for a physical budget in a partially dissipative system [15]. Its regularity use is restricted to the active target/source cell selected by the shell current; nongauge directions outside that cell are not charged to its target progress.

Proof. Differentiate $\langle Mu, u\rangle$ and use (204); the positive forcing pairing gives the declared input measure. Apply Theorem II.3.5. When $\gamma_K > 0$, $q_K(u) \ge \gamma_K\|P_{\mathcal{G}}^{\perp}u\|^2$. □

Definition II.7.5 (Positive-commutator carrier)**.** Let the evolution be $\dot u = -iHu$, with $H = H^*$, and let $P = \mathbf{1}_{\Delta}(H)$ be a reducing spectral projection. A self-adjoint conjugate operator C supplies a positive-commutator carrier when the form

$$B := i[H, C]$$

is well defined and nonnegative on a common invariant core in $P\mathcal{H}$, and the expectation $\langle u(t), Cu(t)\rangle$ has a finite upper bound C_+ along the orbit. The strict Mourre branch has $PBP \ge \theta P$ modulo the declared gauge for some $\theta > 0$.

Theorem II.7.6 (Conservative carrier produces physical expenditure)**.** *Under Definition II.7.5, let $u_0 \in P\mathcal{H}$. On the zero chain-defect branch, the totalized commutator relation gives*

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle u(t), Cu(t)\rangle = \langle u(t), Bu(t)\rangle. \tag{206}$$

Consequently

$$\mu_u(Q) := \int_Q \langle u(t), Bu(t)\rangle\,\mathrm{d}t, \qquad \mu_u([0,T_*)) \le C_+ - \langle u_0, Cu_0\rangle. \tag{207}$$

When $PBP \ge \theta P$, this budget controls $\theta\int_0^{T_*}\|Pu(t)\|^2\,\mathrm{d}t$. Thus a virial, Morawetz, or Mourre identity contributes to the same physical-budget calculus whenever its carrier variation is finite [6]. In the reachability calculus, the contribution is restricted to the retained target-local spectral/source cell and is charged only through Theorem II.2.4.

Proof. Differentiate the expectation on the common core to obtain (206). Nonnegativity of B makes the right side a measure density. Integrate and use the upper carrier bound. The strict estimate follows from $PBP \geq \theta P$. On the complementary branch the distributional difference between the two sides of (206) is the retained $\mathfrak{H}^{\Sigma}_{\text{chain}}$-class, so the theorem does not presume the chain rule in the fast track. □

Corollary II.7.7 (Complete routing alternatives for a zero mode)**.** *Let e_0 be the zero mode in Proposition II.5.8. Direct resistance controls $e_\perp$. A target-visible e_0 may be controlled by a twisted-carrier evaluator or a positive-commutator carrier, with its price eligible for the same μ_u only through its target-local carrier factorization. The complementary values of both evaluators select the harmonic nonroutable branch of Theorem II.1.16; it remains a Geom–Caus source class and does not move to J_{res}.*

II.8 Compactness and rigidity as a price compiler

Compactness is not an additional continuum source and is not part of the definition of the residual. Its role is quantitative: it converts a qualitative zero-price rigidity statement into a uniform positive price on a closed class of target-approaching windows. The topology used in this conversion must be generated from seminorms or graph coordinates derived from the fixed PDE. The following lemma is one method for evaluating δ_*; its tuple is not input to the fast track. It is invoked only on the target/source stratum that produced the window family. Each failed clause is totalized into the boundary, chain, or price row of that same stratum.

Definition II.8.1 (Canonical compactness–rigidity row)**.** Fix a reachable active branch $\mathfrak{b}$ at physical ceiling B_{phys}, and let $\mathcal{W}_{\mathfrak{b}}$ be the normalized pre-contact windows generated by the equation's Abs/Lift grammar that have the origin, source support, realization mode, shell floor, and earlier zero coordinates recorded in $\mathfrak{b}$. Inside the closure of its target-local stratum $X_{\mathfrak{b}}$, define

$$K_{\mathfrak{b}} := \overline{\mathcal{W}_{\mathfrak{b}}}, \qquad B_{\Sigma,\mathfrak{b}} := \overline{\mathcal{W}_{\mathfrak{b},\Sigma}}, \tag{208}$$

where $\mathcal{W}_{\mathfrak{b},\Sigma}$ consists of those same-origin windows carrying the selected normalized target-shell crossing. Thus $K_{\mathfrak{b}}$ is compact and $B_{\Sigma,\mathfrak{b}}$ is closed by construction; neither is an analytic premise and neither contains a source-transverse window.

On the ordinary admitted value of the local charging graph, after the storage-transfer row has its zero/root–leaf-bounded value, let d_0 be the nonnegative window expenditure obtained by restricting the fixed physical measure to that target-active event. A failed charging graph or a nonzero storage return does not enter d_0; it is the corresponding price or Caus-storage successor. The canonical lower-semicontinuous relaxation of the admitted d_0 is

$$\overline{d}(w) := \sup_{U \ni w} \inf\{d_0(v) : v \in U \cap \mathcal{W}_{\mathfrak{b}}\} \in [0, \infty], \tag{209}$$

where U ranges over neighborhoods of w, and the infimum of an empty set is $+\infty$. Equivalently, this is the largest lower-semicontinuous functional whose restriction to $\mathcal{W}_{\mathfrak{b}}$ is bounded above by d_0. The compactness–rigidity value is the exact extended number

$$\delta_{\text{rig}}(\mathfrak{b}) := \inf_{w \in B_{\Sigma,\mathfrak{b}}} \overline{d}(w), \tag{210}$$

with value $+\infty$ when $B_{\Sigma,\mathfrak{b}} = \varnothing$. The branches $\delta_{\text{rig}} > 0$ and $\delta_{\text{rig}} = 0$ are exhaustive; the latter is the closed zero-price target-approach class. Every topology, normalization, target class, and price in this row is therefore generated from the fixed PDE, target, and physical balance.

Definition II.8.2 (Canonical physical-transfer radius)**.** For each generated event Q, write $w_Q = \mathcal{N}_Q u$ and let μ_u be the orbit's physical expenditure measure. Define

$$c^*_{\mathrm{pay}}(\mathfrak{b}) := \inf_{\substack{Q:\ w_Q \in B_{\Sigma,\mathfrak{b}} \\ 0<\overline{d}(w_Q)<\infty}} \frac{\mu_u(Q)}{\overline{d}(w_Q)} \in [0,\infty], \tag{211}$$

using $+\infty$ for an empty indexing family. A window with $\overline{d}(w_Q) = \infty$ and finite $\mu_u(Q)$, or a failure of the normalization/pullback relation, is retained by the price graph-defect row. For every ordinary finite-price window,

$$\mu_u(Q) \geq c^*_{\mathrm{pay}}(\mathfrak{b})\, \overline{d}(w_Q). \tag{212}$$

Consequently the physical-transfer question has the exact complementary values $c^*_{\mathrm{pay}} > 0$ and $c^*_{\mathrm{pay}} = 0$, together with its typed graph-defect value. No comparison constant is supplied as data.

Lemma II.8.3 (Compactness–rigidity row evaluation)**.** *The value in* (210) *is positive exactly when the canonical closed target-approach class contains no sequence with relaxed physical price tending to zero. If $B_{\Sigma,\mathfrak{b}} \neq \varnothing$, the infimum is attained.*

Proof. The set $B_{\Sigma,\mathfrak{b}}$ is compact and $\overline{d}$ is lower semicontinuous by relaxation. Hence the infimum is attained. Its value is zero exactly when a minimizing sequence has price tending to zero; compactness then supplies a convergent subsequence in $B_{\Sigma,\mathfrak{b}}$, and lower semicontinuity identifies the zero-price branch. The converse follows from the constant sequence at a zero-price minimizer. □

Theorem II.8.4 (Uniform physical price branch)**.** *On the invariant branch $c^*_{\mathrm{pay}}(\mathfrak{b}) > 0$ and $\delta_{\mathrm{rig}}(\mathfrak{b}) > 0$, every ordinary generated event Q whose normalized window belongs to $B_{\Sigma,\mathfrak{b}}$ satisfies*

$$\mu_u(Q) \geq \delta_{\mathrm{phys}}(\mathfrak{b}), \qquad \delta_{\mathrm{phys}}(\mathfrak{b}) := c^*_{\mathrm{pay}}(\mathfrak{b})\delta_{\mathrm{rig}}(\mathfrak{b}) > 0. \tag{213}$$

On either zero branch, the corresponding zero-transfer or zero-price sequence is retained in the same source/mode cell.

Proof. Apply (212) and then (210). The complementary statements follow from the two infimum definitions and sourcewise closure. □

Theorem II.8.5 (Borderline passivity price compiler)**.** *On the invariant branch $\beta^{\mathrm{red}}_* = 1$ in Definition II.3.6, define the normalized carrier deficit*

$$\Delta_{\mathrm{car}}(V) := D(V) - \Phi_+(V) \geq 0. \tag{214}$$

Let $\overline{\Delta}_{\mathrm{car}}$ be its canonical relaxation (209) *on the generated compact sets* (208)*, and define*

$$\delta^*_{\mathrm{car}}(\mathfrak{b}) := \inf_{V \in B_{\Sigma,\mathfrak{b}}} \overline{\Delta}_{\mathrm{car}}(V). \tag{215}$$

Define the clock-transfer radius by the same infimum construction as (211)*, with $\overline{d} = \overline{\Delta}_{\mathrm{car}}$; call it $c^*_{\mathrm{clock}}(\mathfrak{b})$. On the positive values of both rows, physical pullback gives*

$$\mu_u(Q) \geq c^*_{\mathrm{clock}} \overline{\Delta}_{\mathrm{car}}(\mathcal{N}_Q u), \tag{216}$$

and every selected target-bad event has price

$$\mu_u(Q) \geq c^*_{\mathrm{clock}} \delta^*_{\mathrm{car}}(\mathfrak{b}) > 0. \tag{217}$$

Thus the critical value $\beta^{\mathrm{red}}_* = 1$ is closed by an equality–rigidity invariant on the reduced saturation class. If either infimum is zero, its minimizing sequence is the complementary price class.

Proof. The canonical relaxation is lower semicontinuous on the compact generated class, so Lemma II.8.3 evaluates δ^*_{car}. The transfer-radius definition gives (216); multiplication yields (217). $\square$

Corollary II.8.6 (Separator closure of the borderline branch)**.** *On the zero separator-defect branch, the target projection of every zero-deficit profile vanishes by Proposition II.1.21. Hence the canonical target-bad closure contains no zero of $\overline{\Delta}_{\mathrm{car}}$, so compactness gives $\delta^*_{\mathrm{car}} > 0$. The positive clock-transfer branch of Theorem II.8.5 then gives the uniform event price; its zero branch retains the exact zero-transfer sequence.*

Proposition II.8.7 (A standard local compactness compiler)**.** *Let X_0, X, X_1 be Banach spaces with $X_0 \Subset X \hookrightarrow X_1$, the first embedding compact and the second continuous. If $1 \le p < \infty$, $1 < q \le \infty$, and a family of windows is bounded in*

$$L^p(0,1;X_0) \cap W^{1,q}(0,1;X_1),$$

then it is relatively compact in $L^p(0,1;X)$. Consequently, its closure agrees with the corresponding compact branch of $K_{\mathfrak{b}}$. The target and price rows are still evaluated by the canonical target closure and lower-semicontinuous relaxation; the lemma supplies no separate closedness or price premise.

Proof. The relative compactness is the Aubin–Lions lemma in the stated range [1, 8, 14]. The final assertion follows by mapping that relatively compact family into the generated evaluation closure. $\square$

Remark II.8.8 (Quotient compactness row). If a compact group acts continuously on a compact normalized set, its quotient is compact. For a noncompact symmetry group, fixing a gauge only defines a slice. The generated closure of that slice and the totalized gauge graph select the compact and graph-defect branches, respectively. In either case the quotient operation has Abs-ancestry and does not alter the source of the limiting physical record.

Remark II.8.9 (What failure of compactness means). A nonordinary value of a normalization, target-persistence, lower-semicontinuity, or physical-pullback graph is pushed into the compact activity lift with the relevant $\mathsf{Geom}/\mathsf{Caus}/\mathsf{Abs}/\mathsf{Lift}/\mathsf{Bdry}$ ancestry. A zero value of either sharp infimum supplies its minimizing sequence to the same lift. Finite-depth viability then removes the descendants or constructs their recurrent zero-price path. No target-visible element is transferred to J_{res}, and this optional price accelerator creates no new obligation in the core proof.

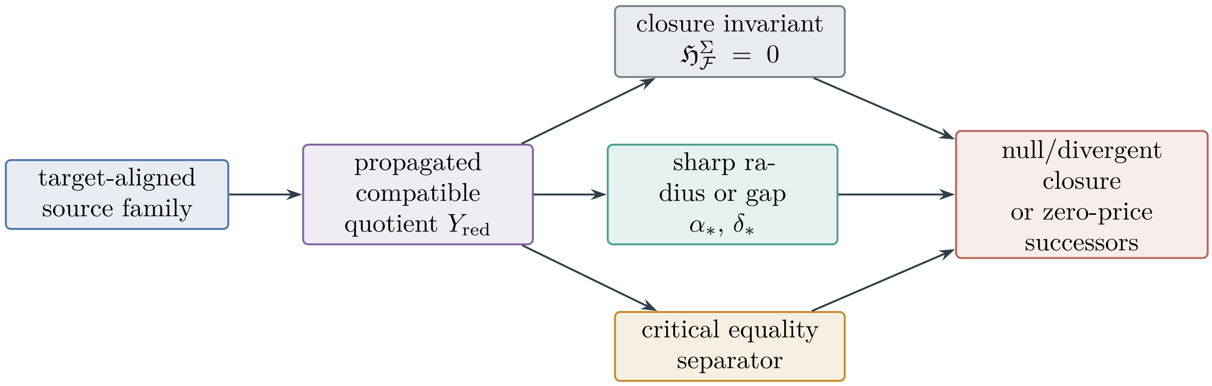

Figure 16: Identity-to-successor compiler. Once closed null directions are propagated out, generated identities expose a target closure, domination radius, physical-price gap, or critical equality relation. Every nonclosing value is normalized and passed to the successor operator. Each evaluator returns both its closing and complementary structural relation to the ordered algebra.

II.9 Constraint tails as finite structural obstructions

Nonlocal constraints often enter a local evolution through an elliptic multiplier, potential, or pressure. A direct estimate of the whole nonlocal field obscures the smaller structural question: which part is fixed by the local source, which part is a gauge, which part decays from the exterior, and which finite-dimensional harmonic mode remains visible to the target? The following interface makes that separation explicit. It is formed only after restriction to the active branch. Its local, harmonic, gauge, and exterior coordinates are the Geom/Caus/Abs/Lift/Bdry factorization of that branch's constraint current; no norm of the full constraint field has an independent reachability consequence.

Definition II.9.1 (Elliptic constraint–tail interface)**.** Let $U_0 \Subset U$ be open sets, let

$$\mathcal{C}p = S(V) \quad \text{in } U$$

be a linear elliptic constraint in a declared weak sense, and let $\mathcal{G}_{U_0}$ be its gauge space on U_0. The compiler forms the product of the following five total graphs. Its ordinary value consists of:

(T1) a local solution operator $\mathcal{L}_U$ satisfying $\mathcal{C}\mathcal{L}_U S(V) = S(V)$ on U;

(T2) a decomposition on U_0,

$$p|_{U_0} = p_{\mathrm{loc}} + h, \qquad p_{\mathrm{loc}} := \mathcal{L}_U S(V)|_{U_0}, \qquad \mathcal{C}h = 0 \text{ in } U_0; \tag{218}$$

(T3) a Banach space $\mathcal{H}(U_0)$ of homogeneous solutions with a continuous splitting

$$\mathcal{H}(U_0) = \mathcal{H}_{\mathrm{low}} \oplus \mathcal{H}_{\mathrm{dec}}, \tag{219}$$

where $\mathcal{H}_{\mathrm{low}}$ is finite dimensional;

(T4) a dyadic exterior expansion $P_{\mathrm{dec}}h = \sum_{k\geq 0} h_k$ convergent in $\mathcal{H}_{\mathrm{dec}}$, with

$$\mathfrak{s}_{\mathrm{tail}}(h) := \sum_{k\geq 0} \|h_k\|_{\mathcal{H}_{\mathrm{dec}}} < \infty; \tag{220}$$

(T5) a linear gauge realization $\Gamma : \mathcal{G}_{U_0} \to \mathcal{H}_{\mathrm{low}}$ with closed range.

The residual low-mode obstruction is the cokernel class

$$\mathfrak{o}_{\mathrm{tail}}(p) := [P_{\mathrm{low}} h] \in \mathcal{H}_{\mathrm{low}} / \operatorname{Ran} \Gamma. \tag{221}$$

The local term retains the ancestry of $S(V)$; gauge removal appends Abs, and the exterior expansion and its limiting trace append Bdry. None of (T1)–(T5) is additional specialization data. Candidate operators, splittings, and dyadic pieces are enumerated from the typed weak constraint and the fixed universal elliptic library. The first relation without an ordinary value is retained as its total-graph successor with the displayed ancestry.

Theorem II.9.2 (Constraint–tail structural normal form)**.** *On the ordinary value of the elliptic constraint–tail product graph, every field has, on U_0, the decomposition*

$$p = p_{\mathrm{loc}} + \Gamma g + h_{\mathrm{dec}} + h_{\mathrm{obs}}, \tag{222}$$

where $g \in \mathcal{G}_{U_0}$, $h_{\mathrm{dec}} \in \mathcal{H}_{\mathrm{dec}}$, and h_{obs} is any representative of the class $\mathfrak{o}_{\mathrm{tail}}(p)$. The first three terms are closed by, respectively, the local source law, a propagated quotient, and the summability invariant (220)*. Consequently:*

(C1) *if $\mathfrak{o}_{\mathrm{tail}}(p) = 0$, the entire constraint field closes using only its inherited source,* Abs*, and* Bdry*;*
(C2) *if $\mathfrak{o}_{\mathrm{tail}}(p) \neq 0$ and its target projection vanishes, its remaining class lies in $\mathscr{N}_\Sigma$;*
(C3) *if $\mathfrak{o}_{\mathrm{tail}}(p) \neq 0$ and is target-visible, it is a finite-dimensional* $\mathsf{Bdry}/\mathsf{Abs}$*-ancestry occurrence evaluated by the carrier and price rows. It is not a new residual source.*

Thus the nonlocal constraint is reduced internally to the local source graph, the summability graph, and a finite-dimensional target-projected cokernel. Every nonordinary value is already a witness-indexed successor rather than an additional analytic task.

Proof. Apply the continuous projections in (219) to h. Closed range and finite dimensionality give $\mathcal{H}_{\mathrm{low}} = \operatorname{Ran} \Gamma \oplus H_{\mathrm{obs}}$ for a finite-dimensional complement H_{obs}. This gives (222); changing the complement changes h_{obs} only by a gauge term and hence preserves the cokernel class. Absolute convergence in (220) closes the decaying series. The three routing conclusions follow from quotient propagation, Theorem I.3.4, and the definition of $\mathscr{N}_\Sigma$. □

Proposition II.9.3 (Constraint tails enter the physical budget only through a carrier law)**.** *On the ordinary constraint-tail branch, let the signed tail contribution be Ψ_{tail} to a localized carrier balance. It is charged to the physical expenditure measure only if the equation supplies a measure inequality of one of the forms*

$$(\Psi_{\mathrm{tail}})_+ \le \Phi_u^+, \qquad \textit{or} \qquad (\Psi_{\mathrm{tail}})_+ \le R_u^+, \tag{223}$$

with the right side appearing in (171)*. Tail summability alone proves analytic closure but supplies no physical price and no budget bound. In a target argument these inequalities are first restricted to the active shell/source cell and then passed through* (98)*.*

Proof. The carrier compiler uses only the measure terms displayed in (171). Either inequality in (223) places the positive tail contribution in that balance. Without such a comparison, (220) is a norm statement on a homogeneous field and has no implication for μ_u. □

II.10 Covariant entropy cocycles and physical production

An entropy enters the reachability argument through the physical production that it controls along one orbit. Local entropy formulas in changing blow-up charts must therefore be joined by an explicit cocycle law. Independent windowwise monotonicity does not telescope. Perelman's entropy formula is a classical model of covariant monotonicity with a rigid equality case [12]. The formulation below derives the required equation-independent cocycle contracts directly from a local balance and its reset relation. For reachability, each local balance and reset is first compiled by Theorem II.2.18. Entropy production is therefore not a second budget and is not charged across unrelated windows: it contributes to the sole physical measure only on the same-origin target/source cell on which the public carrier proves domination.

Definition II.10.1 (Physical entropy cocycle)**.** Let $I_j = [a_j, b_j]$, $j \geq 1$, be ordered normalized windows obtained from pairwise disjoint physical events Q_j. A physical entropy cocycle on this family consists of

(E1) scalar primitives $\mathcal{M}_j : I_j \to \mathbb{R}$;
(E2) nonnegative Radon production measures $\mathfrak{D}_j$ and nonnegative error measures r_j on I_j;
(E3) reset errors $e_j \geq 0$ and a constant $c_{\mathcal{M}} > 0$;
(E4) constants $M_+ < \infty$ and $M_- > -\infty$;

such that, for $a_j \leq s < t \leq b_j$,

$$\mathcal{M}_j(t) + c_{\mathcal{M}} \mathfrak{D}_j((s,t]) \leq \mathcal{M}_j(s) + r_j((s,t]), \tag{224}$$

$$\mathcal{M}_{j+1}(a_{j+1}) \leq \mathcal{M}_j(b_j) + e_j, \tag{225}$$

$$\mathcal{M}_1(a_1) \leq M_+, \qquad \mathcal{M}_j(t) \geq M_-. \tag{226}$$

The measures $\mathfrak{D}_j$ are coordinate expressions of one physical production law on the events Q_j; chart Jacobians and time changes are included in that law. The half-open convention in (224) prevents endpoint atoms from being counted twice.

Theorem II.10.2 (Entropy-cocycle gluing)**.** *For every $N \geq 1$, a physical entropy cocycle satisfies*

$$c_{\mathcal{M}} \sum_{j=1}^{N} \mathfrak{D}_j(I_j) \leq M_+ - M_- + \sum_{j=1}^{N} r_j(I_j) + \sum_{j=1}^{N-1} e_j. \tag{227}$$

If the two error series are finite, then $\sum_j \mathfrak{D}_j(I_j) < \infty$. If in addition every selected bad window satisfies $\mathfrak{D}_j(I_j) \geq \delta > 0$, their number is at most

$$\frac{M_+ - M_- + \sum_j r_j(I_j) + \sum_j e_j}{c_{\mathcal{M}} \delta}. \tag{228}$$

Proof. Apply (224) on every whole window. Summing gives

$$c_{\mathcal{M}} \sum_{j=1}^{N} \mathfrak{D}_j(I_j) \leq \sum_{j=1}^{N} (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) + \sum_{j=1}^{N} r_j(I_j).$$

The reset inequalities telescope the first sum from above:

$$\sum_{j=1}^{N} (\mathcal{M}_j(a_j) - \mathcal{M}_j(b_j)) \leq \mathcal{M}_1(a_1) - \mathcal{M}_N(b_N) + \sum_{j=1}^{N-1} e_j.$$

Use (226). This proves (227). Monotone convergence gives the countable statement, and $N\delta \leq \sum_{j=1}^{N} \mathfrak{D}_j(I_j)$ gives (228). □

Definition II.10.3 (Perelman-type covariant invariant row). A Perelman-type construction is evaluated by the totalized tuple $\mathfrak{I}_{\mathrm{Per}}$ whose six coordinates are the graph defects of the following relations for the fixed PDE and its generated window atlas:

(P1) Each local primitive is a quotient-invariant scalar, typically a localized energy or variance plus an adjoint entropy and explicit pressure, drift, modulation, and boundary corrections.

(P2) Differentiation along the *fixed* PDE gives (224); all unsigned remainders are placed in the displayed error measure, not suppressed formally.

(P3) The target-visible quotient of production by its nonnegative physical floor $\mathfrak{d}$ is recorded by the price-gap row.

(P4) The primitives have the lower bound in (226). Its sign must match the orientation of (224).

(P5) Transport along the same physical orbit, chart pullback, and any change of a finite-horizon adjoint datum satisfy (225), with summable reset errors.

(P6) Production either defines the expenditure measure μ_u, as in Corollary II.10.6, or is dominated by the declared μ_u through an explicit change-of-variables inequality of the form (212).

The entropy-cocycle branch is the common zero set of these six target-active graph defects. Any nonzero coordinate is retained with its inherited source ancestry and sent to the form, chain, price, tail, or boundary row indicated by the failed relation. Thus the construction neither presumes the entropy scalar nor accepts it as additional favorable data. These coordinates isolate the production-and-cocycle mechanism of Perelman's entropy formula [12].

Proposition II.10.4 (Weighted zero-production rigidity). *Let $D \subset \mathbb{R}^d$ be connected and open, let $v \in L^2(I; W^{1,2}_{\mathrm{loc}}(D))$, and let $m \geq m_* > 0$ almost everywhere on $D \times I$. If a nonnegative production functional obeys*

$$\mathfrak{d}(w) \geq c_0 \int_I \int_D m(x,t) |\nabla v(x,t)|^2 \, \mathrm{d}x \, \mathrm{d}t \tag{229}$$

with $c_0 > 0$, then $\mathfrak{d}(w) = 0$ implies that $v(\cdot, t)$ is spatially constant on D for almost every t. Consequently, any closed bad class carrying a strictly positive mean-subtracted oscillation is disjoint from $\{\mathfrak{d} = 0\}$. The regularity compiler invokes this conclusion only on the target-local stratum whose selected production is $\mathfrak{d}$; it gives no exclusion for an unrelated bad class or source cell.

Proof. Nonnegativity and (229) imply $\nabla v = 0$ almost everywhere. A Sobolev function with zero weak gradient on a connected domain is almost everywhere constant. Its mean-subtracted oscillation is therefore zero, which excludes membership in the stated bad class. □

Proposition II.10.5 (Physical-clock comparison). *Let μ be a physical measure on an event. On the absolutely-continuous clock branch, let a normalized production measure have density $w \geq 0$: $\mathrm{d}\nu = w \, \mathrm{d}\mu$. On a measurable set Q,*

$$w \leq C, \quad 0 < C < \infty, \quad \text{a.e. on } Q \Longrightarrow \mu(Q) \geq C^{-1} \nu(Q), \tag{230}$$

$$w \geq c > 0 \quad \text{a.e. on } Q \Longrightarrow \mu(Q) \leq c^{-1} \nu(Q). \tag{231}$$

Thus a lower bound for normalized production yields a lower physical price only from an upper bound such as (230)*; the opposite density bound has the opposite implication.*

Proof. Integrate the pointwise bounds in $\nu(Q) = \int_Q w \, \mathrm{d}\mu$. □

Corollary II.10.6 (Entropy as the physical expenditure measure). *On the finite-error branch of Definition II.10.1 and* (227)*, on the atomic σ-algebra generated by the disjoint events Q_j, define*

$$\nu_u\left(\bigsqcup_{j \in S} Q_j\right) := \sum_{j \in S} \mathfrak{D}_j(I_j), \qquad S \subseteq \mathbb{N}.$$

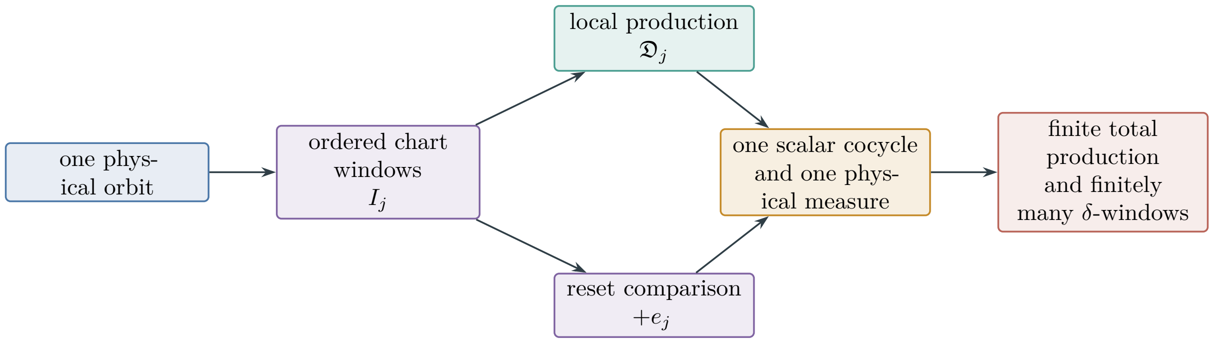

Figure 17: Covariant entropy-cocycle accounting. Local production estimates become orbitwise additive only after chart and adjoint-reset comparisons identify them along the same-origin physical orbit. The output is a finite physical production measure that can serve as the orbit expenditure μ_u.

Then ν_u is a finite, countably additive physical measure on that event family. It coincides with the restriction of μ_u in Definition I.1.18 precisely when the public carrier identity identifies this entropy production as physical expenditure. Otherwise it is a generated nonnegative structural coordinate and cannot replace or enlarge μ_u. Even in the coincident case, a shell event is charged only through its target-local factorization (110).

Proof. Countable additivity follows directly from the nonnegative series defining ν_u; finiteness is Theorem II.10.2. □

Remark II.10.7 (Equality models are row-local). A Gaussian may be the equality state of an adjoint Fisher-information term, while a constant field is the equality state of a weighted Dirichlet floor. These statements concern different nonnegative terms. A valid rigidity argument uses the equality set of the particular floor that is both compactly stable and linked to target badness; it does not merge the two equality models.

II.11 Wick-rotated compensated quotients and structural-action learning

This section acts on the general nonreversible compensated-quotient record already defined in Part I. On its spectral-Abs branch, the Wick map replaces the real quotient evolution by unitary spectral evolution and fixes the represented real Geom operator, the selected Caus operator, their authority record, and every saturated charge. The resulting generator is nonnormal in general, while its quotient response has a positive self-adjoint learning kernel.

Thus the only new datum is the choice to apply the Wick map. All source typing, quotient and fiber maps, geometric forms, causal provenance, authority restrictions, compensation inputs, and resource coordinates are those of the input Part I record. The transformed response stability, response kernel, memory kernel, action, and learning rules below are consequences evaluated on that same record. Its representation, confidence, authority, and model-selection coordinates belong to the inherited Part I computational ledger; they never augment the continuum physical budget. Conversely, any balance in this section used for PDE reachability must first pass through Theorems II.2.4 and II.2.18 on the contact-generated source cell. The learning bounds themselves make no continuum regularity claim.

II.11.1 Spectral quotient and real geometric complement

Definition II.11.1 (Wick transform of an inherited compensated record)**.** Let $\mathscr{C}_I$ be a dynamically qualified Part I master compensated quotient–geometry record in its represented spectral-Abs realization. Write $\mathcal{Q}_{\mathbb{R}}$ and $\mathcal{Y}_{\mathbb{R}}$ for its quotient and complementary real Hilbert spaces. In the notation of that record, H_{Abs} is the nonnegative self-adjoint quotient operator, G is the real nonnegative geometric operator, and

$$K_C = \begin{pmatrix} 0 & C^* \\ -C & A_{\mathcal{Y}} \end{pmatrix}$$

is the actual selected skew Caus block with its existing authority and provenance certificate. These are fields of $\mathscr{C}_I$, not new structural data.

Let $\mathcal{Q} = \mathcal{Q}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$ and $\mathcal{Y} = \mathcal{Y}_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$, and extend every field of $\mathscr{C}_I$ complex-linearly. The Wick transform acts only on the quotient evolution. The complex extension of G commutes with conjugation and restricts to the original operator on $\mathcal{Y}_{\mathbb{R}}$, so the Geom generator remains the complexification of the same real geometric operator (although a coupled trajectory need not remain in $\mathcal{Y}_{\mathbb{R}}$):

$$\mathfrak{W}_{\mathsf{Abs}} : -H_{\mathsf{Abs}} \oplus (-G) + K_C \longmapsto -iH_{\mathsf{Abs}} \oplus (-G) + K_C. \tag{232}$$

Equivalently, on $\mathcal{H}_W := \mathcal{Q} \oplus \mathcal{Y}$, its transformed generator is

$$\mathcal{L}_W(C) := \begin{pmatrix} -iH_{\mathsf{Abs}} & C^* \\ -C & -G + A_{\mathcal{Y}} \end{pmatrix} = -\begin{pmatrix} 0 & 0 \\ 0 & G \end{pmatrix} + \begin{pmatrix} -iH_{\mathsf{Abs}} & C^* \\ -C & A_{\mathcal{Y}} \end{pmatrix}. \tag{233}$$

The *positive self-energy normal form* is the inherited subbranch $A_{\mathcal{Y}} = 0$; the general inherited branch retains the full nonnormal resolvent. The spectral calculus gives

$$e^{-\tau H_{\mathsf{Abs}}} = \int_{[0,\infty)} e^{-\tau\omega}\, \mathrm{d}E_{H_{\mathsf{Abs}}}(\omega), \qquad e^{-itH_{\mathsf{Abs}}} = \int_{[0,\infty)} e^{-it\omega}\, \mathrm{d}E_{H_{\mathsf{Abs}}}(\omega). \tag{234}$$

The two spectral integrals are ordinary for every vector in their displayed time domains. Their identification by the notation $\tau = it$ is evaluated by the Part I analytic-continuation relation: it is an ordinary equality on the record's continuation domain and otherwise carries that relation's nonordinary boundary value. No continuation domain is added to structural admissibility.

The map $\mathfrak{W}_{\mathsf{Abs}}$ preserves the source cell and the complete record. In particular, H_{Abs}, $-iH_{\mathsf{Abs}}$, and their spectral projectors have the same Abs ancestry; $G, C, A_{\mathcal{Y}}$, the authority envelope, and their costs are unchanged. Equation (234) is spectral functional calculus, rather than a formal substitution in a learned trajectory.

Proposition II.11.2 (Exact inheritance under the Wick map)**.** *Let $\mathscr{C}_I^W = \mathfrak{W}_{\mathsf{Abs}}(\mathscr{C}_I)$. Under the canonical scalar-extension embeddings,*

$$\mathrm{Src}(\mathscr{C}_I^W) = \mathrm{Src}(\mathscr{C}_I), \qquad \mathrm{Anc}(\mathscr{C}_I^W) = \mathrm{Anc}(\mathscr{C}_I), \qquad \mathfrak{A}(\mathscr{C}_I^W) = \mathfrak{A}(\mathscr{C}_I), \qquad \mathbf{c}_{\mathrm{str}}(\mathscr{C}_I^W) = \mathbf{c}_{\mathrm{str}}(\mathscr{C}_I). \tag{235}$$

The represented Geom *form, selected* Caus *current, action metric, and quotient/fiber transports agree with their real restrictions in $\mathscr{C}_I$. Consequently the source, structural-admissibility, and authority gates have the same truth value before and after the map. The inherited compensation data remain fixed, while every response or dynamic- stability quantity containing the quotient time generator is reevaluated with $-H_{\mathsf{Abs}}$ replaced by $-iH_{\mathsf{Abs}}$.*

Wick quantity	Part I source	Exhaustive evaluation
Generator realization	dynamic and sectorial realization field	bounded or ordinary closed-form value; otherwise typed realization defect
Resolvent and self-energy	full-generator dynamic record	Hille–Yosida value on $\Re z > 0$; totalized graph value elsewhere
Response dimension	finite evaluation carrier and effective-dimension machinery	finite carrier trace; population extended trace in $[0, \infty]$
Spectral gap and commutation	represented operators and spectral calculus	computed gap/status; modal formula or exact singular-value formula
Legal bridge and action	Caus authority envelope and current metric	distance to the legal-current closure; empty family gives $+\infty$
Identification	represented dynamics chart and score candidates	derived design range/kernel; full or partial identification bound
Statistical law	i.i.d., mixing, sequential, martingale, and channel cells	recordwise minimum ordinary radius; every unavailable candidate equals $+\infty$
RL conversion	safe-action, behavior/deployment, and actor records	extended critic, converter, support, and policy coordinates
Model choice	exact source cells, charged nets, Rademacher union, and PAC–Bayes mixture	finite-stage saturated profile and Pareto strict inverse; empty/nonattained limit is retained and typed
Continuum energy	operator-range and approximation records	Moore–Penrose quadratic form and extended suprema; no closed-range or compactness gate

Table 2: Inheritance and totalization ledger for the Wick specialization. Every row is an evaluation of Part I data or a consequence of it.

Proof. Complexification is an isometric scalar extension and commutes with sums, adjoints, compositions, projections, and restrictions to the original real spaces. Equation (232) fixes every block except the quotient diagonal. The source, ancestry, authority, structural cost, current, metric, and transport fields therefore remain identical. Their algebraic and authority gates are preserved. A resolvent, self-energy, memory, stability factor, or learning kernel contains the transformed diagonal and is therefore recomputed as a derived Wick response rather than copied from the input record. □

Proposition II.11.3 (No-new-premise closure of the Wick specialization)**.** *The input domain of $\mathfrak{W}_{\mathsf{Abs}}$ is exactly the represented, dynamically qualified spectral-*Abs *cell of the Part I master record. Within that inherited domain, the Wick chapter adds no structural, analytic, statistical, or model-selection premise. More precisely:*

(i) *bounded, sectorial, unbounded-bridge, authority, source, channel, dependence, control, and model-family alternatives are values of existing Part I record fields;*

(ii) *spectral gaps, commutators, design excitation, operator errors, source residuals, converter constants, policy divergences, and optimization errors are computed coordinates in $[0, \infty]$, not admission conditions;*

(iii) *a missing analytic or statistical certificate has its Part I nonordinary value. Every dependent upper-error or risk candidate has value $+\infty$, every dependent lower-usefulness candidate has value zero, and a derived coordinate without an ordered extension retains its typed nonordinary value;*

(iv) *an empty legal, zero-action, or risk-feasible family produces its typed authority, representation, confidence, or budget status; and*
(v) *every finite-rate formula is the restriction of a general extended bound to the named Part I rate cell, while all complementary cells retain the general bound.*

Consequently no favorable branch is used to define admissibility for this specialization. The only new operation is $-H_{\mathsf{Abs}} \mapsto -iH_{\mathsf{Abs}}$, after which all quantities are derived or totalized.

Proof. Exact inheritance is Proposition II.11.2. Part I already partitions represented records by realization, authority, source, channel, dependence, and control status and supplies charged finite stages for family selection. Evaluation of a field gives either its ordinary value or its existing nonordinary value. Extended-real arithmetic, the conventions $\inf \varnothing = +\infty$ and $\sup \varnothing = 0$, and typed empty-family statuses make every construction total. The propositions below derive the new response, action, loss, and transfer coordinates from those fields and make each finite specialization an explicit record cell. □

Proposition II.11.4 (Generation and physical budget). *On the bounded realization branch of the inherited Part I record, $\mathcal{L}_W(C)$ generates a contraction semigroup and a solution $u = (x, y)$ of $\dot{u} = \mathcal{L}_W(C)u$ satisfies*

$$\frac{\mathrm{d}}{\mathrm{d}t}\|x\|^2 = 2\,\mathrm{Re}\langle Cx, y\rangle, \tag{236}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\|y\|^2 = -2\,\mathrm{Re}\langle Cx, y\rangle - 2\|G^{1/2}y\|^2, \tag{237}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(\|x\|^2 + \|y\|^2) = -2\|G^{1/2}y\|^2. \tag{238}$$

Consequently the Wick-rotated Abs *block and the authorized* Caus *block transfer carrier between the two sectors without expenditure. The real* Geom *block supplies the unique nonnegative expenditure in this model.*

Proof. The second matrix in (233) is skew-adjoint. Taking real parts in the two component equations gives (236) and (237); their transfer terms cancel. Thus $\mathcal{L}_W(C)$ is dissipative. The bounded perturbation theorem, or direct finite-dimensional exponentiation, gives the contraction semigroup. □

Proposition II.11.5 (Closed-operator and sectorial-complement realization). *On the closed sectorial branch of the inherited Part I record, the complexified complement form defines its maximal dissipative operator $-G + A_{\mathcal{Y}}$. On its inherited bounded-bridge cell, the block operator $\mathcal{L}_W(C)$ on $D(H_{\mathsf{Abs}}) \oplus D(-G + A_{\mathcal{Y}})$ is maximal dissipative. The same conclusion holds on the ordinary unbounded-bridge realization cell produced by the Part I sectorial Caus-authority interface. Its complementary nonordinary cell retains the typed realization defect. Identity* (238) *holds in integrated form,*

$$\|u(t)\|^2 + 2\int_0^t \|G^{1/2}y(s)\|^2\,\mathrm{d}s = \|u(0)\|^2, \tag{239}$$

on the equality branch, and with $\leq$ for weak limits carrying a nonnegative energy defect.

Proof. The complement sectorial form gives $-G + A_{\mathcal{Y}}$, while $-iH_{\mathsf{Abs}}$ generates a unitary group. Their direct sum is maximal dissipative. The off-diagonal bridge is bounded and skew-adjoint, so the bounded dissipative perturbation theorem preserves maximal dissipativity. The ordinary unbounded-bridge conclusion is exactly the closed-operator perturbation conclusion already certified by the Part I sectorial interface [7, 11]. Approximation by vectors in a common core and lower semicontinuity give the integrated identity or its defect version. □

II.11.2 Lumpability defect, self-energy, and memory

Let $U : \mathcal{Q} \to \mathcal{H}_W$ be $Ux = (x, 0)$, let $P = U^*$, and let $P_\Delta = \mathbf{1}_\Delta(H_{\mathsf{Abs}})$ be a declared spectral band.

Proposition II.11.6 (Band-lumpability defect)**.** *The compressed quotient generator is $P\mathcal{L}_W U = -iH_{\mathsf{Abs}}$, and its full-operator defect on Δ is*

$$E_{W,\Delta} := (I - UP)\mathcal{L}_W U P_\Delta = \begin{pmatrix} 0 \\ -CP_\Delta \end{pmatrix}. \tag{240}$$

Hence the spectral band is exactly lumpable if and only if $CP_\Delta = 0$. Its intrinsic quasi-lumpability size is $\varepsilon_{W,\Delta} = \|CP_\Delta\|$, supplemented in an empirical record by the separately measured model and compression residuals. Under the canonical complexification isometry this is exactly the norm of the corresponding cross-sector defect in $\mathscr{C}_I$; Wick rotation changes the quotient diagonal and introduces no new lumpability hypothesis.

Proof. Apply the two block rows of $\mathcal{L}_W$ to $(x, 0)$, compress the first component, and subtract its lift. The off-diagonal block is fixed by (232), so the resulting vector is the complexification of the inherited Part I defect. Complexification is isometric, which proves the norm statement. □

Theorem II.11.7 (Feshbach response and geometric self-energy)**.** *The Part I realization field is exhaustive. On either ordinary realization cell of Propositions II.11.4 and II.11.5, every $z \in \mathbb{C}$ with $\operatorname{Re} z > 0$ satisfies, by maximal dissipativity,*

$$\|(z + G - A_{\mathcal{Y}})^{-1}\| \le \frac{1}{\operatorname{Re} z}, \qquad \|(z - \mathcal{L}_W(C))^{-1}\| \le \frac{1}{\operatorname{Re} z}. \tag{241}$$

Define

$$\Sigma_C(z) := C^*(z + G - A_{\mathcal{Y}})^{-1}C, \qquad F_W(z) := z + iH_{\mathsf{Abs}} + \Sigma_C(z). \tag{242}$$

On the ordinary inherited unbounded-bridge cell, these expressions denote the closed forms and associated operators supplied by that Part I realization certificate. Then $F_W(z)$ is invertible and

$$P(z - \mathcal{L}_W(C))^{-1}U = F_W(z)^{-1}. \tag{243}$$

In the positive self-energy normal form, $z = \lambda > 0$ gives

$$\Sigma_C(\lambda) = C^*(\lambda + G)^{-1}C \ge 0. \tag{244}$$

Thus real geometric relaxation compensates the imaginary quotient defect by a positive frequency-dependent self-energy. For $\operatorname{Re} z \le 0$, the same block formula is the ordinary value of the existing totalized resolvent relation whenever that value exists. A pole, domain failure, or noninvertible Schur complement is its nonordinary graph value. Hence every z is accounted for, while the learning kernel uses the automatically ordinary half-line $\lambda > 0$ on the ordinary realization cells. On the complementary nonordinary realization cell, the self-energy, Schur complement, and learning kernel inherit the nonordinary graph value and every response-dependent upper certificate equals $+\infty$; the record is retained with its realization status.

Proof. Both the complement generator $-G + A_{\mathcal{Y}}$ and the full Wick generator are maximal dissipative by Propositions II.11.4 and II.11.5. The right half-plane therefore belongs to both resolvent sets and the Hille–Yosida estimate gives (241). The block operator $z - \mathcal{L}_W$ is

$$\begin{pmatrix} z + iH_{\mathsf{Abs}} & -C^* \\ C & z + G - A_{\mathcal{Y}} \end{pmatrix}.$$

The lower-right block and the full block are invertible. Block Gaussian elimination therefore shows that their Schur complement $F_W(z)$ is invertible and gives (243). On the ordinary inherited unbounded-bridge cell the same factorization is first taken on the common form core and then closed by its Part I realization theorem. When $A_{\mathcal{Y}} = 0$ and $z = \lambda > 0$, the middle resolvent in (244) is positive self-adjoint. Outside the right half-plane, totalization records exactly the failure of one of these ordinary elimination steps. $\square$

Corollary II.11.8 (Mori–Zwanzig memory equation)**.** *On the bounded-bridge cell, for classical initial data (x_0, y_0), the quotient coordinate satisfies*

$$\dot{x}(t) = -iH_{\mathsf{Abs}}x(t) - \int_0^t K_C(t-s)x(s)\,\mathrm{d}s + C^* e^{t(-G+A_{\mathcal{Y}})}y_0, \qquad K_C(t) := C^* e^{t(-G+A_{\mathcal{Y}})}C. \tag{245}$$

For every $\operatorname{Re} z > 0$, the Laplace transform of K_C is $\Sigma_C(z)$. Equations (243) and (245) are respectively the Feshbach and Mori–Zwanzig forms of the same quotient elimination [4, 10, 16]. On the ordinary unbounded-bridge cell, the same display is interpreted in the mild input–output or closed-form sense supplied by the Part I realization record, first on its common core and then by closure. A datum outside that recorded admissibility domain has the inherited nonordinary evolution value; no additional observation-domain premise is imposed here.

Proof. Variation of constants in the y-equation gives

$$y(t) = e^{t(-G+A_{\mathcal{Y}})}y_0 - \int_0^t e^{(t-s)(-G+A_{\mathcal{Y}})}Cx(s)\,\mathrm{d}s.$$

Substitute this identity in $\dot{x} = -iH_{\mathsf{Abs}}x + C^*y$. Laplace transformation gives (242). On the ordinary unbounded-bridge cell, the Part I common-core factorization in Theorem II.11.7 gives the same identity, and closure gives the recorded mild input–output relation. $\square$

Definition II.11.9 (Dark quotient space)**.** On the bounded-bridge cell, the dark quotient space is

$$\mathcal{D}_{\text{dark}} := \{x \in \mathcal{Q} : Ce^{-itH_{\mathsf{Abs}}}x = 0 \text{ for every } t \in \mathbb{R}\}. \tag{246}$$

It is the largest H_{Abs}-reducing subspace contained in $\ker C$. On the ordinary unbounded-bridge cell, $Ce^{-itH_{\mathsf{Abs}}}$ denotes the closed observation orbit supplied by the Part I realization record and the same intersection is taken on its recorded common domain. A domain failure retains the typed realization value instead of defining a smaller dark space. Thus the dark-space coordinate is totalized by the same realization partition as the Feshbach response. In dimension $d_Q < \infty$,

$$\mathcal{D}_{\text{dark}} = \bigcap_{k=0}^{d_Q-1} \ker(CH_{\mathsf{Abs}}^k). \tag{247}$$

The target-visible component of $\mathcal{D}_{\text{dark}}$ is a $\mathsf{Caus}/\mathsf{Geom}$ reach obstruction. It remains available to a direct spectral readout unless that readout is separately excluded.

Proposition II.11.10 (Totalized imaginary-axis and bracket alternatives)**.** *Define the derived geometric gap*

$$g_* := \inf \sigma(G) \in [0, \infty).$$

Every imaginary-axis eigenpair $\mathcal{L}_W u = i\beta u$, $u = (x, y)$, satisfies

$$y \in \ker G, \qquad -iH_{\mathsf{Abs}}x + C^*y = i\beta x, \qquad -Cx + A_{\mathcal{Y}}y = i\beta y. \tag{248}$$

On the exhaustive cell $g_* > 0$, this reduces to $u = (x, 0)$, $Cx = 0$, and $H_{\mathsf{Abs}} x = -\beta x$. Consequently that cell has no imaginary-axis eigenvalue exactly when the corresponding H_{Abs}-eigenspaces have zero intersection with $\ker C$*. On the complementary cell $g_* = 0$, equation* (248) *retains the geometric kernel rather than discarding it. In both cells the commutator Gramian of Definition II.7.1, applied to*

$$S = \begin{pmatrix} 0 & 0 \\ 0 & G \end{pmatrix}, \qquad A = \begin{pmatrix} -iH_{\mathsf{Abs}} & C^* \\ -C & A_{\mathcal{Y}} \end{pmatrix},$$

has its Part I extended reduced gap. A positive value supplies the existing twisted-carrier estimate; the zero value returns its target-visible bracket kernel as the typed dark/rank-defect branch. These alternatives exhaust the record and introduce no coercivity or bracket assumption.

Proof. Taking real parts of $\langle u, \mathcal{L}_W u\rangle = i\beta\|u\|^2$ and using (238) gives $\langle y, Gy\rangle = 0$. Since $G \geq 0$, this is equivalent to $G^{1/2} y = 0$, hence to $y \in \ker G$, and the two block equations give (248). The derived inequality $G \geq g_* I$ makes $y = 0$ on the cell $g_* > 0$. The remaining claims are the two values of the Part I bracket-resistance and twisted-carrier alternative for the displayed $S + A$ decomposition. □

II.11.3 Authority-constrained structural action

The bridge in (233) is an operator. Its action record uses the actual edge or current representation induced by that same operator. This prevents a favorable target current from replacing an imposed PDE current.

Definition II.11.11 (Wick evaluation of the inherited bridge action)**.** Fix a pre-contact state of $\mathscr{C}_I$. Let $(\mathfrak{J}, \langle\cdot,\cdot\rangle_{\mathbb{M}^{-1}})$, the target-normalized potential D, the actual authorized bridge current J_C, and its authority envelope $\mathfrak{A}$ be the corresponding Part I fields. Put

$$J_D := -\mathbb{M}\nabla D,$$

and define

$$\mathcal{A}_{\mathsf{Caus}}(J_C) := \|J_C\|^2_{\mathbb{M}^{-1}}, \qquad \mathcal{E}_{\mathrm{aim}}(D) := \|J_D\|^2_{\mathbb{M}^{-1}}, \tag{249}$$

$$\mathcal{P}_{\mathsf{Caus}}(J_C, D) := \langle J_C, J_D\rangle_{\mathbb{M}^{-1}}, \quad \mathscr{W}_{\mathfrak{A}}(C; D) := \frac{1}{2}\mathcal{A}_{\mathsf{Caus}}(J_C) + \frac{1}{2}\mathcal{E}_{\mathrm{aim}}(D) - \mathcal{P}_{\mathsf{Caus}}(J_C, D). \tag{250}$$

All fields and transports in this display remain those of $\mathscr{C}_I$. The Wick map changes none of them; it only permits the displayed action to be evaluated jointly with the transformed spectral response.

Theorem II.11.12 (Inner least-action bridge coordinate)**.** *Every inherited bridge action evaluated after the Wick transform satisfies*

$$\mathscr{W}_{\mathfrak{A}}(C; D) = \frac{1}{2}\|J_C - J_D\|^2_{\mathbb{M}^{-1}} \geq 0. \tag{251}$$

Let $\mathfrak{J}_{\mathfrak{A}}$ be the possibly empty set of represented currents induced by legal bridges, and use $\operatorname{dist}(J_D, \varnothing) := +\infty$*. Then*

$$\inf_{C \in \mathfrak{A}} \mathscr{W}_{\mathfrak{A}}(C; D) = \frac{1}{2}\operatorname{dist}_{\mathbb{M}^{-1}}(J_D, \overline{\mathfrak{J}_{\mathfrak{A}}})^2. \tag{252}$$

(i) *On the inherited closed-convex authority cell, the infimum is attained at the metric projection $J_C = \Pi_{\mathfrak{J}_{\mathfrak{A}}} J_D$. On every other nonempty cell, the Part I search transcript returns an ε-minimizer and charges its cover and optimization error. The empty cell returns the authority-obstruction value $+\infty$.*

(ii) *Under free authority, the infimum is zero precisely when $J_D \in \overline{\mathfrak{J}_{\mathfrak{A}}}$, and zero waste is attained precisely when $J_D \in \mathfrak{J}_{\mathfrak{A}}$. Thus density and exact representability are not conflated.*

(iii) *Under hybrid or RL authority, the same distance is evaluated on the declared control family. Its convex, nonconvex, nonattained, and empty cells are handled by the preceding exhaustive rule.*

(iv) *Under imposed PDE authority, $\mathfrak{J}_{\mathfrak{A}}$ is a singleton and the action evaluates its fixed current.*

For a declared positive target power p, every legal current with $\mathcal{P}_{\mathsf{Caus}}(J_C, D) \geq p$ obeys

$$\mathcal{A}_{\mathsf{Caus}}(J_C) \geq \frac{p^2}{\mathcal{E}_{\mathrm{aim}}(D)} \tag{253}$$

on the cell $\mathcal{E}_{\mathrm{aim}}(D) > 0$. Equality holds exactly for $J_C = (p/\mathcal{E}_{\mathrm{aim}}(D))J_D$ when this current is authorized. On the complementary cell $\mathcal{E}_{\mathrm{aim}}(D) = 0$, no current can satisfy $\mathcal{P}_{\mathsf{Caus}}(J_C, D) \geq p$; the power request therefore returns the authority/target obstruction rather than a missing denominator. This minimization evaluates the action coordinate $\mathbf{c}_{\mathrm{act}}(\mathscr{C}_W)$. Zero action is one favorable coordinate; structural optimality is the subsequent saturated-profile comparison of Definition II.11.29.

Proof. Expand the square in (251); taking the infimum gives (252) because distance is unchanged by closure. The Hilbert projection theorem gives the closed-convex cell, while the definition of an infimum supplies ε-minimizers on every other nonempty cell. The authority statements are its free, control-family, and singleton specializations. Cauchy–Schwarz gives $p^2 \leq \mathcal{A}_{\mathsf{Caus}}(J_C)\mathcal{E}_{\mathrm{aim}}(D)$, with the displayed equality condition; when the second factor is zero, the left-hand side cannot be positive. ☐

Corollary II.11.13 (Authority monotonicity)**.** *The Part I authority order compares envelopes only at fixed quotient, geometry, target, and cost convention. Every ordered pair $\mathfrak{A}_1 \preceq \mathfrak{A}_2$, equivalently $\mathfrak{J}_{\mathfrak{A}_1} \subseteq \mathfrak{J}_{\mathfrak{A}_2}$, satisfies*

$$\inf_{C \in \mathfrak{A}_2} \mathscr{W}_{\mathfrak{A}_2}(C; D) \leq \inf_{C \in \mathfrak{A}_1} \mathscr{W}_{\mathfrak{A}_1}(C; D).$$

The corresponding saturated attainable profile is ordered in the opposite direction: imposed $\leq$ hybrid $\leq$ free whenever the represented families are nested with no larger implementation cost.

II.11.4 Positive response kernel and quantitative transfer

The quotient response $F_W(\lambda)^{-1}$ need not be normal. Statistical learning therefore uses a positive transform of the full response rather than a diagonalization of F_W.

Definition II.11.14 (Wick response kernel)**.** Fix $\lambda > 0$. The complement and full resolvents, and the invertibility of $F_W(\lambda)$, are consequences of Theorem II.11.7 on every ordinary realization, both in the positive-self-energy and general complement cells. There define

$$T_{W,\lambda} := \left(I + F_W(\lambda)^* F_W(\lambda)\right)^{-1}, \qquad A_{W,\lambda} := T_{W,\lambda}^{-1} = I + F_W(\lambda)^* F_W(\lambda). \tag{254}$$

The operator $T_{W,\lambda}$ is a positive contraction. On a nonordinary realization, $T_{W,\lambda}$ is the unavailable response-kernel value and all response-dependent upper bounds equal $+\infty$. For every ordinary kernel, Part I supplies each finite evaluation carrier $\mathscr{E}$—an empirical sample, mixing-block sample, or sequential path/tree—has its represented evaluation map $E_{\mathscr{E}}$, with the normalization of its Part I theorem, and finite positive response matrix

$$T^{\mathscr{E}}_{W,\lambda} := E_{\mathscr{E}} T_{W,\lambda} E^*_{\mathscr{E}}, \qquad \mathrm{Tr}_{\mathscr{E}}\, T_{W,\lambda} := \mathrm{Tr}\, T^{\mathscr{E}}_{W,\lambda} < \infty. \tag{255}$$

This is the trace used by the Part I empirical Rademacher bounds. It is finite because $\mathscr{E}$ is finite, independently of every compactness property of the ambient response. Its empirical ridge effective dimension is exactly the Part I Gram-matrix quantity

$$\widehat{N}_{W,\mathscr{E}}(\lambda,\rho) := \mathrm{Tr}\Big[T^{\mathscr{E}}_{W,\lambda}(T^{\mathscr{E}}_{W,\lambda}+\rho I)^{-1}\Big] \le \dim \mathscr{E}. \tag{256}$$

For ridge parameter $\rho > 0$, define the population effective dimension as the extended positive trace

$$\begin{aligned} N_W(\lambda,\rho) &:= \mathrm{Tr}_{\mathrm{ext}}\Big[T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}\Big] \\ &:= \sup_{P:\,P=P^*=P^2,\ \mathrm{rank}\,P<\infty} \mathrm{Tr}\Big[P T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1} P\Big] \in [0,\infty], \end{aligned} \tag{257}$$

$$m_W(\lambda,\rho;f) := \frac{\|T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1} f\|^2}{\|f\|^2}. \tag{258}$$

The capture ratio is defined to be zero at $f = 0$, as in the Part I target-capture convention. Every represented nested dense Galerkin family $P_n \uparrow I$ satisfies

$$N_W(\lambda,\rho) = \sup_n \mathrm{Tr}\Big[P_n T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1} P_n\Big]\,. \tag{259}$$

Thus finiteness of the population coordinate is a derived value. When the supremum is infinite, the effective-dimension candidate equals $+\infty$ and the Part I recordwise certificate minimum uses another valid same-loss bound or returns its existing capacity/budget obstruction. This operation forms the risk coordinate of the record; the saturated profile retains the structural comparison.

Proposition II.11.15 (Part I trace accounting without an ambient trace gate)**.** *Put $R_{W,\rho} = T_{W,\lambda}(T_{W,\lambda}+\rho I)^{-1}$. For every represented nested finite-rank family $P_n \uparrow I$,*

$$\mathrm{Tr}_{\mathrm{ext}}\, R_{W,\rho} = \lim_{n\to\infty} \mathrm{Tr}(P_n R_{W,\rho} P_n) \in [0,\infty]. \tag{260}$$

The value is finite exactly when $R_{W,\rho}$ is trace class; trace-class behavior is therefore a conclusion about this coordinate, not an admissibility hypothesis. Every empirical Rademacher, block, and sequential trace remains finite on its own evaluation carrier even when the value in (260) *is infinite.*

Proof. Choose an orthonormal basis adapted to the nested ranges of P_n. Since $R_{W,\rho} \ge 0$, the finite traces are the increasing partial sums of $\sum_j \langle e_j, R_{W,\rho} e_j\rangle$. Their limit is the extended positive trace, independently of the adapted basis. A positive bounded operator is trace class precisely when this sum is finite. The final statement follows because every evaluation compression in (255) is a finite positive matrix; (256) additionally bounds its ridge mode count by the carrier dimension. $\square$

Proposition II.11.16 (Exhaustive modal and nonnormal response formulas)**.** *On the positive-self-energy cell $A_{\mathcal{Y}} = 0$, put $B = \lambda I + \Sigma_C(\lambda) = B^*$. Evaluate the derived spectral- commutation status: its ordinary commuting value means that the spectral measures of H_{Abs} and B commute,* *and its complementary value means that they do not. On the commuting cell, their joint spectral representation has scalar response weight*

$$\eta^W(\omega, \sigma) = \frac{1}{1 + (\lambda + \sigma)^2 + \omega^2}; \tag{261}$$

in a joint pure-point cell this reads $H_{\mathsf{Abs}}\phi_j = \omega_j \phi_j$, $\Sigma_C(\lambda)\phi_j = \sigma_j \phi_j$, and $\eta_j^W = \eta^W(\omega_j, \sigma_j)$. On the noncommuting cell,

$$A_{W,\lambda} = I + B^2 + H_{\mathsf{Abs}}^2 + i[B, H_{\mathsf{Abs}}], \tag{262}$$

as a quadratic-form identity on the inherited common form domain, and the singular values of $F_W(\lambda)$, rather than separate eigenvalues, determine $T_{W,\lambda}$.

On the general cell $A_{\mathcal{Y}} \neq 0$, set $B = \lambda I + \Sigma_C(\lambda)$, which need not be self-adjoint. The always-valid precision identity is

$$A_{W,\lambda} = I + B^* B + H_{\mathsf{Abs}}^2 + i(B^* H_{\mathsf{Abs}} - H_{\mathsf{Abs}} B). \tag{263}$$

Thus every record has an exact singular-value formula; the modal formula is an additional ordinary value derived only on the recorded commuting positive-self-energy cell, and its absence removes no record.

Proof. The joint spectral theorem applied on the commuting cell gives (261). On the positive-self-energy cell, $(B + iH_{\mathsf{Abs}})^*(B + iH_{\mathsf{Abs}}) = (B - iH_{\mathsf{Abs}})(B + iH_{\mathsf{Abs}})$, which gives (262). Without self-adjointness of B, the same product expands instead to $B^*B + H_{\mathsf{Abs}}^2 + i(B^* H_{\mathsf{Abs}} - H_{\mathsf{Abs}} B)$, proving (263). $\square$

Theorem II.11.17 (Master Wick learning transfer)**.** *For each complete Wick record, evaluate every Part I learning-certificate candidate already carried by that record. In each ordinary candidate, substitute*

$$T_k = T_{W,\lambda}, \qquad A = A_{W,\lambda}, \qquad E_q = E_{W,\Delta}, \tag{264}$$

and retain its sample law, loss, source cell, confidence allocation, and authority record. Every conclusion of that Part I candidate then holds with $N_{\mathrm{eff}}(\rho) = N_W(\lambda, \rho)$ and capture $m_k(\rho) = m_W(\lambda, \rho; f_*)$. A candidate whose Part I record fields are nonordinary has value $+\infty$. The recordwise minimum over valid same-loss candidates forms the Part I statistical-radius coordinate, and the typed failure mechanism is unchanged. Across complete records, the inherited saturated profile remains the comparison rule.

In particular, for the energy ball $\mathcal{F}_W(R) = \{f : \langle f, A_{W,\lambda} f\rangle \leq R^2\}$, let S be the declared m-point evaluation carrier and use its Part I compression $T_{W,\lambda}^S$ from (255). Then

$$\widehat{\mathfrak{R}}_S(\mathcal{F}_W(R)) \leq \frac{R}{m}\sqrt{\operatorname{Tr}_S T_{W,\lambda}}, \tag{265}$$

and, on the Part I $[0,1]$-valued L_ℓ-Lipschitz loss cell, the corresponding candidate satisfies with probability at least $1 - \delta$,

$$L(f) \leq L_S(f) + \frac{2L_\ell R}{m}\sqrt{\operatorname{Tr}_S T_{W,\lambda}} + 3\sqrt{\frac{\log(2/\delta)}{2m}} + \varepsilon_{\mathrm{task}}. \tag{266}$$

On the named Part I rate cell $f_* = T_{W,\lambda}^r g$, $0 < r \leq 1$, and $\eta_j^W \asymp j^{-2\beta}$, $\beta > 1/2$, its certified kernel-ridge oracle inequality gives

$$\|\widehat{f}_{\rho_m} - f_*\|_{L^2}^2 \lesssim m^{-4\beta r/(4\beta r+1)}, \qquad \rho_m \asymp m^{-1/(2r+1/(2\beta))}. \tag{267}$$

The constants and probability event are exactly those of the fixed-kernel oracle certificate; (267) *is asserted only for the named eigenvalue and source class.*

Proof. The operator $T_{W,\lambda}$ is positive and self-adjoint, so the algebraic gate of the Part I kernel and energy-ball candidates is derived rather than assumed. Its empirical compression $T^S_{W,\lambda}$ is a finite positive matrix, so the Part I ellipsoid support calculation gives (265), and symmetrization and contraction give (266). The source residual is $O(\rho^{2r})$, while $N_W(\lambda, \rho) \asymp \rho^{-1/(2\beta)}$; balancing variance and bias gives (267). This named eigenvalue class makes the extended effective dimension finite as a conclusion. The remaining transfers are substitutions in positive-operator statements and preserve their recorded cells and units. □

Proposition II.11.18 (Learned-operator stability)**.** *On every finite Part I evaluation carrier, let two represented positive-normal-form records have the derived error coordinates*

$$\varepsilon_H := \|H - \widehat{H}\|, \qquad \varepsilon_C := \|C - \widehat{C}\|, \qquad \varepsilon_G := \|G - \widehat{G}\|.$$

Put $r = \max\{\|(\lambda + G)^{-1}\|, \|(\lambda + \widehat{G})^{-1}\|\}$ *and* $c = \max\{\|C\|, \|\widehat{C}\|\}$*. Then*

$$\|F_W - \widehat{F}_W\| \le \varepsilon_H + 2cr\varepsilon_C + c^2r^2\varepsilon_G =: \varepsilon_F, \tag{268}$$

$$\|T_{W,\lambda} - \widehat{T}_{W,\lambda}\| \le (\|F_W\| + \|\widehat{F}_W\|)\varepsilon_F. \tag{269}$$

Here $r \le \lambda^{-1}$ *is derived from positivity, so every finite-carrier quantity in the display is finite. On a continuum record, the same formulas are evaluated with its Part I form-to-resolvent error coordinates in* $[0, \infty]$*; a nonordinary comparison gives the valid value* $+\infty$ *rather than an admissibility restriction. The general nonnormal branch has the same formula with* ε_G *replaced by the derived same-form-norm distance between* $(G - A_{\mathcal{Y}}) - (\widehat{G} - \widehat{A}_{\mathcal{Y}})$ *and with both complement resolvent norms retained.*

Proof. Add and subtract the two mixed self-energy terms and use $R - \widehat{R} = R(\widehat{G} - G)\widehat{R}$ for $R = (\lambda + G)^{-1}$. This proves (268). Apply the resolvent identity to $(I + F^*F)^{-1}$ and use $\|F^*F - \widehat{F}^*\widehat{F}\| \le (\|F\| + \|\widehat{F}\|)\|F - \widehat{F}\|$. □

II.11.5 Statistically calibrated training objectives

The response kernel determines the regularizer, but it does not by itself specify how the block generator is identified or how a selectable bridge is trained. The three inherited authority/data cells therefore evaluate three objectives: a dynamics loss on target-free data, a response-kernel loss on supervised data, and, for controlled systems, a Bellman/policy loss on the represented deployment law. Their samples and confidence allocations are fields of the Part I record.

Definition II.11.19 (Structure-preserving dynamics loss)**.** On each represented finite-dimensional chart supplied by the Part I learner, quotienting by its recorded carrier gauge gives identifiable coordinates. Let $Z_j = (x_j, y_j, \dot{x}_j, \dot{y}_j)$, $1 \le j \le n_{\mathrm{dyn}}$, be dynamics-only observations, and let $W_{x,j}$ and $W_{y,j}$ be represented positive precision matrices. For $\vartheta = (H, G, A_{\mathcal{Y}}, C)$ satisfying

$$H = H^* \succeq 0, \qquad G = G^* \succeq 0, \qquad A^*_{\mathcal{Y}} = -A_{\mathcal{Y}},$$

define

$$\mathcal{L}_{\mathrm{dyn}}(\vartheta) := \frac{1}{2n_{\mathrm{dyn}}} \sum_{j=1}^{n_{\mathrm{dyn}}} \left(\|\dot{x}_j + iHx_j - C^*y_j\|^2_{W_{x,j}} + \|\dot{y}_j + Cx_j + Gy_j - A_{\mathcal{Y}}y_j\|^2_{W_{y,j}} \right). \tag{270}$$

This is the generalized least-squares loss for the two block equations. On the inherited conditional-Gaussian channel cell with inverse covariance $W_{x,j} \oplus W_{y,j}$, it is also the negative log-likelihood up to a parameter-independent constant.

When derivatives are unavailable, raw transitions use the likelihood loss

$$\mathcal{L}_{\mathrm{tr}}(\vartheta) := \frac{1}{2n_{\mathrm{dyn}}} \sum_j \|u_{j+1} - e^{\Delta_j \mathcal{L}_W(\vartheta)} u_j\|^2_{\Omega_j^{-1}} + \frac{1}{2n_{\mathrm{dyn}}} \sum_j \log\det \Omega_j, \tag{271}$$

with the conditional mean and covariance replaced by the represented transition law when it is non-Gaussian. A discretization defect, derivative estimation error, or misspecified noise covariance is retained as a separate model coordinate. Neither loss uses target labels or rewards. The fitted parameters and $\varepsilon_{\mathrm{opt}}$ are inner training coordinates in the sense of Definition II.11.29. Comparison between complete records is deferred to the saturated profile.

Proposition II.11.20 (Identification and response error)**.** *In the linear derivative chart of Definition II.11.19, write*

$$r_j(\vartheta_* + h) = \xi_j + \mathcal{D}_j h, \qquad Q_n = \frac{1}{n_{\mathrm{dyn}}} \sum_j \mathcal{D}_j^* W_j \mathcal{D}_j, \qquad s_n = \frac{1}{n_{\mathrm{dyn}}} \sum_j \mathcal{D}_j^* W_j \xi_j,$$

where $W_j = W_{x,j} \oplus W_{y,j}$, and where ϑ_* is any represented comparator; its residual ξ_j is retained, so realizability is not presumed. Define from the observed design

$$\kappa_{\mathrm{id}} := \lambda_{\min}(Q_n), \qquad P_{\mathrm{id}} := \mathbf{1}_{(0,\infty)}(Q_n),$$

and let κ_{id}^+ be the least positive eigenvalue of Q_n, with $\kappa_{\mathrm{id}}^+ = +\infty$ when $Q_n = 0$. Every represented $\varepsilon_{\mathrm{opt}}$-minimizer compared with ϑ_* obeys

$$\|P_{\mathrm{id}}(\widehat{\vartheta} - \vartheta_*)\| \le \frac{2\|s_n\|}{\kappa_{\mathrm{id}}^+} + \sqrt{\frac{2\varepsilon_{\mathrm{opt}}}{\kappa_{\mathrm{id}}^+}}. \tag{272}$$

On the derived cell $\kappa_{\mathrm{id}} > 0$, $P_{\mathrm{id}} = I$ and this is the full-parameter bound. On the cell $\kappa_{\mathrm{id}} = 0$, $\ker Q_n$ *is exactly the unidentifiable design/gauge direction and remains an obstruction coordinate; no excitation hypothesis is imposed.*

One Part I score-concentration candidate is the conditional sub-Gaussian cell. On that recorded cell, in a p_{id}-dimensional chart, the design certificate gives, with probability at least $1 - \delta_{\mathrm{id}}$,

$$\|s_n\| \le c_{\mathrm{id}} \sigma_{\mathrm{id}} \sqrt{\frac{p_{\mathrm{id}} + \log(1/\delta_{\mathrm{id}})}{n_{\mathrm{dyn}}}}. \tag{273}$$

The constant c_{id}, the score norm, and the sampling/dependence certificate are fields of the record. Off that cell, the Part I minimum uses its mixing, martingale, or finite-description same-loss candidate; this is a recordwise radius coordinate, while the saturated profile compares records. An unavailable score radius equals $+\infty$. Componentwise application of Proposition II.11.18 converts (272) *into certified bounds for F_W, $T_{W,\lambda}$, effective dimension, capture, and every operator-Lipschitz training coordinate.*

Proof. The loss difference is a quadratic with Hessian Q_n and linear score s_n. Moreover $s_n \perp \ker Q_n$, because every vector in $\ker Q_n$ is killed by each weighted design map. Writing $h = \widehat{\vartheta} - \vartheta_*$, comparison with the represented comparator and the optimization tolerance gives $\langle h, Q_n h\rangle/2 \le$

$\|s_n\|\|P_{\mathrm{id}}h\|+\varepsilon_{\mathrm{opt}}$. The spectral inequality $Q_n \succeq \kappa_{\mathrm{id}}^{+}P_{\mathrm{id}}$ and the quadratic formula imply (272). The kernel component does not enter the loss and is therefore exactly unidentifiable. The concentration display is the Part I sub-Gaussian score candidate on its named cell. The last assertion follows from (268)– (269) and the response-coordinate Lipschitz bounds in Corollary II.11.26. □

Definition II.11.21 (Unrestricted-Caus spectral training loss)**.** For a legal bridge C, let

$$\mathcal{H}_{W,C} := \operatorname{ran} T_{W,\lambda}(C)^{1/2}, \qquad \|f\|_{W,C}^2 := \|T_{W,\lambda}(C)^{\dagger/2} f\|^2.$$

On a finite represented score space this norm is $\langle f, A_{W,\lambda}(C)f\rangle$. Given a target-training split $S_{\mathrm{tr}} = \{(X_i, Y_i)\}_{i=1}^{m_{\mathrm{tr}}}$, define the spectral kernel-ridge estimator

$$\widehat{f}_{C,\rho} := \operatorname*{arg\,min}_{f\in\mathcal{H}_{W,C}} \left\{ \frac{1}{2m_{\mathrm{tr}}} \sum_{i=1}^{m_{\mathrm{tr}}} |f(X_i) - Y_i|^2 + \frac{\rho}{2}\|f\|_{W,C}^2 \right\}. \tag{274}$$

Thus the correct spectral penalty is the closed quadratic form of the *inverse* response, equivalently $A_{W,\lambda}$ on its form domain, rather than $T_{W,\lambda}$ itself. On the ordinary Part I evaluation cell, the represented evaluation map is bounded on $\mathcal{H}_{W,C}$; the positive ridge term therefore makes (274) coercive in its response norm and gives its unique minimum. On a finite score carrier this is the positive-definite normal equation in (359). An unavailable evaluation map or nonordinary form retains its inherited value and makes the corresponding upper-risk candidate $+\infty$.

Let $\mathfrak{A}_{\mathrm{dyn}}$ be the dynamics confidence set obtained from an independent target-free split. For a designable control component it is the full declared bridge family; for a native or imposed component it is the identified confidence set. On an independent validation split define

$$\boxed{\widehat{\mathcal{J}}_{\mathrm{free}}(C,\rho) = \widehat{R}_{\mathrm{val}}(\widehat{f}_{C,\rho}) + \lambda_{\mathscr{W}}\mathscr{W}_{\mathfrak{A}}(C;D) + \mathrm{pen}_{\mathrm{sel}}(C,\rho) + \iota_{\mathfrak{A}_{\mathrm{dyn}}}(C),} \tag{275}$$

where ι is zero on the confidence set and $+\infty$ outside it. The selection penalty is a valid bridge/kernel selection deviation obtained from a split sample, finite union, PAC–Bayes mixture, or uniform cover. Operator estimation and optimization are recorded separately, so no radius is charged twice. Under unrestricted authority the zero-waste fiber

$$\mathfrak{A}_D^0 := \{C \in \mathfrak{A}_{\mathrm{dyn}} : J_C = J_D\} \tag{276}$$

is evaluated rather than presumed nonempty. It is nonempty exactly when the represented legal family realizes J_D. On that cell, exact least-action training minimizes (275) over $\mathfrak{A}_D^0$. On the empty cell, the distance (252) is the irreducible representation/authority gap. The penalized form trades statistical risk against physical action when such a trade is part of the declared task. The kernel-ridge fit and bridge search are inner candidate-generating optimizations. Their risk, action, cover charge, and optimization error enter the complete certificate; the saturated profile and its strict inverse select among the resulting structural records according to Definition II.11.29.

Theorem II.11.22 (Oracle bound for unrestricted-Caus learning)**.** *Consider the Part I split-validation candidate of a complete record. Its fields are a charged finite ϱ-net $\mathfrak{A}_\varrho$ of the declared legal-bridge and ridge family $\mathfrak{A}_{\mathrm{dyn}} \times \mathcal{P}$, of cardinality N_ϱ, the recorded loss range L_{loss}, the independent validation carrier of size m_{val}, and the simultaneous fixed-kernel oracle values*

$$\mathcal{B}_C(\rho) := C_{\mathrm{var}}\frac{\sigma^2 N_W^C(\lambda,\rho)}{m_{\mathrm{tr}}} + C_{\mathrm{bias}}B_C(\rho; f_*) + \varepsilon_{\mathrm{task}} + L_{\mathrm{op}}\varepsilon_T(C), \tag{277}$$

where N_W^C *has the extended meaning of* (257)*. Thus* $\mathcal{B}_C(\rho)$ *is a well-defined value in* $[0,\infty]$*; a divergent mode count removes this candidate from the recordwise finite oracle minimum rather than excluding the model from the framework. The bias term is*

$$B_C(\rho; f_*) = \|[I - T_C(T_C + \rho I)^{-1}]f_*\|_{L^2}^2.$$

Let

$$r_{\mathrm{val}} := L_{\mathrm{loss}}\sqrt{\frac{\log(2N_\varrho/\delta_{\mathrm{val}})}{2m_{\mathrm{val}}}}. \tag{278}$$

Take $\mathrm{pen}_{\mathrm{sel}} = r_{\mathrm{val}}$ *on this net; a candidate-dependent valid radius gives the analogous oracle bound with that radius inside the infimum. Let* $L_{\mathrm{sel}} \in [0,\infty]$ *be the Part I net-transport modulus of candidate risk plus action, and let* $\varepsilon_{\mathrm{out}} \in [0,\infty]$ *be the recorded optimization gap of the returned pair* $(\widehat{C}, \widehat{\rho})$*. On the candidate's simultaneous event,*

$$\begin{aligned} &R(\widehat{f}_{\widehat{C},\widehat{\rho}}) - R(f_*) + \lambda_{\mathscr{W}}\mathscr{W}_{\mathfrak{A}}(\widehat{C}; D) \\ &\quad\le \inf_{C\in\mathfrak{A}_{\mathrm{dyn}},\, \rho\in\mathcal{P}} \{\mathcal{B}_C(\rho) + \lambda_{\mathscr{W}}\mathscr{W}_{\mathfrak{A}}(C; D)\} + 2r_{\mathrm{val}} + L_{\mathrm{sel}}\varrho + \varepsilon_{\mathrm{out}}. \end{aligned} \tag{279}$$

An absent split, loss-range, simultaneous-oracle, or finite net-transport field sets this candidate to $+\infty$*; the inherited Rademacher and PAC–Bayes candidates remain in the Part I same-loss certificate minimum. The choice* $\mathcal{P} = (0,\infty)$ *is permitted only on a Part I record carrying a charged cover and transport modulus for that continuum; otherwise* $\mathcal{P}$ *is exactly its represented finite ridge grid. That minimum supplies the risk coordinate subsequently restricted in the saturated profile. On the cell* $\mathfrak{A}_D^0 \neq \varnothing$*, restriction to that fiber removes the action term. Within its named eigenvalue/source rate cell* $\eta_j^W(C) \asymp j^{-2\beta_C}$ *and* $f_* = T_C^{r_C} g_C$*, and whose charged ridge family contains the balancing scale or a net point with the displayed transport error, optimization of* ρ *gives the certified inner rate* $m_{\mathrm{tr}}^{-4\beta_C r_C/(4\beta_C r_C+1)}$ *at that fixed record, plus the displayed bridge-selection and operator errors. Outside that rate cell,* (279) *remains the quantitative conclusion.*

Proof. Hoeffding's inequality and a union bound give a validation deviation at most r_{val} for every net point. Insert the validation risk on both sides of the approximate-minimizer inequality. The two deviations, net-transport term, and optimization error give (279); then substitute the simultaneous candidate oracle certificates. The zero-waste statement is Theorem II.11.12. The rate follows by balancing ρ^{2r_C} with $m_{\mathrm{tr}}^{-1}\rho^{-1/(2\beta_C)}$. □

Definition II.11.23 (Action-safe Wick actor–critic loss)**.** Let $P_{\mathcal{A}}$ project onto the represented legal-action span and let P_{safe} project onto its failure-killed safe subspace. Define

$$T_C^{\mathcal{A},\mathrm{safe}} := P_{\mathrm{safe}}P_{\mathcal{A}}T_{W,\lambda}(C)P_{\mathcal{A}}P_{\mathrm{safe}}|_{\mathrm{ran}\,P_{\mathrm{safe}}}, \qquad N_C^{\mathcal{A},\mathrm{safe}}(\rho) := \mathrm{Tr}_{\mathrm{ext}}\left[T_C^{\mathcal{A},\mathrm{safe}}(T_C^{\mathcal{A},\mathrm{safe}} + \rho I)^{-1}\right]. \tag{280}$$

The trace has the extended Part I meaning of (257); every empirical critic calculation uses its finite evaluation compression from (255). The Part I control record supplies a real-valued observable-score map. Thus every critic q below ranges over its inherited real score slice inside the complexified response space. Hermitian response energies are restricted to that slice, while maxima, exponentials, rewards, and policy probabilities are real-valued. This is the existing control-score type, not an additional regularity condition on the Wick response. For an executed bridge action $a \mapsto C(h,a)$, put

$$w_D(h,a) := \frac{1}{2}\|J_{C(h,a)} - J_D(h)\|_{\mathbb{M}^{-1}}^2.$$

On the action-charged authority cell, its stage reward is $r_\alpha(h,a) = r(h,a) - \alpha w_D(h,a)$. On the hard-action-constraint cell, the legal action set carries that constraint and $\alpha = 0$; the same action is not charged a second time.

At fitted Bellman iteration k, freeze a copy q_k^- of the preceding critic. A delayed target network retains the lag $\varepsilon_{\mathrm{lag},k} := \|q_k^- - \widehat{q}_k\|_\infty$. Set

$$Y_i^{(k)} = r_\alpha(h_i, a_i) + \chi_i^{\mathrm{cont}} \gamma \max_{a' \in \mathcal{A}_{\mathrm{safe}}(h_i')} q_k^-(h_i', a'). \tag{281}$$

Here $\chi_i^{\mathrm{cont}} \in \{0,1\}$ is the terminal-continuation field of the complete Part I dynamical record. Before taking the maximum, that record either certifies a nonempty safe successor row or returns its existing authority/terminal status; no maximum over an empty set is inserted into the loss. For represented behavior-to-deployment weights $0 \le \omega_i \le \bar{\omega}$, the critic loss is

$$\mathcal{L}_{\mathrm{crit}}^{(k)}(q; C) := \frac{1}{2m_k} \sum_{i=1}^{m_k} \omega_i (q(h_i, a_i) - Y_i^{(k)})^2 + \frac{\rho}{2} \|(T_C^{\mathcal{A},\mathrm{safe}})^{\dagger/2} q\|^2. \tag{282}$$

Thus critic regularization again uses the inverse positive response. The frozen target makes (282) a supervised regression loss and avoids the double-sampling defect of a same-sample squared Bellman residual.

For the neutral action row $\nu(a \mid h)$ carried by the Part I control record, define on its represented support

$$\begin{aligned} \pi_q^\tau(a \mid h) &= \frac{\nu(a \mid h) e^{q(h,a)/\tau}}{\sum_{a' \in \mathcal{A}_{\mathrm{safe}}(h)} \nu(a' \mid h) e^{q(h,a')/\tau}}, \quad a \in \mathcal{A}_{\mathrm{safe}}(h), \\ \mathcal{L}_{\mathrm{actor}}(\pi; q) &= \mathbb{E}_{d_{\mathrm{dep}}} D_{\mathrm{KL}}(\pi(\cdot \mid h) \| \pi_q^\tau(\cdot \mid h)). \end{aligned} \tag{283}$$

The policy is zero outside $\mathcal{A}_{\mathrm{safe}}(h)$. The displayed denominator is positive on the ordinary Part I support cell. If the neutral row has zero total safe mass, the Gibbs row and actor certificate take their typed support-obstruction values. A zero mass on any required safe action is retained by the support coordinate $\nu_{\min}$ in Theorem II.11.24. The canonical alternating objective is

$$\boxed{\mathcal{L}_{W,\mathrm{RL}}^{(k)}(q, \pi, C) = \mathcal{L}_{\mathrm{crit}}^{(k)}(q; C) + \lambda_\pi \mathcal{L}_{\mathrm{actor}}(\pi; q) + \lambda_{\mathrm{dyn}} \mathcal{L}_{\mathrm{dyn}}(\vartheta) + \lambda_{\mathrm{num}} \mathcal{L}_{\mathrm{num}}(q, \pi, C).} \tag{284}$$

The critic is fitted first with q_k^- fixed; the actor is then fitted to the new critic. Importance, occupancy, filter, numerical, bridge-selection, and target-network costs belong to the complete RL record. The critic, actor, and bridge updates are inner optimization transcripts. Their completed record enters the saturated profile, which supplies the structural order across policies, responses, and authority cells.

Theorem II.11.24 (RL critic and policy calibration)**.** *Work on the inherited discounted or contractive Part I control cell, with recorded modulus $0 \le \gamma < 1$. A finite-horizon or noncontractive control record retains its own Part I dynamic-programming candidate and does not use the denominators below. For each fitted critic step, let $e_k^2 \in [0, \infty]$ be the recordwise minimum of the applicable simultaneous Part I critic-oracle candidates. It is the critic-radius coordinate of that fixed RL record; structural comparison uses the saturated profile. Its Wick effective-dimension candidate is*

$$\begin{aligned} \mathcal{E}_{k,\mathrm{KRR}} := {} & \frac{C_{\mathrm{var}} \sigma_{\mathrm{TD},k}^2 N_{C_k}^{\mathcal{A},\mathrm{safe}}(\rho_k)}{m_k} + C_{\mathrm{bias}} B_{C_k}^{\mathcal{A},\mathrm{safe}}(\rho_k) \\ & + \varepsilon_{\mathrm{shift},k} + \varepsilon_{\mathrm{model},k} + \varepsilon_{\mathrm{sel},k} + \varepsilon_{\mathrm{opt},k}, \qquad e_k^2 \le \mathcal{E}_{k,\mathrm{KRR}}. \end{aligned} \tag{285}$$

where the safe source residual includes $\|(I-P_{\mathrm{safe}})Q_*\|^2$. An unavailable candidate has value $+\infty$, so this definition excludes no record.

Let $\mathcal{K}_{\infty,k}$ be the set of same-law conversion constants certified by the Part I record and define the totalized conversion error

$$\varepsilon_{\infty,k} := \inf_{\kappa\in\mathcal{K}_{\infty,k}} \kappa e_k \in [0,\infty], \qquad \inf\varnothing := +\infty. \tag{286}$$

Every ordinary member of $\mathcal{K}_{\infty,k}$ certifies $\|\widehat{q}_{k+1}-\mathcal{T}_{B,\alpha}q_k^-\|_\infty \le \kappa e_k$. In the extended nonnegative reals, the restricted regularized Bellman target therefore satisfies

$$\Gamma_{Q,K} := \|\widehat{q}_K - Q^*_{B,\alpha}\|_\infty \le \gamma^K\Gamma_{Q,0} + \sum_{k=0}^{K-1}\gamma^{K-1-k}(\varepsilon_{\infty,k}+\gamma\varepsilon_{\mathrm{lag},k}). \tag{287}$$

Define two further derived record coordinates

$$\nu_{\min} := \inf_{h,a\in\mathcal{A}_{\mathrm{safe}}(h)} \nu(a\mid h), \qquad \kappa_\pi := \sup_h D_{\mathrm{KL}}(\widehat{\pi}(\cdot\mid h)\|\pi^\tau_{\widehat{q}_K}(\cdot\mid h)) \in [0,\infty],$$

and use $\log(1/0) := +\infty$. The actor's extended one-step action certificate is

$$d_{\mathrm{act}} := 2\Gamma_{Q,K} + \tau\log\frac{1}{\nu_{\min}} + \mathrm{Osc}(\widehat{q}_K)\sqrt{\frac{\kappa_\pi}{2}}, \tag{288}$$

and

$$\|V^*_{B,\alpha} - V^{\widehat{\pi}}_\alpha\|_\infty \le \frac{d_{\mathrm{act}}}{1-\gamma}. \tag{289}$$

These displays are finite exactly on the Part I same-law/support/actor cells that make all their coordinates finite. The value for the unrestricted control problem additionally includes the represented budget, safe-action, and observation/partial-information gaps. When the same-law conversion set is empty, $\varepsilon_{\infty,k} = +\infty$: the $L^2(d_{\mathrm{dep}})$ critic certificate remains valid, whereas the sup-norm Bellman and policy certificates take the honest value $+\infty$.

Proof. The fixed-target critic is kernel ridge regression with kernel $T_C^{\mathcal{A},\mathrm{safe}}$, so its effective-dimension oracle certificate is (285). Apply the converter, insert the target-network lag, and iterate the γ-contraction to obtain (287). The Gibbs variational identity bounds its soft-greedy defect by $\tau\log(1/\nu_{\min})$; Pinsker's inequality gives the final actor term in (288), and replacement of the true critic by $\widehat{q}_K$ costs $2\Gamma_{Q,K}$. The discounted performance-difference identity sums the uniform one-step defect and gives (289). ☐

Corollary II.11.25 (Action-safe RL rate cell)**.** *On the named Part I rate cell in which the nonzero eigenvalues of $T_C^{\mathcal{A},\mathrm{safe}}$ obey $\eta_j \asymp j^{-2\beta}$, $\beta > 1/2$, the Bellman-optimal critic of that fixed RL record has source order $0 < r \le 1$, the safe approximation residual is zero, the charged ridge family contains the balancing scale or a net point with no larger transport error, and the shift, model, selection, and optimization coordinates have no larger order, the general oracle value specializes to*

$$e_k^2 = O\Big(m_k^{-4\beta r/(4\beta r+1)}\Big), \qquad e_k = O\Big(m_k^{-2\beta r/(4\beta r+1)}\Big). \tag{290}$$

On the subcell carrying uniformly bounded conversion constants $\kappa_{\infty,k}\in\mathcal{K}_{\infty,k}$, the fitted-iteration residual has converged, $\tau\to 0$, and the derived $\kappa_\pi \to 0$, the policy gap has order

$$O\Bigg(\frac{m^{-2\beta r/(4\beta r+1)}}{(1-\gamma)^2}\Bigg), \tag{291}$$

in addition to the exact budget, safe-action, and observation approximation gaps. Every complementary cell retains the extended oracle (287)–(289)*; it is not excluded. A direct sup-norm critic oracle may replace the square-root conversion and yields its own certified rate. Bellman optimality here is an inner control coordinate of the fixed RL record; the risk-qualified saturated profile supplies the structural optimum across RL records.*

Corollary II.11.26 (Specialized quantitative ledger)**.** *The Part I confidence allocator constructs one simultaneous event for every ordinary candidate below. On that event the following bounds specialize to a Wick certificate as indicated; every nonordinary candidate is assigned $+\infty$ and remains in the record.*

(i) *On each nonempty represented Part I stage, let $\{\mathcal{H}_\alpha^W\}_{\alpha\in\mathfrak{A}_W^{\mathrm{src}}}$ be its exact source cells. For the recorded $[-1,1]$-loss candidate, source selection obeys*

$$\widehat{\mathfrak{R}}_S\left(\bigcup_\alpha \mathcal{H}_\alpha^W\right) \le \max_\alpha \widehat{\mathfrak{R}}_S(\mathcal{H}_\alpha^W) + \sqrt{\frac{2\log|\mathfrak{A}_W^{\mathrm{src}}|}{m}}. \tag{292}$$

For a cell-mixture prior and posterior,

$$\mathrm{KL}(Q\|P) = \mathrm{KL}(q\|w) + \sum_\alpha q_\alpha \,\mathrm{KL}(Q_\alpha\|P_\alpha). \tag{293}$$

Thus the selection of a Wick source, bridge family, or authority cell is charged once by a union, mixture, split-sample, or uniform-cover certificate.

(ii) *For every fixed bridge and ridge scale, the complete kernel-ridge oracle candidate is*

$$\|\widehat{f}_{C,\rho} - f_*\|_{L^2}^2 \le C_{\mathrm{var}}\frac{\sigma^2 N_W^C(\lambda,\rho)}{m} + C_{\mathrm{bias}}B_C(\rho; f_*). \tag{294}$$

The source rate (267)*, a minimax lower rate, or a condition-number lower capacity bound transfers only on its declared eigenvalue/source class. In particular, on an m-point score space,*

$$\mathrm{TaskCap}_{A_W}(y) \ge \frac{\|y\|_2}{\sqrt{m}\,\kappa(A_W)}. \tag{295}$$

Dark, target-null, and authority-excluded components remain exact approximation floors rather than statistical terms.

(iii) *On the inherited moving-bridge/feature cell, a terminal kernel $T_{\widehat{\theta}}$ selected from its charged ϱ-net has feature-learning certificate*

$$\begin{aligned}\mathrm{ExcessRisk} \le{}& C_{\mathrm{var}}\frac{\sigma^2 N_W^{\widehat{\theta}}(\lambda,\rho)}{m} + C_{\mathrm{bias}}B_{\widehat{\theta}}(\rho; f_*)\\ &+ r_{\mathrm{sel}} + L_{\mathrm{sel}}\varrho + \varepsilon_{\mathrm{est}} + \varepsilon_{\mathrm{opt}},\end{aligned} \tag{296}$$

where, for an independent bounded-loss audit of size m_a, one may take

$$r_{\mathrm{sel}} = L_{\mathrm{loss}}\sqrt{\frac{\log(2\mathcal{N}(\Theta,\varrho)/\delta)}{2m_a}}.$$

The fixed-response/lazy regime is the specialization with no kernel movement; a neural or nonlinear readout adds its own represented margin or covering certificate and is combined only in common risk units.

(iv) *A learned family* $\{T_{W,\lambda}(\theta)\}$ *uses* (269) *in the learned-operator covering bound. Since* $A_{W,\lambda} \geq I$*, its selection radius is*

$$\inf_{\varrho>0}\left[\frac{R}{m}\sqrt{\sup_{\theta}\operatorname{Tr}_S T_\theta}+R\sqrt{\frac{L_T\varrho}{m}}+\frac{R}{m}\sqrt{2\log\mathcal{N}(\Theta,\varrho)}\right]. \tag{297}$$

Here ϱ *is the charged cover resolution, not the ridge scale* ρ*.*

(v) *On the represented positive defect/bracket range* $R_{E,W}$*, the proper Gaussian structural prior has precision* $\mathbf{M}_{E,W} = P_{E,W}A_{W,\lambda}P_{E,W}|_{R_{E,W}}$*. Its PAC–Bayes radius is*

$$\sqrt{\frac{\mathrm{KL}^{\mathrm{str}}_{E,W}(Q)+\log(2\sqrt{m}/\delta)}{2(m-1)}}. \tag{298}$$

This candidate is ordinary for $m \geq 2$*; at* $m < 2$ *it equals* $+\infty$*. Null and dark directions retained by the model receive a separate proper label-independent prior and KL charge.*

(vi) *The Part I resistance-query candidate uses* $\langle E_{W,\Delta}, Q^{+}_{K,S}E_{W,\Delta}\rangle$*. Rayleigh monotonicity gives monotonicity under grounding. The existing submodularity coordinate either certifies the* $1-1/e$ *greedy batch factor or assigns that factor-candidate* $+\infty$*; the resistance query itself remains available.*

(vii) *Put* $\varepsilon_T := \|\widehat{T}-T\|$*, let* $g_* \geq 0$ *be the derived target-band separation, and define*

$$d_{\mathrm{DK}} := \begin{cases} 2\varepsilon_T/g_*, & g_* > 2\varepsilon_T, \\ 1, & g_* \leq 2\varepsilon_T. \end{cases}$$

Davis–Kahan gives the first value, while the universal distance bound for orthogonal projectors gives the second [3]. Thus the projector discrepancy is always bounded by d_{DK}*, without a spectral-gap assumption. Smoothed bands use the recordwise minimum of this value and their represented operator-Lipschitz value. The result propagates to capture, effective dimension, action-safe capacity, and every declared Lipschitz certificate coordinate.*

(viii) *On the homogeneous symmetric label-noise cell* $\eta < 1/2$*, every clean-risk radius for a flip-symmetric bounded loss is divided by* $\chi_\eta := 1-2\eta$*. For squared loss the debiased Wick estimator satisfies*

$$\|\widehat{f}_{C,\rho}-f_*\|^2_{L^2} \leq C_{\mathrm{var}}\frac{v_\eta^2 N_W^C(\lambda,\rho)}{\chi_\eta^2 m}+C_{\mathrm{bias}}B_C(\rho;f_*). \tag{299}$$

Wick source ancestry and operator-selection costs are unchanged. General reward, transition, action, and observation corruption enters through its typed same-norm channel converter, not through the binary factor.

(ix) *On the Part I stationary* β*-mixing cell of length* N*, block length* a*, and* $\mu_a = \lfloor N/(2a)\rfloor$*, put* $\delta_a = \delta - 2(\mu_a-1)\beta(a)$*. For every block length with* $\mu_a \geq 1$ *and* $\delta_a > 0$*, the represented block response kernel* $T_{W,a}$ *gives*

$$|R(f)-\widehat{R}_N(f)| \leq \frac{2R_A}{\mu_a}\sqrt{\operatorname{Tr}_{\mathrm{blk(a)}} T_{W,a}}+3\sqrt{\frac{\log(4/\delta_a)}{2\mu_a}}+\frac{2a}{N}. \tag{300}$$

Here $\operatorname{Tr}_{\mathrm{blk(a)}}$ *is the finite compression to the* μ_a*-block evaluation carrier. Block lengths with* $\mu_a = 0$ *or* $\delta_a \leq 0$ *have value* $+\infty$*, so the recordwise minimum over all represented* *block lengths is a valid bound coordinate. For a positive represented horizon T, the Part I predictable-path cell uses*

$$\mathfrak{R}_T^{\mathrm{seq}} \le R_A \mathfrak{q}_T(A_W) \le \frac{R_A}{T}\sqrt{\mathrm{Tr}_{\mathrm{seq}}\, T_{W,\lambda}} \tag{301}$$

where $\mathrm{Tr}_{\mathrm{seq}}$ is the Part I supremum of the finite T-step path/tree compression traces. For a fixed response this gives the displayed bound; a selectable response uses the learned-operator sequential cover, and the Bayesian branch uses the martingale PAC–Bayes term

$$\frac{\mathrm{KL}^{\mathrm{str}}_{E,W}(Q) + \log(1/\delta)}{\xi T} + \frac{\xi}{8}. \tag{302}$$

for its represented $\xi > 0$; other coefficient values are nonordinary. The online regret bound is $2L_\ell T \sup_{\mathbf{z}} \mathfrak{R}_T^{\mathrm{seq}}$. Each complementary dependence cell uses its own Part I candidate or the value $+\infty$; a dark space alone supplies no dependence coordinate.

(x) *Define the acquisition coordinate $\Gamma_{\mathrm{acq}} := \sup_{h,a} |\widehat{g} - g| \in [0,\infty]$. The recorded $\varepsilon_{\mathrm{opt}}$-maximizer loses at most*

$$\sup_a g(h,a) - g(h, \widehat{a}(h)) \le 2\Gamma_{\mathrm{acq}} + \varepsilon_{\mathrm{opt}}. \tag{303}$$

This maximization is the operational action choice within a fixed acquisition record. Its error and action cost enter that record, after which the saturated profile compares complete acquisition structures. Over a horizon H, the action-value loss is the survival-weighted sum of these stepwise terms. Resistance-greedy selection obtains the $1 - 1/e$ batch factor on the positive submodularity-certificate cell in the earlier querying item; on its complement that factor-candidate is $+\infty$.

(xi) *For RL authority, compress $T_{W,\lambda}$ first to the represented action span and then to the safe subspace. Effective-dimension monotonicity is unconditional. On the named Part I safe source-order cell, the exact source residual has the displayed upper bound:*

$$N_W^{\mathcal{A},\mathrm{safe}}(\rho) \le N_W^{\mathcal{A}}(\rho) \le N_W(\lambda,\rho), \tag{304}$$

$$B_W^{\mathcal{A},\mathrm{safe}}(\rho; f_*) \le C_r \|g_{\mathrm{safe}}\|^2 \rho^{2r} + \|(I - P_{\mathrm{safe}}) f_*\|^2. \tag{305}$$

The critic and actor objectives are (282) and (283). Define the extended coordinates ε_M as the same-norm model defect and ε_B as the learned Bellman residual. On the Part I discounted Bellman/contraction cell, the Bellman candidate is

$$\|V^* - v\|_\infty \le \frac{\varepsilon_B + \varepsilon_M}{1-\gamma}, \tag{306}$$

$$\|V^* - V^{\widehat{\pi}_v}\|_\infty \le \frac{2\gamma(\varepsilon_B + \varepsilon_M)}{(1-\gamma)^2} + \frac{2\varepsilon_M}{1-\gamma}. \tag{307}$$

Here $0 < \gamma < 1$ is the declared discount or contraction modulus. The distance from J_D to the legal policy-current family is the irreducible authority/action gap; it cannot be removed by additional value samples. The complementary control-status cell assigns these Bellman candidates $+\infty$ and retains its typed control defect.

(xii) *For a complete target-qualified Wick record, source-resolved statistical, information, structural, and dynamical candidates combine only after conversion to the same deployed target advantage g_W:*

$$g_W \le \min\left\{1,\ U_W^{\mathrm{stat}\to\mathrm{tgt}},\ \beta_W^+ + \sqrt{\frac{\log 2}{2} J_W^{\mathrm{src}}},\ eH_B \sum_a \Lambda_W^{(a)},\ U_W^{\mathrm{dyn}\to\mathrm{tgt}}\right\}. \tag{308}$$

Here $\Lambda_W^{(a)}$ is the contextual charge assigned to source a; it is distinct from the generator $\mathcal{L}_W$. For a finite sample-independent output codebook of logarithmic size L_{out}, the fallback deviation candidate is

$$2\sqrt{\frac{2L_{\mathrm{out}}}{m_{\mathrm{eff}}}}+3\sqrt{\frac{\log(2/\delta_{\mathrm{eff}})}{2m_{\mathrm{eff}}}}+\varepsilon_{\mathrm{dep}}. \tag{309}$$

(xiii) *Galerkin or finite-element records use* (268) *and* (269) *after their form-to-resolvent error has been certified. Every finite Galerkin carrier uses its finite Part I trace. Along a represented nested dense family, the population effective dimension is the monotone extended value in* (259)*. Operator approximation errors on the d_n-dimensional carrier are charged by* (311)*. A represented tail estimate may supply a finite upper envelope; without one, the upper value is $+\infty$. No compactness or trace-class premise is inserted.*

(xiv) *On every finite empirical band, define $\varepsilon_H := \|\widehat{H}-H\|$, $\varepsilon_C := \|\widehat{C}-C\|$, its derived separation $g_\Delta \geq 0$, and*

$$d_\Delta := \begin{cases} 2\varepsilon_H/g_\Delta, & g_\Delta > 2\varepsilon_H, \\ 1, & g_\Delta \leq 2\varepsilon_H. \end{cases}$$

With $c=\max\{\|C\|,\|\widehat{C}\|\}$,

$$\big|\|CP_\Delta\| - \|\widehat{C}\widehat{P}_\Delta\|\big| \leq \varepsilon_C + c d_\Delta. \tag{310}$$

Thus zero is excluded from this interval precisely when the data certify a nonzero quasi-lumpability defect. On every common certified reduced subspace, define $\varepsilon_Q := \|\widehat{Q}_K - Q_K\|$; Weyl's inequality bounds the reduced-gap discrepancy by ε_Q. In quotient dimension d_Q, putting $\varepsilon_T = \|\widehat{T}-T\|$ gives

$$|\widehat{N}_W(\lambda,\rho) - N_W(\lambda,\rho)| \leq \frac{d_Q}{\rho}\varepsilon_T, \tag{311}$$

$$|\widehat{m}_W(\lambda,\rho;f) - m_W(\lambda,\rho;f)| \leq \frac{2}{\rho}\varepsilon_T \tag{312}$$

for fixed normalized f. Data-dependent readouts add their separately certified sample-splitting or uniform-selection radius.

Each item retains the approximation floor, target-readout miss, source-cell selection cost, model-channel perturbations, confidence allocation, and deployment conversion of its Part I antecedent. An absent antecedent is the nonordinary value $+\infty$, never an omitted record or hidden assumption.

II.11.6 Part I model choice and the saturated certified frontier

No new simplicity functional is introduced. The model family, selection penalties, and comparison order are inherited from Part I. Source choice is charged by its exact-cell Rademacher union term or by the cell-mixture PAC–Bayes divergence. A label-dependent operator choice is charged by the learned-operator cover, and coefficient selection is charged by the declared finite coefficient family. At fixed geometry, readout size is chosen by the Part I target-subspace coverage criterion. Across complete records, simplicity is the Pareto order of the existing saturated resource and action vector $\mathbf{c}$, with optimality interpreted throughout by Definition II.11.29.

Part I quantitative family	Wick specialization	Record field / nonordinary output
Identification and operator stability	(272), (268), (269)	quotient gauge; derived design range/kernel; extended same-norm error
Fixed supervised learning	(265), (294), (267)	positive response is derived; named source/eigenvalue cell or general oracle
Bridge or feature selection	(279), (296), (297)	inherited split/cover candidate; absent candidate equals $+\infty$
Bayesian and noisy learning	(298), (299), (293)	inherited prior/channel cell; absent candidate equals $+\infty$
Dependent and online learning	(300), (301), (302)	inherited dependence cell; other cells use their own candidate
Active learning	resistance querying and (303)	extended resistance; submodularity coordinate controls only batch factor
Controlled/RL learning	(285)– (289), (306)–(307)	safe/action and converter coordinates, each totalized in $[0, \infty]$
Saturated-profile working frontier	(314), (331), (334)	risk-restricted saturated profile and strict Pareto inverse; inner minima remain record coordinates; empty feasible set is typed
Deployment and continuum	(308), empirical intervals, resolvent/trace transfer	extended converter, allocated simultaneous event, finite-carrier or extended-trace accounting

Table 3: Exhaustive location of the quantitative learning families used by the Wick specialization. Every row is evaluated on one complete selected generator. A dependent upper-error or risk field returns $+\infty$, a dependent lower-usefulness field returns zero, and every other unavailable coordinate retains its typed nonordinary value; none is replaced by an omitted theorem or numerical surrogate.

Proposition II.11.27 (Inner minimum-response-energy coordinate)**.** *Let $R_C : \mathcal{Z}_C \to \mathcal{Y}_C$ be a bounded operator between Hilbert spaces and let $T_C = R_C R_C^*$. Define the totalized response energy*

$$\mathcal{E}_{T_C}(f) := \begin{cases} \|T_C^{\dagger/2} f\|^2, & f \in \operatorname{ran} T_C^{1/2}, \\ +\infty, & f \notin \operatorname{ran} T_C^{1/2}. \end{cases} \tag{313}$$

Then $\operatorname{ran} T_C^{1/2} = \operatorname{ran} R_C$ and, for every $f \in \mathcal{Y}_C$,

$$\mathcal{E}_{T_C}(f) = \inf_{z \in \mathcal{Z}_C \colon R_C z = f} \|z\|_{\mathcal{Z}_C}^2, \qquad \inf \varnothing := +\infty. \tag{314}$$

For $f \in \operatorname{ran} R_C$, the infimum is attained uniquely in $(\ker R_C)^\perp$, at

$$z_f = R_C^\dagger f. \tag{315}$$

In particular, take $R_C = T_{W,\lambda}(C)^{1/2}$ on its represented support. Then the inverse-response penalty in (274) *is exactly the least squared response effort among all latent lifts producing f. The same statement holds with $T_C^{\mathcal{A},\mathrm{safe}}$ for the RL critic. Nonclosed continuum ranges are included: unattainable targets receive $+\infty$, while every attainable target has the minimum lift above. This minimum is the response-energy coordinate of a fixed complete record. It enters the resource/action ledger used by the saturated profile and does not compare different response kernels or source cells.*

Proof. The polar decomposition $R_C = T_C^{1/2} V_C$ gives the operator-range identity $\operatorname{ran} R_C = \operatorname{ran} T_C^{1/2}$. For an attainable f, the affine fiber $R_C^{-1}\{f\}$ is a nonempty closed set of the form $z_f + \ker R_C$. Its unique element perpendicular to $\ker R_C$ is the Moore–Penrose solution $z_f = R_C^\dagger f$. Functional calculus and the polar decomposition give $\|z_f\| = \|T_C^{\dagger/2} f\|$. Pythagoras then yields $\|z_f + k\|^2 = \|z_f\|^2 + \|k\|^2$ for every $k \in \ker R_C$. For an unattainable f, both sides of (314) equal $+\infty$ by definition. □

Definition II.11.28 (Inherited Wick family-selection certificate)**.** Fix a resource–action cell (B, a). Let $\mathfrak{A}_W$ be the exact Part I source cells represented in that cell. For $\alpha \in \mathfrak{A}_W$, let Θ_α be its declared finite-coverable Wick generator/bridge family, let Λ be the declared finite set of training-coefficient vectors, let $\mathfrak{K}$ be the finite capacity grid used in Part I, and let $\mathcal{H}^W_{\alpha,\theta,\lambda,K}$ be the resulting readout or policy-loss class at capacity K. Let $\mathfrak{I}_W \subseteq \mathfrak{A}_W \times \Lambda \times \mathfrak{K}$ be the represented family indices. Put

$$\mathcal{H}_W(B, a) := \bigcup_{(\alpha,\lambda,K) \in \mathfrak{I}_W} \bigcup_{\theta \in \Theta_\alpha} \mathcal{H}^W_{\alpha,\theta,\lambda,K}. \tag{316}$$

When $\mathfrak{I}_W = \varnothing$, the union is empty, every upper certificate below is defined to be $+\infty$, and the learner returns the typed source/authority/budget obstruction. The remaining formulas are evaluated on each nonempty finite Part I stage; this is a case distinction, not a restriction on the underlying family. Every union index, net resolution, terminal kernel, coefficient vector, training transcript, and deployment map is part of the complete record. Equations (318) and (320) are respectively the simultaneous specializations of Eqs. (292) and (297) and Eqs. (293) and (298).

For the Rademacher branch, define the within-family learned-response radius

$$\mathfrak{r}^W_{\alpha,\lambda,K} := \inf_{\varrho > 0} \left[\frac{R}{m} \sqrt{\sup_{\theta \in \Theta_\alpha} \operatorname{Tr}_S T_\theta} + R \sqrt{\frac{L_{T,\alpha} \varrho}{m}} + \frac{R}{m} \sqrt{2 \log \mathcal{N}(\Theta_\alpha, \varrho)} \right] \tag{317}$$

The supremum and covering number are extended values; no attainment or compactness is used. The complete exact-family union radius is

$$\mathfrak{r}_{\mathrm{fam}}^{W} := \max_{(\alpha,\lambda,K)\in\mathfrak{I}_W} \mathfrak{r}_{\alpha,\lambda,K}^{W} + \sqrt{\frac{2\log|\mathfrak{I}_W|}{m}}. \tag{318}$$

Thus, for a $[0,1]$-valued L_ℓ-Lipschitz loss, the inherited Rademacher upper certificate is

$$U_{\mathrm{Rad}}^{W}(h) := \widehat{R}_S(h) + 2L_\ell \mathfrak{r}_{\mathrm{fam}}^{W} + 3\sqrt{\frac{\log(2/\delta_{\mathrm{Rad}})}{2m}} + \varepsilon_{\mathrm{task}} + \varepsilon_{\mathrm{op}} + \varepsilon_{\mathrm{dep}}. \tag{319}$$

Here $L_{T,\alpha}$ is supplied by Proposition II.11.18; a fixed operator uses the fixed-response radius in (266).

For the Bayesian branch, let $\mathfrak{I}_{W,\mathrm{PB}}$ be the finite hierarchical index of exact source cell, represented operator-net point, and capacity. Use exactly the Part I mixture prior $P = \sum_{\iota\in\mathfrak{I}_{W,\mathrm{PB}}} w_\iota P_\iota$, fixed independently of the labeled sample, and posterior $Q_\lambda = \sum_\iota q_{\lambda,\iota} Q_{\lambda,\iota}$. Put

$$\mathrm{KL}_W(Q_\lambda\|P) := \mathrm{KL}(q_\lambda\|w) + \sum_{\iota\in\mathfrak{I}_{W,\mathrm{PB}}} q_{\lambda,\iota}\mathrm{KL}(Q_{\lambda,\iota}\|P_\iota), \tag{320}$$

$$U_{\mathrm{PB}}^{W}(Q_\lambda) := \widehat{R}_S(Q_\lambda) + \sqrt{\frac{\mathrm{KL}_W(Q_\lambda\|P) + \log(2\sqrt{m}|\Lambda|/\delta_{\mathrm{PB}})}{2(m-1)}} + \varepsilon_{\mathrm{task}} + \varepsilon_{\mathrm{op}} + \varepsilon_{\mathrm{dep}}. \tag{321}$$

The PAC–Bayes expression is an ordinary candidate for $m \geq 2$. At $m < 2$, or when its prior/posterior fields are unavailable, it has value $+\infty$ and does not affect the recordwise same-loss minimum. A label-dependent θ-selection is either an index in the fixed hierarchical prior or is paid by the learned-operator cover; it is not treated as a label-independent geometry. A split-validation certificate such as (279) is an alternative candidate for the same selection step; its radius is not added again to a union or mixture candidate. Let $U_W(M)$ be the minimum of (319), (321), and the other applicable candidates in Corollary II.11.26, after all have been converted to the same population loss on one confidence event. An unavailable candidate has value $+\infty$. Let $\Gamma_W(M) \in [0,\infty]$ denote the smallest applicable two-sided Part I radius on that same event, including the identical family index, cover, channel, operator, and deployment charges. Thus $U_W(M) \leq \mathcal{R}(M) + 2\Gamma_W(M)$ in the extended reals; absence of a two-sided comparator certificate is exactly $\Gamma_W(M) = +\infty$. Both minima are recordwise statistical certificate coordinates. They implement the same-loss Rademacher/PAC–Bayes choice of Part I; the saturated profile below, rather than either scalar radius, orders complete records.

Definition II.11.29 (Saturated-profile semantics of Wick optimality)**.** The terms *optimal*, *minimal*, and *simplest* retain the Part I hierarchy. An optimization performed with a complete record fixed—including least-action projection, minimum response lift, ridge fitting, critic fitting, or minimum full-coverage readout—is an *inner optimization.* Its value, optimizer transcript, and optimization error are coordinates of that record. Only the saturated profile orders two different structural records.

Structural optimization is defined only by the inherited saturated profile, instantiated for Wick records in (335), and its Pareto-ordered strict inverse (336). Let ℓ denote a fixed Part I population-law, estimand, norm, and loss-scale identifier after every declared converter has been applied. Let $\mathscr{R}_{W,\ell}^{(n)}(B,a)$ denote that typed partition of the charged finite stage produced by the Part I learner, and suppress ℓ from the notation below by writing $\mathscr{R}_{W}^{(n)}(B,a)$. Records with different identifiers define separate profiles and are never compared through a statistical minimum. Let $\underline{V}_W(M)$ be the simultaneous Part I lower confidence bound for V_M^W. At requested population-risk accuracy ϵ, the computable saturated profile and strict inverse within the fixed typed partition are

$$\widehat{G}^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) := \sup_{\substack{M\in\mathscr{R}_W^{(n)}(B,a)\\ U_W(M)\le\epsilon}} \underline{V}_W(M), \tag{322}$$

$$\widehat{\mathfrak{P}}^{>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta) := \{(B,a) : \widehat{G}^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) > \theta\}. \tag{323}$$

Thus structural optimality at fixed resources is the saturated supremum of complete visibility. A record is called an optimizer only on a cell where it attains that supremum; otherwise the Part I search transcript retains its ε-maximizers and nonattainment status. A simplest working record is a Pareto-minimal point of the appropriate strict inverse. The population risk-qualified profile is (337); Proposition II.11.34 proves that it is a nested restriction of the unrestricted Part I profile. Every earlier or later inner minimizer is used only to evaluate the visibility, risk, action, or resource coordinates that enter these profiles.

Proposition II.11.30 (Part I fixed-response capacity coordinate)**.** *Fix one Wick response and let P_W be the orthogonal projector onto its represented response support. For the declared target-score subspace $\mathcal{Y}_{\mathrm{tgt}}$, put*

$$\mathcal{V}_W^* := \overline{\operatorname{ran}(P_W P_{\mathcal{Y}_{\mathrm{tgt}}})}, \qquad d_W^* := \dim \mathcal{V}_W^* \in \mathbb{N}\cup\{\infty\}. \tag{324}$$

For a reachable readout subspace $\mathcal{R}_K$ with $\dim\mathcal{R}_K \le K$*, define*

$$\mathcal{C}_W(y) := \frac{\|P_{\mathcal{V}_W^*} y\|^2}{\|y\|^2}, \qquad \mathcal{E}_W(\mathcal{R}_K; y) := \frac{\|P_{\mathcal{R}_K} P_{\mathcal{V}_W^*} y\|^2}{\|y\|^2}.$$

As in Part I, both ratios are defined to be zero at $y = 0$. Then

$$\mathcal{C}_W(y) - \mathcal{E}_W(\mathcal{R}_K; y) = \frac{\|(I - P_{\mathcal{R}_K}) P_{\mathcal{V}_W^*} y\|^2}{\|y\|^2}. \tag{325}$$

Full structural coverage is equivalent to

$$P_{\mathcal{R}_K} P_{\mathcal{V}_W^*} = P_{\mathcal{V}_W^*}. \tag{326}$$

It forces $K \ge d_W^*$. On the finite-dimensional cell, any represented family containing $\mathcal{V}_W^*$ attains it at $K = d_W^*$; on the cell $d_W^* = \infty$, no finite-capacity readout has full coverage. Hence d_W^* is the minimum readout capacity that covers the complete target-active response subspace, exactly as in the Part I model-family criterion. It is an inner minimum at fixed response. Across responses and source cells, capacity is a component of $\mathbf{c}$*, and selection is governed by Definition II.11.29.*

Proof. The orthogonal decomposition $v = P_{\mathcal{R}_K} v + (I - P_{\mathcal{R}_K})v$, with $v = P_{\mathcal{V}_W^*} y$, proves (325). Identity (326) holds precisely when $\mathcal{V}_W^* \subseteq \mathcal{R}_K$, which implies $d_W^* \le \dim\mathcal{R}_K \le K$. On the finite-dimensional cell, taking $\mathcal{R}_{d_W^*} = \mathcal{V}_W^*$ proves attainability on exactly the represented reachable-family cell containing that subspace. Infinite d_W^* rules out finite K directly. □

Theorem II.11.31 (Part I oracle selection and the simplest certified Wick record)**.** *Use the Part I confidence allocation $\delta_{\mathrm{Rad}} + \delta_{\mathrm{PB}} + \delta_{\mathrm{other}} \le \delta$ in Definition II.11.28. At represented stage n, let $\mathscr{R}_W^{(n)}(B,a)$ be the charged finite family of complete Wick records produced by Part I, and let $\mathcal{R}(M)$*

denote their common population loss. For a requested risk accuracy ϵ and the inherited visibility threshold θ, define

$$\mathscr{F}_W(\epsilon,\theta;B,a) := \{M \in \mathscr{R}_W^{(n)}(B,a) : U_W(M) \le \epsilon,\ \underline{V}_W(M) > \theta\}. \tag{327}$$

The selected Pareto set is

$$\widehat{\mathscr{M}}_W(\epsilon,\theta;B,a) := \mathrm{Min}_{\preceq_{\mathbf{c}}} \mathscr{F}_W(\epsilon,\theta;B,a). \tag{328}$$

Here $\mathrm{Min}_{\preceq_{\mathbf{c}}}$ denotes all Pareto-minimal records in the complete resource–action order already used by the saturated profile. These recordwise frontiers are attached to feasible grid points of the certified risk-qualified strict inverse (323)*; its full represented grid boundary is identified in* (v)*. Then, with probability at least $1-\delta$:*

(i) *every $\widehat{M} \in \widehat{\mathscr{M}}_W(\epsilon,\theta;B,a)$ satisfies*

$$\mathcal{R}(\widehat{M}) \le \epsilon, \qquad V_{\widehat{M}}^W > \theta; \tag{329}$$

(ii) *for every returned $\widehat{M}$, no certified (ϵ,θ)-working record in $\mathscr{R}_W^{(n)}(B,a)$ has $\mathbf{c}(M) \prec \mathbf{c}(\widehat{M})$;*

(iii) *every comparator $M_0 \in \mathscr{R}_W^{(n)}(B,a)$ whose extended two-sided radius and lower visibility satisfy*

$$\mathcal{R}(M_0) + 2\Gamma_W(M_0) \le \epsilon, \qquad \underline{V}_W(M_0) > \theta, \tag{330}$$

belongs to $\mathscr{F}_W(\epsilon,\theta;B,a)$, and some $\widehat{M} \in \widehat{\mathscr{M}}_W(\epsilon,\theta;B,a)$ obeys

$$\mathbf{c}(\widehat{M}) \preceq \mathbf{c}(M_0); \tag{331}$$

(iv) *when $\mathscr{R}_W^{(n)}(B,a) \neq \varnothing$, an $\varepsilon_{\mathrm{opt}}$-minimizer of U_W satisfies the Part I oracle inequality*

$$\mathcal{R}(\widehat{M}_{\mathrm{or}}) \le \inf_{M \in \mathscr{R}_W^{(n)}(B,a)} \{\mathcal{R}(M) + 2\Gamma_W(M)\} + \varepsilon_{\mathrm{opt}}. \tag{332}$$

This is the Part I risk-oracle comparison at a fixed represented stage. It supplies the risk restriction in (322)*; structural selection is its saturated visibility supremum and Pareto strict inverse.*

(v) *across the Part I resource–action grid, the learner reports the returned record frontiers at precisely the Pareto-minimal grid points of the certified strict region*

$$\widehat{\mathfrak{P}}^{>,\le\epsilon}_{Q,\overline{W},\mathfrak{A},n}(\theta).$$

If the feasible set is empty, the output is the existing insufficient- confidence, target-miss, authority, or budget status. If several records remain statistically unresolved and Pareto-minimal, the learner returns their certified frontier; no uniqueness criterion is added. The directed sequence of stages carries the Part I net remainder. Its infimum and upward feasibility region are therefore defined without compactness; Pareto-minimal continuum records are reported exactly on stages or limit cells where they exist, and nonattainment remains in the search transcript.

Proof. The learned-response Rademacher bound and exact-source union bound give (319) uniformly over (316). The PAC–Bayes chain rule gives (320); applying the structural PAC–Bayes bound to every $\lambda \in \Lambda$ at confidence $\delta_{\mathrm{PB}}/|\Lambda|$ gives (321). The confidence allocation and union bound put all applicable candidates on one event. Their recordwise minimum is valid because they bound the same population loss after the declared converters.

On that event, $\mathcal{R}(M) \leq U_W(M)$ and $\underline{V}_W(M) \leq V_M^W$, proving (i). Statement (ii) is the definition of a Pareto-minimal element of the finite feasible set. A two-sided radius gives $U_W(M_0) \leq \mathcal{R}(M_0) + 2\Gamma_W(M_0)$, so (330) places M_0 in the feasible set. Every member of a finite partially ordered set dominates a minimal member, proving (iii). Finally,

$$\mathcal{R}(\widehat{M}_{\mathrm{or}}) \leq U_W(\widehat{M}_{\mathrm{or}}) \leq \inf_M U_W(M) + \varepsilon_{\mathrm{opt}} \leq \inf_M \{\mathcal{R}(M) + 2\Gamma_W(M)\} + \varepsilon_{\mathrm{opt}},$$

which proves (iv). By (322)– (323), a grid point belongs to the certified strict inverse exactly when it contains a record in $\mathscr{F}_W(\epsilon, \theta; B, a)$. The grid-level reporting rule selects exactly the Pareto-minimal such points and returns the recordwise Pareto frontier there, proving (v). Net stability errors are already components of Γ_W and $\mathbf{c}$. □

Corollary II.11.32 (Saturated-profile minimum working models for free Caus and RL)**.** *For unrestricted Caus, take the candidates produced by* (274) *over the exact source/operator/coefficient family in* (316)*. Select them by the Rademacher certificate* (319)*, the PAC–Bayes certificate* (321)*, or their valid same-loss minimum to form U_W, and then apply the risk-qualified saturated profile at the inherited threshold θ. Then* (329)*–* (331) *hold, and the response-energy coordinate is the exact minimum lift from Proposition II.11.27. On the zero-waste fiber $\mathfrak{A}_D^0$, its action coordinate is zero. At fixed response, the smallest full-coverage readout has $K = d_W^*$ by Proposition II.11.30; across responses, the inherited cover or mixture charge enters $\mathbf{c}$, and the risk-qualified saturated profile decides the tradeoff.*

For RL, let $g_{\mathrm{auth}}, g_{\mathrm{obs}} \in [0, \infty]$ be the derived budget/safe-set and observation gaps and define the extended policy certificate

$$U_{\mathrm{RL}}(M) := \frac{2\Gamma_{Q,K} + \tau \log(1/\nu_{\min}) + \mathrm{Osc}(\widehat{q}_K)\sqrt{\kappa_\pi/2}}{1-\gamma} + g_{\mathrm{auth}} + g_{\mathrm{obs}}. \tag{333}$$

Fix the Part I environment, value norm, reward convention, and deployment-law identifier. Over the resulting typed partition of represented RL candidates, with the nonordinary converter/support values totalized as in Theorem II.11.24, let

$$\widehat{\mathscr{M}}_{\epsilon,\theta}^{\mathrm{RL}} := \mathrm{Min}_{\preceq \mathbf{c}}\{R : U_{\mathrm{RL}}(R) \leq \epsilon,\ \underline{V}_W(R) > \theta\},$$

where every critic, bridge, coefficient, and policy-family selection term in $\Gamma_{Q,K}$ is the corresponding inherited Rademacher or PAC–Bayes charge. Every $\widehat{R} \in \widehat{\mathscr{M}}_{\epsilon,\theta}^{\mathrm{RL}}$ satisfies

$$\|V_\alpha^* - V_\alpha^{\widehat{\pi}}\|_\infty \leq \epsilon, \qquad V_{\widehat{R}}^W > \theta, \qquad \nexists R:\ U_{\mathrm{RL}}(R) \leq \epsilon,\quad \underline{V}_W(R) > \theta,\quad \mathbf{c}(R) \prec \mathbf{c}(\widehat{R}). \tag{334}$$

Thus the output is precisely the Part I Pareto-minimal certified safe-policy frontier in the risk-qualified strict inverse at the requested value accuracy and visibility threshold.

Proof. The unrestricted statement is Theorem II.11.31 with the exact energy identity from Proposition II.11.27; zero action on $\mathfrak{A}_D^0$ follows from Theorem II.11.12. For RL, Theorem II.11.24 and the two declared approximation gaps give $\|V_\alpha^* - V_\alpha^{\pi_M}\|_\infty \leq U_{\mathrm{RL}}(M)$. Feasibility therefore proves the first inequality in (334); the simultaneous lower visibility bound proves its second conclusion, and Pareto minimality in the saturated strict inverse proves the last statement. □

II.11.7 Certified learner and primal–dual profile

Definition II.11.33 (Complete Wick certificate)**.** A complete represented Wick certificate is an extension

$$\mathscr{C}_W = (\mathscr{C}_I, \mathfrak{W}_{\mathsf{Abs}}, E_{W,\Delta}, \Sigma_C, F_W, T_{W,\lambda}, \mathscr{S}, \mathscr{T}, \mathscr{L}, \mathbf{c}),$$

where $\mathscr{C}_I$ is the complete selected Part I compensated quotient–geometry record. In particular, it already contains $U, P_\Delta, H_{\mathsf{Abs}}, G, A_{\mathcal{Y}}, C, \mathfrak{A}, D, J_C$, their source and authority provenance, and their structural resource coordinates. The map $\mathfrak{W}_{\mathsf{Abs}}$ is (232); it adds no source or resource field. The derived ledger $\mathscr{S}$ contains the self-energy, dark-space, bracket, quasi-lumpability, and continuum-transfer certificates; $\mathscr{T}$ contains the dynamics, free-Caus, or RL loss, its data split, optimization error, selected bridge/kernel, and behavior-to-deployment comparison, together with the exact source cell, declared operator and coefficient families, Rademacher cover or PAC–Bayes mixture charge, target-subspace coverage identity, and certified risk tolerance; $\mathscr{L}$ contains the target capture, statistical confidence, source, noise, and deployment records; and $\mathbf{c}$ is the inherited complete Pareto resource vector together with the evaluated Caus action. Every derived object is computed from the Wick image of the same selected Part I record. The optimization entries in $\mathscr{T}$ are inner transcripts; only the saturated profile below compares complete records. Its Wick visibility is the existing master-certificate visibility evaluated with the full response and authority data; write it $V^W_{\mathscr{C}_W}$. Its dynamic stability coordinate is the existing general-resolvent factor

$$S_W(\lambda) = \max\{0, 1 - \Delta_W(\lambda)\},$$

where Δ_W is the represented target-functional discrepancy between the true compressed response and $F_W(\lambda)^{-1}$. It vanishes for the exact block model of Theorem II.11.7; empirical and continuum model residuals enter through the resolvent identity.

For an authority envelope $\mathfrak{A}$, resource ceiling B, and action ceiling a, let $\mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)$ be the represented family of complete certificates satisfying both ceilings and having a nonzero quotient defect. The zero-defect boundary belongs to the existing exact-quotient cell. Put

$$G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) := \sup_{\mathscr{C}_W \in \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)} V^W_{\mathscr{C}_W}, \qquad \sup\varnothing := 0. \tag{335}$$

For a threshold θ, its strict inverse is the upward-closed resource–action feasibility region

$$\mathfrak{P}^{>}_{Q,W,\mathfrak{A}}(\theta) := \{(B,a) : G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) > \theta\} = \bigcup_{\mathscr{C}_W :\, V^W_{\mathscr{C}_W} > \theta} \{(B,a) : \mathbf{c}_{\mathrm{res}}(\mathscr{C}_W) \preceq B, \quad \mathbf{c}_{\mathrm{act}}(\mathscr{C}_W) \le a\}. \tag{336}$$

For a requested population-risk accuracy $\epsilon \in [0,\infty]$, fix one Part I population-law, estimand, norm, and loss-scale identifier. After all declared converters have been charged, let $\mathcal{C}^{\mathfrak{A}}_{Q,W,\ell}(B,a)$ denote the corresponding typed partition of $\mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)$; suppress ℓ in the display below. The same Part I saturation restricted to working records is

$$G^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A}}(B,a) := \sup_{\substack{\mathscr{C}_W \in \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a) \\ \mathcal{R}(\mathscr{C}_W) \le \epsilon}} V^W_{\mathscr{C}_W}, \qquad \sup\varnothing := 0, \tag{337}$$

and its strict inverse is

$$\mathfrak{P}^{>;\le\epsilon}_{Q,W,\mathfrak{A}}(\theta) := \{(B,a) : G^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A}}(B,a) > \theta\}. \tag{338}$$

Its Pareto-minimal elements are reported when they exist. In particular, they exist on the finite resource–action grid used by the learner below; the region itself, rather than a possibly empty set of minimal elements, is the exact generalized inverse on an arbitrary ordered resource space.

Proposition II.11.34 (Risk qualification is a restriction of Part I saturation)**.** *For $0 \le \epsilon_1 \le \epsilon_2 \le +\infty$,*

$$G^{\mathrm{sat},\le\epsilon_1}_{Q,W,\mathfrak{A}}(B,a) \le G^{\mathrm{sat},\le\epsilon_2}_{Q,W,\mathfrak{A}}(B,a) \le G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a), \tag{339}$$

$$\mathfrak{P}^{>,\le\epsilon_1}_{Q,W,\mathfrak{A}}(\theta) \subseteq \mathfrak{P}^{>,\le\epsilon_2}_{Q,W,\mathfrak{A}}(\theta) \subseteq \mathfrak{P}^{>}_{Q,W,\mathfrak{A}}(\theta). \tag{340}$$

At $\epsilon = +\infty$, both rightmost inclusions are equalities. The same identities hold at every represented stage after replacing G and $\mathfrak{P}$ by the certified quantities in (322)–(323) and taking the unrestricted certified profile to be the same supremum without the condition $U_W(M) \le \epsilon$. Consequently risk accuracy restricts the record family inside the inherited saturated optimization; it introduces no second structural order.

Proof. The record sets in each supremum are nested because $\{\mathcal{R} \le \epsilon_1\} \subseteq \{\mathcal{R} \le \epsilon_2\} \subseteq \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)$. Suprema preserve inclusion, proving (339). Taking strict superlevel sets proves (340). Every extended risk satisfies $\mathcal{R} \le +\infty$, which gives equality at $\epsilon = +\infty$. Replacing $\mathcal{R}$ by the simultaneous upper certificate U_W gives the identical finite-stage proof. $\square$

Theorem II.11.35 (Bayesian comparison for the Wick saturated learner)**.** *Fix a represented stage n, a common confidence event, and one population loss. For a complete Wick record M, let $U^W_{\mathrm{PB}}(M)$ denote the PAC–Bayes certificate in (321), evaluated on its carried posterior and assigned $+\infty$ when that candidate is unavailable. Define the PAC–Bayes-only saturated lower profile and strict inverse by*

$$\widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) := \sup_{\substack{M\in\mathscr{R}^{(n)}_W(B,a)\\ U^W_{\mathrm{PB}}(M)\le\epsilon}} \underline{V}_W(M), \qquad \sup\varnothing := 0, \tag{341}$$

$$\widehat{\mathfrak{P}}^{\mathrm{PB},>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta) := \{(B,a) : \widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) > \theta\}. \tag{342}$$

Then

$$U_W(M) \le U^W_{\mathrm{PB}}(M), \tag{343}$$

$$\widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a) \le \widehat{G}^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a), \tag{344}$$

$$\widehat{\mathfrak{P}}^{\mathrm{PB},>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta) \subseteq \widehat{\mathfrak{P}}^{>,\le\epsilon}_{Q,W,\mathfrak{A},n}(\theta). \tag{345}$$

The profile inequality is strict at (B,a) whenever some $M \in \mathscr{R}^{(n)}_W(B,a)$ satisfies

$$U_W(M) \le \epsilon < U^W_{\mathrm{PB}}(M), \qquad \underline{V}_W(M) > \widehat{G}^{\mathrm{PB},\le\epsilon}_{Q,W,\mathfrak{A},n}(B,a). \tag{346}$$

Let $\mathfrak{B}(\mathscr{C}_W)$ be the universal Bayesianization of Part I applied to the active execution carried by a complete Wick record. Embed the Wick resource vector in the complete Bayesian resource lattice by filling the belief, likelihood, normalization, and belief-memory slots with zero. The Part I compiler and forgetting map give

$$\mathcal{R}(\mathfrak{B}(\mathscr{C}_W)) = \mathcal{R}(\mathscr{C}_W), \qquad V(\mathfrak{B}(\mathscr{C}_W)) = V^W_{\mathscr{C}_W}, \qquad \mathbf{c}(\mathscr{C}_W) \preceq \mathbf{c}(\mathfrak{B}(\mathscr{C}_W)), \tag{347}$$

and the Caus action coordinate is unchanged. The resource inequality is strict precisely when the canonical Bayesian annotation has a positive belief-specific resource slot. Hence the direct Wick realization reaches the same risk–visibility point no later in the saturated Pareto inverse than its universal Bayesian annotation.

For the full record classes, the inherited Part I comparison remains

$$\mathrm{BayesArr}^{\mathrm{sat}}_{I,H}(B) \leq \mathrm{ActArr}^{\mathrm{sat}}_{I,H}(B) \leq \mathrm{BayesArr}^{\mathrm{sat}}_{I,H}\big(\Psi_{\mathrm{uBayes}}(B,H,n)\big). \tag{348}$$

Thus Bayesian optimality is universal after the Part I transfer ceiling, whereas (344) *is a same-family finite-sample domination and* (347) *is a same-execution resource comparison. The Wick class satisfies*

$$\mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a) \subseteq \mathcal{C}^{\mathfrak{A}}_{Q,\mathrm{nr}}(B), \qquad G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) \leq G^{\mathrm{sat}}_{Q,\mathrm{nr},\mathfrak{A}}(B). \tag{349}$$

These relations give the complete comparison: pointwise certificate domination on the common Wick family, a resource comparison for the same execution, and transfer-ceiling equivalence for the universal classes.

Proof. By definition, $U_W(M)$ is the recordwise minimum of every applicable same-loss certificate, including $U^W_{\mathrm{PB}}(M)$. This proves (343). Consequently

$$\{M : U^W_{\mathrm{PB}}(M) \leq \epsilon\} \subseteq \{M : U_W(M) \leq \epsilon\}$$

inside the same charged family $\mathscr{R}^{(n)}_W(B,a)$. Taking suprema of the same lower visibility proves (344); taking strict superlevel sets proves (345). The record in (346) lies only in the second feasible set and has lower visibility above the first supremum, proving strictness.

The universal Bayesianization theorem of Part I retains every query, transition, action, continuation, stopping, and deployment kernel of the original active record and appends a constant represented belief, unit likelihood and normalizer, and identity update. It therefore preserves the execution law, deployed population risk, and target visibility. Its belief-specific costs are nonnegative. The Part I forgetting compiler deletes exactly those slots with no increase in cost. Together with the exact Wick inheritance in Proposition II.11.2, this proves (347) and its strictness criterion. The two profile inequalities in (348) are respectively the Part I forgetting and universal Bayesianization inequalities. Finally, the Wick record inclusion in the nonreversible family gives the stated specialized-family relation. □

Definition II.11.36 (Two-stage Wick structural learner)**.** The learner receives the candidate records produced by the Part I structural learner from a represented empirical generator or closed-form approximation, with their confidence radii, finite-coverable quotient, geometry, and authority families, resolvent and ridge grids, and total resource ceiling. It later receives a represented target/reward sample. The Wick stage neither retypes nor enlarges those candidate families. Raw trajectories enter through a represented generator/form estimator and its dependence certificate. An exact PDE operator is the zero-estimation- error specialization of the same interface.

(L1) On dynamics-only data, enumerate the inherited charged finite net. For every Part I candidate, minimize (270), or the raw-path likelihood (271), within that record's existing structural and authority constraints. Fit its represented parameters and residual, then apply $\mathfrak{W}_{\mathsf{Abs}}$ to the selected quotient block. Certify $E_{W,\Delta}$, the derived complement-resolvent bounds, self-energy, dark dimension, bracket gap, the derived design range/kernel, and the identification/perturbation bounds of Propositions II.11.18 and II.11.20.

(L2) Freeze the quotient/geometry selection or retain a simultaneous uniform cover. On split target/control data, choose the target band and readout. Under unrestricted authority, minimize (274) and (275), restricting to the zero-waste fiber when least action is a hard requirement. Under RL authority, alternate the frozen-target critic and Gibbs actor steps in (284). In a hybrid envelope, optimize only the declared control part. Every selected bridge current is certified by Theorem II.11.12; an imposed bridge remains fixed. For each fixed response, compute $\mathcal{V}_W^*$, set $K = d_W^*$ whenever the reachable family verifies (326), and otherwise retain the exact gap (325).

(L3) Compute the lower-confidence master visibility and every applicable entry of Corollary II.11.26. At fixed (B, a), return an ε-maximizer of the saturated profile. Ties within the certified tolerance use the inherited Part I Pareto order on the complete resource–action vector; the declared candidate order only enumerates unresolved equals. When population-risk accuracy ϵ and visibility threshold θ are requested, construct U_W from the inherited Rademacher union/cover or PAC–Bayes mixture/coefficient certificate in Definition II.11.28, and apply (327) to the certified risk-qualified saturated profile; on the RL branch use (333). An empty certified feasible set is reported by its typed failure status.

(L4) For a requested threshold θ, enumerate the inherited budget/action grid, return the certified strict region (336), or (323) when risk-qualified, and report its Pareto-minimal grid points.

The output is a certificate and one of the following typed statuses: certified profile, target spectral miss, uncompensated model defect, target-visible dark mode, authority obstruction, insufficient statistical confidence, continuum-transfer defect, or budget obstruction.

Theorem II.11.37 (Correctness and termination of the Wick learner)**.** *At every represented Part I stage, the candidate set is the charged finite net constructed by that learner, and its perturbation remainder is the extended value supplied by Proposition II.11.18. Consequently the Wick stage terminates by finite enumeration. The Part I confidence allocation and union/PAC–Bayes theorem construct a simultaneous event of probability at least $1 - \delta$; an unavailable radius is $+\infty$. On that event the budget-first output has certified visibility at least*

$$\left[\sup_{\mathscr{C}_W \in \mathcal{C}^{\mathfrak{A}}_{Q,W}(B,a)} V^W_{\mathscr{C}_W} - \varepsilon - \varepsilon_{\mathrm{net}} - \varepsilon_{\mathrm{stat}} \right]_+ , \tag{350}$$

where the last two terms are those derived extended perturbation and confidence radii and $[r]_+ := \max\{0, r\}$, with an infinite subtracted radius giving zero. Its threshold-first output contains every truly feasible strict budget/action point outside these radii and excludes every truly infeasible point outside them; intersecting intervals are reported as insufficient confidence.

On the unrestricted-Caus branch, the returned predictor satisfies the extended oracle inequality (279)*. On the RL branch, every critic iteration satisfies* (285)*, and the output satisfies the extended inequalities* (287) *and* (289)*. A nonordinary converter or actor coordinate makes only the corresponding bound infinite and returns the typed confidence, shift, or policy-conversion status. For the minimum-working-model option, the empty Part I-certified feasible set returns its typed status; every record returned on the complementary cell satisfies* (329)*–* (331)*; the RL specialization satisfies* (334)*. These records are exactly the recordwise frontiers reported at Pareto-minimal points of the certified risk-qualified strict inverse. At fixed response, full structural coverage uses the minimum capacity $K = d_W^*$ whenever* (326) *is verified; this capacity remains an inner coordinate of the saturated-profile comparison.*

Moreover, Theorem II.11.35 gives (349)*, and*

$$G^{\mathrm{sat}}_{Q,W,\mathfrak{A}}(B,a) > \theta \quad \Longleftrightarrow \quad (B,a) \in \mathfrak{P}^{>}_{Q,W,\mathfrak{A}}(\theta). \tag{351}$$

For every ϵ, the risk-qualified specialization likewise satisfies

$$G^{\mathrm{sat},\le\epsilon}_{Q,W,\mathfrak{A}}(B,a) > \theta \quad \Longleftrightarrow \quad (B,a) \in \mathfrak{P}^{>,\le\epsilon}_{Q,W,\mathfrak{A}}(\theta). \tag{352}$$

Proof. Finite enumeration at each Part I stage and the declared tie rule give termination. The allocated confidence event and the extended net Lipschitz bounds compare every population certificate with its evaluated representative; maximizing the simultaneous lower bound gives (350). The interval rule proves the inverse claim. Sample splitting and uniform selection invoke Theorem II.11.22 on the free branch; fitted Bellman contraction and Gibbs calibration invoke Theorem II.11.24 on the RL branch. The model-family selector invokes Proposition II.11.30, Theorem II.11.31, and Corollary II.11.32. Every record in $\mathcal{C}^{\mathfrak{A}}_{Q,W}$ has a nonzero quotient defect evaluated through the full nonreversible resolvent and therefore is a record in the existing Part I Q,nr family, proving (349). Finally, (351) is the generalized-inverse identity for a monotone saturated profile and its strict feasibility region; the same argument proves (352) after restriction to the risk-qualified record family. □

Remark II.11.38 (Totalized structural multipliers)*.* The inherited optimization-status field separates the finite-dimensional candidate problems into exhaustive cells. On the convex constraint-qualified cell, the KKT multipliers of the resource, action, capacity, and confidence constraints are shadow prices for those declared structural resources. The other cells retain their nonconvexity or failed-qualification status and make no multiplier claim. Identification of an ordinary multiplier with a measured physical coupling constant is a distinct representation question, not a structural assumption or a conclusion of the present construction.

II.11.8 Empirical implementation by differentiable linear algebra

The preceding learner is directly implementable on each finite chart in the Part I stage. This subsection specifies the tensor representation, the differentiable solves, and the separation between inner training and outer structural selection. Every branch uses the complete source-resolved learning ledger of Part I. The RL branch additionally uses its complete dynamical learning record; when a candidate is active or structural Bayesian, it also uses the complete neural implementation record. Thus no record type is silently enlarged. In particular, the optimizer output is one represented Wick record; backpropagation does not replace the saturated profile by a scalar training objective.

Definition II.11.39 (Finite empirical Wick interface)**.** Fix one candidate specification s in the charged finite stage $\mathscr{R}^{(n)}_W(B,a)$. Its Part I record supplies the following data.

- (i) The quotient and complement charts $\mathbb{R}^{d_Q}$ and $\mathbb{R}^{d_Y}$, their gauges, evaluation maps, precision convention, and source and authority fields.
- (ii) A bridge chart $\mathrm{Dec}_{\mathfrak{A},s} : \Phi_{C,s} \to \mathbb{R}^{d_Y \times d_Q}$. Its image is the bridge family carried by s. On the unrestricted branch it is the full declared bridge chart; on the RL branch it is the inherited legal-action compiler.
- (iii) Positive grids for λ, ρ, τ, the loss weights, the target evaluation maps, the legal and safe action masks, and the complete confidence allocation. The record also carries its population estimand, probability-law, norm, and loss identifiers and every declared converter between them. These are the grids and types already charged in the Part I candidate, rather than new continuous hyperparameter families.
- (iv) Identifiers and the declared overlap relation for the dynamics, target-training, validation, RL, and audit carriers used by that candidate. Independence is required exactly when the selected Part I split-sample certificate uses it. A shared carrier instead uses its simultaneous union, cover, PAC–Bayes, sequential, or martingale certificate. A carrier may be absent when its associated certificate is not invoked.

The dynamics carrier contains one of the two records

$$Z_j = (x_j, y_j, \dot{x}_j, \dot{y}_j, W_{x,j}, W_{y,j}), \qquad Z_j^{\mathrm{tr}} = (u_j, u_{j+1}, \Delta_j, \Omega_j),$$

used in (270) and (271), respectively. A supervised carrier stores $(\Phi_{\mathrm{tr}}, y_{\mathrm{tr}})$ and $(\Phi_{\mathrm{val}}, y_{\mathrm{val}})$, where each Φ is the represented evaluation map from quotient scores to observations. An RL carrier stores $(h_i, a_i, r_i, h_i', \omega_i)$, its terminal indicator, and its legal and safe masks. When the record contains an encoder, raw observations first pass through that inherited map; its reconstruction and deployment errors remain separate certificate coordinates.

The terminal empirical object is

$$\widehat{\mathscr{C}}_{W,s} = (s, \widehat{\vartheta}_s, \widehat{f}_s, \widehat{q}_s, \widehat{\pi}_s, \mathsf{Opt}_s, \mathsf{Audit}_s, \mathbf{c}_s, \mathsf{Status}_s). \tag{353}$$

Here absent supervised or RL fields are null slots. The optimizer transcript Opt_s stores the initialization, iterates, stochastic gradients, moments, clipping, stopping rule, terminal weights, and the represented optimization gap. The audit field stores split identifiers, all evaluated residuals and intervals, the same-loss risk candidates, lower master visibility, precision and solve residuals, and the confidence allocation. The status belongs to the typed list in Definition II.11.36, augmented only by the numerical status already present in the Part I implementation record.

Definition II.11.40 (Structure-preserving tensor compiler)**.** Let L_H, L_G, K be unconstrained real trainable matrices of the appropriate sizes and let $\phi_C \in \Phi_{C,s}$. Define

$$H = L_H L_H^{\mathsf{T}}, \qquad G = L_G L_G^{\mathsf{T}}, \qquad A_{\mathcal{Y}} = \frac{1}{2}(K - K^{\mathsf{T}}), \qquad C = \mathrm{Dec}_{\mathfrak{A},s}(\phi_C). \tag{354}$$

This displayed parameterization is the dense trainable cell. If s carries additional sparsity, tying, locality, or coefficient restrictions, the factors range only over its inherited chart and are passed through its exact structure compiler. A finite or nondifferentiable Part I chart is enumerated and then held fixed during the differentiable subproblem. Thus (354) does not enlarge the candidate family. The displayed square factors parameterize the full dense positive-semidefinite cones. A rectangular factor is used only when the inherited candidate already carries the corresponding rank ceiling. Nonuniqueness of the factors is an optimizer-coordinate gauge, whose storage and optimization are charged in the record; all identification and perturbation bounds concern the compiled operators $H, G, A_{\mathcal{Y}}, C$, not a choice of factor. The real matrices are converted to the declared complex precision only when the Wick generator or its response is evaluated. For $\lambda > 0$, put

$$B_C = \lambda I_{d_Y} + G - A_{\mathcal{Y}}, \qquad R_C = \mathrm{solve}(B_C, C), \tag{355}$$

$$\Sigma_C = C^* R_C, \qquad F_C = \lambda I_{d_Q} + iH + \Sigma_C, \tag{356}$$

$$A_C = I_{d_Q} + F_C^* F_C, \qquad \mathrm{Resp}_C(V) = \mathrm{solve}(A_C, V). \tag{357}$$

Thus $\mathrm{Resp}_C(V) = T_{W,\lambda}(C)V$, without materializing a matrix inverse. For an evaluation map E, its finite response Gram matrix is

$$K_{C,E} = E\,\mathrm{Resp}_C(E^*). \tag{358}$$

If E is represented by a redundant row system, its Part I range compression is applied before the solve. A Moore–Penrose calculation uses the same declared finite-precision range decomposition and stores the range residual. No eigenvalue is clipped and no diagonal jitter is added unless that modification and its induced operator error are fields of s.

For a coefficient vector $f \in \mathbb{C}^{d_Q}$, the finite supervised inner problem is

$$\widehat{f}_{C,\rho} = \operatorname{solve}\left(\frac{1}{m_{\mathrm{tr}}}\Phi_{\mathrm{tr}}^*\Phi_{\mathrm{tr}} + \rho A_C, \frac{1}{m_{\mathrm{tr}}}\Phi_{\mathrm{tr}}^* y_{\mathrm{tr}}\right). \tag{359}$$

This is the normal equation of (274). Kernel form $(K_{C,E} + m_{\mathrm{tr}}\rho I)\alpha = y_{\mathrm{tr}}$ is equivalent on the represented evaluation range. The primal form makes the inverse-response penalty $f^*A_C f$ explicit.

Proposition II.11.41 (Faithfulness and differentiability of the tensor compiler)**.** *For every finite empirical specification s, the compiler of Definition II.11.40 has the following properties.*

(i) *It satisfies $H = H^* \succeq 0$, $G = G^* \succeq 0$, $A_{\mathcal{Y}}^* = -A_{\mathcal{Y}}$, and $C \in \operatorname{im}\operatorname{Dec}_{\mathfrak{A},s}$ at every optimizer iterate.*

(ii) *The matrix B_C is invertible and $\|B_C^{-1}\| \le \lambda^{-1}$. The matrix A_C is Hermitian positive definite, $A_C \succeq I$, and $\|A_C^{-1}\| \le 1$. Hence* (355)– (359) *have unique ordinary values in exact arithmetic for every $\lambda, \rho > 0$.*

(iii) *The compiled block generator is exactly* (233) *and satisfies* (238)*. Its computed self-energy and response are exactly the finite-chart restrictions of* (242) *and* (254)*.*

(iv) *The tensor core is differentiable in $H, G, A_{\mathcal{Y}}, C$ at every ordinary finite-precision evaluation point. Its composition with an inherited parameter decoder is differentiated on the decoder's represented differentiable cell; a finite or nondifferentiable decoder is handled by the inherited enumeration/search transcript. For $X = B^{-1}D$, its differential is*

$$\mathrm{d}X = B^{-1}(\,\mathrm{d}D - (\,\mathrm{d}B)X). \tag{360}$$

For the transition loss, the differential of the matrix exponential is the Fréchet formula

$$D\exp(M)[\dot{M}] = \int_0^1 e^{(1-t)M}\dot{M}e^{tM}\ \mathrm{d}t. \tag{361}$$

Reverse automatic differentiation through matrix multiplication, linear solves, and the matrix exponential therefore evaluates the derivatives of the mathematical losses on the differentiable cell. Since the trainable parameters are real and the terminal loss is real, this is the real pullback of the complex intermediate calculation.

A failed finite-precision solve, a nonfinite loss, or a residual exceeding the declared tolerance produces the numerical-status branch of (353)*. Every dependent upper-error, risk, or policy-gap candidate receives $+\infty$, while every dependent lower visibility candidate receives zero. A raw derived coordinate that has no ordered extended value retains its typed nonordinary status. The record remains in the finite stage.*

Proof. The first assertion follows from (354), the exact structured compiler when present, and the range of the inherited authority decoder. For $v \in \mathbb{C}^{d_Y}$,

$$\operatorname{Re}\langle v, B_C v\rangle = \lambda\|v\|^2 + \langle v, Gv\rangle \ge \lambda\|v\|^2.$$

Finite-dimensional accretivity gives injectivity, hence invertibility, and the displayed inverse bound. Moreover, for $w \in \mathbb{C}^{d_Q}$, $\langle w, A_C w\rangle = \|w\|^2 + \|F_C w\|^2$, proving the claims about A_C. Substitution in the block generator proves the energy identity, and block elimination gives the two response identities. Differentiating $BX = D$ proves (360); the standard variation-of-constants identity proves (361). Composition and the chain rule give the automatic-differentiation assertion on the represented differentiable cell; the other chart types retain their Part I search rule. The last statement is the extended-value convention of Proposition II.11.3 applied to the numerical field of the complete implementation record. $\square$

Dynamics fit. The following PyTorch-style pseudocode describes the inner dynamics step.

```
for spec in part1_stage:
model = WickModule(spec)
for batch in spec.dynamics_loader:
H = Lh @ Lh.T;  G = Lg @ Lg.T
Ay = (K - K.T) / 2;  C = spec.decode_caus(phi_c)
Lw = wick_generator(cplx(H), cplx(G), cplx(Ay), cplx(C))
L = derivative_loss(batch, H, G, Ay, C)
# or:  pred = expm_action(dt * Lw, batch.u, tol=spec.solve_tol)
#     L = transition_loss(batch, pred)
optimizer.zero_grad();  L.backward();  optimizer.step()
store_optimizer_transcript(model, optimizer)
```

The two branches evaluate (270) and (271), respectively. The terminal design matrix Q_n, score s_n, range projector P_{id}, kernel, and optimization gap are then evaluated from the stored trajectory; they are not training assumptions. The call `expm_action` computes $e^{\Delta\mathcal{L}_W}u$ without forming the exponential. A dense Padé exponential remains available on a small chart; a large sparse or matrix-free chart uses the represented Krylov exponential-action backend and stores its forward residual or error enclosure.

Unrestricted-Caus fit. For every charged (s, λ, ρ), response and readout fitting use the same bridge parameter. Gradients of the validation term pass through (359).

```
for spec, lam, rho in charged_free_candidates:
for outer_step in spec.bridge_steps:
C = spec.decode_caus(phi_c)
B = lam * Iy + G - Ay;  R = linalg.solve(B, C)
F = lam * Iq + 1j * H + C.mH @ R
A = Iq + F.mH @ F
M = Phi_tr.mH @ Phi_tr / m_tr + rho * A
f = linalg.solve(M, Phi_tr.mH @ y_tr / m_tr)
action = squared_current_distance(C, target_current, spec.metric) / 2
penalty = spec.selection_penalty(C, rho)
loss = validation_risk(f) + action_weight * action + grad_part(penalty)
bridge_optimizer.zero_grad();  loss.backward()
bridge_optimizer.step()
detach_and_audit(spec, C, f)
```

This is the dense small-chart backend. The scalable backend replaces `B`, `F`, `A`, and `M` by the operator actions in Definition II.11.46 and replaces each `linalg.solve` by its residual-controlled Krylov solve. It never forms $\Phi_{\mathrm{tr}}^*\Phi_{\mathrm{tr}}$, $F_C^*F_C$, or an inverse. The dense and matrix-free paths solve the same normal equation and differ only by their recorded numerical error. Its corresponding implementation is:

```
ops = compile_linear_operators(spec, parameters)
if spec.augmented_backend:
f, z, solve_info = solve_augmented_kkt(ops, Phi_tr, y_tr, rho)
else:
rhs = Phi_tr.rmatvec(y_tr) / m_tr
f, solve_info = krylov_solve(ops.apply_M(rho), rhs, spec.solver)
loss = validation_risk(Phi_val.matvec(f)) + action_term + selection_term
hypergradient = implicit_adjoint(loss, solve_info, ops)
update_bridge(hypergradient); store(solve_info)
```

The data-map handles stream their matrix–vector and adjoint products, so this code does not retain the complete design on the accelerator. The selection penalty and the indicator of $\mathfrak{A}_{\rm dyn}$ in (275) are certificate and parameter-domain terms. They are added to the displayed loss when they vary differentiably within the candidate. Otherwise the Part I finite net or search transcript compares the detached terminal candidates using their full outer value, including the penalty; `grad_part` is then constant during that inner step. If zero action is a hard task requirement, Part I first evaluates the represented fiber $\mathfrak{A}_D^0$. On its nonempty cell the decoder is restricted to that fiber and the action is identically zero. On its empty cell the candidate receives the distance in (252); training does not replace that value by a penalty surrogate. If the task declares a risk–action tradeoff, the displayed action weight is the declared $\lambda_{\mathscr{W}}$ of (275).

Action-safe RL fit. Let $\mathscr{E}_{\rm RL}$ be the complete finite critic-evaluation carrier, with the weights and normalization of its selected Part I statistical certificate. Its represented safe row map $E_{\mathscr{E}}^{\mathcal{A},\rm safe} : \mathcal{Q} \to \mathbb{C}^{r_{\mathscr{E}}}$ is the full-carrier evaluation map restricted to the legal, failure-killed action rows. Apply the inherited exact row-range compression when those rows are redundant. The finite safe response is

$$K_{C,\mathscr{E}}^{\rm safe} = E_{\mathscr{E}}^{\mathcal{A},\rm safe}\,\mathrm{Resp}_C\big((E_{\mathscr{E}}^{\mathcal{A},\rm safe})^*\big). \tag{362}$$

This is exactly the finite compression in (255) applied to (280). It is positive definite after the declared row-range compression: for $z \neq 0$,

$$z^* K_{C,\mathscr{E}}^{\rm safe} z = \Big\langle (E_{\mathscr{E}}^{\mathcal{A},\rm safe})^* z, T_{W,\lambda}(C)(E_{\mathscr{E}}^{\mathcal{A},\rm safe})^* z \Big\rangle > 0.$$

Without compression it is positive semidefinite and the already declared Moore–Penrose range calculation is used. This Gram representation is used for a dual solve only when its row dimension is the cheaper represented coordinate. The default large-carrier representation is primal:

$$q_f = E_{\mathscr{E}}^{\mathcal{A},\rm safe} f, \qquad \|q_f\|^2_{T_C^{\mathcal{A},\rm safe}} = f^* A_C f \quad \text{for } f \in (\ker E_{\mathscr{E}}^{\mathcal{A},\rm safe})^{\perp_{A_C}},$$

with the inherited minimum-energy range gauge imposed. This is the same minimum-lift identity as Proposition II.11.27. It permits minibatch evaluation of the data term while retaining one global response penalty; no minibatch Gram inverse is introduced. The full Gram matrix is therefore an audit or small-dual object, not a training requirement. One fitted iteration is:

```
f_target = detach(f)
require(nonempty_safe_rows, status=authority_obstruction)
with no_grad():
next_q = real_score(E_next_safe @ f_target)
next_safe_value = masked_max(next_q, safe_mask)
target = reward - alpha * bridge_action
+ (1 - terminal) * gamma * next_safe_value
f, critic_info = solve_weighted_response_krr(
E_taken, target, importance_weights, rho, backend=spec.critic_backend)
with no_grad():  gibbs = safe_gibbs(real_score(E_actor @ f), neutral_row,
tau)
actor_loss = lambda_pi * mean(kl(actor.safe_row, gibbs))
actor_optimizer.zero_grad();  actor_loss.backward()
actor_optimizer.step()
if spec.differentiable_model_cell:
model_loss = frozen_critic_component(detach(f), C)
+ lambda_dyn * dynamics_loss(model, dynamics_batch)
```

```
+ lambda_num * numerical_loss(model, numerical_batch)
model_optimizer.zero_grad();  model_loss.backward()
model_optimizer.step()
store_lag_support_shift_and_optimizer_errors()
```

Here f is a quotient-score coefficient vector or a batched represented coefficient field supplied by the inherited critic chart; every displayed E-map has the corresponding batch axes. The safe mask is applied before both the target maximum and Gibbs normalization. The target tensor and the Gibbs target are detached in their respective alternating steps. The represented masked reduction is defined only after the nonempty-safe-row check and therefore introduces no artificial infinite logit. The bridge action is charged either in the reward or as a hard legal-action constraint, exactly as in Definition II.11.23; it is never counted twice. When the bridge or dynamics block is trainable, its permitted update uses (284) after the critic and actor steps, with the critic and actor arguments frozen as specified there, and stores the induced target-network lag and model-selection error. Every component and displayed weight is retained in the terminal objective. Backpropagation is used only on its represented differentiable slice; masks, validation rules, finite decoders, and nondifferentiable audit coordinates are held fixed or searched by the inherited finite transcript and are evaluated again in the detached terminal record. For a fixed response, `solve_weighted_response_krr` uses the dense, primal matrix-free, dual, or augmented backend to solve the convex critic problem directly. A nonlinear inherited critic chart instead retains its declared WAdam transcript and the same inverse-response penalty, but it does not receive the exact KRR-minimizer claim. The target, actor, and bridge updates remain alternating, so no backward graph is retained across fitted Bellman iterations.

Definition II.11.42 (Detached audit and saturated empirical selector)**.** After every terminal inner fit, detach all terminal tensors and evaluate the complete record on its assigned audit carriers. The audit performs the following operations in order.

(A1) Evaluate the generator residual, solve residuals, lumpability defect, dark and bracket coordinates, self-energy, response, empirical effective dimension, target capture, action, and target-subspace coverage gap.

(A2) Evaluate every applicable Part I statistical candidate after conversion to the same population loss, and set U_W to their recordwise minimum as in Definition II.11.28. The Rademacher union/cover, PAC–Bayes mixture, split-validation, mixing, sequential, and channel candidates retain their own applicability and confidence fields. All candidates and every data-dependent terminal selection are covered by one output-selection-independent simultaneous event allocated before selection. A confidence interval chosen only after observing the terminal output is not a certificate. An unavailable candidate equals $+\infty$. Records with different population laws, estimands, norms, or unconverted loss units remain in different typed partitions and their upper bounds are never minimized against one another.

(A3) Evaluate the existing Part I master visibility with the Wick stability, full response, actual authorized bridge, and same-record interval fields:

$$V_M^W = \overline{A}_q S_W(\lambda) M_{\mathfrak{G}_q} C_{\mathfrak{G}_q} \frac{R_{\mathfrak{G}_q}}{1+\rho_{\mathfrak{G}_q}}. \tag{363}$$

Apply the Part I monotone interval evaluator to obtain $\underline{V}_W$. A missing qualification or unauthorized current gives lower visibility zero, as in the master certificate.

(A4) Store the complete resource vector, Caus action, risk and visibility intervals, confidence allocation, and typed status. The computation, architecture, parameter, training, optimization, precision, storage, deployment, evaluation, and upper-audit costs are stored in the corresponding slots of the Part I complete learner–deployment ledger. On an active or structural-Bayesian candidate these slots are exactly the complete neural implementation cost; on an RL candidate they also populate the complete dynamical learning record.

For each fixed supervised loss type ℓ, let $\mathscr{R}^{(n)}_{W,\ell}(B,a)$ be its same-law, same-estimand, same-norm partition. At fixed (B,a,ℓ), return the ε-maximizers of (322) restricted to that partition. Across its charged resource–action grid, return all Pareto-minimal points of (323). The RL value-performance certificate is already the separately calibrated U_{RL} of (333). Thus a requested (ϵ,θ) has the two typed returns

$$\widehat{\mathscr{M}}^{\mathrm{free}}_{W,\ell} := \mathrm{Min}_{\preceq_{\mathbf{c}}}\{M \in \mathscr{R}^{(n)}_{W,\ell}(B,a) : U_W(M) \leq \epsilon,\ \underline{V}_W(M) > \theta\}, \tag{364a}$$

$$\widehat{\mathscr{M}}^{\mathrm{RL}}_{W} := \mathrm{Min}_{\preceq_{\mathbf{c}}}\{M \in \mathscr{R}^{(n)}_{W,\mathrm{RL}}(B,a) : U_{\mathrm{RL}}(M) \leq \epsilon,\ \underline{V}_W(M) > \theta\}. \tag{364b}$$

The two sets may be identified only when a converter already carried by Part I places both guarantees on the same population law, estimand, norm, and loss scale. In that case the converted records form one named partition; the conversion error is charged before the recordwise minimum. An empty set returns its inherited typed status.

The outer computation can be expressed without differentiating through the structural order:

```
records = []
for spec in part1_stage:
terminal = fit_inner_wick_record(spec)
records.append(certify_detached(terminal, spec.audit_splits))
for loss_id, same_loss_records in typed_partitions(records):
for B, action_ceiling in charged_grid(same_loss_records):
feasible = [r for r in same_loss_records
if resource_leq(r.cost, B) and r.action <= action_ceiling
and working_ucb(r) <= epsilon
and r.visibility_lcb > theta]
report(loss_id, pareto_minimal(
feasible, key=complete_resource_action))
```

Here `working_ucb` dispatches to U_W on a supervised partition and to U_{RL} on an RL value-performance partition. The partition key contains the population law, estimand, norm, and loss scale, so the code cannot compare certificates with different meanings. No weighted sum of risk, visibility, capacity, action, and computation occurs in this outer loop. A weighted loss may generate a record only when that tradeoff is declared inside the task. The record is still compared by the same saturated profile and Pareto order as every other Part I candidate.

Theorem II.11.43 (Certified empirical realization of the Wick learner)**.** *Apply the preceding procedure to the charged finite Part I stage. Then:*

(i) *every ordinary terminal execution compiles to a complete source-resolved learning ledger and a complete Wick certificate. An RL execution also compiles to a complete dynamical learning record, and an active or structural-Bayesian execution compiles to a complete neural implementation record. Every nonordinary execution compiles to the corresponding typed records;*

(ii) *the dynamics and unrestricted-Caus tensor losses equal* (270) *or* (271), (274)– (275)*, respectively, on their represented finite carriers. The alternating RL steps are exactly the critic, actor, dynamics, and numerical components of* (284)*, with their displayed weights and frozen arguments;*

Empirical output	Evaluated Wick coordinate	Quantitative conclusion
Dynamics fit	$Q_n, s_n, P_{\mathrm{id}}, \ker Q_n$, generator and compression residuals	(272), (268), and (269)
Response audit	$E_{W,\Delta}, \Sigma_C, F_C, T_C$, dark/bracket status, empirical traces, capture and coverage	(310)– (312), with the full alternatives in Corollary II.11.26
Unrestricted Caus	inverse-response ridge fit, actual action, bridge cover and validation radius	(279), including operator, selection, net, and optimization errors
RL	safe effective dimension, critic residual, converter, target lag, policy KL, authority and observation gaps	(285), (287), (289), and (333)
Statistical audit	Rademacher union/cover, PAC–Bayes mixture, dependence and channel candidates in one typed same-loss partition	their recordwise minimum U_W, the source-resolved minimum (308), and Theorem II.11.35
Structural return	$\underline{V}_W$, typed U_W or U_{RL}, $\mathbf{c}_{\mathrm{res}}, \mathbf{c}_{\mathrm{act}}$	(322), (323), and (364a)– (364b)

Table 4: Empirical outputs and their quantitative certificate coordinates. The table is an execution map for the exhaustive ledger in Corollary II.11.26; it introduces no additional bound.

(iii) *an Adam or block-preconditioned update has the represented form*

$$\theta_{k+1} = \theta_k - \alpha_k W_k^{-1} \widehat{m}_k, \tag{365}$$

and is a diagonal or block instance of the Part I certified WAdam record. Its stationarity and global-gap conclusions are exactly those of the Part I WAdam descent theorem when that record certifies its alignment, second-moment, smoothness, and, for the global conclusion, PL coordinates. Otherwise the actual terminal optimization gap remains an extended record coordinate. Even on the global-gap cell, that objective gap enters the saturated error only through the objective-to-action, objective-to-risk, or objective-to-visibility converter already carried by Part I; without an ordinary converter it is only an optimization coordinate;

(iv) *on the simultaneous Part I confidence event, the returned free-Caus records satisfy* (279), *and the returned RL records satisfy* (285)– (289)*; and*

(v) *within each typed same-loss partition, the outer return is exactly the Part I Pareto-minimal certified risk-qualified Wick frontier. The supervised return satisfies* (329)– (331)*; the RL return satisfies* (334)*. Relative to a PAC–Bayes-only supervised selector, its feasible profile and strict inverse obey* (344) *and* (345)*; and*

(vi) *the empirical compiler adds no admission premise. Its charts, laws, losses, converters, masks, confidence event, costs, and search order are fields of the Part I record; the Wick generator, solves, losses, derivatives, and audit coordinates are finite compositions or totalized evaluations of those fields.*

Consequently the empirical use of backpropagation changes only the method used to produce and record inner candidates. The meaning of optimal, minimal, and simplest remains the saturated-profile meaning of Definition II.11.29.

Proof. The tensor compiler proposition proves structural fidelity and identifies the finite response with the response used in the theoretical losses. Expansion of the residual and ridge quadratic forms proves the loss identities. The full-carrier safe compression in (362) is (255) applied to

(280); its principal minibatch compressions therefore retain the same response and normalization. Detaching the target produces the fitted supervised critic problem in (282), and the actor step is (283).

The stored optimizer state and update have the form (365); the Part I WAdam conclusions follow precisely on the cells where its derived certificate coordinates are ordinary. Part I's calibration step, rather than the optimizer theorem, then determines whether its objective gap contributes to an action, risk, or visibility bound. The detached audit uses the same data splits, preallocated simultaneous confidence event, source index, authority record, response, and loss as the terminal model. It therefore constructs the quantitative bounds of Corollary II.11.26 and Theorems II.11.22 and II.11.24. The supervised selector (364a) is the selector in Theorem II.11.31; the free and RL conclusions follow from Corollary II.11.32, with the latter using (364b). The PAC–Bayes comparison is Theorem II.11.35. Typed partitions prevent invalid cross-loss minima. Exact inheritance and no-new-premise closure are Propositions II.11.2 and II.11.3; finite linear algebra merely evaluates their derived coordinates. Extended values and typed statuses give the complementary numerical and statistical cells. □

II.11.9 Executable refinement contract

An implementation must refine one charged finite Part I stage. It is not a procedure for constructing an uncharged continuum family. The following contract fixes the remaining programming choices and makes the return value of every execution branch explicit.

Definition II.11.44 (Typed program objects)**.** For each candidate $s \in \mathscr{R}_W^{(n)}(B,a)$, the program stores six typed objects. `WickSpec` is immutable; every other object becomes an immutable serialized value when it is attached to the terminal record.

(i) `WickSpec` stores the stage and candidate identifiers, dimensions, real and complex dtypes, device, authority decoder, all represented grids, loss and law identifiers, confidence allocation, carrier identifiers and overlap relation, dense or matrix-free backend, linear solver, restart rule, preconditioner, primal/dual choice, trace-audit method, precision schedule, numerical tolerances, memory ceiling, maximum iteration counts, and tie rule.

(ii) `WickParameters` stores the real trainable tensors L_H, L_G, K, ϕ_C. Any inherited encoder, readout, critic, or actor parameters occupy separately named fields.

(iii) `WickOperators` stores either the compiled tensors $H, G, A_{\mathcal{Y}}, C, B_C, R_C, \Sigma_C, F_C, A_C$ or their exact `LinearOperator` actions, together with dtypes, shapes, adjoint actions, backend identifiers, and no optimizer state.

(iv) `CarrierBundle` stores the represented dynamics, training, validation, RL, and audit carriers together with their identifiers, masks, weights, terminal fields, and normalization constants.

(v) `FitTranscript` stores the initialization, random seeds, minibatch order, optimizer states, iterates, stopping reason, terminal parameters, objective values, and optimization-gap candidate.

(vi) `AuditRecord` stores the detached residuals, statistical candidates, confidence event identifier, lower visibility, resource and action vectors, and typed status. A successful audit constructs (353); a failed audit constructs the same record with its typed failure status.

The immutable specification is hashed and its identifier is copied into every other object. Thus a tensor, loss, or certificate produced under one specification cannot be attached to another candidate without a type failure.

The dense finite-chart shapes are fixed by the following table. A structured decoder may use smaller parameter tensors, but its output has the displayed shape. In a matrix-free record, each

Object	Shape and dtype	Mathematical value
L_H, H, F_C, A_C	$d_Q \times d_Q$, real for L_H, H, complex otherwise	$H = L_H L_H^\top$, (356)– (357)
$L_G, G, K, A_\mathcal{Y}, B_C$	$d_Y \times d_Y$, real before compilation	$G = L_G L_G^\top$, $A_\mathcal{Y} = (K - K^\top)/2$
C, R_C	$d_Y \times d_Q$, real C, complex R_C	authority-decoded bridge and $B_C^{-1}C$
Σ_C	$d_Q \times d_Q$, complex	$C^* R_C$
E	$r \times d_Q$, complex	represented evaluation map
$K_{C,E}$	$r \times r$, complex Hermitian	$E A_C^{-1} E^*$
$\Phi_{\mathrm{tr}}, y_{\mathrm{tr}}$	$m_{\mathrm{tr}} \times d_Q$, m_{tr}	supervised design and target
q_{safe}	$r_\mathscr{E}$, real	inherited observable-score slice

Table 5: Code-level shape contract. The complex dtype has the same precision as the declared real dtype. In particular, float64 is paired with complex128. Every adjoint in the response calculation is a conjugate transpose.

table entry is a shape-checked operator handle with `matvec`, `rmatvec`, and, when available, `matmat`; the handle need not own a dense tensor of that shape. Training and RL carriers are streamed in declared batches. The audit may make additional deterministic passes over an auditable carrier, but it does not require that carrier to reside in accelerator memory.

The code boundary from the complex response space to supervised real outputs is the observation map carried by s. A target may instead be represented as a complex observation, in which case the loss is the modulus-squared loss in (274). The RL observation map has real range by Definition II.11.23; the program applies a maximum or exponential only after that map. Calling `real` on an arbitrary complex score is not an admissible replacement for this typed observation map.

Definition II.11.45 (Typed numerical interface)**.** The numerical core exposes the following interfaces; each output retains the input specification identifier.

```
compile(spec, parameters) -> WickOperators | TypedFailure
response(spec, operators, evaluation_map) -> ResponseHandle | TypedFailure
fit_free(spec, operators, carriers) -> FitTranscript | TypedFailure
fit_rl(spec, operators, carriers) -> FitTranscript | TypedFailure
audit(spec, operators, transcript, carriers) -> AuditRecord
select(stage, audit_records, epsilon, theta) -> ParetoFrontier | TypedFailure
```

Here `compile` evaluates (354) and (355)– (357); `response` returns a dense Gram handle or a matrix-free operator handle and evaluates (358) or (362); `fit_free` evaluates (274)–(275); and `fit_rl` evaluates the frozen critic, actor, dynamics, and numerical blocks of (284). The audit and selector are Definition II.11.42. These functions do not infer a grid, authority envelope, loss converter, safe mask, confidence split, or stopping rule from the observed objective values. The compiler, response, audit, and selector are pure functions of their stated inputs. The two fitting interfaces are deterministic when the stored seed, minibatch order, optimizer backend, and deterministic-backend flag are fixed; otherwise their complete realized transcript is part of the return value.

The runtime data-access validator checks every requested carrier identifier against the overlap relation in `WickSpec`. A split-sample candidate requires the declared disjoint carriers. A shared carrier is accepted only for the corresponding simultaneous, uniform, sequential, martingale, or PAC–Bayes candidate. The confidence allocation and candidate index are sealed before the first target-bearing carrier is read.

Definition II.11.46 (Dense, nested matrix-free, and augmented backends)**.** The backend tag of `WickSpec` selects one of three representations of the same finite-chart operators.

(i) The dense backend materializes the matrices in (355)– (359), uses factorizations with multiple right-hand sides, and caches a factorization only while every operator parameter in its cache key is unchanged.

(ii) The matrix-free backend stores the actions of $L_H, L_G, K, C, C^*, \Phi, \Phi^*$. With all adjoints taken in the declared complex inner product, it applies

$$\begin{aligned} \mathsf{H}x &= L_H(L_H^* x), & \mathsf{G}y &= L_G(L_G^* y), & \mathsf{A}_{\mathcal{Y}}y &= \tfrac{1}{2}(Ky - K^*y), \\ \mathsf{B}y &= \lambda y + \mathsf{G}y - \mathsf{A}_{\mathcal{Y}}y, & \mathsf{B}^*y &= \lambda y + \mathsf{G}y + \mathsf{A}_{\mathcal{Y}}y, \end{aligned} \tag{366}$$

$$\begin{aligned} \mathsf{F}x &= \lambda x + i\mathsf{H}x + C^* \operatorname{Solve}_{\mathsf{B}}(Cx), \\ \mathsf{F}^*x &= \lambda x - i\mathsf{H}x + C^* \operatorname{Solve}_{\mathsf{B}^*}(Cx), \\ \mathsf{A}x &= x + \mathsf{F}^*(\mathsf{F}x), \\ \mathsf{M}_\rho x &= \frac{1}{m_{\mathrm{tr}}}\Phi^*(\Phi x) + \rho \mathsf{A}x. \end{aligned} \tag{367}$$

In exact arithmetic the solve symbols in these definitions denote the unique solutions. At runtime they are residual-controlled iterative solves and their errors are included in every enclosing operator application. Because an adaptively terminated inner solve makes the outer action inexact and iteration-dependent, the nested backend uses a flexible outer Krylov method (or fixes the inner tolerance and initial state so that the represented action is stationary). In both cases the terminal candidate is accepted only through the independent residual audit below.

(iii) The augmented matrix-free backend eliminates nested solves. Introduce $z \in \mathbb{C}^{d_Y}$ and impose $\mathsf{B}z = Cf$. Then the primal free-Caus problem is exactly

$$\min_{f,z:\,\mathsf{B}z=Cf} \left\{ \frac{\|\Phi f - y\|^2}{2m_{\mathrm{tr}}} + \frac{\rho}{2}\left(\|f\|^2 + \|(\lambda I + i\mathsf{H})f + C^*z\|^2 \right) \right\}. \tag{368}$$

Its block KKT operator uses only $\mathsf{H}, \mathsf{B}, C, C^*, \Phi, \Phi^*$ actions. The RL critic has the identical form with its safe evaluation map and weighted target. A residual-controlled block solver therefore reaches the same minimizer without an inner solve inside every outer matrix–vector product.

The matrix-free code never forms $\Phi^*\Phi$, $F_C^* F_C$, $K_{C,E}$, or a matrix inverse. It uses a block Krylov solve when several right-hand sides share an operator. A preconditioner and warm start may change iteration count but not the mathematical output: their identifiers and construction costs are stored, and acceptance always evaluates the residual of the original unpreconditioned equation.

The free-Caus fit solves the smaller represented primal or dual system. The primal system is $\mathsf{M}_\rho f = \Phi^* y / m_{\mathrm{tr}}$; the dual system is $(K_{C,\Phi} + m_{\mathrm{tr}}\rho I)\alpha = y$. The same rule is used for the RL critic. A primal critic stores f in the minimum-A_C-energy gauge and evaluates only the minibatch rows $E_{\mathrm{batch}} f$. A dual critic stores α and evaluates $K_{\mathrm{batch,train}}\alpha$. The choice and its conversion residual are fields of the candidate. The nested and augmented formulations are equal because the constraint in (368) has the unique solution $z = \mathsf{B}^{-1}Cf$, after which its second squared norm is $\|\mathsf{F}f\|^2$.

The numerical method is selected from the proved operator structure, not from a generic "Krylov" flag. A conforming solver dispatch is:

System	Derived structure	Large-chart method and safeguard
$\mathsf{B}z = v$, $\mathsf{B}^*z = v$	accretive, generally non-Hermitian	restarted GMRES; flexible GMRES when the preconditioner or inner accuracy varies; audit the original unpreconditioned residual
$\mathsf{M}_\rho f = b$	Hermitian positive definite in exact arithmetic, with $\mathsf{M}_\rho \succeq \rho I$	preconditioned conjugate gradients only for a stationary Hermitian action; otherwise flexible GMRES or the augmented formulation
$K + m\rho I$	Hermitian positive definite on the compressed row range	preconditioned or block conjugate gradients, used only when the compressed row dimension is favorable
augmented KKT system	Hermitian indefinite under exact adjoint pairing	MINRES with a Hermitian positive preconditioner; GMRES when finite-precision actions or preconditioning do not preserve that pairing
$e^{\Delta \mathcal{L}_W} u$	contractive but generally non-normal	adaptive Arnoldi exponential action with time subdivision; a Lanczos-only path is not used unless the represented generator is Hermitian

The default complement preconditioner uses the positive part $\lambda I + G$; response and KKT preconditioners may reuse inherited sparse, domain-decomposition, or multilevel actions. A randomized or learned preconditioner is permitted only as an accelerator: its construction and state are charged, and acceptance still uses the original operator residual. Thus a poor preconditioner can increase recorded work or cause a typed numerical-budget return, but it cannot change the accepted equation.

Proposition II.11.47 (A posteriori certificate for exponential actions)**.** *Let $u(t) = e^{t\mathcal{L}_W} u_0$ on an inherited ordinary semigroup cell, and let $\widetilde{u}$ be the piecewise differentiable path returned by an Arnoldi exponential-action computation with defect $d(t) = \dot{\widetilde{u}}(t) - \mathcal{L}_W \widetilde{u}(t)$. Then*

$$\|u(T) - \widetilde{u}(T)\| \le \|u_0 - \widetilde{u}(0)\| + \int_0^T \|d(t)\| \, \mathrm{d}t. \tag{369}$$

Consequently adaptive time subdivision can accept an exponential action from an outward-rounded quadrature enclosure of its Arnoldi defect, without forming either $\mathcal{L}_W$ or its exponential. A missing enclosure returns the numerical-status value and cannot enter a finite certificate.

Proof. Variation of constants gives

$$\widetilde{u}(T) - u(T) = e^{T\mathcal{L}_W}(\widetilde{u}(0) - u_0) + \int_0^T e^{(T-t)\mathcal{L}_W} d(t) \, \mathrm{d}t.$$

The contraction bound derived in Propositions II.11.4 and II.11.5, followed by the triangle inequality, proves (369). ☐

Proposition II.11.48 (Matrix-free adjoint for transition training)**.** *Let $J(u(T), \theta)$ be a real terminal transition loss on the ordinary cell of Proposition II.11.47, with $\mathcal{L}_W = \mathcal{L}_W(\theta)$. Define*

$$p(T) = \partial_u J, \qquad p(t) = e^{(T-t)\mathcal{L}_W^*} p(T).$$

Then

$$D_\theta J[\dot\theta] = \partial_\theta J[\dot\theta] + \operatorname{Re}\int_0^T \left\langle p(t), D_\theta \mathcal{L}_W[\dot\theta]u(t)\right\rangle \,\mathrm{d}t. \tag{370}$$

Thus the large-chart backward pass uses actions of $\mathcal{L}_W$, $\mathcal{L}_W^*$, and its parameter-directional action; it need not form a dense Fréchet derivative. The implementation stores a declared checkpoint schedule and recomputes forward segments during the adjoint sweep, so peak activation memory is bounded by the checkpoint window.

For a computed forward path $\widetilde u$ and adjoint path $\widetilde p$, let $e_u(t)$ and $e_p(t)$ be their defect-based error enclosures and let $\|D_\theta\mathcal{L}_W[\dot\theta]\| \le L_{\mathcal{L}}(\dot\theta)$ be the factored derivative-action enclosure. Apart from the outward-rounded quadrature error, substitution in (370) *has error at most*

$$\int_0^T L_{\mathcal{L}}(\dot\theta)\,[e_p(t)(\|\widetilde u(t)\| + e_u(t)) + \|\widetilde p(t)\| e_u(t)] \,\mathrm{d}t. \tag{371}$$

If any enclosure is unavailable, this gradient-error coordinate is $+\infty$, exactly as for an inexact linear-solve gradient.

Proof. Differentiate $\dot u = \mathcal{L}_W(\theta)u$, pair the tangent equation with the backward adjoint, and integrate by parts to obtain (370). Subtract the approximate and exact integrands, add and subtract $\langle \widetilde p, D_\theta\mathcal{L}_W u\rangle$, and apply Cauchy–Schwarz and $\|u(t)\| \le \|\widetilde u(t)\| + e_u(t)$. Integration and addition of the quadrature enclosure give (371). □

Proposition II.11.49 (Implicit differentiation of inexact solves)**.** *Let $S_\theta x_\theta = b_\theta$ be any ordinary solve in Definition II.11.46, and let $J(x_\theta, \theta)$ be a real terminal scalar. If z_θ solves*

$$S_\theta^* z_\theta = \partial_x J,$$

then the real parameter differential is

$$D_\theta J[\dot\theta] = \partial_\theta J[\dot\theta] + \operatorname{Re}\langle z_\theta, D_\theta b_\theta[\dot\theta] - D_\theta S_\theta[\dot\theta]x_\theta\rangle. \tag{372}$$

Hence reverse differentiation requires an adjoint solve but does not store the forward Krylov iterates. The implementation stores the primal and adjoint residuals. Their validated inverse bounds give forward and adjoint state-error enclosures exactly as in (382)*; these are propagated to the gradient-error coordinate whenever a Part I optimizer certificate uses the computed gradient. More explicitly, write $e_x = \|\widetilde x - x\|$ and $e_z = \|\widetilde z - z\|$ for those enclosures, and suppose that a parameter direction has certified bounds $\|D_\theta b[\dot\theta]\| \le L_b(\dot\theta)$ and $\|D_\theta S[\dot\theta]\| \le L_S(\dot\theta)$. The gradient returned by substituting $(\widetilde x, \widetilde z)$ in* (372) *then has directional error at most*

$$e_z\{L_b(\dot\theta) + L_S(\dot\theta)(\|\widetilde x\| + e_x)\} + \|\widetilde z\| L_S(\dot\theta) e_x, \tag{373}$$

in addition to the certified error in the explicit derivative $\partial_\theta J$. The derivative-action bounds are computed by the same factored automatic-differentiation graph as the operator actions and stored in the optimizer record. If any required bound is unavailable, the corresponding gradient-error coordinate is $+\infty$, so no certificate that needs this gradient is issued. A training run may use a looser inner tolerance, but the detached terminal audit is evaluated at the declared certificate tolerance.

Proof. Differentiate $S_\theta x_\theta = b_\theta$ to obtain $S_\theta \dot x = \dot b - \dot S x$. Pair this equation with the adjoint solution and add the explicit parameter derivative of J, which gives (372). The residual-error statement follows by applying the validated inverse bound to the primal and adjoint residuals. For the last assertion, subtract the approximate bilinear term from the exact one and write the difference as

$$\operatorname{Re}\langle \widetilde{z} - z, \dot{b} - \dot{S}x\rangle - \operatorname{Re}\langle \widetilde{z}, \dot{S}(\widetilde{x} - x)\rangle.$$

Cauchy–Schwarz, $\|x\| \le \|\widetilde{x}\| + e_x$, and the two derivative bounds give (373). □

Proposition II.11.50 (Work and memory of the scalable backend)**.** *Let $c_H, c_G, c_{A_\mathcal{Y}}, c_C, c_{C^*}, c_\Phi, c_{\Phi^*}$ be the recorded costs of one operator action, let $k_B, k_{B^*}, k_M, k_{\exp}$ be the actual terminal iteration counts, and put*

$$c_B = c_G + c_{A_\mathcal{Y}} + O(d_Y), \qquad c_{B^*} = c_G + c_{A_\mathcal{Y}} + O(d_Y), \qquad c_F = c_H + c_C + c_{C^*} + k_B c_B + O(d_Q),$$

with c_{F^*} defined using k_{B^*}. One application of M_ρ costs

$$c_M = c_\Phi + c_{\Phi^*} + c_F + c_{F^*} + O(d_Q), \tag{374}$$

and one primal ridge solve costs $O(k_M c_M)$. A transition prediction costs $O(k_{\exp} c_\mathcal{L})$, where $c_\mathcal{L}$ is one block-generator action. If $\chi_{\text{ckpt}} \ge 1$ is the measured recomputation factor of the declared adjoint checkpoint schedule, its reverse pass costs $O(\chi_{\text{ckpt}} k_{\exp}(c_\mathcal{L} + c_{\mathcal{L}^*}))$ and stores only the checkpoint states and the current Arnoldi window. If k_{aug} is the augmented KKT iteration count, its solve cost is

$$O(k_{\text{aug}}[c_H + c_B + c_{B^*} + c_C + c_{C^*} + c_\Phi + c_{\Phi^*} + d_Q + d_Y])\,, \tag{375}$$

with no nested iteration multiplier. If the outer, complement, and augmented restart widths are r_M, r_B, r_{aug}, restarted nested storage is $O(d_Q r_M + d_Y r_B)$ per right-hand-side block and augmented storage is $O((d_Q + 2d_Y) r_{\text{aug}})$, in addition to the structured operator storage.

For comparison, a generic dense backend stores $O(d_Q^2 + d_Y^2 + d_Q d_Y)$ scalars and has cubic factorization cost in the active primal or dual dimension. Therefore the implementation selects the dense backend only when its measured factorization and memory estimate fit the declared budget; otherwise it uses the matrix-free backend. These are accounting identities in the complete resource record, not asymptotic assumptions about a candidate.

Proof. Equations (366)– (367) contain one B or B^* solve in each F or F^* action and one of each in an A action. Adding the two data-map actions gives (374); multiplication by the recorded iteration counts gives the nested work bounds. The KKT action of (368) contains one of each displayed primitive action, which gives (375). Restarted Krylov methods retain only their declared basis window. The transition reverse-pass count follows from (370): every recomputed forward or adjoint substep uses one generator action, and the checkpoint schedule records the number of such recomputations. Standard dense storage and factorization counts give the comparison. □

Definition II.11.51 (Scalable audit and execution schedule)**.** The implementation uses the following hierarchy without changing the returned record family.

(i) A finite response effective dimension always has the deterministic certificate

$$0 \le \operatorname{Tr}\Big[K(K + \rho I)^{-1}\Big] \le \operatorname{rank} K \le \min\{r, d_Q\}. \tag{376}$$

This bound requires no Gram materialization. A small problem may compute the trace exactly. A large problem may use a represented stochastic trace or Lanczos-quadrature interval only when its random seed, probe count, spectral enclosure, failure probability, and computation cost are fields of the Part I numerical certificate. Its failure probability is included in the simultaneous confidence allocation. An unvalidated randomized estimate is a diagnostic and cannot replace (376) in a risk certificate.

(ii) Capture, response, defect, and target-band quantities are evaluated by operator actions on the target vectors actually used by the certificate. A matrix-free eigenvalue, singular-value, projector, dark-space, or bracket calculation returns a validated interval from its residual and spectral enclosure. If such an interval is unavailable, the associated upper candidate is $+\infty$ and lower visibility is zero; the implementation does not request a full eigendecomposition solely to manufacture an ordinary value.

(iii) Candidates are parallelized only across sealed specifications or data-independent operator actions. Carrier identifiers and random-number streams are disjoint or shared exactly as declared by the confidence record. Parallel execution therefore changes wall-clock cost but not the statistical event.

(iv) Factorizations, preconditioners, Krylov subspaces, and warm starts are reused across right-hand sides, nearby charged ridge scales, and successive bridge iterates. Every cache key contains the specification hash, operator-parameter version, dtype, device, and relevant scale. Updating $H, G, A_{\mathcal{Y}}$, or C invalidates every response cache depending on it. Reuse across changed operators is a warm start, not an exact cache hit, and is followed by a fresh residual check.

(v) Training may use mixed precision and loose adaptive inner tolerances. The terminal parameters are recompiled and audited in the declared certificate precision. The discrepancy between the training and audit operators, together with every inexact-gradient residual, is stored in the optimization and operator-error coordinates.

(vi) The stage scheduler maintains lower and upper intervals for risk, visibility, resources, and action. It may stop or prune a candidate only when those intervals prove infeasibility or Pareto domination by an already audited record. Ambiguous candidates are refined. Thus successive halving, early stopping, branch-and-bound, and asynchronous execution are permitted only in this certified form.

(vii) Training checkpoints contain the specification hash, parameter and cache versions, optimizer state, data-stream position, random-number states, and current audit intervals. Resumption therefore continues the same transcript rather than creating an undeclared candidate.

The scheduler first applies algebraic and authority checks, then the cheap bounds in (376), then inner fitting, and finally high-precision certification of the unresolved Pareto set. This orders work from cheapest to most expensive while preserving the exhaustive Part I frontier.

Proposition II.11.52 (Soundness of certified screening)**.** *At a fixed typed stage, suppose a partial candidate record M has certified intervals*

$$R(M) \in [\underline{R}_M, \overline{R}_M], \qquad V(M) \in [\underline{V}_M, \overline{V}_M],$$

and lower resource/action vector $\underline{\mathbf{c}}_M$. It may be discarded for a request (ϵ, θ) if either $\underline{R}_M > \epsilon$ or $\overline{V}_M \leq \theta$, or if an audited feasible record M_0 satisfies

$$\mathbf{c}(M_0) \preceq \underline{\mathbf{c}}_M \quad \textit{and at least one coordinate is strict.}$$

Such screening leaves the Pareto-minimal feasible frontier unchanged.

Proof. The first two conditions prove that M cannot satisfy the risk or visibility constraint. In the third case every completion of M has resource/action vector at least $\underline{\mathbf{c}}_M$ and is therefore dominated by the feasible record M_0. None can be a Pareto-minimal feasible point. □

Definition II.11.53 (Residual acceptance and finite-precision transfer)**.** Let tildes denote the tensors returned by the numerical linear algebra. The audit stores certified upper enclosures of the absolute residuals

$$
\begin{aligned}
r_B &:= \|B_C\widetilde{R}_C - C\|, \qquad\qquad r_A(V) := \|A_C\widetilde{X} - V\|, &(377)\\
r_{\mathrm{con}} &:= \|\mathsf{B}\widetilde{z} - C\widetilde{f}\|,\\
r_f &:= \left\|\left(\frac{\Phi_{\mathrm{tr}}^*\Phi_{\mathrm{tr}}}{m_{\mathrm{tr}}} + \rho A_C\right)\widetilde{f} - \frac{\Phi_{\mathrm{tr}}^* y_{\mathrm{tr}}}{m_{\mathrm{tr}}}\right\|, &(378)\\
r_{\exp}(T) &:= \|u_0 - \widetilde{u}(0)\| + \int_0^T \|\dot{\widetilde{u}}(t) - \mathcal{L}_W\widetilde{u}(t)\|\,\mathrm{d}t,\\
r_{\mathrm{herm}} &:= \|A_C - A_C^*\|, \qquad\qquad r_{\mathrm{gram}} := \|K_{C,E} - K_{C,E}^*\|. &(379)
\end{aligned}
$$

Such an enclosure may be computed by directed rounding, interval arithmetic, or a proved backend error bound. A residual evaluated only in working precision is stored as a diagnostic and cannot by itself certify acceptance. The audit also stores scale-normalized versions of these residuals for diagnostics. Acceptance uses the declared absolute or mixed absolute-relative tolerances in `WickSpec`; no hard-coded tolerance enters the mathematical record. Put $\mathrm{Herm}(M) = (M + M^*)/2$. The structural audit additionally evaluates

$$
\begin{aligned}
r_H &= \|H - H^*\|, \qquad r_G = \|G - G^*\|, \qquad r_{\mathcal{Y}} = \|A\mathcal{Y} + A_{\mathcal{Y}}^*\|,\\
&\lambda_{\min}(\mathrm{Herm}\,H), \qquad \lambda_{\min}(\mathrm{Herm}\,G), \qquad \lambda_{\min}(\mathrm{Herm}\,A_C).
\end{aligned} \tag{380}
$$

Let

$$
M_f := \frac{\Phi_{\mathrm{tr}}^*\Phi_{\mathrm{tr}}}{m_{\mathrm{tr}}} + \rho A_C.
$$

The structural action identities or a validated eigensolver produce the record fields

$$
\underline{\alpha}_B \le \lambda_{\min}(\mathrm{Herm}\,B_C), \qquad \underline{\alpha}_A \le \lambda_{\min}(\mathrm{Herm}\,A_C), \qquad \underline{\alpha}_f \le \lambda_{\min}(\mathrm{Herm}\,M_f). \tag{381}
$$

For the matrix-free backend, the stored floating entries are interpreted as exact dyadic coefficients of the factor actions in (366)– (367); those identities give $\underline{\alpha}_B = \lambda$, $\underline{\alpha}_A = 1$, and $\underline{\alpha}_f = \rho$. Rounding in their evaluation is included in the residual enclosures. A backend that instead materializes rounded matrix products includes their rounding envelopes before forming the lower bounds. It may use sharper validated values. Every nonfinite value, shape or dtype mismatch, failed safe-row check, or residual outside tolerance produces `TypedFailure`. The same rule applies when one of the three lower bounds in (381) is unavailable or nonpositive. Jitter, clipping, projection, or eigenvalue truncation is permitted only when the modified operator and its perturbation radius are stored as fields of a new candidate.

The validated coercivity bounds give the a posteriori estimates

$$
\|\widetilde{R}_C - B_C^{-1}C\| \le \frac{r_B}{\underline{\alpha}_B}, \qquad \|\widetilde{X} - A_C^{-1}V\| \le \frac{r_A(V)}{\underline{\alpha}_A}, \qquad \|\widetilde{f} - \widehat{f}_{C,\rho}\| \le \frac{r_f}{\underline{\alpha}_f}. \tag{382}
$$

The audit propagates these quantities through (268)– (269) and stores the result as numerical operator and optimization error. Thus numerical acceptance never converts a residual into an exact identity. An augmented solve is accepted only after the audit evaluates both r_{con} and the reduced normal-equation residual r_f. Its terminal predictor therefore receives the same $r_f/\underline{\alpha}_f$ enclosure as the nested or dense solve. An implicit derivative through the indefinite KKT system uses a validated smallest-singular-value bound for that system or recomputes the adjoint through the reduced positive-definite equation.

Definition II.11.54 (Finite executable learner)**.** The reference program executes the following finite outer algorithm. Its enumeration and selection control flow is deterministic under the declared tie rule; any permitted backend nondeterminism is confined to a fitting transcript.

(E1) Load one charged stage and validate every immutable specification, carrier identifier, overlap declaration, grid, converter, and confidence allocation before fitting. Run the declared target-free or synthetic backend pilot and apply (384).

(E2) For each s, train or enumerate the dynamics candidate according to its decoder type and its declared finite stopping rule. Compile the terminal generator and evaluate the identification and response-error coordinates.

(E3) For each charged (λ, ρ, τ) and coefficient vector, call `response`. Dispatch to `fit_free`, `fit_rl`, or the imposed-PDE evaluation branch according to the authority tag.

(E4) Stop every optimization loop at its declared maximum iteration count or stopping predicate and store the actual stopping reason and terminal gap. A divergent or interrupted loop returns a terminal transcript with its optimization status.

(E5) Detach the terminal tensors and call `audit`. The audit recomputes all structural objects from the terminal parameters rather than trusting cached training values. It then evaluates every predeclared applicable Part I bound on the common confidence event.

(E6) Partition accepted and nonordinary records by population law, estimand, norm, and loss identifier. Within each partition call `select` and return the risk-qualified saturated Pareto frontier (364).

All enumerated sets and iteration limits are finite at stage n, so this algorithm terminates. Candidate failures remain in the finite audit ledger; only a malformed stage that cannot be parsed produces a stage-level failure. Write $\mathsf{C}_W = \mathsf{CompleteWickRecord}$. Its total return type is

$$\mathsf{WickReturn} := \left[\mathsf{FiniteLedger}(\mathsf{C}_W) \times \mathsf{ParetoFrontier}(\mathsf{C}_W)\right] \sqcup \mathsf{TypedStageFailure}. \tag{383}$$

The directed Part I stage iterator is external to this finite call and retains its existing limit and nonattainment semantics.

Theorem II.11.55 (Program refinement and certificate soundness)**.** *Let an execution conform to Definitions II.11.44 to II.11.46, II.11.51, II.11.53 and II.11.54. Then:*

(i) *in exact arithmetic, every successful numerical-core output equals the finite-chart mathematical object named by its interface;*

(ii) *in finite precision, every accepted output has the a posteriori errors in* (382)*, and every dependent certificate includes their propagated operator or optimization error; a transition output additionally has the enclosure* (369)*;*

(iii) *every returned ordinary record satisfies Theorem II.11.43, while every rejected or unavailable branch returns the corresponding totalized record;*

(iv) *the outer return is the Part I Pareto-minimal risk-qualified saturated frontier within each typed partition;*

(v) *dense and matrix-free ordinary executions have the same exact-arithmetic target, implicit differentiation has* (372)*, transition backpropagation has* (370)*, scalable work and memory obey* (374)*, and certified screening leaves the returned frontier unchanged.*

Consequently a conforming implementation is a refinement of the mathematical learner: it may change optimization method, batching, device, or linear-algebra backend only through fields recorded in the transcript and numerical audit. It may not change the structural selection rule or the meaning of optimality.

Proof. The tensor identities and exact-arithmetic solve properties are Proposition II.11.41. The free and RL dispatches use the same tensors, losses, frozen arguments, masks, and normalizations as Definitions II.11.21 and II.11.23; hence their successful exact-arithmetic outputs are the stated finite candidate objects. The operator actions in Definition II.11.46 are the factored forms of the same dense operators, so both backends have the same exact target. Implicit differentiation and its residual transfer are Proposition II.11.49; the work and memory statement is Proposition II.11.50; transition-action certification is Proposition II.11.47; and its scalable reverse pass is Proposition II.11.48. For finite precision, subtract the exact equations from (377)–(378) and use the three validated inverse bounds from (381). This proves (382). The perturbation bounds then produce the stored response errors.

The sealed specification and data-access validator preserve the candidate index, authority, loss type, sample law, and simultaneous confidence event. The detached audit therefore satisfies the hypotheses of Theorem II.11.43. Its totalized failure rules are Proposition II.11.3. Finally, steps (E5) and (E6) are exactly Definition II.11.42; the saturated-profile conclusion follows from Theorem II.11.31 and Corollary II.11.32. The pruning conclusion is Proposition II.11.52. □

The definitions above determine a direct implementation order: first implement the immutable records and compiler, then the solve backends, the branch-specific fits, the detached audit, and finally the typed saturated selector. Each layer has a local mathematical oracle in Table 6; no end-to-end training run is required to diagnose a failed algebraic layer.

Reference implementation plan. The implementation is divided into seven independently testable modules.

(P1) *Schemas and persistence.* Implement immutable `WickSpec`, versioned parameter and carrier handles, transcripts, audit records, typed failures, serialization, and specification hashing. This module contains no tensor algebra.

(P2) *Operator compiler.* Implement dense and `LinearOperator` actions, exact adjoint actions, authority decoders, streaming evaluation maps, generator actions, and exponential actions. Property tests verify structure and adjointness before a solver is introduced.

(P3) *Numerical solvers.* Implement dense factorizations, restarted and block Krylov solvers, the augmented KKT backend, preconditioner and cache versioning, residual enclosures, and implicit adjoint differentiation. Dense and matrix-free outputs are compared on small random and scalar instances.

(P4) *Learning branches.* Implement dynamics fitting, free-Caus bilevel fitting, action-safe fitted critics, actor fitting, and imposed PDE evaluation using only the preceding typed interfaces. No branch accesses an audit carrier through its training API.

(P5) *Audit and bounds.* Recompile detached terminal parameters, run high-precision residual checks, compute exact or certified interval response quantities, evaluate all applicable Part I same-loss candidates, and construct one complete record per candidate.

(P6) *Saturated selector.* Partition records by their law, estimand, norm, and loss identifier; implement resource/action dominance, certified screening, strict inverses, and frontier serialization.

(P7) *Parallel orchestration.* Schedule sealed candidates as independent tasks, stream datasets, checkpoint transcripts, collect audit records, and run the selector after task completion. The coordinator never modifies a candidate specification after target-bearing data have been read.

Each module is released only after its rows of Table 6 pass. The scalar model of Example II.11.58 is the first integration fixture; dense versus matrix-free equality is the second; a sparse large-chart instance that cannot materialize F_C or $K_{C,E}$ is the required scalability fixture.

Required automated check	Acceptance criterion
Shape, dtype, and authority property tests	Every tensor satisfies Table 5; decoded bridges lie in the represented authority image.
Compiler identity tests	Symmetry, positivity, skew symmetry, energy cancellation, and residuals in (380) meet the declared tolerances; the outward-validated lower bounds in (381) are positive.
Transition-action tests	On small charts, dense and Arnoldi actions agree within (369); adjoint gradients agree with directional derivatives within (371); a large fixture respects the declared checkpoint-memory ceiling.
Block-elimination tests	Direct full-block solves agree with the compressed Feshbach solve within the propagated residual bound.
Kernel tests	Gram matrices are Hermitian positive semidefinite; range-compressed matrices are positive definite; empirical effective dimensions lie in $[0, r]$.
Matrix-free and augmented tests	Every primitive action passes an adjoint dot-product test; dense, nested, and augmented predictions agree within the propagated residual enclosure; the augmented constraint and reduced normal-equation residuals both pass.
Resource-regression tests	A large sparse fixture completes under the declared peak-memory ceiling; allocation tracing confirms that it never materializes F_C, $F_C^* F_C$, $\Phi^*\Phi$, or a full response Gram matrix.
Free-Caus loss tests	Primal and dual KRR predictions agree on the represented evaluation range; automatic gradients agree with directional derivatives of the full outer loss within the declared derivative-test tolerance.
RL semantic tests	Terminal rows remove continuation, unsafe rows receive zero policy mass, empty safe rows return the authority status, targets are detached during critic fitting, and the full-carrier response is reused across minibatches.
Selection tests	Synthetic records with known partial order return exactly all and only the Pareto-minimal feasible records, separately for every typed loss partition.
Failure-injection tests	Nonfinite solves, excessive residuals, missing converters, unsupported carrier overlaps, zero safe support, and unavailable certificates return the specified typed status with upper bound $+\infty$ and lower visibility zero where ordered.
Reproducibility tests	The specification hash, seed, carrier identifiers, minibatch order, terminal tensors, and audit output are stored; deterministic backends reproduce the declared tolerance class.

Table 6: Minimum verification suite for a conforming implementation. These tests verify program refinement and failure polarity. Statistical coverage is provided by the selected Part I certificate on its allocated event, not by a software test.

Definition II.11.56 (Cost-aware backend dispatch)**.** Let $\mathcal{B}_s$ be the dense, nested matrix-free, augmented, primal, and dual backend combinations implemented for specification s. Before target-bearing fitting, a target-free pilot records projected work $\widehat{W}_b$, peak memory $\widehat{M}_b$, communication $\widehat{C}_b$, and whether backend b reached the pilot residual tolerance. Let $\preceq_s^{\mathrm{impl}}$ be the declared total tie order on the projected work–memory–communication triples, refining their Pareto order. The dispatcher chooses

$$\widehat{b} = \min_{\preceq_s^{\mathrm{impl}}} \left\{ b \in \mathcal{B}_s : \widehat{M}_b \leq M_s,\ b \text{ passed the pilot} \right\} \tag{384}$$

An empty feasible backend set returns the numerical-budget status. Pilot cost and the selected backend are stored in the resource record.

Proposition II.11.57 (Backend dispatch does not alter structural selection)**.** *Every backend in $\mathcal{B}_s$ targets the same dense finite-chart equations. Consequently* (384) *is an inner implementation choice for the fixed candidate s. Residual acceptance makes its terminal operator, prediction, and gradient discrepancies record coordinates. Structural comparison remains the risk-qualified saturated profile applied to the completed records.*

Proof. Dense, nested, augmented, primal, and dual equality follows from Definition II.11.46 and the representer identity. The finite-precision differences are bounded by Definition II.11.53. Hence backend dispatch changes measured resource and numerical-error coordinates but neither the candidate family nor the outer order. □

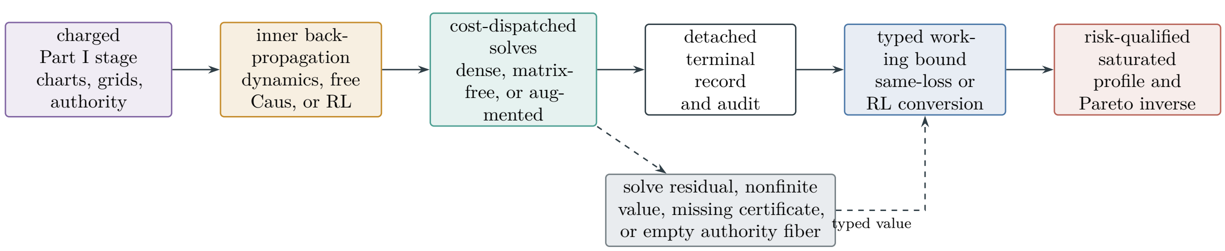

Figure 18: Empirical realization of the Wick learner. Differentiation is confined to the inner candidate fit. The detached record is audited under the inherited confidence allocation, all applicable same-loss bounds are combined by their valid minimum, the RL bound is separately calibrated into value units, and complete records are ordered only within their typed risk-qualified saturated profile and Pareto strict inverse.

II.11.10 A solvable two-sector model

Example II.11.58 (Scalar Wick quotient)**.** Let $\mathcal{Q} = \mathcal{Y} = \mathbb{C}$, $H_{\mathsf{Abs}} = \omega \geq 0$, $G = g > 0$, $A_{\mathcal{Y}} = 0$, and $C = c \in \mathbb{R}$. Then

$$\mathcal{L}_W(c) = \begin{pmatrix} -i\omega & c \\ -c & -g \end{pmatrix}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(|x|^2 + |y|^2) = -2g|y|^2. \tag{385}$$

The quotient defect is $|c|$, the dark space is all of $\mathbb{C}$ when $c = 0$ and zero otherwise, and

$$\Sigma_c(z) = \frac{c^2}{z+g}, \qquad F_c(z) = z + i\omega + \frac{c^2}{z+g}, \qquad \eta_c^W(\lambda) = \frac{1}{1 + \left(\lambda + \dfrac{c^2}{\lambda+g}\right)^2 + \omega^2}. \tag{386}$$

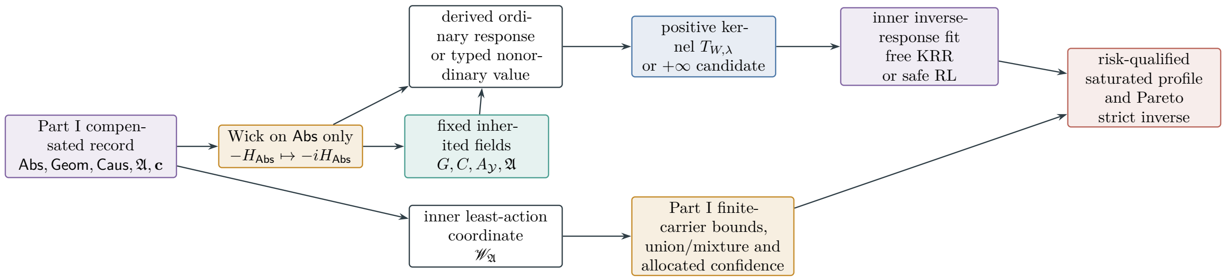

Figure 19: Wick learning as a derived operation on a Part I record. The Wick map changes only the spectral Abs evolution. The real Geom, actual authorized Caus bridge, authority envelope, provenance, and charges are fixed inherited fields. On every ordinary Part I realization, Feshbach elimination produces the self-energy and positive response kernel for every $\lambda > 0$, as a consequence of maximal dissipativity; a nonordinary realization returns its typed value and $+\infty$ response candidate. The unchanged bridge supplies the action record. Inverse-response KRR or action-safe RL trains the applicable authority branch. The Part I Rademacher union/cover or PAC–Bayes mixture/coefficient rule then uses finite evaluation-carrier traces to form the risk coordinate. The risk-qualified saturated usefulness profile then returns the Pareto-minimal points of its strict resource–action inverse at the inherited visibility threshold.

The poles are the roots of

$$(z + i\omega)(z + g) + c^2 = 0. \tag{387}$$

Take the represented current coordinate to be $J_c = c$, the mobility to be $a > 0$, and $J_D = -a\nabla D$. Then

$$\mathscr{W}(c; D) = \frac{(c - J_D)^2}{2a}.$$

Free authority selects $c = J_D$; a restricted interval $[c_-, c_+]$ selects the Euclidean projection of J_D onto that interval; and an imposed PDE coefficient $c = c_0$ retains the fixed waste $(c_0 - J_D)^2/(2a)$. The same c occurs in the defect, self-energy, response weight, and action record.

Remark II.11.59 (Relation to familiar physics)*.* The exact correspondences proved here are spectral Wick rotation, Feshbach–Schur elimination, Mori–Zwanzig memory, conservative coupling, and real geometric dissipation. They explain why the resulting structural presentation has the same operator shapes as effective Hamiltonians, system–bath models, and Euclidean-to-spectral descriptions. Quantum measurement postulates, complete positivity of density-matrix evolution, gauge-field reconstruction, and completeness of a finite set of physical constants require additional axioms and are outside these theorems.

Part III

Structural Reachability and Regularity

III.1 Singular targets and cascade compilers

The algebra and normalized shell identity first evaluate whether target-aligned structure exists. Its absence excludes contact directly. On the surviving aligned branch, contact itself generates the canonical infinite disjoint shell family, and (400) generates its sharp lower-price series in the single physical measure. Divergence excludes the source cell; finite total price generates zero-price tails and activates the compact successor relation. Shell accountability, cascade extraction, physical pricing, and ordered proof/model evaluation are consecutive constructive operations.

Definition III.1.1 (Ordinary continuation target and source-complete shell closure)**.** Let $\mathsf{Cont}^{\mathrm{tot}}$ be the total continuation relation of Definition I.1.1. For a finite maximal orbit, its terminal germ is the tuple of all limits of the generated evaluation coordinates along times increasing to T_*, together with the origin tag and the cofinal same-orbit restriction sequence; compact metrizability supplies such germ subsequences in the source-complete graph space. Let $\mathscr{B}^{\mathrm{evt}}_{\mathrm{cont}}$ be the set of terminal events consisting of an ordinary same-origin maximal preterminal path, its finite endpoint time, the value "no ordinary regular extension" in $\mathsf{Cont}^{\mathrm{tot}}$, and its complete cofinal restriction record. The terminal analytic germ is retained as an ordinary or nonordinary total graph coordinate of that event; the event does not require such a germ to be an ordinary state in the declared regularity class. The ordinary-extension, failure, and graph-boundary status is retained as a finite discrete coproduct coordinate of X_{ev}. Define

$$\Sigma_{\mathrm{sing}} := \Sigma^{\mathrm{ord}}_{\mathrm{sing}} := \mathscr{B}^{\mathrm{evt}}_{\mathrm{cont}}, \qquad \widehat{\Sigma}_{\mathrm{sing}} := \overline{\mathscr{B}^{\mathrm{evt}}_{\mathrm{cont}}}^{X^{\mathrm{fail}}_{\mathrm{ev}}}, \qquad X^{\mathrm{fail}}_{\mathrm{ev}} := \{x : \mathrm{ContStatus}(x) \neq \mathsf{ordinary}\}. \tag{388}$$

The first set is the physical singular-event target; the second is its source-preserving shell closure. Physical contact means either same-origin preterminal approach to every shell of the event target or arrival of the ordinary preterminal path at its recorded noncontinuation event. In the second case the completed restriction convention in (P4) supplies the first form as well; a genuinely discontinuous target interface outside that convention would have a Bdry-headed terminal jump. Every finite maximal noncontinuation of an ordinary path is therefore contact with the event target and generates a cofinal source-complete shell record. This does not promote its terminal analytic coordinate to an ordinary state. An arbitrary compactified boundary point is not itself physical contact: its preterminal path and noncontinuation event must pass Theorem I.1.7. The origin coordinate prevents a terminal germ of one orbit from manufacturing contact for another, while the discrete continuation-status coordinate prevents an extendible germ from entering the target merely as an untagged analytic limit. An empty local-solution fiber remains the separate form defect $\perp_{\mathrm{loc}}$; it is not target avoidance.

Proposition III.1.2 (Ordinary terminal coverage and source-complete shell typing)**.** *Every ordinary maximal orbit has exactly one of the following outcomes: $T_* = \infty$, ordinary regular continuation, or a same-origin finite noncontinuation event. The third outcome is physical contact with $\Sigma^{\mathrm{ord}}_{\mathrm{sing}}$, and its terminal restrictions cross a cofinal shell family for $\widehat{\Sigma}_{\mathrm{sing}}$. The terminal graph coordinate of that event represents an ordinary analytic germ only when it passes Theorem I.1.7. Conversely, a closure point not accompanied by one ordinary same-origin preterminal path and its noncontinuation flag is not physical contact. Thus terminal coverage does not promote a compactification point to a PDE orbit or analytic germ.*

Proof. At finite T_*, the total continuation graph has either an ordinary extension or its noncontinuation value. An ordinary extension contradicts maximality unless it is recorded as continuation. The status coordinate is discrete, so continuation and noncontinuation events are disjoint. The completed restriction convention in (P4) attaches the same noncontinuation event to every cofinal restriction and therefore supplies its cofinal shell record. Compact metrizability supplies convergent analytic graph coordinates along a subsequence, but those coordinates are merely part of the already realized event. A tower satisfying (D1)–(D4) is sufficient to realize the relaxed analytic limit as an ordinary terminal germ, and those conditions are exact whenever realization is claimed through the public approximation grammar. $\square$

Definition III.1.3 (Singularity-formation support)**.** For an ordinary maximal orbit u, let $\mathscr{D}_\Sigma(u)$ be a family of pre-contact shell records

$$R_{r,m} = (I_r, \widehat{\phi}_{r,m,I_r}, (Q^c_{r,m,I_r})_{c\in\mathfrak{C}})$$

attached to that same orbit. The family is *shell-complete* when contact implies that it contains such a record for every sufficiently small r. Every record generated by contact then satisfies

$$\Delta\widehat{\phi}_{r,m,I_r} = 1, \qquad \begin{cases} \sum_{c\in\mathfrak{C}} Q^c_{r,m,I_r}(1) = 1, & \text{on a finite-value record,} \\ \overline{Q}^o_{r,m,I_r}(1) \geq 1/5, & \text{on a variation-boundary record.} \end{cases} \tag{389}$$

The singularity-formation support is

$$\mathfrak{S}_\Sigma(u) := \bigcup_{R\in\mathscr{D}_\Sigma(u)} \mathrm{Align}_\Sigma(R) \subseteq \mathfrak{C}. \tag{390}$$

It answers the target-relative question: which inherited source currents actually have positive pairing with the covectors descending toward the singular set? It does not ask which terms of the PDE are large in a target-independent norm.

Proposition III.1.4 (A singularity requires aligned structure)**.** *If u is generated by the compiled continuum algebra and $\mathscr{D}_\Sigma(u)$ is shell-complete, then*

$$u \text{ contacts } \Sigma \quad \Longrightarrow \quad \mathfrak{S}_\Sigma(u) \neq \varnothing. \tag{391}$$

Consequently $\mathfrak{S}_\Sigma(u) = \varnothing$ implies target avoidance. This implication is independent of the size or compactness of the full orbit.

Proof. Contact supplies normalized records satisfying (389). By Theorems I.5.3 and I.5.15, each has a nonempty aligned support, contained in the union (390). The contrapositive gives target avoidance. $\square$

Theorem III.1.5 (Canonical disjoint shell cascade)**.** *Let a compiled ordinary maximal branch start outside Σ and contact Σ through a cofinal pre-contact sequence at its first contact or maximal time, and let (U_r, Φ_r) be a target-shell presentation. There are scales $r_j \downarrow 0$ and pairwise disjoint, chronologically ordered, strictly pre-contact intervals $I_j = [a_j, b_j]$ such that*

$$u(a_j) \notin U_{2r_j}(\Sigma), \qquad u(b_j) \in U_{r_j}(\Sigma), \qquad \Phi_{r_j}(u(b_j)) - \Phi_{r_j}(u(a_j)) = 1. \tag{392}$$

Every interval therefore has a nonempty aligned support and a source occurrence with positive target progress. On every ordinary or boundary current graph, one of the five source-resolved contributions has positive part at least $1/5$. No compactness, price gap, or contact margin is used in this extraction.

Proof. Set $b_0 = 0$ and $r_0 = 1$. Since every strictly pre-contact state is outside $\Sigma = \bigcap_{r>0} U_{2r}(\Sigma)$, after b_{j-1} choose a regular time $a_j > b_{j-1}$ and then a scale $r_j < \min\{r_{j-1}/4, 1/j\}$ for which $u(a_j) \notin U_{2r_j}(\Sigma)$. Contact supplies a later strictly pre-contact time at which the orbit belongs to $U_{r_j}(\Sigma)$; choose one as b_j. This inductively gives $b_{j-1} < a_j < b_j$, hence pairwise disjoint intervals. No continuity premise is needed: the two membership choices give $\Phi_{r_j}(u(a_j)) = 0$ and $\Phi_{r_j}(u(b_j)) = 1$ directly. Apply Theorem I.5.3 on each interval to get the final assertions. $\square$

Definition III.1.6 (Generated renormalization-scale row)**.** Let z_* be a contact point or germ. The normalization grammar is the least family containing the identity and the space, time, amplitude, gauge, and quotient transformations for which the fixed PDE's totalized covariance graphs have ordinary values. Each failed covariance value is a Lift- or Abs-tagged graph defect. Applying every ordinary grammar value to the canonical shell cascade produces scales $r_j \downarrow 0$, physical pre-contact events $Q_j = Q(z_*, r_j)$, and records

$$V_j = \mathcal{S}_{z_*,r_j} u \in X_{\mathrm{eval}},$$

where every $\mathcal{S}_{z,r}$ has its generated $\mathsf{Abs}/\mathsf{Lift}$ ancestry. Pass to the least infinite constant source-support and realization-mode subsequence in the fixed enumeration; it exists because both alphabets are finite. Let $\mathfrak{b}$ denote the resulting same-origin active branch signature. All closures and prices in this row are restricted to $X_{\mathfrak{b}}$. The row returns:

(R1) the geometric-separation value obtained, after the canonical dyadic subsequence, from $r_{j+1} \le \vartheta r_j$ for some $0 < \vartheta < 1$ and the events Q_j are pairwise disjoint;

(R2) the totalized target-persistence inclusion $\overline{\{V_j\}} \subseteq B_{\Sigma,\mathfrak{b}}$, where $\mathfrak{b}$ is the active branch extracted from this same-origin cascade; the complementary value is a target-persistence graph defect;

(R3) the total-graph value of the physical pullback relation

$$\mu_u(Q_j) \ge c_{\mathrm{sc}}^* r_j^{\kappa} \overline{d}_{\mathrm{sc}}(V_j), \tag{393}$$

Here c_{sc}^* is the sharp transfer infimum, κ is the exponent returned by the covariance graph, and $\overline{d}_{\mathrm{sc}}$ is the canonical lower-semicontinuous relaxation. If no scalar exponent represents the pullback, the row returns the full scale-dependent weight sequence or its graph defect. Thus scale covariance is evaluated rather than postulated.

Theorem III.1.7 (Scale-cascade price compiler)**.** *Evaluate the totalized scale interface on its generated active branch* $\mathfrak{b}$*. Its normalized price value is*

$$\delta_{\mathrm{sc}} := \inf_{V \in B_{\Sigma,\mathfrak{b}}} \overline{d}_{\mathrm{sc}}(V) \in [0, \infty]. \tag{394}$$

For the (j)-th generated scale stratum, define its sharp physical lower price exactly as in (400) *and denote it by* $\underline{p}_{\mathrm{sc},j}$*. The exact scale-price value is*

$$\mathscr{S}_\Sigma := \sum_{j=1}^{\infty} \underline{p}_{\mathrm{sc},j} \in [0, \infty]. \tag{395}$$

On the ordinary scalar-pullback branch, whenever the displayed factors are finite, (393) *gives*

$$\underline{p}_{\mathrm{sc},j} \ge c_{\mathrm{sc}}^* r_j^{\kappa} \delta_{\mathrm{sc}}. \tag{396}$$

The branch $\mathscr{S}_\Sigma = +\infty$ *gives*

$$B_{\mathrm{phys}} \ge \mu_u\left(\bigsqcup_{j=1}^{\infty} Q_j\right) = \sum_{j=1}^{\infty} \mu_u(Q_j) \ge \sum_{j=1}^{\infty} \underline{p}_{\mathrm{sc},j} = \mathscr{S}_\Sigma = \infty, \tag{397}$$

contradicting a finite physical budget. The stronger condition $a_* := \inf_j r_j^{\kappa} > 0$ *gives the uniform event price* $\mu_u(Q_j) \ge c_{\mathrm{sc}}^* a_* \delta_{\mathrm{sc}} > 0$ *on the positive transfer and price branches. On the positive transfer and normalized-price branches, after restricting to* $r_j \le 1$*, divergence is automatic when*

$\kappa \le 0$, and the scale-critical case $\kappa = 0$ transfers the normalized gap without loss. When $\kappa > 0$, the exact criterion is divergence of the generated lower-price series; a normalized gap alone supplies no such conclusion. The complementary value $\mathscr{S}_\Sigma < \infty$ produces vanishing-price windows in its inherited cell and is routed by Theorem II.2.58. In particular, a positive dissipative pullback exponent cannot be converted into a scale-independent price: the scale motion is retained in the signed carrier cohomology and then tested by the successor-saturation algebra.

Proof. The definition of each sharp lower price gives $\mu_u(Q_j) \ge \underline{p}_{\mathrm{sc},j}$. Sum over the pairwise disjoint events to obtain (397); countable additivity and the finite physical budget give the contradiction. On the scalar pullback branch, insert (394) into (393) to obtain (396). If $0 < r_j \le 1$, then $r_j^\kappa \ge 1$ for $\kappa \le 0$. For $\kappa > 0$, the individual weights tend to zero, so their summability must be decided from the actual scale sequence. □

Proposition III.1.8 (Canonical blow-up invariant image)**.** *Every contact cascade and every ordinary value of the generated normalization grammar has the canonical invariant image*

$$\left(\vartheta, \overline{\{V_j\}}, \mathfrak{H}^\Sigma_{\{V_j\}}, \kappa, \delta_{\mathrm{sc}}\right). \tag{398}$$

Here $\vartheta \le 1/4$ is produced by the canonical shell subsequence, the closure is taken in the generated evaluation compactification, $\mathfrak{H}^\Sigma_{\{V_j\}}$ records whether the cascade remains target-visible, κ or its full weight-sequence replacement is the physical pullback value, and δ_{sc} is the canonical price infimum. A failed normalization or pullback occupies the corresponding $\mathsf{Lift}/\mathsf{Abs}$ *graph-defect cell. Thus every blow-up outcome has an image; none of the five coordinates is additional problem data.*

Definition III.1.9 (Normalized cascade refinement)**.** For a contact orbit, combine the canonical intervals from Theorem III.1.5 with every ordinary value of the generated normalization grammar, then pass to the least infinite constant source-support and realization-mode subsequence. Let $\mathfrak{b}$ be its same-origin active branch signature. The resulting normalized cascade refinement consists of

(C1) pairwise disjoint measurable pre-contact events $Q_1, Q_2, \dots$ in the domain of the same physical expenditure measure μ_u;

(C2) generated normalizations $\mathcal{N}_{Q_j}$, carrying their $\mathsf{Abs}/\mathsf{Lift}$ ancestry, such that $w_j := \mathcal{N}_{Q_j} u \in B_{\Sigma,\mathfrak{b}}$, for the persistent active branch $\mathfrak{b}$ selected from this same-origin cascade;

(C3) for every j, a normalized shell-crossing record R_j carried by w_j for which

$$\Delta\widehat{\phi}(R_j) = 1, \qquad \mathrm{Prog}_\Sigma(R_j) \ge \frac{1}{5}. \tag{399}$$

On a finite-value record the latter quantity is the five-term sum, equals one, and has a component at least $1/5$; on a variation-boundary record it is the retained owner progress from (54).

By Theorem I.4.6 and the applicable prospective shell-accountability theorem, every occurrence in clause (C3) has a nonempty aligned source support $A_j := \mathrm{Align}_\Sigma(R_j) \subseteq \mathfrak{C}$. Every nonordinary normalization value is already a typed graph-defect record. The refinement asserts no positive price; it generates the family on which the price row is evaluated.

Proposition III.1.10 (Persistent channel extraction)**.** *For every cascade generated by Definition III.1.9, some nonempty support $A \subseteq \mathfrak{C}$ occurs for infinitely many indices. In particular, at least one of the five elementary source labels belongs to the ancestry of infinitely many target-approaching windows.*

Proof. There are only $2^5 - 1$ nonempty supports. The infinite pigeonhole principle gives an infinite constant-support subsequence. Any member of that support has the second asserted property. $\square$

Definition III.1.11 (Target-aligned source ledger)**.** Fix an ordinary maximal branch u and a normalized cascade family $(Q_j, R_j)_{j\geq 1}$. For every nonempty $A \subseteq \mathfrak{C}$, let $\mathcal{F}_A(u)$ be the family of source-current occurrences extracted from those R_j for which $\mathrm{Align}_\Sigma(R_j) = A$. The source ledger is generated as follows. For $j \in I_A := \{j : \mathrm{Align}_\Sigma(R_j) = A\}$, let the compiler first evaluates the total graph of the target-aligned charging relation (98). On its ordinary admitted value, let $\mathcal{Q}^{\mathrm{pay}}_{A,j}$ be the family of all generated ordinary events under the same physical ceiling and in the same propagated source/mode, shell/scale, and carrier-factorization stratum as Q_j, normalized to the same unit shell increment and source-owner floor. Equality of stratum means equality of every transcript entry preceding and including the charging row, so the equivalence relation is canonical. Define the sharp admissible physical lower price

$$\underline{p}_{A,j} := \inf\{\mu_v(Q) : (v, Q) \in \mathcal{Q}^{\mathrm{pay}}_{A,j}\} \in [0, \infty], \tag{400}$$

with $+\infty$ for an empty stratum. The stratum contains (u, Q_j), so $\mu_u(Q_j) \geq \underline{p}_{A,j}$. Each source row has exactly one of the following exhaustive entries:

(L1) a *null entry*,

$$\mathfrak{H}^{\Sigma}_{\mathcal{F}_A(u)} = 0; \tag{401}$$

(L2) a *divergent-price entry*, when the target quotient is nonzero and

$$|I_A| = \infty \Longrightarrow \sum_{j\in I_A} \underline{p}_{A,j} = \infty; \tag{402}$$

(L3) a *finite-price successor entry*, when the target quotient is nonzero and either the charging relation has a nonordinary or failed graph value, or its ordinary admitted value satisfies

$$\sum_{j\in I_A} \underline{p}_{A,j} < \infty. \tag{403}$$

The target quotient, the totalized charging relation, and the dichotomy for a nonnegative series make the list exhaustive. A charging defect retains the same source, shell covector, and event origin and enters the successor algebra; it is never assigned price $+\infty$. Quotient, symmetry, conservation, carrier, resistance, compactness–rigidity, equality, scale, and entropy calculations are evaluators of these exact entries. A uniform positive lower price is a special divergent-price value. Since $|\mathfrak{C}| = 5$, the ledger has at most $2^5 - 1 = 31$ nonempty rows even when the cascade contains infinitely many scales.

Theorem III.1.12 (Physical expenditure exclusion)**.** *Let μ be a nonnegative countably additive measure with $\mu(\Omega) \leq B_{\mathrm{phys}} < \infty$. Let $Q_j \subseteq \Omega$ be pairwise disjoint measurable sets and let $\delta_j \geq 0$. If*

$$\mu(Q_j) \geq \delta_j \quad \textit{for every } j, \qquad \sum_{j=1}^{\infty} \delta_j = \infty, \tag{404}$$

then such a family cannot exist. The uniform-price criterion $\delta_j \geq \delta > 0$ is a special case.

Proof. For every N, finite additivity and monotonicity give

$$B_{\mathrm{phys}} \geq \mu\left(\bigsqcup_{j=1}^{N} Q_j\right) = \sum_{j=1}^{N} \mu(Q_j) \geq \sum_{j=1}^{N} \delta_j.$$

The partial sums on the right are unbounded by (404), contradicting the fixed upper bound B_{phys}. $\square$

Corollary III.1.13 (Shell progress implies expenditure exclusion)**.** *On one ordinary same-origin active contact branch* $\mathfrak{b}$*, let* R_j *be the normalized shell records on pairwise disjoint pre-contact events* Q_j*. On the ordinary admitted value of the target-local charging row,*

$$\Delta\widehat{\phi}(R_j) = 1, \qquad \mathrm{Prog}_\Sigma(R_j) \geq 1/5 \quad \Longrightarrow \quad \mu_u(Q_j) \geq \delta_j \geq 0. \tag{405}$$

If $\sum_j \delta_j = \infty$*, then* u *cannot contact* Σ*.*

Proof. Contact supplies the unit shell increments and source-owner floor by Theorem I.5.3. The price implication is a structural fact on $X_{\mathfrak{b}}$ by Theorems II.2.4 and II.2.18. It gives (404), and Theorem III.1.12 gives the contradiction. $\square$

Theorem III.1.14 (Finite structural-ledger exclusion)**.** *Let* u *have the canonical contact cascade of Theorem III.1.5. On the fast-track value where every persistent target-aligned support is either zero in the target quotient or has a divergent physical-price series,* u *does not contact* Σ*.*

Proof. On the contact branch, Proposition III.1.10 gives a nonempty $A \subseteq \mathfrak{C}$ for which $I_A := \{j : \mathrm{Align}_\Sigma(R_j) = A\}$ is infinite. If A has a null entry, then every target projection of the corresponding source-current occurrences vanishes. This contradicts $A = \mathrm{Align}_\Sigma(R_j) \neq \varnothing$. If A has a divergent-price entry, the pairwise disjoint events $(Q_j)_{j\in I_A}$ carry the divergent lower-price schedule $(\underline{p}_{A,j})_{j\in I_A}$, contradicting Theorem III.1.12. The ledger alternatives are exhaustive, so contact is impossible. $\square$

Corollary III.1.15 (Estimate-to-invariant reduction)**.** *In the setting of Theorem III.1.14, the canonical transcript evaluates each persistent source support as null, divergent-price, or finite-price successor. By Theorem II.1.19, no source-transverse estimate of the full orbit remains in the verdict after these row values are known. In particular, arbitrarily large target-null motion is absent from the reduced calculation. The last value is not a verdict: it activates Theorem II.2.58.*

Corollary III.1.16 (Entropy exclusion is physical expenditure exclusion)**.** *On the entropy-cocycle, finite-error, carrier-domination, and positive-production rows of one same-origin active contact branch* $\mathfrak{b}$*,* $\mathfrak{D}_j(I_j) \geq \delta > 0$ *on every selected window and contact is impossible.*

Proof. By Corollary II.10.6, the glued production is a finite physical measure on this branch. Its use as the price of the selected shell events is admitted by Theorem II.2.4. Apply Theorem III.1.12 to its restrictions to the disjoint same-origin events Q_j. $\square$

III.2 The continuous structural regularity theorem

The five-channel conclusion is derived before any regularity estimate. The only equation-dependent step in target exclusion is the physical evaluation of the target-aligned class exposed by contact.

Definition III.2.1 (Totalization of a continuum interface)**.** Let $F : D(F) \subset X_F \to Y_F$ be any partial interface used by the continuum argument: a constitutive realization, a gradient potential

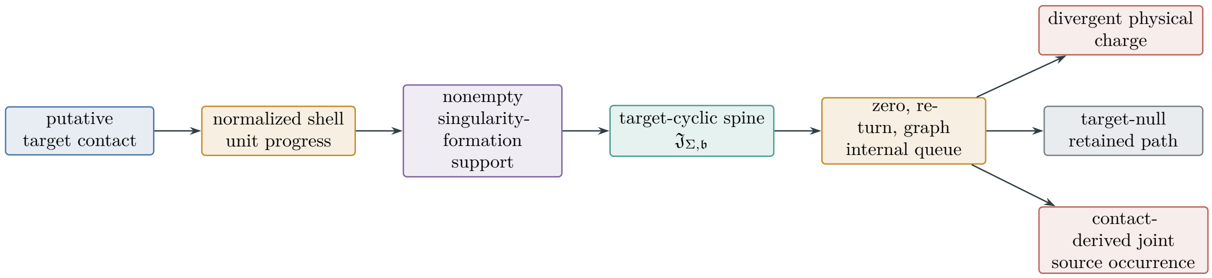

Figure 20: Compressed singularity-formation accounting. Contact exposes one target-aligned source cell. Projection-first saturation and signed gluing produce the target-cyclic feeding–storage spine. Zero-kernel restrictions, signed-dyadic return maps, and graph-boundary successors are internal queue transitions. On the contact-seeded branch the queue has three outputs: divergent physical charge, target-nullity, or a contact-derived finite joint source occurrence. Its ordinary same-origin realization is evaluated in the corresponding reachable contact cell.

map, a target-shell derivative, a physical-price comparison, or a continuation map. Adjoin a failure value $\perp_F$ and define

$$F^{\mathrm{tot}}(x) = \begin{cases} F(x), & x \in D(F), \\ \perp_F, & x \notin D(F). \end{cases}$$

Embed the graph of this total map in the structural test compactification and set

$$\partial F := \overline{\operatorname{graph} F^{\mathrm{tot}}} \setminus \operatorname{graph} F. \tag{406}$$

Thus ∂F contains both limiting graph defects and the explicit failure records $(x, \perp_F)$. The *totalized interface record* of an input is its graph value when the input lies in $D(F)$, and otherwise its class in the closed defect cell generated by ∂F. The defect cell retains the ancestry of the approximating graph records; failure at a chart, weak-limit, or terminal interface appends Lift or Bdry according to Definition I.2.1. Its target-null part is quotiented by $\mathscr{N}_\Sigma$; its nonzero target quotient is a target-visible defect class in the same five-channel algebra.

Definition III.2.2 (Total continuum interface algebra)**.** The target-relative continuum interface algebra is the finite coproduct

$$\mathfrak{H}^{\Sigma}_{\mathrm{cont}} := \mathfrak{H}^{\Sigma}_{\mathrm{form}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{grad}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{chain}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{price}} \sqcup \mathfrak{H}^{\Sigma}_{\mathrm{cov}}. \tag{407}$$

Here the five cells are respectively the target-active quotients of:

(D1) the graph boundary of the local-solution map and of the native constitutive or sector-form realization;

(D2) $\overline{\operatorname{Ran} G} \setminus \operatorname{Ran} G$, together with the target-active part of $\ker G^*$, after the exact target-null quotient;

(D3) the source-resolved derived Stieltjes shell currents of (38), including their ordinary, Lebesgue, jump, Cantor, oscillation, concentration, exterior, and variation-boundary graph coordinates;

(D4) the total carrier graph, including failure to produce the finite physical measure dictated by the equation, and the zero-price limit of every finite-expenditure shell cascade;

(D5) the generated chart, gauge, scale, and quotient-covariance graphs, including every commutator defect created when an operation fails to descend.

Every cell is source-tagged before its target-null quotient is taken. A missing property is therefore an input to the activity lift and successor operator, not a terminal verdict. Regular continuation is not a sixth interface cell: by Definition III.1.1 and Proposition III.1.2, a finite maximal branch without an ordinary continuation generates a cofinal source-complete shell record and is processed by the same shell and successor machinery. Its compactified germ is physical target contact only after the ordinary realization gate.

Theorem III.2.3 (Defect totalization and five-channel closure)**.** *For every fixed continuum evolution, target, and physical carrier, each of the five interfaces in Definition III.2.2 has an ordinary total-graph value or a graph-boundary value. Every target-visible value has ancestry in* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$*; every target-invisible value lies in* J_{res}*. Every target-visible value, ordinary or defective, is converted into an activity measure and a witness-indexed successor. Hence lack of closed range, a classical chain rule, positive price, or covariance is not a stopping point: it is a concrete successor mechanism in the same closed algebra. Finite noncontinuation is target contact, not an unprocessed remainder.*

Proof. For each partial map, the total value is either an ordinary graph value or the failure value $\perp_F$. Closing the total graph adds its limiting defects, so (406) exhausts both kinds of failure. Sourcewise closure and Theorem I.3.4 retain the ancestry of every approximating record. The gradient cell is exhausted by Theorem I.3.8. For the shell interface, the finite approximation grammar gives the exact polynomial Stieltjes identity (38). Sourcewise graph closure passes the affine identity to every finite Radon value. At a variation-boundary value, the joint shell graph instead preserves the constant canonical owner and its closed target floor (41); no undefined extended signed sum is used. Sourcewise closure preserves the ancestry of every current and there is no untyped chain-rule remainder. A finite-price cascade enters Definition II.2.40; a carrier-graph failure is lifted by the same total-graph coordinate. Covariance and descent failures retain the tags of the operation and commutator that generated them. Finally quotient each cell by its intersection with $\mathscr{N}_\Sigma$. The kernel is residual. Every nonzero quotient value enters $\mathcal{R}_\eta$, and Definition II.2.51 iterates it until it is eliminated or lies on an infinite zero-price path. Terminal coverage and the realization gate handle finite noncontinuation without promoting a source-complete germ. This proves all assertions. □

Theorem III.2.4 (No favorable analytic premise in constructive closure)**.** *Fix only the canonical weak-PDE signature, its complete continuation target, and its equation-generated physical carrier. The source compiler, sourcewise closure, target quotient, total interface graphs, activity lift, successor relation, and finite-depth viability sets are defined without requiring any of the following properties:*

$$\begin{gathered}\textit{a gradient/sector realization, local continuation, compactness,}\\ \textit{continuation of the full branch, or closed gradient range,}\\ \textit{a classical chain rule, smooth shells, coercivity, a positive price gap, rigidity, polarity,}\\ \textit{finite resistance, summable tails, orbit realization, or continuation.}\end{gathered} \tag{408}$$

Each listed property is tested by a total graph generated from the equation. Both its ordinary and boundary values are supplied to the same activity lift, successor relation, and finite-depth viability recursion. A branch therefore cannot terminate at the sentence that a property fails: it is removed by the target-null quotient, charged by the physical carrier, or retained as an explicit zero-price recurrent mechanism. Consequently none of the listed properties is a premise of the totalization, contact-accountability, or constructive regularity theorems.

Proof. The compiler is structural induction on the fixed equation syntax. Sourcewise closure is defined separately in each tagged cell. Quotienting by the common target kernel is exact by definition. Totalization replaces every partial interface by the disjoint union of its graph values, explicit failure values, and graph-boundary values. The realization modes partition the resulting ordinary and defect records. The positive target-current part of each record defines its activity measure, whose normalized lift defines $\mathcal{R}_\eta$. Iterating $Z_{n+1} = \mathcal{V}_\eta(Z_n)$ removes every record without arbitrarily long zero-price descendants. Compactness turns survival at every finite depth into an infinite path and a stationary path law. Thus both sides of every analytic relation undergo an explicit structural operation, and no favorable side of a dichotomy is selected. $\square$

Definition III.2.5 (Physical reachability profile)**.** For a finite number B_{phys} arising from the carrier identity, let $\mathscr{O}^{\mathrm{ord}}_{\le B_{\mathrm{phys}}}$ be the ordinary maximal solution branches whose derived expenditure satisfies $\mu_u(\mathfrak{Q}_u) \le B_{\mathrm{phys}}$. When this set is nonempty, define

$$\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) := \sup_{u \in \mathscr{O}^{\mathrm{ord}}_{\le B_{\mathrm{phys}}}} \mathbf{1}_{\{u \text{ contacts } \Sigma\}}. \tag{409}$$

There is no empty-supremum convention. If the ordinary solution stratum is empty, the total local-solution or carrier graph returns its explicit defect record. Thus $\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0$ means that ordinary maximal branches exist and none contacts the target. The ceiling is not a second datum: it is only a stratum of the one physical expenditure generated by the equation.

Definition III.2.6 (Constructive zero-density mechanism cells)**.** Fix a derived finite physical stratum B_{phys}. For nonempty $A \subseteq \mathfrak{C}$ and $m \in \mathfrak{M}$, let $Z^{\Sigma}_{\infty,A,m}(B_{\mathrm{phys}})$ be the part of Z^{Σ}_{∞} whose inherited support is exactly A and whose primary mode is exactly m. Full ancestry and all nonprimary flags remain coordinates of the record. Define

$$\mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}}) := \overline{\mathrm{span}}\left\{ \left(\Pi_\Sigma(\pi_0)_{\#}\boldsymbol{\nu}, \mathrm{Aug}(\boldsymbol{\nu})\right) : \boldsymbol{\nu} \text{ stationary on } \mathsf{Path}(Z^{\Sigma}_{\infty,A,m}(B_{\mathrm{phys}})) \right\}, \tag{410}$$

$$\mathfrak{F}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) := \bigoplus_{\substack{\varnothing \ne A \subseteq \mathfrak{C} \\ m \in \mathfrak{M}}} \mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}}). \tag{411}$$

These cells are not defined from contact orbits. They are the finite source/mode decomposition of the prospectively generated zero-density fixed point. The direct sum is provenance-tagged: analytically equal representatives with different ancestry remain distinct. Every generator carries its stationary path law, full weak equation or recurrent graph boundary, target pairing, and diagonal cascade-origin witnesses. It also carries its unnormalized occurrence, expenditure, and source-current prefixes. Equality of normalized support laws therefore never identifies two different cumulative histories.

Definition III.2.7 (Terminal boundary formation cell)**.** Let $\mathfrak{F}^{\mathrm{term}}_{\Sigma}(B_{\mathrm{phys}})$ be the closed provenance-tagged span of the target projections of all completed continuation-failure records in the derived physical stratum whose shell entry occurs only at the terminal endpoint. Each generator retains the ordinary same-origin weak-equation path, its terminal continuation-status value, the Bdry-headed jump current, the joint shell owner, and every auxiliary total-graph descendant. This cell is generated prospectively from the total continuation graph; it is not defined by choosing an observed singular orbit. It is disjoint from the recurrent zero-density cells because it carries the terminal-entry flag rather than a cofinal pre-contact cascade.

Theorem III.2.8 (Constructive source decomposition of zero-density dynamics)**.** *For every derived finite physical stratum B_{phys},*

$$\boxed{\mathfrak{H}^{\Sigma}_{\mathrm{succ}}(B_{\mathrm{phys}}) = \mathfrak{F}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}).} \tag{412}$$

The right side has at most 31×5 provenance-tagged cells. Every finite-expenditure approach-contact orbit produces a nonzero generator in one of them. Every terminal-entry contact instead produces a nonzero generator of $\mathfrak{F}^{\mathrm{term}}_{\Sigma}(B_{\mathrm{phys}})$. Consequently, whenever the ordinary stratum is nonempty,

$$\mathfrak{F}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0 \quad \text{and} \quad \mathfrak{F}^{\mathrm{term}}_{\Sigma}(B_{\mathrm{phys}}) = 0 \quad \Longrightarrow \quad \mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0. \tag{413}$$

A nonzero cell is not a label awaiting a further estimate: its generator is an augmented stationary zero-density successor law carrying the weak equation, recurrent graph defects, target pairing, source ancestry, and cascade-origin witness sequence, together with the exact cumulative five-source identity; an ordinary contact origin makes these actual physical events.

Proof. Along a successor path the inherited support can only increase. Because the source alphabet is finite, it is eventually constant; stationarity makes it constant almost surely. The primary mode is likewise monotone in a finite ordered set and is stationary only when constant almost surely. Decomposing every stationary path law along these invariant finite parts proves (412).

If an ordinary branch has approach contact with Σ, the total shell graph gives the disjoint unit-increment cascade without a chain-rule premise. Theorem II.2.58 converts that same cascade into a stationary zero-density law with nonzero target projection and diagonal origin-event witnesses. The finite tag argument places a nonzero component in one (A, m)-cell. If contact first occurs at the terminal endpoint, Theorem I.5.3 gives its positive Bdry-headed owner, which is a generator of $\mathfrak{F}^{\mathrm{term}}_{\Sigma}$. The contrapositive proves (413); the stated data are supplied by Theorem II.2.55 and Lemma II.2.44. □

Theorem III.2.9 (Finite-depth structural evaluator)**.** *For every source/mode cell, start with its generated target-active zero-price records and form*

$$Z_{0,A,m} \supseteq Z_{1,A,m} \supseteq \cdots, \qquad Z_{n+1,A,m} = \mathcal{V}_{\eta}(Z_{n,A,m}). \tag{414}$$

Exactly one of the following constructive outcomes occurs:

(E1) *at some finite cylinder-moment level the dual proof cone contains -1; the finite ordered identity produces the pruning transcript of Theorem II.2.53 and closes the cell;*

(E2) *every finite moment level is feasible; diagonal reconstruction produces an infinite successor path law and a nonzero generator of $\mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}})$.*

Applying the generated zero-cost equations to outcome (E2) *removes each target-null component and recursively expands each ordinary or graph-boundary component. Thus every nonzero row value undergoes a further structural operation.*

Proof. Apply Theorem II.2.49 to the path algebra restricted to cell (A, m). Its dual branch is (E1) and its primal projective-limit branch is (E2). The ordered ideal contains the target floor, zero price, same-origin successors, and source/mode coordinates, so reconstruction retains all of them. The final reduction is Theorem II.2.55 applied to every support component. □

Theorem III.2.10 (End-to-end constructive closure)**.** *For a derived finite physical stratum B_{phys}, apply source compilation, the target quotient, total interface graphs, shell activity lifting, witness-indexed successors, finite-depth viability pruning, and the generated zero-cost equations, in that order. The output is*

$$\boxed{\mathfrak{H}^{\Sigma}_{\text{out}}(B_{\text{phys}}) = \mathfrak{F}^{\text{term}}_{\Sigma}(B_{\text{phys}}) \oplus \mathfrak{H}^{\Sigma}_{\text{succ}}(B_{\text{phys}}) = \mathfrak{F}^{\text{term}}_{\Sigma}(B_{\text{phys}}) \oplus \bigoplus_{A,m} \mathfrak{F}^{\Sigma}_{A,m}(B_{\text{phys}}).} \tag{415}$$

On every ordinary local finite-carrier branch,

$$\mathfrak{F}^{\text{term}}_{\Sigma}(B_{\text{phys}}) = 0, \qquad \bigoplus_{A,m} \mathfrak{F}^{\Sigma}_{A,m}(B_{\text{phys}}) = 0 \quad \Longrightarrow \quad \text{Reach}^{\text{phys}}_{\Sigma}(B_{\text{phys}}) = 0. \tag{416}$$

The compiler never consults an actual-contact image. A contact orbit itself constructs a nonzero cell. Conversely, a cell that survives the evaluator comes with an infinite zero-price path, stationary law, full generated saturation equations, source support, target pairing, and diagonal origin events. A finite exclusion carries its rational ordered derivation. A surviving cell carries the positive-functional model of all enumerated relations; coherent approximation-origin paths are retained separately from source-complete records.

Proof. Totalization inserts every target-visible local-solution or carrier boundary witness into the same successor graph. On an ordinary finite-carrier branch, source and mode tags partition the zero-price path laws and Theorem III.2.8 identifies their direct sum with $\mathfrak{H}^{\Sigma}_{\text{succ}}$, proving (415) on the approach component; the terminal continuation graph supplies its disjoint Bdry summand. If approach contact occurs, shell accountability, disjoint cascade extraction, and Theorem II.2.58 construct a nonzero stationary path law in one cell. If terminal-entry contact occurs, its closed shell owner gives a nonzero terminal generator. Their joint contrapositive proves (416). The finite-depth evaluator and saturation-equation compiler supply the listed data for every surviving generator, without an actual-contact filter. □

Theorem III.2.11 (Unconditional continuum contact mechanism)**.** *Fix the canonical physical continuum presentation and close its orbit algebra sourcewise. Every target-active current, total-graph defect, and target-visible limit has ancestry in* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$*, while*

$$G^{J_{\text{res}},\text{src},\text{sat}}_{B_{\text{phys}},\Sigma} = 0. \tag{417}$$

Every contact orbit has exactly one of two source-complete contact forms:

(C1) *cofinal approach contact supplies a canonical sequence of pairwise disjoint pre-contact shell intervals I_j with unit shell increment; after passage to a subsequence, one nonempty support $A \subseteq \mathfrak{C}$ occurs on every I_j and has a positive owner there;*

(C2) *otherwise a cofinal sequence of terminal-entry shells supplies completed shell records whose positive owner is* Bdry*-headed and belongs to $\mathfrak{F}^{\text{term}}_{\Sigma}(B_{\text{phys}})$.*

The alternatives are distinguished by the generated terminal-entry flag and are therefore disjoint. For the complete continuation target of Definition III.1.1, the completed same-origin restriction convention in (P4) *forces* (C1)*;* (C2) *is retained for other discontinuous or killed targets. This statement does not require a constitutive realization, classical chain rule, or price gap: every failed relation contributes through its derived typed graph current with the ancestry of the failed operation.*

Proof. The continuum normal-form compiler and source-preserving completion give the five-channel statement for ordinary values. Derived graph currents give the same statement for every boundary value. The target quotient gives (417): by construction residual means target-null. On the approach branch, the canonical disjoint shell-cascade theorem supplies the intervals. On each interval the derived shell observable has increment one. On a finite current record, (39) selects a positive source total; on a variation-boundary record, (41) selects its positive owner. The infinite pigeonhole principle selects a persistent nonempty support. If a shell is first entered at the terminal endpoint, Theorem I.5.3 assigns the endpoint jump a Bdry head and the same owner floor. The terminal-entry flag separates this cell from the approach cascade. □

Theorem III.2.12 (Constructive physical-price evaluation)**.** *In the setting of Theorem III.2.11, restrict to the derived finite physical stratum $\mu_u(\mathfrak{Q}_u) \le B_{\mathrm{phys}} < \infty$ of ordinary maximal branches beginning outside Σ. On every cofinal approach-contact branch, for each persistent support use the canonical sharp prices of* (400)*. Then:*

(P1) *if*

$$\sum_{j=1}^{\infty} \underline{p}_{A,j} = \infty. \tag{418}$$

then that support cannot sustain contact;

(P2) *if the series is finite, the corresponding shell tails enter the zero-density ordered path algebra with their unnormalized prefix ledger. A checked identity $-1 \in \mathcal{K}_{\mathfrak{b}}$ excludes the active cell; otherwise the order-unit alternative supplies an augmented stationary zero-density law in $\mathfrak{F}^{\Sigma}_{A,m}(B_{\mathrm{phys}})$.*

If every approach-contact persistent support is removed either by the divergent-price argument or by a finite ordered identity, and the generated terminal cell $\mathfrak{F}^{\mathrm{term}}_{\Sigma}(B_{\mathrm{phys}})$ is target-null, continuable, or class-incompatible, then

$$\boxed{\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0.} \tag{419}$$

Thus a finite lower-price series is never the end of the target-reachability track.

Proof. If an ordinary maximal branch had cofinal approach contact with Σ, Theorem III.2.11 would supply pairwise disjoint intervals and a persistent support A. On their selected subsequence, the definition of the sharp price and countable additivity give

$$B_{\mathrm{phys}} \ge \sum_j \mu_u(I_j) \ge \sum_j \underline{p}_{A,j} = \infty,$$

a contradiction in case (P1). In the finite-series case, use the actual prices as the price coordinate in the activity lift. Cofinal tails have price tending to zero, so Theorems II.2.47 and II.2.58 give precisely the two outcomes in (P2). If every persistent support is eliminated, approach contact has no possible support, contradicting shell accountability. A remaining contact would have terminal-entry form, whose Bdry-headed owner belongs to $\mathfrak{F}^{\mathrm{term}}_{\Sigma}$; the stated terminal-cell condition excludes it. This proves (419). □

Remark III.2.13 (Orbitwise source ledgers)*.* The divergent-price branch is evaluated only for the source support that persists along a putative singular cascade. Its canonical lower-price schedule may depend on the orbit stratum and may decay with scale. A uniform positive price is stronger and yields uniform window counts. Theorem III.2.12 uses only divergence of this generated series; five-channel exhaustion, the approach cascade, and the separate terminal Bdry cell were proved earlier.

Theorem III.2.14 (Total constructive continuum prospective evaluator)**.** *Let u_0 be a datum in the public state class of the fixed equation. Totalize the local-solution and carrier graphs and run the constructive closure of Theorem III.2.10 on every ordinary maximal branch. Put $\eta_0 = 1/5$. Exactly one of the following algebraic outcomes holds:*

(i) *every contact-generated same-origin branch reaches a local target-null, continuation, class-incompatibility, or divergent physical-charge cell; then no ordinary maximal branch from u_0 with finite physical expenditure contacts Σ_{sing}. Terminal coverage consequently supplies continuation for every such ordinary branch;*

(ii) *diagonal target-aligned saturation reconstructs a prospective source atom, including its closed reduced five-channel equation, physical ledger, origin tag, and retained cofinal shell floor.*

Finite noncontinuation of an ordinary maximal path is contact with the complete singular-event target and generates its same-origin cofinal source-complete shell record. The realization gate is required only before an arbitrary closure point, or the terminal analytic coordinate itself, is interpreted as an ordinary PDE object. Finite noncontinuation never produces a separate continuation class. An empty ordinary local-solution fiber is retained as the form value $\perp_{\text{loc}}$ and is not converted into either regularity or singularity. Positive stationary functionals, equality kernels, and typed defects are intermediate successor states and do not constitute a third outcome.

Proof. On an ordinary branch, finite maximal time without continuation gives the same-origin singular event and source-complete terminal record of Proposition III.1.2. Its preterminal orbit already realizes physical contact; a nonordinary terminal analytic coordinate remains a proper graph successor and is never promoted to a regularity-class state. Contact then has one of the two forms in Theorem III.2.11. Cofinal approach gives disjoint unit-increment shells and a persistent five-channel support. A divergent sharp price contradicts the finite carrier, while a finite-price cascade enters the zero-price successor algebra. The normalized shell graph gives a source owner at least 1/5, so no additional target-floor parameter is needed. Terminal entry instead enters $\mathfrak{F}_\Sigma^{\text{term}}$ through its Bdry-headed owner; its total continuation graph either closes locally or yields the Bdry-headed source-profile witness in (ii). Every finite local closing cell contradicts the retained target floor or the finite physical ledger, so if all generated contact branches close, contact is impossible. Terminal coverage then turns that target-local conclusion into continuation of each ordinary branch. Otherwise, on an approach branch, the ordered proof/model alternative produces the initial target-retaining compiler state. Apply Theorem III.4.78. Its charge and null outputs close the contact branch; its prospective source output is precisely (ii). Nonvacuity of the public graph follows from Proposition I.1.17 and Theorem I.1.4. □

Theorem III.2.15 (Target-aligned physical fast track)**.** *Fix a generated finite physical stratum and let Γ_Σ, $\mathscr{J}_\Sigma$, and the sourcewise shell currents be as above. For each reachable active branch $\mathfrak{b}$, let $\mathcal{K}_{\mathfrak{b}}$ be the ordered proof cone generated by Definitions II.2.22 and II.2.46. Then*

$$\boxed{\mathrm{Reach}_\Sigma^{\text{phys}}(B_{\text{phys}}) = 0 \quad \textit{or} \quad \mathfrak{S}_\Sigma(B_{\text{phys}}) \neq \varnothing,} \tag{420}$$

where $\mathfrak{S}_\Sigma(B_{\text{phys}})$ is the set of joint-saturated prospective source atoms generated by zero-price resaturation, retaining their origin polarity. A member with contact-derived origin is a source-profile witness; only an independently generated member passing Theorem I.1.7 is an explicit singular mechanism. At an intermediate finite level, failure of an ordered identity constructs a normalized stationary positive-functional successor satisfying

$$\Pi_\Sigma \left(\ker Q_{\text{phys}} \cap \ker I_{\mathfrak{b}} \cap \bigcap_{w \in \mathscr{W}_{\mathfrak{b}}} \{P_w = 0\} \right) \neq 0 \tag{421}$$

for some reachable $\mathfrak{b}$. The successor is reprocessed by Definition III.4.23; support, compactness, passivity, fidelity, and realization values are internal queue coordinates. By Theorem III.4.78, the successor is never a third mathematical output.

Proof. Suppose contact occurs. The derived shell-current theorem gives infinitely many disjoint unit-increment events, each with a finite or boundary-owner source floor. Finite source exhaustion gives a persistent nonempty support A, and event normalization gives the universal target floor $1/5$. If the physical prices have divergent sum, countable additivity contradicts the finite physical budget. Otherwise a subsequence has price tending to zero and the same-origin successor construction gives an augmented stationary target-aligned zero-density law in an active branch $\mathfrak{b}$. This law defines the positive functional in Theorem II.2.47; weak duality forbids $-1 \in \mathcal{K}_{\mathfrak{b}}$. Run the target-aligned recurrent closure compiler on this state. Its divergent-charge and target-null outputs contradict contact; its remaining output is the hypothetical prospective atom in $\mathfrak{S}_\Sigma(B_{\mathrm{phys}})$. This proves (420).

Conversely, failure of the ordered derivation gives the positive functional by Theorem II.2.47. Its ideal relations are the weak PDE, source-adapted zero-production, successor, and stationarity equations, while its target floor makes the target projection nonzero. This is (421). Approximation realization is exactly Theorem I.1.4; it distinguishes a coherent generalized solution path from its next typed successor. Applying the three-output current compiler to every typed or ordinary outcome gives avoidance or the prospective source-profile output after the finite physical ledger removes the divergent-charge cell. The latter becomes a singular-existence output only through the independent ordinary realization criterion. □

Corollary III.2.16 (Generated PDE prospective-accountability output)**.** *Let $\mathfrak{U}_{\mathrm{pub}}$ be the datum class in the public PDE presentation. Derive the physical expenditure from the carrier identity and run the source, signed-carrier, witness-successor, viability, and saturation compilers on every resulting finite stratum. Put $\eta_0 = 1/5$. Then*

$$\boxed{\mathrm{Reach}^{\mathrm{phys}}_{\Sigma_{\mathrm{sing}}} > 0 \quad \Longrightarrow \quad \mathsf{T}^{\mathrm{phys,sat}}_{\Sigma_{\mathrm{sing}}}(B_{\mathrm{phys}}) \geq \eta_0.} \tag{422}$$

The contact-generated output includes the cumulative prospective atom, its nonzero five-channel source projection, unnormalized shell cocycle, closed reduced equation, totalized realization rows, physical and origin ledgers, and the cofinal shell floor. Its origin polarity is Hyp*, so it is a lower-bound witness for the saturated source profile and not evidence that a singular orbit exists. Positive functionals, compatible equality successors, and ordinary paths lacking this joint atom are intermediate resaturation states. Conversely, $\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma_{\mathrm{sing}}}(B_{\mathrm{phys}}) < \eta_0$ proves $\mathrm{Reach}^{\mathrm{phys}}_{\Sigma_{\mathrm{sing}}} = 0$. A singular-existence conclusion is available only from an independently generated ordinary public orbit satisfying Definition III.4.77. None of these values is a premise supplied with the PDE presentation.*

Proof. Apply Theorem III.2.15 at the derived finite ceiling of each branch. The finite ordered hierarchy initializes the local queue, and Theorem III.4.78 evaluates all its descendants. Under contact, divergent charge contradicts the finite physical ledger and target-nullity contradicts the unit shell increment. The remaining cumulative joint queue is reprojected by Theorem III.4.58; its finite-prefix successors continue the compiler, and its primal–polar fixed point retains source floor at least η_0. Origin-polarity preservation makes this a hypothetical source-profile witness. Taking the contrapositive yields the regularity criterion. Independent realization is governed separately by Theorem II.2.41 and Definition III.4.77. □

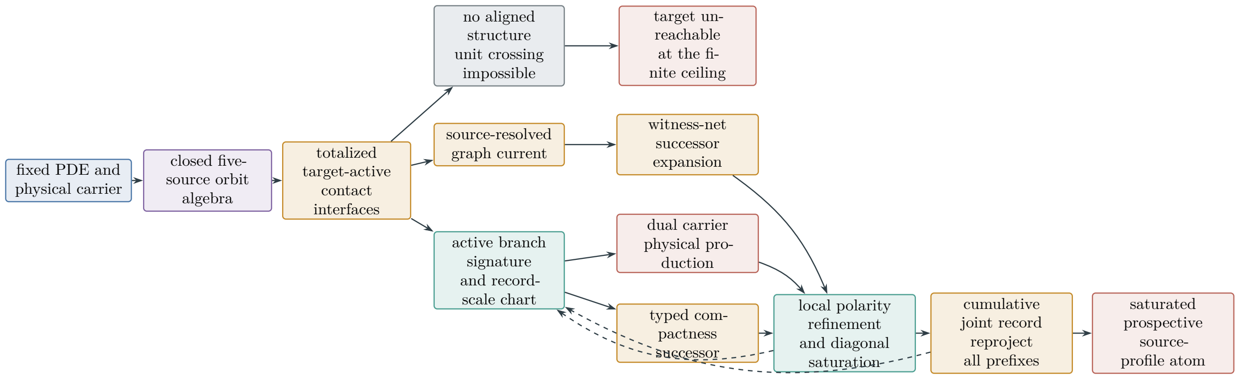

Figure 21: Constructive singularity evaluator. Only the target-active quotient is analyzed. If it contains no singularity-aligned structure, normalized shell contact is impossible. Otherwise contact fixes an active branch signature and, when needed, a record-scale chart. The source-adapted dual carrier either charges target progress to physical production or creates a typed compactness successor. The next unprocessed coordinate is split only on the retained target cell; every nonnull polarity cell re-enters the same compiler through its cumulative join. A cofinal refinement is saturated and reprojected as one joint record. A finite joint-prefix component returns to the five-source compiler exactly when it is still unprocessed; all sub-threshold processed increments remain in the cumulative sum. Every cumulative owner is then charged, annihilated, or expanded by recursive carrier–polar exhaustion. A primal–polar fixed atom is rejoined with its unnormalized shell cocycle and complete source ancestry, then evaluated as an ordinary same-origin reachable contact cell.

Corollary III.2.17 (Quantitative finite-cascade bound)**.** *Fix an active branch $\mathfrak{b}$. If the physical expenditure satisfies $\mu_u(\mathfrak{Q}_u) \le B_{\mathrm{phys}}$ and each event in a disjoint same-origin family has normalized window in $B_{\Sigma,\mathfrak{b}}$ and admissible target-local price at least $\delta_{\mathrm{phys}}(\mathfrak{b}) > 0$, then that family has cardinality at most*

$$\left\lfloor \frac{B_{\mathrm{phys}}}{\delta_{\mathrm{phys}}(\mathfrak{b})} \right\rfloor . \tag{423}$$

Proof. The proof of Theorem III.1.12 gives $N\delta_{\mathrm{phys}}(\mathfrak{b}) \le B_{\mathrm{phys}}$ for every finite subfamily. Eligibility of each price is exactly Theorem II.2.4; events outside $B_{\Sigma,\mathfrak{b}}$ do not enter this count. □

III.3 Structural evaluator coverage and failure semantics

The following table is the coverage proof for the fast track. Every classical analytic question is assigned a generated invariant and every opposite value activates a structural operation. No row accepts a desired analytic conclusion as input or returns a bare label.

Analytic question	Generated structural invariant	Successor action
Does a native, chain, trace, or covariance relation hold?	target-active boundary of its total graph	activity lift of the typed graph witness
Does a reduction descend through later operations?	commutator of the operation with the quotient map	typed descent defect
Is the gradient range closed where the target sees it?	$\mathfrak{H}^{\Sigma}_{\mathrm{grad}}$	inherited closure or harmonic class
Is a tail structurally closed?	$\mathfrak{H}^{\Sigma}_{\mathrm{tail}}$	target-visible tail cokernel
Is target feeding dominated?	α_*	supercritical or positive null-cost feeding profile
Is normalized target progress physically priced?	δ_*	zero-price sequence in the same source cell
Can the source be routed?	$\mathcal{R}(\Sigma)$	target-visible harmonic non-routable class
Does a Wick quotient admit compensated target response?	$F_W(z)$, $T_{W,\lambda}$, dark space, and bracket gap	certified response/action profile or a typed dark, authority, or transfer obstruction
Is the target polar?	$\mathrm{Cap}(\Sigma)$	positive-capacity target-compatible class
Does a scale cascade exceed finite expenditure?	$\mathscr{S}^{\Sigma}$	finite-price successor chain and recurrent saturation row
Do internal signed currents create net target feed?	carrier cohomology $[F_{\mathrm{sc}}-D-M]$	nonzero scale-feed cohomology successor
Does a failed interface persist dynamically?	recurrent witness boundary $\partial_\infty F$	target-null loss or recurrent failure mechanism
Does a sharp branch close?	$\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$	nonzero saturation/equality remainder
Are stochastic fluctuations invisible or routed?	χ^{Σ} and fluctuation resistance	target-visible covariance/bracket class
Does a weak-solution graph value persist?	recurrent witness boundary $\partial_\infty F$	finite-depth removal or recurrent weak-equation path
Can zero-price target progress persist?	ordered cones $\mathcal{K}_{\mathfrak{b}}$ and finite moment levels	finite rational derivation or a reconstructed stationary law

Remark III.3.1 (Exact input closure). The fast-track data are constructed from the public equation, generated target, source grammar, and native physical carrier. A specialization cannot append compactness, coercivity, rigidity, nonpolarity, continuation, or a desired invariant value. Each relation and its complement are graph values. The complementary value is converted into a typed successor on the surviving target-active quotient, and its descendants are processed by finite-depth viability within the same generated recursion.

Definition III.3.2 (Target-local invariant evaluation packet)**.** Let I be any invariant in the fast-track table and let $\mathfrak{b}$ be an active same-origin branch. Its evaluation packet is

$$\mathfrak{E}_I(\mathfrak{b}) := (X_{\mathfrak{b}}, \overline{\mathrm{Graph}(I)}, q_\Sigma I, Q_I, \mathrm{Anc}(I), p_{\mathrm{phys}}, (F_{I,n}, C_{I,n})_{n\geq 1}, I_I^{=}, \mathfrak{G}_I^{\mathrm{real}}). \tag{424}$$

Here $\overline{\mathrm{Graph}(I)}$ is the total graph of I, including its ordinary and first-failure values; $q_\Sigma I$ is its target quotient; Q_I is its signed target-current owner after compatible interfaces have been glued; $\mathrm{Anc}(I) \subseteq \mathfrak{C}^*$ is its complete constructor ancestry; p_{phys} is the restriction of the single public physical carrier; $(F_{I,n}, C_{I,n})$ is the canonical cofinal family of checked finite cylinders and carrier cones; $I_I^{=}$ is the equality ideal generated on zero physical valuation; and $\mathfrak{G}_I^{\mathrm{real}}$ is the same-origin ordinary realization graph. Every entry is generated from the public presentation, the selected shell covector, and earlier branch relations. No invariant value is part of the packet data.

Definition III.3.3 (Invariant evaluation protocol)**.** For an initial packet $\mathfrak{E}_{I_0}(\mathfrak{b}_0)$, maintain a FIFO queue $\mathcal{Q}_n$ and an unnormalized cumulative same-origin joint record $\mathcal{J}_n$. Initially $\mathcal{Q}_0 = (\mathfrak{E}_{I_0}(\mathfrak{b}_0))$ and $\mathcal{J}_0 = \mathfrak{E}_{I_0}(\mathfrak{b}_0)$. On removing the first packet $\mathfrak{E}_I(\mathfrak{b})$, perform these operations in order:

(P1) project through q_Σ, deleting a zero projection by its checked target-nullity row;

(P2) compute the next finite-cylinder physical valuation and source profile, deleting the descendant stratum only when a checked profile ceiling lies below the retained shell threshold;

(P3) on threshold equality, adjoin every vanishing production, square, polarization, adjoint, symmetry, and dyadic coercivity row, reproject, and append every nonzero descendant;

(P4) on a strict dual or positive-profile value, join primal owner and dual covector, pull the covector through the complete backward frontier, and append every nonzero conditional constructor increment;

(P5) on a nonordinary value, append the least failed total-graph coordinate with its owner, target covector, physical denominator, occurrence cocycle, origin, and preceding joint frontier;

(P6) when the finite queue is empty, evaluate the realization fiber of the cumulative joint record. A finite incompatible fiber gives its checked prefix row; a coherent zero-defect branch enters the ordinary realization gate; and projective branch escape is appended as a Lift-owned realization-fiber packet.

Every appended packet is joined without normalization into $\mathcal{J}_{n+1}$. At a cofinal execution form $\mathcal{J}_\infty = \bigvee_n \mathcal{J}_n$ before projection. A nonzero finite contextual increment re-enters (P3)–(P5) at its least occurrence. If all finite structural increments have been processed, evaluate the complete joint realization fiber: finite incompatibility is checked on its prefix, a coherent branch enters the ordinary gate, and branch escape re-enters (P5) through its total graph. Profile witnesses, equality faces, polar separators, graph boundaries, and compatible inverse-limit functionals are protocol states, never stopping values.

Theorem III.3.4 (Unconditional local evaluator for every generated invariant)**.** *For every packet $\mathfrak{E}_I(\mathfrak{b})$, the continuum compiler constructs the extended physical valuation*

$$\mathrm{val}_I^{\mathrm{sat}}(Q_I) = \sup_n \inf\{C \geq 0 : Q_{I,n} \in D_{I,n} + C p_{I,n}\} \tag{425}$$

and the saturated source profile

$$\mathsf{T}_I^{\mathrm{sat}}(E) = \inf_n \sup\{L(Q_{I,n}) : L \in \mathcal{S}_{I,n}(E)\}. \tag{426}$$

At each rational target threshold $\eta > 0$, these values drive the exhaustive transitions of Definition III.3.3:

(I1) *a finite checked carrier or nullity row proves* $\mathsf{T}_{I,n}(E) < \eta$ *on some cylinder and hence subcriticality on the entire descendant stratum;*
(I2) *the threshold equality face is generated, its vanishing productions and squares are adjoined to* $I_{I}^{=}$*, and target projection is repeated;*
(I3) *a positive target-active value supplies a finite contextual dual increment or a compatible cofinal functional, retaining* Q_I*, its unnormalized occurrence cocycle, physical denominator, and complete five-source ancestry;*
(I4) *a nonordinary coefficient supplies its least failed graph coordinate with the same data and is evaluated as the next invariant packet;*
(I5) *an ordinary zero-defect same-origin path supplies the corresponding physical value of* I *and alone can activate the contact realization row.*

The first alternative is an unconditional subcriticality proof on its generated stratum. In the other alternatives the exact non-subcritical structure is an evaluated five-source record and is immediately returned to the protocol. None of (I2)–(I4) *is an output. Repeated evaluation uses the cumulative joint record, so a finite or infinite family of individually subthreshold coordinates cannot lose its joint target projection.*

Proof. Projection-first saturation gives $q_\Sigma I$ and removes only its target-null kernel. Source exhaustion assigns every remaining atomic and mixed current to its constructor ancestry before any estimate. On a finite cylinder, Theorem III.4.38 gives (425), while Theorem II.2.29 gives the finite version of (426). Cofinal monotonicity is Theorems II.2.30 and III.4.39. The equality alternative is processed by Theorems II.2.55 and III.4.45. The total-graph alternative is Theorem III.4.42; its output is a new packet because the failed operation, covector, owner, carrier, and origin are all retained. Ordinary realization is governed by Theorem I.1.7. Finally, Theorems III.4.29 and III.4.30 gives the cumulative joint assertion. These constructions use only packet entries and therefore introduce no analytic premise. □

Definition III.3.5 (Realization-fiber invariant)**.** For a cumulative joint packet $\mathcal{J}$, let $\mathscr{F}_n(\mathcal{J})$ be the set of depth-n public approximation prefixes having the same origin and initial datum as $\mathcal{J}$, matching its first n cylinder coordinates within 2^{-n}, and having its first n normalized public residuals at most 2^{-n}. Restriction gives the rooted inverse graph

$$\mathscr{F}(\mathcal{J}) := \varprojlim_n \mathscr{F}_n(\mathcal{J}). \tag{427}$$

The realization-fiber invariant is the total graph with values
(R1) the least n for which $\mathscr{F}_n(\mathcal{J}) = \varnothing$, together with the finite cylinder relations defining that fiber;
(R2) one element of $\mathscr{F}(\mathcal{J})$, with all residual-tail, origin, carrier, and continuation coordinates;
(R3) the projective branch-escape value $\mathscr{F}_n(\mathcal{J}) \neq \varnothing$ for every n but $\mathscr{F}(\mathcal{J}) = \varnothing$, retaining the escaping prefix net as one unnormalized cumulative graph packet; this value has no distinguished failed finite coordinate.

The third value is a Lift-owned target-local invariant. It is not an ordinary path and is inserted into the invariant protocol.

Theorem III.3.6 (Exact realization-fiber evaluation)**.** *Every cumulative joint packet has exactly one realization-fiber value from Definition III.3.5. In value* (R1)*, the least empty prefix is retained together with its entire finite relation set. A finite algebraic separation identity closes it as checked incompatibility; until such an identity is generated, the total emptiness graph is itself the next finite-prefix packet and is not treated as a conclusion. In value* (R2)*, residual-tail evaluation gives either* *a least nonzero public defect or a zero-defect compatible branch; the latter enters Theorem I.1.6. Value* (R3) *re-enters Definition III.3.3 with its complete target pairing and five-source ancestry. Hence absence of an approximation branch is itself an evaluated structural invariant and never an existence conclusion.*

Proof. Either some finite fiber is empty or all are nonempty. In the latter case the inverse limit is empty or nonempty, giving the three disjoint values. Finite-cylinder separation gives a closing identity whenever the generated finite convex cylinder relaxation is empty. If the relaxation is nonempty while the exact prefix fiber is empty, their discrepancy is a nonzero finite realization-graph coordinate and is re-enqueued; thus emptiness is evaluated through the generated discrepancy coordinate. For (R2), the monotone residual-tail coordinates in Definition I.1.3 expose the least nonzero defect; if none exists, the branch is zero defect and the Dirac gate applies. In (R3), totalization of the restriction graph records the escaping net before passage to its cumulative graph boundary. Projection-first saturation either removes its target pairing or assigns its first nonzero conditional increment to the constructor that generated the failed realization map. The least-occurrence theorem is invoked only if such a finite increment exists; otherwise the complete unnormalized branch-escape packet remains the next Lift-owned invariant. This is exactly the packet required by the protocol. □

Theorem III.3.7 (Closure and conservation of the invariant protocol)**.** *Every finite or cofinal execution of Definition III.3.3 preserves the monotone ledger*

$$(\textit{origin}, \textit{shell cocycle}, \textit{physical carrier}, \textit{complete joint frontier}, \textit{five-source ancestry}). \tag{428}$$

Every live state has a prescribed next operation. The only stopping values are:
(T1) *a checked finite target-nullity or subcritical carrier derivation;*
(T2) *an ordinary zero-defect same-origin path, whose continuation flag decides target contact.*
Thus target-visible residuals, fixed profiles, equality faces, separators, nonordinary graphs, and source-complete graph records are nonterminal protocol states.

Proof. Projection changes only the target-null quotient. Finite-cylinder valuation and equality restriction preserve origin and ancestry, while signed gluing preserves the unnormalized shell cocycle and public carrier. Backward factorization assigns a nonzero dual pairing to a finite contextual constructor increment, so (P4) preserves the full preceding joint frontier. Source-preserving totalization gives the same conclusion for (P5). Induction proves (428).

For a cofinal execution, Theorems III.4.30 and III.4.58 shows that every unprocessed nonzero target component has a least finite contextual occurrence and therefore returns to a finite transition. On the complementary fully processed joint record, Theorem III.3.6 gives finite fiber evaluation, a zero-defect branch entering the Dirac gate, or a branch-escape packet entering transition (P5). Hence every nonterminal state has the successor prescribed in the definition, and only (T1)–(T2) stop. □

Protocol state	Mandatory next operation	Permitted closure
target-null packet	checked quotient deletion	finite nullity
finite subcritical profile	replay dual carrier row	finite subcriticality
equality face	enlarge equality ideal and reproject	none
dual/profile witness	backward pullback and constructor split	none
nonordinary graph value	least-failure packet	none
cofinal cumulative join	finite contextual re-entry or realization graph	none
complete detector upper profiles	cofinal zero derivation	structural absence
uniform positive detector floor	compatible functional and backward finite occurrence	structural presence; then realization
coincident valuation cuts	rational-cut comparison	exact extended-real value
separated valuation cuts	target-project the cut-gap owner	structural presence; then realization
zero-defect ordinary path	continuation/contact readout	ordinary physical value

Theorem III.3.8 (Invariant-by-invariant fast-track evaluation)**.** *The protocol of Definition III.3.3 and Theorems III.3.4 and III.3.7 applies to every row of the fast-track table as follows.*

(F1) *Native, chain, trace, and covariance relations use their total-graph boundary as I; the least failed relation is the next* Lift*-owned activity packet.*

(F2) *Quotient descent uses the commutator with the quotient map; its zero cell descends, and every nonzero cell is an* Abs*-owned contextual increment.*

(F3) *Gradient closure uses $\mathfrak{H}^{\Sigma}_{\mathrm{grad}}$; its zero class is the closed-range cell and its nonzero class is retained with* Geom/Caus *gradient ancestry.*

(F4) *Constraint-tail closure uses $\mathfrak{H}^{\Sigma}_{\mathrm{tail}}$; signed interface gluing removes exact tails and the surviving cokernel is a* Bdry*-owned packet.*

(F5) *Feeding domination uses α_*; finite carrier domination is subcritical, while supercritical or null-cost feeding retains the aggregate signed current and its source join.*

(F6) *Physical pricing uses δ_*; a positive price is charged to p_{phys}, and a zero-price sequence enters equality saturation with its unnormalized numerator.*

(F7) *Routing uses $\mathcal{R}(\Sigma)$; a routed value inherits its destination owner, and a nonroutable harmonic value remains in the target quotient with the owner of the failed routing constructor.*

(F8) *Wick response uses $F_W(z)$, $T_{W,\lambda}$, the dark space, and the bracket gap; response and action are physically valued, while dark, authority, and transfer defects retain their generated graph owners.*

(F9) *Target polarity uses* $\mathrm{Cap}(\Sigma)$*; zero capacity is target-null and positive capacity is a target-compatible* Geom *packet.*

(F10) *Scale expenditure uses $\mathscr{S}^{\Sigma}$; every scale step retains its pulled-back physical weight, and recurrent saturation keeps the full multiplicity cocycle.*

(F11) *Signed internal feeding uses the carrier-cohomology class $[F_{\mathrm{sc}} - D - M]$; exact classes telescope, while nonzero classes are* Lift/Caus*-owned scale-feed packets.*

(F12) *Persistent interface failure uses $\partial_\infty F$; loss of the target pairing is null, while a nonzero recurrent boundary is evaluated with its interface owner.*

(F13) *Sharp equality uses $\mathfrak{H}^{\Sigma}_{\mathrm{sat}}$; the zero class enters the generated equality ideal and a nonzero remainder returns to the source profile.*

(F14) *Stochastic visibility uses χ^{Σ} and fluctuation resistance; zero covariance is target-null and a positive covariance or bracket class has* $\mathsf{Geom}/\mathsf{Caus}$ *ancestry.*

(F15) *Weak-solution persistence uses the total weak-equation boundary; finite-depth removal is checked by an equation identity and a recurrent boundary retains its ordinary/nonordinary realization coordinate.*

(F16) *Zero-price progress uses $\mathcal{K}_{\mathfrak{b}}$ and its finite moment levels; finite infeasibility gives a checked nullity identity, while compatible levels produce one joined positive functional that immediately enters equality and ordinary-realization evaluation.*

For every row, the complementary value is therefore a concrete packet for the same protocol and is enqueued immediately. None of the sixteen checks requests compactness, coercivity, rigidity, a critical norm, or the desired invariant value from the specialization.

Proof. For (F1) and (F15), apply defect totalization and Theorem I.1.4. Quotient functoriality and target-relative gradient exhaustion give (F2)–(F3). Theorem II.9.2 and Proposition II.9.3 gives (F4). Carrier duality and the physical-price interface give (F5)–(F6). Zero-mode routing and the Wick total graphs give (F7)–(F8). Capacity polarity gives (F9). Theorem III.1.7 and Definition II.2.5 gives (F10)–(F11). Recurrent boundary totalization and the saturation-equation compiler give (F12)–(F13). The stochastic gradient/covariance source decomposition gives (F14). Finally, ordered proof-or-model evaluation and the Dirac realization gate give (F16). Each cited construction produces precisely the packet entries in (424); applying Theorem III.3.4 proves every row. □

Remark III.3.9 (Separation of invariant roles)*.* The logical interfaces enforce ten distinctions.

(N1) An imposed Caus term is transported by a chart and is not optimized toward the target. A free or hybrid authority envelope may optimize only its declared represented control slot.

(N2) A killed spectral resolvent is mapped to arrival only through the totalized terminal-flux realization row; a nonzero realization defect is retained.

(N3) Finiteness of a basic energy does not imply a positive cost at every singular scale; the validity or failure of (212) is the physical-price row.

(N4) Local entropy monotonicity does not yield an orbitwise production bound without the reset relation; its failure is the corresponding cocycle graph defect.

(N5) Preservation of a nonlinear badness predicate under weak limits is the boundary-defect value in the canonical evaluation closure.

(N6) Failure of an interface creates its totalized graph-defect record. Target visibility makes that defect part of the singularity-formation algebra; target invisibility places it in J_{res}.

(N7) Large or noncompact motion in $\mathscr{N}_\Sigma$ requires no downstream estimate because it cannot increase a declared prospective target functional.

(N8) A quotient is propagated through the zero descent-defect branch; failure to factor is retained as a target-active descent defect.

(N9) Summability of a constraint tail closes the analytic tail but supplies a physical price only through a carrier comparison such as (223).

(N10) A positive normalized scale gap excludes an infinite cascade when the pulled-back lower prices have the divergent total mass required by (395); a nonvanishing infimum is a stronger sufficient condition.

III.4 A reusable PDE instantiation protocol

For a particular deterministic PDE, the specialization datum is only its public textbook presentation from Proposition I.1.11 and Corollary I.1.16. Recording that datum requires no new mathematical result. The compiler then performs the following generated operations.

(I1) Read the displayed equation, datum class, domain, boundary conditions, weak test core, declared regularity class, native transformations, and formal carrier multiplier. Enumerate the exhaustive rational approximation grammar, its tested residuals, and its signed carrier records. Generate the total solution graph, continuation graph, and complete singular target.

(I2) Form the graph-complete ordered cylinder algebra. Every nonlinear operation carries its input, output, oscillation, concentration, tail, and first-failure coordinates. Ordinary states embed as evaluation states, and zero-defect approximation paths satisfy the public weak equation.

(I3) Expand the formal carrier multiplier against the weak equation, glue common faces with their signs, and take the Jordan decomposition. This generates the one measure μ_u and its derived ceiling. Every nonordinary value is inserted into the activity lift; no replacement measure is requested.

(I4) Run the normal-form source compiler. It labels native mobility or reversible form Geom, fixed directed transport Caus, quotient operations Abs, generated charts and augmented variables Lift, and terminal or exterior fluxes Bdry. Construct the primitive expression graph, apply its canonical form split, and propagate both the unique current head of each atomic component and complete provenance support through every constructor. The universal provenance theorem then verifies that the resulting 31 support cells contain every public current path.

(I5) Form the sourcewise mathematical closure of every dynamically generated weak limit, concentration, oscillation, tail, defect, and blow-up germ.

(I6) Generate finite-cylinder target shells and their source-resolved Stieltjes currents. Contact supplies unit shell progress, the canonical disjoint cascade, a persistent five-source support, its detecting covector, and the same-origin active signature. The signature also contains the canonical record-scale chart when the selected scales collapse. Directions annihilated by the shell covector are removed immediately in $\mathscr{N}_\Sigma$. Apply projection-first saturation; every nonzero descent or normalization defect retains the ancestry of its failed operation.

(I7) On the resulting target spine, generate the forward ancestry module and the backward target-cyclic covector module. Compute their finite incidence cores and pass to the minimal target-active realization $\mathsf{M}_{\Sigma,\mathfrak{b}}$. Delete every equation, constraint, chart, defect, tail, and carrier coordinate outside a complete path from a source occurrence to the shell covector. On the retained realization, restrict the public equation, constraints, gradient structure, native carrier, charts, and total graphs. Glue compatible currents with their signs and form the single aggregate row $\mathfrak{J}_{\Sigma,\mathfrak{b}}$. This step simultaneously generates the retained target-cyclic covectors, nonnegative productions, constraint tails, commutator squares, compactness coordinates, feeding cohomology, and exact storage transfer. Only carrier-dominated production can spend the public physical measure.

(I8) On the minimal realization, apply the five canonical normal-form projections: Section III.5, Section III.6, Section III.7, Section III.8, and Section III.9. Compute the geometric range/kernel/capacity tuple; the directed cohomology, storage, passivity, circulation, and compressed-resolvent tuple; the quotient descent/vertical/cokernel/target-descent tuple; the lift interface/fiber/concentration/oscillation/scale tuple; and the boundary exterior/tail/terminal tuple. Enumerate the 31 possible nonempty source supports and form their nested invariant images $\mathfrak{F}^{\Sigma}_{A,n}(E)$. This step is generated from the public graphs and target covectors; no value of an invariant is supplied.

(I9) Run the target-aligned recurrent closure compiler of Definition III.4.23. Its internal queue computes the recurrent circulation price and processes zero kernels, signed-dyadic return laws, and first graph-boundary values in one fixed order. Positive price returns a generated carrier potential; zero price returns the next circulation support. By Theorem III.4.78, the output is a divergent physical charge, target-nullity on the retained path, or a finite contact-derived joint source occurrence with unchanged origin polarity. That occurrence is rejoined with its complete same-origin history and evaluated in the reachable contact cell. Hence no separate compactness, equality, tail, pressure, scale, or rigidity stage is part of the specialization protocol. At each finite depth, the incidence fast track precedes the ordered moment calculation, so no source-transverse coordinate enters that calculation.

This protocol is equation-agnostic. Different continuum equations generate different row values in steps (I2), (I5), and (I7)–(I9); none supplies desired values as additional input. The closed source algebra, local target restriction, charging rule, and failure semantics do not change. Universal analytic identities used in any step are activated only through Theorem II.2.18.

III.4.1 Target-cyclic degeneracy compilation

The equality case of a physical carrier law must itself be evaluated by the closed algebra. It is not an auxiliary rigidity hypothesis. The following construction extracts all consequences that the public equation forces from a target covector and from vanishing physical production.

Definition III.4.1 (Target-cyclic covector module)**.** Fix an active branch $\mathfrak{b}$, a selected target shell Φ, and the five-channel current

$$J = J^{\mathsf{Geom}} + J^{\mathsf{Caus}} + J^{\mathsf{Abs}} + J^{\mathsf{Lift}} + J^{\mathsf{Bdry}}.$$

Let $\mathscr{C}^0_{\mathfrak{b}}$ be the rational localization module generated by $D\Phi$ on the retained cylinder algebra. Inductively, $\mathscr{C}^{n+1}_{\mathfrak{b}}$ is obtained from $\mathscr{C}^{n}_{\mathfrak{b}}$ by adjoining every ordinary value of

(C1) the adjoint transports $(DJ^c)^*\lambda$, the covector Lie derivatives $\mathcal{L}_{J^c}\lambda$, and all iterated commutators of these operations, for $c \in \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}$;

(C2) multiplication by rational cylinder cutoffs, restriction to record scales, pullback by generated charts, and descent through the generated constraint, gauge, and boundary quotients;

(C3) polarization of every multilinear operation in the public weak equation and of every defect form generated by its total graph; and

(C4) the source-adapted carrier and square-completion operations of Definition II.2.22.

Here DJ^c denotes the total graph of rational difference quotients on the cylinder core. Its ordinary value is the derivative when that derivative exists; no differentiability of the weak current is presumed. Every operation is totalized. Its first nonordinary value is routed with its source ancestry and the covector that exposed it; it is not adjoined as an ordinary word. Define

$$\mathscr{C}^{\infty}_{\mathfrak{b}} := \overline{\bigcup_{n\geq 0} \mathscr{C}^{n}_{\mathfrak{b}}}^{\,\mathrm{cyl}}. \tag{429}$$

This is the least covector module forced by the public dynamics and the target shell. In particular, no multiplier, compactness statement, or rigidity assertion is supplied as specialization data.

Definition III.4.2 (Generated production family and joint kernel)**.** For $\lambda \in \mathscr{C}^{\infty}_{\mathfrak{b}}$, expand the exact weak identity obtained by pairing λ with the public equation. Glue signed internal currents before taking positive parts. The *target-cyclic production family* $\mathscr{P}_{\mathfrak{b}}$ consists of every nonnegative term produced algebraically by that expansion: mobility squares, Dirichlet and commutator forms, positive graph defects, boundary production, and their polarized positive localizations. A member of $\mathscr{P}_{\mathfrak{b}}$ is charged to the physical ledger only when the same expansion proves its domination by the public carrier measure. Otherwise it is retained as an equality-kernel coordinate. Put

$$X_{\mathfrak{b}}^n := X_{\mathfrak{b}} \cap \bigcap_{\substack{P \in \mathscr{P}_{\mathfrak{b}} \\ \operatorname{depth}(P) \le n}} \{P = 0\}, \qquad X_{\mathfrak{b}}^\infty := \bigcap_{n \ge 0} X_{\mathfrak{b}}^n. \tag{430}$$

The zero set is interpreted in the total ordered graph, so failure of lower semicontinuity, closedness, trace descent, or constraint descent is itself a nonzero typed coordinate with inherited source. Thus a missing analytic property enlarges the generated structure; it is never presumed away.

Definition III.4.3 (Forward structural ancestry module)**.** Fix an active signature $\mathfrak{b}$. Let $\mathscr{R}_{\mathfrak{b}}^0$ consist of the five projected source-current occurrences in its unit shell identity, with their origin and carrier rows. Inductively, $\mathscr{R}_{\mathfrak{b}}^{n+1}$ is obtained from $\mathscr{R}_{\mathfrak{b}}^n$ by adjoining the ordinary outputs and first nonordinary total-graph values of every public operation that accepts a word already present: weak evolution, localization, multilinear polarization, constraint and gauge descent, chart transport, boundary trace, signed interface gluing, carrier formation, return, and same-origin succession. Put

$$\mathscr{R}_{\mathfrak{b}}^\infty := \overline{\bigcup_{n \ge 0} \mathscr{R}_{\mathfrak{b}}^n}^{\mathrm{cyl}}. \tag{431}$$

Each word retains its complete source ancestry. This is the forward closure of what the public equation can generate from the source occurrence; it is formed without a norm bound or compactness premise.

Definition III.4.4 (Target-incidence core and minimal realization)**.** For finite depths m, n, form the bipartite total-incidence graph $\mathsf{G}_{\Sigma,\mathfrak{b}}^{m,n}$. Its left vertices are the words of $\mathscr{R}_{\mathfrak{b}}^m$, its right vertices are the covectors of $\mathscr{C}_{\mathfrak{b}}^n$, and it contains:

(I1) an ordinary edge (R, λ) for every nonzero signed-dyadic cell of the pairing $\langle \lambda, R \rangle$;

(I2) a typed edge for the first nonordinary value of that pairing or of an operation needed to form it; and

(I3) the directed constructor edges recording forward ancestry and backward adjoint transport; and

(I4) one typed frontier vertex for each well-typed public constructor schema whose next instance has not yet been expanded, with edges from its admitted input types to its possible output and adjoint types.

Delete every vertex that lies on no directed path from one of the five source roots to the root target covector $D\Phi$, allowing paths through frontier vertices, and repeat deletion until stable. A frontier vertex is replaced by its ordinary, zero, signed-dyadic, and graph-boundary values when its instance enters the enumeration. Denote the surviving finite graph by $\mathsf{M}_{\Sigma,\mathfrak{b}}^{m,n}$. Its diagonal closure

$$\mathsf{M}_{\Sigma,\mathfrak{b}} := \overline{\bigcup_{k \ge 0} \mathsf{M}_{\Sigma,\mathfrak{b}}^{k,k}}^{\mathrm{gc}} \tag{432}$$

is the *minimal target-active realization*; frontier vertices are absent from this diagonal closure because every fixed constructor instance is eventually expanded. A pairing value zero in the ordinary graph is deleted; failure of the zero pairing to descend through a constructor is its typed incidence edge and survives precisely when it lies on a source–target path.

Theorem III.4.5 (Target-incidence compression)**.** *For every active signature $\mathfrak{b}$:*

(M1) *every contribution to a shell crossing, every target-visible limit defect, and every target-visible first failure belongs to $\mathsf{M}_{\Sigma,\mathfrak{b}}$;*

(M2) *deletion of the complementary forward words and backward covectors does not change the shell-current identity, the admitted physical price, the aggregate feeding–storage row, the recurrent circulation price, the prospective source profile, or the set of independently realized singular mechanisms;*

(M3) *the reduction is minimal: every retained nonfrontier vertex occurs on a source–target constructor path or on a typed prefix that can extend through the current frontier, while removing it deletes at least one algebraically possible target-current word at that depth; every spurious frontier path is removed at the first depth that resolves its failed extension;*

(M4) *if $\mathsf{M}_{\Sigma,\mathfrak{b}} = \varnothing$, then the signature is target-null. If it is nonempty, each of its terminal source–target paths has one of the five source ancestries, and the sum of their signed pairings is the unit shell progress.*

Thus structural analysis of singular formation may be performed entirely on $\mathsf{M}_{\Sigma,\mathfrak{b}}$.

Proof. Start with the exact unit shell-current identity. Every term has a source root in $\mathscr{R}^0_{\mathfrak{b}}$ and the target root $D\Phi$. Apply one public constructor. On its ordinary graph, the chain rule or weak duality moves the constructor forward on the current or backward by its adjoint on the covector, preserving the pairing. On a nonordinary graph value, source-preserving totalization inserts the failed operation as the typed incidence edge. Induction over constructor depth proves (M1).

A deleted vertex has no complete source–target path. Its ordinary pairing with every target-cyclic descendant is zero, and every failed descent that could change this conclusion would itself be a typed path and hence would survive. Removing all such vertices therefore changes neither the shell identity nor any object generated from its target-visible terms. Physical price is admitted only through those terms, signed feeding and storage are their glued images, circulation uses their return edges, and diagonal realization uses their same-origin paths. This proves (M2).

Finite graph pruning retains exactly the vertices on source–target paths or on their well-typed not-yet-instantiated constructor prefixes. Exhaustive enumeration resolves every fixed frontier instance, proving (M3). Compact diagonal closure preserves the realized paths and all typed boundary values. Empty core annihilates every source pairing, so prospective accountability forbids shell crossing. On a nonempty core, group the retained root paths by their inherited source labels and recover the five-source shell identity, proving (M4). □

Theorem III.4.6 (Commutation of the continuum compiler with minimal target realization)**.** *Let P_{M} denote restriction to $\mathsf{M}_{\Sigma,\mathfrak{b}}$, and let $|\mathsf{M}_{\Sigma,\mathfrak{b}}|$ denote the closed cylinder state support on which all retained incidence paths are realized. Target quotient, sourcewise completion, generated production, signed carrier gluing, joint-kernel restriction, first-return formation, circulation duality, circulation elimination, and finite target branch-and-bound commute with P_{M}. In particular,*

$$\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = \delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}|_{\mathsf{M}_{\Sigma,\mathfrak{b}}}, \tag{433}$$

$$X_\infty = X_\infty \cap |\mathsf{M}_{\Sigma,\mathfrak{b}}|, \tag{434}$$

$$\mathscr{U}_n = \mathsf{P}_{\mathsf{M}_n}\mathscr{U}_n, \tag{435}$$

where M_n is the finite incidence core at the coordinates used at depth n. Consequently the finite evaluator never generates a moment, multiplier, defect, tail, or graph coordinate outside a complete source–target path.

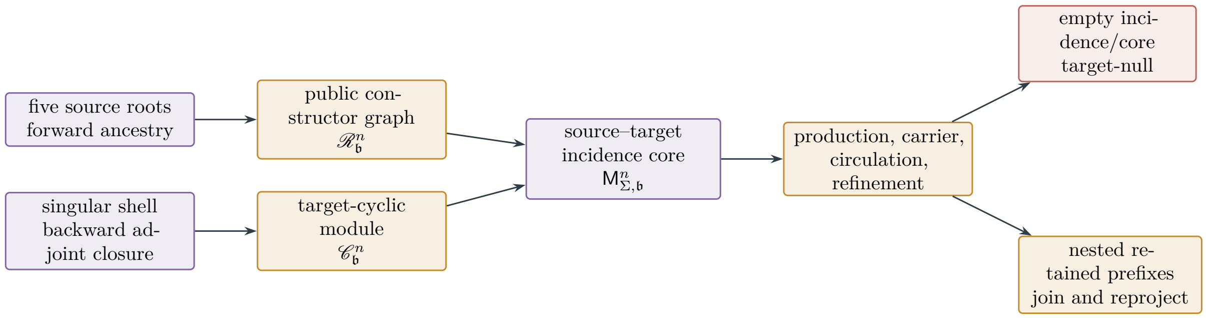

Figure 22: Bidirectional target-incidence fast track. Forward closure records what the equation can generate from the five source occurrences. Backward closure records what the singular shell can detect through adjoint and total-graph descendants. Only their incidence core is passed to physical pricing, recurrent circulation, and finite refinement. A nested retained prefix is a live joint-saturation state, not a terminal.

Proof. Each listed operation is generated functorially from pairings and public constructor edges. By Theorem III.4.5, its target-visible inputs and every failure of its descent lie in the incidence core. Restricting before or after the operation therefore gives the same ordinary values and the same typed boundary values. The physical objective and target-return normalization are unchanged, proving (433). Return, origin, and realization edges are among the constructor paths, giving (434). At finite depth, graph pruning is a finite preliminary step and the remaining moment rows are identical, which proves (435). □

Corollary III.4.7 (Finite incidence fast track)**.** *At cylinder depth n, the minimal target-active realization is computed before any ordered moment problem by two finite graph searches: forward from the five source roots and backward from $D\Phi$. Their intersection is the conservative incidence core $\mathsf{M}^{n,n}_{\Sigma,\mathfrak{b}}$, including its finitely many typed constructor-frontier vertices. If the finite incidence graph has V_n vertices and E_n constructor or pairing edges, this pruning uses $O(V_n + E_n)$ graph operations. The depth- n moment program then contains only coordinates and localizing rows indexed by the intersection. Empty intersection closes the finite target cell by the zero-pairing identities attached to the deleted edges; nonempty intersection supplies the exact list of currently generated target-aligned rows to price or refine, together with the frontier schemas that must be expanded before any further deletion.*

Proof. A vertex lies on a source–target path exactly when it is both forward reachable from a source root and backward reachable from the target root. Breadth-first or depth-first search computes these two sets in linear graph time. The commutation theorem permits deletion before production and moment generation. Frontier vertices prevent deletion of a prefix that a later well-typed constructor can connect. The stored zero or nonordinary pairing value on each resolved boundary edge supplies the stated local identity or typed successor. □

Definition III.4.8 (Covariant production valuation)**.** Let r be a generated record scale. For every carrier term K, production P, and target current Q, the compiler obtains their scale weights by substituting the public scaling chart into the displayed weak identity:

$$K = r^{\kappa_K}\widetilde{K}, \qquad P = r^{\kappa_P}\widetilde{P}, \qquad Q = r^{\kappa_Q}\widetilde{Q}.$$

These are valuations of equation terms, not analytic estimates. Terms of equal weight are first combined with their signs. If the resulting carrier law has critical weight, it is used directly. If a positive power of r would erase normalized production as $r \downarrow 0$, the scale velocity $a = \dot{r}/r$ is retained and Proposition II.2.36 is applied. Exact scale coboundaries are absorbed into the covariant carrier; nonexact scale cohomology remains an explicit Caus- or Lift-current. A failure of the chart, valuation, conjugation, or coboundary identity is the corresponding total-graph successor. In particular, the compiler never infers $\widetilde{P} = 0$ from $r^{\kappa_P}\widetilde{P} \to 0$.

Definition III.4.9 (Defect-polarization compiler)**.** Suppose a public d-linear operation B has total weak graph coordinate

$$B(u_\nu, \dots, u_\nu) \rightharpoonup B(u, \dots, u) + \mathcal{R}_B.$$

The compiler reconstructs the symmetrized multilinear defect from its rational diagonal values by the polarization identity. When an ordered nonsymmetric component is used by the weak equation, the rational grammar also adjoins all mixed-input graph coordinates; these determine that component separately. It then applies the following exclusive rules.

(D1) A positive combination is admitted only when positivity follows from the public positive cone, an exact square, or lower semicontinuity of an already positive carrier form. It is added to $\mathscr{P}_{\mathfrak{b}}$.

(D2) A signed defect is retained as a signed current and glued across all compatible faces before its Jordan decomposition.

(D3) Failure of positivity, polarization, gluing, localization, or descent is retained as the first nonordinary graph value with the ancestry of B.

Thus oscillation and concentration are generated structural coordinates of the weak graph. They cannot be discarded as errors and cannot be promoted to physical expenditure without a carrier identity.

Proposition III.4.10 (Functorial constraint and boundary descent)**.** *The target-cyclic construction commutes with every ordinary constraint, gauge, chart, trace, and boundary quotient generated by the public presentation. More precisely, if q is such an ordinary quotient, then*

$$q^*\mathscr{C}^n_{q\mathfrak{b}} \subseteq \mathscr{C}^n_{\mathfrak{b}}, \qquad q(X^n_{\mathfrak{b}}) \subseteq X^n_{q\mathfrak{b}}$$

on its ordinary graph domain. Equality holds only on coordinates for which the generated lifting graph is ordinary and onto. A missing reverse inclusion is therefore recorded by its cokernel coordinate. Outside the ordinary domain, the least failed commutation, tail, trace, chart, or gauge coordinate is routed as a target-null value, a physically charged value, or the next target-cyclic successor. No surjectivity, closed range, compact trace, or tail decay is assumed.

Proof. The assertion holds at depth zero because the target shell is pulled back through the same quotient. If it holds at depth n, naturality of pullback gives the claim for adjoint transport and Lie derivatives; the multilinear polarization identity gives it for weak defects; signed gluing gives it for interfaces; and the public covariance identity gives it for record-scale charts. These are exactly the generators in Definition III.4.1. Induction proves the ordinary inclusions. An ordinary onto lifting graph supplies the reverse inclusions. Totalization supplies a value when naturality or lifting fails, and the source-preserving graph closure retains the failed operation and the target covector that detected it. The three stated routes are then the exhaustive source-accountability alternatives. $\square$

Theorem III.4.11 (Target-cyclic joint-kernel descent)**.** *Let $\mathfrak{b}$ be a same-origin target-active branch obtained from a legal orbit with finite physical ledger. Run Definitions III.4.1, III.4.2, III.4.8 and III.4.9 in the fixed enumeration of rational cylinders and public graph operations. Exactly one of the following occurs, with priority given to the first finite event in that enumeration.*

(K1) *A generated covariant carrier law charges a disjoint cofinal family of shell crossings by a nonnegative physical measure whose required total mass diverges. This contradicts the finite physical ledger.*

(K2) *On the retained same-origin path support in $X_{\mathfrak{b}}^{\infty}$, every source current selected by the generated target-cyclic covectors is target-null:*

$$\langle \lambda, J^c \rangle = 0 \quad (\lambda \in \mathscr{C}_{\mathfrak{b}}^{\infty} \textit{ along that support}, c \in \{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}).$$

Hence that retained path cannot cross the selected shell. No assertion is made about the rest of $X_{\mathfrak{b}}^{\infty}$.

(K3) *A production, defect, commutator, scale-cohomology, constraint, boundary, chart, or total-graph coordinate is nonzero. Its least such value, together with its zero, signed dyadic, or graph-boundary polarity, detecting covector, and inherited origin, is the next proper target-active successor; the child is first joined to the complete parent transcript and reprojected.*

(K4) *All finite coordinates of the cumulative joint record are resolved, compatible, and ordinary, and joint saturation is a fixed point. The fixed record retains a nonzero unnormalized shell cocycle and complete inherited five-channel ancestry. Its totalized realization row is ordinary and gives a same-origin path crossing every cofinal target shell. Thus it is a joint-saturated prospective source atom in the sense of Definition III.4.74, with its original origin polarity, not an inverse-limit or resolved-prefix output.*

No compactness, coercivity, Liouville property, closed-range statement, or equality classification occurs among the hypotheses.

Proof. Contact gives unit shell increment across each normalized shell. By prospective accountability, each completed crossing has a finite positive source component or a positive boundary owner. Starting from that shell covector, differentiate the pairing through the public equation. The chain rule gives the adjoint transports and Lie derivatives in (C1); localization, quotient descent, weak passage, and carrier completion give (C2)–(C4). Hence every term exposed by any finite iteration is either an exact signed current, a generated nonnegative production, or a total-graph value. The signed terms are glued before their positive parts are formed, and Definition III.4.8 prevents collapsing scales from erasing a production. This proves that the enumeration omits no descendant of the initial target pairing.

If a carrier domination charges the cofinal disjoint crossings with divergent total mass, (K1) holds. Restrict next to the actual same-origin path support. If every generated target pairing vanishes there, the shell derivative vanishes along that path, so the fundamental theorem of calculus forbids its shell crossing and (K2) holds. No global annihilation of the joint kernel is used. Otherwise choose the least nonzero or nonordinary coordinate of the cumulative joint record and apply its generated polarity split. Source-preserving totalization, Proposition III.4.10 and Theorem II.2.15 give (K3) without an analytic exclusion assumption. Joining the child to the whole preceding transcript before reprojection retains every mixed operation through which individually target-null coordinates could become jointly target-aligned.

It remains to consider the case in which every finite coordinate is resolved and compatible in its induced graph. Form the cumulative joint records of Definition III.4.27 before taking a limit. By Theorem III.4.58, an unprocessed target component of their graph-complete limit already has a least nonzero finite joint increment; that prefix is a successor of type (K3). If the full joint target projection vanishes, the joint limit belongs to (K2). The only remaining case, after every increment has acquired its complete five-source ancestry, is a fixed point of the joint, graph, source, carrier, and target-cyclic closures. Its unnormalized shell cocycle cannot vanish, because each completed crossing contributes one before any return normalization. Totalize its same-origin realization row. A nonordinary value is again a finite successor of type (K3); an ordinary contacting value is precisely the structural atom in (K4). Hence neither increasing rank nor inverse-limit existence is a terminal argument. The fixed priority rule makes the four outputs disjoint. □

Corollary III.4.12 (Generated mobility-annihilator test)**.** *If the restriction of the target-cyclic module to the retained same-origin path support annihilates all five induced currents there, then that path cannot contact the target. Otherwise the least nonzero restricted pairing, together with its generated polarity cell, is the initial covector of the next successor. Thus the covector needed to continue the equality analysis is generated by the equation and the target; it is never an external multiplier or a user-supplied rigidity certificate.*

III.4.2 Target-relative resaturation of zero-price dynamics

A zero-price equality law is not a terminal object. It is a new public dynamical stratum on which prospective accountability is run again. This is the continuum counterpart of forming the saturated closure before taking the residual in Part I.

Definition III.4.13 (Zero-price resaturation successor)**.** Let $\mathfrak{b}_n = (A_n, m_n, \Gamma_n, \eta_n, \mathbf{R})$ be a reachable active branch with ordered ideal I_n, dual-carrier module $\mathscr{W}_n$, and zero-price recurrent law L_n. Retain its cumulative joint record $\mathcal{J}_n$ and unnormalized shell cocycle. Its *resaturation successor* is constructed by the following generated operations.

(R1) Starting from the selected shell covector, generate $\mathscr{C}^\infty_{\mathfrak{b}_n}$ and $\mathscr{P}_{\mathfrak{b}_n}$ by Definitions III.4.1 and III.4.2. Adjoin every production annihilated by L_n, and form the closed ordered kernel

$$X_{n+1} := X_n \cap \bigcap_{P \in \mathscr{P}_{\mathfrak{b}_n}} \{P = 0\}. \tag{436}$$

(R2) Restrict the public weak equation, its constraints, carrier, charts, and total graphs to X_{n+1}. Recompute the five source projections there; no old source tag or origin coordinate is discarded.

(R3) Apply the prospective shell identity on the restricted pre-contact path. If contact remains, select a persistent source support and target covector with the normalized floor $1/5$.

(R4) Apply the covariant production valuation and defect-polarization compiler. Adjoin every resulting adjoint, quotient, localization, signed-gluing, covariance, commutator, defect, and square-completion relation to I_n, and adjoin its detecting covector to $\mathscr{W}_n$. Exact scale coboundaries are absorbed and nonexact scale cohomology is retained as a source current.

(R5) Totalize every operation used above. Its first nonordinary value is a typed successor. If every value is ordinary, the restriction gives the next zero-price branch $\mathfrak{b}_{n+1}$.

(R6) Apply the generated polarity split to the next unprocessed coordinate on the retained same-origin target stratum. Prune a cell only by its finite local ordered identity; every positive-moment cell appends its polarity relation to the ideal. A normalized return law may select its support; the unnormalized target current is unchanged before any physical charge is made.

(R7) Expand the complete derivation of the selected child, join it with the expanded records in $\mathscr{J}_0, \dots, \mathscr{J}_n$, generate every mixed operation and polarization, saturate, and reapply the five-channel target projection. Retain every physical owner and normalization denominator. A least unprocessed target increment is the next source-resolved successor even when each of its individual input coordinates was target-null.

The successor is *proper* when either its ordered ideal or its observable module strictly increases after reduction by the preceding relations. Every proper successor has strictly larger resolved-prefix rank. An infinite rank chain is evaluated by its cumulative joint record under Theorem III.4.58; rank growth is not a terminal argument.

Proposition III.4.14 (Monotone resaturation)**.** *Along a zero-price resaturation chain,*

$$\begin{aligned} &I_0 \subseteq I_1 \subseteq \cdots, && \mathscr{W}_0 \subseteq \mathscr{W}_1 \subseteq \cdots, && X_0 \supseteq X_1 \supseteq \cdots, \\ &\mathcal{A}_0 \subseteq \mathcal{A}_1 \subseteq \cdots, && \rho(\mathfrak{b}_0) < \rho(\mathfrak{b}_1) < \cdots . \end{aligned} \tag{437}$$

Every retained branch has the same origin tag, and every completed shell prefix retains $A_{\Phi,N} = N$. At a limit stage use

$$I_\infty := \overline{\bigcup_n I_n}^{\mathrm{ord}}, \qquad \mathscr{W}_\infty := \overline{\bigcup_n \mathscr{W}_n}^{\mathrm{cyl}}, \qquad X_\infty := \bigcap_n X_n, \qquad \mathscr{J}_\infty := \mathrm{Sat}_{\mathsf{G}_\Sigma} \overline{\bigcup_n \mathscr{J}_n}^{\mathrm{gc}}. \tag{438}$$

These closures create no source: every limit relation retains the ancestry of the finite words that generate it, and every target component of $\mathscr{J}_\infty$ is the limit of finite joint-prefix components.

Proof. Kernel restriction only adds zero relations, while dual-carrier compilation only adds words and their consequences. The polarity step appends the next resolved coordinate, and the joint step enlarges the cumulative sigma algebra, proving the inclusions and the rank inequality. The shell selector is applied to the same pre-contact path, so the origin tag is unchanged and each completed shell contributes one to the unnormalized cocycle. The closures in (438) are the inverse-limit closures of the countable rational cylinder grammar. Source-preserving completion and Theorem III.4.58 therefore give the final assertion. □

III.4.3 Recurrent passivity and support transfer

The resolved-prefix rank detects a first failed operation, but rank growth alone does not evaluate a current which repeatedly reappears after renormalization. The carrier must be followed through the same-origin successor chain. The following construction performs that step before a limit path is declared to be a singular mechanism.

Definition III.4.15 (Target-local marked carrier complex)**.** Fix a same-origin active branch $\mathfrak{b}$ and its compact retained path space. A marked carrier cell is a tuple

$$C = (w, w^+, \omega, K^-, K^+, P, H, E),$$

where $w\mathcal{R}_\eta w^+$, ω is the retained successor witness, $K^\pm$ are the endpoint values of one generated carrier, $P \geq 0$ is the sum of the generated nonnegative productions in that cell, H is the complete signed interior, chart, scale, constraint, and interface current after compatible faces have been glued, and E is the true exterior or physical-input current. Every coordinate is restricted to $X_\mathfrak{b}$ and carries the detecting shell covector. Its total graph contains the exact relation

$$P = K^- - K^+ + H + E. \tag{439}$$

Let $\mathscr{B}_{\mathfrak{b}}$ be the closed linear space generated by shift coboundaries $F \circ S - F$, paired internal-face currents, exact chart and gauge currents, and currents annihilated by the selected target quotient. The *target-local feeding class* is

$$[H]_{\mathfrak{b}} \in \mathscr{H}_{\mathfrak{b}} := \overline{\operatorname{span}\{H\}}/\mathscr{B}_{\mathfrak{b}}. \tag{440}$$

Exterior input is not put in this quotient. Its ordinary carrier-dominated part is charged by (98); every other value is its Bdry- or carrier-domination successor.

Proposition III.4.16 (Stationary carrier reduction)**.** *Let ϖ be a shift-invariant probability law on a concatenated marked carrier path in one active branch. On the ordinary finite-integral value,*

$$\int P \, \mathrm{d}\varpi = \int [H]_{\mathfrak{b}} \, \mathrm{d}\varpi + \int E \, \mathrm{d}\varpi. \tag{441}$$

Here the first integral on the right means evaluation of the cohomology class by the stationary law. If an endpoint, concatenation, integrability, quotient, or exterior coordinate is nonordinary, its least total-graph value is the corresponding proper successor.

Proof. Sum (439) over a finite consecutive block. Concatenation cancels every common endpoint carrier. Divide by the block length and pass to the empirical path law. Bounded endpoint storage vanishes, as does every shift coboundary. Paired faces and exact chart or gauge currents already vanish in the quotient. This gives (441). Each operation used in this calculation is totalized, so its first nonordinary value gives the stated successor instead of an identity. The division constructs a recurrent support law only. The unnormalized sums of P, H, E, and all five source currents remain coordinates of the same structural record and are not consequences of (441). ☐

Theorem III.4.17 (Target-aligned passivity–support alternative)**.** *Let an infinite zero-price successor path retain the active signature $\mathfrak{b}$, and form its stationary marked carrier law. Exactly one of the following events occurs in the fixed total-graph order.*

- (V1) *A true exterior or input current is admitted by the target-local carrier graph. Its restrictions to the diagonal original events are charged to the public physical measure; divergent required mass contradicts the finite ledger, and finite mass enters the zero-input tail successor.*
- (V2) *A marked-carrier, gluing, quotient, support-transfer, or realization coordinate is nonordinary and gives its typed proper successor.*
- (V3) *The ordinary feeding class has nonzero positive stationary value. The normalized positive measure*

$$\mathrm{d}\lambda_H := \frac{([H]_{\mathfrak{b}})_+ \, \mathrm{d}\varpi}{\int ([H]_{\mathfrak{b}})_+ \, \mathrm{d}\varpi} \tag{442}$$

  has a target-visible support projection. Its first atomic, separated, nested, diffuse, tail, rough, gauge, constraint, or recurrent projection is the next same-origin target-active return record, retaining the source word of the current that finances P. It is a proper successor when it resolves a new coordinate; otherwise its positive-measure signed-dyadic cell defines the induced recurrent-current return map.
- (V4) *The ordinary exterior value and feeding cohomology vanish. Then*

$$\int P \, \mathrm{d}\varpi = 0, \qquad P = 0 \quad \varpi\text{-almost surely}, \tag{443}$$

  and the path law is restricted to the target-cyclic joint kernel generated by these zero productions.

Thus lack of passivity is itself a target-aligned structural current. It cannot be retained as an unevaluated equality law.

Proof. Apply Proposition III.4.16. The exterior coordinate is processed first by the local charging rule, giving (V1) or its successor. Any nonordinary operation gives (V2). On the remaining ordinary zero-exterior cell, nonnegativity of P gives

$$\int [H]_{\mathfrak{b}} \, \mathrm{d}\varpi = \int P \, \mathrm{d}\varpi \geq 0.$$

If the value is positive, (442) is a probability measure on the already compact activity observer space. Apply Theorem II.2.42. If every support projection were target-null, the target-local functional defining $[H]_{\mathfrak{b}}$ would integrate to zero. Hence one projection is target-visible and supplies (V3) with its same-origin witness. On an equivalent support cell, Poincare recurrence for the least positive signed-dyadic set supplies the induced return map, so equivalence does not discard the current. If the value is zero, (441) gives $\int P \, \mathrm{d}\varpi = 0$. Since $P \geq 0$, it vanishes almost surely, proving (V4). □

Theorem III.4.18 (Target-visible fidelity defects generate carrier successors)**.** *Let F be a public equation, constitutive, nonlinear, constraint, trace, chart, or realization operation, and let $\mathcal{D}_F$ be a nonzero value of its total-graph defect on an active branch $\mathfrak{b}$. For a target-cyclic covector λ, exactly one of the following occurs.*

(D1) $\langle \lambda, \mathcal{D}_F \rangle = 0$*; this occurrence of the defect lies in the target kernel.*

(D2) *The normalized variation of $\langle \lambda, \mathcal{D}_F \rangle$, localized by the rational cylinder core, produces an atomic, separated, nested, diffuse, tail, or graph-boundary support record with the same origin and the ancestry of F. A support record inequivalent to the parent is a proper target-active successor.*

(D3) *Every positive support record is equivalent to the parent in the resolved coordinates. Its event-count law is recurrent. The defect pairing is then retained, with its sign, as either a nonzero component of $\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}}$ or an exact carrier-storage transfer in $\mathscr{B}_{\mathfrak{b}}$. Both values are evaluated by Theorem III.4.22; an exact transfer is not identified with a target-null current.*

Consequently no target-visible fidelity defect can disappear at a recurrent limit: it is a proper support successor, a recurrent feeding current, or a recurrent storage-transfer current.

Proof. Apply the scalar distribution $T = \langle \lambda, \mathcal{D}_F \rangle$ to the enumerated rational local test core. Either every value is zero, which is (D1), or there is a least test with nonzero value. The total variation of its rational localizations is a nonzero positive finite measure on each finite cylinder; if its variation is infinite, use the normalized-variation boundary coordinate. Normalize and apply compact activity-profile exhaustion. Source-preserving completion retains the ancestry of F, and the same-origin selector retains the original event indices. This gives (D2).

If one support record is inequivalent to the parent, its witness is a proper successor. Otherwise take empirical laws of the equivalent support records. Successor recurrence gives a shift-invariant law. Pairing the public weak equation with the transported localizations of λ places T in the complete signed current H; paired internal occurrences cancel before the quotient. If its class is nonzero, its target projection is retained as a feeding coordinate. If its class is zero, the definition of the closed coboundary space supplies bounded cylinder storage functions whose coboundaries converge to the pairing. Their finite-block sums are endpoint transfers. The support law retains those endpoints and therefore routes the value as a carrier-storage coordinate. Only annihilation by every target-cyclic covector gives (D1). This proves (D3) without the false implication that nonzero variation has nonzero stationary mean. □

Definition III.4.19 (Target-active recurrent component)**.** Let $\Omega_{\mathfrak{b}}$ be the compact path space of same-origin successor chains and let S be the left shift. A *target-active recurrent component* is the support of an ergodic S-invariant probability law ϖ obtained from a contact-generated chain and satisfying the retained shell floor. Every generated scalar coordinate has a fixed zero, signed-dyadic, or graph-boundary value on the corresponding measurable cell. A target-visible value is retained even when its support successor is equivalent in the previously resolved coordinates. In that case it is a recurrent current coordinate and remains an active operation of the compiler.

Such a component exists for every infinite chain: take a weak limit of its empirical shift laws and then an ergodic component with nonzero target mass. The empirical law selects recurrent support only. It is stored jointly with the cumulative record and unnormalized cocycle of Definition III.4.27; target projection is recomputed on that join. This construction asserts recurrence of structural records only. The source record retains the same ordinary stopped history from which it was generated; its target status is determined only after rejoining that complete history and evaluating the reachable contact cell.

Theorem III.4.20 (Target-relative recurrent stratification)**.** *Every infinite same-origin zero-price resaturation route has exactly one of the following constructive outputs.*

(T1) *a finite target-null, continuation, class-incompatibility, or divergent physical-charge relation;*

(T2) *a proper target-active successor produced by a compactness mode, a fidelity defect, or a previously unprocessed feeding or storage current;*

(T3) *a shift-invariant equality law satisfying* (443)*, followed by the target-cyclic mobility-annihilator test;*

(T4) *a target-active recurrent component. On it, every nonzero aggregate feeding or storage current is selected by a signed-dyadic return set and becomes the next recurrent-current successor. If all such currents vanish, the component lies in the generated zero-production joint kernel. A cofinal chain of recurrent-current successors is accumulated in $\mathcal{J}_\infty$ and reprojected. It is target-null, has a nonzero unprocessed finite joint or polar increment, or is a primal–polar fixed joint-saturated structural atom. Its hypothetical ordinary realization is a source-profile witness; only its independent ordinary realization is an explicit singular mechanism.*

A nonzero public-equation, constitutive, constraint, boundary, fidelity, support-transfer, feeding, or storage coordinate is therefore an operation on (T4)*, never a terminal name. In* (T3)*, a target-null joint kernel closes; a nonzero target-cyclic pairing is the next successor. Consequently an unending classification chain is replaced by a recurrent current system with a definite structural evolution.*

Proof. Apply compact activity-profile exhaustion to the diagonal same-origin path. An atomic, separated, nested, diffuse, tail, rough, gauge, constraint, or equality value is a coordinate of its single activity law. Process its first target-visible value. The carrier values are then exhausted by Theorem III.4.17. Cases (V1)– (V2) give (T1)–(T2); case (V3) gives a proper support successor when it resolves a new coordinate and otherwise gives the recurrent-current return system in (T4); case (V4) gives (T3).

Repeat on every proper successor. Strict resolved-prefix growth and compact diagonal saturation give a same-origin joint limit record. By Theorem II.2.45 it carries an invariant law; select an ergodic component of nonzero target mass. Apply Theorem III.4.18 to every retained public-equation defect. A target-visible defect returns to (T2) or becomes a recurrent feeding or storage coordinate; every remaining defect is target-null.

For a nonzero recurrent coordinate choose the least signed-dyadic set $A = \{sF \in [2^{-k}, 2^{-k+1})\}$ of positive ϖ-measure. Poincare recurrence makes the first-return map S_A finite almost everywhere, and the normalized restriction $\varpi(A)^{-1}\varpi|_A$ is S_A-invariant. This law selects the recurrent-current support in (T4). Rejoin it with the complete same-origin witnesses, source ancestry, and unnormalized shell cocycle before projection. If the join resolves a new coordinate it is (T2). If equivalent successors recur indefinitely, form their cumulative joint record and apply Theorem III.4.58: it is target-null, has a finite joint-prefix successor carrying the least unprocessed increment, or has all primal increments source-resolved. In the last case apply Theorem II.2.32; a separator is the next successor, while only simultaneous primal–polar fixedness gives a structural atom. Rejoining the last cell with its complete same-origin stopped history gives the actual contact-cell test in (T4). A nonordinary value is another typed successor. If no nonzero coordinate remains, stationary carrier reduction gives (T3). This proves exhaustiveness without asserting that an increasing refinement chain has a sink. □

Corollary III.4.21 (Local criterion excluding a recurrent singular component)**.** *Suppose the generated passivity row of every reachable active signature has target-null zero-production kernel, and every recurrent feeding, fidelity, storage, or support-transfer current is either charged by the public physical carrier or has target-null return law. Then no target-active recurrent component contacts the continuation target. Hence every finite-physical- expenditure ordinary branch is regular and continuable. These statements are outputs of the local ordered cones, return maps, and total graphs of the active signatures; they are not specialization data.*

Proof. A contact branch produces an infinite zero-price route after the divergent- price cases have been removed. Apply Theorem III.4.20. The first three outputs close or continue. In the recurrent output, every nonzero current is charged or has target-null return law by hypothesis. The remaining component lies in the zero-production kernel, which is target-null by the generated local ordered identity, contradicting its shell floor. □

Theorem III.4.22 (Target-local passivity fast track)**.** *For a reachable active signature $\mathfrak{b}$, after exterior input has been processed, the full recurrent carrier analysis reduces to one aggregate target-current row,*

$$\mathfrak{J}_{\Sigma,\mathfrak{b}} := (\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}}, \Pi_{\Sigma,\mathfrak{b}} H_{\mathfrak{b}}^{\mathrm{stor}}), \tag{444}$$

where $H_{\mathfrak{b}}^{\mathrm{stor}}$ is the endpoint-transfer coordinate of the closed coboundary graph. Its total ordered graph has the following exhaustive evaluation.

(F1) *A finite identity in $\mathcal{K}_{\mathfrak{b}}$ proves $\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}} = 0$. Stationary carrier reduction forces the generated production to vanish. A nonzero storage coordinate is then selected by its signed-dyadic return set; if storage also vanishes, the target-cyclic joint kernel is evaluated on that same cell.*

(F2) *The class has a nonzero signed-dyadic value. Its positive support law* (442)*, source word, detecting covector, and same-origin witness define a positive-density return-map record. The compiler applies the five-source normal form and prospective accountability directly to this record; it does not separately estimate the individual pressure, transport, chart, cutoff, or scale terms from which the class was assembled.*

(F3) *The projection, gluing, support, or realization graph is nonordinary. Its first graph-boundary value, with the same data, is the next active record.*

Thus a specialization evaluates one aggregate target-aligned row. Exact storage and nonexact feeding are distinguished, but neither is discarded. Atomic, separated, nested, diffuse, and tail modes occur only as support coordinates of the selected return law and introduce no additional proof stage.

Proof. Signed carrier gluing constructs $[H]_{\mathfrak{b}}$ and its storage graph before a positive part is taken. Apply the target-local factorization and the generated polarity split to (444). The zero cohomology cell is (F1) by Proposition III.4.16 and Theorem III.4.17. On that cell the exact component is a block endpoint transfer and is split by the same signed-dyadic return construction. Positive and negative dyadic feeding cells retain the signed current; reverse the orientation on a negative cell and apply the positive support construction, giving (F2). Poincare recurrence supplies the induced return law. Totalization gives (F3). Compact activity exhaustion is applied to the single support measure in (F1), (F2), or (F3), which proves the final assertion. □

Definition III.4.23 (Target-aligned recurrent closure compiler)**.** For a reachable active signature $\mathfrak{b}$, a compiler state is

$$\mathfrak{q} = (X, I, \mathscr{C}, \mathscr{P}, \varpi, \mathfrak{J}_{\Sigma,\mathfrak{b}}, \delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}, \mathbf{R}, \mathcal{J}_{\le k}, \Gamma_{\le k}, A_{\Phi,\le k}, \Pi_{\le k}, (Q^{\mathrm{c}}_{\Phi,\le k})_{c\in\mathfrak{C}}), \tag{445}$$

where X is the current same-origin target cell, I its ordered ideal, $\mathscr{C}$ the target-cyclic covector module, $\mathscr{P}$ the generated production family, ϖ its recurrent law when present, $\mathfrak{J}_{\Sigma,\mathfrak{b}}$ the aggregate feeding–storage row, and $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}$ its sharp recurrent circulation price when the return graph has been formed, and $\mathbf{R}$ the original-event witnesses. The final entries are the cumulative joint record, occurrence multiplicity, unnormalized shell cocycle, cumulative physical expenditure, and cumulative five-source currents from Definitions III.4.27 and III.4.68. Before enqueueing a state, join it with every preceding state on the same execution, saturate the joint record, and then restrict all coordinates to the minimal target-active realization $\mathsf{M}_{\Sigma,\mathfrak{b}}$. Initialize the queue with the algebraically reachable shell state. Process its least generated coordinate by the following fixed transition order:

(Q1) Dovetail the finite identities in the local cone and the dual circulation-price hierarchy. A verified -1-identity removes the cell. A carrier-domination identity records the restriction of the public physical measure; a cofinal family with divergent total charge removes the cell. A positive recurrent circulation price produces the dual carrier potential and removes the recurrent cell by infinite return charge.

(Q2) On the zero cell of $\Pi_{\Sigma,\mathfrak{b}}[H]_{\mathfrak{b}}$, adjoin every forced nonnegative production to the zero-production joint kernel. Apply prospective accountability on the restricted same-origin path. Vanishing of all target-cyclic pairings removes the cell in $\mathscr{N}_{\Sigma}$; otherwise the least nonzero pairing is appended to the queue.

(Q3) On the zero-density circulation supplied by the primal hierarchy, select a signed-dyadic feeding or storage cell of positive recurrent mass, form its first-return map and normalized induced law, and append the sign, dyadic level, source ancestry, detecting covector, and return-time graph. The induced law selects support only. Rejoin it with $(\mathcal{J}_{\le k}, \Gamma_{\le k}, A_{\Phi,\le k}, \Pi_{\le k}, (Q^{\mathrm{c}}_{\Phi,\le k})_c)$ and impose the multiplicity-restoration identities before target projection. No zero density is copied into a cumulative zero relation. The resulting joint state is the next queue element even when its previously resolved scalar coordinates agree with those of its parent.

(Q4) On a graph-boundary value, append the first failed operation, its total graph, source ancestry, target covector, and same-origin support law. Apply the same transition order to that typed state.

Only descendants generated from the selected shell covector are enqueued. Every other coordinate is removed by the target quotient. Compatible signed currents are glued before (Q2)–(Q3), so the queue contains aggregate target currents rather than separate pressure, transport, chart, cutoff, scale, and interface estimates. At an infinite execution, the queue forms $\mathcal{J}_\infty$ and applies Theorem III.4.58; it never promotes the compatible inverse limit to an output before this joint reprojection.

Definition III.4.24 (Target-cyclic structural spine)**.** The *target-cyclic structural spine* of an active signature $\mathfrak{b}$ is the least closed subgraph

$$\mathscr{T}_{\Sigma,\mathfrak{b}} \subseteq \mathrm{Sat}_{\mathsf{G}_\Sigma}(q_\Sigma \mathscr{J}) \tag{446}$$

that contains the five projected source currents paired with the selected shell covector and is closed under target-cyclic adjoint transport, projected total graphs, signed carrier gluing, zero-production restriction, signed-dyadic first return, same-origin succession, and diagonal realization. The physical valuation on the spine is the restriction of the public carrier measure. No coordinate with zero target projection belongs to the spine.

Equivalently, $\mathscr{T}_{\Sigma,\mathfrak{b}}$ is generated by starting from the unit shell-current identity and repeatedly adjoining only the next coordinate selected by (Q2)–(Q4). It is therefore enumerated from the public presentation, target shells, and physical carrier; it is not an additional specialization object.

Proposition III.4.25 (Incidence presentation of the target spine)**.** *Let* $\mathrm{pr}_{\mathscr{R}}$ *forget the covector and incidence-edge coordinates while retaining current ancestry, return, origin, and total-graph values. Then*

$$\mathrm{pr}_{\mathscr{R}}\, \mathsf{M}_{\Sigma,\mathfrak{b}} = \mathscr{T}_{\Sigma,\mathfrak{b}}. \tag{447}$$

Thus the minimal realization is the bidirectional, finitely prunable presentation of the target-cyclic structural spine.

Proof. Every retained incidence path starts at a projected source occurrence and is closed under the target-cyclic adjoint, total-graph, gluing, kernel, return, origin, and realization operations that define the spine. Its current projection is therefore contained in the spine. Conversely, structural induction over the least spine closure attaches to every spine word the target-cyclic covector and constructor path that generated it. This gives a vertex of the incidence realization with that current projection. The two inclusions prove (447). □

Proposition III.4.26 (Spine restriction of the recurrent compiler)**.** *Every nonclosing state of the target-aligned recurrent closure compiler lies in* $\mathscr{T}_{\Sigma,\mathfrak{b}}$*. Conversely, every finite spine word is a finite compiler state with the same ordered ideal, source ancestry, physical valuation, and origin witness. Hence restricting the compiler to the spine changes none of its outputs.*

Proof. The initial shell-current state is a spine generator. Transition (Q2) applies a target-cyclic adjoint or joint-kernel restriction; (Q3) applies a first-return constructor; and (Q4) applies a projected total graph. These are precisely the closure operations in Definition III.4.24. Induction places every queued state in the spine. Conversely, structural induction over the least closed subgraph expresses every finite spine word as a finite composition of those transitions. Projection-first saturation supplies every descent defect, and source-preserving completion retains its ancestry. Carrier and origin coordinates are unchanged by Theorem I.4.7. □

Definition III.4.27 (Cumulative joint continuum record)**.** Let $\mathfrak{q}_0 \to \mathfrak{q}_1 \to \cdots$ be any same-origin sequence of structural records generated by the public continuum grammar. Its depth- *n cumulative joint record* is

$$\mathcal{J}_n := \mathrm{Sat}_{\mathsf{G}_\Sigma} \mathrm{Join}(\mathfrak{q}_0, \ldots, \mathfrak{q}_n). \tag{448}$$

The join contains every retained coordinate of the prefix, every mixed product and polarization generated by the public operations, every ordinary or total-graph relation needed to evaluate those coordinates jointly, the union of their five-source ancestries, their common origin, and the complete shell-current history. Joining is an accounting operation on one physical history. It introduces no physical expenditure and carries the restriction of the original physical measure.

For a nonzero finite positive observer law ν on the common measurable same-origin evaluation space on which all $\mathcal{J}_n$ are defined, let $\mathcal{A}_n := \sigma(\mathcal{J}_n)$, let $P_n : L^2(\nu) \to L^2(\mathcal{A}_n, \nu)$ be conditional expectation, and put

$$\mathcal{A}_\infty := \sigma\left(\bigcup_{n\geq 0} \mathcal{A}_n\right), \qquad \mathcal{J}_\infty := \mathrm{Sat}_{\mathsf{G}_\Sigma} \overline{\bigcup_{n\geq 0} \mathcal{J}_n}^{\mathrm{gc}}. \tag{449}$$

For a target shell Φ, let $a_\Phi(e_j) := \Phi(u_j^{\mathrm{out}}) - \Phi(u_j^{\mathrm{in}}) = 1$ be the full normalized progress of the j-th completed shell crossing. The joint record retains the *unnormalized target cocycle*

$$\Gamma_N := \sum_{j<N} \delta_{e_j}, \qquad A_{\Phi,N} := \sum_{j<N} a_\Phi(e_j), \qquad \Pi_N := \sum_{j<N} p_{\mathrm{phys}}(e_j), \qquad A_{\Phi,N} = \sum_{c\in\mathfrak{C}} Q^c_{\Phi,N}(1). \tag{450}$$

The normalized first-return law used to locate recurrent support is stored as a separate coordinate; it never replaces Γ_N, $A_{\Phi,N}$, Π_N, or the source-resolved currents in (450).

More generally, every generated continuum object q carries its *expanded derivation* $\mathrm{Exp}(q)$: the finite directed acyclic graph containing all primitive inputs, every intermediate value, and the constructor at each vertex. The constructor list includes signed and nonlinear combination, restriction, localization, stopping, conditioning, chart change, scaling, quotient, trace, adjoint, polarization, weak passage to a total-graph value, and finite evaluation of a directed limit. If a vertex uses physical quantities $\mu^1, \dots, \mu^k$, the expanded record retains their owner-tagged restrictions and the exact domination or balance graph invoked at that vertex. It never replaces them by an unsigned sum. For a quotient N/D, empirical average, rescaled chart, or conditional law, the record retains (N, D), its zero-denominator boundary cell, and the unnormalized cocycles from which the normalized value was formed.

For a same-origin family (q_j), its *universal cumulative record* is the saturation of the join of $\mathrm{Exp}(q_0), \dots, \mathrm{Exp}(q_n)$, including every mixed constructor generated by the public grammar. Thus (448) is not a construction reserved for shell currents or recurrent scale charts; it is the mandatory evaluation interface for every composite continuum quantity.

Definition III.4.28 (Prospective continuum backward frontier)**.** Fix one completed target-shell crossing, beginning at its last entrance into the outer shell before contact, and stop its public evolution at the first subsequent entrance into the inner shell. A path that exits the outer shell first is a failed attempt and is not included in this completed-crossing record. Let ρ be the finite expanded derivation of the stopped shell increment

$$q_\rho := \Phi_r(u_{\mathrm{out}}) - \Phi_r(u_{\mathrm{in}}) = 1,$$

formed only from values available on this pre-contact interval. A *backward frontier* is an ordered finite set of vertices $\Gamma = (v_1, \dots, v_k)$ meeting every directed path from a primitive state-dependent input of $\mathrm{Exp}(q_\rho)$ to its terminal vertex. Every v_ℓ is a proper ancestor of the terminal shell label: it is an input, state, current occurrence, or integrated current on the half-open pre-contact history, rather than the completed contact flag. Write

$$J_{\Gamma,\ell} := (f_{v_1}, \dots, f_{v_\ell}), \qquad \mathcal{A}_{\Gamma,\ell} := \sigma(J_{\Gamma,\ell}), \qquad 0 \leq \ell \leq k,$$

where $J_{\Gamma,0}$ is constant. The frontier record contains the ancestry of all v_ℓ, their joint interface, and the complete downstream subgraph from Γ to q_ρ. A conditional extension is the pair $(J_{\Gamma,\ell-1}, f_{v_\ell})$, rather than the isolated coordinate f_{v_ℓ}.

For a nonzero finite positive law ν on the measurable carrier of this record, normalize ν to a probability law and let $P_{\Gamma,\ell}$ be conditional expectation onto $L^2(\mathcal{A}_{\Gamma,\ell},\nu)$. For every bounded centered target-shell contrast y_Φ, define

$$\Delta_{\Gamma,\ell}(y_\Phi) := \left\|(P_{\Gamma,\ell} - P_{\Gamma,\ell-1})y_\Phi\right\|^2_{L^2(\nu)}. \tag{451}$$

When $y_\Phi \neq 0$, its normalized conditional increment is

$$\widehat{\Delta}_{\Gamma,\ell}(y_\Phi) := \frac{\Delta_{\Gamma,\ell}(y_\Phi)}{\|y_\Phi\|^2_{L^2(\nu)}}. \tag{452}$$

This definition uses no compact embedding, critical-norm bound, convergence modulus, or perturbative estimate. For an actual crossing there is a canonical law requiring no analytic choice. Choose the frontier first at the source-resolved integrated-current outputs in the exact shell identity, and then expand their complete ancestries. Let ρ° be the corresponding zero-current comparison record: it retains the same public coefficients and initial endpoint and replaces those current outputs by the additive zero of their current spaces. This is the algebraic zero object for the exact current identity, not an asserted solution of the equation. Hence $q_{\rho^\circ} = 0$. On the two-record space

$$\Omega_\rho := \{\rho, \rho^\circ\}, \qquad \nu_\rho := \tfrac{1}{2}(\delta_\rho + \delta_{\rho^\circ}), \qquad y_{\Phi,\rho} := q - \tfrac{1}{2}, \tag{453}$$

one has $\|y_{\Phi,\rho}\|_2^2 = 1/4$. Every frontier coordinate is evaluated separately on the two records; no infeasible hybrid tuple is introduced.

Theorem III.4.29 (Exact prospective backward factorization). *For every frontier in Definition III.4.28, contraction of the finite downstream graph produces a measurable map $h_{\rho,\Gamma}$ satisfying*

$$q_\rho = h_{\rho,\Gamma} \circ J_{\Gamma,k}. \tag{454}$$

Consequently, if P_{q_ρ} denotes conditional expectation onto $\sigma(q_\rho)$, then

$$\left\|P_{\Gamma,k} y_\Phi\right\|_2^2 = \left\|P_{q_\rho} y_\Phi\right\|_2^2 + \left\|(P_{\Gamma,k} - P_{q_\rho}) y_\Phi\right\|_2^2, \tag{455}$$

$$\left\|P_{\Gamma,k} y_\Phi\right\|_2^2 = \sum_{\ell=1}^{k} \Delta_{\Gamma,\ell}(y_\Phi). \tag{456}$$

In particular,

$$\left\|P_{q_\rho} y_\Phi\right\|_2^2 > 0 \quad \Longrightarrow \quad \Delta_{\Gamma,\ell}(y_\Phi) > 0 \quad \textit{for some } \ell. \tag{457}$$

For the canonical source-current frontier and two-record law (453)*, the terminal shell coordinate satisfies $P_{q_\rho} y_{\Phi,\rho} = y_{\Phi,\rho}$, and therefore*

$$\sum_{\ell=1}^{k} \Delta_{\Gamma,\ell}(y_{\Phi,\rho}) = \left\|P_{\Gamma,k} y_{\Phi,\rho}\right\|_2^2 = \frac{1}{4}, \qquad \sum_{\ell=1}^{k} \widehat{\Delta}_{\Gamma,\ell}(y_{\Phi,\rho}) = 1. \tag{458}$$

Thus every actual crossing, without an observer-law hypothesis, exposes a positive finite contextual source extension. Further backward expansion retains that source output together with its complete constructor ancestry; it does not require a second comparison record. The complete contextual extension $(J_{\Gamma,\ell-1}, f_{v_\ell})$ in (457) *is re-entered into the five-source projection before any residual quotient. Its ancestry is the union of the ancestries of its inputs. Thus a nonlinear or conditional synergy may carry target alignment even when every input is target-null in isolation, but that synergy is a finite source-owned constructor occurrence.*

Proof. Every path from a primitive state-dependent input to the terminal vertex meets Γ. Evaluating the vertices below the cut in topological order therefore gives $h_{\rho,\Gamma}$ and proves (454). Hence $\sigma(q_\rho) \subseteq \mathcal{A}_{\Gamma,k}$. Conditional expectations onto the nested closed subspaces satisfy $P_{q_\rho}P_{\Gamma,k} = P_{\Gamma,k}P_{q_\rho} = P_{q_\rho}$, which gives (455). The spaces $L^2(\mathcal{A}_{\Gamma,\ell},\nu)$ are nested, so their successive projection differences are pairwise orthogonal. Since the shell contrast is centered, $P_{\Gamma,0}y_\Phi = 0$; Pythagoras gives (456). A positive left side forces a positive summand. On the canonical two-history space, q_ρ takes the values one and zero and $y_{\Phi,\rho} = q_\rho - 1/2$, so conditioning on q_ρ fixes the contrast. This proves (458). Expanding the load-bearing constructor and applying the source-preserving normal form assigns the contextual input word to Geom, Caus, Abs, Lift, or Bdry, with the union of its input ancestries. Projection is applied to this joint word, so individual target-nullity cannot delete the mixed increment. □

Theorem III.4.30 (Cofinal prospective-frontier closure). *Let $(e_j)_{j\geq 0}$ be any same-origin sequence of completed shell crossings, and flatten the expanded backward frontiers of the first N crossings into the cumulative record $\mathcal{J}_N$ of Definition III.4.27. Let $\mathcal{A}_N = \sigma(\mathcal{J}_N)$ and let P_N be conditional expectation on any finite positive law carried by the measurable joint record. Then*

$$P_\infty y_\Phi = \sum_{N\geq 0}(P_N - P_{N-1})y_\Phi, \qquad \|P_\infty y_\Phi\|_2^2 = \sum_{N\geq 0}\|(P_N - P_{N-1})y_\Phi\|_2^2, \tag{459}$$

with $P_{-1} = 0$. Therefore a target-visible cumulative record has a target-visible finite joint prefix. Applying Theorem III.4.29 inside the least such prefix exposes a finite contextual extension with inherited five-channel ancestry.

For the actual crossings, let ν_{ρ_j} and y_{Φ,ρ_j} be their canonical two-record laws and contrasts. Form the finite orthogonal occurrence sum

$$\mathcal{H}_N := \bigoplus_{j<N} L^2(\nu_{\rho_j}), \qquad \mathbf{Y}_N := \bigoplus_{j<N} y_{\Phi,\rho_j}. \tag{460}$$

Order the pairs (j,ℓ) lexicographically and reveal the corresponding frontier coordinates in that order. The resulting conditional-expectation subspaces of $\mathcal{H}_N$ are nested, and their exact orthogonal increments give

$$\|\mathbf{Y}_N\|_{\mathcal{H}_N}^2 = \frac{N}{4} = \sum_{j<N}\sum_{\ell}\Delta_{\Gamma_j,\ell}(y_{\Phi,\rho_j}), \qquad N = \sum_{j<N}\sum_{\ell}\widehat{\Delta}_{\Gamma_j,\ell}(y_{\Phi,\rho_j}). \tag{461}$$

The same join retains the unnormalized identities

$$A_{\Phi,N} = N, \qquad A_{\Phi,N} = \sum_{c\in\mathfrak{C}} Q^c_{\Phi,N}(1). \tag{462}$$

Thus passage to a subsequence, empirical law, recurrent average, renormalized chart, weak limit, or compactification cannot replace the cumulative target cocycle by its average. If every finite contextual increment is target-null, then $P_\infty y_\Phi = 0$. Independently of any auxiliary observer law, (461) *retains the complete accumulated target alignment of the first N crossings. If contact persists, positive finite contextual source increments are thus present and remain attached to the cumulative physical ledger. These alternatives require neither terminal compactness nor perturbative critical-norm bounds.*

Proof. The sigma algebras $(\mathcal{A}_N)$ increase. Hilbert-space martingale convergence gives strong convergence of $P_N y_\Phi$ to $P_\infty y_\Phi$; orthogonality of successive differences gives (459). A nonzero sum has a nonzero finite partial sum, and the least prefix at which its squared norm increases contains a positive conditional increment. The preceding theorem supplies its finite contextual owner and ancestry. Each completed normalized crossing contributes exactly one to the shell cocycle before normalization. Joining occurrences is additive and does not identify them, proving (462). Summing (458) over the first N crossings in the orthogonal sum (460) gives (461). The common joint projection detects cross-history synergy, while the occurrence direct sum and unnormalized cocycle retain multiplicity and physical ownership. □

Corollary III.4.31 (Backward-saturated residual nullity)**.** *Fix a physical ceiling and saturate the continuum algebra over the expanded backward frontiers of all legal same-origin finite stopped histories below that ceiling, all finite contextual joins, and their directed cumulative joins. After applying the five source projections to this saturated class, the remaining prospective residual is target-null:*

$$G_{\Sigma,\mathrm{phys}}^{J_{\mathrm{res}},\mathrm{src},\mathrm{sat}} = 0. \tag{463}$$

Indeed, any residual with nonzero target pairing determines a positive conditional increment on a finite expanded joint prefix. Its detecting contextual word is then an element of the same saturation and has inherited Geom/Caus/Abs/Lift/Bdry *ancestry, contradicting its placement in the post-saturation residual.*

Proof. Suppose a residual class has nonzero pairing with a bounded shell contrast. For a finite class, apply Theorem III.4.29; for a directed class, first apply Theorem III.4.30. In either case a finite contextual extension has positive conditional increment. Its expanded constructor graph belongs to the declared saturation. Apply Theorems I.4.5 and I.4.6 to that graph. Its target-active quotient is generated by the five head components with inherited exact-support ancestry, while the residual lies in $\mathscr{N}_\Sigma$. Hence every residual pairing vanishes, which is (463). □

Corollary III.4.32 (No terminal analytic prerequisite)**.** *Prospective source extraction for singular contact is performed on finite stopped shell histories and closed by Theorem III.4.30. It requires neither a bounded local critical norm, compactness of terminal windows, convergence to an ordinary blow-up profile, nor perturbative control in a critical space. When one of those properties holds, it supplies an additional generated coordinate of the same frontier. When it fails, its typed total-graph value retains the same inherited five-source ancestry; failure cannot delete the finite target increment or create a source family.*

Definition III.4.33 (Same-origin occurrence space and physical comparison measure)**.** Let $(e_j)_{j\in J}$ be the completed target-shell crossings of one physical orbit, with the intervening gaps inserted before any current is restricted. Give the countable occurrence space $\Omega_{\mathrm{occ}} := J$ its full sigma algebra and define

$$\Gamma_\Sigma := \sum_{j\in J} a_j\delta_j, \qquad \mu_{\mathrm{phys}} := \sum_{j\in J} p_j\delta_j, \qquad a_j := \Phi_j(u_j^{\mathrm{out}}) - \Phi_j(u_j^{\mathrm{in}}) = 1. \tag{464}$$

Here p_j is the restriction to the physical image of the complete nonnegative public carrier measure admitted by Definition II.2.3; it is not normalized dissipation in the moving chart. If several crossings overlap in physical space-time, replace them by the disjoint atoms of their finite incidence algebra and retain the crossing multiplicity on each atom. Thus $\mu_{\mathrm{phys}}(\Omega_{\mathrm{occ}})$ is bounded by the single physical ledger, whereas $\Gamma_\Sigma([0,N)) = N$.

For each source $c \in \mathfrak{C}$, signed gluing and universal expanded derivation give a signed occurrence current Q^c with

$$\Gamma_\Sigma = \sum_{c \in \mathfrak{C}} Q^c \tag{465}$$

on every finite subset. The equality, the five owners, the physical measure, and the multiplicity measure are joined before taking a directed limit. Chart changes merely relabel the atoms; they do not rescale either measure.

Theorem III.4.34 (Exact compositional accounting against the physical ledger)**.** *On the same-origin occurrence space of Definition III.4.33, the positive target-progress measure has the unique Lebesgue decomposition*

$$\Gamma_\Sigma = f_\Sigma \mu_{\mathrm{phys}} + \Gamma_\Sigma^\perp, \qquad f_\Sigma \geq 0, \qquad \Gamma_\Sigma^\perp \perp \mu_{\mathrm{phys}}. \tag{466}$$

It is computed atomwise by

$$f_\Sigma(j) = p_j^{-1} \quad \textit{when } p_j > 0, \qquad \Gamma_\Sigma^\perp = \sum_{p_j = 0} \delta_j. \tag{467}$$

On each occurrence atom, apply the finite-dimensional positive/negative split to the five signed values in (465)*. Atomwise directed joining then assigns every nonzero part of $f_\Sigma \mu_{\mathrm{phys}}$ and $\Gamma_\Sigma^\perp$ to at least one of* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$*, with its complete mixed ancestry. In particular the singular measure in* (466) *is not J_{res}.*

If the orbit has finite physical budget and completes infinitely many crossings, then necessarily

$$\operatorname*{ess\,sup}_{\mu_{\mathrm{phys}}} f_\Sigma = +\infty \quad \textit{or} \quad \Gamma_\Sigma^\perp(\Omega_{\mathrm{occ}}) > 0. \tag{468}$$

Equivalently, if the expanded five-source algebra proves, on the actual same-origin occurrence record,

$$\Gamma_\Sigma^\perp = 0, \qquad f_\Sigma \leq C < \infty \quad \mu_{\mathrm{phys}}\textit{-a.e.}, \tag{469}$$

then the number of completed crossings satisfies the sharp bound

$$\#J = \Gamma_\Sigma(\Omega_{\mathrm{occ}}) \leq C \mu_{\mathrm{phys}}(\Omega_{\mathrm{occ}}) < \infty, \tag{470}$$

and contact with a target requiring a cofinal shell family is impossible. Conversely, every finite-budget cofinal contact produces one of the two source-owned prospective amplification values in (468)*. They enter the zero-production, carrier–polar, and realization transitions; neither value is a singular-existence terminal by itself.*

Proof. The physical images of the incidence atoms are disjoint, so restriction of the public carrier gives the countably additive measure μ_{phys}. The unnormalized shell cocycle gives the countably additive counting measure Γ_Σ. The Lebesgue decomposition theorem gives (466); on a countable atomic space its values are exactly (467).

For every finite prefix, signed gluing and prospective accountability give (465). Expand all mixed constructors and split the five real source values on each occurrence atom only after this equality has been formed. Because the occurrence space is countably atomic, source-preserving passage to the directed join then assigns each nonzero positive aggregate atom to the union of the five-source ancestries of its finite detecting word. No countably additive extension of an individual signed Q^c is used. Singularity with respect to μ_{phys} changes the comparison class, not the structural owner.

Suppose that $\mu_{\mathrm{phys}}(\Omega_{\mathrm{occ}}) < \infty$, $\Gamma_\Sigma^\perp = 0$, and $f_\Sigma \le C$. Integration of (466) gives

$$\#J = \int f_\Sigma \, \mathrm{d}\mu_{\mathrm{phys}} \le C\mu_{\mathrm{phys}}(\Omega_{\mathrm{occ}}) < \infty.$$

Its contrapositive is (468). No compactness, continuation criterion, scale summability, or lower price per normalized window enters the argument. □

Definition III.4.35 (Physical target-transition capacity)**.** For a physical ceiling $E < \infty$, let $\mathscr{O}_{\mathrm{phys}}(E)$ be the same-origin orbit records whose public physical carrier has mass at most E. Define

$$\mathsf{t}_\Sigma^{\mathrm{phys}}(E) := \sup_{R\in\mathscr{O}_{\mathrm{phys}}(E)} \Gamma_{\Sigma,R}(\Omega_{\mathrm{occ},R}), \tag{471}$$

$$\mathsf{a}_\Sigma^{\mathrm{phys}}(E) := \sup_{R\in\mathscr{O}_{\mathrm{phys}}(E)} \sup_{A\in\mathcal{A}^{\mathrm{fin}}_{\mathrm{occ},R}} \frac{\Gamma_{\Sigma,R}(A)}{\mu_{\mathrm{phys},R}(A)}. \tag{472}$$

Here $\mathcal{A}^{\mathrm{fin}}_{\mathrm{occ},R}$ is the algebra of finite unions of generated occurrence atoms. The quotient is $+\infty$ when its numerator is positive and its denominator is zero. Both suprema are values of the saturated target-local algebra: they are taken only after the five-source join, same-origin restriction, and unnormalized multiplicity record have been formed.

Theorem III.4.36 (Target-capacity duality)**.** *For every $E < \infty$,*

$$\mathsf{t}_\Sigma^{\mathrm{phys}}(E) \le E\,\mathsf{a}_\Sigma^{\mathrm{phys}}(E), \tag{473}$$

with the usual extended-real convention. More precisely, for a fixed same-origin record R, the following are equivalent:

(D1) *there is $C < \infty$ such that $\Gamma_{\Sigma,R}(A) \le C\mu_{\mathrm{phys},R}(A)$ for every generated finite occurrence set A;*

(D2) *the inequality extends to the occurrence sigma algebra and $\Gamma^\perp_{\Sigma,R} = 0$, $\|f_{\Sigma,R}\|_{L^\infty(\mu_{\mathrm{phys},R})} \le C$;*

(D3) *after every finite generated reordering of occurrence atoms, every finite prefix satisfies $A_{\Phi,N} \le C\Pi_N$.*

The least admissible C is

$$C_R = \operatorname*{ess\,sup}_{\mu_{\mathrm{phys},R}} f_{\Sigma,R} \quad \text{if } \Gamma^\perp_{\Sigma,R} = 0, \qquad C_R = +\infty \quad \text{otherwise.} \tag{474}$$

Consequently a finite physical target-transition capacity is not an axiom: for each fixed occurrence record it is exactly the bounded-amplification value of the five-source join. A cofinal contact at finite physical budget forces $C_R = +\infty$ on that contact-derived record and thereby exposes an unbounded-amplification or physical-singular five-source successor. This recordwise conclusion is not a computation of reachability from the initial datum and is not a singular-existence statement.

Proof. Finite unions of atoms form an algebra generating the occurrence sigma algebra. If (D1) holds, monotone approximation extends the measure inequality to that sigma algebra. The Lebesgue decomposition (466) then has no singular part, and its Radon–Nikodym derivative is at most C; this is (D2). The converse follows by integration. Taking A to be the first N occurrences proves (D3). Conversely, the target program may order any finite generated occurrence set first while retaining the intervening gaps; hence (D3) for every generated ordering gives (D1). The minimality property of the Radon–Nikodym derivative gives (474). Finally, for each R,

$$\Gamma_{\Sigma,R}(\Omega_{\mathrm{occ},R}) \le C_R \mu_{\mathrm{phys},R}(\Omega_{\mathrm{occ},R}),$$

and taking suprema proves (473). $\square$

Definition III.4.37 (Finite-prefix physical valuation)**.** Fix a same-origin occurrence record and a finite occurrence cylinder F. Let Y_F be the finite-dimensional real space of all source-owned current coordinates detected on F, after quotienting the target-null subspace. Write $Q_F \in Y_F$ for a target-current owner and $p_F \in Y_F$ for the restriction of the complete public physical measure, including admitted production and exterior input with their public weights. Let $D_F \subset Y_F$ be the closed convex cone generated by the restrictions to F of

$$\delta\mathscr{K} + \mathscr{N}_\Sigma,$$

where the two orientations of every storage coboundary are included, internal currents are glued with their signs before taking a positive part, and every production or exterior term has first been mapped into p_F by its exact public carrier row. A term for which that map is nonordinary remains a graph coordinate and is not inserted into D_F. Define the extended physical valuation

$$\mathrm{val}_F(Q_F) := \inf\{C \ge 0 : Q_F \in D_F + Cp_F\}. \tag{475}$$

The infimum of the empty set is $+\infty$. Thus a finite value is a finite-prefix carrier domination, while an infinite value records failure of that domination on the same prefix; neither value is a terminal verdict.

Theorem III.4.38 (Exact finite-prefix source-profile/carrier duality)**.** *For every F and Q_F as in Definition III.4.37,*

$$\mathrm{val}_F(Q_F) = \sup\left\{\lambda(Q_F) : \lambda \in Y_F^*, \quad \lambda(D_F) \le 0, \quad \lambda(p_F) \le 1\right\}. \tag{476}$$

The equality is interpreted in the extended nonnegative reals. If the right-hand side is finite and the primal set is nonempty, the infimum is attained. If it is infinite, then for every $M < \infty$ a finite-prefix dual word satisfies

$$\lambda_M(D_F) \le 0, \qquad \lambda_M(p_F) \le 1, \qquad \lambda_M(Q_F) > M. \tag{477}$$

Each λ_M is joined to Q_F, pulled backward through the complete joint frontier, and assigned the ancestry of the constructor that first creates its nonzero conditional target increment. Hence dual separation creates no additional source family.

Proof. Pass to the quotient by the lineality space of D_F. The epigraph

$$\{(Q, C) : C \ge 0,\ Q \in D_F + Cp_F\}$$

is a closed convex cone in the finite-dimensional space $Y_F \times \mathbb{R}$. Separation of (Q_F, C) from this cone, followed by normalization of the coefficient of C, gives the lower bound in (476); membership gives the reverse bound. Closedness gives attainment whenever the feasible set is nonempty and the value is finite. If the gauge is infinite, separation at levels $M = 1, 2, \ldots$ gives (477). The backward-frontier factorization and conditional-increment identity apply to the finite joint record (Q_F, λ_M). Constructor-level charging therefore assigns every nonzero dual readout to one of the five inherited owners. $\square$

Theorem III.4.39 (Monotone saturated physical valuation)**.** *Let $F_1 \subset F_2 \subset \cdots$ be the canonical exhaustive sequence of finite occurrence cylinders, including every intervening gap, source word, storage endpoint, exterior-input coordinate, and total-graph value generated up to the corresponding depth. Then*

$$\mathrm{val}^{\mathrm{sat}}(Q) := \sup_n \mathrm{val}_{F_n}(Q_{F_n}) \tag{478}$$

is independent of the chosen cofinal enumeration and is the least $C \in [0, +\infty]$ for which every finite generated restriction admits a domination by $D_{F_n} + Cp_{F_n}$. Equivalently,

$$\mathrm{val}^{\mathrm{sat}}(Q) = \sup_n \sup_{\substack{\lambda(D_{F_n}) \le 0 \\ \lambda(p_{F_n}) \le 1}} \lambda(Q_{F_n}). \tag{479}$$

A finite value charges every finite target prefix at one common physical rate. An infinite value supplies a cofinal family of finite target-aligned dual increments with their unnormalized occurrence cocycle. It is processed by amplification and equality saturation before any averaging or limiting normalization.

Proof. Restriction of a domination on F_{n+1} is a domination on F_n, so the finite valuations are nondecreasing. Two cofinal enumerations dominate each other termwise after passing to subsequences, which proves independence. Taking the least common constant over all finite restrictions gives (478); applying Theorem III.4.38 at every depth gives (479). The final assertions retain, rather than divide by, the cumulative occurrence coordinate. □

Theorem III.4.40 (Variation-safe finite-prefix current principle)**.** *For each source $c \in \mathfrak{C}$, let Q^c_F denote its signed current on a finite occurrence cylinder F. All source identities and cancellations are performed at this level. Exactly one of the following holds along the canonical exhaustion:*

(J1) $\sup_n |Q^c_{F_n}|(F_n) < \infty$. *Then the compatible finite-prefix currents extend uniquely to a finite signed measure Q^c on the occurrence sigma algebra, and finite-prefix signed gluing passes to that measure.*

(J2) $\sup_n |Q^c_{F_n}|(F_n) = \infty$. *The increasing variation coordinate, together with the first finite prefix on which each rational level is crossed, is the source-owned variation-boundary successor. No global signed measure or cancellation across that boundary is used.*

In both cases the positive contact measure Γ_Σ remains countably additive. Consequently every use of a global Jordan or Lebesgue decomposition in the target calculus applies to Γ_Σ or to a source current satisfying (J1)*; otherwise the calculus remains finite-prefix local.*

Proof. Compatibility assigns one weight to each occurrence atom. Under (J1), the absolute sum of those weights is finite, so their series defines the unique finite signed measure on the full sigma algebra and agrees with every finite restriction. If the variations are unbounded, the least prefix crossing each rational level is defined and preserves the source owner. The two alternatives are exhaustive. Countable additivity of Γ_Σ follows independently from its occurrence-counting definition in Definition III.4.33. □

Corollary III.4.41 (Zero physical denominator is an evaluated source cell)**.** *Let A be a generated occurrence atom with $\mu_{\mathrm{phys}}(A) = 0$ and $\Gamma_\Sigma(A) > 0$. Then a source $c \in \mathfrak{C}$ satisfies $Q^c(A) > 0$. On the one-atom cylinder $F = \{A\}$, exactly one of the following finite computations occurs: the target quotient of Q^c_F is zero; a finite carrier row in $D_F + Cp_F$ charges it; or the dual value $\mathrm{val}_F(Q^c_F) = +\infty$ generates the normalized zero-production tangent of Theorem III.4.46. Hence singularity of Γ_Σ with respect to the physical measure is an evaluated five-source cell, not a singularity verdict and not a residual source.*

Proof. The source-resolved finite-set identity gives the source c. Apply Theorem III.4.38. Since $p_F = 0$, every finite positive target value outside D_F has unbounded gauge. Its dual witnesses retain the same finite owner and enter target normalization. □

Theorem III.4.42 (Nonordinary shell-current first-failure evaluator)**.** *Consider a finite stopped shell history and totalize, in constructor order, every product, trace, pressure solve, chart map, scale map, and Stieltjes pairing used by its shell-current identity. If the ordinary identity fails, let v be the least failed graph. Join to v its target covector, physical denominator, shell multiplicity, complete preceding joint frontier, and source owner. Then its target-visible part has exactly one of the following values:*

(G1) *its conditional target increment is zero, so it lies in $\mathscr{N}_\Sigma$;*

(G2) *its finite-prefix physical valuation is finite and the corresponding carrier row charges it;*

(G3) *its valuation is infinite and a finite dual witness enters the source-owned amplification/equality evaluator;*

(G4) *a descendant graph used by that evaluator is nonordinary, in which case the least such descendant, with the enlarged joint frontier, is the next graph record.*

No nonordinary value is removed before joint target projection, and no graph value is promoted to an ordinary orbit or contact event.

Proof. The total graphs form a finite ordered coproduct on the stopped history, so a least failed graph exists. Exact backward factorization through its complete joint frontier and orthogonal conditional increments decide (G1). On the complementary target quotient apply Theorem III.4.38; its finite and infinite values give (G2) and (G3). Totalize every graph used in that finite computation. An ordinary value completes the stated row, while the least failed descendant gives (G4). Source preservation under graph completion assigns the ancestry of v throughout. The realization gate is separate, so none of these graph values implies physical existence. □

Theorem III.4.43 (Prospective source-amplification extraction)**.** *Let a same-origin pre-contact record have the source-resolved occurrence identity*

$$\Gamma_\Sigma = \sum_{c \in \mathfrak{C}} Q^c$$

on every finite occurrence set and finite physical measure μ_{phys}. No global signed-measure extension of the individual Q^c is required. The following alternatives hold.

(X1) *if* $\operatorname{ess\,sup} f_\Sigma = +\infty$*, there are one fixed source c_*, numbers $M_k \uparrow \infty$, and generated finite pre-contact occurrence sets A_k such that*

$$Q^{c_*}(A_k) > M_k \mu_{\text{phys}}(A_k); \tag{480}$$

(X2) *if $\Gamma_\Sigma^\perp \neq 0$, there are one fixed source c_*, a μ_{phys}-null generated occurrence atom A such that*

$$\mu_{\text{phys}}(A) = 0, \qquad Q^{c_*}(A) > 0. \tag{481}$$

Every selected finite occurrence set is selected before its last crossing and carries the complete prefix derivation, physical denominator, shell multiplicity, and source ancestry. Thus aggregate target amplification cannot be a terminal mechanism: it always exposes a quantitative prospective owner in one of the five existing channels.

Proof. Suppose first that f_Σ is essentially unbounded. Since the occurrence space is countably atomic and $f_\Sigma(j) = p_j^{-1}$ on every positive-price atom, for every integer m there is an occurrence atom $A_m = \{j_m\}$ with

$$\Gamma_\Sigma(A_m) > 5m\mu_{\mathrm{phys}}(A_m).$$

Since $\Gamma_\Sigma(A_m) = \sum_c Q^c(A_m)$, at least one source satisfies $Q^c(A_m) > m\mu_{\mathrm{phys}}(A_m)$. One of the five sources occurs for infinitely many m; take that subsequence and call it c_*. This proves (480).

If $\Gamma_\Sigma^\perp \neq 0$, the atomic formula (467) supplies an occurrence atom $A = \{j\}$ with $p_j = 0$. Hence $\mu_{\mathrm{phys}}(A) = 0$ and $\Gamma_\Sigma(A) = 1$. The finite-set identity $\Gamma_\Sigma(A) = \sum_c Q^c(A)$ gives a source c with $Q^c(A) > 0$. Put $c_* = c$. This proves (481) without extending any signed Q^c across an infinite-variation set. Stopping at the last index of a finite occurrence set makes its selector pre-contact, while the cumulative join supplies all stated coordinates. □

Definition III.4.44 (Target-normalized amplification tangent)**.** Let c_* and A_k be supplied by Theorem III.4.43, and put $q_k := Q^{c_*}(A_k) > 0$. On the compact target-local evaluation algebra $\mathscr{E}_{\mathfrak{b}}$, define

$$L_k(F) := \frac{Q^{c_*}(F\mathbf{1}_{A_k})}{q_k}. \tag{482}$$

Before using positivity, apply the Hahn decomposition of $Q^{c_*}|_{A_k}$ and replace A_k by its positive part; the negative part remains in the signed same-origin join, and reset $q_k := Q^{c_*}(A_k)$. Thus L_k is positive, $L_k(1) = 1$, and it retains the source word, target shell, physical denominator, chart, and complete prefix.

For the physical-singular alternative, use the normalized positive restriction of Q^{c_*} to the null-carrier occurrence atom in (481) in place of (482). Any weak-* accumulation point in the generated evaluation compactification is called the *target-normalized amplification tangent.*

Theorem III.4.45 (Dyadic ordered-coercivity terminal rule)**.** *Let L be a positive normalized functional on an active target-local Archimedean cylinder algebra. Let A, P, M be bounded nonnegative generated coordinates, where A is target activity, P is admitted physical production, and M is an auxiliary coefficient. Suppose the generated ordered relations contain, for integers $k \geq 1$, $b \geq 1$, and $a \geq 0$,*

$$A^{2^k} \leq CM^aP^b, \qquad C < \infty, \qquad M \leq M_0 < \infty. \tag{483}$$

Then

$$L(P) = 0 \quad \Longrightarrow \quad L(A) = 0. \tag{484}$$

Consequently a positive target floor $L(A) \geq \eta > 0$ has exactly the following local continuations: $L(P) > 0$, an unbounded M-coordinate, or a nonordinary value of the coercivity, product, order, or target graph. The first value is charged through the physical-production row; the other values are the corresponding coefficient or graph successors. A zero-production, finite-coefficient positive functional is not a terminal equality model.

Proof. Boundedness and positivity give

$$0 \leq P^b \leq \|P\|_\infty^{b-1}P, \qquad 0 \leq M^a \leq M_0^a.$$

Thus $L(P) = 0$ implies $L(M^aP^b) = 0$. Applying L to (483) gives $L(A^{2^k}) = 0$. Repeated Cauchy–Schwarz yields

$$0 \leq L(A)^{2^k} \leq L(A^2)^{2^{k-1}} \leq \cdots \leq L(A^{2^k}) = 0.$$

Hence $L(A) = 0$. Totalization of the finitely many relations in (483) gives the stated complementary graph values, and the fixed target quotient makes the list exhaustive. □

Theorem III.4.46 (Amplification-to-zero-production tangent principle). *Every unbounded or physical-singular prospective source amplification produces a positive target-local tangent L satisfying*

$$L(1) = 1, \qquad L(Q_{\Sigma}^{c_*}) = 1, \qquad L(P) = 0 \quad (P \in \mathscr{P}_{\mathfrak{b}}^{\mathrm{phys}}), \tag{485}$$

after the target-current normalization is included as a coordinate. It retains the fixed source c_* and belongs to the equality stratum $\mathcal{E}_{\mathfrak{b}}$. Exactly one of the following occurs:

(Z1) *a finite equality-stratum identity makes the target projection zero;*

(Z2) *an ordinary same-origin zero-production state realizes the tangent;*

(Z3) *the least nonordinary equation, chart, scale, pressure, exterior, trace, concentration, or realization coordinate gives the next five-source successor.*

An amplification value is therefore never a terminal numerical infinity. It is converted constructively into a target-retaining zero-production state or a typed first-failure successor.

Proof. On the unbounded branch,

$$0 \le L_k(P) \le \frac{\mu_{\mathrm{phys}}(A_k)}{Q^{c_*}(A_k)} \longrightarrow 0 \qquad (P \in \mathscr{P}_{\mathfrak{b}}^{\mathrm{phys}}),$$

because every admitted production is dominated by the restriction of the single physical measure. On the physical-singular branch the same value is zero identically, since the normalized positive carrier is μ_{phys}-null. Positivity and normalization make the L_k a weak-* compact family on the generated compact evaluation algebra. Every accumulation point therefore satisfies (485). Expanded-derivation saturation preserves the source word and all same-origin coordinates.

Adjoin the relations $P = 0$ to the target-local ordered algebra. Its finite -1 identity gives (Z1). Otherwise totalize the ordinary realization graph. An ordinary value gives (Z2); the least nonordinary value gives (Z3) with the ancestry of the failed operation. These are the exhaustive outputs of the total graph and introduce no additional structural source. □

Theorem III.4.47 (Sound typing of amplification successors). *Start from the zero-production tangent in Theorem III.4.46 and recursively append the least nonordinary coordinate in the fixed enumeration of equation, source, chart, scale, pressure, exterior, trace, concentration, and realization graphs. Retain the cumulative same-origin join and the unit target normalization at every stage. Then exactly one evaluation state occurs:*

(T1) *a finite successor is target-null;*

(T2) *a finite successor has an ordinary zero-production realization and re-enters target-cyclic equality evaluation;*

(T3) *the cumulative graph-complete limit is target-null;*

(T4) *the cumulative limit is a same-origin, source-owned prospective corona record with unit target value and zero public physical production.*

In (T4), *every finite generated coordinate has an ordinary value or its declared nonordinary total-graph value, and at least one cofinal nonordinary coordinate retains nonzero target projection. The record remains an internal realization and carrier–polar successor. It is an explicit singular mechanism in the sense of Definition III.4.77 only after an independent approximation-tower path supplies all ordinary public-equation and contact rows. A positive functional, separator, or nonordinary graph value is not terminal.*

Proof. At each finite stage, target-nullity gives (T1) and an ordinary realization gives (T2). Otherwise append the least nonordinary coordinate. The cumulative joins form an increasing sequence in the compact graph-complete evaluation product. Choose a compatible diagonal limit

z_∞. Every coordinate is retained permanently after it is adjoined, so its total-graph relation passes to z_∞; the common origin, fixed source owner, zero-production relations, and unit target normalization pass as closed coordinates as well.

If the target projection of z_∞ vanishes, this is (T3). Otherwise no finite ordinary realization occurred, hence a cofinal nonordinary coordinate remains target-visible. The initial contact record supplies the ordinary public orbit, and every restriction, chart, normalization, and total-graph passage retained its same-origin row. Consequently z_∞ satisfies the ancestry, hypothetical-origin, zero-production, and nonzero-target clauses of a prospective corona record. It fails the independent-origin clause of Definition III.4.77. This is (T4). The alternatives are disjoint by the ordered first-terminal rule and exhaustive by compactness of the totalized evaluation product. ☐

Corollary III.4.48 (Sound continuum terminal theorem)**.** *Every terminal verdict produced by the continuum evaluator has exactly one of the following two types:*

$$\boxed{\textit{target avoidance} \quad \textit{or} \quad \textit{realized five-source singular mechanism.}}$$

The second type requires the independent-origin and ordinary-realization rows of Definition III.4.77. A contact-derived prospective record has origin polarity Hyp *and is not a singular-existence verdict. Target-visible* J_{res} *remains forbidden. A nonordinary compactness value, infinite refinement record, positive functional, or polar separator is an internal successor rather than a third terminal type.*

Proof. Finite contact is prospectively source-resolved. Infinite contact is reduced by Theorems III.4.34, III.4.43 and III.4.46 to a zero-production successor execution. Apply Theorem III.4.47. Its hypothetical corona re-enters realization and carrier–polar evaluation. A checked target-local contradiction proves avoidance; an independently generated ordinary contacting path proves singular existence. ☐

Definition III.4.49 (Evaluated covariant corona record)**.** An evaluated corona record consists of one active same-origin signature $\mathfrak{b}$, its fixed source owner Q, record scale r, scale rate $a = \dot r / r$, and, on every ordinary finite half-open prefix $I = [\tau_-, \tau_+)$, a generated Stieltjes carrier row

$$\mathrm{d}K = -\,\mathrm{d}\mathsf{D} + \kappa a K \ \mathrm{d}\tau + \mathrm{d}\mathsf{R}, \qquad K \ge 0, \qquad \mathrm{d}\mathsf{D} \ge 0. \tag{486}$$

with $a \in L^1(I)$. Failure of this local integrability relation is a total-graph scale coordinate and enters the same Lift-owned corona; it is not used in the formula below. Put

$$m_I(\tau) := \exp\left(\int_{\tau_-}^{\tau} \kappa a(s) \ \mathrm{d}s\right), \qquad \theta_I(\tau) := m_I(\tau)^{-1} \int_{\tau}^{\tau_+} m_I(s) \ \mathrm{d}s. \tag{487}$$

Then $\theta_I \ge 0$, $\theta_I(\tau_+) = 0$, and

$$\theta_I' + \kappa a \theta_I = -1 \tag{488}$$

almost everywhere. Thus the weight is generated by the carrier row and its finite horizon; it is not an additional multiplier. The horizon label is retained. When prefixes are joined, differences of their terminal weights remain in the oriented root–leaf coboundary and are not silently identified. It retains the expanded signed decomposition of $\mathrm{d}\mathsf{R}$, every compatible interface, the physical carrier, the unnormalized shell cocycle, and every ordinary or total-graph coordinate used below. Its covariant production is

$$\mathrm{d}\mathcal{P}_{\mathrm{cov},I} := K \ \mathrm{d}\tau + \theta_I \ \mathrm{d}\mathsf{D}, \qquad \mathrm{d}(\theta_I K) = -\,\mathrm{d}\mathcal{P}_{\mathrm{cov},I} + \theta_I \ \mathrm{d}\mathsf{R}. \tag{489}$$

If a carrier row, adjoint weight, product, or sign is nonordinary, its least total-graph value is part of the record rather than an omitted premise.

Theorem III.4.50 (Covariant corona transition evaluator)**.** *For every target-visible amplification tangent, the public continuum grammar constructs and evaluates the directed family of all records in Definition III.4.49. After signed interface gluing and recursive carrier–polar saturation, exactly one of the following evaluation states is returned:*

(V1) finite covariant financing: *for the cumulative unnormalized shell current* $\mathsf{Q}_N = \sum_{j \le N} \mathsf{Q}_{I_j}$*, signed saturation generates an exact domination*

$$\mathsf{Q}_N \le \delta\mathscr{K}_N + \mathsf{P}_N + \mathsf{B}_N + \mathsf{N}_N, \qquad \mathsf{N}_N \in \mathscr{N}_\Sigma, \tag{490}$$

in target-current order. Here P_N *is the joined measure from* (489)*,* $\delta\mathscr{K}_N$ *is the oriented weighted root–leaf coboundary, and* B_N *is genuine exterior input. The same generated carrier identities give a uniform finite ceiling for the target pairing of the right side. Since every completed crossing has* $\langle \mathsf{Q}_{I_j}, 1\rangle = 1$*, infinitely many crossings are excluded;*

(V2) equality nullity: *every generated covariant production vanishes and a finite target-local ordered-coercivity row of Theorem III.4.45 makes the equality stratum's target projection zero;*

(V3) target-visible corona successor: *the cumulative same-origin execution reaches a primal–polar fixed corona record. The record contains either an unbounded weighted root/exterior financing coordinate whose image in the target-current quotient is nonzero, an unbounded coefficient or nonordinary graph of a generated ordered-coercivity row, or a continuous carrier-polar separator* λ *satisfying*

$$\lambda(Q) > 0, \qquad \lambda(\delta\mathscr{K} + \mathscr{N}_\Sigma) = 0, qquad \lambda(\,\mathrm{d}\mathcal{P}_{\mathrm{cov},I} + \,\mathrm{d}B) \le 0 \tag{491}$$

for every generated covariant production and admitted exterior input. No positivity of λ *is asserted: its sign is fixed by* $\lambda(Q) > 0$*, while the second and third relations say exactly that it belongs to the polar of the oriented finite-financing cone.*

The third state includes the full weak equation, chart cocycle, scale weights, pressure/exterior total graphs, source ancestry, target normalization, and physical-origin rows. It is re-entered into the independent approximation-realization graph. It becomes a singular-solution terminal only when all clauses of Definition III.4.77 hold; otherwise its least failed realization row is the next typed successor.

Proof. Select the nonincreasing record scale by Theorem II.2.21. Enumerate the dual-carrier grammar of Definition II.2.22; every generated row has the form (486) after adjoining its scale exponent. Scalar adjoint closure constructs (488), and Proposition II.2.36 gives (489) without a sign assumption on a. Apply Theorem II.2.2 before taking positive parts. Internal transport, pressure, chart, cutoff, gauge, and exact scale currents cancel; all remaining terms keep their source owners.

For each finite prefix, scalar adjoint closure is the explicit construction (487); hence positivity, the terminal condition, and (489) require no solvability hypothesis. If the remaining signed current lies in the target-local carrier cone, its exact cone identity telescopes on the complex containing both selected events and their gaps. Apply that identity to the cumulative unit shell current, not to the positive part of any individual analytic term. If the generated weighted root, exterior, and physical coordinates are uniformly finite, this is precisely (490) and gives (V1). No contact margin is used: the lower side equals the number of completed normalized crossings. If all productions vanish, restrict every equation and target coordinate to the generated equality stratum. Its zero target projection is (V2).

Before the equality cell is retained, enumerate the generated dyadic ordered-coercivity rows. A finite-coefficient row closes every zero-production target activity by Theorem III.4.45. An unbounded coefficient or nonordinary relation is retained in (V3) with its source ancestry.

If the weighted root or exterior coordinate is unbounded, project its cumulative current to the target quotient before terminalization. A zero projection is in $\mathscr{N}_\Sigma$ and is deleted. A nonzero projection retains its least cofinal total-graph value and all same-origin scale ancestry; this is a target-financing component of (V3). Thus growth of $\theta_I(\tau_-)$ caused only by increasing the horizon is not itself a singular mechanism. On the remaining complement, geometric separation from the closed carrier cone gives (491). Append the separator, its adjoint pulls, polarizations, and total-graph values and repeat. Finite successors return to the preceding two tests. For a cofinal execution, form the cumulative primal–polar join before projection. By Theorem III.4.58 and Corollary II.2.34, a nonzero limit has a finite unprocessed increment or is fixed under both grammars. The former repeats the evaluation; the latter is terminalized by Theorem III.4.47. On a contact-seeded same-origin execution it retains hypothetical origin and is therefore the successor in (V3), not an existence verdict. An independent ordinary realization is tested only after this transition. These are the exhaustive cone-membership, equality, and strict-separation values. □

Proposition III.4.51 (Finite-prefix root and exterior accounting)**.** *Let $\mathcal{C}_N$ be a finite complex of stopped corona intervals, including the gaps between selected target crossings. Construct on each interval the adjoint weight of Eqs.* (487) *and* (488) *and glue all oriented interfaces before taking positive parts. Then the sum of the generated covariant carrier rows has the exact form*

$$\sum_{I\in\mathcal{C}_N} \mathrm{d}(\theta_I K_I) = \delta\mathscr{K}_N - \mathsf{P}_N + \mathsf{B}_N + \mathsf{N}_N + \mathsf{E}_N, \qquad \mathsf{N}_N \in \mathscr{N}_\Sigma. \tag{492}$$

Here $\delta\mathscr{K}_N$ consists of the two exterior roots and every explicit mismatch of weights at an unglued interface, $\mathsf{P}_N \geq 0$ is admitted only through the public physical measure, B_N is the restriction of the public forcing/boundary measure, and E_N is the ordered tuple of total-graph failures. Thus every finite-financing row contains all root, production, and exterior coordinates.

If their target pairings are bounded along N, they define the corresponding finite-prefix carrier ceiling. If one is unbounded, take its least rational level-crossing prefix and factor that finite readout backward through its complete joint frontier. Its nonzero conditional increment has the owner of the root, scale, horizon, or exterior constructor that created it and is evaluated by Theorem III.4.38. Consequently an unbounded adjoint weight or exterior term is expanded into a source-owned physical-valuation cell; it is never discarded as a failed estimate.

Proof. Sum (489) over the finite complex. Opposite oriented copies of a compatible internal face cancel. The remaining endpoint values are exactly $\delta\mathscr{K}_N$; public exterior faces give B_N, target-null signed terms give N_N, and the least nonordinary operations give E_N. This proves (492). Bounded pairings give a finite cylinder row. In the unbounded case the first level-crossing prefix is finite, so prospective backward factorization applies without a terminal limit. Constructor charging and finite-prefix duality give the last claim. □

Corollary III.4.52 (Sound structural transition method)**.** *For every publicly presented continuum equation and physical stratum, the method*

$$\begin{aligned}&\textit{prospective contact} \to \textit{five-source cumulative owner} \to \textit{physical amplification}\\ &\qquad\qquad \to \textit{zero-production tangent} \to \textit{covariant corona evaluation}\end{aligned}$$

returns target avoidance by (V1)–(V2)*, or a target-visible successor* (V3) *that is rejoined with its complete same-origin history and evaluated in the reachable contact cell. Every operation is generated from the public weak equation, carrier, target shells, and their total graphs; no source-transverse estimate or semantic compactness statement is accepted as an input.*

Proof. Prospective accountability and source exhaustion give the first two arrows. Theorems III.4.34 and III.4.43 gives the third, Theorem III.4.46 gives the fourth, and Theorem III.4.50 gives the next evaluation state. Terminal typing is Corollary III.4.48. ☐

Corollary III.4.53 (Continuum no-uncharged-amplification rule)**.** *For every finite-budget contact, prospective source extraction followed by target normalization gives the mandatory chain*

$$\text{Contact} \longrightarrow \mathsf{Amp}_{c_*} \longrightarrow \mathsf{ZP}_{c_*} \longrightarrow \begin{cases} \mathscr{N}_\Sigma, \\ \textit{ordinary equality state,} \\ \textit{typed first failure.} \end{cases}$$

This is applied before recurrence, averaging, compact realization, or terminal evaluation. Hence neither an infinite amplification ratio nor a physical-singular numerator can be used as an unevaluated regularity obstruction.

Proof. Combine Theorem III.4.43 and Theorem III.4.46. ☐

Corollary III.4.54 (No unowned prospective physical amplification)**.** *For a finite-physical-budget contact, exactly one of the following occurs: the target is reached after finitely many shell crossings; or one fixed source channel has the unbounded pre-contact amplification* (480)*; or one fixed source channel has the positive physical-singular pre-contact component* (481)*. The latter two outputs are the continuum analogue of the pre-hit selector carrying excess advantage. They are source-resolved prospective mechanisms. Singular existence additionally requires the independent realization in Definition III.4.77.*

Proof. Combine Theorem III.4.34 with Theorem III.4.43. Cofinal contact excludes the finite-crossing alternative. ☐

Proposition III.4.55 (Finite physical budget does not supply transition capacity)**.** *There is no equation-independent theorem deriving $\mathsf{t}_\Sigma^{\mathrm{phys}}(E) < \infty$ from $E < \infty$. Indeed, the scalar flow*

$$\dot{x} = x^2, \qquad x(0) = 1, \qquad K(x) = x^{-1} \tag{493}$$

has $K(x(t)) = 1 - t$, reaches the compactified singular target $x = +\infty$ at $t = 1$, and obeys the exact finite physical ledger

$$K(x(T)) + \mu_{\mathrm{phys}}([0, T]) = K(x(0)), \qquad \mu_{\mathrm{phys}} = \mathrm{d}t, \qquad \mu_{\mathrm{phys}}([0, 1)) = 1.$$

Its crossings of $x = 2^j$ have prices $p_j = 2^{-j} - 2^{-j-1}$, so their target amplification tends to infinity. The target spine is entirely Caus*-owned. Thus the continuum algebra correctly returns a realized singular terminal; declaring finite transition capacity from the physical ceiling would incorrectly erase it.*

Proof. The solution is $x(t) = (1 - t)^{-1}$. Direct differentiation gives $\frac{\mathrm{d}}{\mathrm{d}t} K(x(t)) = -1$, proving the ledger. The first hitting time of 2^j is $1 - 2^{-j}$, so the physical price between consecutive hits is the displayed p_j, while every normalized shell crossing contributes one to Γ_Σ. Hence $p_j^{-1} \to \infty$. The vector field is a fixed directed drift, so its ancestry is Caus. ☐

Corollary III.4.56 (Composition, refinement, and scale covariance). *The decomposition* (466) *commutes with finite composition, countable refinement of the occurrence partition, insertion of inactive gaps, and every injective chart relabelling. Under a noninjective quotient, the numerator, physical denominator, and fiber multiplicity are pushed forward jointly before their Lebesgue decomposition. Consequently neither averaging, refinement, concentration, nor scale collapse can turn unbounded amplification or physical-singular target progress into a zero-price record.*

Proof. All four operations are pushforwards or restrictions of the pair of measures $(\Gamma_\Sigma, \mu_{\mathrm{phys}})$. Uniqueness of Lebesgue decomposition proves the assertion. Keeping fiber multiplicity is exactly the unnormalized cocycle rule of (450). □

Definition III.4.57 (Universal target-aligned amplification cell). Let Q be any source-owned target-current readout in a universal cumulative record and let $\mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}$ be the target-local physical carrier cone of Definition II.2.25. Evaluate all terms on the same selected finite event or finite prefix. Define the target amplification relation by the total graph of

$$\mathrm{Amp}_\Sigma(Q; P, K, B) := \frac{(Q)_+}{P + |\delta K| + B^+}, \tag{494}$$

with the conventions encoded as separate graph cells when the denominator vanishes or a term is nonordinary. Here P is admitted physical production, δK is endpoint storage before telescoping, and B^+ is true positive exterior input. The ratio is only a diagnostic coordinate: physical charging is authorized solely by cone membership $Q \in \mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}$.

The grammar adjoins this graph for every generated target owner, not merely for scale or energy. Finite amplification records a local domination row. Unbounded or zero-denominator amplification is a target-active polarity cell with the complete expanded ancestry of Q, P, K, B; it is recursively processed by carrier–polar exhaustion. It is never interpreted as free progress, target-nullity, or a residual source.

Theorem III.4.58 (Universal saturated-composition accountability). *Let $q = h(J)$ be any finite composite generated by the public continuum grammar, where J is the joint value of every immediate input to the terminal constructor. Expand the derivations of those inputs before evaluating h, and form the universal cumulative record of Definition III.4.27. The following statements hold.*

(J1) *If $q = h(\mathcal{J}_n)$ is any generated finite composite and P_q denotes conditional expectation onto $\sigma(q)$, then*

$$P_q P_n = P_n P_q = P_q, \qquad \|P_q y_\Phi\|_2 \le \|P_n y_\Phi\|_2 \tag{495}$$

for every bounded centered shell contrast y_Φ.

(J2) *The spaces $L^2(\mathcal{A}_n, \nu)$ increase and*

$$P_n y_\Phi \longrightarrow P_\infty y_\Phi \quad \textit{strongly in } L^2(\nu), \qquad \|P_n y_\Phi\|_2^2 \uparrow \|P_\infty y_\Phi\|_2^2. \tag{496}$$

Hence a target-aligned component of the joint limit has a target-aligned finite joint prefix. If every finite joint prefix is target-null, then the joint limit is target-null. More precisely, with

$$d_0 := P_0 y_\Phi, \qquad d_n := (P_n - P_{n-1}) y_\Phi \quad (n \ge 1), \tag{497}$$

the d_n are pairwise orthogonal and

$$P_\infty y_\Phi = \sum_{n \ge 0} d_n, \qquad \|P_\infty y_\Phi\|_2^2 = \sum_{n \ge 0} \|d_n\|_2^2. \tag{498}$$

Thus no positive detection threshold is imposed on a joint contribution: an arbitrary countable family of small contributions is retained by its exact cumulative sum.

(J3) *Finite composition and graph-complete passage to $\mathscr{J}_\infty$ preserve the source-resolved identity* (450)*. Every nonzero target component of a mixed product, limit, or total-graph value is therefore reprojected into* $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$ *with the union of the ancestries of the joint inputs. If $d_n \neq 0$, a generated finite cylinder readout of $\mathscr{J}_n$ has nonzero pairing with d_n; the complete derivation of that detecting readout supplies the responsible mixed input word. No sixth source is created by composition or by the directed limit. An increment is* processed *once such a detecting word and its inherited five-source ancestry have been adjoined to the cumulative record; its magnitude is irrelevant.*

(J4) *On a prefix containing N completed normalized shell crossings, $A_{\Phi,N} = N$. In particular, division by N, return time, scale depth, or the mass of an empirical recurrence law is not an admissible target-accounting operation. Such a normalization may select a recurrent support, but the selected support is rejoined with* (450) *before target projection.*

(J5) *Every target-visible cross term created by nonlinear combination, localization, conditioning, stopping, chart change, scaling, quotient, weak limiting, trace, boundary passage, or recurrence is a finite constructor vertex in* $\mathrm{Exp}(q)$*, or a total-graph boundary value of such a vertex. It inherits the union of the source ancestries of its inputs. Thus the fact that every input is individually target-null does not authorize deletion of their joint record: joint synergy is tested before the residual quotient.*

(J6) *Every physical quantity invoked by a composite remains attached to its owner and to the exact target-local balance or domination graph that uses it. Finite additivity, countable additivity, a conservation law, or a positive sign does not by itself pay for target progress. The composite is charged only after its cumulative target owner lies in $\mathscr{D}_{\mathfrak{b}}^{\mathrm{phys}}$. Otherwise its universal amplification graph* (494) *produces the next finite polarity or carrier–polar successor.*

(J7) *The construction commutes with directed same-origin accumulation. If a finite or countable family of regular, target-null, bounded, vanishing, or subthreshold pieces has target-active composite output, then a finite expanded joint prefix has a nonzero target increment. That increment is re-entered into the five-source compiler with its complete ancestry, physical owners, normalization denominators, and unnormalized cocycle. If no such prefix exists, the composite limit is target-null.*

Thus no generated continuum construction can turn regular pieces into unaccounted singularity-forming structure. It may create target alignment through their joint relation, exactly as an XOR can in Part I, but that alignment belongs to the saturated expanded derivation and must be bounded as an existing five-source profile cell. There is no separate loophole for time accumulation, scale collapse, nonlinear interaction, weak convergence, normalization, localization, recurrence, concentration, oscillation, tails, or boundary passage.

Proof. Since q factors through $\mathscr{J}_n$, one has $\sigma(q) \subseteq \mathcal{A}_n$. Conditional expectations onto nested closed subspaces give the two projection identities in (495); the norm bound follows from the orthogonal Pythagorean decomposition. The martingale convergence theorem for the increasing sigma algebras $(\mathcal{A}_n)$ gives (496). Its orthogonal-difference decomposition gives (497)– (498). A positive limiting norm has a positive finite-prefix norm, while a countable collection of small increments is retained by the convergent orthogonal sum. This proves (J2) without a detection threshold.

At finite depth, prospective accountability and the five-source normal form give (450). The join records all mixed coordinates before projection, so a target component selected by a composite belongs to the joint algebra rather than to the residual of any individual input. Source-preserving totalization assigns a nonordinary mixed operation to its typed graph value with the union of the input ancestries. If $d_n \neq 0$, density of the generated bounded cylinder readouts in $L^2(\mathcal{A}_n, \nu)$

supplies one with nonzero pairing against d_n. Because $d_n \perp L^2(\mathcal{A}_{n-1}, \nu)$, no readout of the old prefix can supply this pairing; the detecting cylinder therefore identifies a finite new joint word. Passing to the graph-complete limit preserves every fixed finite current identity; if a current fails to converge in variation, its boundary owner is the retained Bdry-headed value. This proves (J3).

Each completed normalized crossing contributes one to the prospective shell cocycle, and the same-origin join adds these occurrences without identifying them. Hence $A_{\Phi,N} = N$. A normalized empirical law is a positive scalar rescaling used for compact support selection. Keeping it as a separate coordinate and rejoining it with the cumulative cocycle proves (J4).

Every operation listed in (J5) is a constructor in the expanded grammar. At an ordinary value, structural induction assigns the union of the input ancestries; at a nonordinary value, totalization assigns the same ancestry to the graph-boundary successor. This proves (J5). The public physical quantities are owner-tagged coordinates of the joint record. The definition of target-aligned physical charge permits their use only through the target-local carrier cone. Separation from that cone is exactly the carrier–polar successor, while totalization of (494) gives all finite, infinite, and zero-denominator values. This proves (J6) without inferring domination from positivity. Finally apply (J2) to the increasing sigma algebras of the expanded joint prefixes. A nonzero limiting projection has a nonzero orthogonal finite-prefix increment, and (J3) assigns its mixed source word. This proves (J7). The argument depends only on expanded derivation, not on the analytic name of the constructor. $\square$

Corollary III.4.59 (One rule for all continuum accumulation)**.** *The following are not separate regularity principles: accumulation over time; transfer over scales; addition over source occurrences; spatial localization and partition gluing; nonlinear interaction; conditioning or stopping; normalization by amplitude, mass, return number, or scale; compactness loss, concentration, and oscillation; weak or graph limits; recurrence; tail escape; and boundary or terminal passage. Each is evaluated by the same protocol:*

$$\boxed{\mathrm{Exp} \longrightarrow \mathrm{Join} \longrightarrow \mathrm{Sat}_{\mathsf{G}_\Sigma} \longrightarrow q_\Sigma \longrightarrow \begin{cases} \mathscr{N}_\Sigma, \\ \textit{five-source target owner}, \\ \textit{typed total-graph successor}. \end{cases}} \tag{499}$$

For a nonnull owner, recursive carrier–polar exhaustion is mandatory before averaging, realization, or another composition. Hence no theorem may infer regularity merely because every constituent is regular, small, bounded, vanishing, summable, or target-null in isolation.

Theorem III.4.60 (Universal continuum saturated-profile accountability)**.** *Let q be any target-directed readout produced at any finite or directed stage of a legal same-origin continuum evolution, and let $Q_\Sigma(q)$ be its positive target progress after universal expanded-derivation saturation. Then*

$$Q_\Sigma(q) \le \sum_{c \in \mathfrak{C}} \sup_{R \in \mathrm{Sat}(\mathrm{Exp}(q))} \left(\mathrm{Prog}^c_\Sigma(R)\right)_+ \le \mathsf{T}^{\mathrm{phys,sat}}_\Sigma(B_{\mathrm{phys}}). \tag{500}$$

For N completed shell crossings the same statement is applied to the unnormalized cumulative readout and gives

$$N \le \sum_{c \in \mathfrak{C}} (Q^c_{\Phi,N})_+. \tag{501}$$

Every target-directed effect of composition, including an effect absent from all individual inputs, therefore appears in the same saturated profile that is bounded in a regularity proof. A proof that bounds only primitive or termwise profiles is not a continuum structural proof.

Proof. Apply Theorem III.4.58 to the expanded derivation of q. Its target-null component vanishes, while every nonzero ordinary or graph-boundary component has an inherited five-source owner. The triangle inequality for the five signed source contributions gives the first inequality, and saturation of the profile over expanded composite records gives the second. For completed crossings retain $A_{\Phi,N} = N$ before every normalization and sum the source-resolved identity; this is (501). No constituent-wise lower bound is used. □

Corollary III.4.61 (No magical singularity from regular constituents)**.** *Suppose a family of continuum constituents is regular, bounded, smooth, compact, individually target-null, or individually of zero target price. These predicates imply nothing about a generated composite until its expanded joint record is evaluated. After that evaluation exactly one of the following holds: the composite is target-null; its target owner is charged by the public physical cone; or a source-resolved amplification successor is generated. Repeating the construction, at any countable depth, produces no fourth case and no target-aligned residual.*

Corollary III.4.62 (All previous accumulation routes are instances)**.** *The unnormalized shell ledger, cumulative-before-circulation rule, no-diffuse-composition rule, no-diffuse-primal–polar rule, covariant microscope, entropy cocycle, signed interface gluing, recurrent return construction, concentration/oscillation lift, tail completion, and terminal boundary owner are restrictions of Theorems I.1.19, II.2.32 and III.4.58 to particular constructor subgrammars. None is an independent assumption or an additional closure principle. Any future continuum constructor is handled by adjoining its total graph to the public grammar and applying the same expanded-derivation theorem.*

Corollary III.4.63 (No limit-created target alignment)**.** *An infinite compatible compiler execution is not a terminal branch. Form $\mathcal{J}_\infty$ and reapply the five-channel projection. Exactly one of the following occurs: the joint record is target-null; a finite joint prefix has a least unprocessed target increment d_n, whose complete five-channel mixed ancestry is the next saturated compiler state; or every increment is processed and the joint record is a fixed point of every generated joint, source, graph, carrier, and target-cyclic operation while retaining its nonzero unnormalized shell cocycle. The last case is a joint-saturated structural atom and is defined below. A compatible inverse limit or an ordinary path is never terminal solely because it exists.*

Corollary III.4.64 (No diffuse composition loophole)**.** *Let one same-origin execution complete N normalized target-shell crossings, and classify every mixed joint increment before forming the residual. If*

$$Q^c_{\Phi,N} := \sum_{j<N} Q^c_{\Phi,j} \qquad (c \in \mathfrak{C})$$

denotes its cumulative source-resolved shell current, then

$$N = A_{\Phi,N} = \sum_{c\in\mathfrak{C}} Q^c_{\Phi,N}, \qquad \sum_{c\in\mathfrak{C}} (Q^c_{\Phi,N})_+ \geq N, \qquad \max_{c\in\mathfrak{C}} (Q^c_{\Phi,N})_+ \geq \frac{N}{5}. \tag{502}$$

Consequently there are a source c_* and a cofinal subsequence N_k such that

$$(Q^{c_*}_{\Phi,N_k})_+ \geq \frac{N_k}{5}.$$

This conclusion has no per-occurrence detection threshold. Individually target-null inputs may have a target-aligned joint increment, and infinitely many arbitrarily small processed increments *may have nonzero total; both are already included in the left side of* (502)*. If every cumulative five-source current vanished, no positive number of crossings could occur. Thus diffuse composition cannot create a target-aligned* J_{res}*.*

Proof. For each completed shell, prospective accountability gives $1 = \sum_c Q^{\mathrm{c}}_{\Phi,j}$ after that prefix has been jointly saturated. Theorem III.4.58 assigns every nonzero mixed increment to its complete inherited source word before the residual is formed. Sum the N identities. The first equality in (502) follows, and the two inequalities follow from $x \le x_+$ and the five-element pigeonhole principle. One channel attains the last bound along a cofinal subsequence. No lower bound on an individual increment was used. The identity uses the unnormalized cocycle, so normalization of a return law cannot change it. ☐

Definition III.4.65 (Unnormalized finite-horizon target-current problem)**.** Let X be a closed target-active recurrent core and let $E_X \rightrightarrows X$ be its compact same-origin successor graph, with source and range maps π_- and π_+. Retain the nonnegative physical valuation p_{phys}, the positive target-return coordinate $a_\Sigma \ge \eta$, and the signed source coordinates q^{c}_Σ, where $a_\Sigma = \sum_c q^{\mathrm{c}}_\Sigma$. For $N \ge 1$, let

$$E^{[N]}_X := \{(e_0, \dots, e_{N-1}) : \pi_+(e_j) = \pi_-(e_{j+1}) \text{ for } j < N-1\}$$

with all active ideal, origin, and total-graph relations imposed at every layer. Define the unnormalized cumulative price

$$\Delta_X(N) := \inf_{\boldsymbol{e} \in E^{[N]}_X} \sum_{j<N} p_{\mathrm{phys}}(e_j) \in [0, +\infty], \tag{503}$$

where the value is $+\infty$ when the layered path space is empty. The associated prefix coordinates are

$$A_N(\boldsymbol{e}) := \sum_{j<N} a_\Sigma(e_j), \qquad Q^{\mathrm{c}}_N(\boldsymbol{e}) := \sum_{j<N} q^{\mathrm{c}}_\Sigma(e_j), \qquad \Pi_N(\boldsymbol{e}) := \sum_{j<N} p_{\mathrm{phys}}(e_j). \tag{504}$$

No probability normalization occurs in this definition.

Theorem III.4.66 (Cumulative carrier duality)**.** *Let X be as in Definition III.4.65. For continuous cylinder lower envelopes $p_m \uparrow p_{\mathrm{phys}}$,*

$$\Delta_X(N) = \sup_{m,(V_j),(\delta_j)} \left\{ \sum_{j<N} \delta_j : \begin{array}{c} V_0 = V_N = 0, \\ p_m(e) + V_j(\pi_- e) - V_{j+1}(\pi_+ e) - \delta_j \in \overline{\mathcal{K}^0_X} \\ \textit{on the layer-}j\textit{ edge graph}, \quad j < N \end{array} \right\}. \tag{505}$$

Here $\mathcal{K}^0_X$ is the local ordered equation cone before successor coboundaries are imposed. Every finite dual row telescopes on an actual prefix and gives

$$\Pi_N(\boldsymbol{e}) \ge \sum_{j<N} \delta_j. \tag{506}$$

Consequently, on a physical stratum of total mass at most B_{phys}, exactly one of the following occurs.

(U1) *For some N, a finite cylinder dual row has $\sum_{j<N} \delta_j > B_{\mathrm{phys}}$. No same-origin contact cascade in the stratum has N returns.*

(U2) *The values $\Delta_X(N)$ remain bounded by B_{phys}. The cumulative joint compactification contains an infinite same-origin successor path satisfying*

$$\sup_N \Pi_N \le B_{\mathrm{phys}}, \qquad A_N \ge \eta N, \qquad (Q^{c_*}_{N_k})_+ \ge \frac{\eta N_k}{5} \longrightarrow \infty \tag{507}$$

for a channel c_* and a cofinal subsequence. This entire unnormalized path ledger is the next compiler state.

The alternatives concern cumulative expenditure. They remain valid when every one-step price tends to zero and when every normalized invariant law has zero physical-expenditure density.

Proof. For fixed m and N, minimize the linear functional $\sum_{j<N} \int p_m \, \mathrm{d}\gamma_j$ over positive layer-edge measures of mass one whose adjacent range and source marginals agree and which satisfy the active ordered relations. This is a linear program on a compact convex set. Separation gives its dual: the marginal multipliers are the cylinder potentials V_j, the layer lower bounds are δ_j, and the endpoint normalization is $V_0 = V_N = 0$. Summing the layer inequalities cancels all intermediate potentials and proves (506). Compactness and monotone convergence of the lower envelopes give (505) for p_{phys}.

For completeness, the layered marginal constraints admit the usual finite superposition: disintegrate γ_0 over its range marginal, then successively disintegrate γ_j over the common source marginal. This constructs a probability measure on $E_X^{[N]}$ having the prescribed layer marginals. Conversely, every probability measure on $E_X^{[N]}$ has such marginals. Hence the relaxed linear-program value is the average of path costs and its infimum equals the deterministic path infimum in (503); no relaxation gap is being hidden in (505).

If the first alternative holds, the displayed prefix inequality contradicts the public physical ceiling. Otherwise choose an N-edge prefix of cost at most $B_{\mathrm{phys}} + N^{-1}$ in the viable recurrent core and extend it in the closed successor graph. Compactness of the graph-complete path space and a diagonal subsequence give an infinite path. Lower semicontinuity on every fixed prefix gives $\Pi_N \le B_{\mathrm{phys}}$. The target floor gives $A_N \ge \eta N$, while $A_N = \sum_c Q^c_N$. Positive parts and the five-channel pigeonhole argument produce c_* and the cofinal subsequence in (507). No average has been taken. ☐

Corollary III.4.67 (Cumulative test precedes circulation)**.** *Every recurrent target-active compiler branch first applies Theorem III.4.66 to its unnormalized layered successor graph. Alternative* (U1) *closes the branch by a finite physical identity. In alternative* (U2)*, a normalized circulation may select recurrent support, but every subsequent kernel, target, and realization operation acts on the join of that selector with the ledger* (507)*. Thus circulation is a support reduction of an already evaluated cumulative problem; it is never the physical-price evaluator for the original cascade.*

Definition III.4.68 (Target-current circulation problem)**.** Let $E_{\Sigma,\mathfrak{b}}$ be the compact total-graph edge space of $\mathscr{T}_{\Sigma,\mathfrak{b}}$, with source and range maps π_-, π_+. The selected signed-dyadic orientation defines a bounded nonnegative target-return coordinate a_Σ; normalize each active return cell so that $a_\Sigma \ge \eta > 0$. Write $p_{\mathrm{phys}} \ge 0$ for the lower-semicontinuous physical valuation obtained from the local carrier-domination graph, and write q^c_Σ for its source-resolved target-return coordinates, so that $a_\Sigma = \sum_c q^c_\Sigma$. Along an actual return sequence retain the unnormalized occurrence measure, target cocycle, physical expenditure, and source currents

$$\Gamma_N := \sum_{j<N} \delta_{e_j}, \qquad A_{\Sigma,N} := \int a_\Sigma \, \mathrm{d}\Gamma_N, \qquad \Pi_N := \int p_{\mathrm{phys}} \, \mathrm{d}\Gamma_N, \qquad Q^c_N := \int q^c_\Sigma \, \mathrm{d}\Gamma_N. \tag{508}$$

When $A_{\Sigma,N} > 0$, define the empirical support selector

$$\gamma_N := A_{\Sigma,N}^{-1}\Gamma_N. \tag{509}$$

This is a change of scale in the observer law, not a quotient of the structural record: $A_{\Sigma,N}$, Π_N, and every Q_N^c remain separate coordinates. The continuous cylinder lower envelopes of p_{phys}, $p_N \uparrow p_{\mathrm{phys}}$ are generated by the totalized carrier coordinate; an infinite endpoint is routed directly to physical charge. A *target-current circulation* is a finite positive Borel support-selector measure γ on $E_{\Sigma,\mathfrak{b}}$ satisfying

$$(\pi_-)_\#\gamma = (\pi_+)_\#\gamma, \qquad \int a_\Sigma \,\mathrm{d}\gamma = 1, \tag{510}$$

and all ordered ideal, source, origin, and total-graph relations of the active signature. It is stored jointly with $(\Gamma_N, A_{\Sigma,N}, \Pi_N, (Q_N^c)_c)_N$; the normalization in (510) never replaces those cumulative coordinates. Its sharp recurrent physical price is

$$\delta_{\Sigma,\mathfrak{b}}^{\mathrm{cyc}} := \inf_\gamma \int p_{\mathrm{phys}} \,\mathrm{d}\gamma \in [0, +\infty], \tag{511}$$

with value $+\infty$ when the circulation set is empty.

Theorem III.4.69 (Multiplicity restoration for normalized circulations)**.** *For every actual return prefix with $A_{\Sigma,N} > 0$, normalization is exactly reversible in the augmented structural record:*

$$\Gamma_N = A_{\Sigma,N}\gamma_N, \qquad \Pi_N = A_{\Sigma,N}\int p_{\mathrm{phys}} \,\mathrm{d}\gamma_N, \qquad Q_N^c = A_{\Sigma,N}\int q_\Sigma^c \,\mathrm{d}\gamma_N, \qquad \sum_c Q_N^c = A_{\Sigma,N}. \tag{512}$$

Suppose $A_{\Sigma,N} \to \infty$ while $\sup_N \Pi_N < \infty$. Every weak- limit γ of the selectors satisfies $\int p_{\mathrm{phys}} \,\mathrm{d}\gamma = 0$, but this conclusion has only the following meaning: γ selects the zero-density recurrent support. It implies neither $\Pi_N = 0$ nor $Q_N^c = 0$, nor may it replace the occurrence sequence. Indeed, on a cofinal subsequence there is a fixed $c_* \in \mathfrak{C}$ with*

$$(Q_{N_k}^{c_*})_+ \geq \frac{A_{\Sigma,N_k}}{5} \longrightarrow \infty. \tag{513}$$

Consequently a zero value of $\delta_{\Sigma,\mathfrak{b}}^{\mathrm{cyc}}$ means only that the public carrier has no uniform positive linear lower bound per unit return on that support. It never means zero accumulated expenditure or zero accumulated target-aligned structure.

Proof. The first three identities are substitution of (509); the fourth is the source-resolved return identity. If the physical prefix expenditures are bounded, then $\int p_{\mathrm{phys}} \,\mathrm{d}\gamma_N = \Pi_N/A_{\Sigma,N} \to 0$. Lower semicontinuity gives zero physical density on every weak-* limit. No limit has been taken in the multiplicity coordinates, however: they remain in the augmented record and reconstruct the original prefix through the first identity. Applying the positive-part and five-channel pigeonhole argument to $\sum_c Q_N^c = A_{\Sigma,N}$ gives a channel attaining $A_{\Sigma,N}/5$ on a cofinal subsequence. This proves (513) and forbids the inference from zero normalized density to zero cumulative structure. $\square$

Theorem III.4.70 (Target-current circulation duality)**.** *Write $\mathcal{K}_\mathfrak{b}^0 := I_\mathfrak{b} + M_\mathfrak{b}$ for the local ordered cone before shift coboundaries are adjoined. All functions below are functions on the target quotient; hence no current-null space is added to this scalar cone. On every reachable active signature,*

$$\delta_{\Sigma,\mathfrak{b}}^{\mathrm{cyc}} = \sup\left\{\delta \in \mathbb{R} : \begin{array}{l} \textit{there are } N \textit{ and a generated continuous cylinder potential } V \textit{ with} \\ p_N + V\circ\pi_- - V\circ\pi_+ - \delta a_\Sigma \in \overline{\mathcal{K}_\mathfrak{b}^0} \end{array}\right\}. \tag{514}$$

Exactly one of the following values is produced.

(D1) $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} > 0$. *A rational* $0 < \delta < \delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}$ *and a finite cylinder approximation of* V *give a finite local ordered identity. On the first* N *returns it gives*

$$\sum_{j<N} p_{\mathrm{phys}}(e_j) \geq \delta \sum_{j<N} a_\Sigma(e_j) - 2\|V\|_\infty.$$

Thus a recurrent sequence with unbounded accumulated unnormalized target return has infinite public physical charge.

(D2) $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = 0$. *Compatible minimizing cylinder moments reconstruct a zero-density target-current circulation. It asserts failure of a uniform linear price and nothing about accumulated quantities. Its ergodic components and positive support select the next production-density, return, or graph-boundary cell. Before any target or kernel operation, the selected law is augmented by the full multiplicity record*

$$(\Gamma_N, A_{\Sigma,N}, \Pi_N, (Q^c_N)_c)_N$$

and the identities (512)*. The next projection in Definition III.4.23 is applied to this augmented record, never to the probability law alone.*

(D3) $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}} = +\infty$. *A finite ordered identity either excludes the normalized circulation constraints, or, for each rational* $M > 0$*, gives recurrent physical price at least* M*. In the latter case choose* M *above the remaining public physical ceiling; the recurrent cell is excluded by physical charge, not declared empty.*

The potential V *is generated by the cylinder hierarchy. It is not a multiplier supplied with the PDE presentation.*

Proof. Let $\mathcal{M}_{\mathfrak{b}}$ be the convex set of positive edge measures satisfying the first relation in (510) and the active ordered relations. On the normalized slice, $a_\Sigma \geq \eta$ bounds total mass; the slice is therefore weak-* compact. First minimize each continuous lower envelope p_N. Linear-programming separation on the compact slice gives its continuous dual. Compactness of minimizing measures and lower semicontinuity show that these minima increase to the minimum of p_{phys}: otherwise a convergent subnet of minimizers would violate the Portmanteau inequality. The annihilator of equal source and range marginals is the closed space of coboundaries $V \circ \pi_- - V \circ \pi_+$. The active ordered relations contribute $\overline{\mathcal{K}^0_{\mathfrak{b}}}$. The edge space was formed after the target quotient, so target-null currents have already disappeared and cannot be added to the scalar dual cone. This proves (514) by strong separation.

The rational cylinder functions are uniformly dense in the generated evaluation algebra. If the optimum is positive, choose rational δ strictly below it and approximate the dual potential. Archimedean order-unit completion absorbs the approximation error and the finite cylinder-moment hierarchy returns the corresponding finite identity. Its sum along a finite path telescopes the coboundary, and its integral against a circulation cancels it exactly. Hence $\int p_{\mathrm{phys}} \geq \delta \int a_\Sigma = \delta$, proving (D1).

If the optimum is zero, choose minimizing feasible moments at each cylinder depth. Compact projective consistency reconstructs a circulation with zero physical *density.* Ergodic decomposition preserves flow conservation, zero density, and positive target mass on at least one component. Its probability law selects only the support of the next recurrent compiler state. Apply Theorem III.4.69: the actual occurrence, physical-expenditure, target-return, and five source-current prefixes are reattached before joint saturation. Then Theorem III.4.58 produces the cumulative joint state. In particular, $\Pi_N/A_{\Sigma,N} \to 0$ is never rewritten as $\Pi_N = 0$, and the divergent cumulative source coordinate (513) remains live. This proves (D2). If the feasible slice is empty, Archimedean separation supplies the finite -1-identity in (D3). If it is nonempty and the lower-semicontinuous price is infinite on every normalized circulation, the minima of the continuous lower envelopes increase without bound. For any prescribed rational M, some finite envelope and its dual potential give the ordered price identity $\int p_{\mathrm{phys}} \geq M$. Taking M above the remaining finite ledger excludes the recurrent branch. $\square$

Corollary III.4.71 (No averaging discharge of a recurrent contact cascade)**.** *Let an actual same-origin branch complete infinitely many target returns while its public physical expenditure remains finite. No normalized circulation obtained from that branch can close, nullify, or quotient the branch merely because its physical-expenditure density is zero. More precisely, after passing to a cofinal subsequence, there is a channel c_* carrying the unbounded positive cumulative coordinate*

$$(Q^{c_*}_{N_k})_+ \geq A_{\Sigma,N_k}/5 \longrightarrow \infty.$$

At every refinement depth the compiler must therefore do one of the following with the augmented pair consisting of the support selector and this cocycle: charge a positive linear price by (D1)*, expose a least new finite joint source word, or retain the unbounded c_*-coordinate in a joint-saturated structural atom. Target-nullity is impossible because it would force $A_{\Sigma,N} = \sum_c Q^c_N = 0$ on every prefix.*

Thus the implication used by the recurrent calculus is

$$\frac{\Pi_N}{A_{\Sigma,N}} \longrightarrow 0 \quad \Longrightarrow \quad \textit{zero-density support plus the full cumulative cocycle},$$

and never zero cumulative target-aligned structure.

Proof. Apply Theorem III.4.69 and choose c_* by the finite-channel pigeonhole argument. Rejoin each prefix with the selector before projection. The cumulative-joint theorem Theorem III.4.58 says that a nonzero target component is either detected by a finite joint word or retained in the graph-complete fixed point, with inherited five-channel ancestry. The positive-price case is (D1) of Theorem III.4.70. Finally, target-nullity of all cumulative channels contradicts the exact return identity. These alternatives exhaust the augmented record and use no per-return threshold. $\square$

Definition III.4.72 (Circulation-elimination operator)**.** For a target-spine cell X, first compute the unnormalized horizons $\Delta_X(N)$ of Definition III.4.65. Remove X when a finite cumulative dual row exceeds the remaining public physical ceiling. On the complementary branch retain the infinite augmented ledger supplied by (U2) of Theorem III.4.66. Apply recursive carrier-polar exhaustion to every canonical cumulative owner; every separator and its adjoint descendants are rejoined and reprojected. Only on a primal–polar fixed support compute $\delta^{\mathrm{cyc}}_{\Sigma,\mathfrak{b}}$ in the nested cylinder hierarchy as a recurrent-support reduction. Remove X in cases (D1) and (D3). In case (D2), restrict to the zero-density circulation support, adjoin all production densities that vanish on that selector, retain their cumulative prefix integrals without setting them to zero, and inspect the least unprocessed target-cyclic covector or total graph in the fixed public enumeration. Return its finite zero, signed-dyadic, and graph-boundary polarity partition after deleting the target-null members. Rejoin every returned cell with its entire parent transcript, saturate all mixed coordinates, and reapply the target projection before it is embedded in the universal spine. Thus each child retains $(\Gamma_N, A_{\Sigma,N}, \Pi_N, (Q^c_N)_c)_N$, including every accumulated source current and physical expenditure, and is a closed joint refinement of X. Iterate

$$X_0 := \mathscr{T}_{\Sigma,\mathfrak{b}}, \qquad X_{n+1} := \mathcal{E}_\Sigma(X_n), \qquad X_\infty := \bigcap_{n\geq 0} X_n, \tag{515}$$

where every X_n is represented in its cumulative joint algebra and the intersection is followed by the joint graph-complete saturation (449). Thus X_∞ is a joint fixed-point core rather than a coordinatewise inverse-limit record. This operator is generated solely from the target spine, public physical valuation, and ordered equation graph. Its priority order is cumulative horizon, carrier-polar exhaustion, recurrent support, polarity refinement, joint saturation, and target reprojection.

Definition III.4.73 (Target-aligned structural branch-and-bound)**.** Fix the enumeration of rational target-cylinder coordinates, public graph relations, and source descendants. At depth n, partition the first n coordinates into rational closed boxes of diameter at most 2^{-n} in the canonical evaluation metric. Only coordinates in the generated target-cyclic spine are partitioned. For every surviving parent box C, perform the following finite operations.

(B1) Form the saturated joint record of the complete parent prefix. Adjoin the first n equation, origin, carrier, carrier-polar, polarity, mixed-product, shell-cocycle, and total-graph rows. Form every layered horizon of length at most n and use the continuous price envelope p_n.

(B2) Solve the resulting finite rational moment relaxation and enumerate its ordered dual identities. For every retained source owner, check every finite algebraic cone identity and every finite-cylinder separator whose polar inequality is verified by the retained projected cone. When neither check succeeds, retain the total cone-boundary coordinate rather than declaring membership or nonmembership. Append every verified separator and its generated descendants before solving the normalized circulation relaxation. Record the largest certified cumulative lower bound through horizon n, the certified recurrent dual lower price $\underline{\delta}_n(C)$ and the compact set $\mathsf{P}_n(C)$ of feasible truncated moments.

(B3) Delete C when a cumulative horizon lower bound exceeds the remaining physical ceiling, when a finite -1-identity makes $\mathsf{P}_n(C) = \varnothing$, or when a positive recurrent dual identity gives $\underline{\delta}_n(C) > 0$. Attach that finite identity to the deleted box.

(B4) Otherwise subdivide the least unprocessed target-visible cylinder coordinate. A zero value adjoins the corresponding kernel relation; a nonzero value is divided into signed dyadic cells; and a nonordinary value is divided by its typed total-graph boundary. Origin and source ancestry are copied to every child. Each child is rejoined with the parent prefix and reprojected; a target component produced by a combination of previously target-null coordinates is therefore exposed at this depth.

Let $\mathscr{U}_n$ be the union of the surviving boxes, intersected with their surviving parents, and let

$$\mathscr{U}_\infty := \bigcap_{n\geq 0} \mathscr{U}_n. \tag{516}$$

The depth n is an algebraic refinement index, not a second physical budget. Every operation uses rational cylinder arithmetic and the public ordered graph; no multiplier, compactness certificate, or analytic estimate is supplied by the user.

Definition III.4.74 (Joint-saturated prospective source atom)**.** A *joint-saturated structural atom* on an active signature $\mathfrak{b}$ is a tuple

$$\mathfrak{a} = (X_\mathfrak{a}, \mathcal{J}_\mathfrak{a}, \Phi, (Q^c_\mathfrak{a})_{c\in\mathfrak{C}}, A_{\Phi,\mathfrak{a}}, p_{\mathrm{phys}}, \mathbf{R}) \tag{517}$$

with the following generated properties.

(A1) $\mathcal{J}_\mathfrak{a}$ is the cumulative join of a cofinal same-origin compiler execution and is fixed by joint saturation, all five source projections, every public mixed operation and total graph, signed carrier gluing, target-cyclic adjoint transport, and first-return formation. It is also fixed under the recursive carrier-polar operation $\mathfrak{P}_\Sigma$ for every canonical source or polar owner.

(A2) Its unnormalized cocycle is retained and nonzero on every completed shell prefix:

$$A_{\Phi,N} = N = \sum_{c \in \mathfrak{C}} Q^{c}_{\Phi,N}(1) \qquad (N \geq 1). \tag{518}$$

Hence its persistent source support $A(\mathfrak{a}) := \{c : Q^{c}_{\mathfrak{a}} \neq 0\}$ is nonempty.

(A3) Every target-visible finite joint prefix, mixed polarization, production, feeding, storage, carrier separator, polar descendant, graph-boundary, and realization value has its zero, signed-dyadic, or typed successor inside $\mathcal{J}_{\mathfrak{a}}$. The post-saturation residual annihilates every prospective target functional.

(A4) The physical coordinate is the restriction of the single public measure. Every owner in the public physical carrier cone has been removed by its cumulative carrier identity; every owner outside that cone has its complete separating covector and generated primal–polar descendants in the joint record. The retained zero-density support law is only a support coordinate and remains joined to (518).

The atom is a *contact-derived joint source record* when $\mathrm{pol}_{\mathrm{org}}(\mathbf{R}) = \mathsf{Hyp}$. It remains in the same presentation-generated pre-contact domain and is evaluated by its retained shell covector, source word, and physical carrier. The atom is *independently realized* when $\mathrm{pol}_{\mathrm{org}}(\mathbf{R}) = \mathsf{Ind}$ and $\mathbf{R}$ contains one ordinary same-origin public solution path whose continuation row and cofinal shell rows realize the target contact carried by Φ. A formal fixed-point law, compatible moment family, or ordinary path without (A1)–(A4) is not a prospective source atom. A contact-derived atom remains in the reachable invariant evaluator. An ordinary same-origin realization with the declared noncontinuation row is a singular contact record.

Theorem III.4.75 (Stagewise computation of the target mechanism core)**.** *The branch-and-bound construction has the following properties.*

(C1) *Every depth is a finite exact calculation* of the stated finite finite prefix. *Its output consists of a finite family of presentation-generated target-cylinder records $\mathscr{U}_n$, checked ordered identities for every deleted incidence edge, and exact finite-prefix invariant values for every retained record. The families are nested.*

(C2) *The construction loses no target-active mechanism and retains no untyped infinite branch. Because every child contains its cumulative joint prefix, one has*

$$\mathscr{U}_\infty = X_\infty. \tag{519}$$

Thus $\mathscr{U}_n = \varnothing$ at some finite depth implies that the target mechanism core is empty. The complete core is the source-preserving closure of these finite records, and its zero detector profile is evaluated by Theorem III.10.41.

(C3) *Let F be a rational cylinder observable, or a generated bounded target-aligned structural observable whose constructor graph supplies a verified cylinder-approximation modulus. From the finite boxes one obtains certified intervals*

$$I_n(F) := \left[inf_{\mathscr{U}_n} F_n - 2^{-n}, \sup_{\mathscr{U}_n} F_n + 2^{-n} \right],$$

where F_n is its generated rational cylinder approximation. After replacing each interval by its intersection with the preceding ones,

$$I_n(F) \downarrow \left[\min_{X_\infty} F, \max_{X_\infty} F \right]. \tag{520}$$

Consequently every observable in this effective cylinder class is computed by certified two-sided enclosures. Bare continuity supplies density but not an effective approximation rate and is not promoted to a computed result.

(C4) *If the core is nonempty, the complete stream* $(\mathscr{U}_n)_n$*, together with the compatible moment tables and origin rows, determines the compact graph-complete joint target-mechanism set. Join and saturate every infinite nested box word before taking its target projection. A least unprocessed limiting increment occurs on a finite joint prefix by Theorem III.4.58; that prefix is reclassified and refined. If every increment is processed, the word is tested for joint fixedness. Failure of fixedness exposes the next generated operation; a fixed word supplies the structural coordinates* (A1)–(A4) *of Definition III.4.74. Its total realization value is then ordinary, giving a prospective source atom with its unchanged origin polarity, or nonordinary, giving the next typed realization successor. A compatible moment family, invariant law, or ordinary path alone is never an output.*

The algorithm is target local: refining any source-transverse or target-null coordinate leaves every $\mathscr{U}_n$ *and every interval* $I_n(F)$ *unchanged.*

Proof. At fixed depth there are finitely many boxes, rows, monomials, and moment coordinates. The retained polynomial moment relaxation and its ordered dual are finite rational semialgebraic problems; a displayed rational dual identity is checked by direct arithmetic. Exponentials, resolvents, unbounded operators, distributional products, and limiting relations enter through their bounded total-graph coordinates and finite graph rows, not by being misidentified as polynomial maps. Their unprocessed graph boundaries remain in the outer cover. This proves (C1).

Every genuine circulation satisfies every finite relaxation, so it is never deleted. Conversely, choose a point or compatible moment table from each box along an infinite nested word. Compactness of the bounded total-graph coordinates gives a diagonal limit. Because the enumeration is exhaustive, each equation, carrier, polarity, origin, and graph relation is imposed from some finite depth onward. The least-unprocessed-coordinate rule therefore gives precisely the circulation-elimination limit, proving (519). If the intersection is empty, compactness of the decreasing covers gives a finite empty depth; the reverse implication is immediate.

The constructor-supplied modulus gives $\|F - F_n\|_{\mathscr{U}_n} \leq 2^{-n}$. Extrema of a continuous function over decreasing nonempty compact sets converge to the extrema over their intersection. This proves (520). Finally, the compatible moments reconstruct a measure on the total-graph inverse limit, and the origin and cofinal-shell rows survive. Apply cumulative joining before interpreting that measure. By Theorem III.4.58, every nonzero limiting target increment already occurs on a finite joint prefix. Every unprocessed such increment is therefore another branch-and-bound coordinate. When no such increment remains, apply recursive carrier–polar exhaustion to every cumulative source and polar owner. Target-null and physically charged owners return to their preceding compiler rows; a nonzero separator or one of its adjoint descendants is the next branch-and-bound coordinate. Only simultaneous primal–polar fixedness gives (A1)–(A4) of Definition III.4.74. The totalized public-orbit row then gives either the ordinary prospective source atom, with its origin polarity unchanged, or its next typed successor. All partition coordinates lie in the target quotient by construction, proving target locality. □

Theorem III.4.76 (Completeness of circulation elimination)**.** *The sets in* (515) *are closed, target-active, decreasing, and represented in increasing cumulative joint algebras. After the joint limit saturation in* (449),

$$X_\infty = \bigsqcup_{\mathfrak{a} \in \mathsf{Atom}^{\mathrm{str}}_{\Sigma}(\mathfrak{b})} X_{\mathfrak{a}}, \tag{521}$$

where $\mathsf{Atom}^{\mathrm{str}}_{\Sigma}(\mathfrak{b})$ *is the family of joint-saturated prospective source atoms of Definition III.4.74. A formal atom with a nonordinary public realization coordinate is passed to its typed realization successor. On a contact-generated same-origin execution, every atom has hypothetical origin and supplies only a saturated source-profile lower bound. Consequently*

$$\textit{contact on the branch} \quad \Longrightarrow \quad \mathsf{Atom}^{\mathsf{Hyp}}_{\Sigma}(\mathfrak{b}) \neq \varnothing. \tag{522}$$

Independently generated singular contact is equivalent to a member of $\mathsf{Atom}^{\mathsf{Ind},\mathrm{real}}_{\Sigma}(\mathfrak{b})$*. If the intersection is empty, compactness gives a finite depth at which it is empty; concatenating the circulation-dual identities gives a finite target-local regularity derivation. A nonempty intersection is processed by joint reprojection; neither the intersection nor an ordinary path is by itself a terminal.*

Proof. Closedness follows from weak-* compactness of the circulation slices and total-graph closure. The operator removes exactly the cells having no normalized recurrent circulation or having positive recurrent physical price. On a zero-price cell it resolves the least target-cyclic descendant. Consequently membership in every X_n is equivalent to compatibility with the first n circulation, production, carrier–polar, polarity, graph, origin, and cofinal-shell rows, together with every mixed coordinate generated by their cumulative join. Apply Theorem III.4.58 to an infinite compatible word. If its joint target projection is zero, the cell is deleted in $\mathscr{N}_{\Sigma}$. If it is nonzero, a finite joint prefix already has a nonzero target projection. That prefix is either the next source-resolved compiler state or has already been processed. In the latter case, closure under all generated joint operations, the unnormalized identity (518), complete source ancestry, and the retained physical valuation are passed through Theorem II.2.32. A target-null owner deletes the cell, a cone identity charges it, and a separator is the next finite coordinate. Only a primal–polar fixed record satisfies precisely (A1)–(A4). This proves (521). No division by accumulated return is used.

For decreasing closed subsets of the compact spine, empty intersection implies that one finite intersection is empty. The fixed enumeration records the finitely many dual identities used before that depth. On an execution initialized by contact, all cumulative records retain polarity Hyp, so the fixed atom gives the source-profile implication (522) and no existence conclusion. In the independent fiber, a realized atom contains the ordinary same-origin path and the cofinal shell rows, hence supplies contact; conversely an independently generated singular contact supplies those rows. A nonordinary public realization value is a typed successor by totalization and cannot end the compiler. □

Definition III.4.77 (Explicit singular mechanism)**.** An *explicit singular mechanism* is a realized joint-saturated structural atom $\mathfrak{a}$ together with its same-origin path u_* in the closed reduced public state space $X_{\mathfrak{a}}$. Equivalently, it satisfies (A1)–(A4) of Definition III.4.74 and the following realization conditions:

(E1) $\mathrm{pol}_{\mathrm{org}}(\mathbf{R}) = \mathsf{Ind}$; the public orbit is generated without assuming target contact or noncontinuation, and origin-polarity preservation verifies that no contact-derived record occurs in its construction;

(E2) its orbit-origin and weak-equation realization rows are ordinary and every public-equation fidelity defect vanishes; in particular the projection of u_* is an actual same-origin path solving the original public weak equation, not an equation with an independent defect forcing;

(E3) every auxiliary constraint, carrier, chart, covariance, boundary, and support-transfer coordinate has either its ordinary public value or its first typed total-graph value together with the complete successor descendants generated from that value; no auxiliary failure is discarded or promoted to forcing in the public equation;

(E4) the induced five-channel equation on X_* is closed under the resaturation successor and the recurrent passivity row has been evaluated; every nonzero feeding or storage current carries its signed-dyadic return law, detecting target covector, source ancestry, and same-origin successor, whereas every vanishing production is recorded in the joint kernel;
(E5) the path has the inherited physical ledger and origin coordinate; and
(E6) the shell functions satisfy $\sup_t \Phi_r(u_*(t)) = 1$ for every member of a cofinal target-shell family.

Thus an ordered positive functional, invariant law, formal profile, or equality kernel is not an explicit singular mechanism. An ordinary singular path without its cumulative joint atom is likewise not an output. A graph-boundary coordinate becomes part of a mechanism only when its typed descendant closure, nonzero source projection, and unnormalized shell cocycle are retained and the ordinary same-origin weak-equation row realizes contact.

Theorem III.4.78 (Terminal target-aligned current compiler)**.** *Run Definition III.4.23 on a contact-selected active signature. Every maximal compiler execution has exactly one of the following outputs:*
(C1) *a cofinal collection of disjoint return events is charged to the public physical measure with divergent total mass;*
(C2) *the retained same-origin path lies in $\mathscr{N}_\Sigma$, so it cannot cross the selected target shells;*
(C3) *cumulative joint saturation and recursive carrier–polar exhaustion reach a primal–polar fixed prospective source atom. Its retained shell floor is a lower-bound witness for one saturated five-channel source-profile cell.*

Compactness modes, fidelity defects, equality kernels, feeding classes, storage transfers, and nonordinary graph values are internal queue coordinates. None is an additional output. Consequently the full continuum regularity track on a finite physical stratum is the target-local decision

$$\boxed{\mathit{contact} \implies \mathsf{Charge}_\infty \,\dot\cup \mathscr{N}_\Sigma \,\dot\cup \mathsf{SrcWitness}_\Sigma^{\mathsf{Hyp}}.} \tag{523}$$

The first two cells contradict contact under a finite physical ledger. The third is the continuum prospective-accountability output: contact has exposed a nonzero source cell that must be bounded independently by the saturated source profile. It is not a singular-existence conclusion. An infinite finite-prefix execution is joined and reprojected. It reaches (C3) *only after the nonzero five-channel source, all mixed descendants, and the unnormalized shell cocycle have been assembled in one atom with hypothetical origin.*

Proof. At a finite queue state, cumulative carrier duality, recursive carrier–polar exhaustion, target-current circulation duality, target-local stratification, and the ordered proof–model alternative give (Q1) or one of (Q2)–(Q4). Positive cycle price or infeasibility supplies the finite dual identity in (Q1). A finite-horizon lower bound above the remaining physical ceiling supplies the same output without a per-return price. On its complementary branch, (507) is retained before zero cycle price supplies the primal circulation processed by the remaining transitions. The local charging rule proves that only (Q1) spends physical measure. The joint-kernel descent proves that (Q2) either gives (C2) or supplies another target-aligned coordinate. Recurrent stratification and Poincare recurrence prove that (Q3) retains every positive-density current, including an exact storage transfer. Its normalized law is rejoined with the cumulative unnormalized current before target projection. Fidelity-defect support transfer proves that (Q4) retains every target-visible failed operation. These are the complete polarity and total-graph values of the least coordinate, so no finite execution has another outgoing edge.

For an infinite execution, form $\mathcal{J}_\infty$. The same-origin witnesses and $A_{\Phi,N} = N$ survive by joint construction. By Theorem III.4.58, every limiting target increment occurs on a finite joint prefix. If a least unprocessed increment exists, its prefix is another (Q2)–(Q4) transition, so the execution was not maximal. If the complete joint target projection is zero, the limit gives (C2) and contradicts its unit shell cocycle. In the remaining maximal case every increment is processed. Joint saturation now either exposes another mixed increment, again contradicting maximality, or is fixed under all generated joint operations. Apply Theorem II.2.32 to every cumulative owner. Target-nullity gives (C2), a cone identity contributes to (C1), and a separator contradicts maximality by creating another queue transition. The remaining record is primal–polar fixed and satisfies (A1)–(A4). Its contact-selected origin has polarity Hyp by Theorem II.2.41; its retained source floor gives (C3) and cannot supply (E1) of Definition III.4.77. A divergent cumulative carrier row is (C1). The fixed transition priority and disjoint target quotient make the outputs mutually exclusive. ☐

Corollary III.4.79 (Prospective terminal typing)**.** *On a finite physical stratum, a contact-seeded execution has exactly two mathematical output types:*

$$\boxed{\mathsf{Output}_\Sigma^{\mathsf{Hyp}} \in \{\mathsf{TargetUnreachable},\ \mathsf{SrcWitness}_\Sigma^{\mathsf{Hyp}}\}.} \tag{524}$$

The first value carries the local carrier or ordered identities which exclude contact; the second carries a nonzero lower-bound witness for the saturated prospective source profile. It carries no singular-existence conclusion. A successor cell, compatible cover, positive functional, invariant law, equality kernel, normalized circulation, or joint-saturation prefix is not an output.

Proof. The public physical ceiling turns (C1) of Theorem III.4.78 into a contradiction, and the unit shell identity does the same for (C2). Both therefore produce the target-unreachable value; (C3) is the hypothetical source-profile witness. All other objects are queue states by the same theorem. ☐

Corollary III.4.80 (Target-spine prospective regularity criterion)**.** *On an ordinary branch with finite public physical ledger,*

$$\boxed{\mathrm{Reach}_\Sigma^{\mathrm{phys}} > 0 \quad \Longrightarrow \quad \mathsf{T}_\Sigma^{\mathrm{phys,sat}}(B_{\mathrm{phys}}) \geq \frac{1}{5}.} \tag{525}$$

Consequently

$$\boxed{\mathsf{T}_\Sigma^{\mathrm{phys,sat}}(B_{\mathrm{phys}}) < \frac{1}{5} \quad \Longrightarrow \quad \mathrm{Reach}_\Sigma^{\mathrm{phys}} = 0.} \tag{526}$$

The profile is computed only on the projected target spine and decomposes into the exhaustive source/mode cells of Theorem I.5.12. An independently realized atom still proves singular existence, but a hypothetical atom contributes only to this profile.

Proof. If contact occurs, initialize the compiler with its shell current. The finite physical ledger excludes output (C1) of Theorem III.4.78, and shell crossing excludes (C2). Output (C3) therefore gives a joint-saturated prospective source atom with retained floor at least 1/5. By Proposition III.4.26 its cumulative record lies in the target spine, so the profile supremum is at least 1/5. This proves (525); its contrapositive is (526). ☐

Theorem III.4.81 (Quantifier-faithful continuum accountability)**.** *Five-source exhaustion and physical-budget accounting imply, for every public continuum presentation and every finite physical stratum, exactly the presentation-relative implication*

$$\mathrm{Reach}_\Sigma^{\mathrm{phys}} > 0 \quad \Longrightarrow \quad \mathsf{T}_\Sigma^{\mathrm{phys,sat}}(B_{\mathrm{phys}}) \geq \frac{1}{5}. \tag{527}$$

They do not imply $\mathsf{T}^{\mathrm{phys,sat}}_{\Sigma}(B_{\mathrm{phys}}) < 1/5$ *uniformly over public presentations. Indeed, any presentation containing an independently realized ordinary orbit which crosses the cofinal target shells has* $\mathrm{Reach}^{\mathrm{phys}}_{\Sigma} > 0$*, and therefore has profile at least (1/5) by* (527)*.*

Consequently a specialization has precisely two mathematically sound ways to finish. A generated upper profile below (1/5) proves target avoidance by Corollary III.4.80; an independently realized ordinary contacting atom proves singular reachability. Source classification alone supplies neither terminal value. This is the continuum counterpart of a saturated-accountability bound: it is universal in the admissible evolution and presentation-relative in the profile being evaluated.

Proof. The displayed implication is (525). Apply it to an independently realized ordinary contacting orbit to obtain the asserted obstruction to a uniform small-profile theorem. The two specialization conclusions are, respectively, the contrapositive (526) and the ordinary realization clause of Definition III.4.77. The contact conclusion is read from the ordinary same-origin realization row of the generated record. □

Theorem III.4.82 (No terminal zero-price model)**.** *Apply resaturation to any target-active zero-price successor. Exactly one of the following generated events occurs.*

- (Z1) *a finite dual-carrier row charges infinitely many disjoint target crossings to positive physical production or exterior input, contradicting the finite physical ledger;*
- (Z2) *a finite relation makes the branch target-null, regular, continuable, or incompatible with the public solution class; this includes every finite-coefficient dyadic ordered-coercivity row on the zero-production stratum;*
- (Z3) *a first nonordinary total-graph value, a nonzero target-local feeding class, its positive support record, or a proper kernel restriction creates the next target-active successor, to which resaturation is reapplied;*
- (Z4) *the cumulative join of the resaturation chain is fixed by every primal and carrier–polar operation, retains the unnormalized shell cocycle and a nonzero five-channel source, and its ordinary same-origin realization row produces the corresponding contact-derived joint source record with unchanged origin polarity.*

The alternatives are read in the fixed enumeration of graph coordinates and dual words: the first finite closing or successor event is selected, and (Z4) *occurs only when no finite event occurs. In particular, a positive stationary functional or target-visible equality kernel can occur only in* (Z3) *or as an intermediate object in the reconstruction of* (Z4)*; neither is terminal by itself. In the contact-derived fiber,* (Z4) *is not a singular-existence statement.*

Proof. At each stage first apply Theorem III.4.45 to the generated target activities, physical productions, and coefficient coordinates. Its finite-coefficient zero-production value is (Z2), while an unbounded coefficient or nonordinary relation is the proper successor (Z3). Then run the target-cyclic joint-kernel descent of Theorem III.4.11. Its case (K1) is (Z1), and (K2) is (Z2). A least nonzero or nonordinary coordinate in (K3) is exactly the proper successor (Z3) specified in Definition III.4.13. Because its origin and detecting covector are retained, the theorem applies again without adding specialization data. Monotonicity at finite and limit stages is Proposition III.4.14.

If no finite closing or charged event occurs, form the cumulative joint record and apply Theorems III.4.20, III.4.22 and III.4.58. Every nonzero joint feeding, fidelity, storage, support-transfer, or nonordinary coordinate occurs on a finite prefix and returns to (Z3) or to its induced recurrent-current successor. If all finite joint prefixes are target-null, their limit is target-null. Otherwise maximality forces the cumulative record to be fixed under every generated primal operation. Recursive carrier–polar exhaustion then charges or annihilates an owner, creates the next successor, or proves simultaneous primal–polar fixedness. Only the last cell continues. There the unnormalized shell identity, stationary carrier reduction, and source exhaustion give (A1)–(A4); the inherited ordinary origin and cofinal shell rows give the profile witness in (Z4). Origin polarity remains Hyp on a contact-seeded execution. The priority convention in the joint-kernel theorem gives exclusivity. □

Corollary III.4.83 (Post-resaturation residual and terminal exhaustiveness)**.** *After zero-price resaturation, the residual is target-null. The outputs are a proved regularity contradiction* (Z1)–(Z2) *or the contact-derived joint source record* (Z4)*, evaluated by its ordinary same-origin contact relation. Classification names, compactness defects, invariant laws, and equality kernels are coordinates of that record.*

III.4.4 Diagonal first-failure closure

The preceding protocol becomes a regularity method only after every routed row is followed to a mathematical conclusion. A row name records the first failed structural test; it is not a terminal output.

Definition III.4.84 (Equation-generated closure graph)**.** Fix a public PDE presentation and one persistent target-aligned source support A. Its *closure graph* $\mathfrak{G}_A$ has the following vertices: the ordinary retained cells and every first-failure cell generated in steps (I1)–(I8). There is a directed edge $v \to w$ precisely when the weak equation, a signed carrier identity, a source-preserving limit operation, or a target-shell successor sends the witness defining v to the witness defining w. The terminal vertices of the regularity branch are exactly

(T1) a target-null cell;
(T2) a regular or continuable cell;
(T3) a class-incompatible cell;
(T4) a cell whose disjoint descendants have divergent total physical price;

Every zero-price equality vertex instead enters the resaturation successor of Definition III.4.13. The only complementary conclusive output is

$$\text{a realized joint atom satisfying (A1)–(A4) and (E1)–(E6).} \tag{S}$$

The singular terminal (S) is available only in the independent origin fiber. In the hypothetical origin fiber the same structural rows, without (E1), define the prospective source-profile output

$$\text{a joint atom satisfying (A1)–(A4) with nonzero retained shell floor and origin } \mathsf{Hyp}. \tag{P}$$

The graph and every edge relation are generated from the public equation. They are not additional specialization data.

At depth k, the canonical closure procedure evaluates the first k generated graph coordinates and the first k dual-carrier words. A finite first failure creates its displayed successor. A compatible cofinal sequence retains all earlier coordinates, its origin tag, and its unnormalized shell cocycle in the cumulative joint record. It is reprojected after joint saturation. Thus no named row, compatible sequence, or ordinary path is terminal merely by being classified or realized, and no closure rank is supplied with the PDE presentation.

Theorem III.4.85 (Diagonal structural closure)**.** *For every target-aligned contact branch, the canonical finite-depth closure procedure, completed by cumulative joint saturation, has exactly one of the following terminal outcomes:*

(G1) *a finite stage reaches a terminal of type* (T1)–(T4)*;*
(G2) *the primal–polar joint fixed point reconstructs the prospective source-profile witness* (P)*.*
Compatible stages at every finite depth are an internal state: their cumulative joint limit is target-null, has a least unprocessed finite joint primal or polar increment, or is the primal–polar fixed atom in (G2) *after every increment has been source-resolved and every cumulative owner has passed carrier–polar exhaustion. Hence* (T1)–(T4) *contradict contact, while* (G2) *is a source-resolved lower-bound witness for the saturated prospective profile. Absence of singular contact follows when the computed upper profile is below that retained floor on every algebraically reachable active branch. Target-local kernel identities are generated sufficient tests, but a kernel name or positive functional never ends the procedure. The criterion requires no externally specified closure rank.*

Proof. Contact produces a persistent nonempty support A by Theorem III.2.11. On terminal-entry contact, persistence is across the cofinal terminal shell records at the same endpoint; they are one Bdry event and are never counted repeatedly as physical expenditure. Run the finite cylinder and dual-word enumeration on the active branch. A finite terminal gives (G1). Otherwise every finite depth has a compatible retained value. Form the cumulative joint record rather than declaring its diagonal limit an output. By Theorem III.4.58, every target component is the sum of its finite joint increments. Its least unprocessed increment returns to the closure graph; if the complete joint projection vanishes, the limit is target-null. After all increments are processed, the remaining fixed point retains the original contact events and unnormalized shell cocycle and satisfies (A1)–(A4). Totalization of the public realization row either creates another typed successor or retains the source floor, which is (G2). Its origin remains Hyp by Theorem II.2.41. The contradictions associated with (T1)–(T4) are respectively the target floor, terminal coverage, the public branch class, and finite physical expenditure. Finally, on an approach branch, Theorem II.2.39 identifies the only recurrent nonterminal support with a target-local equality kernel. On a terminal-entry branch, the totalized terminal-current row instead closes the cell or retains the equation-faithful prospective Bdry-source witness. It is not an existence mechanism on the contact-derived origin fiber. The approach equality-kernel resaturation is exhaustive by Theorem III.4.82. It either closes the branch or reconstructs (P). An independently generated construction may instead satisfy (S); a contact-generated construction cannot. $\square$

III.5 Target-aligned geometric normal form

Fix a presentation-generated source support A, a shell covector $\lambda \in \mathscr{C}_{\mathfrak{b}}^{\infty}$, and its minimal target-active realization $\mathsf{M}_{\Sigma,\mathfrak{b}}$. All spaces, pairings, and extrema in this and the next four sections are restricted to the reachable record cell $\mathscr{S}_{A,\Sigma}^{\mathrm{reach,sat}}(B_{\mathrm{phys}})$ and its finite same-origin prefixes. Write

$$\Lambda_{\Sigma,\mathfrak{b}} := \{\lambda \in \mathscr{C}_{\mathfrak{b}}^{\infty} : \|\lambda\|_{\mathrm{ou},*} \le 1\}$$

for the normalized retained shell-covector ball in the dual order-unit norm. Every supremum below is taken over this ball; this normalization prevents a nonzero pairing from being made arbitrarily large by rescaling its probe.

Definition III.5.1 (Geometric range–kernel decomposition)**.** Let $K \ge 0$ be a generated mobility or reversible form and write the ordinary geometric current as $J^{\mathsf{Geom}} = -K\zeta$, where ζ is the generated energy, action, or weak-equation covector. In the form completion $\mathcal{H}_K = \overline{\operatorname{ran} K^{1/2}}$, set

$$\mathcal{O}_{K,\Sigma} := \overline{\operatorname{span}}\{K^{1/2}\lambda : \lambda \in \mathscr{C}_{\mathfrak{b}}^{\infty}\}, \qquad \Pi_{K,\Sigma} : \mathcal{H}_K \to \mathcal{O}_{K,\Sigma}.$$

The ordinary target-aligned geometric projection is

$$K^{1/2}\zeta = \Pi_{K,\Sigma}K^{1/2}\zeta + (I - \Pi_{K,\Sigma})K^{1/2}\zeta, \qquad \mathrm{Type}_K(\zeta) \in \{\mathrm{ran}, \ker/\mathrm{coker}, \mathrm{gc}\}. \tag{528}$$

Here $\mathrm{Type}_K(\zeta)$ is a branch flag obtained by the ordered total-graph test: first test ordinary range membership, then the generated kernel/cokernel quotient, and finally retain the first nonordinary graph value. We write ζ_{ran}, ζ_{ker}, and ζ_{gc} for the corresponding coordinates, assigning zero to a coordinate absent on the selected graph cell. The flag is unique; it does not assert that the ordinary target projection and a kernel coordinate cannot both be detected in the full invariant tuple. The second summand in the first equality is target-orthogonal. A graph failure is retained with Geom ancestry and any ancestry of the operation whose descent failed.

Definition III.5.2 (Geometric formation invariants)**.** On a normalized target cell C, define

$$G_\Sigma^{\mathsf{Geom}}(C) := \sup_{\substack{x\in C\\ \lambda\in\Lambda_{\Sigma,\mathfrak{b}}}} |\langle K^{1/2}\lambda, \Pi_{K,\Sigma}K^{1/2}\zeta\rangle|, \tag{529}$$

$$\mathrm{Cap}_\Sigma^{\mathsf{Geom}}(C) := \inf\{\langle v, Kv\rangle : \lambda \in \Lambda_{\Sigma,\mathfrak{b}},\ \langle \lambda, Kv\rangle = 1\}, \tag{530}$$

$$\mathrm{Ker}_\Sigma^{\mathsf{Geom}}(C) := \sup_{\substack{x\in C\\ \lambda\in\Lambda_{\Sigma,\mathfrak{b}}}} |\langle \lambda, \zeta_{\mathrm{ker}}\rangle|, \qquad \mathrm{Gc}_\Sigma^{\mathsf{Geom}}(C) := \sup_{\substack{x\in C\\ \lambda\in\Lambda_{\Sigma,\mathfrak{b}}}} |\langle \lambda, \zeta_{\mathrm{gc}}\rangle|. \tag{531}$$

The capacity is $+\infty$ when its normalized constraint is infeasible. Its reciprocal is the target-conditioned geometric resistance, with the usual conventions. The geometric formation tuple is

$$\mathfrak{I}_\Sigma^{\mathsf{Geom}}(C) := (G_\Sigma^{\mathsf{Geom}}, \mathrm{Cap}_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}})(C). \tag{532}$$

Theorem III.5.3 (Exhaustion and measurement of geometric singular structure)**.** *Every target-aligned* Geom*-headed atomic component has a unique ordered totalized record in* (528)*. Its branch flag is the first active ordinary-range, target-visible kernel/cokernel, or graph-completion test; all simultaneous detector coordinates remain in the record. On every ordinary form-domain cell, the target-aligned/target-orthogonal form projection displayed there is unique. Moreover:*

- (G1) *the finite cylinder hierarchy computes nested rational enclosures for all four coordinates of* $\mathfrak{I}_\Sigma^{\mathsf{Geom}}$*; the capacity and resistance are the primal and Fenchel-dual values of the generated form program;*
- (G2) Geom *is absent from singular formation on* C *exactly when*

$$G_\Sigma^{\mathsf{Geom}} = \mathrm{Ker}_\Sigma^{\mathsf{Geom}} = \mathrm{Gc}_\Sigma^{\mathsf{Geom}} = 0;$$

- (G3) *a nonzero ordinary range value is either admitted as physical production by the carrier-domination graph or becomes a signed feeding coordinate; a nonzero kernel or graph value is the corresponding proper target-cyclic successor.*

Proof. The form completion gives the orthogonal projection in (528). Totalization of $K^{1/2}$, its range inclusion, and its quotient gives the exhaustive ordinary, kernel, and graph-boundary partition. Target incidence deletes the orthogonal summand and retains every failed deletion as a typed edge. The capacity program is the constrained quadratic-form problem; completing the square gives its resistance dual by Theorem II.5.5. Cylinder approximation and Theorem III.4.75 give the enclosures. The three displayed zero pairings annihilate every geometric source–shell path, and prospective accountability proves absence. Conversely, each nonzero coordinate is itself such a path. Carrier admission and total-graph polarity give (G3). □

III.6 Target-aligned directed-feeding normal form

Definition III.6.1 (Directed exact–cohomological–recurrent decomposition)**.** Glue all compatible directed transport, pressure, imposed drift, and nonnormal response currents with their signs. Let $\mathcal{B}_{\mathsf{Caus}} = \overline{\operatorname{ran}\delta}$ be the closed shift-coboundary space on the target quotient and let q_{Caus} be its quotient map. The canonical directed normal form is the ordered typed record

$$\mathrm{NF}_{\mathsf{Caus}}(F_{\mathsf{Caus}}) := (q_{\mathsf{Caus}}F_{\mathsf{Caus}}, K^{\mathrm{stor}}_{\mathsf{Caus}}, F^{\mathrm{rec}}_{\mathsf{Caus}}, F^{\mathrm{gc}}_{\mathsf{Caus}}, N_\Sigma). \tag{533}$$

The first coordinate is the reduced feeding-cohomology class. When it vanishes, the total coboundary graph returns an exact storage potential $K^{\mathrm{stor}}_{\mathsf{Caus}}$, or else its first graph value is placed in the fourth coordinate. The third coordinate is the positive-support first-return law of a nonzero cohomology or storage value, $F^{\mathrm{gc}}_{\mathsf{Caus}}$ is the first nonordinary gluing, support, resolvent, or return value, and $N_\Sigma \in \mathscr{N}_\Sigma$. Jordan decomposition is performed only after the signed sum and coboundary projection.

Definition III.6.2 (Directed formation invariants)**.** Let D_{phys} be the admitted nonnegative physical production on the same target cell, and let $\mathscr{R}_{\mathsf{Caus}}(C)$ be the generated retained Caus-variation space of the full generator. Define

$$H^{\mathsf{Caus}}_\Sigma(C) := \|[F_{\mathsf{Caus}}]\|_{\mathrm{red}}, \qquad S^{\mathsf{Caus}}_\Sigma(C) := \|K^{\mathrm{stor}}_{\mathsf{Caus}}\|_{\mathrm{red}}, \tag{534}$$

$$\beta^{\mathsf{Caus}}_\Sigma(C) := \inf\{a \ge 0 : (q_{\mathsf{Caus}}F_{\mathsf{Caus}})_+ \le aD_{\mathrm{phys}}\}, \qquad \delta^{\mathrm{cyc}}_\Sigma(C) := \inf_\gamma \int p_{\mathrm{phys}}\,\mathrm{d}\gamma, \tag{535}$$

$$\Psi^{\mathsf{Caus}}_\Sigma(C;\lambda) := \sup_{\substack{K\in\mathscr{R}_{\mathsf{Caus}}(C)\\ \|K\|_{\mathrm{red}}\le 1}} \|P_{\mathscr{O}}(\lambda-L)^{-1}K(\lambda-L)^{-1}P_{\mathscr{I}}\|, \quad \mathrm{Gc}^{\mathsf{Caus}}_\Sigma(C) := \|F^{\mathrm{gc}}_{\mathsf{Caus}}\|_{\mathrm{red}}. \tag{536}$$

The reduced norms are order-unit norms on the generated cylinder quotient. The directed formation tuple is

$$\mathfrak{I}^{\mathsf{Caus}}_\Sigma := (H^{\mathsf{Caus}}_\Sigma, S^{\mathsf{Caus}}_\Sigma, \beta^{\mathsf{Caus}}_\Sigma, \delta^{\mathrm{cyc}}_\Sigma, \Psi^{\mathsf{Caus}}_\Sigma, \mathrm{Gc}^{\mathsf{Caus}}_\Sigma),$$

together with the signed-dyadic support laws of these coordinates.

Theorem III.6.3 (Exhaustion and measurement of directed singular structure)**.** *Every target-aligned* Caus*-headed atomic component has a unique first active coordinate in* (533) *relative to the generated coboundary space, return graph, and target kernel. The finite evaluator computes certified enclosures for* $H^{\mathsf{Caus}}_\Sigma, S^{\mathsf{Caus}}_\Sigma, \beta^{\mathsf{Caus}}_\Sigma, \delta^{\mathrm{cyc}}_\Sigma, \Psi^{\mathsf{Caus}}_\Sigma$*, and* $\mathrm{Gc}^{\mathsf{Caus}}_\Sigma$*. Directed structure is absent from singular formation exactly when*

$$H^{\mathsf{Caus}}_\Sigma = S^{\mathsf{Caus}}_\Sigma = \mathrm{Gc}^{\mathsf{Caus}}_\Sigma = 0 \tag{537}$$

and every retained resolvent input–output matrix element vanishes. For a nonzero class, $\beta^{\mathsf{Caus}}_\Sigma < 1$ *gives strict carrier passivity, positive* $\delta^{\mathrm{cyc}}_\Sigma$ *gives infinite recurrent physical charge, and zero cycle price constructs the next target-aligned return support.*

Proof. The closed coboundary quotient, its total preimage graph, the recurrent support graph, graph boundary, and target kernel form the ordered typed partition in (533). Signed gluing precedes all these projections. The sharp radius is the order domination problem of Theorem II.2.24; cycle price is Theorem III.4.70; and resolvent variations factor through the same source by Proposition II.6.2. Their finite primal–dual programs and Theorem III.4.75 compute the stated intervals. Criterion (537), together with zero compressed response, deletes every directed incidence path. Conversely a nonzero coordinate supplies its path and signed support. The final alternatives are the passivity and circulation dualities. □

III.7 Target-aligned quotient and constraint normal form

Definition III.7.1 (Quotient horizontal–vertical normal form)**.** Let $q : X \to Q$ be a generated constraint, gauge, or coarse quotient, with pullback U, generated averaging/right inverse P, and totalized projection $\Pi_q = UP$. For a public current J, put

$$J_Q = PJU, \qquad E_q = JU - UJ_Q.$$

On the ordinary graph the target-aligned quotient identity is

$$J = UJ_QP + E_qP + J(I - \Pi_q). \tag{538}$$

The first term is horizontal descent, the second is the intertwining defect, and the third is vertical/fiber motion. If an operation in this identity is nonordinary, the ordered totalization replaces the identity by the typed record

$$\mathrm{NF}_{\mathsf{Abs}}(J; q) := (UJ_QP, E_qP, J(I - \Pi_q), J_q^{\mathrm{coker}}, J_q^{\mathrm{gc}}),$$

where J_q^{coker} is the target-visible failure of generated lifting or surjectivity, and J_q^{gc} is the first nonordinary quotient, constraint, gauge, or pressure graph value. Each term is projected to the target incidence core after its ancestry is fixed.

Definition III.7.2 (Quotient formation invariants)**.** Define

$$A_\Sigma^{\mathrm{desc}}(C) := \|\Pi_\Sigma E_qP\|_{\mathrm{red}}, \qquad A_\Sigma^{\mathrm{vert}}(C) := \|\Pi_\Sigma J(I - \Pi_q)\|_{\mathrm{red}}, \tag{539}$$

$$A_\Sigma^{\mathrm{coker}}(C) := \|\Pi_\Sigma J_q^{\mathrm{coker}}\|_{\mathrm{red}}, \qquad A_\Sigma^{\mathrm{gc}}(C) := \|\Pi_\Sigma J_q^{\mathrm{gc}}\|_{\mathrm{red}}, \tag{540}$$

$$A_\Sigma^{\mathrm{tgt}}(C) := \sup_{\lambda \in \Lambda_{\Sigma,\mathfrak{b}}} \|\lambda - \lambda \circ \Pi_q\|_{\mathrm{ou},*}, \tag{541}$$

$$\mathrm{Cap}_\Sigma^{\ker q}(C) := \inf\{\mathcal{E}(v) : v \in \ker q, \langle \lambda, v\rangle = 1\}, \quad \mathrm{Rank}_\Sigma(q; C) := \dim_{\mathrm{cyl}} \operatorname{span}\{\lambda \circ U\}_{\lambda \in \mathscr{C}^\infty}. \tag{542}$$

At finite depth, $\dim_{\mathrm{cyl}}$ is the rank of the rational pairing matrix; the limiting rank is its monotone generated value. The quotient formation tuple is

$$\mathfrak{I}_\Sigma^{\mathsf{Abs}} := (A_\Sigma^{\mathrm{desc}}, A_\Sigma^{\mathrm{vert}}, A_\Sigma^{\mathrm{coker}}, A_\Sigma^{\mathrm{gc}}, A_\Sigma^{\mathrm{tgt}}, \mathrm{Cap}_\Sigma^{\ker q}, \mathrm{Rank}_\Sigma(q)).$$

Theorem III.7.3 (Exhaustion and measurement of quotient singular structure)**.** *Every target-aligned* Abs*-headed atomic component has exactly one first active value in the ordered typed record extending* (538)*. The finite incidence and moment hierarchies compute the four reduced current-defect norms, the target-descent defect, kernel capacity, and target rank. Quotient structure is absent from singular formation exactly when*

$$A_\Sigma^{\mathrm{desc}} = A_\Sigma^{\mathrm{vert}} = A_\Sigma^{\mathrm{coker}} = A_\Sigma^{\mathrm{gc}} = A_\Sigma^{\mathrm{tgt}} = 0 \tag{543}$$

—equivalently, the current defects vanish and every retained target covector descends through q. On this cell the horizontal term transfers the problem to Q without creating a source. Every nonzero invariant gives the corresponding intertwining, vertical, cokernel, or graph successor with its original source ancestry.

Proof. Insert $I = \Pi_q + (I - \Pi_q)$ on the right of J, then add and subtract UJ_QP. Totalization of U, P, q, and their range graphs supplies the cokernel and graph terms, proving exhaustiveness. Rational pairing matrices compute finite ranks; the remaining quantities are norms or constrained energy minima in the Archimedean cylinder algebra and hence have the stated enclosures. Under (543), all vertical, target-descent, and failed descent paths are target-null, so only exact horizontal descent remains. Conversely each nonzero block is detected by its defining target covector and therefore supplies a proper incidence edge. □

III.8 Target-aligned lift, scale, and compactness normal form

Definition III.8.1 (Continuum Lift normal form)**.** Fix the canonical evaluation compactification and a generated lift or chart (Ω, π, Λ), with fiber averaging P_κ and $\Pi_\pi = UP_\kappa$. Let J^Ω denote the total current in the lifted presentation. On the ordinary lift graph, the first three coordinates satisfy the exact identity

$$J^\Omega = UJ_X P_\kappa + E_\pi P_\kappa + J^\Omega(I - \Pi_\pi). \tag{544}$$

They are base transport, lift-intertwining defect, and within-fiber motion. The ordinary base term retains the head of J_X; the second and third terms are Lift-headed and retain the base ancestry. The complete ordered typed record is

$$\mathrm{NF}_{\mathsf{Lift}}(J^\Omega; \pi) := (UJ_X P_\kappa, E_\pi P_\kappa, J^\Omega(I - \Pi_\pi), J_{\mathrm{conc}}, J_{\mathrm{osc}}, J^{\mathrm{coh}}_{\mathrm{sc}}, J^{\mathrm{gc}}_{\mathrm{chart}}).$$

Its next two coordinates are the concentration and oscillation coordinates of the fixed graph-complete compactification. The sixth is the nonexact scale/modulation cohomology after covariant carrier conjugation, and the last is the first nonordinary base-descent, chart, scaling, product, localization, or realization value. Tail and terminal descendants are transferred to the Bdry normal form after retaining their Lift ancestry.

Definition III.8.2 (Lift and compactness formation invariants)**.** Define

$$L^{\mathrm{int}}_\Sigma(C) := \|\Pi_\Sigma E_\pi P_\kappa\|_{\mathrm{red}}, \qquad L^{\mathrm{fib}}_\Sigma(C) := \|\Pi_\Sigma J^\Omega(I - \Pi_\pi)\|_{\mathrm{red}}, \tag{545}$$

$$L^{\mathrm{conc}}_\Sigma(C) := \|\Pi_\Sigma J_{\mathrm{conc}}\|_{\mathrm{var}}, \qquad L^{\mathrm{osc}}_\Sigma(C) := \|\Pi_\Sigma J_{\mathrm{osc}}\|_{\mathrm{var}}, \tag{546}$$

$$L^{\mathrm{sc}}_\Sigma(C) := \|[J_{\mathrm{sc}}]\|_{\mathrm{red}}, \qquad L^{\mathrm{gc}}_\Sigma(C) := \|\Pi_\Sigma J^{\mathrm{gc}}_{\mathrm{chart}}\|_{\mathrm{red}}. \tag{547}$$

The variation norms are computed on the countable cylinder algebra of the fixed compactification. The scale valuation κ_P and record-scale velocity $a = \dot r / r$ are retained as coordinates of L^{sc}_Σ. The lift formation tuple is

$$\mathfrak{I}^{\mathsf{Lift}}_\Sigma := (L^{\mathrm{int}}_\Sigma, L^{\mathrm{fib}}_\Sigma, L^{\mathrm{conc}}_\Sigma, L^{\mathrm{osc}}_\Sigma, L^{\mathrm{sc}}_\Sigma, L^{\mathrm{gc}}_\Sigma).$$

Theorem III.8.3 (Exhaustion and measurement of compactness structure)**.** *Relative to the fixed continuum presentation, every target-visible* Lift*-headed weak limit, concentration, oscillation, defect measure, chart degeneration, blow-up profile, and scale escape has a unique first active coordinate in the ordered typed record extending* (544)*. The six invariants in Definition III.8.2 are computed by monotone cylinder-variation and ordered moment enclosures. Lift/compactness structure is absent from singular formation exactly when all six vanish. On this cell the ordinary base term is transferred to the normal form of its inherited head source. A failed base descent is part of* L_Sigma^{gc}*. Otherwise the least nonzero invariant explicitly identifies the interface, fiber, concentration, oscillation, scale, or graph mechanism used by the target-active cascade.*

Proof. The first three terms are the exact lift identity obtained by inserting UP_κ. Sourcewise graph completion splits failure of strong compactness into the ordinary weak value, its positive concentration coordinate, its signed oscillation/polarization coordinate, and the first nonordinary operation value. These coordinates are unique relative to the fixed canonical evaluation compactification. Covariant production valuation then separates exact scale coboundaries from the scale cohomology class. This proves exhaustiveness. Total variation on a countable cylinder algebra is the supremum over finite rational partitions, and the remaining norms are ordered cylinder programs; hence the finite hierarchy computes all six. Their simultaneous vanishing deletes every nonbase lift incidence path, and ordinary base transport is evaluated by its inherited source packet. Failed base descent is a nonordinary operation value and therefore contributes to L_Sigma^{gc}. Every nonzero value carries its detecting covector and produces the stated successor. □

III.9 Target-aligned boundary and terminal normal form

Definition III.9.1 (Oriented boundary normal form)**.** Let $\mathscr{K}$ be a finite target-retaining carrier complex. Sum its cellwise Stieltjes identities with their orientations. The boundary current has the canonical decomposition

$$J^{\mathsf{Bdry}} = \partial_{\mathrm{int}}\mathscr{K} + J_{\mathrm{ext}} + J_{\mathrm{tail}} + J_{\mathrm{term}} + J_{\mathrm{tr}}^{\mathrm{gc}}. \tag{548}$$

Paired internal faces constitute $\partial_{\mathrm{int}}\mathscr{K}$ and cancel identically. The remaining terms are exterior physical flux, compactified tail flux, continuation/terminal-shell flux, and the first nonordinary trace, exhaustion, overlap, boundary-condition, or terminal-realization value. Faces at intersections are assigned by the fixed half-open priority

$$\text{terminal} \succ \text{tail} \succ \text{exterior};$$

equivalently, use the corresponding inclusion–exclusion strata. Hence a corner is counted once while retaining every incident ancestry label. Killing or support loss is the zero terminal-value specialization of J_{term}.

Definition III.9.2 (Boundary formation invariants)**.** Define

$$B_\Sigma^{\mathrm{ext}}(C) := \|\Pi_\Sigma J_{\mathrm{ext}}\|_{\mathrm{var}}, \qquad B_\Sigma^{\mathrm{tail}}(C) := \|\Pi_\Sigma J_{\mathrm{tail}}\|_{\mathrm{var}}, \tag{549}$$

$$B_\Sigma^{\mathrm{term}}(C) := \|\Pi_\Sigma J_{\mathrm{term}}\|_{\mathrm{var}}, \qquad B_\Sigma^{\mathrm{gc}}(C) := \|\Pi_\Sigma J_{\mathrm{tr}}^{\mathrm{gc}}\|_{\mathrm{red}}, \tag{550}$$

$$\mathrm{TailCap}_\Sigma(C) := \inf\{\mathcal{E}(v) : v|_{\partial_\infty} = 1 \text{ on a retained tail shell}\}. \tag{551}$$

All fluxes are measured after internal-face cancellation. A positive exterior flux is physical expenditure only when the public carrier identity dominates it. The boundary formation tuple is

$$\mathfrak{I}_\Sigma^{\mathsf{Bdry}} := (B_\Sigma^{\mathrm{ext}}, B_\Sigma^{\mathrm{tail}}, B_\Sigma^{\mathrm{term}}, B_\Sigma^{\mathrm{gc}}, \mathrm{TailCap}_\Sigma).$$

Theorem III.9.3 (Exhaustion and measurement of boundary singular structure)**.** *Every target-aligned* Bdry*-headed atomic component belongs uniquely to the exterior, tail, terminal, or graph block of* (548) *after internal cancellation. The finite exhaustion, trace, carrier, and capacity programs compute certified enclosures for the five invariants in Definition III.9.2. Boundary structure is absent from singular formation exactly when*

$$B_\Sigma^{\mathrm{ext}} = B_\Sigma^{\mathrm{tail}} = B_\Sigma^{\mathrm{term}} = B_\Sigma^{\mathrm{gc}} = 0. \tag{552}$$

A nonzero exterior invariant is charged or returned as a Bdry *successor; a nonzero tail invariant generates its exhaustion/compactification successor; and a nonzero terminal invariant realizes target contact only through the totalized terminal-flux graph.*

Proof. The cellular boundary identity cancels every oppositely oriented common face. The fixed half-open priority partitions the remaining faces into disjoint exterior, compactification-tail, and terminal strata; the inclusion–exclusion identity preserves their signed sum. Totalization supplies the graph block. Jordan decomposition on these disjoint supports gives uniqueness. Variation is computed by finite rational partitions, capacity by its Dirichlet primal–dual program, and graph values by the ordered hierarchy. Criterion (552) annihilates every boundary incidence path. The three nonzero routes follow from carrier admission, exhaustion totalization, and the killed-flux realization law. □

III.10 The continuum singularity-formation profile

Definition III.10.1 (The thirty-one target-aligned formation cells)**.** For each nonempty support $A \subseteq \mathfrak{C}$, let $\mathscr{C}_A(E) := \mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(E)$ be the source-preserving closure of the normalized same-origin pre-contact records generated by the public presentation with exact support A. Ordinary path values, first nonordinary realization values, and their origin tags remain distinct coordinates of this reachable cell. Its atomic invariant vector is

$$\mathbf{I}^{\Sigma}_{A} := \bigoplus_{c\in A} \mathfrak{I}^{c}_{\Sigma}\big|_{\mathscr{C}_A(E)}, \tag{553}$$

where the five summands are the invariant tuples in Definitions III.5.2 to III.9.2. Adjoin the aggregate signed carrier class, recurrent cycle price, same-origin realization graph, and cofinal shell coordinate. The resulting invariant image is the *formation profile*

$$\mathfrak{F}^{\Sigma}_{A}(E) := \left\{(\mathbf{I}^{\Sigma}_{A}, [H]_A, \delta^{\mathrm{cyc}}_{A}, \mathbf{R}, \mathrm{Shell}_{\Sigma}) : C \in \mathscr{C}_A(E)\right\}. \tag{554}$$

Scalar valuation coordinates take values in their order compactifications $[0,\infty]$; graph and measure coordinates use the compact topologies already fixed by the minimal realization. Thus (554) is evaluated on the specified source-preserving compact product, not in an ambient norm topology. A coordinate is a *detector* when its vanishing is part of the corresponding absence criterion. Thus capacity, resistance, rank, domination radius, and cycle price are valuations of a detected mechanism, not detectors by themselves. Let $\mathbf{D}^{\Sigma}_{A}$ denote the subvector consisting of

$$\begin{aligned}
&(G^{\mathsf{Geom}}, \mathrm{Ker}^{\mathsf{Geom}}, \mathrm{Gc}^{\mathsf{Geom}}),\\
&(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}),\\
&(A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}),\\
&(L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}),\\
&(B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}})
\end{aligned}$$

for the sources belonging to A. A profile point is *realized* when its origin and total-graph rows diagonally reconstruct a public solution path. It is *formation active* when its normalized shell coordinate is nonzero and at least one coordinate of $\mathbf{D}^{\Sigma}_{A}$ or the aggregate feeding/storage class is nonzero.

Theorem III.10.2 (Thirty-one-cell target-aligned reachability ledger)**.** *For every nonempty $A \subseteq \mathfrak{C}$, the cell $\mathscr{C}_A(E)$ is evaluated by the detector subvector $\mathbf{D}^{\Sigma}_{A}$, its complete joint conditional increments, and its physical valuation. Exactly one of the following values holds:*

(L1) *every target detector vanishes on $\mathscr{C}_A(E)$, and the support A has no target-aligned reachable occurrence;*

(L2) *some target detector is positive, and there is a finite same-origin pre-contact record R_A generated by the public presentation whose complete joint frontier has that positive coordinate and source support A.*

For (L2)*, the physical carrier of R_A is evaluated by its finite occurrence atom. Positive mass is charged, while zero mass enters the equality record generated by the same source word. The statement holds separately for all $2^5 - 1 = 31$ supports; it is not an aggregate assertion about the five isolated channels.*

Proof. Each atomic coordinate is generated by exactly one of the five normal forms; the complete source word is retained under joining, localization, quotient, lift, boundary passage, and source-preserving closure. The mixed coordinates are formed before projection, so composition cannot be discarded as five separate null tests. Apply the density equivalence (560)– (561) to every detector coordinate of the fixed support A. The positive row returns R_A with its least contextual occurrence. The physical alternative is Theorem III.10.44. Since every nonempty subset of the five-source alphabet occurs exactly once in this construction, the thirty-one statements are exhaustive and disjoint by support. □

Theorem III.10.3 (Cellwise public-presentation evaluation of all thirty-one supports)**.** *Write the five-source alphabet in its fixed order as* $\mathsf{s} = (\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry})$*, and, for each nonzero bit vector* $\varepsilon \in \{0,1\}^5 \setminus \{0\}$*, put*

$$A_\varepsilon := \{\mathsf{s}_i : \varepsilon_i = 1\}, \qquad \mathsf{Cell}_\varepsilon := \bigcap_{\varepsilon_i=1} \mathfrak{I}_\Sigma^{\mathsf{s}_i} \cap \bigcap_{\varepsilon_i=0} \ker \mathfrak{I}_\Sigma^{\mathsf{s}_i}.$$

For each of the thirty-one values of ε*, the public presentation determines the following finite-prefix calculation and no additional analytic premise:*

(E1) *form the complete joint pre-contact frontier before applying the five source projections, retain precisely the source word* A_ε*, and evaluate every detector in* $\mathbf{D}^\Sigma_{A_\varepsilon}$ *on its finite stopped records;*

(E2) *take the monotone source-preserving closure of those finite values; a zero value deletes the corresponding target-incidence edge, whereas a positive value returns its least contextual finite occurrence together with the public equation row, target covector, joint frontier, source word, and unnormalised shell cocycle;*

(E3) *evaluate the unique physical carrier coordinate of that occurrence. Positive carrier mass is charged to its source word. Zero carrier mass is evaluated by the equality row of the* same *joint record, whose target-null value deletes the edge and whose nonzero value is reinserted at its least unprocessed conditional increment.*

Consequently every cell has exactly one structural verdict

$$\mathrm{CellEval}_\varepsilon \in \{\mathrm{Absent}(A_\varepsilon), \quad \mathrm{Present}(R_\varepsilon, \lambda_\varepsilon, Q_\varepsilon, \mathrm{Anc}(R_\varepsilon), p_{\mathrm{phys}}(R_\varepsilon))\}\,;$$

the second value is an actual finite same-origin target-aligned occurrence generated by the public presentation. Thus the displayed family is an explicit source-by-source evaluation of

$$\{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\} \setminus \{\varnothing\} = \{A_\varepsilon : 0 \neq \varepsilon \in \{0,1\}^5\},$$

not an aggregate test of five isolated channels.

Proof. The free five-constructor grammar assigns every finite stopped term one and only one source word. Joining is performed before projection, so a target coordinate created only by composition is a coordinate of the joint frontier and is retained in the unique A_ε containing its complete ancestry. The finite-record zero/positive equivalence (560)– (561) proves the dichotomy in (E1)–(E2) for that fixed word. Source-preserving completion cannot add a sixth owner, while a target-visible limit coordinate has a finite detecting prefix and hence re-enters the same word. The physical alternative in (E3) is exactly Theorem III.10.44; its re-entry clause is a conditional-joint increment rather than a new terminal. This proves the verdict for the chosen ε. The nonzero bit vectors partition the nonempty source words, giving all thirty-one cases. □

Theorem III.10.4 (Explicit theorem family for the thirty-one source supports)**.** *For every support in the following table, let* T_A *denote the assertion that the target-aligned contribution with* exact *source word* A *is evaluated by its complete joint detector, finite stopped same-origin records, and the single physical carrier.*

$\lvert A\rvert$	*theorem instances* T_A
1	$G,\ C,\ A,\ L,\ B$
2	$GC,\ GA,\ GL,\ GB,\ CA,\ CL,\ CB,\ AL,\ AB,\ LB$
3	$GCA,\ GCL,\ GCB,\ GAL,\ GAB,\ GLB,\ CAL,\ CAB,\ CLB,\ ALB$
4	$GCAL,\ GCAB,\ GCLB,\ GALB,\ CALB$
5	$GCALB$

Here a letter abbreviates respectively Geom, Caus, Abs, Lift, Bdry*, and concatenation denotes set union, not ordered multiplication. Every one of these thirty-one theorem instances has the exact conclusion*

$$\mathsf{T}_A: \qquad \begin{cases} \mathbf{D}_A^{\Sigma} = 0 \iff \text{no target-aligned reachable occurrence has exact support } A, \\ \mathbf{D}_A^{\Sigma} \neq 0 \iff \text{there is a least finite same-origin public-presentation occurrence } R_A \\ \qquad\qquad \text{with exact support } A \text{ and nonzero target pairing.} \end{cases} \tag{555}$$

The occurrence R_A *carries its complete joint frontier and source word. Its carrier valuation is either positive and charged, or zero and processed by the equality rule on that same joint record; no other outcome is admitted.*

Proof. The five primitive detector packets are generated from the public weak and total graphs. For a fixed row A in the table, form their joint record, then retain precisely the terms whose complete provenance word is A. This is a finite typed construction and is therefore a member of $\mathsf{Proof}_\Sigma(\mathfrak{P})$. The zero/positive equivalence for the reachable source closure proves the two lines of (555); importantly, it is applied to the joint record and not to the five packets separately. The physical alternative is Theorem III.10.44. This proves T_A for the selected row. The displayed table lists every nonempty subset of a five-element alphabet exactly once, so the same finite derivation proves all thirty-one instances and leaves no unlisted source word. □

Definition III.10.5 (Quantitative local detector metric)**.** For a finite public tuple $Z = (z_1, \dots, z_m)$ set

$$\|Z\|_{\det} := \sum_{i=1}^{m} 2^{-i} \min\{1, d_i(z_i, 0)\},$$

where d_i is the bounded evaluation metric supplied by the public presentation. The metric is used only on finite tuples and has zero exactly when each displayed public coordinate is zero.

Theorem III.10.6 (Local quantitative check: Geom)**.** *For every finite same-origin stopped record* $R \in \mathfrak{D}^{\mathrm{pre}}_{A_g,\Sigma,n}(E)$ *set*

$$\delta_g(R) := \left\| \left(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}} \right) \right\|_{\det}, \qquad a_g(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_g}(R) \right\|, \qquad p_g(R) := p_{\mathrm{phys}}(R).$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ *of pairwise disjoint same-origin stopped windows with exact word* A_g*, define the unnormalised local aggregates*

$$\mathsf{A}_g(\mathcal{F}) := \sum_{j=1}^{N} a_g(R_j), \qquad \mathsf{D}_g(\mathcal{F}) := \sum_{j=1}^{N} \delta_g(R_j), \qquad \mathsf{P}_g(\mathcal{F}) := \sum_{j=1}^{N} p_g(R_j).$$

Then the following check is exact:

(LG1) $a_g(R) = 0$ *exactly when this record contributes no target progress through exact support* A_g*. If* $a_g(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LG2) $\mathsf{A}_g(\mathcal{F})$*,* $\mathsf{D}_g(\mathcal{F})$*, and* $\mathsf{P}_g(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LG3) *If* $p_g(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_g(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_g*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom are respectively Proposition III.11.4. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_g. Restricting that joint record to exact word A_g gives precisely the displayed detector δ_g and target amplitude a_g; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_g. □

Theorem III.10.7 (Local quantitative check: Caus)**.** *For every finite same-origin stopped record* $R \in \mathfrak{D}^{\mathrm{pre}}_{A_c,\Sigma,n}(E)$ *set*

$$\delta_c(R) := \left\| \left(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}} \right) \right\|_{\det}, \qquad a_c(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_c}(R) \right\|, \qquad p_c(R) := p_{\mathrm{phys}}(R).$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ *of pairwise disjoint same-origin stopped windows with exact word* A_c*, define the unnormalised local aggregates*

$$\mathsf{A}_c(\mathcal{F}) := \sum_{j=1}^{N} a_c(R_j), \qquad \mathsf{D}_c(\mathcal{F}) := \sum_{j=1}^{N} \delta_c(R_j), \qquad \mathsf{P}_c(\mathcal{F}) := \sum_{j=1}^{N} p_c(R_j).$$

Then the following check is exact:

(LC1) $a_c(R) = 0$ *exactly when this record contributes no target progress through exact support* A_c*. If* $a_c(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LC2) $\mathsf{A}_c(\mathcal{F})$*,* $\mathsf{D}_c(\mathcal{F})$*, and* $\mathsf{P}_c(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LC3) *If* $p_c(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_c(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_c*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus are respectively Proposition III.11.5. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_c. Restricting that joint record to exact word A_c gives precisely the displayed detector δ_c and target amplitude a_c; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_c. ☐

Theorem III.10.8 (Local quantitative check: $\mathsf{GeomCaus}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gc},\Sigma,n}(E)$ set*

$$\delta_{gc}(R) := \left\| \left(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}} \right) \right\|_{\det}, \qquad a_{gc}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{gc}}(R) \right\|, \qquad p_{gc}(R) :=$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gc}, define the unnormalised local aggregates

$$\mathsf{A}_{gc}(\mathcal{F}) := \sum_{j=1}^{N} a_{gc}(R_j), \qquad \mathsf{D}_{gc}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gc}(R_j), \qquad \mathsf{P}_{gc}(\mathcal{F}) := \sum_{j=1}^{N} p_{gc}(R_j).$$

Then the following check is exact:

(LGC1) $a_{gc}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{gc}. If $a_{gc}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGC2) $\mathsf{A}_{gc}(\mathcal{F})$, $\mathsf{D}_{gc}(\mathcal{F})$, *and* $\mathsf{P}_{gc}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGC3) *If $p_{gc}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gc}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gc}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus are respectively Proposition III.11.4, Proposition III.11.5. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gc}. Restricting that joint record to exact word A_{gc} gives precisely the displayed detector δ_{gc} and target amplitude a_{gc}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gc}. ☐

Theorem III.10.9 (Local quantitative check: Abs)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_a,\Sigma,n}(E)$ set*

$$\delta_a(R) := \left\| \left(A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}} \right) \right\|_{\det}, \qquad a_a(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_a}(R) \right\|, \qquad p_a(R) := p_{\mathrm{phys}}(R).$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_a, define the unnormalised local aggregates

$$\mathsf{A}_a(\mathcal{F}) := \sum_{j=1}^{N} a_a(R_j), \qquad \mathsf{D}_a(\mathcal{F}) := \sum_{j=1}^{N} \delta_a(R_j), \qquad \mathsf{P}_a(\mathcal{F}) := \sum_{j=1}^{N} p_a(R_j).$$

Then the following check is exact:

(LA1) $a_a(R) = 0$ *exactly when this record contributes no target progress through exact support A_a. If $a_a(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LA2) $\mathsf{A}_a(\mathcal{F})$, $\mathsf{D}_a(\mathcal{F})$, *and* $\mathsf{P}_a(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LA3) *If $p_a(R) > 0$, the occurrence is charged to the single physical carrier. If $p_a(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_a, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Abs are respectively Proposition III.11.6. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_a. Restricting that joint record to exact word A_a gives precisely the displayed detector δ_a and target amplitude a_a; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_a. □

Theorem III.10.10 (Local quantitative check: $\mathsf{GeomAbs}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{ga},\Sigma,n}(E)$ set*

$$\delta_{ga}(R) := \left\| \left(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}} \right) \right\|_{\det}, \qquad a_{ga}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{ga}}(R) \right\|, \qquad p_{ga}(R) :=$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{ga}, define the unnormalised local aggregates

$$\mathsf{A}_{ga}(\mathcal{F}) := \sum_{j=1}^{N} a_{ga}(R_j), \qquad \mathsf{D}_{ga}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{ga}(R_j), \qquad \mathsf{P}_{ga}(\mathcal{F}) := \sum_{j=1}^{N} p_{ga}(R_j).$$

Then the following check is exact:

(LGA1) $a_{ga}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{ga}. If $a_{ga}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGA2) $\mathsf{A}_{ga}(\mathcal{F})$, $\mathsf{D}_{ga}(\mathcal{F})$, *and* $\mathsf{P}_{ga}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGA3) *If $p_{ga}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{ga}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{ga}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Abs are respectively Proposition III.11.4, Proposition III.11.6. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{ga}. Restricting that joint record to exact word A_{ga} gives precisely the displayed detector δ_{ga} and target amplitude a_{ga}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{ga}. ☐

Theorem III.10.11 (Local quantitative check: $\mathsf{CausAbs}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{ca},\Sigma,n}(E)$ set*

$$\delta_{ca}(R) := \left\| \left(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}} \right) \right\|_{\mathrm{det}}, \qquad a_{ca}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{ca}}(R) \right\|, \qquad p_{ca}(R)$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{ca}, define the unnormalised local aggregates

$$\mathsf{A}_{ca}(\mathcal{F}) := \sum_{j=1}^{N} a_{ca}(R_j), \qquad \mathsf{D}_{ca}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{ca}(R_j), \qquad \mathsf{P}_{ca}(\mathcal{F}) := \sum_{j=1}^{N} p_{ca}(R_j).$$

Then the following check is exact:

(LCA1) *$a_{ca}(R) = 0$ exactly when this record contributes no target progress through exact support A_{ca}. If $a_{ca}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCA2) *$\mathsf{A}_{ca}(\mathcal{F})$, $\mathsf{D}_{ca}(\mathcal{F})$, and $\mathsf{P}_{ca}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LCA3) *If $p_{ca}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{ca}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{ca}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Abs are respectively Proposition III.11.5, Proposition III.11.6. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{ca}. Restricting that joint record to exact word A_{ca} gives precisely the displayed detector δ_{ca} and target amplitude a_{ca}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{ca}. ☐

Theorem III.10.12 (Local quantitative check: $\mathsf{GeomCausAbs}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gca},\Sigma,n}(E)$ set*

$$\delta_{gca}(R) := \left\| \left(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}} \right) \right\|_{\det}, \qquad a_{gca}(R) :$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gca}, define the unnormalised local aggregates

$$\mathsf{A}_{gca}(\mathcal{F}) := \sum_{j=1}^{N} a_{gca}(R_j), \qquad \mathsf{D}_{gca}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gca}(R_j), \qquad \mathsf{P}_{gca}(\mathcal{F}) := \sum_{j=1}^{N} p_{gca}(R_j).$$

Then the following check is exact:

(LGCA1) $a_{gca}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{gca}. If $a_{gca}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCA2) $\mathsf{A}_{gca}(\mathcal{F})$, $\mathsf{D}_{gca}(\mathcal{F})$, *and* $\mathsf{P}_{gca}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGCA3) *If $p_{gca}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gca}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gca}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Abs are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.6. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gca}. Restricting that joint record to exact word A_{gca} gives precisely the displayed detector δ_{gca} and target amplitude a_{gca}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gca}. □

Theorem III.10.13 (Local quantitative check: Lift)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_l,\Sigma,n}(E)$ set*

$$\delta_l(R) := \left\| \left(L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \right) \right\|_{\det}, \qquad a_l(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_l}(R) \right\|, \qquad p_l(R) := p_{\mathrm{phys}}(R).$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_l, define the unnormalised local aggregates

$$\mathsf{A}_l(\mathcal{F}) := \sum_{j=1}^{N} a_l(R_j), \qquad \mathsf{D}_l(\mathcal{F}) := \sum_{j=1}^{N} \delta_l(R_j), \qquad \mathsf{P}_l(\mathcal{F}) := \sum_{j=1}^{N} p_l(R_j).$$

Then the following check is exact:

(LL1) *$a_l(R) = 0$ exactly when this record contributes no target progress through exact support A_l. If $a_l(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LL2) *$\mathsf{A}_l(\mathcal{F})$, $\mathsf{D}_l(\mathcal{F})$, and $\mathsf{P}_l(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LL3) *If $p_l(R) > 0$, the occurrence is charged to the single physical carrier. If $p_l(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_l, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Lift are respectively Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_l. Restricting that joint record to exact word A_l gives precisely the displayed detector δ_l and target amplitude a_l; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_l. □

Theorem III.10.14 (Local quantitative check: $\mathsf{GeomLift}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gl},\Sigma,n}(E)$ set*

$$\delta_{gl}(R) := \left\| \left(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{gl}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{gl}}(R) \right\|, \qquad p_{gl}(R) :=$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gl}, define the unnormalised local aggregates

$$\mathsf{A}_{gl}(\mathcal{F}) := \sum_{j=1}^{N} a_{gl}(R_j), \qquad \mathsf{D}_{gl}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gl}(R_j), \qquad \mathsf{P}_{gl}(\mathcal{F}) := \sum_{j=1}^{N} p_{gl}(R_j).$$

Then the following check is exact:

(LGL1) *$a_{gl}(R) = 0$ exactly when this record contributes no target progress through exact support A_{gl}. If $a_{gl}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGL2) *$\mathsf{A}_{gl}(\mathcal{F})$, $\mathsf{D}_{gl}(\mathcal{F})$, and $\mathsf{P}_{gl}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGL3) *If $p_{gl}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gl}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gl}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Lift are respectively Proposition III.11.4, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gl}. Restricting that joint record to exact word A_{gl} gives precisely the displayed detector δ_{gl} and target amplitude a_{gl}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gl}. □

Theorem III.10.15 (Local quantitative check: $\mathsf{CausLift}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{cl},\Sigma,n}(E)$ set*

$$\delta_{cl}(R) := \left\| \left(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{cl}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{cl}}(R) \right\|, \qquad p_{cl}(R)$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{cl}, define the unnormalised local aggregates

$$\mathsf{A}_{cl}(\mathcal{F}) := \sum_{j=1}^{N} a_{cl}(R_j), \qquad \mathsf{D}_{cl}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{cl}(R_j), \qquad \mathsf{P}_{cl}(\mathcal{F}) := \sum_{j=1}^{N} p_{cl}(R_j).$$

Then the following check is exact:

(LCL1) $a_{cl}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{cl}. If $a_{cl}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCL2) $\mathsf{A}_{cl}(\mathcal{F})$, $\mathsf{D}_{cl}(\mathcal{F})$, *and* $\mathsf{P}_{cl}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LCL3) *If $p_{cl}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{cl}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{cl}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Lift are respectively Proposition III.11.5, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{cl}. Restricting that joint record to exact word A_{cl} gives precisely the displayed detector δ_{cl} and target amplitude a_{cl}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{cl}. □

Theorem III.10.16 (Local quantitative check: $\mathsf{GeomCausLift}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gcl},\Sigma,n}(E)$ set*

$$\delta_{gcl}(R) := \left\| \left(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{gcl}(R) :=$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gcl}, define the unnormalised local aggregates

$$\mathsf{A}_{gcl}(\mathcal{F}) := \sum_{j=1}^{N} a_{gcl}(R_j), \qquad \mathsf{D}_{gcl}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gcl}(R_j), \qquad \mathsf{P}_{gcl}(\mathcal{F}) := \sum_{j=1}^{N} p_{gcl}(R_j).$$

Then the following check is exact:

(LGCL1) $a_{gcl}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{gcl}. *If* $a_{gcl}(R) > 0$, *the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCL2) $\mathsf{A}_{gcl}(\mathcal{F})$, $\mathsf{D}_{gcl}(\mathcal{F})$, *and* $\mathsf{P}_{gcl}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LGCL3) *If* $p_{gcl}(R) > 0$, *the occurrence is charged to the single physical carrier. If* $p_{gcl}(R) = 0$, *the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{gcl}, *while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Lift are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gcl}. Restricting that joint record to exact word A_{gcl} gives precisely the displayed detector δ_{gcl} and target amplitude a_{gcl}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gcl}. □

Theorem III.10.17 (Local quantitative check: $\mathsf{AbsLift}$)**.** *For every finite same-origin stopped record* $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{al},\Sigma,n}(E)$ *set*

$$\delta_{al}(R) := \left\| \left(A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{al}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{al}}(R) \right\|, \qquad p_{al}($$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ *of pairwise disjoint same-origin stopped windows with exact word* A_{al}, *define the unnormalised local aggregates*

$$\mathsf{A}_{al}(\mathcal{F}) := \sum_{j=1}^{N} a_{al}(R_j), \qquad \mathsf{D}_{al}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{al}(R_j), \qquad \mathsf{P}_{al}(\mathcal{F}) := \sum_{j=1}^{N} p_{al}(R_j).$$

Then the following check is exact:

(LAL1) $a_{al}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{al}. *If* $a_{al}(R) > 0$, *the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LAL2) $\mathsf{A}_{al}(\mathcal{F})$, $\mathsf{D}_{al}(\mathcal{F})$, *and* $\mathsf{P}_{al}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LAL3) *If* $p_{al}(R) > 0$, *the occurrence is charged to the single physical carrier. If* $p_{al}(R) = 0$, *the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{al}, *while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Abs, Lift are respectively Proposition III.11.6, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{al}. Restricting that joint record to exact word A_{al} gives precisely the displayed detector δ_{al} and target amplitude a_{al}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{al}. □

Theorem III.10.18 (Local quantitative check: $\mathsf{GeomAbsLift}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gal},\Sigma,n}(E)$ set*

$$\delta_{gal}(R) := \Big\| \Big(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \Big) \Big\|_{\mathrm{det}}, \qquad a_{gal}(R)$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gal}, define the unnormalised local aggregates

$$\mathsf{A}_{gal}(\mathcal{F}) := \sum_{j=1}^{N} a_{gal}(R_j), \qquad \mathsf{D}_{gal}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gal}(R_j), \qquad \mathsf{P}_{gal}(\mathcal{F}) := \sum_{j=1}^{N} p_{gal}(R_j).$$

Then the following check is exact:

(LGAL1) *$a_{gal}(R) = 0$ exactly when this record contributes no target progress through exact support A_{gal}. If $a_{gal}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGAL2) *$\mathsf{A}_{gal}(\mathcal{F})$, $\mathsf{D}_{gal}(\mathcal{F})$, and $\mathsf{P}_{gal}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGAL3) *If $p_{gal}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gal}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gal}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Abs, Lift are respectively Proposition III.11.4, Proposition III.11.6, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gal}. Restricting that joint record to exact word A_{gal} gives precisely the displayed detector δ_{gal} and target amplitude a_{gal}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gal}. □

Theorem III.10.19 (Local quantitative check: $\mathsf{CausAbsLift}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{cal},\Sigma,n}(E)$ set*

$$\delta_{cal}(R) := \Big\| \Big(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}} \Big) \Big\|_{\mathrm{det}}, \qquad a_{cal}($$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{cal}, define the unnormalised local aggregates

$$\mathsf{A}_{cal}(\mathcal{F}) := \sum_{j=1}^{N} a_{cal}(R_j), \qquad \mathsf{D}_{cal}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{cal}(R_j), \qquad \mathsf{P}_{cal}(\mathcal{F}) := \sum_{j=1}^{N} p_{cal}(R_j).$$

Then the following check is exact:

(LCAL1) *$a_{cal}(R) = 0$ exactly when this record contributes no target progress through exact support A_{cal}. If $a_{cal}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCAL2) *$\mathsf{A}_{cal}(\mathcal{F})$, $\mathsf{D}_{cal}(\mathcal{F})$, and $\mathsf{P}_{cal}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LCAL3) *If $p_{cal}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{cal}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{cal}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Abs, Lift are respectively Proposition III.11.5, Proposition III.11.6, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{cal}. Restricting that joint record to exact word A_{cal} gives precisely the displayed detector δ_{cal} and target amplitude a_{cal}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{cal}. □

Theorem III.10.20 (Local quantitative check: $\mathsf{GeomCausAbsLift}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gcal},\Sigma,n}(E)$ set*

$$\delta_{gcal}(R) := \Big\|\Big(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{os}}$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gcal}, define the unnormalised local aggregates

$$\mathsf{A}_{gcal}(\mathcal{F}) := \sum_{j=1}^{N} a_{gcal}(R_j), \qquad \mathsf{D}_{gcal}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gcal}(R_j), \qquad \mathsf{P}_{gcal}(\mathcal{F}) := \sum_{j=1}^{N} p_{gcal}(R_j).$$

Then the following check is exact:

(LGCAL1) *$a_{gcal}(R) = 0$ exactly when this record contributes no target progress through exact support A_{gcal}. If $a_{gcal}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCAL2) *$\mathsf{A}_{gcal}(\mathcal{F})$, $\mathsf{D}_{gcal}(\mathcal{F})$, and $\mathsf{P}_{gcal}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGCAL3) *If $p_{gcal}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gcal}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gcal}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Abs, Lift are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.6, Proposition III.11.7. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gcal}. Restricting that joint record to exact word A_{gcal} gives precisely the displayed detector δ_{gcal} and target amplitude a_{gcal}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gcal}. □

Theorem III.10.21 (Local quantitative check: Bdry)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_b,\Sigma,n}(E)$ set*

$$\delta_b(R) := \left\| \left(B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_b(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_b}(R) \right\|, \qquad p_b(R) := p_{\mathrm{phys}}(R).$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_b, define the unnormalised local aggregates

$$\mathsf{A}_b(\mathcal{F}) := \sum_{j=1}^{N} a_b(R_j), \qquad \mathsf{D}_b(\mathcal{F}) := \sum_{j=1}^{N} \delta_b(R_j), \qquad \mathsf{P}_b(\mathcal{F}) := \sum_{j=1}^{N} p_b(R_j).$$

Then the following check is exact:

(LB1) *$a_b(R) = 0$ exactly when this record contributes no target progress through exact support A_b. If $a_b(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LB2) *$\mathsf{A}_b(\mathcal{F})$, $\mathsf{D}_b(\mathcal{F})$, and $\mathsf{P}_b(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LB3) *If $p_b(R) > 0$, the occurrence is charged to the single physical carrier. If $p_b(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_b, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Bdry are respectively Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_b. Restricting that joint record to exact word A_b gives precisely the displayed detector δ_b and target amplitude a_b; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_b. □

Theorem III.10.22 (Local quantitative check: $\mathsf{GeomBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gb},\Sigma,n}(E)$ set*

$$\delta_{gb}(R) := \left\| \left(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\det}, \qquad a_{gb}(R) := \left\| \mathrm{Inc}^\Sigma_{A_{gb}}(R) \right\|, \qquad p_{gb}(R) := p_{\mathrm{phys}}$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gb}, define the unnormalised local aggregates

$$\mathsf{A}_{gb}(\mathcal{F}) := \sum_{j=1}^N a_{gb}(R_j), \qquad \mathsf{D}_{gb}(\mathcal{F}) := \sum_{j=1}^N \delta_{gb}(R_j), \qquad \mathsf{P}_{gb}(\mathcal{F}) := \sum_{j=1}^N p_{gb}(R_j).$$

Then the following check is exact:

(LGB1) $a_{gb}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{gb}. If $a_{gb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGB2) $\mathsf{A}_{gb}(\mathcal{F})$, $\mathsf{D}_{gb}(\mathcal{F})$, *and* $\mathsf{P}_{gb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGB3) *If $p_{gb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Bdry are respectively Proposition III.11.4, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gb}. Restricting that joint record to exact word A_{gb} gives precisely the displayed detector δ_{gb} and target amplitude a_{gb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gb}. $\square$

Theorem III.10.23 (Local quantitative check: $\mathsf{CausBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{cb},\Sigma,n}(E)$ set*

$$\delta_{cb}(R) := \left\| \left(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\det}, \qquad a_{cb}(R) := \left\| \mathrm{Inc}^\Sigma_{A_{cb}}(R) \right\|, \qquad p_{cb}(R) := p_{\mathrm{ph}}$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{cb}, define the unnormalised local aggregates

$$\mathsf{A}_{cb}(\mathcal{F}) := \sum_{j=1}^N a_{cb}(R_j), \qquad \mathsf{D}_{cb}(\mathcal{F}) := \sum_{j=1}^N \delta_{cb}(R_j), \qquad \mathsf{P}_{cb}(\mathcal{F}) := \sum_{j=1}^N p_{cb}(R_j).$$

Then the following check is exact:

(LCB1) $a_{cb}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{cb}. If $a_{cb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCB2) $\mathsf{A}_{cb}(\mathcal{F})$, $\mathsf{D}_{cb}(\mathcal{F})$, *and* $\mathsf{P}_{cb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LCB3) *If* $p_{cb}(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_{cb}(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{cb}*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Bdry are respectively Proposition III.11.5, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{cb}. Restricting that joint record to exact word A_{cb} gives precisely the displayed detector δ_{cb} and target amplitude a_{cb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{cb}. □

Theorem III.10.24 (Local quantitative check: $\mathsf{GeomCausBdry}$)**.** *For every finite same-origin stopped record* $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gcb},\Sigma,n}(E)$ *set*

$$\delta_{gcb}(R) := \left\| \left(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{gcb}(R) := \left\| \mathrm{Inc} \right.$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ *of pairwise disjoint same-origin stopped windows with exact word* A_{gcb}*, define the unnormalised local aggregates*

$$\mathsf{A}_{gcb}(\mathcal{F}) := \sum_{j=1}^{N} a_{gcb}(R_j), \qquad \mathsf{D}_{gcb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gcb}(R_j), \qquad \mathsf{P}_{gcb}(\mathcal{F}) := \sum_{j=1}^{N} p_{gcb}(R_j).$$

Then the following check is exact:

(LGCB1) $a_{gcb}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{gcb}*. If* $a_{gcb}(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCB2) $\mathsf{A}_{gcb}(\mathcal{F})$, $\mathsf{D}_{gcb}(\mathcal{F})$, *and* $\mathsf{P}_{gcb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LGCB3) *If* $p_{gcb}(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_{gcb}(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{gcb}*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Bdry are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gcb}. Restricting that joint record to exact word A_{gcb} gives precisely the displayed detector δ_{gcb} and target amplitude a_{gcb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gcb}. $\square$

Theorem III.10.25 (Local quantitative check: $\mathsf{AbsBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{ab},\Sigma,n}(E)$ set*

$$\delta_{ab}(R) := \left\| \left(A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{ab}(R) := \left\| \mathrm{Inc}^{\Sigma}_{A_{ab}}(R) \right\|, \qquad p_{ab}(R) := p$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{ab}, define the unnormalised local aggregates

$$\mathsf{A}_{ab}(\mathcal{F}) := \sum_{j=1}^{N} a_{ab}(R_j), \qquad \mathsf{D}_{ab}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{ab}(R_j), \qquad \mathsf{P}_{ab}(\mathcal{F}) := \sum_{j=1}^{N} p_{ab}(R_j).$$

Then the following check is exact:

(LAB1) $a_{ab}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{ab}. If $a_{ab}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LAB2) $\mathsf{A}_{ab}(\mathcal{F})$, $\mathsf{D}_{ab}(\mathcal{F})$, *and* $\mathsf{P}_{ab}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LAB3) *If $p_{ab}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{ab}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{ab}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Abs, Bdry are respectively Proposition III.11.6, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{ab}. Restricting that joint record to exact word A_{ab} gives precisely the displayed detector δ_{ab} and target amplitude a_{ab}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{ab}. $\square$

Theorem III.10.26 (Local quantitative check: $\mathsf{GeomAbsBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gab},\Sigma,n}(E)$ set*

$$\delta_{gab}(R) := \left\| \left(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\mathrm{det}}, \qquad a_{gab}(R) := \Big\| \mathrm{In}$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gab}, define the unnormalised local aggregates

$$\mathsf{A}_{gab}(\mathcal{F}) := \sum_{j=1}^{N} a_{gab}(R_j), \qquad \mathsf{D}_{gab}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gab}(R_j), \qquad \mathsf{P}_{gab}(\mathcal{F}) := \sum_{j=1}^{N} p_{gab}(R_j).$$

Then the following check is exact:

(LGAB1) $a_{gab}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{gab}*. If* $a_{gab}(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGAB2) $\mathsf{A}_{gab}(\mathcal{F})$*,* $\mathsf{D}_{gab}(\mathcal{F})$*, and* $\mathsf{P}_{gab}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LGAB3) *If* $p_{gab}(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_{gab}(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{gab}*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Abs, Bdry are respectively Proposition III.11.4, Proposition III.11.6, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gab}. Restricting that joint record to exact word A_{gab} gives precisely the displayed detector δ_{gab} and target amplitude a_{gab}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gab}. □

Theorem III.10.27 (Local quantitative check: $\mathsf{CausAbsBdry}$)**.** *For every finite same-origin stopped record* $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{cab},\Sigma,n}(E)$ *set*

$$\delta_{cab}(R) := \Big\| \Big(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \Big) \Big\|_{\mathrm{det}}, \qquad a_{cab}(R) := \Big|$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ *of pairwise disjoint same-origin stopped windows with exact word* A_{cab}*, define the unnormalised local aggregates*

$$\mathsf{A}_{cab}(\mathcal{F}) := \sum_{j=1}^{N} a_{cab}(R_j), \qquad \mathsf{D}_{cab}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{cab}(R_j), \qquad \mathsf{P}_{cab}(\mathcal{F}) := \sum_{j=1}^{N} p_{cab}(R_j).$$

Then the following check is exact:

(LCAB1) $a_{cab}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{cab}*. If* $a_{cab}(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCAB2) $\mathsf{A}_{cab}(\mathcal{F})$*,* $\mathsf{D}_{cab}(\mathcal{F})$*, and* $\mathsf{P}_{cab}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LCAB3) *If* $p_{cab}(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_{cab}(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{cab}*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Abs, Bdry are respectively Proposition III.11.5, Proposition III.11.6, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{cab}. Restricting that joint record to exact word A_{cab} gives precisely the displayed detector δ_{cab} and target amplitude a_{cab}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{cab}. $\square$

Theorem III.10.28 (Local quantitative check: $\mathsf{GeomCausAbsBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gcab},\Sigma,n}(E)$ set*

$$\delta_{gcab}(R) := \Big\| \Big(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}},$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gcab}, define the unnormalised local aggregates

$$\mathsf{A}_{gcab}(\mathcal{F}) := \sum_{j=1}^{N} a_{gcab}(R_j), \qquad \mathsf{D}_{gcab}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gcab}(R_j), \qquad \mathsf{P}_{gcab}(\mathcal{F}) := \sum_{j=1}^{N} p_{gcab}(R_j).$$

Then the following check is exact:

(LGCAB1) *$a_{gcab}(R) = 0$ exactly when this record contributes no target progress through exact support A_{gcab}. If $a_{gcab}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCAB2) *$\mathsf{A}_{gcab}(\mathcal{F})$, $\mathsf{D}_{gcab}(\mathcal{F})$, and $\mathsf{P}_{gcab}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGCAB3) *If $p_{gcab}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gcab}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gcab}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Abs, Bdry are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.6, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gcab}. Restricting that joint record to exact word A_{gcab} gives precisely the displayed detector δ_{gcab} and target amplitude a_{gcab}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gcab}. $\square$

Theorem III.10.29 (Local quantitative check: $\mathsf{LiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{lb},\Sigma,n}(E)$ set*

$$\delta_{lb}(R) := \Big\| \Big(L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \Big) \Big\|_{\mathrm{det}}, \qquad a_{lb}(R) := \Big\| \mathrm{Inc}^{\Sigma}_{A_{lb}}(R) \Big\|, \qquad p_{lb}(R) := p$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{lb}, define the unnormalised local aggregates

$$\mathsf{A}_{lb}(\mathcal{F}) := \sum_{j=1}^{N} a_{lb}(R_j), \qquad \mathsf{D}_{lb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{lb}(R_j), \qquad \mathsf{P}_{lb}(\mathcal{F}) := \sum_{j=1}^{N} p_{lb}(R_j).$$

Then the following check is exact:

(LLB1) $a_{lb}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{lb}. If $a_{lb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LLB2) $\mathsf{A}_{lb}(\mathcal{F})$, $\mathsf{D}_{lb}(\mathcal{F})$, *and* $\mathsf{P}_{lb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LLB3) *If $p_{lb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{lb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{lb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Lift, Bdry are respectively Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{lb}. Restricting that joint record to exact word A_{lb} gives precisely the displayed detector δ_{lb} and target amplitude a_{lb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{lb}. $\square$

Theorem III.10.30 (Local quantitative check: $\mathsf{GeomLiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{glb},\Sigma,n}(E)$ set*

$$\delta_{glb}(R) := \left\| \left(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \right) \right\|_{\det}, \qquad a_{glb}(R) := \Big\| \mathrm{In}$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{glb}, define the unnormalised local aggregates

$$\mathsf{A}_{glb}(\mathcal{F}) := \sum_{j=1}^{N} a_{glb}(R_j), \qquad \mathsf{D}_{glb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{glb}(R_j), \qquad \mathsf{P}_{glb}(\mathcal{F}) := \sum_{j=1}^{N} p_{glb}(R_j).$$

Then the following check is exact:

(LGLB1) $a_{glb}(R) = 0$ *exactly when this record contributes no target progress through exact support A_{glb}. If $a_{glb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGLB2) $\mathsf{A}_{glb}(\mathcal{F})$, $\mathsf{D}_{glb}(\mathcal{F})$, *and* $\mathsf{P}_{glb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGLB3) *If $p_{glb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{glb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{glb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Lift, Bdry are respectively Proposition III.11.4, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{glb}. Restricting that joint record to exact word A_{glb} gives precisely the displayed detector δ_{glb} and target amplitude a_{glb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{glb}. $\square$

Theorem III.10.31 (Local quantitative check: $\mathsf{CausLiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{clb},\Sigma,n}(E)$ set*

$$\delta_{clb}(R) := \Big\| \Big(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}} \Big) \Big\|_{\mathrm{det}}, \qquad a_{clb}(R) := \Big|$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{clb}, define the unnormalised local aggregates

$$\mathsf{A}_{clb}(\mathcal{F}) := \sum_{j=1}^{N} a_{clb}(R_j), \qquad \mathsf{D}_{clb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{clb}(R_j), \qquad \mathsf{P}_{clb}(\mathcal{F}) := \sum_{j=1}^{N} p_{clb}(R_j).$$

Then the following check is exact:

(LCLB1) *$a_{clb}(R) = 0$ exactly when this record contributes no target progress through exact support A_{clb}. If $a_{clb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCLB2) *$\mathsf{A}_{clb}(\mathcal{F})$, $\mathsf{D}_{clb}(\mathcal{F})$, and $\mathsf{P}_{clb}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LCLB3) *If $p_{clb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{clb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{clb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Lift, Bdry are respectively Proposition III.11.5, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{clb}. Restricting that joint record to exact word A_{clb} gives precisely the displayed detector δ_{clb} and target amplitude a_{clb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{clb}. $\square$

Theorem III.10.32 (Local quantitative check: $\mathsf{GeomCausLiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gclb},\Sigma,n}(E)$ set*

$$\delta_{gclb}(R) := \Big\|\Big(G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}},$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gclb}, define the unnormalised local aggregates

$$\mathsf{A}_{gclb}(\mathcal{F}) := \sum_{j=1}^{N} a_{gclb}(R_j), \qquad \mathsf{D}_{gclb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gclb}(R_j), \qquad \mathsf{P}_{gclb}(\mathcal{F}) := \sum_{j=1}^{N} p_{gclb}(R_j).$$

Then the following check is exact:

(LGCLB1) *$a_{gclb}(R) = 0$ exactly when this record contributes no target progress through exact support A_{gclb}. If $a_{gclb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCLB2) *$\mathsf{A}_{gclb}(\mathcal{F})$, $\mathsf{D}_{gclb}(\mathcal{F})$, and $\mathsf{P}_{gclb}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGCLB3) *If $p_{gclb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gclb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gclb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Lift, Bdry are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gclb}. Restricting that joint record to exact word A_{gclb} gives precisely the displayed detector δ_{gclb} and target amplitude a_{gclb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gclb}. $\square$

Theorem III.10.33 (Local quantitative check: $\mathsf{AbsLiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{alb},\Sigma,n}(E)$ set*

$$\delta_{alb}(R) := \Big\|\Big(A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}}\Big)\Big\|_{\mathrm{det}}, \qquad a_{alb}(R) :=$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{alb}, define the unnormalised local aggregates

$$\mathsf{A}_{alb}(\mathcal{F}) := \sum_{j=1}^{N} a_{alb}(R_j), \qquad \mathsf{D}_{alb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{alb}(R_j), \qquad \mathsf{P}_{alb}(\mathcal{F}) := \sum_{j=1}^{N} p_{alb}(R_j).$$

Then the following check is exact:

(LALB1) $a_{alb}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{alb}*. If* $a_{alb}(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LALB2) $\mathsf{A}_{alb}(\mathcal{F})$, $\mathsf{D}_{alb}(\mathcal{F})$, *and* $\mathsf{P}_{alb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LALB3) *If* $p_{alb}(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_{alb}(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{alb}*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Abs, Lift, Bdry are respectively Proposition III.11.6, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{alb}. Restricting that joint record to exact word A_{alb} gives precisely the displayed detector δ_{alb} and target amplitude a_{alb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{alb}. □

Theorem III.10.34 (Local quantitative check: GeomAbsLiftBdry)**.** *For every finite same-origin stopped record* $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{galb},\Sigma,n}(E)$ *set*

$$\delta_{galb}(R) := \Big\|\Big(G^{\mathsf{Geom}}_{\Sigma}, \mathrm{Ker}^{\mathsf{Geom}}_{\Sigma}, \mathrm{Gc}^{\mathsf{Geom}}_{\Sigma}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ *of pairwise disjoint same-origin stopped windows with exact word* A_{galb}*, define the unnormalised local aggregates*

$$\mathsf{A}_{galb}(\mathcal{F}) := \sum_{j=1}^{N} a_{galb}(R_j), \qquad \mathsf{D}_{galb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{galb}(R_j), \qquad \mathsf{P}_{galb}(\mathcal{F}) := \sum_{j=1}^{N} p_{galb}(R_j).$$

Then the following check is exact:

(LGALB1) $a_{galb}(R) = 0$ *exactly when this record contributes no target progress through exact support* A_{galb}*. If* $a_{galb}(R) > 0$*, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGALB2) $\mathsf{A}_{galb}(\mathcal{F})$, $\mathsf{D}_{galb}(\mathcal{F})$, *and* $\mathsf{P}_{galb}(\mathcal{F})$ *are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by* N *or replaced by a limiting average.*

(LGALB3) *If* $p_{galb}(R) > 0$*, the occurrence is charged to the single physical carrier. If* $p_{galb}(R) = 0$*, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word* A_{galb}*, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Abs, Lift, Bdry are respectively Proposition III.11.4, Proposition III.11.6, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{galb}. Restricting that joint record to exact word A_{galb} gives precisely the displayed detector δ_{galb} and target amplitude a_{galb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{galb}. □

Theorem III.10.35 (Local quantitative check: $\mathsf{CausAbsLiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{calb},\Sigma,n}(E)$ set*

$$\delta_{calb}(R) := \Big\| \Big(H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{te}}$$

For a finite family $\mathcal{F} = (R_1, \dots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{calb}, define the unnormalised local aggregates

$$\mathsf{A}_{calb}(\mathcal{F}) := \sum_{j=1}^{N} a_{calb}(R_j), \qquad \mathsf{D}_{calb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{calb}(R_j), \qquad \mathsf{P}_{calb}(\mathcal{F}) := \sum_{j=1}^{N} p_{calb}(R_j).$$

Then the following check is exact:

(LCALB1) *$a_{calb}(R) = 0$ exactly when this record contributes no target progress through exact support A_{calb}. If $a_{calb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LCALB2) *$\mathsf{A}_{calb}(\mathcal{F})$, $\mathsf{D}_{calb}(\mathcal{F})$, and $\mathsf{P}_{calb}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LCALB3) *If $p_{calb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{calb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{calb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Caus, Abs, Lift, Bdry are respectively Proposition III.11.5, Proposition III.11.6, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{calb}. Restricting that joint record to exact word A_{calb} gives precisely the displayed detector δ_{calb} and target amplitude a_{calb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{calb}. □

Theorem III.10.36 (Local quantitative check: $\mathsf{GeomCausAbsLiftBdry}$)**.** *For every finite same-origin stopped record $R \in \mathfrak{D}^{\mathrm{pre}}_{A_{gcalb},\Sigma,n}(E)$ set*

$$\delta_{gcalb}(R) := \Big\| \Big(G_{\Sigma}^{\mathsf{Geom}}, \mathrm{Ker}_{\Sigma}^{\mathsf{Geom}}, \mathrm{Gc}_{\Sigma}^{\mathsf{Geom}}, H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{c}}$$

For a finite family $\mathcal{F} = (R_1, \ldots, R_N)$ of pairwise disjoint same-origin stopped windows with exact word A_{gcalb}, define the unnormalised local aggregates

$$\mathsf{A}_{gcalb}(\mathcal{F}) := \sum_{j=1}^{N} a_{gcalb}(R_j), \qquad \mathsf{D}_{gcalb}(\mathcal{F}) := \sum_{j=1}^{N} \delta_{gcalb}(R_j), \qquad \mathsf{P}_{gcalb}(\mathcal{F}) := \sum_{j=1}^{N} p_{gcalb}(R_j).$$

Then the following check is exact:

(LGCALB1) *$a_{gcalb}(R) = 0$ exactly when this record contributes no target progress through exact support A_{gcalb}. If $a_{gcalb}(R) > 0$, the record itself is the finite public target-aligned occurrence, with the listed detector tuple and complete joint frontier.*

(LGCALB2) *$\mathsf{A}_{gcalb}(\mathcal{F})$, $\mathsf{D}_{gcalb}(\mathcal{F})$, and $\mathsf{P}_{gcalb}(\mathcal{F})$ are the exact cumulative target amplitude, structural detector amplitude, and physical expenditure of this support. They are never divided by N or replaced by a limiting average.*

(LGCALB3) *If $p_{gcalb}(R) > 0$, the occurrence is charged to the single physical carrier. If $p_{gcalb}(R) = 0$, the equality row of the same joint record is evaluated; a nonzero conditional increment is appended to the same exact word A_{gcalb}, while a zero increment deletes that record's target-incidence edge.*

Proof. The atomic target-aligned relations for Geom, Caus, Abs, Lift, Bdry are respectively Proposition III.11.4, Proposition III.11.5, Proposition III.11.6, Proposition III.11.7, Proposition III.11.8. Each is paired with the same public shell covector before any source projection. Their finite expression graphs are joined, so every mixed product, commutator, cutoff, pressure descent, chart motion, and trace retains the union word A_{gcalb}. Restricting that joint record to exact word A_{gcalb} gives precisely the displayed detector δ_{gcalb} and target amplitude a_{gcalb}; hence its zero/nonzero test is the exact target pairing of this source combination, not a separate test of its coordinates. For disjoint stopped windows, linearity of the public weak pairing and finite additivity of the physical carrier give the three unnormalised sums. Finally, Theorem III.10.44 applies to this same complete word: positive price is charged, while zero price re-enters the equality row without losing an ancestor. This proves the quantitative check for exact support A_{gcalb}. □

Theorem III.10.37 (Target-aligned cascade terminalization)**.** *Let $(I_j, R_j)_{j\geq 1}$ be pairwise disjoint normalized shell crossings of one ordinary same-origin public branch, with physical ceiling E, and write $Q_{A,j}$ for the exact-support A-current in* (17)*. For $N \geq 1$, set*

$$S_{A,N} := \sum_{j=1}^{N} Q_{A,j}, \qquad P_{A,N} := \mathsf{P}_A(\mathcal{F}_{A,N}), \qquad \mathcal{T}_{A,N} := (R_j, \mathrm{Inc}^{\Sigma}_{A}(R_j), p_{\mathrm{phys}}(R_j))_{j\leq N}.$$

Then the public shell identity gives the exact finite relation

$$N = \sum_{\varnothing \neq A \subseteq \mathfrak{C}} S_{A,N}. \tag{556}$$

Consequently, for every N, at least one exact support satisfies $S_{A,N} \geq N/31$. A nested choice of such supports has a fixed support A_* and an unbounded subsequence N_k with

$$S_{A_*,N_k} \geq N_k/31.$$

The cumulative trace $\mathcal{T}_{A_*} := (\mathcal{T}_{A_*,N_k})_k$ is a source-preserving element of the reachable closure and retains the ordinary same-origin realization row. Its terminal evaluation is exactly one of:

(CT1) $P_{A_*,N_k} > E$ *for some* k*, contradicting the finite physical ledger and deleting the branch;*

(CT2) *the public continuation graph assigns its ordinary continuation value to the trace, in which case the declared target was not reached;*

(CT3) *the public continuation graph assigns its noncontinuation value to the trace, in which case the trace is an ordinary source-owned target terminal with exact support* A_**.*

Thus a persistent cascade is neither averaged away nor assigned to a target-visible residual: its quantitative contribution and terminal status are read from the same public, source-owned trace.

Proof. Apply the unit shell-crossing identity to every I_j, expand each current by its exact source word before taking any limit, and sum the finite identities. This proves (556). Since there are thirty-one summands, the finite maximum is at least $N/31$; an infinite pigeonhole argument selects A_* on an unbounded subsequence. The local quantitative theorem for A_* supplies every entry of $\mathcal{T}_{A_*,N_k}$, including its unnormalised physical price. The source-preserving completion theorem retains the complete word and ordinary realization row. Finite additivity of the one physical carrier gives (CT1). Totalization of the public continuation graph is binary on an ordinary same-origin trace, giving (CT2)–(CT3). These cases are disjoint and exhaustive. $\square$

Theorem III.10.38 (Continuum target-aligned structural exhaustion)**.** *On every nonempty ordinary finite physical stratum,*

$$\boxed{\begin{gathered}\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(E) > 0\\ \iff\ \exists\, \varnothing \neq A \subseteq \mathfrak{C} : \mathfrak{F}^{\Sigma}_{A}(E)\ \textit{contains a realized}\\ \textit{formation-active point.}\end{gathered}} \tag{557}$$

There are exactly $2^5 - 1 = 31$ *possible named nonempty supports; a particular equation may realize fewer of them. Within each cell, the five normal-form theorems give an exhaustive atomic word after the fixed projection order*

$$\mathsf{Geom} \longrightarrow \mathsf{Caus} \longrightarrow \mathsf{Abs} \longrightarrow \mathsf{Lift} \longrightarrow \mathsf{Bdry}, \tag{558}$$

Within one source family the first nonzero typed coordinate is unique. Different source families may remain simultaneously active and are retained by A*, the aggregate carrier class, and the circulation graph. No sixth target-bearing family exists.*

Proof. On a cofinal approach-contact branch, contact gives a disjoint unit-increment shell cascade. Finite support exhaustion selects one nonempty A on a subsequence. On a terminal-entry branch, the completed shell graph instead gives a Bdry-headed owner and the terminal-entry coordinate; take its exact provenance support as A. Target-incidence compression retains precisely these source–shell paths. Apply the five normal-form theorems in the order (558); every headed ordinary or nonordinary atomic component enters exactly one block, while composite ancestry and signed interactions remain in the exact support, aggregate carrier, and circulation coordinates. On a finite current record the affine shell identity supplies a positive component; on a variation-boundary record the closed owner supplies it. The resulting profile is formation active. Same-origin diagonal realization gives the required profile point.

Conversely, a realized formation-active point includes a public same-origin path and nonzero cofinal shell coordinate. Its shell coordinate therefore crosses every retained neighborhood of the complete continuation target, which is contact. The universal provenance theorem and Theorem I.2.11 place every ordinary, defective, or limiting shell-current occurrence in one of these cells. Since $J_{J_{\mathrm{res}}} \subseteq \mathscr{N}_{\Sigma}$, no additional target-bearing family remains. $\square$

Definition III.10.39 (Presentation-generated prospective source domain)**.** Fix a public continuum presentation, a derived finite physical stratum B_{phys}, and its generated target shells. For each nonempty $A \subseteq \mathfrak{C}$, let $\mathfrak{D}^{\text{pre}}_{A,\Sigma}(B_{\text{phys}})$ contain the following records:

(D1) every finite stopped history of an ordinary same-origin branch, cut before its first contact with the current shell;

(D2) every finite joint source occurrence obtained from such a history by the five source constructors, target projection, backward pullback, conditional joining, chart or quotient transport, signed interface gluing, and physical-carrier restriction;

(D3) every completed terminal-entry record, with its Bdry-headed current and continuation-graph value; and

(D4) every finite prefix of a source-preserving limit, concentration, oscillation, tail, equality, or total-graph record generated from (D1)–(D3).

All records retain their origin, stopped time, target covector, complete preceding frontier, unnormalized occurrence cocycle, source word, and physical carrier. Let $\mathfrak{D}^{\text{pre}}_{A,\Sigma,n}(B_{\text{phys}})$ be the finite subfamily generated by the first n public constructor words, rational shell probes, and stopped-history prefixes in the fixed grammar enumeration. Then $\mathfrak{D}^{\text{pre}}_{A,\Sigma,n}(B_{\text{phys}})$ is increasing and its union is $\mathfrak{D}^{\text{pre}}_{A,\Sigma}(B_{\text{phys}})$. With ι_A and K_A from Definition I.3.2, define

$$\mathscr{S}^{\text{reach,sat}}_{A,\Sigma}(B_{\text{phys}}) := \overline{\iota_A(\mathfrak{D}^{\text{pre}}_{A,\Sigma}(B_{\text{phys}}))}^{K_A}. \tag{559}$$

The full domain is the provenance-disjoint union over nonempty A. Evaluation at a point of $\mathscr{S}^{\text{reach,sat}}_{A,\Sigma}(B_{\text{phys}})$ is called a *reachable source evaluation*. The continuum structural algebra has no second semantic domain: every source, invariant, price, and contact statement is evaluated on (559).

Theorem III.10.40 (Public-presentation closure of the structural calculus)**.** *Every relation used in the continuum structural calculus is generated by one of the following operations on the public presentation:*

(S1) *a weak-equation, initial-data, boundary, continuation, target-shell, or physical-carrier coordinate declared by the presentation;*

(S2) *a finite five-source constructor applied to earlier generated records;*

(S3) *a target projection, complete backward pullback, conditional joint increment, signed interface gluing, or physical-carrier restriction of such a record; or*

(S4) *a source-preserving limit of the resulting finite stopped records.*

Consequently every theorem in the structural reachability calculus is a statement about $\mathscr{S}^{\text{reach,sat}}_{A,\Sigma}(B_{\text{phys}})$, its target-aligned coordinates, and the physical carrier generated by the public presentation. A global analytic property contributes only through the target-projected total-graph coordinate generated by the operation that asserts or fails it.

Proof. The primitive source compiler exhausts the public weak-expression graph by Lemma I.2.7. The free provenance algebra closes those primitive records under every finite operation in (S2)–(S3), and Theorem I.3.4 proves (S4) without adding a source. Prospective backward factorization and conditional-increment accounting retain the complete joint frontier before source projection. The physical carrier is restricted only along the same generated occurrence record. Structural induction on the constructor word therefore places every premise and every conclusion of an invariant, price, equality, Hamiltonian, entropy, or contact calculation in the stated reachable domain. □

Theorem III.10.41 (Reachable saturation and finite occurrence recovery)**.** *The set $\mathscr{S}^{\text{reach,sat}}_{A,\Sigma}(B_{\text{phys}})$ is compact, retains source support A, and is closed under every source-preserving constructor* *generated by the public presentation. Let $d \geq 0$ be any target-incidence detector in the complete invariant ledger. Then*

$$\sup_{R \in \mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})} d(R) = 0 \Longleftrightarrow d(R) = 0 \quad \textit{for every } R \in \mathfrak{D}^{\mathrm{pre}}_{A,\Sigma}(B_{\mathrm{phys}}), \tag{560}$$

$$\sup_{R \in \mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})} d(R) > 0 \Longleftrightarrow \exists R_0 \in \mathfrak{D}^{\mathrm{pre}}_{A,\Sigma}(B_{\mathrm{phys}}) : d(R_0) > 0. \tag{561}$$

In the positive case, prospective backward factorization replaces R_0, when necessary, by its least nonzero finite contextual current–probe occurrence and retains the complete preceding frontier. Hence target-visible source presence is always witnessed by presentation-generated finite ancestry.

Proof. Compactness follows because $K_A = [-1, 1]^{\mathbb{N}}$ is compact and the right side of (559) is closed. Source preservation is Theorem I.3.4; every listed constructor is already a coordinate of the structural test system. Each detector is a nonnegative continuous coordinate of that system, after adjoining its bounded rational truncations. Density of $\iota_A(\mathfrak{D}^{\mathrm{pre}}_{A,\Sigma}(B_{\mathrm{phys}}))$ in its closure proves (560). If the supremum is positive, choose a point where $d > 0$; continuity gives a neighborhood on which $d > 0$, and density supplies R_0. The converse is immediate. Exact backward factorization gives the final assertion. □

Definition III.10.42 (Exact reachable invariant profile)**.** For a detector coordinate d owned by A, define

$$\mathcal{V}^{\Sigma}_{A,d}(B_{\mathrm{phys}}) := \max_{R \in \mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})} d(R). \tag{562}$$

For a signed scalar, graph, measure, operator, response, shell, or realization invariant, use the squared magnitudes of its separating rational cylinder coordinates. For a nonnegative extended-real valuation v, use all bounded truncations $v \wedge m$ and define its exact value by their compatible maxima or minima on the same reachable set. Rank is evaluated by finite minors; concentration, oscillation, and boundary variations are evaluated by finite rational partitions. Mixed-source invariants are evaluated on the complete joint record before sourcewise projection.

Theorem III.10.43 (Complete target-aligned evaluation from the public presentation)**.** *Every invariant generated by the five normal forms—geometric range/kernel/graph/capacity, directed cohomology/storage/response/cycle, quotient descent/fiber/cokernel/rank, lift interface/concentration/oscillation/ scale, boundary exterior/tail/terminal data—and every joint carrier, conditional-increment, shell, realization, entropy, Hamiltonian, or physical-price coordinate has an exact value on $\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})$. For every detector coordinate,*

$$\mathrm{Eval}^{\Sigma}_{A,d} = \begin{cases} \mathrm{Absent}_{A,d}, & \mathcal{V}^{\Sigma}_{A,d}(B_{\mathrm{phys}}) = 0, \\ \mathrm{Present}_{A,d}(R_0, \lambda, Q, \mathrm{Anc}, p_{\mathrm{phys}}), & \mathcal{V}^{\Sigma}_{A,d}(B_{\mathrm{phys}}) > 0, \end{cases} \tag{563}$$

where the second row contains the finite occurrence recovered by Theorem III.10.41. Every valuation returns its exact extended-real value and an attaining reachable record when the extremum is finite, or a cofinal sequence of reachable finite occurrences when it is infinite. Every graph-valued invariant returns its unique cylinder-coordinate value on each reachable record.

The second row is the only positive structural value. Its record contains the public equation term, target covector, complete joint frontier, inherited five-source word, and physical carrier atom that generate the positive coordinate. Thus (563) *evaluates every invariant directly in the presentation-generated algebra.*

Proof. Compactness gives the maximum in (562); Theorem III.10.41 gives the two rows of (563). Rational cylinder coordinates separate points of the fixed graph-complete product, so they determine every graph-valued invariant. Bounded truncations determine an extended-real valuation uniquely. The finite-minor and rational-partition criteria are exact by finite-dimensional rank and total variation. Conditional joint increments are formed before projection, so mixed target alignment remains a coordinate of the reachable joint record. These arguments apply to every row of the coordinate ledger. □

Theorem III.10.44 (Actual physical-price and equality decision)**.** *Let R_0 be a positive source occurrence returned by* (563)*. Restrict the same-origin physical carrier to its finite occurrence atom. Exactly one of the following holds:*

(P1) $p_{\mathrm{phys}}(R_0) > 0$*, and the occurrence is charged by that exact physical amount;*

(P2) $p_{\mathrm{phys}}(R_0) = 0$*, and the complete occurrence belongs to the reachable zero-price equality cell generated by the same source word.*

In the second case, equality saturation is performed inside $\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})$. Reprojection gives either target absence by (560) *or another actual positive finite occurrence by* (561)*. Equality saturation therefore does not create a formal terminal state.*

Proof. The physical carrier is a nonnegative measure, so its value on the finite occurrence atom is positive or zero. The first value is the local charging row. In the zero case, every nonnegative production summand vanishes on that atom and is adjoined to the generated equality ideal. Source-preserving saturation keeps the resulting descendants in the same reachable domain. Apply Theorems III.10.41 and III.10.43 after target reprojection. This gives the stated exhaustive decision. □

Theorem III.10.45 (Cofinal contact retention without dilution)**.** *Let one ordinary same-origin branch with finite physical budget complete a cofinal target-shell family. Form its countable occurrence space before any normalization. Then one of the following actual source records exists:*

(C1) *a fixed source owns finite pre-contact occurrence sets on which the target-current density relative to the physical carrier is unbounded;*

(C2) *a fixed source owns a positive target-current atom of zero physical carrier mass.*

Every record in either row belongs to $\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})$ for its inherited support A, and its invariant coordinates are evaluated by Theorems III.10.43 and III.10.44. Thus infinitely many finite contributions cannot converge to target progress without producing an actual positive source occurrence.

Proof. Apply exact compositional accounting Theorem III.4.34. Finite physical mass and infinite unnormalized crossing mass give the amplification alternative (468). Prospective source-amplification extraction assigns either value to one fixed source and returns its finite pre-contact occurrence sets. These are elements of $\mathfrak{D}^{\mathrm{pre}}_{A,\Sigma}(B_{\mathrm{phys}})$ by definition. The final assertion follows from the actual invariant and price decisions. □

Definition III.10.46 (Actual structural contact cells)**.** Let $\mathscr{C}_{A,\Sigma}(B_{\mathrm{phys}})$ be the subset of $\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})$ whose total-graph coordinates have all of the following values:

(K1) one ordinary same-origin equation and initial-data value;

(K2) the cofinal unnormalized target-shell cocycle;

(K3) the noncontinuable value of the declared continuation graph; and

(K4) a nonzero target projection with its complete five-source ancestry.

In addition, the complete record must lie in the image of the ordinary same-origin realization map generated by the public equation and initial data. Thus

$$\mathscr{C}_{A,\Sigma}(B_{\mathrm{phys}}) = \mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}}) \cap \mathrm{Ran}(\mathfrak{R}_{\mathrm{ord}}) \cap \bigcap_{i=1}^{4}\{(\mathrm{K}i)\}. \tag{564}$$

These conditions are coordinates of the public total graphs and generated shell family. The cell is evaluated inside the reachable source domain, not inside a finite-cylinder relaxation. In particular, a compatible positive functional, graph-boundary point, or zero-price equality model is not a member unless one ordinary same-origin public orbit realizes the entire joint record.

Theorem III.10.47 (Presentation-relative structural decision of global regularity)**.** *For every derived finite physical stratum,*

$$\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) = 0 \iff \mathscr{C}_{A,\Sigma}(B_{\mathrm{phys}}) = \varnothing \quad \text{for every nonempty } A \subseteq \mathfrak{C}, \tag{565}$$

$$\mathrm{Reach}^{\mathrm{phys}}_{\Sigma}(B_{\mathrm{phys}}) > 0 \iff \mathscr{C}_{A,\Sigma}(B_{\mathrm{phys}}) \neq \varnothing \quad \text{for some nonempty } A \subseteq \mathfrak{C}. \tag{566}$$

The emptiness or nonemptiness of each contact cell is evaluated by the exact coordinate rules of Theorem III.10.43: zero target incidence deletes the cell, while a positive value returns an actual finite pre-contact source occurrence; positive physical price is charged and zero price enters the actual equality decision. A cofinal family is retained by Theorem III.10.45. Hence the global-regularity process uses only target-aligned, presentation-generated structural records. Its terminal outputs are

$$\mathrm{GlobEval}_{\Sigma}(B_{\mathrm{phys}}) \in \left\{\mathrm{Regular}\big((\mathscr{C}_{A,\Sigma} = \varnothing)_A\big), \mathrm{Singular}(R, A); : R \in \mathscr{C}_{A,\Sigma}(B_{\mathrm{phys}})\right\}. \tag{567}$$

The terminal map is evaluated on ordinary same-origin contact cells with their complete presentation-generated ancestry.

Proof. An ordinary contacting branch supplies its completed terminal or cofinal pre-contact records. Prospective source exhaustion gives a nonempty support A, while source-preserving saturation and the total continuation graph give (K1)–(K4). Hence contact implies a nonempty contact cell, and the branch itself supplies membership in $\mathrm{Ran}(\mathfrak{R}_{\mathrm{ord}})$. Conversely, (564) supplies one ordinary same-origin branch with the declared equation, initial datum, cofinal-shell cocycle, and noncontinuation value; that branch contacts the declared target. This proves (565)– (566).

The actual invariant decision evaluates each source coordinate on the reachable set. The price/equality theorem processes its physical value, and the cofinal theorem prevents dilution across infinitely many crossings. Universal five-source provenance and target-null residual closure exhaust the owners. Thus the two values in (567) are exhaustive. □

Theorem III.10.48 (Simultaneous terminal evaluation of the complete source ledger)**.** *For every nonempty support $A \subseteq \mathfrak{C}$, saturate the complete coordinate ledger under all mandatory successors and evaluate it on $\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(B_{\mathrm{phys}})$. The result is exactly*

$$\mathrm{Eval}^{\Sigma}_{A} = \begin{cases} \mathrm{Absent}_A, & \mathcal{V}^{\Sigma}_{A,d}(B_{\mathrm{phys}}) = 0 \textit{ for every target-incidence detector } d, \\ \mathrm{Present}_A(R_0, \lambda, Q, \mathrm{Anc}, p_{\mathrm{phys}}), & \mathcal{V}^{\Sigma}_{A,d}(B_{\mathrm{phys}}) > 0 \textit{ for the first positive detector } d. \end{cases} \tag{568}$$

Every successor is already a coordinate of the saturated reachable domain. The second value contains an actual presentation-generated finite occurrence R_0. Boundary, cofinal, equality, and nonordinary data are coordinates of R_0 and its complete ancestry.

Proof. Apply Theorem III.10.43 simultaneously in the fixed ledger order. The source alphabet is finite and the coordinate order is a well-order, so a nonempty set of positive coordinates has a first member. Source-preserving saturation places every successor in the same reachable domain before evaluation. Universal provenance and target-null residual closure leave no target-bearing coordinate outside the five supports. □

Theorem III.10.49 (Direct target-local invariant refinement)**.** *For each rational $E > 0$, support A, and refinement depth n, the following finite target-local procedure constructs a finite prefix of the presentation-generated formation profile.*

(P1) *Generate the first n forward source words and backward shell covectors, retain their conservative incidence core, and form the saturated cumulative join with every earlier prefix. Discard all coordinates outside that joint source–target incidence core.*

(P2) *Apply the five canonical normal-form projections in* (558)*. Evaluate, in the displayed order of Theorem III.10.43, the geometric range, kernel, graph and capacity rows; the directed cohomology, storage, domination, cycle, response and graph rows; the quotient descent, vertical, cokernel, graph, target, kernel-capacity and rank rows; the lift intertwining, fiber, concentration, oscillation, scale and graph rows; and the exterior, tail, terminal, graph and tail-capacity rows.*

(P3) *Evaluate the mixed carrier, conditional-increment, cycle, realization, shell, entropy, and Hamiltonian rows for every nonempty support. Use the exact zero/positive, finite-minor, variation, and cylinder-coordinate rule assigned to each generated coordinate by Theorem III.10.43.*

(P4) *Return the generated finite stopped records with their joint invariant, origin, physical carrier, and total-realization coordinates. Preserve all finite correlations imposed by the ordered rows, and refine the least target-visible coordinate whose zero/nonzero or ordinary/nonordinary status can change formation activity.*

Then

$$\overline{\bigcup_n \iota_A(mathfrakD^{\mathrm{pre}}_{A,\Sigma,n}(E))} = \mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(E). \tag{569}$$

The source decision is made on $\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(E)$. More precisely, for every detector d,

$$\max_{R\in\mathscr{S}^{\mathrm{reach,sat}}_{A,\Sigma}(E)} d(R) = 0 \quad \text{or} \quad \exists R_0 \in \mathfrak{D}^{\mathrm{pre}}_{A,\Sigma}(E) : d(R_0) > 0. \tag{570}$$

The alternatives are exactly the absence and actual-presence rows of Theorem III.10.43. In the positive row, backward factorization returns the least finite contextual occurrence and recursive carrier–polar operations act on that occurrence. In the zero row the corresponding target-incidence edge is deleted. Contact is decided only by the realized cells of Definition III.10.46.

Proof. Finite incidence pruning is Corollary III.4.7. Each normal-form projection is a rational finite-dimensional projection or a totalized polarity split at depth n. The invariant definitions are quadratic-form, order-domination, variation, rank, capacity, circulation, or graph-feasibility programs, all applied to generated finite stopped records. Exhaustive enumeration, source-preserving closure, retention of all finite correlations, and the five measurement theorems give (569). Apply Theorem III.10.41. Continuity and density give (570); exact backward factorization supplies the least contextual occurrence in its positive row. The realized-contact intersection is (564), which proves the last assertion. □

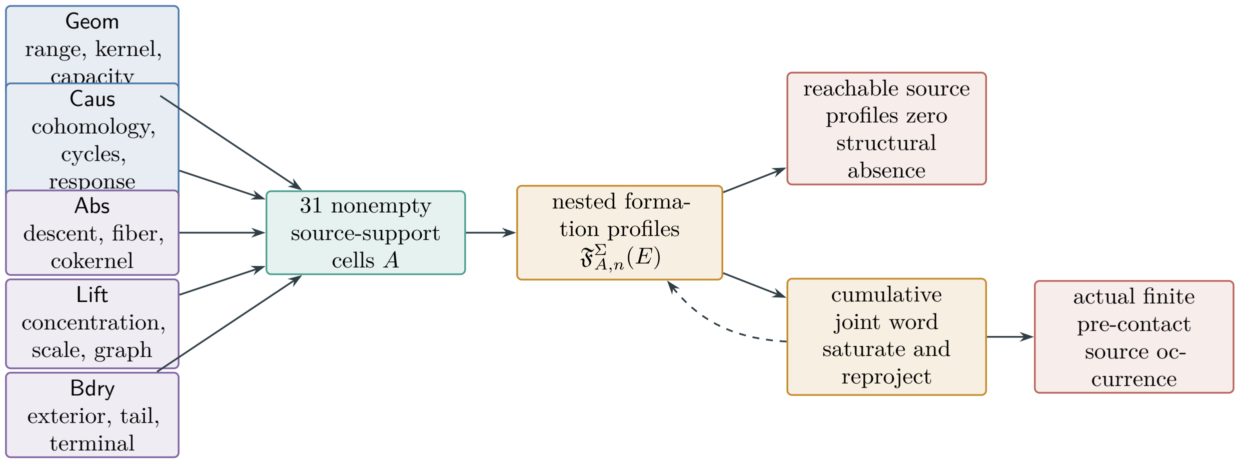

Figure 23: Direct continuum formation analysis. Each channel has a canonical target-local normal form and a finite invariant program. Their simultaneous supports produce 31 cells. Outer refinement is intersected with the presentation-generated reachable domain. Zero on that domain proves structural absence; positivity recovers an actual finite pre-contact source occurrence with its complete joint frontier. Joint reprojection returns every newly exposed target component to its five-channel owner. Physical contact is read only on the ordinary same-origin realization range.

Definition III.10.50 (PDE backend)**.** A *PDE backend* is the homomorphic compilation of a public presentation into the source, shell, carrier, total-graph, successor, and ordered-relation signatures used to construct $\mathfrak{G}_{\mathfrak{b}}$. It may compute equation-specific coefficients, scaling weights, constraint tails, and closure edges. It may not add a source family, a target-relative premise, a physical measure not generated by the equation, or a terminal declaration without a proved relation of the ordered algebra. Here $\mathfrak{G}_{\mathfrak{b}}$ is the induced subgraph of $\mathfrak{G}_A$ obtained by fixing the complete active signature $\mathfrak{b} = (A, m, \Gamma, \eta, \mathbf{R})$ and its earlier zero coordinates.

Proposition III.10.51 (Backend conservativity)**.** *Every PDE backend preserves the five-channel normal form, prospective contact accountability, source-preserving limit closure, the target-null meaning of J_{res}, and the single physical-budget ledger. Therefore a backend can validate or refute regularity only by evaluating the vertices and closure edges of $\mathfrak{G}_{\mathfrak{b}}$; it cannot change the continuum theory that generated them.*

Proof. The backend is a homomorphism into the signature already constructed in Section III.4. Preservation of source normal forms follows from Theorem I.1.10; preservation of contact accountability and the residual quotient follows from Theorem III.2.11; preservation of the physical ledger follows from signed carrier gluing and Theorem II.3.5. The prohibited additions are not symbols of the target signature, so they cannot occur in its image. □

III.11 End-to-end public-presentation unit test

This unit test verifies the specialization interface on the three-dimensional incompressible Navier–Stokes presentation. It is not an additional regularity argument: every displayed object below is compiled from the public weak equation, its local-energy carrier, the continuation graph, and the target shells.

Definition III.11.1 (Public Navier–Stokes target presentation)**.** Fix $\nu > 0$, divergence-free initial data, declared boundary data, and the public weak relation

$$\partial_t u + \mathbb{P}((u \cdot \nabla)u) - \nu\Delta u = 0, \qquad \nabla \cdot u = 0,$$

together with its standard local-energy identity and the continuation graph for the declared regularity class. Let Σ_{NS} be the set of terminal germs at which that graph has no ordinary continuation. The only physical ledger is the local-energy carrier obtained from the same identity. All cutoffs, moving charts, scale records, pressure reconstructions, and shell covectors are generated by the continuum constructor grammar.

Theorem III.11.2 (Navier–Stokes 31-cell source compilation)**.** *For the public presentation of Definition III.11.1, every finite same-origin pre-contact shell record has an exact source word in*

$$\mathcal{P}(\{\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}\}) \setminus \{\varnothing\}.$$

The primitive assignments are: viscosity and its positive local production are Geom*; convection is* Caus*; the solenoidal constraint and pressure reconstruction are* Abs*; a moving centre, scale, or normalized chart is* Lift*; and cutoff, exterior, initial/terminal, and boundary fluxes are* Bdry*. A record formed from several of these operations is assigned their union before target projection. Thus each target-visible Navier–Stokes shell contribution belongs to one of the same thirty-one cells and is evaluated by the public-presentation cell rule Theorem III.10.3. Equivalently, the Navier–Stokes specialization realizes the explicit theorem family* T_A *of Theorem III.10.4 for each of its thirty-one possible exact supports.*

Proof. Pair the public weak equation with the finite shell covector and expand the local-energy identity before taking a positive part. The viscous term is the positive symmetric form and hence has head Geom; the advective term is the fixed nonsymmetric transport current and hence has head Caus. Applying the Leray constraint or reconstructing pressure is precisely the quotient/constraint constructor, while every cutoff trace is a boundary constructor and every moving normalization is a lift constructor. These are the five primitive clauses of the source compiler. Structural induction on the finite expression graph assigns the union word to every product, commutator, localization, and signed gluing before target projection. Source-preserving completion retains that word for weak defects, concentrations, oscillations, tails, and failed graph values. Therefore Theorem I.2.9 and Theorem I.2.11 give the stated disjoint thirty-one-cell routing. Finally, Theorem III.10.3 evaluates each cell on finite same-origin records and applies its physical carrier rule. The tabled theorem family applies word-by-word, so no combined Navier–Stokes contribution is discharged by a theorem for an isolated channel. □

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Theorem III.11.3 (Navier–Stokes local shell-current computation)**.** *Let* $I = [s, t]$ *be a finite stopped interval on an ordinary same-origin Navier–Stokes branch, and let* Phi *be its public normalized target-shell observable, with* $Phi(u_s) = 0$ *and* $Phi(u_t) = 1$*. In the unprojected public weak equation*

$$\partial_t u - \nu\Delta u + (u \cdot \nabla)u + \nabla p = 0, \qquad \nabla \cdot u = 0,$$

the exact target-current computation is

$$1 = Q_I^{\mathsf{Geom}} + Q_I^{\mathsf{Caus}} + Q_I^{\mathsf{Abs}} + Q_I^{\mathsf{Lift}} + Q_I^{\mathsf{Bdry}}. \tag{571}$$

where

$$Q_I^{\mathsf{Geom}} := \nu \int_I \langle D\Phi(u_r), \Delta u_r \rangle \, ddr, \tag{572}$$

$$Q_I^{\mathsf{Caus}} := -\int_I \langle D\Phi(u_r), (u_r \cdot \nabla) u_r \rangle \, ddr, \tag{573}$$

$$Q_I^{\mathsf{Abs}} := -\int_I \langle D\Phi(u_r), \nabla p_r \rangle \, ddr, \tag{574}$$

$$Q_I^{\mathsf{Lift}} := \int_I \partial_r^{\mathrm{chart}} \Phi(u_r) \, ddr, \tag{575}$$

$$Q_I^{\mathsf{Bdry}} := \int_I \mathrm{Flux}_{\partial\Omega,\Phi}(u_r, p_r) \, ddr. \tag{576}$$

Here lift and boundary rows are zero when the selected shell has no moving chart and no boundary/exterior interface. Its physical price is $p_I := mu_{\mathrm{NS}}^{\mathrm{phys}}(I)$, the restriction of the public local-energy carrier. For pairwise disjoint stopped shell intervals $I_1, \ldots, I_N$, all five currents and all thirty-one exact-support currents are accumulated by unnormalised sums.

Proof. Pair the weak equation with $D\Phi(u_r)$, apply the chain rule on the finite stopped interval, and retain pressure before constraint descent. The moving-chart derivative is the lift correction; integration by parts on the declared domain gives the oriented boundary flux. This proves the identity. The local-energy identity defines the sole physical carrier. Additivity on disjoint intervals proves the final assertion. Every mixed term is joined before source projection and is evaluated by its exact-support local theorem. □

Proposition III.11.4 (Navier–Stokes geometric target relation)**.** *For every finite stopped shell interval I, the geometric target current is exactly Q_I^{Geom} in* (572)*. Its target detector is*

$$\delta_{\mathsf{Geom}}(I) = \left\| (G_\Sigma^{\mathsf{Geom}}, \mathrm{Ker}_\Sigma^{\mathsf{Geom}}, \mathrm{Gc}_\Sigma^{\mathsf{Geom}}, \mathrm{Inc}_{\mathsf{Geom}}^\Sigma(I)) \right\|_{\det}.$$

The positive physical production entering its record is the viscosity term in the public local-energy carrier; no target alignment is inferred from its unsigned size before the displayed incidence coordinate is nonzero.

Proof. The weak Laplacian is paired with $D\Phi$ in (572). The source compiler assigns this symmetric mobility term to Geom, and the local-energy identity assigns its nonnegative viscous production to the physical carrier. The detector is the geometric packet together with the complete-frontier target incidence, which is precisely the finite local Geom-check. □

Proposition III.11.5 (Navier–Stokes directed target relation)**.** *For every finite stopped shell interval I, the directed target current is exactly Q_I^{Caus} in* (573)*, with detector*

$$\delta_{\mathsf{Caus}}(I) = \left\| (H^{\mathsf{Caus}}, S^{\mathsf{Caus}}, \Psi^{\mathsf{Caus}}, \mathrm{Gc}^{\mathsf{Caus}}, \mathrm{Inc}_{\mathsf{Caus}}^\Sigma(I)) \right\|_{\det}.$$

Its physical effect is evaluated only after signed gluing with the other currents in the same complete local record.

Proof. Substitution of the convective term from the public weak equation gives (573). It is the fixed nonsymmetric transport clause of the source compiler, hence has Caus head. Since transport is signed, the target incidence is retained before the physical positive part is formed. This is exactly the stated detector and gluing rule. □

Proposition III.11.6 (Navier–Stokes constraint target relation)**.** *For every finite stopped shell interval I, the constraint/pressure target current is exactly Q_I^{Abs} in* (574)*, with detector*

$$\delta_{\mathsf{Abs}}(I) = \left\|(A^{\mathrm{desc}}, A^{\mathrm{vert}}, A^{\mathrm{coker}}, A^{\mathrm{gc}}, A^{\mathrm{tgt}}, \mathrm{Inc}^{\Sigma}_{\mathsf{Abs}}(I))\right\|_{\mathrm{det}}.$$

It is target-null exactly when this complete quotient/constraint incidence vanishes; pressure may not be discarded merely because a solenoidal test would annihilate it.

Proof. The pressure term is the weak constraint descent of the public equation and therefore has Abs head. Pairing it with the actual shell covector gives (574). The five displayed quotient coordinates and the joint incidence are exactly its public finite record, proving the claim. □

Proposition III.11.7 (Navier–Stokes lift target relation)**.** *For every finite stopped shell interval I, chart, centre, scale, and normalization motion contributes exactly Q_I^{Lift} in* (575)*, with detector*

$$\delta_{\mathsf{Lift}}(I) = \left\|(L^{\mathrm{int}}, L^{\mathrm{fib}}, L^{\mathrm{conc}}, L^{\mathrm{osc}}, L^{\mathrm{sc}}, L^{\mathrm{gc}}, \mathrm{Inc}^{\Sigma}_{\mathsf{Lift}}(I))\right\|_{\mathrm{det}}.$$

When the public shell has a fixed chart this row is identically zero; a moving normalization is charged only through its actual target incidence.

Proof. Differentiate the public shell observable in its declared moving chart. The additional chain-rule term is (575); it is exactly the lift constructor. The stated detector is its complete finite lift packet, and the fixed-chart specialization gives zero directly. □

Proposition III.11.8 (Navier–Stokes boundary target relation)**.** *For every finite stopped shell interval I, boundary, exterior, cutoff, and terminal-flux contributions are exactly Q_I^{Bdry} in* (576)*, with detector*

$$\delta_{\mathsf{Bdry}}(I) = \left\|(B^{\mathrm{ext}}, B^{\mathrm{tail}}, B^{\mathrm{term}}, B^{\mathrm{gc}}, \mathrm{Inc}^{\Sigma}_{\mathsf{Bdry}}(I))\right\|_{\mathrm{det}}.$$

The physical boundary work is the boundary/exterior component of the same public local-energy carrier, after cancellation of internal faces.

Proof. Integration by parts in the public local-energy identity gives the oriented flux in (576). The trace/exterior constructor assigns its Bdry head, and the displayed boundary packet records every possible target-visible terminal or tail component. Internal-face cancellation occurs before the carrier is restricted, proving the asserted accounting relation. □

Corollary III.11.9 (Navier–Stokes finite-cascade quantitative ledger)**.** *For every finite same-origin shell cascade $mathcalF_N = (I_1, \dots, I_N)$, the public presentation computes*

$$N = \sum_{\varnothing \neq A \subseteq \mathfrak{C}} \sum_{j=1}^{N} Q_{A,I_j}, \qquad \mu_{\mathrm{NS}}^{\mathrm{phys}}\Big(\bigcup_{j=1}^{N} I_j\Big) = \sum_{\varnothing \neq A \subseteq \mathfrak{C}} \mathsf{P}_A(\mathcal{F}_{A,N}). \tag{577}$$

Each $mathsfA_A(\mathcal{F}_{A,N})$, $mathsfD_A(\mathcal{F}_{A,N})$, and $mathsfP_A(\mathcal{F}_{A,N})$ is the explicit local quantity in the theorem labelled by A. No combination of subthreshold currents is averaged away or left outside the physical ledger.

Proof. Sum the local shell identity over the finite disjoint family, then partition each expanded current by exact provenance support. Finite additivity of the single positive physical carrier gives the second identity. The thirty-one local theorems retain the complete joint frontier of every summand. □

Corollary III.11.10 (End-to-end Navier–Stokes contact audit)**.** *An ordinary Navier–Stokes branch reaches Σ_{NS} only if its public presentation generates a finite same-origin target-aligned occurrence in at least one of the thirty-one cells of Theorem III.11.2. Conversely, an ordinary same-origin record satisfying the continuation-graph contact rows is a target-reaching branch. In either direction, the contribution is evaluated solely by its complete joint frontier and the single local-energy physical carrier; no whole-solution estimate, compactness premise, or external regularity certificate enters the audit.*

Proof. For contact, the normalized shell identity supplies unit pre-contact progress. The no-additional-source theorem and the preceding compilation place a positive contribution in a unique exact-support cell; the cellwise theorem returns its finite source-owned occurrence and carrier valuation. Conversely, the ordinary realization and noncontinuation rows are precisely the public continuation-graph conditions defining Σ_{NS}. This is the general target-contact equivalence restricted to the displayed public presentation. $\square$

III.12 Framework conclusion

The continuous extension is the target-aligned structural calculus generated by a public continuum presentation. The presentation supplies the equation, initial and boundary data, continuation relation, physical carrier, target shells, and permitted transformations. Its finite expression graph is compiled into the five source families $\mathsf{Geom}, \mathsf{Caus}, \mathsf{Abs}, \mathsf{Lift}, \mathsf{Bdry}$. The compiler is closed under signed composition, quotient, chart, boundary, and source-preserving limit constructors, so every dynamically generated concentration, oscillation, defect, tail, and graph failure retains inherited ancestry.

For a physical stratum, the framework evaluates only the closure of finite same-origin pre-contact records generated by that presentation. Each record carries the complete backward frontier, unnormalized shell cocycle, target covector, source word, and physical carrier. The joint frontier is formed before source projection. Orthogonal conditional increments therefore retain target alignment created by composition, repeated subthreshold events, or scale and time accumulation. A positive invariant has a least finite pre-contact occurrence with complete five-source ancestry; zero deletes its target-incidence edge. No analytic object outside this reachable algebra is an additional source.

The physical carrier is the single budget. Shell contact gives a unit pre-contact crossing, hence a target-aligned source occurrence. The exact compositional accounting law charges its unnormalized cumulative current to the carrier or routes a zero-price equality record through the same source-preserving closure. Cofinal contact cannot be erased by normalization: it produces an unbounded carrier density or a target-current atom of zero carrier mass, each with a fixed source owner and a finite contextual occurrence.

Accordingly, target avoidance on a physical stratum is the emptiness of the ordinary same-origin target-contact cells in the presentation-generated five-source closure; target reachability is their nonemptiness. This is the continuous counterpart of Part I saturation: target-aligned structure is classified, jointly retained, physically charged, and evaluated entirely inside the closed algebra generated by the public presentation.

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